

cardona-qft-methods: formula evidence

The page, the confidence and the picture of an inline formula are its HOST LINE’s — a formula has none of its own. A line’s confidence is not a formula’s.

Inline formulas (first occurrence)

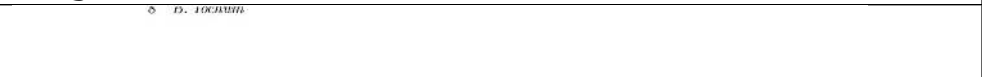
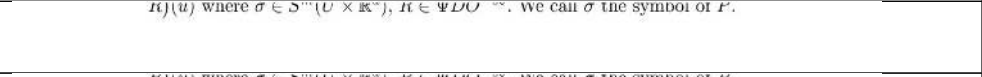
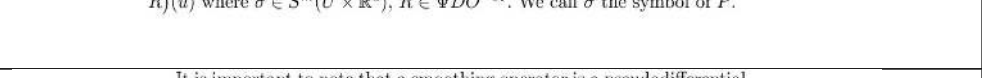
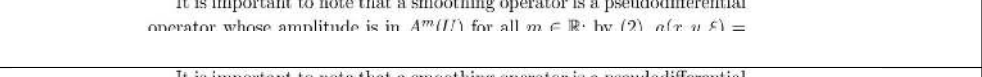
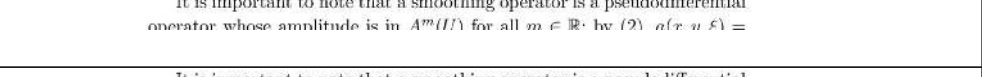
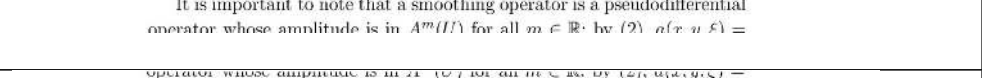
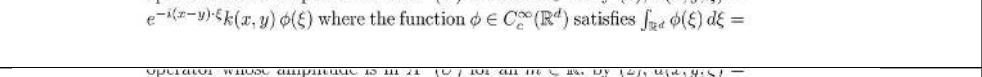
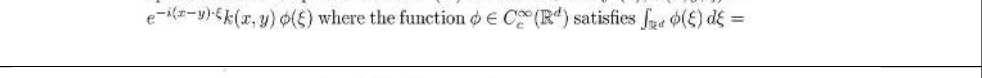
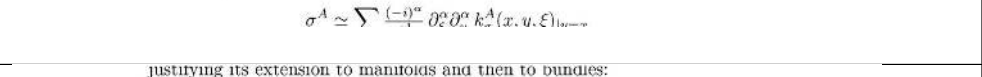
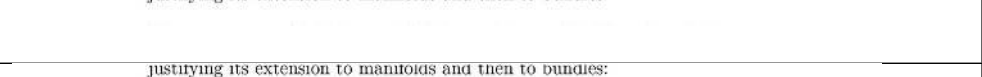
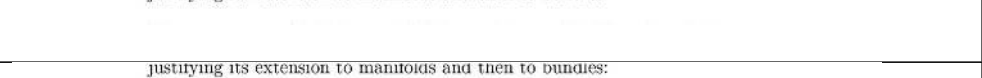
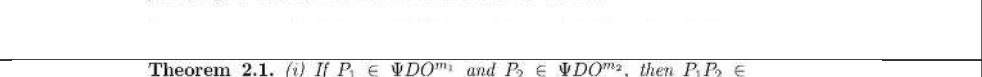
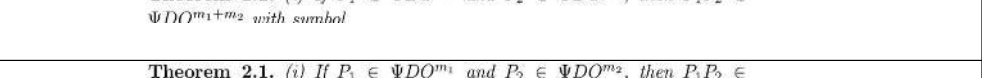
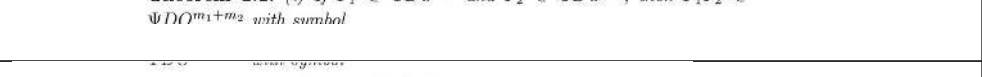
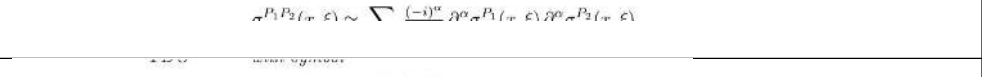
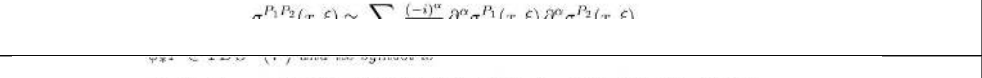
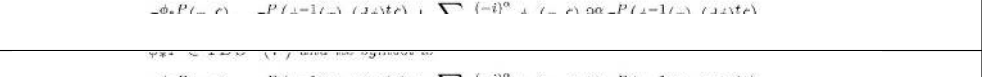
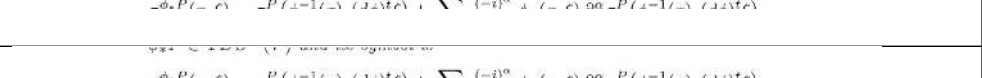
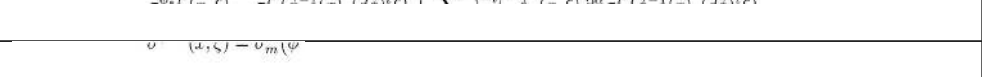
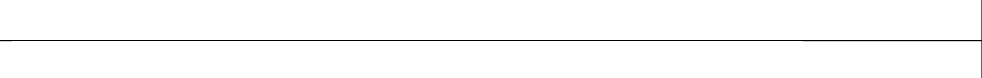
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|----------------------------|------|-------|--------------------------------------------------------------------------------|--------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| cardona-qft-methods_F00001 | 8    | 1.000 | A d S / C F T                                                                  | $AdS/CFT$                                                                      |                                                                                                                                                                                                                   |
| cardona-qft-methods_F00002 | 8    | 1.000 | $\mathcal{N}=4$                                                                | $\mathcal{N} = 4$                                                              | Yang Baxter equations presented as an algebraic integrability condition by                                                                                                                                        |
| cardona-qft-methods_F00003 | 10   | 1.000 | G                                                                              | $G$                                                                            | Feynman Periods                                                                                                                                                                                                   |
| cardona-qft-methods_F00004 | 14   | 1.000 | A                                                                              | $A$                                                                            |                                                                                                                                                                                                                   |
| cardona-qft-methods_F00005 | 14   | 0.992 | $A=\int_{a\in\mathrm{Sp}(a)}\mathrm{ad}P_a$                                    | $A=\int_{a\in\mathrm{Sp}(a)}adP_a$                                             |                                                                                                                                                                                                                   |
| cardona-qft-methods_F00006 | 14   | 0.992 | $\mathrm{Sp}(A)$                                                               | $\mathrm{Sp}(A)$                                                               |                                                                                                                                                                                                                   |
| cardona-qft-methods_F00007 | 15   | 0.523 | $\psi(\vec{x})$                                                                | $\psi(\vec{x})$                                                                | especially in the Heisenberg picture where the tools are the observables                                                                                                                                          |
| cardona-qft-methods_F00008 | 15   | 0.523 | $\vec{x}\in\mathbb{R}^4$                                                       | $\vec{x}\in\mathbb{R}^4$                                                       | especially in the Heisenberg picture where the tools are the observables                                                                                                                                          |
| cardona-qft-methods_F00009 | 15   | 0.523 | E                                                                              | $E$                                                                            | especially in the Heisenberg picture where the tools are the observables                                                                                                                                          |
| cardona-qft-methods_F00010 | 15   | 0.523 | $B\in$                                                                         | $B\in$                                                                         | especially in the Heisenberg picture where the tools are the observables                                                                                                                                          |
| cardona-qft-methods_F00011 | 15   | 0.803 | $\mathrm{Function}\left(M=\mathbb{R}^4,\mathbb{R}^3\right)$                    | $\mathrm{Function}\left(M=\mathbb{R}^4,\mathbb{R}^3\right)$                    | namey are senuajoint operators.<br>But this notion is also useful in geometry. In special relativity, we con-                                                                                                     |
| cardona-qft-methods_F00012 | 15   | 1.000 | $x^\mu$                                                                        | $x^\mu$                                                                        | function( $M=\mathbb{R}^4,\mathbb{R}^3$ ). Einstein introduced in 1915 the gravitational field and the equation of motion of matter. But a problem appeared: what is                                              |
| cardona-qft-methods_F00013 | 15   | 1.000 | $g_{\mu\nu}(x)$                                                                | $g_{\mu\nu}(x)$                                                                | the physical meaning of coordinates $x^\mu$ and equations of fields? Assume                                                                                                                                       |
| cardona-qft-methods_F00014 | 15   | 1.000 | $e_{\mu}^I(x)$                                                                 | $e_{\mu}^I(x)$                                                                 | the physical meaning of coordinates $x^\mu$ and equations of fields? Assume                                                                                                                                       |
| cardona-qft-methods_F00015 | 15   | 1.000 | I                                                                              | $I$                                                                            | the general covariance of field equations. If $g_{\mu\nu}(x)$ or the tetradfield $e_{\mu}^I(x)$ is a solution (where $I$ is a local inertial reference frame) then for any dif.                                   |
| cardona-qft-methods_F00016 | 15   | 1.000 | $\phi$                                                                         | $\phi$                                                                         | feomorphism $\phi$ of $M$ which is active or passive (i.e. change of coordinates),                                                                                                                                |
| cardona-qft-methods_F00017 | 15   | 1.000 | M                                                                              | $M$                                                                            | feomorphism $\phi$ of $M$ which is active or passive (i.e. change of coordinates),                                                                                                                                |
| cardona-qft-methods_F00018 | 15   | 1.000 | $e_{\nu}^{\prime I}(x)=\frac{\partial x^\mu}{\partial\phi(x)^\nu}e_{\mu}^I(x)$ | $e_{\nu}^{\prime I}(x)=\frac{\partial x^\mu}{\partial\phi(x)^\nu}e_{\mu}^I(x)$ | $e_{\nu}^{\prime I}(x)=\frac{\partial x^\mu}{\partial\phi(x)^\nu}e_{\mu}^I(x)$ is also a solution. As a consequence, when relativity became general, the points disappeared and only fields over fields remained. |
| cardona-qft-methods_F00019 | 15   | 1.000 | $\left(\square+m^2\right)\psi(\vec{x})=s_b(\vec{x})$                           | $(\square+m^2)\psi(\vec{x})=s_b(\vec{x})$                                      | is based on a spectral triple $(\mathcal{A},\mathcal{H},\mathcal{L})$ where the $\ast$ -algebra $\mathcal{A}$ generalizes smooth functions on space-time $M$ (or the coordinates) with pointwise                  |
| cardona-qft-methods_F00020 | 15   | 1.000 | $(i\not\partial-$                                                              | $(i\not\partial-$                                                              | is based on a spectral triple $(\mathcal{A},\mathcal{H},\mathcal{L})$ where the $\ast$ -algebra $\mathcal{A}$ generalizes smooth functions on space-time $M$ (or the coordinates) with pointwise                  |
| cardona-qft-methods_F00021 | 15   | 1.000 | $m)\psi(\vec{x})=s_f(\vec{x})$                                                 | $m)\psi(\vec{x})=s_f(\vec{x})$                                                 | product, $\mathcal{H}$ generalizes the Hilbert space of above quoted spinors $\psi$ and                                                                                                                           |
| cardona-qft-methods_F00022 | 15   | 1.000 | $s_b, s_f$                                                                     | $s_b, s_f$                                                                     | product, $\mathcal{H}$ generalizes the Hilbert space of above quoted spinors $\psi$ and                                                                                                                           |

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| cardona-qft-methods_F00023 | 15   | ■ 1.000 | <code>\not \partial</code>                                                          | $\not{\partial}$                                                         | product, $\mathcal{H}$ generalizes the Hilbert space of above quoted spinors $\psi$ and                                                                                                                                             |
| cardona-qft-methods_F00024 | 15   | ■ 0.999 | <code>{ }^{24,30}</code>                                                            | 24,30                                                                    | for the phase space since it gives an uncertainty principle: above a certain scale, phase-space points are indistinguishable. Below the scale $\Lambda^{-1}$ , a cer-                                                               |
| cardona-qft-methods_F00025 | 15   | ■ 0.945 | <code>(\mathcal{A}, \mathcal{H}, \mathcal{D})</code>                                | $(\mathcal{A}, \mathcal{H}, \mathcal{D})$                                |                                                                                                                                                                                                                                     |
| cardona-qft-methods_F00026 | 15   | ■ 0.945 | <code>*</code>                                                                      | $*$                                                                      |                                                                                                                                                                                                                                     |
| cardona-qft-methods_F00027 | 15   | ■ 0.945 | <code>\mathcal{A}</code>                                                            | $\mathcal{A}$                                                            |                                                                                                                                                                                                                                     |
| cardona-qft-methods_F00028 | 15   | ■ 1.000 | <code>\mathcal{H}</code>                                                            | $\mathcal{H}$                                                            |                                                                                                                                                                                                                                     |
| cardona-qft-methods_F00029 | 15   | ■ 1.000 | <code>\psi</code>                                                                   | $\psi$                                                                   |                                                                                                                                                                                                                                     |
| cardona-qft-methods_F00030 | 15   | ■ 1.000 | <code>\mathcal{D}</code>                                                            | $\mathcal{D}$                                                            |                                                                                                                                                                                                                                     |
| cardona-qft-methods_F00031 | 15   | ■ 1.000 | <code>\Lambda^{-1}</code>                                                           | $\Lambda^{-1}$                                                           |                                                                                                                                                                                                                                     |
| cardona-qft-methods_F00032 | 16   | ■ 1.000 | <code>S(\mathcal{D}, \Lambda, f)=\operatorname{Tr}(f(\mathcal{D} / \Lambda))</code> | $S(\mathcal{D}, \Lambda, f) = \operatorname{Tr}(f(\mathcal{D}/\Lambda))$ | plays an essential role in physics, for instance, the Einstein-Hilbert action in gravity or the Yang-Mills-Higgs action in particle physics. So here,                                                                               |
| cardona-qft-methods_F00033 | 16   | ■ 1.000 | <code>\Lambda \in \mathbb{R}^{+}</code>                                             | $\Lambda \in \mathbb{R}^{+}$                                             | given the data $(\mathcal{A}, \mathcal{H}, \mathcal{D})$ , the appropriate notion of action was introduced                                                                                                                          |
| cardona-qft-methods_F00034 | 16   | ■ 1.000 | <code>f</code>                                                                      | $f$                                                                      | given the data $(\mathcal{A}, \mathcal{H}, \mathcal{D})$ , the appropriate notion of action was introduced                                                                                                                          |
| cardona-qft-methods_F00035 | 16   | ■ 1.000 | <code>\Lambda \rightarrow \infty</code>                                             | $\Lambda \rightarrow \infty$                                             | by Chamseddine and Connes <sup>11</sup> and defined as $S(\mathcal{D}, \Lambda, f) = \operatorname{Tr}(f(\mathcal{D}/\Lambda))$ where $\Lambda \in \mathbb{R}^{+}$ plays the role of a cut-off and $f$ is a positive even function. |
| cardona-qft-methods_F00036 | 16   | ■ 1.000 | <code>M \times F</code>                                                             | $M \times F$                                                             | The asymptotic series in $\Lambda \rightarrow \infty$ yields to an effective theory. For instance,                                                                                                                                  |
| cardona-qft-methods_F00037 | 16   | ■ 1.000 | <code>F</code>                                                                      | $F$                                                                      | this action applied to a noncommutative model of space-time $M \times F$ with a fine structure for fermions encoded in a finite geometry $F$ gives rise from                                                                        |
| cardona-qft-methods_F00038 | 16   | ■ 1.000 | <code>{ }^{11,20,30}</code>                                                         | 11,20,30                                                                 | where we recover the standard action of pure gravity. <sup>11,20,30</sup> From pure gravity to the standard model coupled with gravity. <sup>11,20,30</sup>                                                                         |
| cardona-qft-methods_F00039 | 16   | ■ 0.997 | <code>\Psi</code>                                                                   | $\Psi$                                                                   | portant example of elliptic operators. Then follows the notion of Wodzicki                                                                                                                                                          |
| cardona-qft-methods_F00040 | 16   | ■ 1.000 | <code>\mathcal{L}^{1, \infty}(\mathcal{H})</code>                                   | $\mathcal{L}^{1, \infty}(\mathcal{H})$                                   | this was generalized to a class of compactness of order $\infty$ which eventually turns out to be an “asset” because it is precisely the Dixmier trace and                                                                          |
| cardona-qft-methods_F00041 | 16   | ■ 1.000 | <code>\sum_{n=1}^{\infty} n^{-1}</code>                                             | $\sum_{n=1}^{\infty} n^{-1}$                                             | zero function. Thus, they control the spectral aspects of a manifold and secondly they can be generalized to any spectral triple.                                                                                                   |
| cardona-qft-methods_F00042 | 16   | ■ 0.987 | <code>(M, g)</code>                                                                 | $(M, g)$                                                                 |                                                                                                                                                                                                                                     |
| cardona-qft-methods_F00043 | 16   | ■ 1.000 | <code>E=\mathcal{C} \ell^* M</code>                                                 | $E = \mathcal{C}\ell^* M$                                                |                                                                                                                                                                                                                                     |
| cardona-qft-methods_F00044 | 16   | ■ 1.000 | <code>\mathcal{C} \ell T_x^* M</code>                                               | $\mathcal{C}\ell T_x^* M$                                                |                                                                                                                                                                                                                                     |
| cardona-qft-methods_F00045 | 16   | ■ 1.000 | <code>x \in M</code>                                                                | $x \in M$                                                                |                                                                                                                                                                                                                                     |
| cardona-qft-methods_F00046 | 16   | ■ 0.999 | <code>\nabla^S</code>                                                               | $\nabla^S$                                                               |                                                                                                                                                                                                                                     |
| cardona-qft-methods_F00047 | 16   | ■ 0.999 | <code>\not D=-i c \circ \nabla^S</code>                                             | $\not{D} = -ic \circ \nabla^S$                                           |                                                                                                                                                                                                                                     |
| cardona-qft-methods_F00048 | 16   | ■ 0.999 | <code>c</code>                                                                      | $c$                                                                      |                                                                                                                                                                                                                                     |
| cardona-qft-methods_F00049 | 16   | ■ 1.000 | <code>g</code>                                                                      | $g$                                                                      |                                                                                                                                                                                                                                     |



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| cardona-qft-methods_F00075 | 17   | ■ 1.000 | $\xi \in \mathbb{R}^d,  \xi  = \left( \sum_{k=1}^d  \xi_k ^2 \right)^{1/2}$                                                    |          | on the set $\Psi DO(M)$ of pseudodifferential operators on a compact manifold $M$ of dimension $d$ .                                                                                                                                                                 |
| cardona-qft-methods_F00076 | 17   | ■ 1.000 | $\mathcal{B}(\mathcal{H}), \mathcal{K}(\mathcal{H}), \mathcal{L}^p(\mathcal{H})$                                               |          | References for this section: Classical books are <sup>99,102</sup> For an orientation                                                                                                                                                                                |
| cardona-qft-methods_F00077 | 17   | ■ 1.000 | $p$                                                                                                                            |          | more in the spirit of noncommutative geometry here we follow <sup>3,33</sup> based on <sup>3,33</sup> see also the excellent books <sup>49,82,83,104,105</sup>                                                                                                       |
| cardona-qft-methods_F00078 | 17   | ■ 1.000 | $\mathcal{L}^1(\mathcal{H})$                                                                                                   |          | more in the spirit of noncommutative geometry here we follow <sup>3,33</sup> based on <sup>3,33</sup> see also the excellent books <sup>49,82,83,104,105</sup>                                                                                                       |
| cardona-qft-methods_F00079 | 17   | ■ 0.998 | $\Psi DO(M)$                                                                                                                   |          |                                                                                                                                                                                                                                                                      |
| cardona-qft-methods_F00080 | 17   | ■ 1.000 | $\{ \}^{99,102}$                                                                                                               |          |                                                                                                                                                                                                                                                                      |
| cardona-qft-methods_F00081 | 17   | ■ 1.000 | $\{ \}^{86,87}$                                                                                                                |          |                                                                                                                                                                                                                                                                      |
| cardona-qft-methods_F00082 | 17   | ■ 1.000 | $\{ \}^{3,33}$                                                                                                                 |          |                                                                                                                                                                                                                                                                      |
| cardona-qft-methods_F00083 | 17   | ■ 1.000 | $\{ \}^{49,82,83,104,105}$                                                                                                     |          |                                                                                                                                                                                                                                                                      |
| cardona-qft-methods_F00084 | 18   | ■ 1.000 | $m \in \mathbb{C}$                                                                                                             |          |                                                                                                                                                                                                                                                                      |
| cardona-qft-methods_F00085 | 18   | ■ 1.000 | $\sigma(x, \xi)$                                                                                                               |          | 2.1. A quick overview on pseudodifferential operators                                                                                                                                                                                                                |
| cardona-qft-methods_F00086 | 18   | ■ 1.000 | $m$                                                                                                                            |          | 2.1. A quick overview on pseudodifferential operators                                                                                                                                                                                                                |
| cardona-qft-methods_F00087 | 18   | ■ 1.000 | $C^\infty$                                                                                                                     |          | 2.1. A quick overview on pseudodifferential operators                                                                                                                                                                                                                |
| cardona-qft-methods_F00088 | 18   | ■ 1.000 | $(x, \xi) \in U \times \mathbb{R}^d \rightarrow \mathbb{C}$                                                                    |          | 2.1. A quick overview on pseudodifferential operators                                                                                                                                                                                                                |
| cardona-qft-methods_F00089 | 18   | ■ 0.915 | $ \partial_x^\alpha \partial_\xi^\beta \sigma(x, \xi)  \leq C_{\alpha\beta}(x)(1 +  \xi )^{\Re(m) -  \beta }, C_{\alpha\beta}$ |          | DEFINITION 2.1. In the following, $m \in \mathbb{C}$ .<br>A symbol $\sigma(x, \xi)$ of order $m$ is a $C^\infty$ function: $(x, \xi) \in U \times \mathbb{R}^d \rightarrow \mathbb{C}$                                                                               |
| cardona-qft-methods_F00090 | 18   | ■ 0.915 | $U$                                                                                                                            |          | DEFINITION 2.1. In the following, $m \in \mathbb{C}$ .<br>A symbol $\sigma(x, \xi)$ of order $m$ is a $C^\infty$ function: $(x, \xi) \in U \times \mathbb{R}^d \rightarrow \mathbb{C}$                                                                               |
| cardona-qft-methods_F00091 | 18   | ■ 1.000 | $\sigma(x, \xi) \simeq \sum_{j \geq 0} \sigma_{m-j}(x, \xi)$                                                                   |          | satisfying<br>(i) $ \partial_x^\alpha \partial_\xi^\beta \sigma(x, \xi)  < C_{\alpha\beta}(x)(1 +  \xi )^{\Re(m) -  \beta }$ , $C_{\alpha\beta}$ bounded on $U$ .                                                                                                    |
| cardona-qft-methods_F00092 | 18   | ■ 1.000 | $\sigma_k$                                                                                                                     |          | satisfying<br>(i) $ \partial_x^\alpha \partial_\xi^\beta \sigma(x, \xi)  < C_{\alpha\beta}(x)(1 +  \xi )^{\Re(m) -  \beta }$ , $C_{\alpha\beta}$ bounded on $U$ .                                                                                                    |
| cardona-qft-methods_F00093 | 18   | ■ 1.000 | $k$                                                                                                                            |          | (i) $ \partial_x^\alpha \partial_\xi^\beta \sigma(x, \xi)  < C_{\alpha\beta}(x)(1 +  \xi )^{\Re(m) -  \beta }$ , $C_{\alpha\beta}$ bounded on $U$ .<br>(ii) We suppose that $\sigma(x, \xi) \simeq \sum_{j \geq 0} \sigma_{m-j}(x, \xi)$ where $\sigma_k$ is homoge- |
| cardona-qft-methods_F00094 | 18   | ■ 1.000 | $\xi$                                                                                                                          |          | (i) $ \partial_x^\alpha \partial_\xi^\beta \sigma(x, \xi)  < C_{\alpha\beta}(x)(1 +  \xi )^{\Re(m) -  \beta }$ , $C_{\alpha\beta}$ bounded on $U$ .<br>(ii) We suppose that $\sigma(x, \xi) \simeq \sum_{j \geq 0} \sigma_{m-j}(x, \xi)$ where $\sigma_k$ is homoge- |
| cardona-qft-methods_F00095 | 18   | ■ 1.000 | $\simeq$                                                                                                                       |          | (i) $ \partial_x^\alpha \partial_\xi^\beta \sigma(x, \xi)  < C_{\alpha\beta}(x)(1 +  \xi )^{\Re(m) -  \beta }$ , $C_{\alpha\beta}$ bounded on $U$ .<br>(ii) We suppose that $\sigma(x, \xi) \simeq \sum_{j \geq 0} \sigma_{m-j}(x, \xi)$ where $\sigma_k$ is homoge- |
| cardona-qft-methods_F00096 | 18   | ■ 1.000 | $ \xi  \geq 1$                                                                                                                 |          | for $ \xi  \geq 1$ , with $C_{N\alpha\beta}$ bounded on $U$ . The set of symbols of order $m$ is denoted by $\mathcal{S}^m(U \times \mathbb{R}^d)$                                                                                                                   |
| cardona-qft-methods_F00097 | 18   | ■ 1.000 | $C_{N\alpha\beta}$                                                                                                             |          | for $ \xi  \geq 1$ , with $C_{N\alpha\beta}$ bounded on $U$ . The set of symbols of order $m$ is denoted by $\mathcal{S}^m(U \times \mathbb{R}^d)$                                                                                                                   |
| cardona-qft-methods_F00098 | 18   | ■ 1.000 | $S^m(U \times \mathbb{R}^d)$                                                                                                   |          | denoted by $\mathcal{S}^m(U \times \mathbb{R}^d)$ .<br>A function $a \in \ell$                                                                                                                                                                                       |
| cardona-qft-methods_F00099 | 18   | ■ 1.000 | $a \in C^\infty(U \times U \times \mathbb{R}^d)$                                                                               |          | compact $K \subset U$ and any $\alpha, \beta, \gamma \in \mathbb{N}^d$ there exists a constant $C_{K\alpha\beta\gamma}$ such                                                                                                                                         |

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| cardona-qft-methods_F00100 | 18   | 1.000 | $K \subset U$                                                       | $K \subset U$                                                       |       |
| cardona-qft-methods_F00101 | 18   | 1.000 | $\alpha, \beta, \gamma \in \mathbb{N}^d$                            | $\alpha, \beta, \gamma \in \mathbb{N}^d$                            |       |
| cardona-qft-methods_F00102 | 18   | 1.000 | $C_{K \alpha \beta \gamma}$                                         | $C_{K \alpha \beta \gamma}$                                         |       |
| cardona-qft-methods_F00103 | 18   | 1.000 | $A^m(U)$                                                            | $A^m(U)$                                                            |       |
| cardona-qft-methods_F00104 | 18   | 1.000 | $\sigma \in S^m \left( U \times \mathbb{R}^d \right)$               | $\sigma \in S^m \left( U \times \mathbb{R}^d \right)$               |       |
| cardona-qft-methods_F00105 | 18   | 1.000 | $\sigma(\cdot, D): u \in C_c^\infty(U) \rightarrow$                 | $\sigma(\cdot, D): u \in C_c^\infty(U) \rightarrow$                 |       |
| cardona-qft-methods_F00106 | 18   | 1.000 | $C^\infty(U)$                                                       | $C^\infty(U)$                                                       |       |
| cardona-qft-methods_F00107 | 18   | 0.994 | $\sigma(\cdot, D)$                                                  | $\sigma(\cdot, D)$                                                  |       |
| cardona-qft-methods_F00108 | 18   | 0.848 | $Op(\sigma)$                                                        | $Op(\sigma)$                                                        |       |
| cardona-qft-methods_F00109 | 18   | 0.848 | $\sigma(x, \xi) = \sum_{\alpha} a_{\alpha}(x) \xi^{\alpha}$         | $\sigma(x, \xi) = \sum_{\alpha} a_{\alpha}(x) \xi^{\alpha}$         |       |
| cardona-qft-methods_F00110 | 18   | 0.848 | $\sigma(x, D) =$                                                    | $\sigma(x, D) =$                                                    |       |
| cardona-qft-methods_F00111 | 18   | 1.000 | $\sum_{\alpha} a_{\alpha}(x) D_x^{\alpha}$                          | $\sum_{\alpha} a_{\alpha}(x) D_x^{\alpha}$                          |       |
| cardona-qft-methods_F00112 | 18   | 1.000 | $D_x = -i \partial_x$                                               | $D_x = -i \partial_x$                                               |       |
| cardona-qft-methods_F00113 | 18   | 1.000 | $\mathcal{D}'_c(U)$                                                 | $\mathcal{D}'_c(U)$                                                 |       |
| cardona-qft-methods_F00114 | 18   | 0.999 | $\mathcal{D}'(U)$                                                   | $\mathcal{D}'(U)$                                                   |       |
| cardona-qft-methods_F00115 | 18   | 1.000 | $ \alpha  = m$                                                      | $ \alpha  = m$                                                      |       |
| cardona-qft-methods_F00116 | 18   | 1.000 | $\sigma(x, D)$                                                      | $\sigma(x, D)$                                                      |       |
| cardona-qft-methods_F00117 | 18   | 0.907 | $Op(a)$                                                             | $Op(a)$                                                             |       |
| cardona-qft-methods_F00118 | 18   | 0.907 | $a$                                                                 | $a$                                                                 |       |
| cardona-qft-methods_F00119 | 18   | 0.997 | $P: C_c^\infty(U) \rightarrow C^\infty(U)$                          | $P: C_c^\infty(U) \rightarrow C^\infty(U)$                          |       |
| cardona-qft-methods_F00120 | 18   | 1.000 | $C^\infty(U \times U)$                                              | $C^\infty(U \times U)$                                              |       |
| cardona-qft-methods_F00121 | 18   | 1.000 | $\Psi DO^{-\infty}(U)$                                              | $\Psi DO^{-\infty}(U)$                                              |       |
| cardona-qft-methods_F00122 | 19   | 1.000 | $\Psi DO^m(U)$                                                      | $\Psi DO^m(U)$                                                      |       |
| cardona-qft-methods_F00123 | 19   | 1.000 | $P$                                                                 | $P$                                                                 |       |
| cardona-qft-methods_F00124 | 19   | 1.000 | $P: C_c^\infty(U) \rightarrow C^\infty(U), Pu(x) = (\sigma(x, D) +$ | $P: C_c^\infty(U) \rightarrow C^\infty(U), Pu(x) = (\sigma(x, D) +$ |       |
| cardona-qft-methods_F00125 | 19   | 0.914 | $R)(u)$                                                             | $R)(u)$                                                             |       |

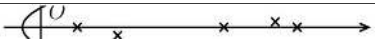
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| cardona-qft-methods_F00126 | 19   | ■ 0.914 | $\backslash\sigma\in S^m\left(U\times\mathbb{R}^d\right), R\in\Psi D^{-\infty}$          | $\sigma\in S^m\left(U\times\mathbb{R}^d\right), R\in\Psi DO^{-\infty}$                     |    |
| cardona-qft-methods_F00127 | 19   | ■ 0.914 | $\backslash\sigma$                                                                       | $\sigma$                                                                                   |    |
| cardona-qft-methods_F00128 | 19   | ■ 0.861 | $m\in\mathbb{R}$                                                                         | $m\in\mathbb{R}$                                                                           |    |
| cardona-qft-methods_F00129 | 19   | ■ 0.861 | $(2), a(x, y, \xi)=$                                                                     | $(2), a(x, y, \xi) =$                                                                      |    |
| cardona-qft-methods_F00130 | 19   | ■ 1.000 | $e^{-i(x-y)\cdot\xi}k(x, y)\phi(\xi)$                                                    | $e^{-i(x-y)\cdot\xi}k(x, y)\phi(\xi)$                                                      |    |
| cardona-qft-methods_F00131 | 19   | ■ 1.000 | $\phi\in C_c^\infty\left(\mathbb{R}^d\right)$                                            | $\phi\in C_c^\infty\left(\mathbb{R}^d\right)$                                              |    |
| cardona-qft-methods_F00132 | 19   | ■ 1.000 | $\int_{\mathbb{R}^d}\phi(\xi)d\xi=$                                                      | $\int_{\mathbb{R}^d}\phi(\xi)d\xi=$                                                        |    |
| cardona-qft-methods_F00133 | 19   | ■ 1.000 | $(2\pi)^d$                                                                               | $(2\pi)^d$                                                                                 |    |
| cardona-qft-methods_F00134 | 19   | ■ 1.000 | $k^{\sigma(x, D)}$                                                                       | $k^{\sigma(x, D)}$                                                                         |    |
| cardona-qft-methods_F00135 | 19   | ■ 1.000 | $\sigma^A(x, \xi)=e^{-ix\cdot\xi}A\left(x\rightarrow e^{ix\cdot\xi}\right)$              | $\sigma^A(x, \xi) = e^{-ix\cdot\xi}A\left(x\rightarrow e^{ix\cdot\xi}\right)$              |    |
| cardona-qft-methods_F00136 | 19   | ■ 1.000 | $k^A(x, y, \xi)$                                                                         | $k^A(x, y, \xi)$                                                                           |    |
| cardona-qft-methods_F00137 | 19   | ■ 1.000 | $k_\sigma^A(x, y)$                                                                       | $k_\sigma^A(x, y)$                                                                         |   |
| cardona-qft-methods_F00138 | 19   | ■ 1.000 | $\sigma^A(x, \xi)=$                                                                      | $\sigma^A(x, \xi) =$                                                                       |  |
| cardona-qft-methods_F00139 | 19   | ■ 1.000 | $e^{iD_\xi D_y}k^A(x, y, \xi) _{y=x}$                                                    | $e^{iD_\xi D_y}k^A(x, y, \xi) _{y=x}$                                                      |  |
| cardona-qft-methods_F00140 | 19   | ■ 1.000 | $e^{iD_\xi D_y}=1+iD_\xi D_y-\frac{1}{2}(D_\xi D_y)^2+\cdots$                            | $e^{iD_\xi D_y}=1+iD_\xi D_y-\frac{1}{2}(D_\xi D_y)^2+\cdots$                              |  |
| cardona-qft-methods_F00141 | 19   | ■ 1.000 | $A=Op\left(\sigma^A\right)+R$                                                            | $A=Op\left(\sigma^A\right)+R$                                                              |  |
| cardona-qft-methods_F00142 | 19   | ■ 1.000 | $R$                                                                                      | $R$                                                                                        |  |
| cardona-qft-methods_F00143 | 19   | ■ 1.000 | $P_1\in\Psi D^{m_1}$                                                                     | $P_1\in\Psi DO^{m_1}$                                                                      |  |
| cardona-qft-methods_F00144 | 19   | ■ 1.000 | $P_2\in\Psi D^{m_2}$                                                                     | $P_2\in\Psi DO^{m_2}$                                                                      |  |
| cardona-qft-methods_F00145 | 19   | ■ 1.000 | $P_1P_2\in$                                                                              | $P_1P_2\in$                                                                                |  |
| cardona-qft-methods_F00146 | 19   | ■ 1.000 | $\Psi D^{m_1+m_2}$                                                                       | $\Psi DO^{m_1+m_2}$                                                                        |  |
| cardona-qft-methods_F00147 | 19   | ■ 1.000 | $P_1P_2$                                                                                 | $P_1P_2$                                                                                   |  |
| cardona-qft-methods_F00148 | 19   | ■ 1.000 | $\sigma_{m_1+m_2}^{P_1P_2}(x, \xi)=\sigma_{m_1}^{P_1}(x, \xi)\sigma_{m_2}^{P_2}(x, \xi)$ | $\sigma_{m_1+m_2}^{P_1P_2}(x, \xi) = \sigma_{m_1}^{P_1}(x, \xi)\sigma_{m_2}^{P_2}(x, \xi)$ |  |
| cardona-qft-methods_F00149 | 19   | ■ 1.000 | $P\in\Psi D^m(U)$                                                                        | $P\in\Psi DO^m(U)$                                                                         |  |
| cardona-qft-methods_F00150 | 19   | ■ 1.000 | $\phi\in\operatorname{Diff}(U, V)$                                                       | $\phi\in\operatorname{Diff}(U, V)$                                                         |  |

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| cardona-qft-methods_F00151 | 19   | ■ 1.000 | $V$                                                                                           | $V$                                                                                           |                                                                                                                                                                                 |
| cardona-qft-methods_F00152 | 19   | ■ 1.000 | $\phi_*P: f \in C^\infty(V) \rightarrow P(f \circ \phi) \circ \phi^{-1}$                      | $\phi_*P: f \in C^\infty(V) \rightarrow P(f \circ \phi) \circ \phi^{-1}$                      |                                                                                                                                                                                 |
| cardona-qft-methods_F00153 | 19   | ■ 1.000 | $\phi_*P \in \Psi DO^m(V)$                                                                    | $\phi_*P \in \Psi DO^m(V)$                                                                    |                                                                                                                                                                                 |
| cardona-qft-methods_F00154 | 20   | ■ 1.000 | $\phi_\alpha$                                                                                 | $\phi_\alpha$                                                                                 |                                                                                                                                                                                 |
| cardona-qft-methods_F00155 | 20   | ■ 1.000 | $\alpha$                                                                                      | $\alpha$                                                                                      |                                                                                                                                                                                 |
| cardona-qft-methods_F00156 | 20   | ■ 1.000 | $\sigma^{\phi_*P}_m = \phi_*\sigma^P_m$                                                       | $\sigma^{\phi_*P}_m = \phi_*\sigma^P_m$                                                       |                                                                                                                                                                                 |
| cardona-qft-methods_F00157 | 20   | ■ 1.000 | $\sigma^P(x, \xi) = e^{-ix \cdot \xi} P(x \rightarrow e^{ix \cdot \xi})$                      | $\sigma^P(x, \xi) = e^{-ix \cdot \xi} P(x \rightarrow e^{ix \cdot \xi})$                      | asymptotic behavior requires some care, see, e.g., [10]. An interesting remark is in order: $\sigma^P(x, \xi) = e^{-ix \cdot \xi} P(x \rightarrow e^{ix \cdot \xi})$ , thus     |
| cardona-qft-methods_F00158 | 20   | ■ 1.000 | $\xi \rightarrow t\xi$                                                                        | $\xi \rightarrow t\xi$                                                                        | the dilation $\xi \rightarrow t\xi$ with $t > 0$ gives $t^{-m} e^{-itx \cdot \xi} P e^{itx \cdot \xi} = t^{-m} \sigma^P(x, t\xi) \simeq$                                        |
| cardona-qft-methods_F00159 | 20   | ■ 1.000 | $t > 0$                                                                                       | $t > 0$                                                                                       | the dilation $\xi \rightarrow t\xi$ with $t > 0$ gives $t^{-m} e^{-itx \cdot \xi} P e^{itx \cdot \xi} = t^{-m} \sigma^P(x, t\xi) \simeq$                                        |
| cardona-qft-methods_F00160 | 20   | ■ 1.000 | $t^{-m} e^{-itx \cdot \xi} P e^{itx \cdot \xi} = t^{-m} \sigma^P(x, t\xi) \simeq$             | $t^{-m} e^{-itx \cdot \xi} P e^{itx \cdot \xi} = t^{-m} \sigma^P(x, t\xi) \simeq$             | the dilation $\xi \rightarrow t\xi$ with $t > 0$ gives $t^{-m} e^{-itx \cdot \xi} P e^{itx \cdot \xi} = t^{-m} \sigma^P(x, t\xi) \simeq$                                        |
| cardona-qft-methods_F00161 | 20   | ■ 1.000 | $t^{-m} \sum_{j \geq 0} \sigma_{m-j}^P(x, t\xi) = \sigma_m^P(x, \xi) + o(t^{-1})$             | $t^{-m} \sum_{j \geq 0} \sigma_{m-j}^P(x, t\xi) = \sigma_m^P(x, \xi) + o(t^{-1})$             | $t^{-m} \sum_{j \geq 0} \sigma_{m-j}^P(x, t\xi) = \sigma_m^P(x, \xi) + o(t^{-1})$ . Thus, if $P \in \Psi DO^m(U)$ with $m \geq 0$ ,                                             |
| cardona-qft-methods_F00162 | 20   | ■ 1.000 | $m \geq 0$                                                                                    | $m \geq 0$                                                                                    |                                                                                                                                                                                 |
| cardona-qft-methods_F00163 | 20   | ■ 1.000 | $h \in C^\infty(U)$                                                                           | $h \in C^\infty(U)$                                                                           | <b>2.2. Case of manifolds</b>                                                                                                                                                   |
| cardona-qft-methods_F00164 | 20   | ■ 1.000 | $dh(x) = \xi$                                                                                 | $dh(x) = \xi$                                                                                 | <b>2.2. Case of manifolds</b>                                                                                                                                                   |
| cardona-qft-methods_F00165 | 20   | ■ 1.000 | $\Psi DO^m(M)$                                                                                | $\Psi DO^m(M)$                                                                                | Of course, this can be generalized: Given a vector bundle $E$ over $M$ , $\Psi DO^m(M, E) \rightarrow \Psi DO^m(M, E)$ is the $\Psi DO^m(M, E)$ which is $LP$                   |
| cardona-qft-methods_F00166 | 20   | ■ 1.000 | $C_c^\infty(M) \rightarrow C^\infty(M)$                                                       | $C_c^\infty(M) \rightarrow C^\infty(M)$                                                       | is smooth off the diagonal.                                                                                                                                                     |
| cardona-qft-methods_F00167 | 20   | ■ 1.000 | $k^P \in C^\infty(M \times M)$                                                                | $k^P \in C^\infty(M \times M)$                                                                | valued symbols.                                                                                                                                                                 |
| cardona-qft-methods_F00168 | 20   | ■ 1.000 | $f \in C_c^\infty(\phi(U)) \rightarrow P(f \circ \phi) \circ \phi^{-1} \in C^\infty(\phi(U))$ | $f \in C_c^\infty(\phi(U)) \rightarrow P(f \circ \phi) \circ \phi^{-1} \in C^\infty(\phi(U))$ | The covariance formula implies that $\sigma_m^P$ is independent of the chosen                                                                                                   |
| cardona-qft-methods_F00169 | 20   | ■ 0.999 | $\Psi DO^m(\phi(U))$                                                                          | $\Psi DO^m(\phi(U))$                                                                          | local chart so is globally defined on the bundle $T^*M \rightarrow M$ and $\sigma_m^P$ is i for every $P \in \Psi DO^m$ using overlapping charts and matching with $\sigma_m^P$ |
| cardona-qft-methods_F00170 | 20   | ■ 0.999 | $(U, \phi: U \rightarrow \mathbb{R}^d)$                                                       | $(U, \phi: U \rightarrow \mathbb{R}^d)$                                                       | local chart so is globally defined on the bundle $T^*M \rightarrow M$ and $\sigma_m^P$ is i for every $P \in \Psi DO^m$ using overlapping charts and matching with $\sigma_m^P$ |
| cardona-qft-methods_F00171 | 20   | ■ 1.000 | $P: \Gamma_c^\infty(M, E) \rightarrow \Gamma^\infty(M, E)$                                    | $P: \Gamma_c^\infty(M, E) \rightarrow \Gamma^\infty(M, E)$                                    | An important class of pseudodifferential operators are those which are invertible modulo regularizing ones.                                                                     |
| cardona-qft-methods_F00172 | 20   | ■ 1.000 | $\Psi DO^m(M, E)$                                                                             | $\Psi DO^m(M, E)$                                                                             | An important class of pseudodifferential operators are those which are invertible modulo regularizing ones.                                                                     |
| cardona-qft-methods_F00173 | 20   | ■ 1.000 | $k^P$                                                                                         | $k^P$                                                                                         | An important class of pseudodifferential operators are those which are invertible modulo regularizing ones.                                                                     |
| cardona-qft-methods_F00174 | 20   | ■ 1.000 | $\Psi DO$                                                                                     | $\Psi DO$                                                                                     | $\Psi DO = \Psi DO^0(M, E) + \Psi DO^{-1}(M, E) + \dots$                                                                                                                        |
| cardona-qft-methods_F00175 | 20   | ■ 1.000 | $\sigma_m^P$                                                                                  | $\sigma_m^P$                                                                                  |                                                                                                                                                                                 |
| cardona-qft-methods_F00176 | 20   | ■ 1.000 | $T^*M \rightarrow M$                                                                          | $T^*M \rightarrow M$                                                                          |                                                                                                                                                                                 |

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| cardona-qft-methods_F00177 | 20   | ■ 1.000 | $P \in \Psi D 0^{\{m\}}$                                              | $P \in \Psi DO^m$                                              |                                                                                                                                                                                           |
| cardona-qft-methods_F00178 | 20   | ■ 1.000 | $P \in \Psi D 0^{\{m\}}(M, E)$                                        | $P \in \Psi DO^m(M, E)$                                        |                                                                                                                                                                                           |
| cardona-qft-methods_F00179 | 20   | ■ 1.000 | $\sigma_m^P(x, \xi)$                                                  | $\sigma_m^P(x, \xi)$                                           |                                                                                                                                                                                           |
| cardona-qft-methods_F00180 | 20   | ■ 1.000 | $0 \neq \xi \in TM_x^*$                                               | $0 \neq \xi \in TM_x^*$                                        |                                                                                                                                                                                           |
| cardona-qft-methods_F00181 | 21   | ■ 1.000 | $ \left  \sigma^P(x, \xi) \right  \geq c_1(x)  \xi ^m$                | $ \sigma^P(x, \xi)  \geq c_1(x)  \xi ^m$                       |                                                                                                                                                                                           |
| cardona-qft-methods_F00182 | 21   | ■ 1.000 | $ \xi  \geq c_2(x), x \in U$                                          | $ \xi  \geq c_2(x), x \in U$                                   |                                                                                                                                                                                           |
| cardona-qft-methods_F00183 | 21   | ■ 1.000 | $c_{\{1\}}, c_{\{2\}}$                                                | $c_1, c_2$                                                     |                                                                                                                                                                                           |
| cardona-qft-methods_F00184 | 21   | ■ 0.967 | $Op(\sigma) \in \Psi D 0^{\{m\}}(U)$                                  | $Op(\sigma) \in \Psi DO^m(U)$                                  | exists a parametriz:                                                                                                                                                                      |
| cardona-qft-methods_F00185 | 21   | ■ 1.000 | $\sigma^{\prime} \in S^{-m} \left( U \times \mathbb{R}^d \right)$     | $\sigma' \in S^{-m}(U \times \mathbb{R}^d)$                    | <b>Lemma 2.1.</b> The following are equivalent:<br>(i) $Op(\sigma) \in \Psi DO^m(U)$ is elliptic                                                                                          |
| cardona-qft-methods_F00186 | 21   | ■ 1.000 | $\sigma \circ \sigma^{\prime} = 1$                                    | $\sigma \circ \sigma' = 1$                                     | <b>Lemma 2.1.</b> The following are equivalent:<br>(i) $Op(\sigma) \in \Psi DO^m(U)$ is elliptic                                                                                          |
| cardona-qft-methods_F00187 | 21   | ■ 1.000 | $\sigma^{\prime} \circ \sigma = 1$                                    | $\sigma' \circ \sigma = 1$                                     | <b>Lemma 2.1.</b> The following are equivalent:<br>(i) $Op(\sigma) \in \Psi DO^m(U)$ is elliptic                                                                                          |
| cardona-qft-methods_F00188 | 21   | ■ 0.998 | $Op(\sigma) Op(\sigma^{\prime}) = Op(\sigma^{\prime}) Op(\sigma) = 1$ | $Op(\sigma) Op(\sigma') = Op(\sigma') Op(\sigma) = 1$          | (i) $Op(\sigma) \in \Psi DO^m(U)$ is elliptic, (ii) There exist $\sigma' \in S^{-m}(U \times \mathbb{R}^d)$ such that $\sigma \circ \sigma' = 1$ or $\sigma' \circ \sigma = 1$ .          |
| cardona-qft-methods_F00189 | 21   | ■ 1.000 | $Op(\sigma^{\prime}) \in \Psi DO^{-m}(U)$                             | $Op(\sigma') \in \Psi DO^{-m}(U)$                              | (iii) $Op(\sigma) Op(\sigma') = Op(\sigma') Op(\sigma) = 1$ modulo $\Psi DO^{-m}(U)$                                                                                                      |
| cardona-qft-methods_F00190 | 21   | ■ 1.000 | $P \in \Psi D 0(M, E)$                                                | $P \in \Psi DO(M, E)$                                          | Remark that any $P \in \Psi DO(M, E)$ can be extended to a bounded operator                                                                                                               |
| cardona-qft-methods_F00191 | 21   | ■ 1.000 | $L^2(M, E)$                                                           | $L^2(M, E)$                                                    | on $L^2(M, E)$ when $\Re(m) \leq 0$ but this needs an existing scalar product for linear sections on $M$ and $E$                                                                          |
| cardona-qft-methods_F00192 | 21   | ■ 1.000 | $\Re(m) \leq 0$                                                       | $\Re(m) \leq 0$                                                | on $L^2(M, E)$ when $\Re(m) \leq 0$ but this needs an existing scalar product for linear sections on $M$ and $E$                                                                          |
| cardona-qft-methods_F00193 | 21   | ■ 1.000 | $P \in \Psi D 0^{-m}(M, E)$                                           | $P \in \Psi DO^{-m}(M, E)$                                     | 0, its spectrum is discrete when $M$ is compact.                                                                                                                                          |
| cardona-qft-methods_F00194 | 21   | ■ 1.000 | $\Re(m) >$                                                            | $\Re(m) >$                                                     | 0, its spectrum is discrete when $M$ is compact.                                                                                                                                          |
| cardona-qft-methods_F00195 | 21   | ■ 1.000 | $r$                                                                   | $r$                                                            | $\{U, \phi\}$ , these operators are $r \times r$ matrices of pseudodifferential operators                                                                                                 |
| cardona-qft-methods_F00196 | 21   | ■ 1.000 | $s \in \Gamma^{\infty}(M, E), t^* \in \Gamma^{\infty}(M, E^*)$        | $s \in \Gamma^{\infty}(M, E), t^* \in \Gamma^{\infty}(M, E^*)$ | of order $-m$ . The total symbol is in $C^{\infty}(T^*U) \otimes End(E)$ with $End(E) \simeq M(l^*)$ . The principal symbol can be globally defined: $\sigma^P(x, \xi) : E_x \rightarrow$ |
| cardona-qft-methods_F00197 | 21   | ■ 1.000 | $f \in C^{\infty}(M) \rightarrow$                                     | $f \in C^{\infty}(M) \rightarrow$                              | of order $-m$ . The total symbol is in $C^{\infty}(T^*U) \otimes End(E)$ with $End(E) \simeq M(l^*)$ . The principal symbol can be globally defined: $\sigma^P(x, \xi) : E_x \rightarrow$ |
| cardona-qft-methods_F00198 | 21   | ■ 1.000 | $\langle t^*, P(fs) \rangle \in C^{\infty}(M)$                        | $\langle t^*, P(fs) \rangle \in C^{\infty}(M)$                 | $E_x$ for $x \in M$ and $\xi \in T_x^*M$ , can be seen as a smooth homomorphism                                                                                                           |
| cardona-qft-methods_F00199 | 21   | ■ 1.000 | $(U, \phi)$                                                           | $(U, \phi)$                                                    | homogeneous of degree $-m$ on all fibers of $T^*M$ . We get the simple formula which could be seen as a definition of the principal symbol                                                |
| cardona-qft-methods_F00200 | 21   | ■ 1.000 | $r \times r$                                                          | $r \times r$                                                   | homogeneous of degree $-m$ on all fibers of $T^*M$ . We get the simple formula which could be seen as a definition of the principal symbol                                                |
| cardona-qft-methods_F00201 | 21   | ■ 1.000 | $-m$                                                                  | $-m$                                                           | which could be seen as a definition of the principal symbol:<br>$\sigma^P(x, \xi) = \lim_{t \rightarrow 0} t^{-m} (\rho^{-ith}, P, \rho^{ith})(x)$ for $x \in M, \xi \in T_x^*M$ (3)      |
| cardona-qft-methods_F00202 | 21   | ■ 1.000 | $C^{\infty}(T^*U) \otimes End(E)$                                     | $C^{\infty}(T^*U) \otimes End(E)$                              | which could be seen as a definition of the principal symbol:<br>$\sigma^P(x, \xi) = \lim_{t \rightarrow 0} t^{-m} (\rho^{-ith}, P, \rho^{ith})(x)$ for $x \in M, \xi \in T_x^*M$ (3)      |

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| cardona-qft-methods_F00203 | 21   | ■ 1.000 | <code>\operatorname{End}(E) \simeq</code>                                                                                | $\operatorname{End}(E) \simeq$                                                          |       |
| cardona-qft-methods_F00204 | 21   | ■ 1.000 | <code>M_{\{r\}}(\mathbb{C})</code>                                                                                       | $M_r(\mathbb{C})$                                                                       |       |
| cardona-qft-methods_F00205 | 21   | ■ 1.000 | <code>\sigma_{-m}^{\{P\}}(x, \xi): E_{\{x\}} \rightarrow</code>                                                          | $\sigma_{-m}^P(x, \xi): E_x \rightarrow$                                                |       |
| cardona-qft-methods_F00206 | 21   | ■ 1.000 | <code>E_{\{x\}}</code>                                                                                                   | $E_x$                                                                                   |       |
| cardona-qft-methods_F00207 | 21   | ■ 1.000 | <code>\xi \in T_{\{x\}}^* M</code>                                                                                       | $\xi \in T_x^* M$                                                                       |       |
| cardona-qft-methods_F00208 | 21   | ■ 1.000 | <code>T^{\{*\}} M</code>                                                                                                 | $T^* M$                                                                                 |       |
| cardona-qft-methods_F00209 | 21   | ■ 1.000 | <code>h \in C^{\infty}(M)</code>                                                                                         | $h \in C^\infty(M)$                                                                     |       |
| cardona-qft-methods_F00210 | 21   | ■ 1.000 | <code>d_{\{x\}} h=\xi</code>                                                                                             | $d_x h = \xi$                                                                           |       |
| cardona-qft-methods_F00211 | 21   | ■ 1.000 | <code>\mathbb{R}^d \backslash \{0\}</code>                                                                               | $\mathbb{R}^d \setminus \{0\}$                                                          |       |
| cardona-qft-methods_F00212 | 21   | ■ 1.000 | <code>\mathcal{S}</code>                                                                                                 | $\mathcal{S}$                                                                           |       |
| cardona-qft-methods_F00213 | 21   | ■ 1.000 | <code>\mathcal{S}^{\prime}</code>                                                                                        | $\mathcal{S}'$                                                                          |       |
| cardona-qft-methods_F00214 | 21   | ■ 1.000 | <code>f_{\{\lambda\}}(\xi)=f(\lambda \xi), \lambda \in \mathbb{R}_+^*</code>                                             | $f_\lambda(\xi) = f(\lambda \xi), \lambda \in \mathbb{R}_+^*$                           |       |
| cardona-qft-methods_F00215 | 21   | ■ 1.000 | <code>\tau \in \mathcal{S}^{\prime} \rightarrow \tau_\lambda</code>                                                      | $\tau \in \mathcal{S}' \rightarrow \tau_\lambda$                                        |       |
| cardona-qft-methods_F00216 | 21   | ■ 0.974 | <code>\left\langle \tau_\lambda, f \right\rangle = \lambda^{-d} \left\langle \tau, f_{\lambda^{-1}} \right\rangle</code> | $\langle \tau_\lambda, f \rangle = \lambda^{-d} \langle \tau, f_{\lambda^{-1}} \rangle$ |       |
| cardona-qft-methods_F00217 | 21   | ■ 0.974 | <code>f \in \mathcal{S}</code>                                                                                           | $f \in \mathcal{S}$                                                                     |       |
| cardona-qft-methods_F00218 | 21   | ■ 0.974 | <code>\tau \in \mathcal{S}^{\prime}</code>                                                                               | $\tau \in \mathcal{S}'$                                                                 |       |
| cardona-qft-methods_F00219 | 21   | ■ 1.000 | <code>\tau_\lambda=\lambda^m \tau</code>                                                                                 | $\tau_\lambda = \lambda^m \tau$                                                         |       |
| cardona-qft-methods_F00220 | 22   | ■ 1.000 | <code>\sigma \in C^{\infty}(\mathbb{R}^d \backslash \{0\})</code>                                                        | $\sigma \in C^\infty(\mathbb{R}^d \setminus \{0\})$                                     |       |
| cardona-qft-methods_F00221 | 22   | ■ 1.000 | <code>k \in \mathbb{Z}</code>                                                                                            | $k \in \mathbb{Z}$                                                                      |       |
| cardona-qft-methods_F00222 | 22   | ■ 1.000 | <code>k&gt;-d</code>                                                                                                     | $k > -d$                                                                                |       |
| cardona-qft-methods_F00223 | 22   | ■ 1.000 | <code>k=-d</code>                                                                                                        | $k = -d$                                                                                |       |
| cardona-qft-methods_F00224 | 22   | ■ 1.000 | <code>c_{\{\sigma\}}=\int_{\mathbb{S}^{d-1}} \sigma(\xi) d \xi</code>                                                    | $c_\sigma = \int_{\mathbb{S}^{d-1}} \sigma(\xi) d\xi$                                   |       |
| cardona-qft-methods_F00225 | 22   | ■ 1.000 | <code>\tau_\lambda=\lambda^{-d} \left( \tau+c_{\{\sigma\}} \log \left( \lambda \right) \delta_0 \right)</code>           | $\tau_\lambda = \lambda^{-d} (\tau + c_\sigma \log(\lambda) \delta_0)$                  |       |
| cardona-qft-methods_F00226 | 22   | ■ 0.807 | <code>\operatorname{kernel} k^P</code>                                                                                   | <b>kernel</b> $k^P$                                                                     |       |
| cardona-qft-methods_F00227 | 22   | ■ 0.807 | <code>P \in \Psi D 0</code>                                                                                              | $P \in \Psi DO$                                                                         |       |
| cardona-qft-methods_F00228 | 22   | ■ 0.954 | <code>\tau=\phi \circ \tau+(1-\phi) \circ \tau</code>                                                                    | $\tau = \phi \circ \tau + (1 - \phi) \circ \tau$                                        |       |

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| cardona-qft-methods_F00229 | 22   | ■ 0.954 | $\backslash\mathrm{phi}=1$                                                                                                                                                                                       | $\phi = 1$                                               | of the kernel $k^P$ for $P \in \Psi DO$ . For any $\tau \in \mathcal{S}'$ , we choose the decomposition $\phi \tau = \phi \tau + (1 - \phi) \tau$ where $\phi \in C^\infty(\mathbb{R}^d)$ and $\phi = 1$ near 0. We can look at |
| cardona-qft-methods_F00230 | 22   | ■ 1.000 | $\backslash\mathrm{tau}$                                                                                                                                                                                         | $\tau$                                                   | the infrared behavior of $\tau$ near the origin and its ultraviolet behavior near                                                                                                                                               |
| cardona-qft-methods_F00231 | 22   | ■ 0.919 | $\backslash\mathrm{phi} \ \backslash\mathrm{circ} \ \backslash\mathrm{tau}$                                                                                                                                      | $\phi \circ \tau$                                        | infinity. Remark first that, since $\phi \circ \tau$ has a compact support, $(\phi \circ \tau)^\vee \in \mathcal{S}'$ ,                                                                                                         |
| cardona-qft-methods_F00232 | 22   | ■ 0.919 | $(\backslash\mathrm{phi} \ \backslash\mathrm{circ} \ \backslash\mathrm{tau})^{\wedge\{\vee\}} \ \backslash\mathrm{in} \ \mathrm{\mathcal{S}}^{\wedge\{\mathrm{prime}\}}$                                         | $(\phi \circ \tau)^\vee \in \mathcal{S}'$                | infinity. Remark first that, since $\phi \circ \tau$ has a compact support, $(\phi \circ \tau)^\vee \in \mathcal{S}'$ ,                                                                                                         |
| cardona-qft-methods_F00233 | 22   | ■ 0.529 | $\backslash\mathrm{tau}^{\wedge\{\vee\}}$                                                                                                                                                                        | $\tau^\vee$                                              | so the regularity of $\tau$ depends only of its ultraviolet part $((1 - \phi) \circ \tau)^\vee$ .                                                                                                                               |
| cardona-qft-methods_F00234 | 22   | ■ 0.529 | $((1\text{-}\backslash\mathrm{phi}) \ \backslash\mathrm{circ} \ \backslash\mathrm{tau})^{\wedge\{\vee\}}$                                                                                                        | $((1 - \phi) \circ \tau)^\vee$                           | so the regularity of $\tau$ depends only of its ultraviolet part $((1 - \phi) \circ \tau)^\vee$ .                                                                                                                               |
| cardona-qft-methods_F00235 | 22   | ■ 0.979 | $P \ \backslash\mathrm{in} \ \Psi \mathrm{D} \ 0^{\wedge\{m\}}(U), \ m \ \backslash\mathrm{in} \ \mathrm{\mathbb{Z}}$                                                                                            | $P \in \Psi DO^m(U), m \in \mathbb{Z}$                   | diagonal,                                                                                                                                                                                                                       |
| cardona-qft-methods_F00236 | 22   | ■ 1.000 | $\mathrm{a}_{\{j\}}(x, \ y) \ \backslash\mathrm{in} \ C^{\wedge\{\infty\}}(U \ \backslash\mathrm{times} \ U \ \backslash\mathrm{backslash}\{x\})$                                                                | $a_j(x, y) \in C^\infty(U \times U \setminus \{x\})$     | $c_P(x, y) = \frac{1}{(2\pi)^d} \int_{\mathbb{S}^{d-1}} e^{-id(x, y) \cdot \zeta} d\zeta, \quad (4)$                                                                                                                            |
| cardona-qft-methods_F00237 | 22   | ■ 1.000 | $j$                                                                                                                                                                                                              | $j$                                                      | $c_P(x, y) = \frac{1}{(2\pi)^d} \int_{\mathbb{S}^{d-1}} e^{-id(x, y) \cdot \zeta} d\zeta, \quad (4)$                                                                                                                            |
| cardona-qft-methods_F00238 | 22   | ■ 1.000 | $y$                                                                                                                                                                                                              | $y$                                                      | $c_P(x, y) = \frac{1}{(2\pi)^d} \int_{\mathbb{S}^{d-1}} e^{-id(x, y) \cdot \zeta} d\zeta, \quad (4)$                                                                                                                            |
| cardona-qft-methods_F00239 | 22   | ■ 1.000 | $\mathrm{c}_{\{P\}}(x) \ \backslash\mathrm{in} \ C^{\wedge\{\infty\}}(U)$                                                                                                                                        | $c_P(x) \in C^\infty(U)$                                 | We deduce rea                                                                                                                                                                                                                   |
| cardona-qft-methods_F00240 | 22   | ■ 1.000 | $P \ \backslash\mathrm{in} \ \Psi \mathrm{D} \ 0^{\wedge\{m\}}(M, \ E), \ m \ \backslash\mathrm{in} \ \mathrm{\mathbb{Z}}$                                                                                       | $P \in \Psi DO^m(M, E), m \in \mathbb{Z}$                | by                                                                                                                                                                                                                              |
| cardona-qft-methods_F00241 | 22   | ■ 1.000 | $\mathrm{a}_{\{j\}}$                                                                                                                                                                                             | $a_j$                                                    | The main difficulty is of course to prove that $c_P$ is well defined.                                                                                                                                                           |
| cardona-qft-methods_F00242 | 22   | ■ 1.000 | $y, \ \mathrm{c}_{\{P\}}$                                                                                                                                                                                        | $y, c_P$                                                 | The main difficulty is of course to prove that $c_P$ is well defined.                                                                                                                                                           |
| cardona-qft-methods_F00243 | 22   | ■ 0.990 | $\mathrm{c}_{\{P\}}(x)  d \ x $                                                                                                                                                                                  | $c_P(x)  dx $                                            |                                                                                                                                                                                                                                 |
| cardona-qft-methods_F00244 | 22   | ■ 1.000 | $\mathrm{c}_{\{P\}}$                                                                                                                                                                                             | $c_P$                                                    |                                                                                                                                                                                                                                 |
| cardona-qft-methods_F00245 | 23   | ■ 1.000 | $\int_{\mathrm{M}} \mathrm{c}_{\{P\}}(x)  d \ x $                                                                                                                                                                | $\int_M c_P(x)  dx $                                     | 12 B. Iochum                                                                                                                                                                                                                    |
| cardona-qft-methods_F00246 | 23   | ■ 1.000 | $\mathrm{\mathbb{C}}$                                                                                                                                                                                            | $\mathbb{C}$                                             | 3.4. Weyl's lemma                                                                                                                                                                                                               |
| cardona-qft-methods_F00247 | 23   | ■ 1.000 | $m \ \backslash\mathrm{in} \ \mathrm{\mathbb{C}} \ \backslash\mathrm{backslash} \ \mathrm{\mathbb{Z}}$                                                                                                           | $m \in \mathbb{C} \setminus \mathbb{Z}$                  | in $\mathbb{C}$ in the same spirit as in Proposition 3.1, one obtains the following                                                                                                                                             |
| cardona-qft-methods_F00248 | 23   | ■ 1.000 | $\backslash\mathrm{Omega}$                                                                                                                                                                                       | $\Omega$                                                 | $m \in \mathbb{C} \setminus \mathbb{Z}$ can be uniquely extended to a homogeneous distribution.                                                                                                                                 |
| cardona-qft-methods_F00249 | 23   | ■ 1.000 | $\backslash\mathrm{sigma}: \ \backslash\mathrm{Omega} \ \rightarrow \mathrm{S}^{\wedge\{m\}} \backslash\mathrm{left}(U \ \backslash\mathrm{times} \ \mathrm{\mathbb{R}}^{\wedge\{d\}} \backslash\mathrm{right})$ | $\sigma : \Omega \rightarrow S^m(U \times \mathbb{R}^d)$ | <b>Definition 2.6.</b> Let $U$ be an open set in $\mathbb{R}^d$ and $\Omega$ be a domain in $\mathbb{C}$ .<br>A map $\sigma : \Omega \rightarrow \mathcal{S}'(U \times \mathbb{R}^d)$ is said to be holomorphic when            |
| cardona-qft-methods_F00250 | 23   | ■ 1.000 | $z \ \backslash\mathrm{in} \ \backslash\mathrm{Omega} \ \rightarrow \backslash\mathrm{sigma}(z)(x, \ \backslash\mathrm{xi})$                                                                                     | $z \in \Omega \rightarrow \sigma(z)(x, \xi)$             | the map $z \in \Omega \rightarrow \sigma(z)(x, \xi)$ is analytic for all $x \in U, \xi \in \mathbb{R}^d$ ,                                                                                                                      |
| cardona-qft-methods_F00251 | 23   | ■ 1.000 | $x \ \backslash\mathrm{in} \ U, \ \backslash\mathrm{xi} \ \backslash\mathrm{in} \ \mathrm{\mathbb{R}}^{\wedge\{d\}}$                                                                                             | $x \in U, \xi \in \mathbb{R}^d$                          | the map $z \in \Omega \rightarrow \sigma(z)(x, \xi)$ is analytic for all $x \in U, \xi \in \mathbb{R}^d$ ,                                                                                                                      |
| cardona-qft-methods_F00252 | 23   | ■ 1.000 | $m(z)$                                                                                                                                                                                                           | $m(z)$                                                   | the order $m(z)$ of $\sigma(z)$ is analytic on $\Omega$ ,                                                                                                                                                                       |
| cardona-qft-methods_F00253 | 23   | ■ 1.000 | $\backslash\mathrm{sigma}(z)$                                                                                                                                                                                    | $\sigma(z)$                                              | the order $m(z)$ of $\sigma(z)$ is analytic on $\Omega$ ,                                                                                                                                                                       |
| cardona-qft-methods_F00254 | 23   | ■ 1.000 | $\backslash\mathrm{sigma}(z) \ \backslash\mathrm{simeq} \ \sum_{\{j\}} \backslash\mathrm{sigma}_{\{m(z)-j\}}(z)$                                                                                                 | $\sigma(z) \simeq \sum_j \sigma_{m(z)-j}(z)$             | This hypothesis is sufficient to get that the map $z \rightarrow$                                                                                                                                                               |

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| cardona-qft-methods_F00255 | 23   | ■ 1.000 | $z$                                                                                         | $z$                                                                                         | This hypothesis is sufficient to get that the map $z \rightarrow$                                                                                                                                                       |
| cardona-qft-methods_F00256 | 23   | ■ 1.000 | $z \rightarrow \sigma_{m(z)-j}(z)$                                                          | $z \rightarrow \sigma_{m(z)-j}(z)$                                                          | symbol on $U \times \mathbb{R}^d$ such that $\partial_z \sigma(z)(x, \xi) \simeq \sum_{j \geq 0} \partial_z \sigma_{m(z)-j}(z)(x, \xi)$ .                                                                               |
| cardona-qft-methods_F00257 | 23   | ■ 1.000 | $C^{\infty}(\infty)\left(U \times \mathbb{R}^d \backslash \{0\}\right)$                     | $C^{\infty}(U \times \mathbb{R}^d \backslash \{0\})$                                        | <b>Definition 2.7.</b> The map $P : \Omega \subset \mathbb{C} \rightarrow \Psi DO(U)$ is said to be holomorphic                                                                                                         |
| cardona-qft-methods_F00258 | 23   | ■ 1.000 | $\partial_z \sigma(z)(x, \xi)$                                                              | $\partial_z \sigma(z)(x, \xi)$                                                              | <b>Definition 2.7.</b> The map $P : \Omega \subset \mathbb{C} \rightarrow \Psi DO(U)$ is said to be holomorphic                                                                                                         |
| cardona-qft-methods_F00259 | 23   | ■ 1.000 | $U \times \mathbb{R}^d$                                                                     | $U \times \mathbb{R}^d$                                                                     | if it has the decomposition $P(z) = \sigma(z)(\cdot, D) + R(z)$ (see definition (1)) where $\sigma \in \mathcal{S}(U \times \mathbb{R}^d)$ and $R \in \mathcal{O}(\mathcal{S}(U \times \mathbb{R}^d))$ are holomorphic. |
| cardona-qft-methods_F00260 | 23   | ■ 1.000 | $\partial_z \sigma(z)(x, \xi) \simeq \sum_{j \geq 0} \partial_z \sigma_{m(z)-j}(z)(x, \xi)$ | $\partial_z \sigma(z)(x, \xi) \simeq \sum_{j \geq 0} \partial_z \sigma_{m(z)-j}(z)(x, \xi)$ | if it has the decomposition $P(z) = \sigma(z)(\cdot, D) + R(z)$ (see definition (1)) where $\sigma \in \mathcal{S}(U \times \mathbb{R}^d)$ and $R \in \mathcal{O}(\mathcal{S}(U \times \mathbb{R}^d))$ are holomorphic. |
| cardona-qft-methods_F00261 | 23   | ■ 1.000 | $P: \Omega \subset \mathbb{C} \rightarrow \Psi D^0(U)$                                      | $P : \Omega \subset \mathbb{C} \rightarrow \Psi DO(U)$                                      | As a consequence, there exists a holomorphic map from $\Omega$ into                                                                                                                                                     |
| cardona-qft-methods_F00262 | 23   | ■ 1.000 | $P(z)=\sigma(z)(\cdot, D)+R(z)$                                                             | $P(z) = \sigma(z)(\cdot, D) + R(z)$                                                         | $\Psi DO(M, E)$ with a holomorphic product (when $M$ is compact).                                                                                                                                                       |
| cardona-qft-methods_F00263 | 23   | ■ 1.000 | $\sigma: \Omega \rightarrow S(\infty)\left(U \times \mathbb{R}^d\right)$                    | $\sigma : \Omega \rightarrow S(U \times \mathbb{R}^d)$                                      | <i>Elliptic operators.</i> Recall that $P \in \Psi DO^{-m}(U)$ , $m \in \mathbb{Z}$ , is elliptic essentially means that $P$ is invertible modulo smoothing operators. More gener-                                      |
| cardona-qft-methods_F00264 | 23   | ■ 1.000 | $R: \Omega \rightarrow C^{\infty}(U \times U)$                                              | $R : \Omega \rightarrow C^{\infty}(U \times U)$                                             | <i>Elliptic operators.</i> Recall that $P \in \Psi DO^{-m}(U)$ , $m \in \mathbb{Z}$ , is elliptic essentially means that $P$ is invertible modulo smoothing operators. More gener-                                      |
| cardona-qft-methods_F00265 | 23   | ■ 1.000 | $\Psi D^0(M, E)$                                                                            | $\Psi DO(M, E)$                                                                             | Let $Q \in \Psi DO^m(M, E)$ with $\Re(m) > 0$ . We assume that $M$ is compact and $O$ is elliptic. Thus $O$ has a discrete spectrum. We assume $\text{Spectrum}(O) \cap$                                                |
| cardona-qft-methods_F00266 | 23   | ■ 1.000 | $P \in \Psi D^{0^m}(U), m \in \mathbb{C}$                                                   | $P \in \Psi DO^m(U), m \in \mathbb{C}$                                                      | $\mathbb{R}^- = \emptyset$ and the existence of a curve $\Gamma$ whose interior contains the spectrum                                                                                                                   |
| cardona-qft-methods_F00267 | 23   | ■ 1.000 | $Q \in \Psi D^{0^m}(M, E)$                                                                  | $Q \in \Psi DO^m(M, E)$                                                                     |                                                                                                                                     |
| cardona-qft-methods_F00268 | 23   | ■ 1.000 | $\Re(m) > 0$                                                                                | $\Re(m) > 0$                                                                                |                                                                                                                                    |
| cardona-qft-methods_F00269 | 23   | ■ 0.994 | $Q$                                                                                         | $Q$                                                                                         |                                                                                                                                    |
| cardona-qft-methods_F00270 | 23   | ■ 0.994 | $\operatorname{Spectrum}(Q) \cap$                                                           | $\text{Spectrum}(Q) \cap$                                                                   |                                                                                                                                    |
| cardona-qft-methods_F00271 | 23   | ■ 1.000 | $\mathbb{R}^- = \emptyset$                                                                  | $\mathbb{R}^- = \emptyset$                                                                  |                                                                                                                                    |
| cardona-qft-methods_F00272 | 23   | ■ 1.000 | $\Gamma$                                                                                    | $\Gamma$                                                                                    |                                                                                                                                    |
| cardona-qft-methods_F00273 | 23   | ■ 1.000 | $\lambda^z$                                                                                 | $\lambda^z$                                                                                 |                                                                                                                                                                                                                         |
| cardona-qft-methods_F00274 | 23   | ■ 1.000 | $z=0$                                                                                       | $z = 0$                                                                                     |                                                                                                                                                                                                                         |
| cardona-qft-methods_F00275 | 24   | ■ 1.000 | $\Re(s) < 0, Q^s = \frac{1}{i2\pi} \int_{\Gamma} \lambda^s (\lambda - Q)^{-1} d\lambda$     | $\Re(s) < 0, Q^s = \frac{1}{i2\pi} \int_{\Gamma} \lambda^s (\lambda - Q)^{-1} d\lambda$     |                                                                                                                                                                                                                         |
| cardona-qft-methods_F00276 | 24   | ■ 1.000 | $s \rightarrow Q^s$                                                                         | $s \rightarrow Q^s$                                                                         | <i>spectral groupoid</i> $g \rightarrow \lambda g$                                                                                                                                                                      |
| cardona-qft-methods_F00277 | 24   | ■ 1.000 | $Q^0$                                                                                       | $Q^0$                                                                                       | <i>spectral groupoid</i> $g \rightarrow \lambda g$                                                                                                                                                                      |
| cardona-qft-methods_F00278 | 24   | ■ 1.000 | $Q^1$                                                                                       | $Q^1$                                                                                       | <i>spectral groupoid</i> $g \rightarrow \lambda g$                                                                                                                                                                      |
| cardona-qft-methods_F00279 | 24   | ■ 1.000 | $\Re(s) \leq 0$                                                                             | $\Re(s) \leq 0$                                                                             | When $\Re(s) < 0$ , $Q^s = \frac{1}{i2\pi} \int_{\Gamma} \lambda^s (\lambda - Q)^{-1} d\lambda$ makes sense as operator on $\mathcal{D}(Q) \subset \mathcal{H}$ .                                                       |
| cardona-qft-methods_F00280 | 24   | ■ 0.997 | $S_{\text{int}}$                                                                            | $S_{\text{int}}$                                                                            | <i>As <math>\lambda \rightarrow \lambda^*</math></i><br>The map $s \rightarrow Q^s$ is a one-parameter group containing $Q^0$ and $Q^1$ which                                                                           |

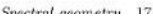
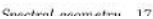
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| cardona-qft-methods_F00281 | 24   | 1.000 | $L: \sigma \in S_{\text{int}}^{\mathbb{Z}}(\mathbb{R}^d) \rightarrow L(\sigma) = \check{\sigma}(0) = \frac{1}{(2\pi)^d} \int_{\mathbb{R}^d} \sigma(\xi) d\xi$            |                                                                                                                                                                          | Proposition 2.1 and Lemma 2.2, one proves                                                                                                                                                                                                                                                           |
| cardona-qft-methods_F00282 | 24   | 0.223 | $L$                                                                                                                                                                      | $L$                                                                                                                                                                      | <b>Prop 2.3.</b> Let $L: \sigma \in \mathcal{D}'_{\text{int}}(\mathbb{R}^n) \rightarrow L(\sigma) = \sigma(0) = \frac{1}{(2\pi)^d} \int_{\mathbb{R}^d} \sigma(\xi) d\xi$ . Then $L$ has a unique holomorphic extension $\tilde{L}$ on $\mathcal{S}'(\mathbb{C}/\mathbb{Z}/\mathbb{R}^d)$ . Moreover |
| cardona-qft-methods_F00283 | 24   | 0.223 | $\widetilde{L}$                                                                                                                                                          | $\tilde{L}$                                                                                                                                                              | <b>Prop 2.3.</b> Let $L: \sigma \in \mathcal{D}'_{\text{int}}(\mathbb{R}^n) \rightarrow L(\sigma) = \sigma(0) = \frac{1}{(2\pi)^d} \int_{\mathbb{R}^d} \sigma(\xi) d\xi$ . Then $L$ has a unique holomorphic extension $\tilde{L}$ on $\mathcal{S}'(\mathbb{C}/\mathbb{Z}/\mathbb{R}^d)$ . Moreover |
| cardona-qft-methods_F00284 | 24   | 0.223 | $\left. S^{\mathbb{C}\mathbb{Z}}(\mathbb{R}^d) \right $                                                                                                                  | $S^{\mathbb{C}\mathbb{Z}}(\mathbb{R}^d)$                                                                                                                                 | <b>Prop 2.3.</b> Let $L: \sigma \in \mathcal{D}'_{\text{int}}(\mathbb{R}^n) \rightarrow L(\sigma) = \sigma(0) = \frac{1}{(2\pi)^d} \int_{\mathbb{R}^d} \sigma(\xi) d\xi$ . Then $L$ has a unique holomorphic extension $\tilde{L}$ on $\mathcal{S}'(\mathbb{C}/\mathbb{Z}/\mathbb{R}^d)$ . Moreover |
| cardona-qft-methods_F00285 | 24   | 0.748 | $\sigma(\xi) \simeq \sum_j \sigma_{m-j}(\xi), m \in \mathbb{C} \setminus \mathbb{Z}, \tilde{L}(\sigma) = \left( \sigma - \sum_{j \leq N} \tau_{m-j} \right)^{\sim}(0) =$ | $\sigma(\xi) \simeq \sum_j \sigma_{m-j}(\xi), m \in \mathbb{C} \setminus \mathbb{Z}, \tilde{L}(\sigma) = \left( \sigma - \sum_{j \leq N} \tau_{m-j} \right)^{\sim}(0) =$ | then $L$ has a unique holomorphic extension $\tilde{L}$ on $\mathcal{S}'(\mathbb{C}/\mathbb{Z}/\mathbb{R}^d)$ , when $\sigma(\xi) \simeq \sum_j \sigma_{m-j}(\xi), m \in \mathbb{C} \setminus \mathbb{Z}, \tilde{L}(\sigma) = \left( \sigma - \sum_{j \leq N} \tau_{m-j} \right)^{\sim}(0) =$       |
| cardona-qft-methods_F00286 | 24   | 1.000 | $\frac{1}{(2\pi)^d} \int_{\mathbb{R}^d} \left( \sigma - \sum_{j \leq N} \tau_{m-j} \right)(\xi) d\xi$                                                                    | $\frac{1}{(2\pi)^d} \int_{\mathbb{R}^d} \left( \sigma - \sum_{j \leq N} \tau_{m-j} \right)(\xi) d\xi$                                                                    | $\frac{1}{(2\pi)^d} \int_{\mathbb{R}^d} (\sigma - \sum_{j \leq N} \tau_{m-j})(\xi) d\xi$ where $m$ is the order of $\sigma$ , $N$ is an integer with $N > \Re(m) + d$ and $\tau_{m-j}$ is the extension of $\sigma_{m-j}$ of Lemma 2.2.                                                             |
| cardona-qft-methods_F00287 | 24   | 1.000 | $\sigma, N$                                                                                                                                                              | $\sigma, N$                                                                                                                                                              | $\frac{1}{(2\pi)^d} \int_{\mathbb{R}^d} (\sigma - \sum_{j \leq N} \tau_{m-j})(\xi) d\xi$ where $m$ is the order of $\sigma$ , $N$ is an integer with $N > \Re(m) + d$ and $\tau_{m-j}$ is the extension of $\sigma_{m-j}$ of Lemma 2.2.                                                             |
| cardona-qft-methods_F00288 | 24   | 1.000 | $N > \Re(m) + d$                                                                                                                                                         | $N > \Re(m) + d$                                                                                                                                                         |                                                                                                                                                                                                                                                                                                     |
| cardona-qft-methods_F00289 | 24   | 1.000 | $\tau_{m-j}$                                                                                                                                                             | $\tau_{m-j}$                                                                                                                                                             |                                                                                                                                                                                                                                                                                                     |
| cardona-qft-methods_F00290 | 24   | 1.000 | $\sigma_{m-j}$                                                                                                                                                           | $\sigma_{m-j}$                                                                                                                                                           |                                                                                                                                                                                                                                                                                                     |
| cardona-qft-methods_F00291 | 24   | 0.976 | $\sigma: \mathbb{C} \rightarrow S(\mathbb{R}^d)$                                                                                                                         | $\sigma: \mathbb{C} \rightarrow S(\mathbb{R}^d)$                                                                                                                         | then $L(\sigma(s))$ is meromorphic with simple poles on $\mathbb{Z}$ and for $p \in \mathbb{Z}$ , $\text{Res} \tilde{L}(\sigma(s)) = -\frac{1}{(2\pi)^d} \int_{\mathbb{R}^d} \sigma(p)(\xi) d\xi$                                                                                                   |
| cardona-qft-methods_F00292 | 24   | 0.976 | $(\sigma(s)) = s$                                                                                                                                                        | $(\sigma(s)) = s$                                                                                                                                                        | then $L(\sigma(s))$ is meromorphic with simple poles on $\mathbb{Z}$ and for $p \in \mathbb{Z}$ , $\text{Res} \tilde{L}(\sigma(s)) = -\frac{1}{(2\pi)^d} \int_{\mathbb{R}^d} \sigma(p)(\xi) d\xi$                                                                                                   |
| cardona-qft-methods_F00293 | 24   | 1.000 | $\tilde{L}(\sigma(s))$                                                                                                                                                   | $\tilde{L}(\sigma(s))$                                                                                                                                                   | $\frac{1}{(2\pi)^d} \int_{\mathbb{R}^d} \sigma(p)(\xi) d\xi = -\frac{1}{(2\pi)^d} \int_{\mathbb{R}^d} \sigma(p)(\xi) d\xi$                                                                                                                                                                          |
| cardona-qft-methods_F00294 | 24   | 1.000 | $\mathbb{Z}$                                                                                                                                                             | $\mathbb{Z}$                                                                                                                                                             | $\frac{1}{(2\pi)^d} \int_{\mathbb{R}^d} \sigma(p)(\xi) d\xi = -\frac{1}{(2\pi)^d} \int_{\mathbb{R}^d} \sigma(p)(\xi) d\xi$                                                                                                                                                                          |
| cardona-qft-methods_F00295 | 24   | 1.000 | $p \in \mathbb{Z}$                                                                                                                                                       | $p \in \mathbb{Z}$                                                                                                                                                       | $\frac{1}{(2\pi)^d} \int_{\mathbb{R}^d} \sigma(p)(\xi) d\xi = -\frac{1}{(2\pi)^d} \int_{\mathbb{R}^d} \sigma(p)(\xi) d\xi$                                                                                                                                                                          |
| cardona-qft-methods_F00296 | 24   | 0.997 | $\text{Res}_{s=p} \tilde{L}(\sigma(s)) = -\frac{1}{(2\pi)^d} \int_{\mathbb{R}^d} \sigma(p)(\xi) d\xi$                                                                    | $\text{Res}_{s=p} \tilde{L}(\sigma(s)) = -\frac{1}{(2\pi)^d} \int_{\mathbb{R}^d} \sigma(p)(\xi) d\xi$                                                                    | We are now ready to get the main result Wodzicki. <sup>109,110</sup>                                                                                                                                                                                                                                |
| cardona-qft-methods_F00297 | 24   | 1.000 | $\{109,110\}$                                                                                                                                                            | 109,110                                                                                                                                                                  | a vect                                                                                                                                                                                                                                                                                              |
| cardona-qft-methods_F00298 | 24   | 1.000 | $\mathcal{D} \in \Psi DO(M, E)$                                                                                                                                          | $\mathcal{D} \in \Psi DO(M, E)$                                                                                                                                          | the class of pseudodifferential operators whose symbols are in $\mathcal{O}_{\text{int}}$ , i.e. integrable in the $\xi$ -variable.                                                                                                                                                                 |
| cardona-qft-methods_F00299 | 24   | 0.935 | $\Psi DO_{\text{int}}(M, E) = \{Q \in \Psi DO^{\mathbb{C}}(M, E) \mid \Re(\text{order}(Q)) < -d\}$                                                                       | $\Psi DO_{\text{int}}(M, E) = \{Q \in \Psi DO^{\mathbb{C}}(M, E) \mid \Re(\text{order}(Q)) < -d\}$                                                                       | $\text{WRes} P = \text{Res} \text{Tr}(P \mathcal{D} ^{-s}). \quad (7)$                                                                                                                                                                                                                              |
| cardona-qft-methods_F00300 | 24   | 1.000 | $P \in \Psi DO_{\text{int}}(M, E)$                                                                                                                                       | $P \in \Psi DO_{\text{int}}(M, E)$                                                                                                                                       | <b>Theorem 2.4.</b> (i) The map $P \in \Psi DO_{\text{int}}(M, E) \rightarrow \text{Tr}(P) \in \mathbb{C}$ has a                                                                                                                                                                                    |
| cardona-qft-methods_F00301 | 24   | 1.000 | $k^P(x, x)$                                                                                                                                                              | $k^P(x, x)$                                                                                                                                                              | <b>Theorem 2.4.</b> (i) The map $P \in \Psi DO_{\text{int}}(M, E) \rightarrow \text{Tr}(P) \in \mathbb{C}$ has a                                                                                                                                                                                    |
| cardona-qft-methods_F00302 | 24   | 1.000 | $M \times M$                                                                                                                                                             | $M \times M$                                                                                                                                                             | unique analytic extension on $\Psi DO^{-s}(M, E)$ .<br>(ii) If $P \in \Psi DO^{\mathbb{C}}(M, E)$ , the map: $s \in \mathbb{C} \rightarrow \text{Tr}(P \mathcal{D} ^{-s})$ has a                                                                                                                    |

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| cardona-qft-methods_F00303 | 24   | ■ 1.000 | $\operatorname{End}(E)$                                                                    | $\operatorname{End}(E)$                                                                    | <i>unique analytic extension on <math>\Psi DO^{\mathbb{Z}}(M, E)</math>.<br/>(ii) If <math>P \in \Psi DO^{\mathbb{Z}}(M, E)</math>, the map: <math>s \in \mathbb{C} \rightarrow \operatorname{Tr}(P \mathcal{D} ^{-s})</math> has at</i>                                                                       |
| cardona-qft-methods_F00304 | 24   | ■ 1.000 | $P \in \Psi DO^{\mathbb{Z}}(M, E)$                                                         | $P \in \Psi DO^{\mathbb{Z}}(M, E)$                                                         | <i>simple poles on <math>\mathbb{Z}</math> and</i>                                                                                                                                                                                                                                                             |
| cardona-qft-methods_F00305 | 24   | ■ 1.000 | $P \in \Psi DO_{\operatorname{int}}(M, E) \rightarrow \operatorname{Tr}(P) \in \mathbb{C}$ | $P \in \Psi DO_{\operatorname{int}}(M, E) \rightarrow \operatorname{Tr}(P) \in \mathbb{C}$ |                                                                                                                                                                                                                                                                                                                |
| cardona-qft-methods_F00306 | 24   | ■ 1.000 | $\Psi DO^{\mathbb{C}\backslash\mathbb{Z}}(M, E)$                                           | $\Psi DO^{\mathbb{C}\backslash\mathbb{Z}}(M, E)$                                           |                                                                                                                                                                                                                                                                                                                |
| cardona-qft-methods_F00307 | 24   | ■ 1.000 | $s \in \mathbb{C} \rightarrow \operatorname{Tr}(P \mathcal{D} ^{-s})$                      | $s \in \mathbb{C} \rightarrow \operatorname{Tr}(P \mathcal{D} ^{-s})$                      |                                                                                                                                                                                                                                                                                                                |
| cardona-qft-methods_F00308 | 25   | ■ 1.000 | $c_P(x) =$                                                                                 | $c_P(x) =$                                                                                 |                                                                                                                                                                                                                                                                                                                |
| cardona-qft-methods_F00309 | 25   | ■ 1.000 | $\frac{1}{(2\pi)^d} \int_{\mathbb{S}^{d-1}} \operatorname{tr}(\sigma_{-d}^P(x, \xi)) d\xi$ | $\frac{1}{(2\pi)^d} \int_{\mathbb{S}^{d-1}} \operatorname{tr}(\sigma_{-d}^P(x, \xi)) d\xi$ | <i>1d - R Isosium</i>                                                                                                                                                                                                                                                                                          |
| cardona-qft-methods_F00310 | 25   | ■ 1.000 | $\Psi DO^{\mathbb{Z}}(M, E)$                                                               | $\Psi DO^{\mathbb{Z}}(M, E)$                                                               |                                                                                                                                                                                                                                                                                                                |
| cardona-qft-methods_F00311 | 25   | ■ 1.000 | $s \rightarrow \operatorname{Tr}(P \mathcal{D} ^{-s})$                                     | $s \rightarrow \operatorname{Tr}(P \mathcal{D} ^{-s})$                                     | $\frac{1}{(2\pi)^d} \int_{\mathbb{S}^{d-1}} \operatorname{tr}(\sigma_{-d}^P(x, \xi)) d\xi$ .<br>(iii) $WRes$ is a trace on the algebra $\Psi DO^{\mathbb{Z}}(M, E)$ .                                                                                                                                          |
| cardona-qft-methods_F00312 | 25   | ■ 1.000 | $P \in \Psi DO^{\mathbb{C}\backslash\mathbb{Z}}(M, E)$                                     | $P \in \Psi DO^{\mathbb{C}\backslash\mathbb{Z}}(M, E)$                                     | <i>(iii) <math>WRes</math> is a trace on the algebra <math>\Psi DO^{\mathbb{Z}}(M, E)</math>.</i>                                                                                                                                                                                                              |
| cardona-qft-methods_F00313 | 25   | ■ 1.000 | $\Psi DO_{\operatorname{int}}(M, E)$                                                       | $\Psi DO_{\operatorname{int}}(M, E)$                                                       | <i>(iii) <math>WRes</math> is a trace on the algebra <math>\Psi DO^{\mathbb{Z}}(M, E)</math>.</i>                                                                                                                                                                                                              |
| cardona-qft-methods_F00314 | 25   | ■ 1.000 | $s=0$                                                                                      | $s=0$                                                                                      | (2) First, fix $s=0$ . We are interested in the function $L_\phi(\sigma) = \operatorname{Tr}(\phi\sigma(x, D))$ with $\sigma \in \mathcal{S}'(T \times \mathbb{R}^d)$ and $\phi \in C^\infty(T)$ . For instance, if                                                                                            |
| cardona-qft-methods_F00315 | 25   | ■ 1.000 | $L_\phi(\sigma) =$                                                                         | $L_\phi(\sigma) =$                                                                         | (2) First, fix $s=0$ . We are interested in the function $L_\phi(\sigma) = \operatorname{Tr}(\phi\sigma(x, D))$ with $\sigma \in \mathcal{S}'(T \times \mathbb{R}^d)$ and $\phi \in C^\infty(T)$ . For instance, if                                                                                            |
| cardona-qft-methods_F00316 | 25   | ■ 0.999 | $\operatorname{Tr}(\phi\sigma(x, D))$                                                      | $\operatorname{Tr}(\phi\sigma(x, D))$                                                      | $\mathbb{H}\{\psi\psi(x, D)\}$ with $\psi \in \mathcal{D}'_{\operatorname{int}}(U \times \mathbb{R}^d)$ and $\psi \in \mathcal{C}'(U)$ . For instance, if $P = \sigma(\cdot, D)$ ,                                                                                                                             |
| cardona-qft-methods_F00317 | 25   | ■ 0.999 | $\sigma \in S_{\operatorname{int}}(U \times \mathbb{R}^d)$                                 | $\sigma \in S_{\operatorname{int}}(U \times \mathbb{R}^d)$                                 | $\mathbb{H}\{\psi\psi(x, D)\}$ with $\psi \in \mathcal{D}'_{\operatorname{int}}(U \times \mathbb{R}^d)$ and $\psi \in \mathcal{C}'(U)$ . For instance, if $P = \sigma(\cdot, D)$ ,                                                                                                                             |
| cardona-qft-methods_F00318 | 25   | ■ 0.999 | $\phi \in C^\infty(U)$                                                                     | $\phi \in C^\infty(U)$                                                                     | $\mathbb{H}\{\psi\psi(x, D)\}$ with $\psi \in \mathcal{D}'_{\operatorname{int}}(U \times \mathbb{R}^d)$ and $\psi \in \mathcal{C}'(U)$ . For instance, if $P = \sigma(\cdot, D)$ ,                                                                                                                             |
| cardona-qft-methods_F00319 | 25   | ■ 1.000 | $P = \sigma(\cdot, D)$                                                                     | $P = \sigma(\cdot, D)$                                                                     |                                                                                                                                                                                                                                                                                                                |
| cardona-qft-methods_F00320 | 25   | ■ 0.999 | $L_\phi$                                                                                   | $L_\phi$                                                                                   | uniform bounds in $x$ , thus we get, via Lemma 2.2 applied to $L_\phi(\sigma(s)(x, \cdot))$ uniformly in $x$ , yielding a natural extension to $\tilde{L}_\phi(\sigma(s))$ which is holomor-                                                                                                                   |
| cardona-qft-methods_F00321 | 25   | ■ 0.999 | $S^{\mathbb{C}\backslash\mathbb{Z}}(U \times \mathbb{R}^d)$                                | $S^{\mathbb{C}\backslash\mathbb{Z}}(U \times \mathbb{R}^d)$                                | uniform bounds in $x$ , thus we get, via Lemma 2.2 applied to $L_\phi(\sigma(s)(x, \cdot))$ uniformly in $x$ , yielding a natural extension to $\tilde{L}_\phi(\sigma(s))$ which is holomor-                                                                                                                   |
| cardona-qft-methods_F00322 | 25   | ■ 0.999 | $\tilde{L}_\phi(\sigma) =$                                                                 | $\tilde{L}_\phi(\sigma) =$                                                                 | uniform bounds in $x$ , thus we get, via Lemma 2.2 applied to $L_\phi(\sigma(s)(x, \cdot))$ uniformly in $x$ , yielding a natural extension to $\tilde{L}_\phi(\sigma(s))$ which is holomor-                                                                                                                   |
| cardona-qft-methods_F00323 | 25   | ■ 0.998 | $\int_U \phi(x) \tilde{L}_\phi(\sigma(x, \cdot))  dx $                                     | $\int_U \phi(x) \tilde{L}_\phi(\sigma(x, \cdot))  dx $                                     | <i>phic on <math>\mathbb{C}\backslash\mathbb{Z}</math>.<br/>When <math>\operatorname{order}(\sigma(s)) = s</math>,</i>                                                                                                                                                                                         |
| cardona-qft-methods_F00324 | 25   | ■ 1.000 | $\sigma(x, \xi) = \sigma(s)(x, \xi)$                                                       | $\sigma(x, \xi) = \sigma(s)(x, \xi)$                                                       | When $\operatorname{order}(\sigma(s)) = s$ , the map $L_\phi(\sigma(s))$ has at most simple poles on $\mathbb{Z}$ and for each $n \in \mathbb{Z}$ , $\operatorname{Res} \tilde{L}_\phi(\sigma(s)) = -\frac{1}{2\pi i} \int_{\mathbb{S}^{d-1}} \phi(x) \sigma_{-d}(x, \xi) d\xi  dx  =$                         |
| cardona-qft-methods_F00325 | 25   | ■ 1.000 | $s$                                                                                        | $s$                                                                                        | When $\operatorname{order}(\sigma(s)) = s$ , the map $L_\phi(\sigma(s))$ has at most simple poles on $\mathbb{Z}$ and for each $n \in \mathbb{Z}$ , $\operatorname{Res} \tilde{L}_\phi(\sigma(s)) = -\frac{1}{2\pi i} \int_{\mathbb{S}^{d-1}} \phi(x) \sigma_{-d}(x, \xi) d\xi  dx  =$                         |
| cardona-qft-methods_F00326 | 25   | ■ 1.000 | $x$                                                                                        | $x$                                                                                        | for each $P \in \mathcal{D}'_{\operatorname{int}}$ , $\operatorname{Res} L_\phi(\sigma(s)) = -\frac{1}{(2\pi)^d} \int_U \int_{\mathbb{S}^{d-1}} \psi(x) \psi(-d(P)(x, \xi)) \omega_{\mathbb{S}^{d-1}}  dx  = -\int_{\cdot, \cdot} \phi(x) c_{P_n}(x)  dx $ where we used (5) with $P = Op(\sigma_n(x, \xi))$ . |
| cardona-qft-methods_F00327 | 25   | ■ 1.000 | $\tilde{L}_\phi(\sigma(s)(x, \cdot))$                                                      | $\tilde{L}_\phi(\sigma(s)(x, \cdot))$                                                      | for each $P \in \mathcal{D}'_{\operatorname{int}}$ , $\operatorname{Res} L_\phi(\sigma(s)) = -\frac{1}{(2\pi)^d} \int_U \int_{\mathbb{S}^{d-1}} \psi(x) \psi(-d(P)(x, \xi)) \omega_{\mathbb{S}^{d-1}}  dx  = -\int_{\cdot, \cdot} \phi(x) c_{P_n}(x)  dx $ where we used (5) with $P = Op(\sigma_n(x, \xi))$ . |

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| cardona-qft-methods_F00328 | 25   | ■ 1.000 | $\widetilde{L}_{\phi}(\sigma(s))$                                                                                                                                            | $\widetilde{L}_{\phi}(\sigma(s))$                                                                                                                                            | (4) In the general case, we get a unique meromorphic extension of the                                                                                                                                                                                                                                                                                   |
| cardona-qft-methods_F00329 | 25   | ■ 1.000 | $\mathbb{C} \backslash \mathbb{Z}$                                                                                                                                           | $\mathbb{C} \backslash \mathbb{Z}$                                                                                                                                           | usi                                                                                                                                                                                                                                                                                                                                                     |
| cardona-qft-methods_F00330 | 25   | ■ 1.000 | $\operatorname{order}(\sigma(s))=s$                                                                                                                                          | $\operatorname{order}(\sigma(s)) = s$                                                                                                                                        | $s$ , then $\operatorname{Tr}(P(s))$ has at most poles on $\mathbb{Z}$ and $\operatorname{Res}_{s=p} \operatorname{Tr}(P(s)) =$                                                                                                                                                                                                                         |
| cardona-qft-methods_F00331 | 25   | ■ 1.000 | $p \in \mathbb{Z}, \operatorname{Res}_{s=p} \widetilde{L}_{\phi}(\sigma(s)) = -\frac{1}{(2\pi)^d} \int_U \int_{\mathbb{S}^{d-1}} \phi(x) \sigma_{-d}(p)(x, \xi) d\xi  dx  =$ | $p \in \mathbb{Z}, \operatorname{Res}_{s=p} \widetilde{L}_{\phi}(\sigma(s)) = -\frac{1}{(2\pi)^d} \int_U \int_{\mathbb{S}^{d-1}} \phi(x) \sigma_{-d}(p)(x, \xi) d\xi  dx  =$ | $-\int_M c_{P(p)}(x)  dx $ for $p \in \mathbb{Z}$ . So we get the claim for the family $P(s) = P \mathcal{D} ^{-s}$ .                                                                                                                                                                                                                                   |
| cardona-qft-methods_F00332 | 25   | ■ 1.000 | $\int_U \phi(x) c_{P_p}(x)  dx $                                                                                                                                             | $\int_U \phi(x) c_{P_p}(x)  dx $                                                                                                                                             | (iii) Let $P_1, P_2 \in \Psi DO^{\mathbb{Z}}(M, E)$ . Since $\operatorname{Tr}$ is a trace on $\Psi DO^{\mathbb{C} \backslash \mathbb{Z}}(M, E)$ , we get by (i) $\operatorname{Tr}(P_1 P_2  \mathcal{D} ^{-s}) = \operatorname{Tr}(P_2  \mathcal{D} ^{-s} P_1)$ . Moreover                                                                             |
| cardona-qft-methods_F00333 | 25   | ■ 1.000 | $P = Op(\sigma_p(x, \xi))$                                                                                                                                                   | $P = Op(\sigma_p(x, \xi))$                                                                                                                                                   | (iii) Let $P_1, P_2 \in \Psi DO^{\mathbb{Z}}(M, E)$ . Since $\operatorname{Tr}$ is a trace on $\Psi DO^{\mathbb{C} \backslash \mathbb{Z}}(M, E)$ , we get by (i) $\operatorname{Tr}(P_1 P_2  \mathcal{D} ^{-s}) = \operatorname{Tr}(P_2  \mathcal{D} ^{-s} P_1)$ . Moreover                                                                             |
| cardona-qft-methods_F00334 | 25   | ■ 1.000 | $\operatorname{Tr}$                                                                                                                                                          | $\operatorname{Tr}$                                                                                                                                                          | $\operatorname{Tr} \operatorname{Res}_{s=0} (1+2) = \operatorname{Res}_{s=0} \operatorname{Tr} (1+2)  \mathcal{D} ^{-s} = \operatorname{Tr} \operatorname{Res}_{s=0} (1+2)  \mathcal{D} ^{-s} = \operatorname{Tr} \operatorname{Res}_{s=0} (1+2) = \operatorname{Tr} \operatorname{Res}_{s=0} (1+2) = \operatorname{Tr} \operatorname{Res}_{s=0} (1+2)$ |
| cardona-qft-methods_F00335 | 25   | ■ 1.000 | $P: \mathbb{C} \rightarrow \Psi DO^{\mathbb{Z}}(M, E)$                                                                                                                       | $P: \mathbb{C} \rightarrow \Psi DO^{\mathbb{Z}}(M, E)$                                                                                                                       | where for the second equality we used (8) so the residue depends only of the value of $P(s)$ at $s = 0$ . $\square$                                                                                                                                                                                                                                     |
| cardona-qft-methods_F00336 | 25   | ■ 1.000 | $\operatorname{order}(P(s)) =$                                                                                                                                               | $\operatorname{order}(P(s)) =$                                                                                                                                               | where for the second equality we used (8) so the residue depends only of the value of $P(s)$ at $s = 0$ . $\square$                                                                                                                                                                                                                                     |
| cardona-qft-methods_F00337 | 25   | ■ 0.814 | $\operatorname{Tr}(P(s))$                                                                                                                                                    | $\operatorname{Tr}(P(s))$                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                                                         |
| cardona-qft-methods_F00338 | 25   | ■ 0.814 | $\operatorname{Res}_{s=p}^{\operatorname{Tr}}(P(s)) =$                                                                                                                       | $\operatorname{Res}_{s=p}^{\operatorname{Tr}}(P(s)) =$                                                                                                                       |                                                                                                                                                                                                                                                                                                                                                         |
| cardona-qft-methods_F00339 | 25   | ■ 0.987 | $-\int_M c_{P(p)}(x)  dx $                                                                                                                                                   | $-\int_M c_{P(p)}(x)  dx $                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                                                         |
| cardona-qft-methods_F00340 | 25   | ■ 0.987 | $P(s) =$                                                                                                                                                                     | $P(s) =$                                                                                                                                                                     |                                                                                                                                                                                                                                                                                                                                                         |
| cardona-qft-methods_F00341 | 25   | ■ 1.000 | $P \mathcal{D} ^{-s}$                                                                                                                                                        | $P \mathcal{D} ^{-s}$                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                         |
| cardona-qft-methods_F00342 | 25   | ■ 0.997 | $P_1, P_2 \in \Psi DO^{\mathbb{Z}}(M, E)$                                                                                                                                    | $P_1, P_2 \in \Psi DO^{\mathbb{Z}}(M, E)$                                                                                                                                    |                                                                                                                                                                                                                                                                                                                                                         |
| cardona-qft-methods_F00343 | 25   | ■ 0.956 | (i), $\operatorname{Tr}(P_1 P_2  \mathcal{D} ^{-s}) = \operatorname{Tr}(P_2  \mathcal{D} ^{-s} P_1)$                                                                         | $(i), \operatorname{Tr}(P_1 P_2  \mathcal{D} ^{-s}) = \operatorname{Tr}(P_2  \mathcal{D} ^{-s} P_1)$                                                                         |                                                                                                                                                                                                                                                                                                                                                         |
| cardona-qft-methods_F00344 | 25   | ■ 1.000 | $P(s)$                                                                                                                                                                       | $P(s)$                                                                                                                                                                       |                                                                                                                                                                                                                                                                                                                                                         |
| cardona-qft-methods_F00345 | 25   | —       | $\square$                                                                                                                                                                    | $\square$                                                                                                                                                                    | —                                                                                                                                                                                                                                                                                                                                                       |
| cardona-qft-methods_F00346 | 26   | ■ 0.852 | $\Psi DO^{-m}(M, E), m \in \mathbb{N}$                                                                                                                                       | $\Psi DO^{-m}(M, E), m \in \mathbb{N}$                                                                                                                                       | <b>Corollary 2.2.</b> The Wodzicki residue $WRes$ is the only trace (up to mul.                                                                                                                                                                                                                                                                         |
| cardona-qft-methods_F00347 | 26   | ■ 1.000 | $d \geq 2$                                                                                                                                                                   | $d \geq 2$                                                                                                                                                                   | multiplication by a $\alpha$                                                                                                                                                                                                                                                                                                                            |
| cardona-qft-methods_F00348 | 26   | ■ 1.000 | $d=1, T^*M$                                                                                                                                                                  | $d = 1, T^*M$                                                                                                                                                                | $d = 1, T^*M$ is disconnected and they are two residues.                                                                                                                                                                                                                                                                                                |
| cardona-qft-methods_F00349 | 26   | ■ 1.000 | $\sigma_{-d}^P = 0$                                                                                                                                                          | $\sigma_{-d}^P = 0$                                                                                                                                                          | $\sum_{k=1}^d \xi_k \partial_{\xi_k} \sigma_{m-j}^P = (m-j) \sigma_{m-j}^P$ (this is false for $m = j!$ ) and                                                                                                                                                                                                                                           |
| cardona-qft-methods_F00350 | 26   | ■ 1.000 | $\sigma^P(x, \xi)$                                                                                                                                                           | $\sigma^P(x, \xi)$                                                                                                                                                           | $\sum_{k=1}^d \xi_k \partial_{\xi_k} \sigma_{m-j}^P = (m-j) \sigma_{m-j}^P$ (this is false for $m = j!$ ) and                                                                                                                                                                                                                                           |
| cardona-qft-methods_F00351 | 26   | ■ 0.998 | $\sigma^P \simeq \sum_j \sigma_{m-j}^P$                                                                                                                                      | $\sigma^P \simeq \sum_j \sigma_{m-j}^P$                                                                                                                                      | $\sum_{k=1}^d [\xi_k, \xi_k \sigma_{m-j}^P] = i \sum_{k=1}^d \partial_{\xi_k} \xi_k \sigma_{m-j}^P = i(m-j+d) \sigma_{m-j}^P$ . So $\sigma^P =$                                                                                                                                                                                                         |

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| cardona-qft-methods_F00352 | 26   | ■ 0.998 | $m=\operatorname{order}(P)$                                                                                                                                                                                                                                | $m = \operatorname{order}(P)$                                                                                               | $\sum_{k=1}^a [x^k, \xi_k \sigma_{m-j}^P] = i \sum_{k=1}^a \partial_{\xi_k} \xi_k \sigma_{m-j}^P = i(m-j+d) \sigma_{m-j}^P$ . So $\sigma^P =$                                                                                                                                                                                  |
| cardona-qft-methods_F00353 | 26   | ■ 1.000 | $\sum_{k=1}^d \xi_k \partial_{\xi_k} \sigma_{m-j}^P = (m-j) \sigma_{m-j}^P$<br>$\sigma_{m-j}^P = (m-j) \sigma_{m-j}^P$                                                                                                                                     | $\sum_{k=1}^d \xi_k \partial_{\xi_k} \sigma_{m-j}^P = (m-j) \sigma_{m-j}^P$                                                 | $\sum_{k=1}^d \xi_k \partial_{\xi_k} \sigma_{m-j}^P$ .<br>Let $T$ be another trace on $\Psi DO^{\mathbb{Z}}(M, E)$ . Then $T(P)$ depends only on                                                                                                                                                                               |
| cardona-qft-methods_F00354 | 26   | ■ 1.000 | $m=j!$                                                                                                                                                                                                                                                     | $m = j!$                                                                                                                    | $\sum_{k=1}^d \xi_k \partial_{\xi_k} \sigma_{m-j}^P$ .<br>Let $T$ be another trace on $\Psi DO^{\mathbb{Z}}(M, E)$ . Then $T(P)$ depends only on                                                                                                                                                                               |
| cardona-qft-methods_F00355 | 26   | ■ 1.000 | $\sum_{k=1}^d \left[ x^k, \xi_k \sigma_{m-j}^P \right] = i \sum_{k=1}^d \partial_{\xi_k} \xi_k \sigma_{m-j}^P = i(m-j+d) \sigma_{m-j}^P$<br>$\sum_{k=1}^d \left[ x^k, \xi_k \sigma_{m-j}^P \right] = i \sum_{k=1}^d \partial_{\xi_k} \xi_k \sigma_{m-j}^P$ | $\sum_{k=1}^d [x^k, \xi_k \sigma_{m-j}^P] = i \sum_{k=1}^d \partial_{\xi_k} \xi_k \sigma_{m-j}^P = i(m-j+d) \sigma_{m-j}^P$ | $\sigma_{-d}^P$ because $T([\cdot, \cdot]) = 0$ .<br>3) We have $\int_{\mathbb{S}^{d-1}} \sigma_{-d}^P(x, \xi) d \xi  = 0$ if and only if $\sigma_{-d}^P$ is sum of deriva-                                                                                                                                                    |
| cardona-qft-methods_F00356 | 26   | ■ 1.000 | $\sigma^P =$                                                                                                                                                                                                                                               | $\sigma^P =$                                                                                                                | $\sigma_{-d}^P$ because $T([\cdot, \cdot]) = 0$ .<br>3) We have $\int_{\mathbb{S}^{d-1}} \sigma_{-d}^P(x, \xi) d \xi  = 0$ if and only if $\sigma_{-d}^P$ is sum of deriva-                                                                                                                                                    |
| cardona-qft-methods_F00357 | 26   | ■ 1.000 | $\sum_{k=1}^d [\xi_k \tau, x^k]$                                                                                                                                                                                                                           | $\sum_{k=1}^d [\xi_k \tau, x^k]$                                                                                            | $\sum_{k=1}^d [\xi_k \tau, x^k]$                                                                                                                                                                                                                                                                                               |
| cardona-qft-methods_F00358 | 26   | ■ 1.000 | $T$                                                                                                                                                                                                                                                        | $T$                                                                                                                         | The $u$ part is direct. Only $u$ part: $\sigma_{-d}$ is orthogonal to constant functions on the sphere $\mathbb{S}^{d-1}$ and these are kernels of the Laplacian: $\Delta_{\mathbb{S}} f = 0 \iff$                                                                                                                             |
| cardona-qft-methods_F00359 | 26   | ■ 1.000 | $T(P)$                                                                                                                                                                                                                                                     | $T(P)$                                                                                                                      | The $u$ part is direct. Only $u$ part: $\sigma_{-d}$ is orthogonal to constant functions on the sphere $\mathbb{S}^{d-1}$ and these are kernels of the Laplacian: $\Delta_{\mathbb{S}} f = 0 \iff$                                                                                                                             |
| cardona-qft-methods_F00360 | 26   | ■ 1.000 | $\sigma_{-d}^P$                                                                                                                                                                                                                                            | $\sigma_{-d}^P$                                                                                                             | $df = 0 \iff f = cst$<br>$\int_{\mathbb{S}^{d-1}} \sigma_{-d}^P(x, \xi) d \xi  = 0$                                                                                                                                                                                                                                            |
| cardona-qft-methods_F00361 | 26   | ■ 1.000 | $T([\cdot, \cdot])=0$                                                                                                                                                                                                                                      | $T([\cdot, \cdot]) = 0$                                                                                                     | $df = 0 \iff f = cst$<br>$\int_{\mathbb{S}^{d-1}} \sigma_{-d}^P(x, \xi) d \xi  = 0$                                                                                                                                                                                                                                            |
| cardona-qft-methods_F00362 | 26   | ■ 1.000 | $\int_{\mathbb{S}^{d-1}} \sigma_{-d}^P(x, \xi) d \xi  = 0$                                                                                                                                                                                                 | $\int_{\mathbb{S}^{d-1}} \sigma_{-d}^P(x, \xi) d \xi  = 0$                                                                  | $h(\xi) =  \xi ^{-d+2} h(\frac{\xi}{ \xi })$ is its extension to $\mathbb{R}^d \setminus \{0\}$ , then we get $\Delta_{\mathbb{R}^d} h(\xi) =  \xi  \sigma_{-d}^P(\frac{\xi}{ \xi }) = \sigma_{-d}^P(\xi)$ because $\Delta_{\mathbb{R}^d} = r^{1-d} \partial_r (r^{d-1} \partial_r) + r^{-2} \Delta_{\mathbb{S}^{d-1}}$ . This |
| cardona-qft-methods_F00363 | 26   | ■ 1.000 | $\Delta_{\mathbb{S}} f = 0 \iff$                                                                                                                                                                                                                           | $\Delta_{\mathbb{S}} f = 0 \iff$                                                                                            | $\Delta_{\mathbb{S}} f = 0 \iff$                                                                                                                                                                                                                                                                                               |
| cardona-qft-methods_F00364 | 26   | ■ 0.987 | $df = 0 \iff f = cst$                                                                                                                                                                                                                                      | $df = 0 \iff f = cst$                                                                                                       | $\sigma_{-d} = \sum_{k=1}^d \partial_{\xi_k} \mu - \sum_{k=1}^d \partial_{\xi_k} \mu, \sigma_{-d}$                                                                                                                                                                                                                             |
| cardona-qft-methods_F00365 | 26   | ■ 0.987 | $\Delta_{\mathbb{S}^{d-1}} h = \sigma_{-d \mathbb{S}^{d-1}}^P$                                                                                                                                                                                             | $\Delta_{\mathbb{S}^{d-1}} h = \sigma_{-d \mathbb{S}^{d-1}}^P$                                                              | $\sigma_{-d} = \sum_{k=1}^d \partial_{\xi_k} \mu - \sum_{k=1}^d \partial_{\xi_k} \mu, \sigma_{-d}$                                                                                                                                                                                                                             |
| cardona-qft-methods_F00366 | 26   | ■ 0.987 | $h$                                                                                                                                                                                                                                                        | $h$                                                                                                                         | $\sigma_{-d} = \sum_{k=1}^d \partial_{\xi_k} \mu - \sum_{k=1}^d \partial_{\xi_k} \mu, \sigma_{-d}$                                                                                                                                                                                                                             |
| cardona-qft-methods_F00367 | 26   | ■ 1.000 | $\tilde{h}(\xi) =  \xi ^{-d+2} h\left(\frac{\xi}{ \xi }\right)$                                                                                                                                                                                            | $\tilde{h}(\xi) =  \xi ^{-d+2} h\left(\frac{\xi}{ \xi }\right)$                                                             | is a sum of commutators.<br>$\int_{\mathbb{S}^{d-1}} \sigma_{-d}^P(x, \xi) d \xi  = 0$                                                                                                                                                                                                                                         |
| cardona-qft-methods_F00368 | 26   | ■ 1.000 | $\Delta_{\mathbb{R}^d} \tilde{h}(\xi) =$                                                                                                                                                                                                                   | $\Delta_{\mathbb{R}^d} \tilde{h}(\xi) =$                                                                                    | is a sum of commutators.<br>$\int_{\mathbb{S}^{d-1}} \sigma_{-d}^P(x, \xi) d \xi  = 0$                                                                                                                                                                                                                                         |
| cardona-qft-methods_F00369 | 26   | ■ 1.000 | $ \xi  \sigma_{-d}^P\left(\frac{\xi}{ \xi }\right) = \sigma_{-d}^P(\xi)$                                                                                                                                                                                   | $ \xi  \sigma_{-d}^P\left(\frac{\xi}{ \xi }\right) = \sigma_{-d}^P(\xi)$                                                    | 4) End of proof: the symbol $\sigma_{-d}^P(x, \xi) = \frac{1}{\sqrt{d(d+1)}} c_P(x)$ is of order $-d$ with zero integral, thus is a sum of commutators by 3) and $T(P) =$                                                                                                                                                      |
| cardona-qft-methods_F00370 | 26   | ■ 1.000 | $\Delta_{\mathbb{R}^d} = r^{1-d} \partial_r (r^{d-1} \partial_r) + r^{-2} \Delta_{\mathbb{S}^{d-1}}$                                                                                                                                                       | $\Delta_{\mathbb{R}^d} = r^{1-d} \partial_r (r^{d-1} \partial_r) + r^{-2} \Delta_{\mathbb{S}^{d-1}}$                        | 4) End of proof: the symbol $\sigma_{-d}^P(x, \xi) = \frac{1}{\sqrt{d(d+1)}} c_P(x)$ is of order $-d$ with zero integral, thus is a sum of commutators by 3) and $T(P) =$                                                                                                                                                      |
| cardona-qft-methods_F00371 | 26   | ■ 1.000 | $\tilde{h}$                                                                                                                                                                                                                                                | $\tilde{h}$                                                                                                                 | $T(Op( \xi ^{-d} c_P(x)), \forall T \in \Psi DO^{\mathbb{Z}}(M, E)$ . So the map $\mu : f \in C_c^\infty(U) \rightarrow T(Op(f) \xi ^{-d})$ is linear, continuous and satisfies $\mu(\partial_r f) = 0$ because                                                                                                                |
| cardona-qft-methods_F00372 | 26   | ■ 1.000 | $d-2$                                                                                                                                                                                                                                                      | $d-2$                                                                                                                       | $T(Op( \xi ^{-d} c_P(x)), \forall T \in \Psi DO^{\mathbb{Z}}(M, E)$ . So the map $\mu : f \in C_c^\infty(U) \rightarrow T(Op(f) \xi ^{-d})$ is linear, continuous and satisfies $\mu(\partial_r f) = 0$ because                                                                                                                |
| cardona-qft-methods_F00373 | 26   | ■ 1.000 | $\partial_{\xi} \tilde{h}$                                                                                                                                                                                                                                 | $\partial_{\xi} \tilde{h}$                                                                                                  | $T(Op( \xi ^{-d} c_P(x)), \forall T \in \Psi DO^{\mathbb{Z}}(M, E)$ . So the map $\mu : f \in C_c^\infty(U) \rightarrow T(Op(f) \xi ^{-d})$ is linear, continuous and satisfies $\mu(\partial_r f) = 0$ because                                                                                                                |
| cardona-qft-methods_F00374 | 26   | ■ 1.000 | $d-1$                                                                                                                                                                                                                                                      | $d-1$                                                                                                                       | $T(Op( \xi ^{-d} c_P(x)), \forall T \in \Psi DO^{\mathbb{Z}}(M, E)$ . So the map $\mu : f \in C_c^\infty(U) \rightarrow T(Op(f) \xi ^{-d})$ is linear, continuous and satisfies $\mu(\partial_r f) = 0$ because                                                                                                                |

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| cardona-qft-methods_F08375 | 26   | ■ 1.000 | $\sigma_{-d}^{\{P\}}(x,\xi) = \frac{ \xi ^{-d}}{\text{Vol}(\mathbb{S}^{d-1})} c_P(x)$                | $\sigma_{-d}^P(x,\xi) = \frac{ \xi ^{-d}}{\text{Vol}(\mathbb{S}^{d-1})} c_P(x)$                  |                                                                                                                                                                                                                                           |
| cardona-qft-methods_F08376 | 26   | ■ 1.000 | $-d$                                                                                                 | $-d$                                                                                             |                                                                                                                                                                                                                                           |
| cardona-qft-methods_F08377 | 26   | ■ 1.000 | $T(P)=$                                                                                              | $T(P) =$                                                                                         |                                                                                                                                                                                                                                           |
| cardona-qft-methods_F08378 | 26   | ■ 1.000 | $T\left(0\leq\left \xi\right ^{-d}c_p(x)\right),\forall T\in\Psi\mathcal{D}^0(M,E)$                  | $T\left(0\leq\left \xi\right ^{-d}c_p(x)\right),\forall T\in\Psi DO^{\mathbb{Z}}(M,E)$           |                                                                                                                                                                                                                                           |
| cardona-qft-methods_F08379 | 26   | ■ 1.000 | $\mu: f\in C_c^\infty(U)\rightarrow$                                                                 | $\mu: f\in C_c^\infty(U)\rightarrow$                                                             |                                                                                                                                                                                                                                           |
| cardona-qft-methods_F08380 | 26   | ■ 1.000 | $T\left(0\leq\left f\right \left \xi\right ^{-d}\right)$                                             | $T\left(0\leq\left f\right \left \xi\right ^{-d}\right)$                                         |                                                                                                                                                                                                                                           |
| cardona-qft-methods_F08381 | 26   | ■ 1.000 | $\mu\left(\partial_{x^k}f\right)=0$                                                                  | $\mu\left(\partial_{x^k}f\right)=0$                                                              |                                                                                                                                                                                                                                           |
| cardona-qft-methods_F08382 | 27   | ■ 1.000 | $\partial_{x^k}(f) \xi ^{-d}$                                                                        | $\partial_{x^k}(f) \xi ^{-d}$                                                                    |                                                                                                                                                                                                                                           |
| cardona-qft-methods_F08383 | 27   | ■ 1.000 | $\mu$                                                                                                | $\mu$                                                                                            |                                                                                                                                                                                                                                           |
| cardona-qft-methods_F08384 | 27   | ■ 1.000 | $(1+\Delta)^{-d/2}\in\Psi\mathcal{D}^0(M)$                                                           | $(1+\Delta)^{-d/2}\in\Psi DO(M)$                                                                 | <b>PROOF.</b> $(1+\Delta)^{-d/2}\in\Psi DO(M)$ has order $-d$ and its principal symbol $\sigma_{-d}(1+\Delta)^{-d/2}$ satisfies $\sigma_{-d}(1+\Delta)^{-d/2}(x,\xi)=-\left(g_x^{ij}\xi_i\xi_j\right)^{-d/2}=-\ \xi\ _x^{-d}$ . $\square$ |
| cardona-qft-methods_F08385 | 27   | ■ 1.000 | $\sigma_{-d}^{\{(1+\Delta)^{-d/2}\}}$                                                                | $\sigma_{-d}^{(1+\Delta)^{-d/2}}$                                                                | $\sigma_{-d}$ satisfies $\sigma_{-d}(x,\xi)=-\left(g_x^{ij}\xi_i\xi_j\right)^{-d/2}=-\ \xi\ _x^{-d}$ . $\square$ gives                                                                                                                    |
| cardona-qft-methods_F08386 | 27   | ■ 1.000 | $\sigma_{-d}^{\{(1+\Delta)^{-d/2}\}}(x,\xi)=-\left(g_x^{ij}\xi_i\xi_j\right)^{-d/2}=-\ \xi\ _x^{-d}$ | $\sigma_{-d}^{(1+\Delta)^{-d/2}}(x,\xi)=-\left(g_x^{ij}\xi_i\xi_j\right)^{-d/2}=-\ \xi\ _x^{-d}$ | $\sigma_{-d}$ satisfies $\sigma_{-d}(x,\xi)=-\left(g_x^{ij}\xi_i\xi_j\right)^{-d/2}=-\ \xi\ _x^{-d}$ . $\square$ gives                                                                                                                    |
| cardona-qft-methods_F08387 | 27   | ■ 0.675 | $\{ \}^{\{33,49,68,87,104,105\}}$                                                                    | 33,49,68,87,104,105                                                                              | additive map from $\mathcal{M}^+=\{aa^+\mid a\in\mathcal{M}\}$ to $\mathcal{M}$ . A state is a weight $\omega\in\mathcal{M}^*(\mathcal{M})$ s.t. $\omega(a)\geq 0$ , $\forall a\in\mathcal{M}^+$ .                                        |
| cardona-qft-methods_F08388 | 27   | ■ 1.000 | $\mathcal{B}(\mathcal{H})$                                                                           | $\mathcal{B}(\mathcal{H})$                                                                       | an increasing bounded net. This is equivalent to say that $\omega$ is lower semi-                                                                                                                                                         |
| cardona-qft-methods_F08389 | 27   | ■ 1.000 | $\omega$                                                                                             | $\omega$                                                                                         | continuous with respect to the $\sigma$ -weak topology.<br>In particular, the usual trace $\text{Tr}$ is normal on $\mathcal{B}(\mathcal{H})$ . Remark that the                                                                           |
| cardona-qft-methods_F08390 | 27   | ■ 1.000 | $\mathcal{M}$                                                                                        | $\mathcal{M}$                                                                                    | continuous with respect to the $\sigma$ -weak topology.<br>In particular, the usual trace $\text{Tr}$ is normal on $\mathcal{B}(\mathcal{H})$ . Remark that the                                                                           |
| cardona-qft-methods_F08391 | 27   | ■ 1.000 | $\mathcal{M}^+=\{a\,a^*\mid a\in\mathcal{M}\}$                                                       | $\mathcal{M}^+=\{aa^*\mid a\in\mathcal{M}\}$                                                     | net $(a_\alpha)_\alpha$ converges in $\mathcal{B}(\mathcal{H})$ and this property looks innoc                                                                                                                                             |
| cardona-qft-methods_F08392 | 27   | ■ 1.000 | $[0,\infty]$                                                                                         | $[0,\infty]$                                                                                     | net $(a_\alpha)_\alpha$ converges in $\mathcal{B}(\mathcal{H})$ and this property looks innoc                                                                                                                                             |
| cardona-qft-methods_F08393 | 27   | ■ 0.788 | $\omega\in\mathcal{M}^*$                                                                             | $\omega\in\mathcal{M}^*$                                                                         | preserves positivity.<br>Nevertheless it is natural to address the question: are all traces (in particular                                                                                                                                |
| cardona-qft-methods_F08394 | 27   | ■ 0.788 | $\omega(a)<\infty,\forall a\in\mathcal{M}$                                                           | $\omega(a)<\infty,\forall a\in\mathcal{M}$                                                       | preserves positivity.<br>Nevertheless it is natural to address the question: are all traces (in particular                                                                                                                                |
| cardona-qft-methods_F08395 | 27   | ■ 0.788 | $\omega(1)=1$                                                                                        | $\omega(1)=1$                                                                                    | preserves positivity.<br>Nevertheless it is natural to address the question: are all traces (in particular                                                                                                                                |
| cardona-qft-methods_F08396 | 27   | ■ 1.000 | $\omega(a\,a^*)=\omega(a^*\,a)$                                                                      | $\omega\left(aa^*\right)=\omega\left(a^*a\right)$                                                | is satisfied. It is natural to address the question: are all traces (as put forward on an arbitrary von Neumann algebra) normal? In 1966, Dixmier answer                                                                                  |
| cardona-qft-methods_F08397 | 27   | ■ 1.000 | $a\in\mathcal{M}$                                                                                    | $a\in\mathcal{M}$                                                                                | is satisfied. It is natural to address the question: are all traces (as put forward on an arbitrary von Neumann algebra) normal? In 1966, Dixmier answer                                                                                  |
| cardona-qft-methods_F08398 | 27   | ■ 0.993 | $\omega\left(\sup_{\alpha}a_{\alpha}\right)=\sup_{\alpha}\omega\left(a_{\alpha}\right)$              | $\omega\left(\sup_{\alpha}a_{\alpha}\right)=\sup_{\alpha}\omega\left(a_{\alpha}\right)$          | by the negative <sup>34</sup> by exhibiting non-normal, say singular, traces. Actually, his motivation was to answer the following related question: is any trace $\omega$                                                                |
| cardona-qft-methods_F08399 | 27   | ■ 0.993 | $\left(a_{\alpha}\right)\subset\mathcal{M}^+$                                                        | $(a_\alpha)\subset\mathcal{M}^+$                                                                 | by the negative <sup>34</sup> by exhibiting non-normal, say singular, traces. Actually, his motivation was to answer the following related question: is any trace $\omega$                                                                |

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| cardona-qft-methods_F00400 | 27   | ■ 1.000 | $\left(a_{\alpha}\right)_{\alpha}$                                                                         | $(a_{\alpha})_{\alpha}$                                                                                    |                                                                                                                                                                                 |
| cardona-qft-methods_F00401 | 28   | ■ 0.628 | $\operatorname{Tr}_{\mathrm{Dix}}(T) = \lim_{N \rightarrow \infty} \frac{1}{\log N} \sum_{n=0}^N \mu_n(T)$ | $\operatorname{Tr}_{\mathrm{Dix}}(T) = \lim_{N \rightarrow \infty} \frac{1}{\log N} \sum_{n=0}^N \mu_n(T)$ |                                                                                              |
| cardona-qft-methods_F00402 | 28   | ■ 0.628 | $\mu_n(T)$                                                                                                 | $\mu_n(T)$                                                                                                 |                                                                                              |
| cardona-qft-methods_F00403 | 28   | ■ 0.988 | $P \in \Psi D O^{-d}(M)$                                                                                   | $P \in \Psi D O^{-d}(M)$                                                                                   | this to the Wodzicki trace. It is a non-normal trace on some set that we                                                                                                        |
| cardona-qft-methods_F00404 | 28   | ■ 1.000 | $S^*M$                                                                                                     | $S^*M$                                                                                                     | where $S^*M$ is the cosphere bundle on $M$ .                                                                                                                                    |
| cardona-qft-methods_F00405 | 28   | ■ 1.000 | $\sum_{n \in \mathbb{N}^*} \frac{1}{n}$                                                                    | $\sum_{n \in \mathbb{N}^*} \frac{1}{n}$                                                                    | The physical motivation is quite essential: we know now $\sum_{n \in \mathbb{N}^*} \frac{1}{n}$ diverges and this is related to the fact the electromagnetic or Newton gravita- |
| cardona-qft-methods_F00406 | 28   | ■ 1.000 | $\frac{1}{r}$                                                                                              | $\frac{1}{r}$                                                                                              | as previous series). Actually, this (logarithmic-type) divergence appears everywhere in physics and explains the widely use of the Riemann zeta func-                           |
| cardona-qft-methods_F00407 | 28   | ■ 1.000 | $\zeta: s \in \mathbb{C} \rightarrow \sum_{n \in \mathbb{N}^*} \frac{1}{n^s}$                              | $\zeta: s \in \mathbb{C} \rightarrow \sum_{n \in \mathbb{N}^*} \frac{1}{n^s}$                              | We now have a quick review on the main properties of singular values                                                                                                            |
| cardona-qft-methods_F00408 | 28   | ■ 1.000 | $\zeta_P(s) = \operatorname{Tr} (P \mathcal{D} ^{-s})$                                                     | $\zeta_P(s) = \operatorname{Tr} (P \mathcal{D} ^{-s})$                                                     | <b>3.1 Singular values of compact operators</b>                                                                                                                                 |
| cardona-qft-methods_F00409 | 28   | ■ 1.000 | $T \in \mathcal{K}(\mathcal{H})$                                                                           | $T \in \mathcal{K}(\mathcal{H})$                                                                           | Moreover, we have following properties:                                                                                                                                         |
| cardona-qft-methods_F00410 | 28   | ■ 1.000 | $\mu_n(T) =$                                                                                               | $\mu_n(T) =$                                                                                               | Moreover, we have following properties:                                                                                                                                         |
| cardona-qft-methods_F00411 | 28   | ■ 0.707 | $\inf \left\{ \ T_{ E^\perp}\  \mid E \right\}$                                                            | $\inf \left\{ \ T_{ E^\perp}\  \mid E \right\}$                                                            | $\mu_n(T) = \mu_n(T^*) = \mu_n( T ),$<br>$T \in \mathcal{K}(\mathcal{H}) \implies \ T\  = \ T^*\  \implies \ T\  = \ T^*\ $                                                     |
| cardona-qft-methods_F00412 | 28   | ■ 0.707 | $\left\{ \dim(E) = n \right\}, n \in \mathbb{N}$                                                           | $\dim(E) = n, n \in \mathbb{N}$                                                                            | $\mu_n(T) = \mu_n(T^*) = \mu_n( T ),$<br>$T \in \mathcal{K}(\mathcal{H}) \implies \ T\  = \ T^*\  \implies \ T\  = \ T^*\ $                                                     |
| cardona-qft-methods_F00413 | 28   | ■ 1.000 | $(n+1)$                                                                                                    | $(n+1)$                                                                                                    | $\mu_N(UTU^*) = \mu_N(T)$ when $U$ is a unitary.                                                                                                                                |
| cardona-qft-methods_F00414 | 28   | ■ 1.000 | $ T $                                                                                                      | $ T $                                                                                                      | $\mu_N(UTU^*) = \mu_N(T)$ when $U$ is a unitary.                                                                                                                                |
| cardona-qft-methods_F00415 | 28   | ■ 1.000 | $\lim_{n \rightarrow \infty} \mu_n(T) = 0$                                                                 | $\lim_{n \rightarrow \infty} \mu_n(T) = 0$                                                                 | <b>Definition 3.1.</b> For $T \in \mathcal{K}(\mathcal{H})$ , the partial trace of order $N \in \mathbb{N}$ is                                                                  |
| cardona-qft-methods_F00416 | 28   | ■ 1.000 | $\epsilon > 0$                                                                                             | $\epsilon > 0$                                                                                             | <b>Definition 3.1.</b> For $T \in \mathcal{K}(\mathcal{H})$ , the partial trace of order $N \in \mathbb{N}$ is                                                                  |
| cardona-qft-methods_F00417 | 28   | ■ 0.933 | $E_{\epsilon}$                                                                                             | $E_{\epsilon}$                                                                                             | $\sigma_N(T) = \sum_{n=0}^{N-1} \mu_n(T).$                                                                                                                                      |
| cardona-qft-methods_F00418 | 28   | ■ 0.933 | $\left\  T_{ E_{\epsilon}^\perp} \right\  < \epsilon$                                                      | $\left\  T_{ E_{\epsilon}^\perp} \right\  < \epsilon$                                                      | $\sigma_N(T) = \sum_{n=0}^{N-1} \mu_n(T).$                                                                                                                                      |
| cardona-qft-methods_F00419 | 28   | ■ 1.000 | $N \in \mathbb{N}$                                                                                         | $N \in \mathbb{N}$                                                                                         |                                                                                                                                                                                 |
| cardona-qft-methods_F00420 | 28   | ■ 1.000 | $\sigma_N(T) = \sum_{n=0}^N \mu_n(T)$                                                                      | $\sigma_N(T) = \sum_{n=0}^N \mu_n(T)$                                                                      |                                                                                                                                                                                 |
| cardona-qft-methods_F00421 | 29   | ■ 0.999 | $\ T\  \leq \sigma_N(T) \leq N\ T\ $                                                                       | $\ T\  \leq \sigma_N(T) \leq N\ T\ $                                                                       |                                                                                                                                                                                 |
| cardona-qft-methods_F00422 | 29   | ■ 0.999 | $\sigma_n \simeq \ \cdot\ $                                                                                | $\sigma_n \simeq \ \cdot\ $                                                                                |                                                                                                                                                                                 |
| cardona-qft-methods_F00423 | 29   | ■ 0.999 | $\mathcal{K}(\mathcal{H})$                                                                                 | $\mathcal{K}(\mathcal{H})$                                                                                 |                                                                                                                                                                                 |
| cardona-qft-methods_F00424 | 29   | ■ 1.000 | $\sigma_N$                                                                                                 | $\sigma_N$                                                                                                 | $\sigma_N(T)$                                                                                                                                                                   |

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|----------------------------|------|---------|---------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| cardona-qft-methods_F00425 | 29   | ■ 1.000 | $\lambda \in \mathbb{R}^+$                                                                                    | $\lambda \in \mathbb{R}^+$                                                                                    | <b>Definition 3.2</b> The partial trace of $T$ of order $\lambda \in \mathbb{R}^+$ is                                                                                                                                                 |
| cardona-qft-methods_F00426 | 29   | ■ 1.000 | $\sigma_{\lambda_1}(T_1) + \sigma_{\lambda_2}(T_2) = \sigma_{\lambda_1 + \lambda_2}(T_1 + T_2)$               | $\sigma_{\lambda_1}(T_1) + \sigma_{\lambda_2}(T_2) = \sigma_{\lambda_1 + \lambda_2}(T_1 + T_2)$               | $T_1, T_2 \in \mathcal{K}(\mathcal{H})$ . We define a real interpolate space between $\mathcal{L}^1(\mathcal{H})$ and $\mathcal{K}(\mathcal{H})$                                                                                      |
| cardona-qft-methods_F00427 | 29   | ■ 1.000 | $\lambda_1, \lambda_2 \in \mathbb{R}^+, 0 \leq$                                                               | $\lambda_1, \lambda_2 \in \mathbb{R}^+, 0 \leq$                                                               | $T_1, T_2 \in \mathcal{K}(\mathcal{H})$ . We define a real interpolate space between $\mathcal{L}^1(\mathcal{H})$ and $\mathcal{K}(\mathcal{H})$                                                                                      |
| cardona-qft-methods_F00428 | 29   | ■ 0.996 | $T_1, T_2 \in \mathcal{K}(\mathcal{H})$                                                                       | $T_1, T_2 \in \mathcal{K}(\mathcal{H})$                                                                       | by                                                                                                                                                                                                                                    |
| cardona-qft-methods_F00429 | 29   | ■ 1.000 | $\mathcal{L}^p(\mathcal{H})$                                                                                  | $\mathcal{L}^p(\mathcal{H})$                                                                                  | <b>Lemma 9.1</b> $\mathcal{L}^{1,\infty}$ is a $C^*$ -ideal of $\mathcal{B}(\mathcal{H})$ for the norm $\ \cdot\ _{1,\infty}$ . Moreover,                                                                                             |
| cardona-qft-methods_F00430 | 29   | ■ 1.000 | $\mathrm{Tr}( T ^p) < \infty$                                                                                 | $\mathrm{Tr}( T ^p) < \infty$                                                                                 | <b>Lemma 9.1</b> $\mathcal{L}^{1,\infty}$ is a $C^*$ -ideal of $\mathcal{B}(\mathcal{H})$ for the norm $\ \cdot\ _{1,\infty}$ . Moreover,                                                                                             |
| cardona-qft-methods_F00431 | 29   | ■ 1.000 | $\sigma_\lambda(T) =$                                                                                         | $\sigma_\lambda(T) =$                                                                                         | <b>Lemma 9.1</b> $\mathcal{L}^{1,\infty}$ is a $C^*$ -ideal of $\mathcal{B}(\mathcal{H})$ for the norm $\ \cdot\ _{1,\infty}$ . Moreover,                                                                                             |
| cardona-qft-methods_F00432 | 29   | ■ 1.000 | $\mathcal{O}(\lambda^{1-1/p})$                                                                                | $\mathcal{O}(\lambda^{1-1/p})$                                                                                | <b>Lemma 9.1</b> $\mathcal{L}^{1,\infty}$ is a $C^*$ -ideal of $\mathcal{B}(\mathcal{H})$ for the norm $\ \cdot\ _{1,\infty}$ . Moreover,                                                                                             |
| cardona-qft-methods_F00433 | 29   | ■ 1.000 | $\mathcal{L}^{1,\infty}$                                                                                      | $\mathcal{L}^{1,\infty}$                                                                                      | <b>3.2. Diemier trace</b>                                                                                                                                                                                                             |
| cardona-qft-methods_F00434 | 29   | ■ 1.000 | $C^*$                                                                                                         | $C^*$                                                                                                         | <b>3.2. Diemier trace</b>                                                                                                                                                                                                             |
| cardona-qft-methods_F00435 | 29   | ■ 1.000 | $\ \cdot\ _{1,\infty}$                                                                                        | $\ \cdot\ _{1,\infty}$                                                                                        | <b>3.2. Diemier trace</b>                                                                                                                                                                                                             |
| cardona-qft-methods_F00436 | 29   | ■ 1.000 | $\frac{\sigma_\rho(T)}{\log \rho}$                                                                            | $\frac{\sigma_\rho(T)}{\log \rho}$                                                                            | $\tau_\lambda(T_1 + T_2) - \tau_\lambda(T_1) - \tau_\lambda(T_2) \underset{\lambda \rightarrow \infty}{\simeq} \mathcal{O}\left(\frac{\log(\log \lambda)}{\log \lambda}\right)$ , when $0 \leq T_1, T_2 \in \mathcal{L}^{1,\infty}$ . |
| cardona-qft-methods_F00437 | 29   | ■ 1.000 | $\mathbb{R}_+^*$                                                                                              | $\mathbb{R}_+^*$                                                                                              | <b>Len</b>                                                                                                                                                                                                                            |
| cardona-qft-methods_F00438 | 29   | ■ 0.997 | $\lambda \geq e$                                                                                              | $\lambda \geq e$                                                                                              | $ \tau_\lambda(x_1 + x_2) - \tau_\lambda(x_1) - \tau_\lambda(x_2)  \leq \frac{1}{\log \lambda} \ x_1 + x_2\ _{1,\infty}$ ,<br>when $T_1, T_2 \in \mathcal{L}_+^{1,\infty}$ .                                                          |
| cardona-qft-methods_F00439 | 29   | ■ 0.997 | $\tau_\lambda(T) = \frac{1}{\log \lambda} \int_e^\lambda \frac{\sigma_\rho(T)}{\log \rho} \frac{d\rho}{\rho}$ | $\tau_\lambda(T) = \frac{1}{\log \lambda} \int_e^\lambda \frac{\sigma_\rho(T)}{\log \rho} \frac{d\rho}{\rho}$ | $ \tau_\lambda(x_1 + x_2) - \tau_\lambda(x_1) - \tau_\lambda(x_2)  \leq \frac{1}{\log \lambda} \ x_1 + x_2\ _{1,\infty}$ ,<br>when $T_1, T_2 \in \mathcal{L}_+^{1,\infty}$ .                                                          |
| cardona-qft-methods_F00440 | 29   | ■ 1.000 | $\sigma_\rho(T) \leq \log \rho \ T\ _{1,\infty}$                                                              | $\sigma_\rho(T) \leq \log \rho \ T\ _{1,\infty}$                                                              |                                                                                                                                                                                                                                       |
| cardona-qft-methods_F00441 | 29   | ■ 1.000 | $\tau_\lambda(T) \leq \ T\ _{1,\infty}$                                                                       | $\tau_\lambda(T) \leq \ T\ _{1,\infty}$                                                                       |                                                                                                                                                                                                                                       |
| cardona-qft-methods_F00442 | 29   | ■ 1.000 | $\lambda \rightarrow$                                                                                         | $\lambda \rightarrow$                                                                                         |                                                                                                                                                                                                                                       |
| cardona-qft-methods_F00443 | 29   | ■ 1.000 | $\tau_\lambda(T)$                                                                                             | $\tau_\lambda(T)$                                                                                             |                                                                                                                                                                                                                                       |
| cardona-qft-methods_F00444 | 29   | ■ 1.000 | $C_b([e, \infty])$                                                                                            | $C_b([e, \infty])$                                                                                            |                                                                                                                                                                                                                                       |
| cardona-qft-methods_F00445 | 29   | ■ 0.905 | $T_1, T_2 \in \mathcal{L}_+^{1,\infty}$                                                                       | $T_1, T_2 \in \mathcal{L}_+^{1,\infty}$                                                                       |                                                                                                                                                                                                                                       |
| cardona-qft-methods_F00446 | 30   | ■ 1.000 | $\lambda \rightarrow \tau_\lambda(T)$                                                                         | $\lambda \rightarrow \tau_\lambda(T)$                                                                         |                                                                                                                                                                                                                                       |
| cardona-qft-methods_F00447 | 30   | ■ 1.000 | $\lambda \rightarrow \left( \frac{\log 2(2 + \log \log \lambda)}{\log \lambda} \right)$                       | $\lambda \rightarrow \left( \frac{\log 2(2 + \log \log \lambda)}{\log \lambda} \right)$                       | <i>Spectral asymptotics</i> 19                                                                                                                                                                                                        |
| cardona-qft-methods_F00448 | 30   | ■ 1.000 | $C_0([e, \infty])$                                                                                            | $C_0([e, \infty])$                                                                                            | <i>Spectral asymptotics</i> 19                                                                                                                                                                                                        |

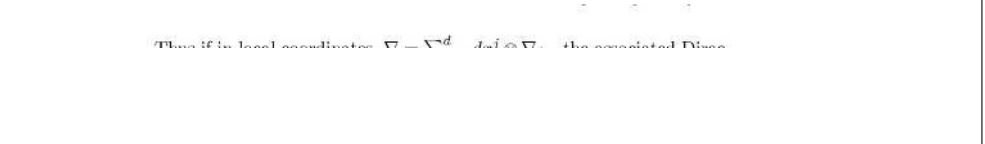
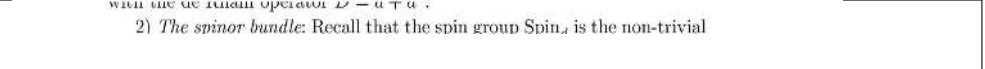
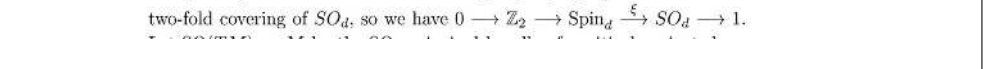
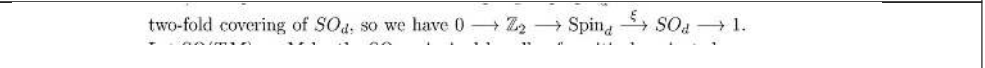
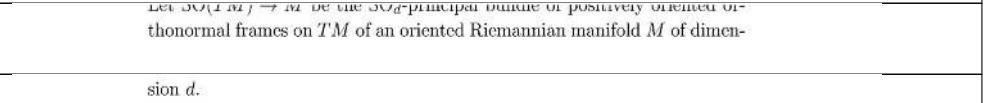
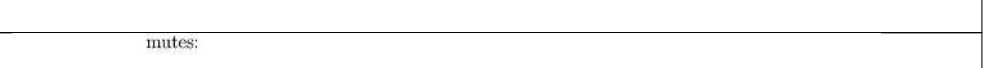
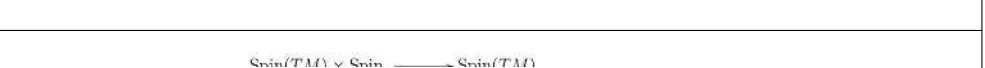
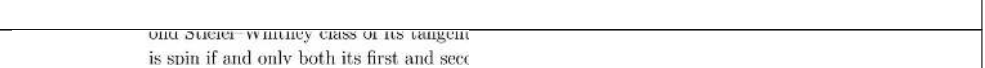
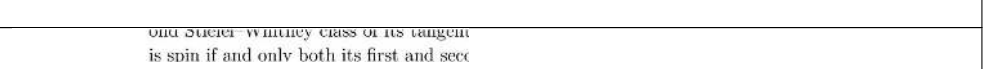
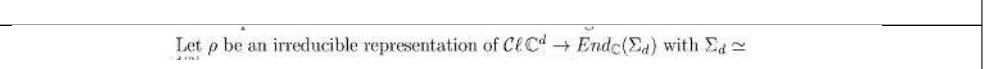
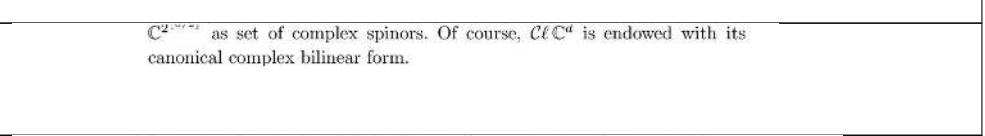
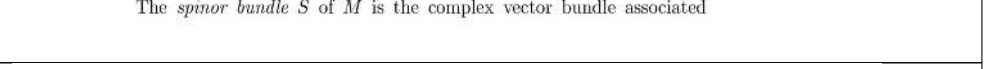
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| cardona-qft-methods_F00449 | 30   | ■ 1.000 | $\mathcal{A} = C_b([e, \infty)) / C_0([e, \infty))$                    | $\mathcal{A} = C_b([e, \infty))/C_0([e, \infty))$                      |                                                                                                                                                                                                                                         |
| cardona-qft-methods_F00450 | 30   | ■ 1.000 | $[\tau(T)]$                                                            | $[\tau(T)]$                                                            | The Dixmier's idea was to force additivity: since the map $\lambda \rightarrow \tau_\lambda(T)$                                                                                                                                         |
| cardona-qft-methods_F00451 | 30   | ■ 1.000 | $[\tau]: T \rightarrow [\tau(T)]$                                      | $[\tau]: T \rightarrow [\tau(T)]$                                      | is in $C_b([e, \infty))$ and $\lambda \rightarrow \{\frac{-\frac{1}{\log \lambda} - \frac{1}{\log \lambda}}{\log \lambda}\}$ is in $C_0([e, \infty))$ , consider the $C^*$ -algebra $\mathcal{A} = C_b([e, \infty))/C_0([e, \infty))$ . |
| cardona-qft-methods_F00452 | 30   | ■ 1.000 | $\mathcal{L}_+^{1,\infty}$                                             | $\mathcal{L}_+^{1,\infty}$                                             | is in $C_b([e, \infty))$ and $\lambda \rightarrow \{\frac{-\frac{1}{\log \lambda} - \frac{1}{\log \lambda}}{\log \lambda}\}$ is in $C_0([e, \infty))$ , consider the $C^*$ -algebra $\mathcal{A} = C_b([e, \infty))/C_0([e, \infty))$ . |
| cardona-qft-methods_F00453 | 30   | ■ 1.000 | $[\tau(UTU^*)] = [\tau(T)]$                                            | $[\tau(UTU^*)] = [\tau(T)]$                                            | If $[\tau(T)]$ is the class of the map $\lambda \rightarrow \tau_\lambda(T)$ in $\mathcal{A}$ , previo                                                                                                                                  |
| cardona-qft-methods_F00454 | 30   | ■ 0.902 | $\omega \circ [\tau(\cdot)]$                                           | $\omega \circ [\tau(\cdot)]$                                           | Now let $\omega$ be a state on $\mathcal{A}$ , namely a positive linear form on $\mathcal{A}$ with                                                                                                                                      |
| cardona-qft-methods_F00455 | 30   | ■ 0.902 | $C^*$ -                                                                | $C^*$ -                                                                | Now let $\omega$ be a state on $\mathcal{A}$ , namely a positive linear form on $\mathcal{A}$ with                                                                                                                                      |
| cardona-qft-methods_F00456 | 30   | ■ 0.915 | $\omega \circ [\tau(\cdot)]: T \in \mathcal{L}^{1,\infty} \rightarrow$ | $\omega \circ [\tau(\cdot)]: T \in \mathcal{L}^{1,\infty} \rightarrow$ | elements, and this map can be extended to a map $\omega \circ [\tau(\cdot)]: T \in \mathcal{L}^{1,\infty} \rightarrow$                                                                                                                  |
| cardona-qft-methods_F00457 | 30   | ■ 0.976 | $\omega([\tau(T)]) \in \mathbb{C}$                                     | $\omega([\tau(T)]) \in \mathbb{C}$                                     | $\omega([\tau(T)]) \in \mathbb{C}$ such that $\omega([\tau(T_1T_2)]) = \omega([\tau(T_2T_1)])$ for $T_1, T_2 \in \mathcal{L}^{1,\infty}$ . This leads to the following definition and result:                                           |
| cardona-qft-methods_F00458 | 30   | ■ 0.976 | $\omega([\tau(T_1T_2)]) = \omega([\tau(T_2T_1)])$                      | $\omega([\tau(T_1T_2)]) = \omega([\tau(T_2T_1)])$                      | $\omega([\tau(T)]) \in \mathbb{C}$ such that $\omega([\tau(T_1T_2)]) = \omega([\tau(T_2T_1)])$ for $T_1, T_2 \in \mathcal{L}^{1,\infty}$ . This leads to the following definition and result:                                           |
| cardona-qft-methods_F00459 | 30   | ■ 0.976 | $T_1, T_2 \in \mathcal{L}^{1,\infty}$                                  | $T_1, T_2 \in \mathcal{L}^{1,\infty}$                                  | $\omega([\tau(T)]) \in \mathbb{C}$ such that $\omega([\tau(T_1T_2)]) = \omega([\tau(T_2T_1)])$ for $T_1, T_2 \in \mathcal{L}^{1,\infty}$ . This leads to the following definition and result:                                           |
| cardona-qft-methods_F00460 | 30   | ■ 1.000 | $\text{Tr}_\omega$                                                     | $\text{Tr}_\omega$                                                     | $\text{Tr}_\omega(\cdot) = \omega \circ [\tau(\cdot)]$ .                                                                                                                                                                                |
| cardona-qft-methods_F00461 | 30   | ■ 1.000 | $\text{Tr}_\omega(\cdot) = \omega \circ [\tau(\cdot)]$                 | $\text{Tr}_\omega(\cdot) = \omega \circ [\tau(\cdot)]$                 | <b>Theorem</b><br>conver for                                                                                                                                                                                                            |
| cardona-qft-methods_F00462 | 30   | ■ 1.000 | $\text{Tr}_\omega(T) = 0$                                              | $\text{Tr}_\omega(T) = 0$                                              | 2/ The $C^*$ -algebra $\mathcal{A}$ is not separable, so it is impossible to exhaust any state $\omega$ ! Despite (10) and the fact that the $\mathcal{L}^p(\mathcal{H})$ are separable ideals for                                      |
| cardona-qft-methods_F00463 | 30   | ■ 1.000 | $T \in \mathcal{L}^1(\mathcal{H})$                                     | $T \in \mathcal{L}^1(\mathcal{H})$                                     | 2/ The $C^*$ -algebra $\mathcal{A}$ is not separable, so it is impossible to exhaust any state $\omega$ ! Despite (10) and the fact that the $\mathcal{L}^p(\mathcal{H})$ are separable ideals for                                      |
| cardona-qft-methods_F00464 | 30   | ■ 1.000 | $\ \cdot\ _{1,\infty}$                                                 | $\ \cdot\ _{1,\infty}$                                                 | $p \geq 1$ , $\mathcal{L}^{1,\infty}$ is not a separable.<br>Moreover, as for Lebesgue integral, there are subadditive and not measurable                                                                                               |
| cardona-qft-methods_F00465 | 30   | ■ 1.000 | $p \geq 1, \mathcal{L}^{1,\infty}$                                     | $p \geq 1, \mathcal{L}^{1,\infty}$                                     | is independent of $\omega$ . In this                                                                                                                                                                                                    |
| cardona-qft-methods_F00466 | 30   | ■ 1.000 | $f \in C_b([e, \infty))$                                               | $f \in C_b([e, \infty))$                                               | Along of the map $f \in C_b([e, \infty)) \rightarrow \tau_\lambda(f) \in C_b([e, \infty))$ converges to 0 as $\lambda \rightarrow \infty$ .<br>A Bos-Dixmier simulas if a non-normal tensor have been double inner                      |
| cardona-qft-methods_F00467 | 30   | ■ 1.000 | $\ell = \lim_{\lambda \rightarrow \infty} f(\lambda)$                  | $\ell = \lim_{\lambda \rightarrow \infty} f(\lambda)$                  | Along of the map $f \in C_b([e, \infty)) \rightarrow \tau_\lambda(f) \in C_b([e, \infty))$ converges to 0 as $\lambda \rightarrow \infty$ .<br>A Bos-Dixmier simulas if a non-normal tensor have been double inner                      |
| cardona-qft-methods_F00468 | 30   | ■ 1.000 | $\ell = \omega(f)$                                                     | $\ell = \omega(f)$                                                     | 2/ Bos-Dixmier simulas if a non-normal tensor have been double inner<br>tigated, see for instance, <sup>72,74</sup>                                                                                                                     |
| cardona-qft-methods_F00469 | 30   | ■ 1.000 | $T \in \mathcal{L}^{1,\infty}$                                         | $T \in \mathcal{L}^{1,\infty}$                                         | If $T \in \mathcal{K}_+(\mathcal{H})$ , then $T$ is measurable if and only if                                                                                                                                                           |
| cardona-qft-methods_F00470 | 30   | ■ 1.000 | $\text{Tr}_\omega(T)$                                                  | $\text{Tr}_\omega(T)$                                                  | If $T \in \mathcal{K}_+(\mathcal{H})$ , then $T$ is measurable if and only if                                                                                                                                                           |
| cardona-qft-methods_F00471 | 30   | ■ 1.000 | $\text{Tr}_{\text{Dix}}$                                               | $\text{Tr}_{\text{Dix}}$                                               | $\lim_{N \rightarrow \infty} \frac{1}{\log N} \sum_{n=0}^N \mu_n(T)$ exists.                                                                                                                                                            |
| cardona-qft-methods_F00472 | 30   | ■ 1.000 | $\text{Tr}_\omega(T) = \ell$                                           | $\text{Tr}_\omega(T) = \ell$                                           | on the torus:                                                                                                                                                                                                                           |
| cardona-qft-methods_F00473 | 30   | ■ 1.000 | $\lambda \in \mathbb{R}^+ \rightarrow \tau_\lambda(T) \in \mathcal{A}$ | $\lambda \in \mathbb{R}^+ \rightarrow \tau_\lambda(T) \in \mathcal{A}$ |                                                                                                                                                                                                                                         |

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| cardona-qft-methods_F00474 | 30   | ■ 1.000 | $\ell$                                                                                                                                           | $\ell$                                                                                                                                           |                                                                                                                                                                                                                                                                                                                                |
| cardona-qft-methods_F00475 | 30   | ■ 1.000 | $\{ \}^{72,74,75}$                                                                                                                               | 72,74,75                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                |
| cardona-qft-methods_F00476 | 30   | ■ 1.000 | $T \in \mathcal{K}_+(\mathcal{H})$                                                                                                               | $T \in \mathcal{K}_+(\mathcal{H})$                                                                                                               |                                                                                                                                                                                                                                                                                                                                |
| cardona-qft-methods_F00477 | 30   | ■ 1.000 | $\lim_{N \rightarrow \infty} \frac{1}{\log N} \sum_{n=0}^N \mu_n(T)$                                                                             | $\lim_{N \rightarrow \infty} \frac{1}{\log N} \sum_{n=0}^N \mu_n(T)$                                                                             |                                                                                                                                                                                                                                                                                                                                |
| cardona-qft-methods_F00478 | 31   | ■ 1.000 | $\mathbb{T}^d = \mathbb{R}^d / 2\pi \mathbb{Z}^d$                                                                                                | $\mathbb{T}^d = \mathbb{R}^d / 2\pi \mathbb{Z}^d$                                                                                                |                                                                                                                                                                                                                                                                                                                                |
| cardona-qft-methods_F00479 | 31   | ■ 1.000 | $\Delta = -\sum_{i=1}^d \partial_{x_i}^2$                                                                                                        | $\Delta = -\sum_{i=1}^d \partial_{x_i}^2$                                                                                                        |                                                                                                                                                                                                                                                                                                                                |
| cardona-qft-methods_F00480 | 31   | ■ 0.997 | $\mathcal{H} = L^2(\mathbb{T}^d)$                                                                                                                | $\mathcal{H} = L^2(\mathbb{T}^d)$                                                                                                                | 90 R. Iachum                                                                                                                                                                                                                                                                                                                   |
| cardona-qft-methods_F00481 | 31   | ■ 1.000 | $\mathrm{Tr}_\omega((1+\Delta)^{-p})$                                                                                                            | $\mathrm{Tr}_\omega((1+\Delta)^{-p})$                                                                                                            |                                                                                                                                                                                                                                                                                                                                |
| cardona-qft-methods_F00482 | 31   | ■ 1.000 | $\frac{d}{2} \leq p \in \mathbb{N}^*$                                                                                                            | $\frac{d}{2} \leq p \in \mathbb{N}^*$                                                                                                            |                                                                                                                                                                                                                                                                                                                                |
| cardona-qft-methods_F00483 | 31   | ■ 1.000 | $1+\Delta$                                                                                                                                       | $1+\Delta$                                                                                                                                       |                                                                                                                                                                                                                                                                                                                                |
| cardona-qft-methods_F00484 | 31   | ■ 1.000 | $\epsilon$                                                                                                                                       | $\epsilon$                                                                                                                                       | compute $\mathrm{Tr}_\omega((1+\Delta)^{-p})$ for $\frac{d}{2} \leq p \in \mathbb{N}^*$ . We use $1+\Delta$ to                                                                                                                                                                                                                 |
| cardona-qft-methods_F00485 | 31   | ■ 1.000 | $e_k(x) = \frac{1}{2\pi} e^{ik \cdot x}$                                                                                                         | $e_k(x) = \frac{1}{2\pi} e^{ik \cdot x}$                                                                                                         | problem with the inverse. As the following proof shows, 1 can be replaced by any $\epsilon > 0$ and the result does not depends on $\epsilon$ .                                                                                                                                                                                |
| cardona-qft-methods_F00486 | 31   | ■ 1.000 | $x \in \mathbb{T}^d, k \in \mathbb{Z}^d =$                                                                                                       | $x \in \mathbb{T}^d, k \in \mathbb{Z}^d =$                                                                                                       | problem with the inverse. As the following proof shows, 1 can be replaced by any $\epsilon > 0$ and the result does not depends on $\epsilon$ .                                                                                                                                                                                |
| cardona-qft-methods_F00487 | 31   | ■ 1.000 | $(\mathbb{T}^d)^*$                                                                                                                               | $(\mathbb{T}^d)^*$                                                                                                                               | by any $\epsilon > 0$ and the result does not depends on $\epsilon$ .<br>Notice that the functions $e_k(x) = \frac{1}{2\pi} e^{ik \cdot x}$ with $x \in \mathbb{T}^d, k \in \mathbb{Z}^d =$                                                                                                                                    |
| cardona-qft-methods_F00488 | 31   | ■ 1.000 | $\Delta e_k =  k ^2 e_k$                                                                                                                         | $\Delta e_k =  k ^2 e_k$                                                                                                                         | by any $\epsilon > 0$ and the result does not depends on $\epsilon$ .<br>Notice that the functions $e_k(x) = \frac{1}{2\pi} e^{ik \cdot x}$ with $x \in \mathbb{T}^d, k \in \mathbb{Z}^d =$                                                                                                                                    |
| cardona-qft-methods_F00489 | 31   | ■ 1.000 | $t \in$                                                                                                                                          | $t \in$                                                                                                                                          | by any $\epsilon > 0$ and the result does not depends on $\epsilon$ .<br>Notice that the functions $e_k(x) = \frac{1}{2\pi} e^{ik \cdot x}$ with $x \in \mathbb{T}^d, k \in \mathbb{Z}^d =$                                                                                                                                    |
| cardona-qft-methods_F00490 | 31   | ■ 1.000 | $\mathbb{R}_+^*, e^t \mathrm{Tr}(e^{-t(1+\Delta)}) = \sum_{k \in \mathbb{Z}^d} e^{-t k ^2} = \left( \sum_{k \in \mathbb{Z}} e^{-tk^2} \right)^d$ | $\mathbb{R}_+^*, e^t \mathrm{Tr}(e^{-t(1+\Delta)}) = \sum_{k \in \mathbb{Z}^d} e^{-t k ^2} = \left( \sum_{k \in \mathbb{Z}} e^{-tk^2} \right)^d$ | $(\mathbb{T}^d)^*$ form a basis of $\mathcal{H}$ of eigenvectors: $\Delta e_k =  k ^2 e_k$ . For $t \in \mathbb{R}_+^*, \mathrm{Tr}(e^{-t(1+\Delta)}) = \sum_{k \in \mathbb{Z}^d} e^{-t k ^2} = \left( \sum_{k \in \mathbb{Z}} e^{-tk^2} \right)^d$ . We know that                                                             |
| cardona-qft-methods_F00491 | 31   | ■ 1.000 | $\left  \int_{-\infty}^{\infty} e^{-tx^2} dx - \sum_{k \in \mathbb{Z}} e^{-tk^2} \right  \leq 1$                                                 | $\left  \int_{-\infty}^{\infty} e^{-tx^2} dx - \sum_{k \in \mathbb{Z}} e^{-tk^2} \right  \leq 1$                                                 | $\mathbb{R}_+^*, e^t \mathrm{Tr}(e^{-t(1+\Delta)}) = \sum_{k \in \mathbb{Z}^d} e^{-t k ^2} = \left( \sum_{k \in \mathbb{Z}} e^{-tk^2} \right)^d$ . We know that $\left  \int_{-\infty}^{\infty} e^{-tx^2} dx - \sum_{k \in \mathbb{Z}} e^{-tk^2} \right  \leq 1$ , and since the first integral is $\sqrt{\frac{\pi}{t}}$ , we |
| cardona-qft-methods_F00492 | 31   | ■ 1.000 | $\sqrt{\frac{\pi}{t}}$                                                                                                                           | $\sqrt{\frac{\pi}{t}}$                                                                                                                           | $\mathbb{R}_+^*, e^t \mathrm{Tr}(e^{-t(1+\Delta)}) = \sum_{k \in \mathbb{Z}^d} e^{-t k ^2} = \left( \sum_{k \in \mathbb{Z}} e^{-tk^2} \right)^d$ . We know that $\left  \int_{-\infty}^{\infty} e^{-tx^2} dx - \sum_{k \in \mathbb{Z}} e^{-tk^2} \right  \leq 1$ , and since the first integral is $\sqrt{\frac{\pi}{t}}$ , we |
| cardona-qft-methods_F00493 | 31   | ■ 0.953 | $e^t \mathrm{Tr}(e^{-t(1+\Delta)}) \underset{t \downarrow 0^+}{\sim} \left( \frac{\pi}{t} \right)^{d/2} = \alpha t^{-d/2}$                       | $e^t \mathrm{Tr}(e^{-t(1+\Delta)}) \underset{t \downarrow 0^+}{\sim} \left( \frac{\pi}{t} \right)^{d/2} = \alpha t^{-d/2}$                       | get $e^t \mathrm{Tr}(e^{-t(1+\Delta)}) \underset{t \downarrow 0^+}{\sim} \left( \frac{\pi}{t} \right)^{d/2} = \alpha t^{-d/2}$ .<br>We will use a Tauberian theorem in [14].                                                                                                                                                   |
| cardona-qft-methods_F00494 | 31   | ■ 0.995 | $\mu_n((1+\Delta)^{-d/2}) \underset{n \rightarrow \infty}{\simeq} \left( \alpha \frac{1}{\Gamma(d/2+1)} \right) \frac{1}{n}$                     | $\mu_n((1+\Delta)^{-d/2}) \underset{n \rightarrow \infty}{\simeq} \left( \alpha \frac{1}{\Gamma(d/2+1)} \right) \frac{1}{n}$                     | We will use a Tauberian theorem: $\mu_n((1+\Delta)^{-d/2}) \underset{n \rightarrow \infty}{\simeq} \left( \alpha \frac{1}{\Gamma(d/2+1)} \right) \frac{1}{n}$ , see, 49,54. Thus                                                                                                                                               |
| cardona-qft-methods_F00495 | 31   | ■ 1.000 | $\{ \}^{49,54}$                                                                                                                                  | 49,54                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                |

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| cardona-qft-methods_F08496 | 31   | ■ 1.000 | $(1+\Delta)^{-d/2}$                                                                           | $(1 + \Delta)^{-d/2}$                                                              | $\Delta)^{-p}) = 0.$                                                                                                                                                                              |
| cardona-qft-methods_F08497 | 31   | ■ 1.000 | $\left.\text{Tr}_{\text{Dix}}\right((1+\Delta)^{-d/2})=\text{Tr}_{\omega}((1+\Delta)^{-d/2})$ | $\text{Tr}_{\text{Dix}}\left((1 + \Delta)^{-d/2}\right) = \text{Tr}_{\omega}((1 +$ | $\Delta)^{-p}) = 0.$                                                                                                                                                                              |
| cardona-qft-methods_F08498 | 31   | ■ 0.997 | $\left.\text{Tr}_{\text{Dix}}\right((1+\Delta)^{-d/2})=\frac{\pi^{d/2}}{\Gamma(d/2+1)}$       | $\Delta)^{-d/2}) = \frac{\pi^{d/2}}{\Gamma(d/2 + 1)}$                              | This result has been generalized in Connes' trace theorem: <sup>29</sup> since $W\text{Res}$ and $\text{Tr}_{Dix}$ are traces on $\Psi DO^{-m}(M, E)$ , $m \in \mathbb{N}$ , we get the following |
| cardona-qft-methods_F08499 | 31   | ■ 0.997 | $(1+\Delta)^{-p}$                                                                             | $(1 + \Delta)^{-p}$                                                                | This result has been generalized in Connes' trace theorem: <sup>29</sup> since $W\text{Res}$ and $\text{Tr}_{Dix}$ are traces on $\Psi DO^{-m}(M, E)$ , $m \in \mathbb{N}$ , we get the following |
| cardona-qft-methods_F08500 | 31   | ■ 0.997 | $p>\frac{d}{2}, \text{Tr}_{\text{Dix}}((1+\Delta)^{-p})=0$                                    | $p > \frac{d}{2}, \text{Tr}_{\text{Dix}}((1 +$                                     | This result has been generalized in Connes' trace theorem: <sup>29</sup> since $W\text{Res}$ and $\text{Tr}_{Dix}$ are traces on $\Psi DO^{-m}(M, E)$ , $m \in \mathbb{N}$ , we get the following |
| cardona-qft-methods_F08501 | 31   | ■ 0.843 | $\left.\text{Tr}_{\text{Dix}}\right((1+\Delta)^{-p})=0$                                       | $\Delta)^{-p}) = 0$                                                                |                                                                                                                                                                                                   |
| cardona-qft-methods_F08502 | 31   | ■ 1.000 | $P \in \Psi D^0(M, E)$                                                                        | $P \in \Psi DO^{-d}(M, E)$                                                         | fine Clifford algebras, their representations, the notion of Clifford modules,                                                                                                                    |
| cardona-qft-methods_F08503 | 31   | ■ 1.000 | $P \in \mathcal{L}^{1, \infty}$                                                               | $P \in \mathcal{L}^{1, \infty}$                                                    | fine Clifford algebras, their representations, the notion of Clifford modules,                                                                                                                    |
| cardona-qft-methods_F08504 | 31   | ■ 1.000 | $\text{Tr}_{\text{Dix}}(P)=\frac{1}{d}W\text{Res}(P)$                                         | $\text{Tr}_{\text{Dix}}(P) = \frac{1}{d}W\text{Res}(P)$                            | $\text{spin}^c$ structures on orientable manifolds $M$ between the $C^*$ -algebras $C(M)$ and $\Gamma(\mathcal{C}\ell M)$                                                                         |
| cardona-qft-methods_F08505 | 31   | ■ 1.000 | $\{ \}^c$                                                                                     | $^c$                                                                               | <b>4.1. Definition and main properties</b>                                                                                                                                                        |
| cardona-qft-methods_F08506 | 31   | ■ 1.000 | $C(M)$                                                                                        | $C(M)$                                                                             | Let $(M, g)$ be a compact Riemannian manifold with metric $g$ , of dimension $d$ and $E$ be a vector bundle over $M$ . An example is the $(\text{Clifford})$ bundle.                              |
| cardona-qft-methods_F08507 | 31   | ■ 1.000 | $\Gamma(\mathcal{C}\ell M)$                                                                   | $\Gamma(\mathcal{C}\ell M)$                                                        | Let $(M, g)$ be a compact Riemannian manifold with metric $g$ , of dimension $d$ and $E$ be a vector bundle over $M$ . An example is the $(\text{Clifford})$ bundle.                              |
| cardona-qft-methods_F08508 | 32   | ■ 1.000 | $E=\mathcal{C}\ell T^*M$                                                                      | $E = \mathcal{C}\ell T^*M$                                                         |                                                                                                                                                                                                   |
| cardona-qft-methods_F08509 | 32   | ■ 1.000 | $T_x^*M$                                                                                      | $T_x^*M$                                                                           |                                                                                                                                                                                                   |
| cardona-qft-methods_F08510 | 32   | ■ 0.611 | $m \in \mathbb{N}$                                                                            | $m \in \mathbb{N}$                                                                 | vector space $\mathcal{L}_x M$ for $x \in M$ endowed with the nondegenerate quadratic form $g$ .                                                                                                  |
| cardona-qft-methods_F08511 | 32   | ■ 0.611 | $\text{Diff}^m(M, E)=\Gamma(M, \text{End}(E))$                                                | $\text{Diff}^m(M, E) = \Gamma(M, \text{End}(E)).$                                  | vector space $\mathcal{L}_x M$ for $x \in M$ endowed with the nondegenerate quadratic form $g$ .                                                                                                  |
| cardona-qft-methods_F08512 | 32   | ■ 0.989 | $\{\nabla_{X_1} \cdots \nabla_{X_j} \mid X_j \in \Gamma(M, TM), j \leq m\}$                   | $\{\nabla_{X_1} \cdots \nabla_{X_j} \mid X_j \in \Gamma(M, TM), j \leq m\}$        | Given a connection $\nabla$ on $E$ , a differential                                                                                                                                               |
| cardona-qft-methods_F08513 | 32   | ■ 1.000 | $\text{Diff}^m(M, E)$                                                                         | $\text{Diff}^m(M, E)$                                                              | $m \in \mathbb{N}$ on $E$ is an element of $\text{Diff}^m(M, E) = \Gamma(M, \text{End}(E))$ .                                                                                                     |
| cardona-qft-methods_F08514 | 32   | ■ 1.000 | $\text{End}(\Gamma(M, E))$                                                                    | $\text{End}(\Gamma(M, E))$                                                         | $m \in \mathbb{N}$ on $E$ is an element of $\text{Diff}^m(M, E) = \Gamma(M, \text{End}(E))$ .                                                                                                     |
| cardona-qft-methods_F08515 | 32   | ■ 0.980 | $\Gamma(T^*M, \pi^* \text{End}(E))$                                                           | $\Gamma(T^*M, \pi^* \text{End}(E))$                                                | In particular, $\text{Diff}^m(M, E)$ is a subalgebra of $\text{End}(\Gamma(M, E))$ and the                                                                                                        |
| cardona-qft-methods_F08516 | 32   | ■ 0.980 | $\pi$                                                                                         | $\pi$                                                                              | In particular, $\text{Diff}^m(M, E)$ is a subalgebra of $\text{End}(\Gamma(M, E))$ and the                                                                                                        |
| cardona-qft-methods_F08517 | 32   | ■ 1.000 | $E=\wedge T^*M$                                                                               | $E = \wedge T^*M$                                                                  | <i>Example:</i> Let $E = \wedge T^*M$ . The exterior product and the contraction given                                                                                                            |
| cardona-qft-methods_F08518 | 32   | ■ 1.000 | $\omega, \omega_j \in E$                                                                      | $\omega, \omega_j \in E$                                                           | on $\omega, \omega$                                                                                                                                                                               |
| cardona-qft-methods_F08519 | 32   | ■ 1.000 | $c(\omega)=\epsilon(\omega)+\iota(\omega)$                                                    | $c(\omega) = \epsilon(\omega) + \iota(\omega)$                                     | $E$ has a natural scalar product: if $e_1, \cdots, e_d$ is an orthonormal basis of $T_x^*M$                                                                                                       |
| cardona-qft-methods_F08520 | 32   | ■ 1.000 | $e_1, \cdots, e_d$                                                                            | $e_1, \cdots, e_d$                                                                 | scalar product on $\Gamma(M, E)$ ; then their principal symbols are                                                                                                                               |

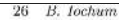
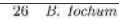
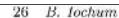
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| cardona-qft-methods_F00521 | 32   | ■ 1.000 | $e_{\{1\}} \wedge \cdots \wedge e_{\{p\}}$                                                | $e_{i_1} \wedge \cdots \wedge e_{i_p}$                                       |                                                                                                               |
| cardona-qft-methods_F00522 | 32   | ■ 1.000 | $i_{\{1\}} < \cdots < i_{\{p\}}$                                                          | $i_1 < \cdots < i_p$                                                         |                                                                                                               |
| cardona-qft-methods_F00523 | 32   | ■ 1.000 | $d \in$                                                                                   | $d \in$                                                                      |                                                                                                               |
| cardona-qft-methods_F00524 | 32   | ■ 1.000 | $\wedge^{\{1\}}$                                                                          | 1                                                                            |                                                                                                               |
| cardona-qft-methods_F00525 | 32   | ■ 1.000 | $d^{\{*\}}$                                                                               | $d^*$                                                                        |                                                                                                               |
| cardona-qft-methods_F00526 | 32   | ■ 1.000 | $\Gamma(M, E)$                                                                            | $\Gamma(M, E)$                                                               | where $h$ is such that $d_x h = \xi$ , so $\sigma_1^d(x, \xi) = i \xi$ , and similarly for $\sigma_1^{d^*}$ . |
| cardona-qft-methods_F00527 | 32   | ■ 1.000 | $\sigma_{1}^d(x,\xi)=i\,\xi$                                                              | $\sigma_1^d(x,\xi) = i \xi$                                                  |                                                                                                               |
| cardona-qft-methods_F00528 | 32   | ■ 1.000 | $\sigma_{1}^{d^*}$                                                                        | $\sigma_1^{d^*}$                                                             |                                                                                                               |
| cardona-qft-methods_F00529 | 32   | ■ 1.000 | $P \in \operatorname{Diff}^m(M)$                                                          | $P \in \operatorname{Diff}^m(M)$                                             |                                                                                                               |
| cardona-qft-methods_F00530 | 32   | ■ 1.000 | $\sigma_m^P(dh)=\frac{1}{i^mm!}(adh)^m(P)$                                                | $\sigma_m^P(dh) = \frac{1}{i^m m!} (adh)^m(P)$                               |                                                                                                               |
| cardona-qft-methods_F00531 | 32   | ■ 0.901 | $adh=[\cdot,h]$                                                                           | $adh = [\cdot, h]$                                                           |                                                                                                               |
| cardona-qft-methods_F00532 | 32   | ■ 0.901 | $\sigma_m^{P^*}(\omega)=\sigma_m^P(\omega)^*$                                             | $\sigma_m^{P^*}(\omega) = \sigma_m^P(\omega)^*$                              |                                                                                                               |
| cardona-qft-methods_F00533 | 32   | ■ 0.901 | $P^*$                                                                                     | $P^*$                                                                        |                                                                                                               |
| cardona-qft-methods_F00534 | 32   | ■ 1.000 | $E:\left\langle\psi,\psi'\right\rangle,\psi^{\prime}\left\langle\psi,\psi'\right\rangle=$ | $E:\langle\psi,\psi'\rangle =$                                               |                                                                                                               |
| cardona-qft-methods_F00535 | 32   | ■ 1.000 | $\int_M\left\langle\psi(x),\psi'(x)\right\rangle_x dx $                                   | $\int_M \langle \psi(x), \psi'(x) \rangle_x  dx $                            |                                                                                                               |
| cardona-qft-methods_F00536 | 32   | ■ 1.000 | $P \in \operatorname{Diff}^2(M, E)$                                                       | $P \in \operatorname{Diff}^2(M, E)$                                          |                                                                                                               |
| cardona-qft-methods_F00537 | 32   | ■ 1.000 | $\sigma_2^P(x,\xi)= \xi _x^2id_{E_x}$                                                     | $\sigma_2^P(x,\xi) =  \xi _x^2 id_{E_x}$                                     |                                                                                                               |
| cardona-qft-methods_F00538 | 32   | ■ 1.000 | $x \in M, \xi \in$                                                                        | $x \in M, \xi \in$                                                           |                                                                                                               |
| cardona-qft-methods_F00539 | 33   | ■ 1.000 | $P=$                                                                                      | $P =$                                                                        |                                                                                                               |
| cardona-qft-methods_F00540 | 33   | ■ 1.000 | $-\sum_{i,j}g^{ij}(x)\partial_{x^i}\partial_{x^j}+b^j(x)\partial_{x^j}+c(x)$              | $-\sum_{i,j}g^{ij}(x)\partial_{x^i}\partial_{x^j}+b^j(x)\partial_{x^j}+c(x)$ |                                                                                                               |
| cardona-qft-methods_F00541 | 33   | ■ 1.000 | $b^{\{j\}}$                                                                               | $b^j$                                                                        |                                                                                                               |
| cardona-qft-methods_F00542 | 33   | ■ 0.981 | $\Gamma(M, \operatorname{End}(E))$                                                        | $\Gamma(M, \operatorname{End}(E))$                                           |                                                                                                               |
| cardona-qft-methods_F00543 | 33   | ■ 1.000 | $E=E^+\oplus E^-$                                                                         | $E = E^+ \oplus E^-$                                                         |                                                                                                               |
| cardona-qft-methods_F00544 | 33   | ■ 1.000 | $\mathbb{Z}_2$                                                                            | $\mathbb{Z}_2$                                                               |                                                                                                               |
| cardona-qft-methods_F00545 | 33   | ■ 1.000 | $D \in \operatorname{Diff}^1(M, E)$                                                       | $D \in \operatorname{Diff}^1(M, E)$                                          | <br><b>Definition 4.2.</b> Assume that $E = E^+ \oplus E^-$ is a $\mathbb{Z}_2$ -graded vector bundle.        |

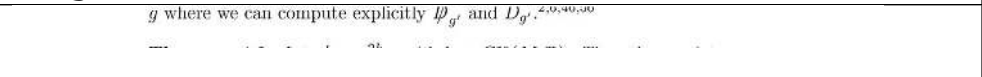
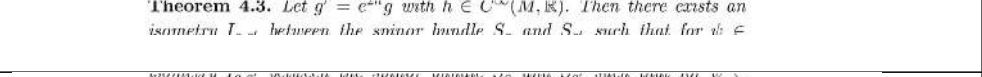
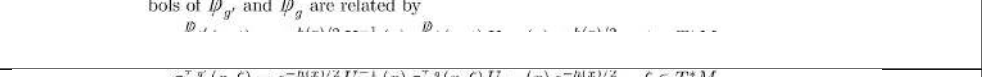
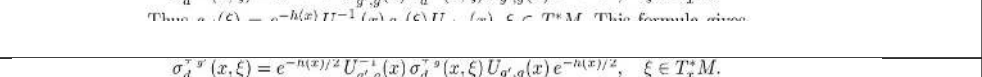
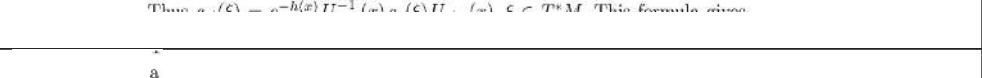
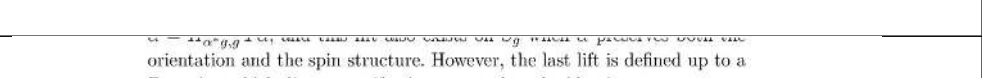
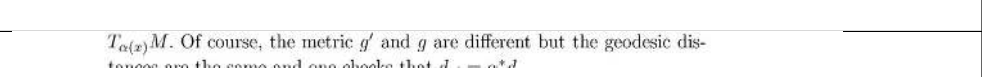
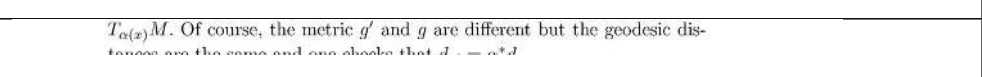
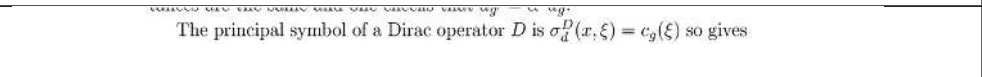
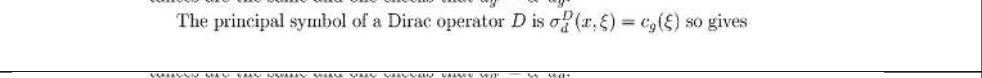
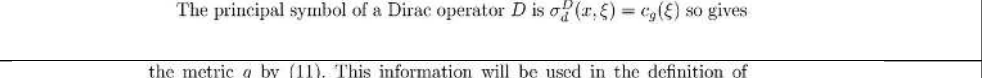
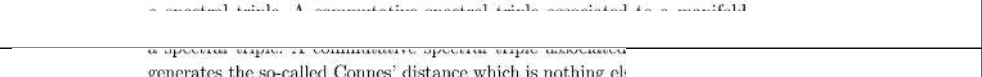
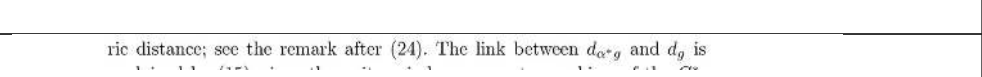
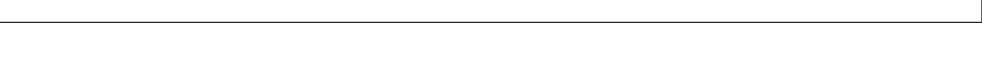
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| cardona-qft-methods_F08546 | 33   | ■ 1.000 | $D=\left(\begin{array}{cc}0 & D^+ \\ D^- & 0\end{array}\right)$                 | $D=\left(\begin{array}{cc}0 & D^+ \\ D^- & 0\end{array}\right)$                 | <p><math>\mathrm{in}\,\,1\left(M,E\mathrm{ad}(E)\right)</math>.</p> <p><b>Definition 4.2.</b> Assume that <math>E=E^+\oplus E^-</math> is a <math>\mathbb{Z}_2</math>-graded vector bundle.</p>                              |
| cardona-qft-methods_F08547 | 33   | ■ 1.000 | $D$                                                                             | $D$                                                                             | <p><math>\mathrm{in}\,\,1\left(M,E\mathrm{ad}(E)\right)</math>.</p> <p><b>Definition 4.2.</b> Assume that <math>E=E^+\oplus E^-</math> is a <math>\mathbb{Z}_2</math>-graded vector bundle.</p>                              |
| cardona-qft-methods_F08548 | 33   | ■ 0.992 | $D^{\pm}:\Gamma\left(M,E^{\mp}\right)\rightarrow\Gamma\left(M,E^{\pm}\right),D$ | $D^{\pm}:\Gamma\left(M,E^{\mp}\right)\rightarrow\Gamma\left(M,E^{\pm}\right),D$ | <p>When <math>D\in\mathrm{Diff}^1(M,E)</math> and <math>D=\left(\begin{array}{cc}0 &amp; D^+ \\ D^- &amp; 0\end{array}\right)</math> (i.e. <math>D</math> is odd) where</p>                                                  |
| cardona-qft-methods_F08549 | 33   | ■ 0.992 | $D^2=$                                                                          | $D^2=$                                                                          | <p>When <math>D\in\mathrm{Diff}^1(M,E)</math> and <math>D=\left(\begin{array}{cc}0 &amp; D^+ \\ D^- &amp; 0\end{array}\right)</math> (i.e. <math>D</math> is odd) where</p>                                                  |
| cardona-qft-methods_F08550 | 33   | ■ 1.000 | $\left(\begin{array}{cc}D^-D^+ & 0 \\ 0 & D^+D^-\end{array}\right)$             | $\left(\begin{array}{cc}D^-D^+ & 0 \\ 0 & D^+D^-\end{array}\right)$             | <p><math>D^{\pm}:\Gamma(M,E^{\mp})\rightarrow\Gamma(M,E^{\pm})</math>, <math>D</math> is called <math>\left(\begin{array}{cc}D^-D^+ &amp; 0 \\ 0 &amp; 0\end{array}\right)</math> is a generalized Laplacian.</p>            |
| cardona-qft-methods_F08551 | 33   | ■ 0.370 | $E=\wedge T^*M=\wedge^{\text{even}}T^*M\oplus\bigwedge^{\text{odd}}T^*M$        | $E=\wedge T^*M=\wedge^{\text{even}}T^*M\oplus\bigwedge^{\text{odd}}T^*M$        | <p>A good example is given by <math>E=\wedge T^*M=\wedge^{\text{even}}T^*M\oplus\wedge^{\text{odd}}T^*M</math> and the</p>                                                                                                   |
| cardona-qft-methods_F08552 | 33   | ■ 1.000 | $D=d+d^*$                                                                       | $D=d+d^*$                                                                       | <p>de Rham operator <math>D=d+d^*</math>. It is a Dirac operator since <math>D^2=dd^*+d^*d</math> is a generalized Laplacian according to (19). <math>D^2</math> is also called the Laplace</p>                              |
| cardona-qft-methods_F08553 | 33   | ■ 1.000 | $D^2=d\,d^*+d^*\,d$                                                             | $D^2=dd^*+d^*d$                                                                 | <p>de Rham operator <math>D=d+d^*</math>. It is a Dirac operator since <math>D^2=dd^*+d^*d</math> is a generalized Laplacian according to (19). <math>D^2</math> is also called the Laplace</p>                              |
| cardona-qft-methods_F08554 | 33   | ■ 1.000 | $D^2$                                                                           | $D^2$                                                                           | <p>Beltrami operator.</p>                                                                                                                                                                                                    |
| cardona-qft-methods_F08555 | 33   | ■ 1.000 | $\mathcal{C}\ell M$                                                             | $\mathcal{C}\ell M$                                                             | <p>isomorphism <math>X\in TM\leftrightarrow X^b\in T^*M</math>.</p>                                                                                                                                                          |
| cardona-qft-methods_F08556 | 33   | ■ 1.000 | $\mathcal{C}\ell T_xM$                                                          | $\mathcal{C}\ell T_xM$                                                          | <p>A bundle <math>E</math> is called a Clifford bundle over <math>M</math> when there exists a <math>\mathbb{Z}_2</math>-graduate action <math>c:\Gamma(M,\mathcal{C}\ell M)\rightarrow\mathrm{End}(\Gamma(M,E))</math>.</p> |
| cardona-qft-methods_F08557 | 33   | ■ 1.000 | $X\in TM\rightarrow X^b\in T^*M$                                                | $X\in TM\leftrightarrow X^b\in T^*M$                                            | <p>This bundle also satisfies defining this as</p>                                                                                                                                                                           |
| cardona-qft-methods_F08558 | 33   | ■ 1.000 | $c:\Gamma(M,\mathcal{C}\ell M)\rightarrow\mathrm{End}(\Gamma(M,E))$             | $c:\Gamma(M,\mathcal{C}\ell M)\rightarrow\mathrm{End}(\Gamma(M,E))$             | <p><b>Prop 4.1.</b> If <math>E</math> is a Clifford module, every odd <math>D\in\mathrm{Diff}</math></p>                                                                                                                     |
| cardona-qft-methods_F08559 | 33   | ■ 0.997 | $D\in\mathrm{Diff}^1$                                                           | $D\in\mathrm{Diff}^1$                                                           | <p>Clifford module for <math>c=i(\epsilon+\iota)</math> coming from the Dirac operator <math>D=d+d^*</math>:</p>                                                                                                             |
| cardona-qft-methods_F08560 | 33   | ■ 1.000 | $[D,f]=i\,c(df)$                                                                | $[D,f]=ic(df)$                                                                  | <p><math>i[D,f]=i[d+d^*,f]=i(i\sigma_1^d(df)-i\sigma_1^{d^*}(df))=-i(\epsilon</math></p>                                                                                                                                     |
| cardona-qft-methods_F08561 | 33   | ■ 1.000 | $f\in C^\infty(M)$                                                              | $f\in C^\infty(M)$                                                              | <p><math>i[D,f]=i[d+d^*,f]=i(i\sigma_1^d(df)-i\sigma_1^{d^*}(df))=-i(\epsilon</math></p>                                                                                                                                     |
| cardona-qft-methods_F08562 | 33   | ■ 1.000 | $c(df)=-i[D,f]$                                                                 | $c(df)=-i[D,f]$                                                                 | <p><math>E</math> is a <math>\mathbb{C}</math></p>                                                                                                                                                                           |
| cardona-qft-methods_F08563 | 33   | ■ 1.000 | $c=i(\epsilon+\iota)$                                                           | $c=i(\epsilon+\iota)$                                                           | <p><math>D_{\nabla}=-ic\circ\nabla,\quad\Gamma(M,E)\xrightarrow{\nabla}\Gamma(M,T^*M\otimes E)\xrightarrow{c\otimes 1}\Gamma(M,E),</math></p>                                                                                |
| cardona-qft-methods_F08564 | 33   | ■ 1.000 | $a\in\Gamma(M,\mathcal{C}\ell M)$                                               | $a\in\Gamma(M,\mathcal{C}\ell M)$                                               |                                                                                                                                                                                                                              |
| cardona-qft-methods_F08565 | 33   | ■ 1.000 | $X\in\Gamma(M, TM)$                                                             | $X\in\Gamma(M, TM)$                                                             |                                                                                                                                                                                                                              |
| cardona-qft-methods_F08566 | 33   | ■ 1.000 | $\left[\nabla_X,c(a)\right]=c\left(\nabla_X^{LC}a\right)$                       | $[\nabla_X,c(a)]=c\left(\nabla_X^{LC}a\right)$                                  |                                                                                                                                                                                                                              |
| cardona-qft-methods_F08567 | 33   | ■ 1.000 | $\nabla_X^{LC}$                                                                 | $\nabla_X^{LC}$                                                                 |                                                                                                                                                                                                                              |
| cardona-qft-methods_F08568 | 33   | ■ 1.000 | $D_{\nabla}$                                                                    | $D_{\nabla}$                                                                    |                                                                                                                                                                                                                              |
| cardona-qft-methods_F08569 | 33   | ■ 1.000 | $c\otimes 1$                                                                    | $c\otimes 1$                                                                    |                                                                                                                                                                                                                              |
| cardona-qft-methods_F08570 | 34   | ■ 1.000 | $\nabla=\sum_{j=1}^d dx^j\otimes\nabla_{\partial_j}$                            | $\nabla=\sum_{j=1}^d dx^j\otimes\nabla_{\partial_j}$                            |                                                                                                                                                                                                                              |

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| cardona-qft-methods_F00571 | 34   | ■ 1.000 | $D_{\nabla}=-i\sum_{j}c\left(d x^j\right)\nabla_{\partial_i}\nabla_{\partial_j}$                                                                   | $D_{\nabla}=-i\sum_jc\left(dx^j\right)\nabla_{\partial_i}$                                                                                  |    |
| cardona-qft-methods_F00572 | 34   | ■ 0.529 | $\left[D_{\nabla},f\right]=-i\sum_{i=1}^dc\left(dx^i\right)\left[\nabla_{\partial_j},f\right]=\sum_{j=1}^d-ic\left(dx^j\right)\partial_jf=-ic(df)$ | $[D_{\nabla},fid_E]=-i\sum_{i=1}^dc\left(dx^i\right)\left[\nabla_{\partial_j},f\right]=\sum_{j=1}^d-ic\left(dx^j\right)\partial_jf=-ic(df)$ |    |
| cardona-qft-methods_F00573 | 34   | ■ 1.000 | $\operatorname{Spin}_d$                                                                                                                            | $\operatorname{Spin}_d$                                                                                                                     |    |
| cardona-qft-methods_F00574 | 34   | ■ 1.000 | $SO_d$                                                                                                                                             | $SO_d$                                                                                                                                      |    |
| cardona-qft-methods_F00575 | 34   | ■ 1.000 | $0\longrightarrow\mathbb{Z}_2\longrightarrow\operatorname{Spin}_d\overset{\xi}{\longrightarrow}SO_d\longrightarrow1$                               | $0\longrightarrow\mathbb{Z}_2\longrightarrow\operatorname{Spin}_d\overset{\xi}{\longrightarrow}SO_d\longrightarrow1$                        |    |
| cardona-qft-methods_F00576 | 34   | ■ 0.999 | $\operatorname{SO}(TM)\rightarrow M$                                                                                                               | $SO(TM)\rightarrow M$                                                                                                                       |    |
| cardona-qft-methods_F00577 | 34   | ■ 0.999 | $\operatorname{SO}_d$                                                                                                                              | $SO_d$                                                                                                                                      |    |
| cardona-qft-methods_F00578 | 34   | ■ 1.000 | $TM$                                                                                                                                               | $TM$                                                                                                                                        |    |
| cardona-qft-methods_F00579 | 34   | ■ 0.901 | $\operatorname{Spin}(TM)\overset{\pi}{\rightarrow}M$                                                                                               | $\operatorname{Spin}(TM)\overset{\pi}{\rightarrow}M$                                                                                        |    |
| cardona-qft-methods_F00580 | 34   | ■ 0.608 | $\operatorname{Spin}(TM)\overset{\eta}{\rightarrow}\operatorname{SO}(TM)$                                                                          | $\operatorname{Spin}(TM)\overset{\eta}{\rightarrow}SO(TM)$                                                                                  |    |
| cardona-qft-methods_F00581 | 34   | ■ 0.999 | $\operatorname{Spin}(TM)$                                                                                                                          | $\operatorname{Spin}(TM)$                                                                                                                   |   |
| cardona-qft-methods_F00582 | 34   | ■ 0.999 | $\operatorname{SO}(TM)$                                                                                                                            | $SO(TM)$                                                                                                                                    |  |
| cardona-qft-methods_F00583 | 34   | ■ 1.000 | $\operatorname{Spin}_d^c$                                                                                                                          | $\operatorname{Spin}_d^c$                                                                                                                   |  |
| cardona-qft-methods_F00584 | 34   | ■ 0.999 | $\mathbb{T}:0\longrightarrow\mathbb{T}\longrightarrow\operatorname{Spin}_d^c\overset{\xi}{\longrightarrow}SO_d\longrightarrow1$                    | $\mathbb{T}:0\longrightarrow\mathbb{T}\longrightarrow\operatorname{Spin}_d^c\overset{\xi}{\longrightarrow}SO_d\longrightarrow1$             |  |
| cardona-qft-methods_F00585 | 34   | ■ 0.972 | $M,g$                                                                                                                                              | $M,g$                                                                                                                                       |  |
| cardona-qft-methods_F00586 | 34   | ■ 1.000 | $H^1\left(M,\mathbb{Z}_2\right)$                                                                                                                   | $H^1\left(M,\mathbb{Z}_2\right)$                                                                                                            |  |
| cardona-qft-methods_F00587 | 34   | ■ 1.000 | $\rho$                                                                                                                                             | $\rho$                                                                                                                                      |  |
| cardona-qft-methods_F00588 | 34   | ■ 1.000 | $\mathcal{CL}\mathbb{C}^d\rightarrow\operatorname{End}_{\mathbb{C}}\left(\Sigma_d\right)$                                                          | $\mathcal{CL}\mathbb{C}^d\rightarrow\operatorname{End}_{\mathbb{C}}\left(\Sigma_d\right)$                                                   |  |
| cardona-qft-methods_F00589 | 34   | ■ 1.000 | $\Sigma_d\simeq$                                                                                                                                   | $\Sigma_d\simeq$                                                                                                                            |  |
| cardona-qft-methods_F00590 | 34   | ■ 0.999 | $\mathbb{C}^{2^{\lfloor d/2 \rfloor}}$                                                                                                             | $\mathbb{C}^{2^{\lfloor d/2 \rfloor}}$                                                                                                      |  |
| cardona-qft-methods_F00591 | 34   | ■ 0.999 | $\mathcal{CL}\mathbb{C}^d$                                                                                                                         | $\mathcal{CL}\mathbb{C}^d$                                                                                                                  |  |
| cardona-qft-methods_F00592 | 34   | ■ 1.000 | $S$                                                                                                                                                | $S$                                                                                                                                         |  |
| cardona-qft-methods_F00593 | 35   | ■ 1.000 | $S=\operatorname{Spin}(TM)\times_{\rho_d}\Sigma_d$                                                                                                 | $S=\operatorname{Spin}(TM)\times_{\rho_d}\Sigma_d$                                                                                          |  |
| cardona-qft-methods_F00594 | 35   | ■ 1.000 | $\rho_d$                                                                                                                                           | $\rho_d$                                                                                                                                    |  |

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| cardona-qft-methods_F08595 | 35   | ■ 1.000 | $\operatorname{Aut}(\Sigma_d)$                                                                       | $\operatorname{Aut}(\Sigma_d)$                                                                       |                                                                                                                                                                                                                                     |
| cardona-qft-methods_F08596 | 35   | ■ 1.000 | $d=2\ m$                                                                                             | $d=2m$                                                                                               | to the principal bundle $\operatorname{Spin}(TM)$ with the spinor representation, namely                                                                                                                                            |
| cardona-qft-methods_F08597 | 35   | ■ 1.000 | $\rho_d=\rho^++\rho^-$                                                                               | $\rho_d=\rho^++\rho^-$                                                                               | to the principal bundle $\operatorname{Spin}(TM)$ with the spinor representation, namely                                                                                                                                            |
| cardona-qft-methods_F08598 | 35   | ■ 1.000 | $\rho^\pm$                                                                                           | $\rho^\pm$                                                                                           | to the principal bundle $\operatorname{Spin}(TM)$ with the spinor representation, namely                                                                                                                                            |
| cardona-qft-methods_F08599 | 35   | ■ 1.000 | $\operatorname{Spin}_{2\ m}$                                                                         | $\operatorname{Spin}_{2m}$                                                                           | $\sigma=\operatorname{Spin}(TM)\wedge\rho_d\ \omega_d$ , here $\rho_d$ is a representation of $\operatorname{Spin}_d$ on $\operatorname{Aut}(\omega_d)$ which is the restriction of $\rho$ .                                        |
| cardona-qft-methods_F08600 | 35   | ■ 1.000 | $\Sigma_{2\ m}=\phantom{0}$                                                                          | $\Sigma_{2m}=\phantom{0}$                                                                            | $\sigma=\operatorname{Spin}(TM)\wedge\rho_d\ \omega_d$ , here $\rho_d$ is a representation of $\operatorname{Spin}_d$ on $\operatorname{Aut}(\omega_d)$ which is the restriction of $\rho$ .                                        |
| cardona-qft-methods_F08601 | 35   | ■ 1.000 | $\Sigma_{2\ m}^+\oplus\Sigma_{2\ m}^-$                                                               | $\Sigma_{2m}^+\oplus\Sigma_{2m}^-$                                                                   | More precisely, if $d=2m$ is even, $\rho_d=\rho^++\rho^-$ where $\rho^\pm$ are two                                                                                                                                                  |
| cardona-qft-methods_F08602 | 35   | ■ 1.000 | $d=2\ m+1$                                                                                           | $d=2m+1$                                                                                             | More precisely, if $d=2m$ is even, $\rho_d=\rho^++\rho^-$ where $\rho^\pm$ are two                                                                                                                                                  |
| cardona-qft-methods_F08603 | 35   | ■ 0.798 | $S=S^+\oplus S^-$                                                                                    | $S=S^+\oplus S^-$                                                                                    | In practice, $M$ is a spin manifold means that there exists a Clifford bundle $S=S^+\oplus S^-$ such that $S\sim\wedge T^*M$ . Due to the dimension of $M$ , the                                                                    |
| cardona-qft-methods_F08604 | 35   | ■ 0.798 | $S\simeq\bigwedge T^*M$                                                                              | $S\simeq\bigwedge T^*M$                                                                              | In practice, $M$ is a spin manifold means that there exists a Clifford bundle $S=S^+\oplus S^-$ such that $S\sim\wedge T^*M$ . Due to the dimension of $M$ , the                                                                    |
| cardona-qft-methods_F08605 | 35   | ■ 1.000 | $S\simeq M\times\mathbb{C}^{d/2}$                                                                    | $S\simeq M\times\mathbb{C}^{d/2}$                                                                    | A spin connection $\nabla:\Gamma(\omega,\sigma)\rightarrow\Gamma(\omega,\sigma)\otimes\mathbb{R}$ connection which is compatible with Clifford action: $[\nabla^i$                                                                  |
| cardona-qft-methods_F08606 | 35   | ■ 0.974 | $\nabla^S:\Gamma^\infty(M,S)\rightarrow\Gamma^\infty(M,S)\otimes\Gamma^\infty(M,T^*M)$               | $\nabla^S:\Gamma^\infty(M,S)\rightarrow\Gamma^\infty(M,S)\otimes\Gamma^\infty(M,T^*M)$               | It is uniquely determined by the choice of a spin structure on $M$ (once an                                                                                                                                                         |
| cardona-qft-methods_F08607 | 35   | ■ 0.977 | $[\nabla^S,c(\cdot)]=c(\nabla^{LC}\cdot)$                                                            | $[\nabla^S,c(\cdot)]=c(\nabla^{LC}\cdot)$                                                            | orientation of $M$ is chosen).                                                                                                                                                                                                      |
| cardona-qft-methods_F08608 | 35   | ■ 1.000 | $\not{D}=-i\ c\left(dx^j\right)\left(\partial_j-\omega_j(x)\right)$                                  | $\not{D}=-ic\left(dx^j\right)\left(\partial_j-\omega_j(x)\right)$                                    | around $x_0$ ,                                                                                                                                                                                                                      |
| cardona-qft-methods_F08609 | 35   | ■ 1.000 | $\omega_j$                                                                                           | $\omega_j$                                                                                           | around $x_0$ ,                                                                                                                                                                                                                      |
| cardona-qft-methods_F08610 | 35   | ■ 1.000 | $\omega_j=\frac{1}{4}\left(\Gamma_{ji}^kg_{kl}-\right)$                                              | $\omega_j=\frac{1}{4}\left(\Gamma_{ji}^kg_{kl}-\right)$                                              | $c(dx^i)(x_0)=\gamma^i$ , $\sigma_1^-(x_0,\xi)=c(\xi)=\gamma^i\xi_j$ where the $\gamma$ 's are constant hermitean matrices.                                                                                                         |
| cardona-qft-methods_F08611 | 35   | ■ 1.000 | $\partial_i\left(h_j^\alpha\delta_{\alpha\beta}h_l^\beta\right)c\left(dx^i\right)c\left(dx^l\right)$ | $\partial_i\left(h_j^\alpha\delta_{\alpha\beta}h_l^\beta\right)c\left(dx^i\right)c\left(dx^l\right)$ | The Hilbert space of spinors is                                                                                                                                                                                                     |
| cardona-qft-methods_F08612 | 35   | ■ 1.000 | $H=\left[h_j^\alpha\right]$                                                                          | $H=\left[h_j^\alpha\right]$                                                                          | The Hilbert space of spinors is                                                                                                                                                                                                     |
| cardona-qft-methods_F08613 | 35   | ■ 1.000 | $H^tH=$                                                                                              | $H^tH=$                                                                                              | The Hilbert space of spinors is                                                                                                                                                                                                     |
| cardona-qft-methods_F08614 | 35   | ■ 1.000 | $[g_{ij}]$                                                                                           | $[g_{ij}]$                                                                                           | $\mathcal{H}=L^2((M,g),S))=\{\psi\in\Gamma^\infty(M,S) \int\langle\psi,\psi\rangle_xdvol_g(x)<\infty\}$ (14)                                                                                                                        |
| cardona-qft-methods_F08615 | 35   | ■ 1.000 | $\sigma_1^D(x,\xi)=c(\xi)+ic\left(dx^j\right)\omega_j(x)$                                            | $\sigma_1^D(x,\xi)=c(\xi)+ic\left(dx^j\right)\omega_j(x)$                                            | where we have a scalar product which is $\mathbb{C}^{\infty}(M)$ -valued. On its domain $\Gamma^\infty(M,S)$ , $\not{D}$ is symmetric: $\langle\psi,\not{D}\phi\rangle=\langle\not{D}\psi,\phi\rangle$ . Moreover, it has a selfad- |
| cardona-qft-methods_F08616 | 35   | ■ 1.000 | $x_{\{0\}}$                                                                                          | $x_0$                                                                                                |                                                                                                                                                                                                                                     |
| cardona-qft-methods_F08617 | 35   | ■ 1.000 | $\gamma$                                                                                             | $\gamma$                                                                                             |                                                                                                                                                                                                                                     |
| cardona-qft-methods_F08618 | 35   | ■ 0.999 | $C^\infty(M)$                                                                                        | $C^\infty(M)$                                                                                        |                                                                                                                                                                                                                                     |
| cardona-qft-methods_F08619 | 35   | ■ 1.000 | $\Gamma^\infty(M,S),\not{D}$                                                                         | $\Gamma^\infty(M,S),\not{D}$                                                                         |                                                                                                                                                                                                                                     |

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| cardona-qft-methods_F00620 | 35   | ■ 1.000 | $\backslash \mathrm{angle} \psi, \backslash \mathrm{not} D \backslash \phi \backslash \mathrm{rangle} = \backslash \mathrm{angle} \backslash \mathrm{not} D \backslash \psi, \backslash \phi \backslash \mathrm{rangle}$                                                                    | $\langle \psi, \not{D} \phi \rangle = \langle \not{D} \psi, \phi \rangle$                                                                           |                                                                                                                                                                                 |
| cardona-qft-methods_F00621 | 35   | ■ 0.719 | $\backslash \mathrm{not} D^{\wedge \{ * \} }$                                                                                                                                                                                                                                               | $\not{D}^{**}$                                                                                                                                      |                                                                                                                                                                                 |
| cardona-qft-methods_F00622 | 36   | ■ 0.919 | $\backslash \mathrm{mathcal} \{ H \}, \backslash \mathrm{not} D$                                                                                                                                                                                                                            | $\mathcal{H}, \not{D}$                                                                                                                              |                                                                                                                                                                                 |
| cardona-qft-methods_F00623 | 36   | ■ 1.000 | $\backslash \Gamma^{\infty} \{ M, S \}$                                                                                                                                                                                                                                                     | $\Gamma^{\infty}(M, S)$                                                                                                                             |                                                                                                                                                                                 |
| cardona-qft-methods_F00624 | 36   | ■ 1.000 | $\backslash \mathrm{not} D$                                                                                                                                                                                                                                                                 | $\not{D}$                                                                                                                                           | <i>essentially selfadjoint on its original domain <math>\mathcal{D}^{\infty}(M, S)</math>. It is a differential (unbounded) operator of order one which is elliptic</i>         |
| cardona-qft-methods_F00625 | 36   | ■ 1.000 | $R \in \backslash \Gamma^{\infty} \left( M, \bigwedge^2 T^{\wedge \{ * \} } M \otimes \backslash \mathrm{otimes} \backslash \mathrm{right} .\right.$                                                                                                                                        | $R \in \Gamma^{\infty} \left( M, \bigwedge^2 T^* M \otimes$                                                                                         | Before giving it, we need to fix a few notations: let $R \in \Gamma^{\infty}(M, \Lambda^2 T^* M \otimes$                                                                        |
| cardona-qft-methods_F00626 | 36   | ■ 0.785 | $\backslash \mathrm{operatorname} \{ \mathrm{End} \} (T \ M))$                                                                                                                                                                                                                              | $\mathrm{End}(TM))$                                                                                                                                 | $\mathrm{End}(TM))$ be the <i>Riemann curvature tensor</i> with components $R_{ijkl} =$                                                                                         |
| cardona-qft-methods_F00627 | 36   | ■ 0.785 | $R_{\{ i \ j \ k \ l \} } =$                                                                                                                                                                                                                                                                | $R_{ijkl} =$                                                                                                                                        | $\mathrm{End}(TM))$ be the <i>Riemann curvature tensor</i> with components $R_{ijkl} =$                                                                                         |
| cardona-qft-methods_F00628 | 36   | ■ 1.000 | $g \backslash \mathrm{left} ( \backslash \mathrm{partial} _{ \{ i \} } , R \backslash \mathrm{left} ( \backslash \mathrm{partial} _{ \{ k \} } , \backslash \mathrm{partial} _{ \{ l \} } \backslash \mathrm{right} ) \backslash \mathrm{partial} _{ \{ j \} } \backslash \mathrm{right} )$ | $g(\partial_i, R(\partial_k, \partial_l)\partial_j)$                                                                                                | $g(\partial_i, R(\partial_k, \partial_l)\partial_j)$ , the <i>Ricci tensor</i> components are $R_{jl} = g^{kl}R_{ijkl}$ and the <i>scalar curvature</i> is $s = g^{jl}R_{jl}$ . |
| cardona-qft-methods_F00629 | 36   | ■ 1.000 | $R_{\{ j \ l \} } = g^{\{ i \ k \} } T_{\{ i \ j \ k \ l \} }$                                                                                                                                                                                                                              | $R_{jl} = g^{ik}T_{ijkl}$                                                                                                                           | $g(\partial_i, R(\partial_k, \partial_l)\partial_j)$ , the <i>Ricci tensor</i> components are $R_{jl} = g^{kl}R_{ijkl}$ and the <i>scalar curvature</i> is $s = g^{jl}R_{jl}$ . |
| cardona-qft-methods_F00630 | 36   | ■ 1.000 | $s = g^{\{ j \ l \} } R_{\{ j \ l \} }$                                                                                                                                                                                                                                                     | $s = g^{jl}R_{jl}$                                                                                                                                  | <b>Prop 4.2</b> (Schrödinger–Li                                                                                                                                                 |
| cardona-qft-methods_F00631 | 36   | ■ 1.000 | $M, \backslash \mathrm{not} D^{\wedge \{ 2 \} } = \Delta^S + \frac{1}{4} s$                                                                                                                                                                                                                 | $M, \not{D}^2 = \Delta^S + \frac{1}{4}s$                                                                                                            | The proof is ju<br>We already kno                                                                                                                                               |
| cardona-qft-methods_F00632 | 36   | ■ 1.000 | $\backslash \mathrm{not} D^{\wedge \{ -1 \} }$                                                                                                                                                                                                                                              | $\not{D}^{-1}$                                                                                                                                      | creasing order including multiplicity (and in increasing order for an un-                                                                                                       |
| cardona-qft-methods_F00633 | 36   | ■ 1.000 | $\backslash \mathrm{left} \{ \backslash \lambda_{\{ n \} } (T) \backslash \mathrm{right} \} _{n \in \backslash \mathrm{mathbb} \{ N \} }$                                                                                                                                                   | $\{ \lambda_n(T) \}_{n \in \mathbb{N}}$                                                                                                             | <b>Theorem 4.2.</b> <i>With same hypothesis, the asymptotics of the Dirac op-</i>                                                                                               |
| cardona-qft-methods_F00634 | 36   | ■ 1.000 | $T^{\wedge \{ -1 \} }$                                                                                                                                                                                                                                                                      | $T^{-1}$                                                                                                                                            | $\mathrm{Vol}(M) = \int_{\mathrm{M}} \mathrm{dvol}.$                                                                                                                            |
| cardona-qft-methods_F00635 | 36   | ■ 1.000 | $N_{\{ T \} } (\backslash \lambda) =$                                                                                                                                                                                                                                                       | $N_T(\lambda) =$                                                                                                                                    | $\mathrm{Vol}(M) = \int_{\mathrm{M}} \mathrm{dvol}.$                                                                                                                            |
| cardona-qft-methods_F00636 | 36   | ■ 0.996 | $\backslash \mathrm{left} \{ \backslash \lambda_{\{ n \} } (T) \mid \backslash \lambda_{\{ n \} } \leq \backslash \lambda \} \backslash \mathrm{right} \}$                                                                                                                                  | $\# \{ \lambda_n(T) \mid \lambda_n \leq \lambda \}$                                                                                                 | We already encounter such computation in $\mathbb{E}$                                                                                                                           |
| cardona-qft-methods_F00637 | 36   | ■ 0.999 | $N_{ \not{D} } (\backslash \lambda) \underset{\backslash \lambda \rightarrow \infty}{\sim} \frac{2^d \mathrm{Vol} \left( \mathbb{S}^{d-1} \right)}{d (2 \pi)^d} \mathrm{Vol} (M) \backslash \lambda^d$                                                                                      | $N_{ \not{D} }(\lambda) \underset{\lambda \rightarrow \infty}{\sim} \frac{2^d \mathrm{Vol}(\mathbb{S}^{d-1})}{d(2\pi)^d} \mathrm{Vol}(M) \lambda^d$ | Recall that the spinor bundle $S_g$ and square integrable spinors $\mathcal{H}_g$ defined in (14) depends on the chosen metric $g$ , so we note $M_g$ instead of $M$ and        |
| cardona-qft-methods_F00638 | 36   | ■ 0.600 | $\backslash \mathrm{operatorname} \{ \mathrm{Vol} \} (\backslash \mathrm{mathrm} \{ M \} ) = \int_{\backslash \mathrm{mathrm} \{ M \} } \backslash \mathrm{mathrm} \{ \mathrm{dvol} \}$                                                                                                     | $\mathrm{Vol}(M) = \int_M \mathrm{dvol}$                                                                                                            | $\mathcal{H}_g := L^2_{\mathrm{compact}}$                                                                                                                                       |
| cardona-qft-methods_F00639 | 36   | ■ 1.000 | $S_{\{ g \} }$                                                                                                                                                                                                                                                                              | $S_g$                                                                                                                                               |                                                                                                                                                                                 |
| cardona-qft-methods_F00640 | 36   | ■ 1.000 | $\backslash \mathrm{mathcal} \{ H \} _{ \{ g \} }$                                                                                                                                                                                                                                          | $\mathcal{H}_g$                                                                                                                                     |                                                                                                                                                                                 |
| cardona-qft-methods_F00641 | 36   | ■ 1.000 | $M_{\{ g \} }$                                                                                                                                                                                                                                                                              | $M_g$                                                                                                                                               |                                                                                                                                                                                 |
| cardona-qft-methods_F00642 | 36   | ■ 1.000 | $\backslash \mathrm{mathcal} \{ H \} _{ \{ g \} } := L^{\wedge \{ 2 \} } \backslash \mathrm{left} ( M_{\{ g \} }, S_{\{ g \} } \backslash \mathrm{right} )$                                                                                                                                 | $\mathcal{H}_g := L^2(M_g, S_g)$                                                                                                                    |                                                                                                                                                                                 |
| cardona-qft-methods_F00643 | 36   | ■ 1.000 | $g^{\wedge \{ \mathrm{prime} \} }$                                                                                                                                                                                                                                                          | $g'$                                                                                                                                                |                                                                                                                                                                                 |
| cardona-qft-methods_F00644 | 36   | ■ 1.000 | $f_{\{ g, g^{\wedge \{ \mathrm{prime} \} } \} } : M \rightarrow \backslash \mathrm{mathbb} \{ R \} ^{\wedge \{ + \} }$                                                                                                                                                                      | $f_{g,g'} : M \rightarrow \mathbb{R}^+$                                                                                                             |                                                                                                                                                                                 |

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| cardona-qft-methods_F00645 | 36   | ■ 0.945 | $d \, \mathrm{vol}_{g^{\prime}} = f_{g^{\prime},g}$                                            | $dvol_{g'} = f_{g',g}$                                  |                                                                                                                                                                                    |
| cardona-qft-methods_F00646 | 36   | ■ 1.000 | $I_{\{g, \, g^{\prime}\}}(x) \colon S_{\{g\}} \rightarrow S_{\{g^{\prime}\}}$                  | $I_{g,g'}(x) : S_g \rightarrow S_{g'}$                  |                                                                                                                                                                                    |
| cardona-qft-methods_F00647 | 36   | ■ 1.000 | $\left I_{\{g, \, g^{\prime}\}}(x) \, \psi(x)\right _{g^{\prime}} =$                           | $ I_{g,g'}(x)\psi(x) _{g'} =$                           |                                                                                                                                                                                    |
| cardona-qft-methods_F00648 | 37   | ■ 1.000 | $ \psi(x) _{g}$                                                                                | $ \psi(x) _g$                                           |                                                                                                                                                                                    |
| cardona-qft-methods_F00649 | 37   | ■ 1.000 | $H_{\{g, \, g^{\prime}\}}$                                                                     | $H_{g,g'}$                                              |                                                                                                 |
| cardona-qft-methods_F00650 | 37   | ■ 1.000 | $2^{\lfloor d/2 \rfloor}$                                                                      | $2^{\lfloor d/2 \rfloor}$                               |                                                                                                 |
| cardona-qft-methods_F00651 | 37   | ■ 1.000 | $g^{\prime}(X, \, Y) = g\left(H_{\{g, \, g^{\prime}\}} X, \, Y\right)$                         | $g'(X,Y) = g(H_{g,g'}X,Y)$                              |                                                                                                 |
| cardona-qft-methods_F00652 | 37   | ■ 1.000 | $X, \, Y \in TM$                                                                               | $X,Y \in TM$                                            |                                                                                                 |
| cardona-qft-methods_F00653 | 37   | ■ 1.000 | $\iota_{\{g, \, g^{\prime}\}} X = H_{\{g, \, g^{\prime}\}}^{-1/2} X$                           | $\iota_{g,g'}X = H_{g,g'}^{-1/2}X$                      |                                                                                                 |
| cardona-qft-methods_F00654 | 37   | ■ 1.000 | $\iota_{\{g, \, g^{\prime}\}}$                                                                 | $\iota_{g,g'}$                                          | $ \psi(x) _g$ . Let us first see its construction: there always exists a $g$ -symmetric automorphism $H_{-}$ of the $\lfloor d/2 \rfloor$ -dimensional vector space $TM$ such that |
| cardona-qft-methods_F00655 | 37   | ■ 1.000 | $O_{\{d\}}$                                                                                    | $O_d$                                                   | $ \psi(x) _g$ . Let us first see its construction: there always exists a $g$ -symmetric automorphism $H_{-}$ of the $\lfloor d/2 \rfloor$ -dimensional vector space $TM$ such that |
| cardona-qft-methods_F00656 | 37   | ■ 0.637 | $_{\{d\}}$                                                                                     | $d$                                                     | $g'(X,Y) = g(H_{g,g'}X,Y)$ for $X,Y \in TM$ so define $\iota_{g,g'}X = H_{g,g'}^{-1/2}X$ .                                                                                         |
| cardona-qft-methods_F00657 | 37   | ■ 0.963 | $I_{\{g, \, g^{\prime}\}}\left(\right.$                                                        | $I_{g,g'} ($                                            | Note that $\iota_{g,g'}$ commutes with right action of the orthogonal group $O_d$                                                                                                  |
| cardona-qft-methods_F00658 | 37   | ■ 0.963 | $\left.^{\{6\}}\right)$                                                                        | $\left.^{\{6\}}\right)$                                 | Note that $\iota_{g,g'}$ commutes with right action of the orthogonal group $O_d$                                                                                                  |
| cardona-qft-methods_F00659 | 37   | ■ 1.000 | $I_{\{g, \, g^{\prime}\}} \colon \mathcal{H}_{\{g\}} \rightarrow \mathcal{H}_{\{g^{\prime}\}}$ | $I_{g,g'} : \mathcal{H}_g \rightarrow \mathcal{H}_{g'}$ | and can be lifted up to a diffeomorphism $Fin_d$ -equivariant on the spin structures associated to $g$ and $g'$ and this lift is denoted by $I_{g,g'}$ (see <sup>6</sup> ). This   |
| cardona-qft-methods_F00660 | 37   | ■ 0.999 | $\left(I_{\{g, \, g^{\prime}\}} \, \psi\right)(x) := I_{\{g, \, g^{\prime}\}}(x) \, \psi(x)$   | $(I_{g,g'}\psi)(x) := I_{g,g'}(x)\psi(x)$               | isometry is extended as operat                                                                                                                                                     |
| cardona-qft-methods_F00661 | 37   | ■ 1.000 | $U_{\{g^{\prime}\},g}$                                                                         | $U_{g',g}$                                              | Then by construction, $U_{g',g}$ is a unitary operator from $\mathcal{H}_{g'}$ onto $\mathcal{H}_g$ : for $\psi' \in \mathcal{H}_{g'}$ ,                                           |
| cardona-qft-methods_F00662 | 37   | ■ 1.000 | $\mathcal{H}_{\{g^{\prime}\}}$                                                                 | $\mathcal{H}_{g'}$                                      | Then by construction, $U_{g',g}$ is a unitary operator from $\mathcal{H}_{g'}$ onto $\mathcal{H}_g$ : for $\psi' \in \mathcal{H}_{g'}$ ,                                           |
| cardona-qft-methods_F00663 | 37   | ■ 1.000 | $\psi^{\prime} \in \mathcal{H}_{\{g^{\prime}\}}$                                               | $\psi' \in \mathcal{H}_{g'}$                            |                                                                                                                                                                                    |
| cardona-qft-methods_F00664 | 37   | ■ 1.000 | $\not D_{\{g^{\prime}\}}$                                                                      | $\not D_{g'}$                                           | This is an unbounded operator on $\mathcal{H}_g$ which has the same eigenvalue $\not D_{-}$ .                                                                                      |
| cardona-qft-methods_F00665 | 37   | ■ 1.000 | $D_{\{g^{\prime}\}}$                                                                           | $D_{g'}$                                                | This is an unbounded operator on $\mathcal{H}_g$ which has the same eigenvalue $\not D_{-}$ .                                                                                      |
| cardona-qft-methods_F00666 | 37   | ■ 1.000 | $H^k\left(M_{\{g\}}, \, S_{\{g\}}\right)$                                                      | $H^k(M_g,S_g)$                                          | particular, (after the transport map $\phi_g$ ), the domain of $\not D_{g'}$ and $\not D_{g'}$ are the same.                                                                       |
| cardona-qft-methods_F00667 | 37   | ■ 1.000 | $\Gamma^{\infty}\left(M_{\{g\}}, \, S_{\{g\}}\right)$                                          | $\Gamma^{\infty}(M_g,S_g)$                              | A nice example of this situation is when $g'$ is in the conformal class of $\frac{1}{2}g$ (see e.g. [2, 3, 4, 5, 6]).                                                              |
| cardona-qft-methods_F00668 | 37   | ■ 1.000 | $\ \psi\ _{\{k\}}^2 =$                                                                         | $\ \psi\ _k^2 =$                                        | A nice example of this situation is when $g'$ is in the conformal class of $\frac{1}{2}g$ (see e.g. [2, 3, 4, 5, 6]).                                                              |
| cardona-qft-methods_F00669 | 37   | ■ 0.922 | $\sum_{j=0}^k \int_M \left \nabla^j \psi(x)\right ^2 \, dx$                                    | $\sum_{j=0}^k \int_M  \nabla^j \psi(x) ^2 dx$           | $g$ where we can compute explicitly $\not D_{g'}$ and $D_{g',\mathcal{A}^{0,0,0,0}}$ .                                                                                             |

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| cardona-qft-methods_F00670 | 37   | ■ 0.922 | $\nabla \psi$                                                                    | $\nabla \psi$                                                                    |    |
| cardona-qft-methods_F00671 | 37   | ■ 1.000 | $T^*M_g \otimes S_g$                                                             | $T^*M_g \otimes S_g$                                                             |    |
| cardona-qft-methods_F00672 | 37   | ■ 1.000 | $U_{g,g'}: H^k(M_g, S_g) \rightarrow H^k(M_{g'}, S_{g'})$                        | $U_{g,g'}: H^k(M_g, S_g) \rightarrow H^k(M_{g'}, S_{g'})$                        |    |
| cardona-qft-methods_F00673 | 37   | ■ 0.801 | $\{2, 6, 46, 56\}$                                                               | $2, 6, 46, 56$                                                                   |    |
| cardona-qft-methods_F00674 | 37   | ■ 1.000 | $g^{\prime} = e^{2h} g$                                                          | $g' = e^{2h} g$                                                                  |    |
| cardona-qft-methods_F00675 | 37   | ■ 1.000 | $h \in C^{\infty}(M, \mathbb{R})$                                                | $h \in C^{\infty}(M, \mathbb{R})$                                                |    |
| cardona-qft-methods_F00676 | 37   | ■ 1.000 | $I_{g,g'}$                                                                       | $I_{g,g'}$                                                                       |    |
| cardona-qft-methods_F00677 | 37   | ■ 1.000 | $S_{g'}$                                                                         | $S_{g'}$                                                                         |    |
| cardona-qft-methods_F00678 | 37   | ■ 1.000 | $\psi \in \Gamma^{\infty}(M, S_g)$                                               | $\psi \in \Gamma^{\infty}(M, S_g)$                                               |    |
| cardona-qft-methods_F00679 | 37   | ■ 0.987 | $\Gamma^{\infty}(M, S_g)$                                                        | $\Gamma^{\infty}(M, S_g)$                                                        |    |
| cardona-qft-methods_F00680 | 38   | ■ 1.000 | $\sigma^{D_{g'}}(x, \xi) = e^{-h(x)} c_g(\xi)$                                   | $\sigma^{D_{g'}}(x, \xi) = e^{-h(x)} c_g(\xi)$                                   |   |
| cardona-qft-methods_F00681 | 38   | ■ 0.965 | $\not{D}_g$                                                                      | $\not{D}_g$                                                                      |  |
| cardona-qft-methods_F00682 | 38   | ■ 1.000 | $c_{g'}(\xi) = e^{-h(x)} U_{g',g}^{-1}(x) c_g(\xi) U_{g',g}(x), \xi \in T_x^* M$ | $c_{g'}(\xi) = e^{-h(x)} U_{g',g}^{-1}(x) c_g(\xi) U_{g',g}(x), \xi \in T_x^* M$ |  |
| cardona-qft-methods_F00683 | 38   | ■ 1.000 | $g'(\xi, \eta) = e^{-2h} g(\xi, \eta)$                                           | $g'(\xi, \eta) = e^{-2h} g(\xi, \eta)$                                           |  |
| cardona-qft-methods_F00684 | 38   | ■ 1.000 | $c_g(\xi) c_g(\eta) + c_g(\eta) c_g(\xi) =$                                      | $c_g(\xi) c_g(\eta) + c_g(\eta) c_g(\xi) =$                                      |  |
| cardona-qft-methods_F00685 | 38   | ■ 0.947 | $2g(\xi, \eta) \mathrm{id}_{S_g}$                                                | $2g(\xi, \eta) \mathrm{id}_{S_g}$                                                |  |
| cardona-qft-methods_F00686 | 38   | ■ 1.000 | $\tilde{\alpha} = H_{\alpha^*g,g}^{-1/2} T\alpha$                                | $\tilde{\alpha} = H_{\alpha^*g,g}^{-1/2} T\alpha$                                |  |
| cardona-qft-methods_F00687 | 38   | ■ 0.999 | $g' = \alpha^* g$                                                                | $g' = \alpha^* g$                                                                |  |
| cardona-qft-methods_F00688 | 38   | ■ 0.999 | $(\alpha^* g)_x(\xi, \eta) =$                                                    | $(\alpha^* g)_x(\xi, \eta) =$                                                    |  |
| cardona-qft-methods_F00689 | 38   | ■ 0.997 | $g_{\alpha(x)}(\alpha_*(\xi), \alpha_*(\eta)), x \in M$                          | $g_{\alpha(x)}(\alpha_*(\xi), \alpha_*(\eta)), x \in M$                          |  |
| cardona-qft-methods_F00690 | 38   | ■ 0.997 | $\alpha_*$                                                                       | $\alpha_*$                                                                       |  |
| cardona-qft-methods_F00691 | 38   | ■ 0.997 | $T_x M \rightarrow$                                                              | $T_x M \rightarrow$                                                              |  |
| cardona-qft-methods_F00692 | 38   | ■ 0.998 | $T_{\alpha(x)} M$                                                                | $T_{\alpha(x)} M$                                                                |  |
| cardona-qft-methods_F00693 | 38   | ■ 1.000 | $d_{g'} = \alpha^* d_g$                                                          | $d_{g'} = \alpha^* d_g$                                                          |  |
| cardona-qft-methods_F00694 | 38   | ■ 1.000 | $\sigma_d^D(x, \xi) = c_g(\xi)$                                                  | $\sigma_d^D(x, \xi) = c_g(\xi)$                                                  |  |

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| cardona-qft-methods_F00695 | 38   | ■ 1.000 | $d_{\{\alpha^*\} \, g}$                                                    | $d_{\alpha^* g}$                                                           | References for this section: <sup>[43, 44]</sup> and especially, <sup>[40]</sup><br>The heat kernel is a Green function of the heat operator $e^{t\Delta}$ (recall that                                                      |
| cardona-qft-methods_F00696 | 38   | ■ 1.000 | $d_{\{g\}}$                                                                | $d_g$                                                                      | References for this section: <sup>[43, 44]</sup> and especially, <sup>[40]</sup><br>The heat kernel is a Green function of the heat operator $e^{t\Delta}$ (recall that                                                      |
| cardona-qft-methods_F00697 | 38   | ■ 1.000 | $\{ \}^{\{4, 43, 44\}}$                                                    | 4, 43, 44                                                                  | $\partial_t k_t(x, y) = \Delta_x k_t(x, y), \quad \forall t > 0, x, y \in \mathbb{R}$                                                                                                                                        |
| cardona-qft-methods_F00698 | 38   | ■ 1.000 | $e^{\{t \, \Delta\}}$                                                      | $e^{t\Delta}$                                                              | with initial condition $\lim_{t \downarrow 0} k_t(x, y) = \delta(x - y)$ . Actually, $k_t(x, y) =$<br>$\frac{1}{(4\pi t)^{d/2}} e^{-\frac{ x-y ^2}{4t}}$                                                                     |
| cardona-qft-methods_F00699 | 38   | ■ 1.000 | $-\Delta$                                                                  | $-\Delta$                                                                  | $\frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-ts^2} e^{is(x-y)} ds$                                                                                                                                                            |
| cardona-qft-methods_F00700 | 38   | ■ 1.000 | $\lim_{t \downarrow 0} k_t(x, y) = \delta(x - y)$                          | $\lim_{t \downarrow 0} k_t(x, y) = \delta(x - y)$                          |                                                                                                                                                                                                                              |
| cardona-qft-methods_F00701 | 38   | ■ 1.000 | $k_{\{t\}}(x, y) =$                                                        | $k_t(x, y) =$                                                              |                                                                                                                                                                                                                              |
| cardona-qft-methods_F00702 | 38   | ■ 1.000 | $\frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-ts^2} e^{is(x-y)} ds$          | $\frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-ts^2} e^{is(x-y)} ds$          |                                                                                                                                                                                                                              |
| cardona-qft-methods_F00703 | 38   | ■ 1.000 | $d=1$                                                                      | $d = 1$                                                                    |                                                                                                                                                                                                                              |
| cardona-qft-methods_F00704 | 39   | ■ 1.000 | $f \in \mathcal{D}(\mathbb{R}^d)$                                          | $f \in \mathcal{D}(\mathbb{R}^d)$                                          |                                                                                                                                                                                                                              |
| cardona-qft-methods_F00705 | 39   | ■ 1.000 | $\lim_{t \downarrow 0} \int_{\mathbb{R}^d} k_t(x, y) f(y) dy = f(x)$       | $\lim_{t \downarrow 0} \int_{\mathbb{R}^d} k_t(x, y) f(y) dy = f(x)$       |                                                                                                                                                                                                                              |
| cardona-qft-methods_F00706 | 39   | ■ 1.000 | $\lambda_n$                                                                | $\lambda_n$                                                                |                                                                                                                                                                                                                              |
| cardona-qft-methods_F00707 | 39   | ■ 1.000 | $\psi_n \in L^2(U)$                                                        | $\psi_n \in L^2(U)$                                                        | cian<br>$\psi_n = 0$ on $\partial U$ .                                                                                                                                                                                       |
| cardona-qft-methods_F00708 | 39   | ■ 1.000 | $\Delta^{-1}$                                                              | $\Delta^{-1}$                                                              | $-\Delta\phi = \lambda\psi$ in $U$ , $\psi = 0$ on $\partial U$ .<br>If $\psi_n \in L^2(U)$ are the normalized eigenfunctions, the inverse Dirichlet                                                                         |
| cardona-qft-methods_F00709 | 39   | ■ 1.000 | $0 \leq \lambda_1 \leq \lambda_2 \leq \lambda_3 \leq$                      | $0 \leq \lambda_1 \leq \lambda_2 \leq \lambda_3 \leq$                      | $-\Delta\phi = \lambda\psi$ in $U$ , $\psi = 0$ on $\partial U$ .<br>If $\psi_n \in L^2(U)$ are the normalized eigenfunctions, the inverse Dirichlet                                                                         |
| cardona-qft-methods_F00710 | 39   | ■ 1.000 | $\dots, \lambda_n \rightarrow \infty$                                      | $\dots, \lambda_n \rightarrow \infty$                                      | $\lambda_1$                                                                                                                                                                                                                  |
| cardona-qft-methods_F00711 | 39   | ■ 1.000 | $f(x) = \int_0^\infty dt e^{-tx} \phi(x)$                                  | $f(x) = \int_0^\infty dt e^{-tx} \phi(x)$                                  | $\dots, \lambda_n \rightarrow \infty$ .<br>The interest for the heat kernel is that, if $f(x) = \int_0^\infty dt e^{-tx} \phi(x)$ is the                                                                                     |
| cardona-qft-methods_F00712 | 39   | ■ 1.000 | $\text{Tr}(f(-\Delta)) = \int_0^\infty dt \phi(t) \text{Tr}(e^{t\Delta})$  | $\text{Tr}(f(-\Delta)) = \int_0^\infty dt \phi(t) \text{Tr}(e^{t\Delta})$  | Laplace transform of $\phi$ , then $\text{Tr}(f(-\Delta)) = \int_0^\infty dt \phi(t) \text{Tr}(e^{t\Delta})$ (if every-                                                                                                      |
| cardona-qft-methods_F00713 | 39   | ■ 0.986 | $\text{Tr}(e^{t\Delta}) = \int_M d\text{vol}(x) \text{tr}_{V_x} k_t(x, x)$ | $\text{Tr}(e^{t\Delta}) = \int_M d\text{vol}(x) \text{tr}_{V_x} k_t(x, x)$ | thing makes sense) is controlled by $\text{Tr}(e^{t\Delta}) = \int_M d\text{vol}(x) \text{tr}_{V_x} k_t(x, x)$ since<br>$\text{Tr}(e^{t\Delta}) = \sum_n e^{t\lambda_n}$ and                                                 |
| cardona-qft-methods_F00714 | 39   | ■ 1.000 | $\text{Tr}(e^{t\Delta}) = \sum_{n=1}^\infty e^{t\lambda_n}$                | $\text{Tr}(e^{t\Delta}) = \sum_{n=1}^\infty e^{t\lambda_n}$                | $\text{Tr}(e^{t\Delta}) = \sum_{n=1}^\infty e^{t\lambda_n}$                                                                                                                                                                  |
| cardona-qft-methods_F00715 | 39   | ■ 1.000 | $k_{\{t\}}$                                                                | $k_t$                                                                      | <b>3.1. The asymptotics of heat kernels</b><br>Let now $M$ be a smooth comp                                                                                                                                                  |
| cardona-qft-methods_F00716 | 39   | ■ 1.000 | $t=0$                                                                      | $t = 0$                                                                    | $V$ be a vector bundle over $M$                                                                                                                                                                                              |
| cardona-qft-methods_F00717 | 39   | ■ 1.000 | $P \in \Psi D O^m(M, V)$                                                   | $P \in \Psi D O^m(M, V)$                                                   | which means that $ k_t(x, x) - \sum_{k \leq k(n)} a_k(x) t^{(-d+k)/m} _{\infty, n} < c_n t^n$ for $0 <$                                                                                                                      |
| cardona-qft-methods_F00718 | 39   | ■ 1.000 | $m>0$                                                                      | $m > 0$                                                                    | $t < 1$ where we used $ f _{\infty, n} := \sup_{x \in M} \sum_{ \alpha  \leq n}  \partial_x^\alpha f $ (since $P$ is elliptic,<br>$k_t(x, u)$ is a smooth function of $(t, x, u)$ for $t > 0$ , see [43, section 1.6, 1.7]). |
| cardona-qft-methods_F00719 | 39   | ■ 1.000 | $k_{\{t\}}(x, y)$                                                          | $k_t(x, y)$                                                                | $t < 1$ where we used $ f _{\infty, n} := \sup_{x \in M} \sum_{ \alpha  \leq n}  \partial_x^\alpha f $ (since $P$ is elliptic,<br>$k_t(x, u)$ is a smooth function of $(t, x, u)$ for $t > 0$ , see [43, section 1.6, 1.7]). |
| cardona-qft-methods_F00720 | 39   | ■ 1.000 | $e^{-tP}$                                                                  | $e^{-tP}$                                                                  | $t < 1$ where we used $ f _{\infty, n} := \sup_{x \in M} \sum_{ \alpha  \leq n}  \partial_x^\alpha f $ (since $P$ is elliptic,<br>$k_t(x, u)$ is a smooth function of $(t, x, u)$ for $t > 0$ , see [43, section 1.6, 1.7]). |

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| cardona-qft-methods_F00721 | 39   | ■ 1.000 | $\left k_{\{t\}}(x, x)-\sum_{\{k \leq k(n)\}} a_{\{k\}}(x) t^{\{(-d+k) / m\}}\right _{\{\infty, n\}} < c_{\{n\}} t^{\{n\}}$ | $k_t(x, x)-\sum_{k \leq k(n)} a_k(x) t^{(-d+k) / m} < c_n t^n$                              |                                                                                                                                                                                                                            |
| cardona-qft-methods_F00722 | 39   | ■ 1.000 | $0 <$                                                                                                                       | $0 <$                                                                                       |                                                                                                                                                                                                                            |
| cardona-qft-methods_F00723 | 39   | ■ 1.000 | $t < 1$                                                                                                                     | $t < 1$                                                                                     |                                                                                                                                                                                                                            |
| cardona-qft-methods_F00724 | 39   | ■ 1.000 | $ f _{\{\infty, n\}}:=\sup _{\{x \in M\}} \sum_{\{ \alpha  \leq n\}}\left \partial_{\{x\}}^{\alpha} f\right $               | $ f _{\infty, n}:=\sup _{x \in M} \sum_{ \alpha  \leq n}\left \partial_x^{\alpha} f\right $ |                                                                                                                                                                                                                            |
| cardona-qft-methods_F00725 | 39   | ■ 0.665 | $(t, x, y)$                                                                                                                 | $(t, x, y)$                                                                                 |                                                                                                                                                                                                                            |
| cardona-qft-methods_F00726 | 39   | ■ 1.000 | $k(t, f, P)=\operatorname{Tr}\left(f e^{-t P}\right)$                                                                       | $k(t, f, P)=\operatorname{Tr}\left(f e^{-t P}\right)$                                       |                                                                                                                                                                                                                            |
| cardona-qft-methods_F00727 | 40   | ■ 1.000 | $a_{\{k\}}$                                                                                                                 | $a_k$                                                                                       |                                                                                                                                                                                                                            |
| cardona-qft-methods_F00728 | 40   | ■ 1.000 | $\{ \} ^{\{43,44\}}$                                                                                                        | $43,44$                                                                                     | The utility of function $f$ will appear later for the computation of coefficie                                                                                                                                             |
| cardona-qft-methods_F00729 | 40   | ■ 1.000 | $a_{\{2 k\}}(f, P)$                                                                                                         | $a_{2 k}(f, P)$                                                                             | $a_k$ .<br>The following points are of importance:                                                                                                                                                                         |
| cardona-qft-methods_F00730 | 40   | ■ 1.000 | $a_{\{2 k+1\}}(f, P)=0$                                                                                                     | $a_{2 k+1}(f, P)=0$                                                                         | By the odd coefficients are zero, $a_{2 k+1}(f, P)=0$ .<br>For instance, let us assume from now on that $P$ is a Laplace ty                                                                                                |
| cardona-qft-methods_F00731 | 40   | ■ 1.000 | $\left(g^{\mu \nu}\right)_{1 \leq \mu, \nu \leq d}$                                                                         | $\left(g^{\mu \nu}\right)_{1 \leq \mu, \nu \leq d}$                                         | $\nabla$ on $V$ and a unique endomorphism $E$ such that                                                                                                                                                                    |
| cardona-qft-methods_F00732 | 40   | ■ 1.000 | $\mathbb{A}^{\mu}$                                                                                                          | $\mathbb{A}^{\mu}$                                                                          | $F=-\left(\operatorname{Tr}_g \nabla^*+E\right), \quad \nabla^*(X, Y):=\left[\nabla X, \nabla Y\right]-\nabla_{\nabla_X^L C} Y,$<br>$X, Y$ are vector fields on $M$ and $\nabla^{LC}$ is the Levi-Civita connection on $M$ |
| cardona-qft-methods_F00733 | 40   | ■ 1.000 | $\mathbb{B}$                                                                                                                | $\mathbb{B}$                                                                                | $F=-\left(\operatorname{Tr}_g \nabla^*+E\right), \quad \nabla^*(X, Y):=\left[\nabla X, \nabla Y\right]-\nabla_{\nabla_X^L C} Y,$<br>$X, Y$ are vector fields on $M$ and $\nabla^{LC}$ is the Levi-Civita connection on $M$ |
| cardona-qft-methods_F00734 | 40   | ■ 1.000 | $L(V)$                                                                                                                      | $L(V)$                                                                                      | $F=-\left(\operatorname{Tr}_g \nabla^*+E\right), \quad \nabla^*(X, Y):=\left[\nabla X, \nabla Y\right]-\nabla_{\nabla_X^L C} Y,$<br>$X, Y$ are vector fields on $M$ and $\nabla^{LC}$ is the Levi-Civita connection on $M$ |
| cardona-qft-methods_F00735 | 40   | ■ 1.000 | $X, Y$                                                                                                                      | $X, Y$                                                                                      | and $\mathbb{B}$ through                                                                                                                                                                                                   |
| cardona-qft-methods_F00736 | 40   | ■ 1.000 | $\nabla^{\{L C\}}$                                                                                                          | $\nabla^{LC}$                                                                               | and $\mathbb{B}$ through                                                                                                                                                                                                   |
| cardona-qft-methods_F00737 | 40   | ■ 1.000 | $\Gamma_{\mu \nu}^{\rho}$                                                                                                   | $\Gamma_{\mu \nu}^{\rho}$                                                                   | In this case, the coefficients $a_k(f, P)=\int_M d v o l_g \operatorname{tr}_E(f(x) a_k(P)(x))$ and the $a_i(P)=c_i \alpha^i(P)$ are linear combination with constants $c_i$ of all possible                               |
| cardona-qft-methods_F00738 | 40   | ■ 0.993 | $\nabla =d x^{\mu} \otimes\left(\partial_{\mu}+\omega_{\mu}\right)$                                                         | $\nabla=d x^{\mu} \otimes\left(\partial_{\mu}+\omega_{\mu}\right)$                          | independent invariants $\alpha_k^i(P)$ of dimension $k$ constructed from $E, \Omega, R$ and                                                                                                                                |
| cardona-qft-methods_F00739 | 40   | ■ 0.993 | $g^{\mu \nu}, \mathbb{A}^{\mu}$                                                                                             | $g^{\mu \nu}, \mathbb{A}^{\mu}$                                                             | independent invariants $\alpha_k^i(P)$ of dimension $k$ constructed from $E, \Omega, R$ and                                                                                                                                |
| cardona-qft-methods_F00740 | 40   | ■ 0.969 | $a_{\{k\}}(f, P)=\int_M$                                                                                                    | $a_k(f, P)=\int_M$                                                                          |                                                                                                                                                                                                                            |
| cardona-qft-methods_F00741 | 40   | ■ 0.969 | $_{\{g\}} \operatorname{tr}_{\{E\}}\left(f(x) a_{\{k\}}(P)(x)\right)$                                                       | $_g \operatorname{tr}_E(f(x) a_k(P)(x))$                                                    |                                                                                                                                                                                                                            |
| cardona-qft-methods_F00742 | 40   | ■ 1.000 | $a_{\{k\}}(P)=c_{\{i\}} \alpha_{\{k\}}^{\{i\}}(P)$                                                                          | $a_k(P)=c_i \alpha_k^i(P)$                                                                  |                                                                                                                                                                                                                            |
| cardona-qft-methods_F00743 | 40   | ■ 1.000 | $c_{\{i\}}$                                                                                                                 | $c_i$                                                                                       |                                                                                                                                                                                                                            |
| cardona-qft-methods_F00744 | 40   | ■ 1.000 | $\alpha_{\{k\}}^{\{i\}}(P)$                                                                                                 | $\alpha_k^i(P)$                                                                             |                                                                                                                                                                                                                            |
| cardona-qft-methods_F00745 | 40   | ■ 1.000 | $E, \Omega, R$                                                                                                              | $E, \Omega, R$                                                                              |                                                                                                                                                                                                                            |
| cardona-qft-methods_F00746 | 40   | ■ 1.000 | $k=2, E$                                                                                                                    | $k=2, E$                                                                                    |                                                                                                                                                                                                                            |



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| cardona-qft-methods_F00770 | 42   | 0.953 | $p=d, W \operatorname{Res} (D^{-d})=\frac{2}{\Gamma(p/2)} a_0(D^2)=\frac{2}{\Gamma(p/2)} \frac{\operatorname{Rank}(E)}{(4\pi)^{d/2}} \operatorname{Vol}(M)$ | $p = d, W \operatorname{Res} (D^{-d}) = \frac{2}{\Gamma(p/2)} a_0(D^2) = \frac{2}{\Gamma(p/2)} \frac{\operatorname{Rank}(E)}{(4\pi)^{d/2}} \operatorname{Vol}(M)$ | A few remarks are in order:                                                                                                                                                                                                                                                                                   |
| cardona-qft-methods_F00771 | 42   | 1.000 | $\operatorname{Tr} (e^{-t D^2}) \underset{t \downarrow 0^+}{\sim} a_0(D^2) t^{-d/2}$                                                                        | $\operatorname{Tr} \left( e^{-t D^2} \right) \underset{t \downarrow 0^+}{\sim} a_0(D^2) t^{-d/2}$                                                                 | 1) If $p = d$ , $W \operatorname{Res}(D^{-d}) = \frac{2}{\Gamma(p/2)} a_0(D^2) = \frac{2}{\Gamma(p/2)} \frac{\operatorname{Rank}(E)}{(4\pi)^{d/2}} \operatorname{Vol}(M)$ . Since $\operatorname{Tr}(e^{-t D^2}) \underset{t \downarrow 0^+}{\sim} a_0(D^2) t^{-d/2}$ , the Tauberian theorem used in Example |
| cardona-qft-methods_F00772 | 42   | 1.000 | $D^{-d} = (D^{-2})^{d/2}$                                                                                                                                   | $D^{-d} = (D^{-2})^{d/2}$                                                                                                                                         | 3.1 implies that $D^{-d} = (D^{-2})^{d/2}$ is measurable and we obtain Connes' trace                                                                                                                                                                                                                          |
| cardona-qft-methods_F00773 | 42   | 1.000 | $D \neq \not D$                                                                                                                                             | $D = \not D$                                                                                                                                                      | $a_2(D^2)$ (see (19) with $f = 1$ ) can be easily computed (see <sup>43,49</sup> ): if $s$ is the                                                                                                                                                                                                             |
| cardona-qft-methods_F00774 | 42   | 0.648 | $a_2(D^2)$                                                                                                                                                  | $a_2(D^2)$                                                                                                                                                        | SCHRIFF CURVATURE,                                                                                                                                                                                                                                                                                            |
| cardona-qft-methods_F00775 | 42   | 0.648 | (19)                                                                                                                                                        | (19)                                                                                                                                                              | SCHRIFF CURVATURE,                                                                                                                                                                                                                                                                                            |
| cardona-qft-methods_F00776 | 42   | 0.648 | $f=1$                                                                                                                                                       | $f = 1$                                                                                                                                                           | SCHRIFF CURVATURE,                                                                                                                                                                                                                                                                                            |
| cardona-qft-methods_F00777 | 42   | 0.648 | { }^{43,49}                                                                                                                                                 | 43,49                                                                                                                                                             | SCHRIFF CURVATURE,                                                                                                                                                                                                                                                                                            |
| cardona-qft-methods_F00778 | 42   | 0.480 | $W \operatorname{Res} (\not D^{-d+2})=\frac{2}{\Gamma(d/2-1)} a_2(D^2)=c \int_M s(x) d \operatorname{vol}_g(x)$                                             | $W \operatorname{Res} (\not D^{-d+2}) = \frac{2}{\Gamma(d/2-1)} a_2(D^2) = c \int_M s(x) d \operatorname{vol}_g(x)$                                               | isolipsums, see (9).                                                                                                                                                                                                                                                                                          |
| cardona-qft-methods_F00779 | 42   | 1.000 | $D \in \Psi D O^1(M, E)$                                                                                                                                    | $D \in \Psi D O^1(M, E)$                                                                                                                                          | The main properties of a compact spin Riemannian manifold $M$ can be                                                                                                                                                                                                                                          |
| cardona-qft-methods_F00780 | 42   | 1.000 | $\zeta_D^P(s) = \operatorname{Tr} (P  D ^{-s})$                                                                                                             | $\zeta_D^P(s) = \operatorname{Tr} (P  D ^{-s})$                                                                                                                   | recaptured using the following triple $(\mathcal{A} = C^\infty(M), \mathcal{H} = L^2(M, S), \not D)$ . The coordinates $x = (x^1, \dots, x^d)$ are endowed with the algebra $C^\infty(M)$ .                                                                                                                   |
| cardona-qft-methods_F00781 | 42   | 0.992 | $W \operatorname{Res} P = \operatorname{Res}_{s=0} \zeta_D^P(s) = - \int_M c_P(x)  dx $                                                                     | $W \operatorname{Res} P = \operatorname{Res}_{s=0} \zeta_D^P(s) = - \int_M c_P(x)  dx $                                                                           | The coordinates $x = (x^1, \dots, x^d)$ are exchanged with the algebra $C^\infty(M)$ , the Dirac operator $\not D$ gives the dimension $d$ as seen in Theorem 4.2, but                                                                                                                                        |
| cardona-qft-methods_F00782 | 42   | 1.000 | { }^{24,30,33,36,49}                                                                                                                                        | 24,30,33,36,49                                                                                                                                                    |                                                                                                                                                                                                                                                                                                               |
| cardona-qft-methods_F00783 | 42   | 0.982 | $\mathcal{A} = C^\infty(M), \mathcal{H} = L^2(M, S), \not D$                                                                                                | $\mathcal{A} = C^\infty(M), \mathcal{H} = L^2(M, S), \not D$                                                                                                      |                                                                                                                                                                                                                                                                                                               |
| cardona-qft-methods_F00784 | 42   | 1.000 | $x = (x^1, \dots, x^d)$                                                                                                                                     | $x = (x^1, \dots, x^d)$                                                                                                                                           |                                                                                                                                                                                                                                                                                                               |
| cardona-qft-methods_F00785 | 43   | 0.985 | $K O$                                                                                                                                                       | $K O$                                                                                                                                                             | quantized calculus. The idea is to forget about the commutativity of the                                                                                                                                                                                                                                      |
| cardona-qft-methods_F00786 | 43   | 0.997 | $\mathcal{A}, \mathcal{H}, \mathcal{D}$                                                                                                                     | $\mathcal{A}, \mathcal{H}, \mathcal{D}$                                                                                                                           | orientability, Poincaré duality, Hochschild (co)homology etc.                                                                                                                                                                                                                                                 |
| cardona-qft-methods_F00787 | 43   | 1.000 | $[\mathcal{D}, \pi(a)]$                                                                                                                                     | $[\mathcal{D}, \pi(a)]$                                                                                                                                           | and a selfadjoint operator $\mathcal{D}$ with compact resolvent (thus with spectrum) such that $[\mathcal{D}, \pi(a)]$ is bounded for any $a \in \mathcal{A}$ .                                                                                                                                               |
| cardona-qft-methods_F00788 | 43   | 1.000 | $a \in \mathcal{A}$                                                                                                                                         | $a \in \mathcal{A}$                                                                                                                                               | and a selfadjoint operator $\mathcal{D}$ with compact resolvent (thus with spectrum) such that $[\mathcal{D}, \pi(a)]$ is bounded for any $a \in \mathcal{A}$ .                                                                                                                                               |
| cardona-qft-methods_F00789 | 43   | 1.000 | $\mathcal{A} :=$                                                                                                                                            | $\mathcal{A} :=$                                                                                                                                                  | We could impose the existence of a $C^*$ -algebra $A$ where $\mathcal{A} := \{a \in A   \exists \mathcal{D} \text{ as above}\}$ is bounded. In some cases in $A$ as $A$ is a new $C^*$ -algebra.                                                                                                              |
| cardona-qft-methods_F00790 | 43   | 0.980 | $\{a \in A \mid [\mathcal{D}, \pi(a)]$                                                                                                                      | $\{a \in A \mid [\mathcal{D}, \pi(a)]$                                                                                                                            | stable by holomorphic calculus.                                                                                                                                                                                                                                                                               |
| cardona-qft-methods_F00791 | 43   | 0.980 | $\}$                                                                                                                                                        | $\}$                                                                                                                                                              | stable by holomorphic calculus.                                                                                                                                                                                                                                                                               |
| cardona-qft-methods_F00792 | 43   | 1.000 | $\pi(a)$                                                                                                                                                    | $\pi(a)$                                                                                                                                                          | We now give useful lemmias:                                                                                                                                                                                                                                                                                   |
| cardona-qft-methods_F00793 | 43   | 0.994 | $\chi$                                                                                                                                                      | $\chi$                                                                                                                                                            | $J : \mathcal{H} \rightarrow \mathcal{H}$ such that $J \mathcal{D} = \epsilon \mathcal{D} J, J^2 = \epsilon', J \chi = \epsilon'' \chi J$ with the following                                                                                                                                                  |

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|-----------------------------|------|---------|-------------------------------------------------------------------------------|-------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| cardona-qft-methods_ F00794 | 43   | ■ 0.994 | $\chi=\chi^{\ast},[\chi,\pi(a)]=0,\forall a\in$                               | $\chi=\chi^{\ast},[\chi,\pi(a)]=0,\forall a\in$                               | $J:\mathcal{H}\rightarrow\mathcal{H}$ such that $J\mathcal{D}=\epsilon\mathcal{D}J, J^2=\epsilon', J\chi=\epsilon''\chi J$ with the following                                                                                                                                                                                                                                                                         |
| cardona-qft-methods_ F00795 | 43   | ■ 1.000 | $\mathcal{A},\mathcal{D}\chi=-\chi\mathcal{D}$                                | $\mathcal{A},\mathcal{D}\chi=-\chi\mathcal{D}$                                |                                                                                                                                                                                                                                                                                                                                                                                                                       |
| cardona-qft-methods_ F00796 | 43   | ■ 1.000 | $d\in\mathbb{Z}/8$                                                            | $d\in\mathbb{Z}/8$                                                            | $\begin{array}{c cccccccc} d & 0 & 1 & 2 & 3 & 4 & 5 & 6 & 7 \\ \hline \epsilon & 1 & -1 & 1 & 1 & 1 & -1 & 1 & 1 \end{array}$                                                                                                                                                                                                                                                                                        |
| cardona-qft-methods_ F00797 | 43   | ■ 1.000 | $J:\mathcal{H}\rightarrow\mathcal{H}$                                         | $J:\mathcal{H}\rightarrow\mathcal{H}$                                         | $\begin{array}{c cccccccc} \epsilon & 1 & 1 & -1 & -1 & -1 & -1 & 1 & 1 \end{array}$                                                                                                                                                                                                                                                                                                                                  |
| cardona-qft-methods_ F00798 | 43   | ■ 1.000 | $J\mathcal{D}=\epsilon\mathcal{D}J, J^2=\epsilon', J\chi=\epsilon''\chi J$    | $J\mathcal{D}=\epsilon\mathcal{D}J, J^2=\epsilon', J\chi=\epsilon''\chi J$    | $\begin{array}{c cccccccc} \epsilon & 1 & 1 & -1 & -1 & -1 & -1 & 1 & 1 \end{array}$                                                                                                                                                                                                                                                                                                                                  |
| cardona-qft-methods_ F00799 | 43   | ■ 1.000 | $\epsilon,\epsilon',\epsilon''$                                               | $\epsilon,\epsilon',\epsilon''$                                               | $\epsilon''$                                                                                                                                                                                                                                                                                                                                                                                                          |
| cardona-qft-methods_ F00800 | 43   | ■ 1.000 | $\epsilon'$                                                                   | $\epsilon'$                                                                   | $[\pi(a),\pi(b)^{\circ}]=0,\quad [[\mathcal{D},\pi(a)],\pi(b)^{\circ}]=0,\forall a,b\in\mathcal{A}$<br>$(a)^{\circ}=J\pi(a^*)J^{-1}$ is a representation of the opposite alge<br>$l$ -summable (or has metric dimension $d$ ) if the singular val<br>ike $\mu_n(\mathcal{D}^{-1})=\mathcal{O}(n^{-1/d})$ .<br>regular if $\mathcal{A}$ and $[\mathcal{D},\mathcal{A}]$ are in the domain of $\delta^n$ for all $n\in$ |
| cardona-qft-methods_ F00801 | 43   | ■ 1.000 | $\epsilon''$                                                                  | $\epsilon''$                                                                  | $[\pi(a),\pi(b)^{\circ}]=0,\quad [[\mathcal{D},\pi(a)],\pi(b)^{\circ}]=0,\forall a,b\in\mathcal{A}$<br>$(a)^{\circ}=J\pi(a^*)J^{-1}$ is a representation of the opposite alge<br>$l$ -summable (or has metric dimension $d$ ) if the singular val<br>ike $\mu_n(\mathcal{D}^{-1})=\mathcal{O}(n^{-1/d})$ .<br>regular if $\mathcal{A}$ and $[\mathcal{D},\mathcal{A}]$ are in the domain of $\delta^n$ for all $n\in$ |
| cardona-qft-methods_ F00802 | 43   | ■ 1.000 | $\pi(a)^{\circ}=J\pi(a^*)J^{-1}$                                              | $\pi(a)^{\circ}=J\pi(a^*)J^{-1}$                                              | $\pi(a)\pi(b)^{\circ}[\mathcal{D},\pi(a_1)]\cdots[\mathcal{D},\pi(a_d)]$ and $d$ is its metric dimension.                                                                                                                                                                                                                                                                                                             |
| cardona-qft-methods_ F00803 | 43   | ■ 1.000 | $\mathcal{A}^{\circ}$                                                         | $\mathcal{A}^{\circ}$                                                         | $\pi(a)\pi(b)^{\circ}[\mathcal{D},\pi(a_1)]\cdots[\mathcal{D},\pi(a_d)]$ and $d$ is its metric dimension.                                                                                                                                                                                                                                                                                                             |
| cardona-qft-methods_ F00804 | 43   | ■ 1.000 | $\mu_n(\mathcal{D}^{-1})=\mathcal{O}(n^{-1/d})$                               | $\mu_n(\mathcal{D}^{-1})=\mathcal{O}(n^{-1/d})$                               |                                                                                                                                                                                                                                                                                                                                                                                                                       |
| cardona-qft-methods_ F00805 | 43   | ■ 1.000 | $[\mathcal{D},\mathcal{A}]$                                                   | $[\mathcal{D},\mathcal{A}]$                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                       |
| cardona-qft-methods_ F00806 | 43   | ■ 1.000 | $\delta^n$                                                                    | $\delta^n$                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                       |
| cardona-qft-methods_ F00807 | 43   | ■ 1.000 | $n\in\mathbb{N}$                                                              | $n\in\mathbb{N}$                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                       |
| cardona-qft-methods_ F00808 | 43   | ■ 1.000 | $\delta(T)=[ \mathcal{D} ,T]$                                                 | $\delta(T)=[ \mathcal{D} ,T]$                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                       |
| cardona-qft-methods_ F00809 | 43   | ■ 1.000 | $\mathcal{H}^{\infty} =$                                                      | $\mathcal{H}^{\infty} =$                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                       |
| cardona-qft-methods_ F00810 | 43   | ■ 1.000 | $\bigcap_k\mathcal{D}^k$                                                      | $\bigcap_k\mathcal{D}^k$                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                       |
| cardona-qft-methods_ F00811 | 43   | ■ 1.000 | $c\in$                                                                        | $c\in$                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                       |
| cardona-qft-methods_ F00812 | 43   | ■ 1.000 | $Z_d(\mathcal{A},\mathcal{A}\otimes\mathcal{A}^{\circ})$                      | $Z_d(\mathcal{A},\mathcal{A}\otimes\mathcal{A}^{\circ})$                      |                                                                                                                                                                                                                                                                                                                                                                                                                       |
| cardona-qft-methods_ F00813 | 43   | ■ 1.000 | $\pi_{\mathcal{D}}(c)=\chi$                                                   | $\pi_{\mathcal{D}}(c)=\chi$                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                       |
| cardona-qft-methods_ F00814 | 43   | ■ 1.000 | $\pi_{\mathcal{D}}((a\otimes b^{\circ})\otimes a_1\otimes\cdots\otimes a_d)=$ | $\pi_{\mathcal{D}}((a\otimes b^{\circ})\otimes a_1\otimes\cdots\otimes a_d)=$ |                                                                                                                                                                                                                                                                                                                                                                                                                       |
| cardona-qft-methods_ F00815 | 43   | ■ 1.000 | $\pi(a)\pi(b)^{\circ}[\mathcal{D},\pi(a_1)]\cdots[\mathcal{D},\pi(a_d)]$      | $\pi(a)\pi(b)^{\circ}[\mathcal{D},\pi(a_1)]\cdots[\mathcal{D},\pi(a_d)]$      |                                                                                                                                                                                                                                                                                                                                                                                                                       |
| cardona-qft-methods_ F00816 | 44   | ■ 1.000 | $\{ \}^{\{20,27,30\}}$                                                        | $20,27,30$                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                       |

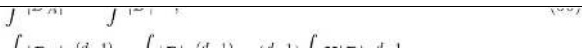
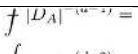
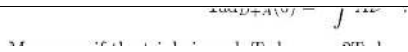
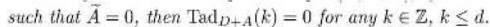
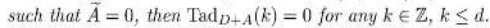
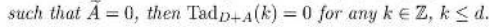
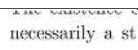
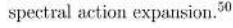
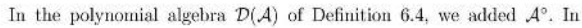
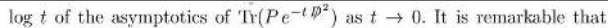
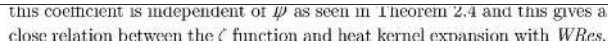
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| cardona-qft-methods_F00817 | 44   | ■ 1.000 | J                                                                                            | $J$                                                                                          | dimension is given by the finite part of the model. <sup>20,27,30</sup>                                                                                                       |
| cardona-qft-methods_F00818 | 44   | ■ 1.000 | $\mathcal{A} \simeq C^\infty(M)$                                                             | $\mathcal{A} \simeq C^\infty(M)$                                                             | $\mathcal{A} \simeq C^\infty(M)$ and $D$ is a Dirac operator.                                                                                                                 |
| cardona-qft-methods_F00819 | 44   | ■ 1.000 | $M = \operatorname{Sp}(\mathcal{A}) = \operatorname{Sp}(A)$                                  | $M = \operatorname{Sp}(\mathcal{A}) = \operatorname{Sp}(A)$                                  | is known only via its principal symbol, so is not unique. $J$ encodes the                                                                                                     |
| cardona-qft-methods_F00820 | 44   | ■ 1.000 | $\phi_1, \phi_2$                                                                             | $\phi_1, \phi_2$                                                                             | spectrum $\mathcal{A}$ .<br>It is known that one cannot hear the shape of a drum since the knowl-                                                                             |
| cardona-qft-methods_F00821 | 44   | ■ 1.000 | K                                                                                            | $K$                                                                                          | triple) since the knowledge of the spectrum of the Dirac operator and the                                                                                                     |
| cardona-qft-methods_F00822 | 44   | ■ 1.000 | $t \in \mathbb{R}$                                                                           | $t \in \mathbb{R}$                                                                           |                                                                                                                                                                               |
| cardona-qft-methods_F00823 | 44   | ■ 1.000 | $F_t: T \in \mathcal{B}(\mathcal{H}) \rightarrow e^{it \mathcal{D} } T e^{-it \mathcal{D} }$ | $F_t: T \in \mathcal{B}(\mathcal{H}) \rightarrow e^{it \mathcal{D} } T e^{-it \mathcal{D} }$ |                                                                                                                                                                               |
| cardona-qft-methods_F00824 | 44   | ■ 1.000 | $\alpha \in \mathbb{R}$                                                                      | $\alpha \in \mathbb{R}$                                                                      |                                                                                                                                                                               |
| cardona-qft-methods_F00825 | 44   | ■ 0.777 | $OP^0 = \{T \mid t \rightarrow F_t(T) \in C^\infty(\mathbb{R}, \mathcal{B}(\mathcal{H}))\}$  | $OP^0 = \{T \mid t \rightarrow F_t(T) \in C^\infty(\mathbb{R}, \mathcal{B}(\mathcal{H}))\}$  |                                                                                                                                                                               |
| cardona-qft-methods_F00826 | 44   | ■ 0.777 | $\leq 0$                                                                                     | $\leq 0$                                                                                     |                                                                                                                                                                               |
| cardona-qft-methods_F00827 | 44   | ■ 1.000 | $OP^\alpha = \{T \mid  T ^{ \mathcal{D} ^{-\alpha}} \in OP^0\}$                              | $OP^\alpha = \{T \mid  T ^{ \mathcal{D} ^{-\alpha}} \in OP^0\}$                              |                                                                                                                                                                               |
| cardona-qft-methods_F00828 | 44   | ■ 1.000 | $\leq \alpha$                                                                                | $\leq \alpha$                                                                                |                                                                                                                                                                               |
| cardona-qft-methods_F00829 | 45   | ■ 1.000 | $\delta(T) = [ \mathcal{D} , T], \quad \nabla(T) = [\mathcal{D}^2, T]$                       | $\delta(T) = [ \mathcal{D} , T], \quad \nabla(T) = [\mathcal{D}^2, T]$                       |                                                                                                                                                                               |
| cardona-qft-methods_F00830 | 45   | ■ 1.000 | $C^\infty(M) = OP^0 \bigcap L^\infty(M)$                                                     | $C^\infty(M) = OP^0 \bigcap L^\infty(M)$                                                     | <sup>34</sup> B. Iochum                                                                                                                                                       |
| cardona-qft-methods_F00831 | 45   | ■ 1.000 | $L^\infty(M)$                                                                                | $L^\infty(M)$                                                                                | <sup>34</sup> B. Iochum                                                                                                                                                       |
| cardona-qft-methods_F00832 | 45   | ■ 1.000 | $\mathcal{A} = C^\infty(M)$                                                                  | $\mathcal{A} = C^\infty(M)$                                                                  | Moreover we set $\delta(T) = [ \mathcal{D} , T] \quad \nabla(T) = [\mathcal{D}^2, T]$                                                                                         |
| cardona-qft-methods_F00833 | 45   | ■ 1.000 | $OP^\alpha$                                                                                  | $OP^\alpha$                                                                                  | Moreover we set $\delta(T) = [ \mathcal{D} , T] \quad \nabla(T) = [\mathcal{D}^2, T]$                                                                                         |
| cardona-qft-methods_F00834 | 45   | ■ 0.983 | $\mathcal{A} \subset OP^0 = \bigcap_{k \geq 0} \delta^k \subset$                             | $\mathcal{A} \subset OP^0 = \bigcap_{k \geq 0} \delta^k \subset$                             | mann algebra generated by $\mathcal{A} = C^\infty(M)$ . The spaces $OP^\alpha$ have the expected properties:                                                                  |
| cardona-qft-methods_F00835 | 45   | ■ 1.000 | $\alpha, \beta \in \mathbb{R}$                                                               | $\alpha, \beta \in \mathbb{R}$                                                               | <b>Prop 6.1.</b> Assume that $\ell$ .                                                                                                                                         |
| cardona-qft-methods_F00836 | 45   | ■ 1.000 | $X = a \mathcal{D} [\mathcal{D}, b]\mathcal{D}^{-3}$                                         | $X = a \mathcal{D} [\mathcal{D}, b]\mathcal{D}^{-3}$                                         | order of $a$ is 0, of $ \mathcal{D} $ is 1, of $[\mathcal{D}, b]$ is 0 and of $\mathcal{D}^{-3}$ is -3, we get $X \in OP^{-2}$ .                                              |
| cardona-qft-methods_F00837 | 45   | ■ 0.842 | $[\mathcal{D}, b]$                                                                           | $[\mathcal{D}, b]$                                                                           | <b>Definition 6.4.</b> Let $(\mathcal{A}, \mathcal{H}, \mathcal{D})$ be a spectral triple and $\mathcal{D}(\mathcal{A})$ be the poly-                                         |
| cardona-qft-methods_F00838 | 45   | ■ 0.842 | $\mathcal{D}^{-3}$                                                                           | $\mathcal{D}^{-3}$                                                                           | <b>Definition 6.4.</b> Let $(\mathcal{A}, \mathcal{H}, \mathcal{D})$ be a spectral triple and $\mathcal{D}(\mathcal{A})$ be the poly-                                         |
| cardona-qft-methods_F00839 | 45   | ■ 0.842 | $X \in OP^{-2}$                                                                              | $X \in OP^{-2}$                                                                              | <b>Definition 6.4.</b> Let $(\mathcal{A}, \mathcal{H}, \mathcal{D})$ be a spectral triple and $\mathcal{D}(\mathcal{A})$ be the poly-                                         |
| cardona-qft-methods_F00840 | 45   | ■ 0.838 | $\mathcal{D}(\mathcal{A})$                                                                   | $\mathcal{D}(\mathcal{A})$                                                                   | ifferential operators as                                                                                                                                                      |
| cardona-qft-methods_F00841 | 45   | ■ 1.000 | $\mathcal{A}, \mathcal{A}^\circ, \mathcal{D}$                                                | $\mathcal{A}, \mathcal{A}^\circ, \mathcal{D}$                                                | $\Psi(\mathcal{A}) = \{T \mid \forall N \in \mathbb{N} \exists P \in \mathcal{D}(\mathcal{A}), R \in OP^{-N}, p \in \mathbb{N} \text{ s.t. } T = P \mathcal{D} ^{-p} + R\}$ . |
| cardona-qft-methods_F00842 | 45   | ■ 1.000 | $OP^{-\infty}$                                                                               | $OP^{-\infty}$                                                                               | is a pseudo-differential operator of order $\leq -\infty$ .                                                                                                                   |

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| cardona-qft-methods_F00843 | 45   | ■ 1.000 | N                                                                                            | $N$                                                                                          | 6.3. Zeta-functions and dimension spectrum                                                                                                                                                                                                           |
| cardona-qft-methods_F00844 | 45   | ■ 1.000 | $\mathcal{D} \in \operatorname{Diff}^1(M, E)$                                                | $\mathcal{D} \in \operatorname{Diff}^1(M, E)$                                                | 6.3. Zeta-functions and dimension spectrum                                                                                                                                                                                                           |
| cardona-qft-methods_F00845 | 45   | ■ 1.000 | $\Psi(\left(C^{\infty}(M)\right) \subset \Psi D^0(M, E)$                                     | $\Psi\left(C^{\infty}(M)\right) \subset \Psi D O(M, E)$                                      | Definition 6.5. For $P \in \Psi^*(A)$ , we define the ze                                                                                                                                                                                             |
| cardona-qft-methods_F00846 | 45   | ■ 1.000 | $\{ \}^{\{30,33,49\}}$                                                                       | 30,33,49                                                                                     | $\zeta_{\mathcal{D}}: s \in \mathbb{C} \rightarrow \operatorname{Tr}\left(F \mathcal{L} ^{-s}\right) \quad (20)$<br>which makes sense since for $\Re(s) \gg 1$ , $\mathcal{D} \mathcal{D} -s \in \mathcal{L}^1(\mathcal{H})$                         |
| cardona-qft-methods_F00847 | 45   | ■ 1.000 | $P \in \Psi^*(\mathcal{A})$                                                                  | $P \in \Psi^*(\mathcal{A})$                                                                  | In (25), we assume $\mathcal{D}$ invertible since otherwise, one can replace $\mathcal{D}$ by the                                                                                                                                                    |
| cardona-qft-methods_F00848 | 45   | ■ 1.000 | $P($                                                                                         | $P($                                                                                         | inve                                                                                                                                                                                                                                                 |
| cardona-qft-methods_F00849 | 45   | ■ 1.000 | $\mathcal{D})$                                                                               | $\mathcal{D})$                                                                               | inve                                                                                                                                                                                                                                                 |
| cardona-qft-methods_F00850 | 45   | ■ 1.000 | $\Re(s) \gg 1, P \mathcal{D} ^{-s} \in \mathcal{L}^1(\mathcal{H})$                           | $\Re(s) \gg 1, P \mathcal{D} ^{-s} \in \mathcal{L}^1(\mathcal{H})$                           |                                                                                                                                                                                                                                                      |
| cardona-qft-methods_F00851 | 45   | ■ 0.743 | $S d$                                                                                        | $Sd$                                                                                         |                                                                                                                                                                                                                                                      |
| cardona-qft-methods_F00852 | 45   | ■ 0.743 | $\zeta_{\mathcal{D}}^P(s) \mid P \in$                                                        | $\zeta_{\mathcal{D}}^P(s) \mid P \in$                                                        |                                                                                                                                                                                                                                                      |
| cardona-qft-methods_F00853 | 45   | ■ 1.000 | $\left.\Psi(\mathcal{A}) \cap O P^0\right\}$                                                 | $\Psi(\mathcal{A}) \cap O P^0\}$                                                             |                                                                                                                                                                                                                                                      |
| cardona-qft-methods_F00854 | 45   | ■ 1.000 | $f P=\operatorname{Res}_{s=0} \zeta_{\mathcal{D}}^P(s)$                                      | $f P=\operatorname{Res}_{s=0} \zeta_{\mathcal{D}}^P(s)$                                      |                                                                                                                                                                                                                                                      |
| cardona-qft-methods_F00855 | 45   | ■ 1.000 | $\mathcal{D}+P, P$                                                                           | $\mathcal{D}+P, P$                                                                           |                                                                                                                                                                                                                                                      |
| cardona-qft-methods_F00856 | 45   | ■ 0.928 | $f X=0$                                                                                      | $f X=0$                                                                                      |                                                                                                                                                                                                                                                      |
| cardona-qft-methods_F00857 | 45   | ■ 1.000 | $X$                                                                                          | $X$                                                                                          |                                                                                                                                                                                                                                                      |
| cardona-qft-methods_F00858 | 46   | ■ 1.000 | $\mathcal{U}(\mathcal{A})$                                                                   | $\mathcal{U}(\mathcal{A})$                                                                   | 6.4. One-forms and fluctuations of $\mathcal{D}$ One-forms and                                                                                                                                                                                       |
| cardona-qft-methods_F00859 | 46   | ■ 1.000 | $\alpha_u$                                                                                   | $\alpha_u$                                                                                   | 6.4. One-forms and fluctuations of $\mathcal{D}$ One-forms and                                                                                                                                                                                       |
| cardona-qft-methods_F00860 | 46   | ■ 1.000 | $a \in \mathcal{A} \rightarrow u a u^* \in \mathcal{A}$                                      | $a \in \mathcal{A} \rightarrow u a u^* \in \mathcal{A}$                                      | that one forms the fluctuations of $\mathcal{D}$ one forms and fluctuations of $D$                                                                                                                                                                   |
| cardona-qft-methods_F00861 | 46   | ■ 0.999 | $\operatorname{Inn}(\mathcal{A})$                                                            | $\operatorname{Inn}(\mathcal{A})$                                                            | The unitary group $\mathcal{U}(\mathcal{A})$ of $\mathcal{A}$ gives rise to the automorphism $\alpha_u :$                                                                                                                                            |
| cardona-qft-methods_F00862 | 46   | ■ 0.999 | $\operatorname{Aut}(\mathcal{A})=$                                                           | $\operatorname{Aut}(\mathcal{A})=$                                                           | The unitary group $\mathcal{U}(\mathcal{A})$ of $\mathcal{A}$ gives rise to the automorphism $\alpha_u :$                                                                                                                                            |
| cardona-qft-methods_F00863 | 46   | ■ 1.000 | $\{\alpha \in \operatorname{Aut}(A) \mid \alpha(\mathcal{A}) \subset \mathcal{A}\}$          | $\{\alpha \in \operatorname{Aut}(A) \mid \alpha(\mathcal{A}) \subset \mathcal{A}\}$          | $a \in \mathcal{A} \rightarrow u a u^* \in \mathcal{A}$ . This defines the inner automorphisms group $\operatorname{Inn}(\mathcal{A})$ which is a normal subgroup of the automorphisms $\operatorname{Aut}(\mathcal{A})$                             |
| cardona-qft-methods_F00864 | 46   | ■ 1.000 | $\mathcal{A}=C^{\infty}(M, M_n(\mathbb{C})) \simeq C^{\infty}(M) \otimes M_n(\mathbb{C})$    | $\mathcal{A}=C^{\infty}(M, M_n(\mathbb{C})) \simeq C^{\infty}(M) \otimes M_n(\mathbb{C})$    | $\operatorname{Inn}(\mathcal{A})$ which is a normal subgroup of the automorphisms $\operatorname{Aut}(\mathcal{A})=\{\alpha \in \operatorname{Aut}(A) \mid \alpha(\mathcal{A}) \subset \mathcal{A}\}$ . For instance, in case of a gauge theory, the |
| cardona-qft-methods_F00865 | 46   | ■ 0.997 | $\mathcal{G}=C^{\infty}(M, P S U(n))$                                                        | $\mathcal{G}=C^{\infty}(M, P S U(n))$                                                        | algebra $\mathcal{A}=C^{\infty}(M, M_n(\mathbb{C})) \simeq C^{\infty}(M) \otimes M_n(\mathbb{C})$ is typically used.                                                                                                                                 |
| cardona-qft-methods_F00866 | 46   | ■ 1.000 | $\operatorname{Aut}\left(C^{\infty}(M)\right) \simeq \operatorname{Diff}(M)$                 | $\operatorname{Aut}\left(C^{\infty}(M)\right) \simeq \operatorname{Diff}(M)$                 | Then, $\operatorname{Inn}(\mathcal{A})$ is locally isomorphic to $\mathcal{G}=C^{\infty}(M, P S U(n))$ . Since $\operatorname{Aut}\left(C^{\infty}(M)\right) \simeq \operatorname{Diff}(M)$ , we get a complete parallel analogy between fol-        |
| cardona-qft-methods_F00867 | 46   | ■ 0.998 | $\{ \}^{\{24,49\}}$                                                                          | 24,49                                                                                        | One-forms are defined as $\Omega_{\mathcal{D}}^1(\mathcal{A})=\operatorname{span}\{a d$                                                                                                                                                              |
| cardona-qft-methods_F00868 | 46   | ■ 0.852 | $\Omega_{\mathcal{D}}^1(\mathcal{A})=\operatorname{span}\{a d b \mid a, b \in \mathcal{A}\}$ | $\Omega_{\mathcal{D}}^1(\mathcal{A})=\operatorname{span}\{a d b \mid a, b \in \mathcal{A}\}$ | the Hilbert space $\mathcal{H}$ , gives a natural hermitean fluctuation of $\mathcal{D}$ : $\mathcal{D} \rightarrow \mathcal{D}_A =$                                                                                                                 |

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| cardona-qft-methods_F00869 | 46   | 0.852 | $d b=[\mathrm{d}, b]$                                                    | $db = [\mathcal{D}, b]$                                                  | the Hilbert space $\mathcal{H}$ , gives a natural hermitean fluctuation of $\mathcal{D}$ : $\mathcal{D} \rightarrow \mathcal{D}_A =$ |
| cardona-qft-methods_F00870 | 46   | 0.998 | $\mathrm{d} : \mathrm{d} \rightarrow \mathrm{d}_A =$                     | $\mathcal{D} : \mathcal{D} \rightarrow \mathcal{D}_A =$                  | $\mathcal{D}_{\tilde{A}} = \mathcal{D} + \tilde{A}, \quad \tilde{A} = A + \epsilon J A J^{-1}, \quad A = A^*. \quad (26)$            |
| cardona-qft-methods_F00871 | 46   | 1.000 | $\mathrm{d} + A$                                                         | $\mathcal{D} + A$                                                        | The next two results show that, with the same algebra $\mathcal{A}$ and Hilbert                                                      |
| cardona-qft-methods_F00872 | 46   | 1.000 | $A = A^* \in \Omega_{\mathcal{D}}^1(\mathcal{A})$                        | $A = A^* \in \Omega_{\mathcal{D}}^1(\mathcal{A})$                        | The next two results show that, with the same algebra $\mathcal{A}$ and Hilbert                                                      |
| cardona-qft-methods_F00873 | 46   | 0.552 | $\Omega_{\mathcal{D}}^1(C^\infty(M)) = \{c(da) \mid a \in C^\infty(M)\}$ | $\Omega_{\mathcal{D}}^1(C^\infty(M)) = \{c(da) \mid a \in C^\infty(M)\}$ | space $\mathcal{H}$ , a fluctuation of $\mathcal{D}$ still give rise                                                                 |
| cardona-qft-methods_F00874 | 46   | 1.000 | $\mathcal{D}_A J = \epsilon J \mathcal{D}_A$                             | $\mathcal{D}_A J = \epsilon J \mathcal{D}_A$                             | $(\mathcal{A}, \mathcal{H}, \mathcal{D}_A)$                                                                                          |
| cardona-qft-methods_F00875 | 46   | 0.957 | $(\mathcal{A}, \mathcal{H}, \mathcal{D}_A)$                              | $(\mathcal{A}, \mathcal{H}, \mathcal{D}_A)$                              |                                                                                                                                      |
| cardona-qft-methods_F00876 | 46   | 0.990 | $(\mathcal{A}, \mathcal{H}, \mathcal{D}_{\tilde{A}})$                    | $(\mathcal{A}, \mathcal{H}, \mathcal{D}_{\tilde{A}})$                    |                                                                                                                                      |
| cardona-qft-methods_F00877 | 46   | 0.967 | $A \in \Omega_{\mathcal{D}}^1$                                           | $A \in \Omega_{\mathcal{D}}^1$                                           |                                                                                                                                      |
| cardona-qft-methods_F00878 | 46   | 0.967 | $\mathcal{D}_A$                                                          | $\mathcal{D}_A$                                                          |                                                                                                                                      |
| cardona-qft-methods_F00879 | 46   | 0.967 | $\mathcal{D}_{\tilde{A}}$                                                | $\mathcal{D}_{\tilde{A}}$                                                |                                                                                                                                      |
| cardona-qft-methods_F00880 | 46   | 1.000 | $\mathcal{D} \rightarrow \mathcal{D}_u = u \mathcal{D} u^*$              | $\mathcal{D} \rightarrow \mathcal{D}_u = u \mathcal{D} u^*$              |                                                                                                                                      |
| cardona-qft-methods_F00881 | 46   | 1.000 | $\mathcal{D}_u = \mathcal{D} + u [\mathcal{D}, u^*]$                     | $\mathcal{D}_u = \mathcal{D} + u [\mathcal{D}, u^*]$                     |                                                                                                                                      |
| cardona-qft-methods_F00882 | 47   | 1.000 | $\mathcal{D}_A = 0$                                                      | $\mathcal{D}_A = 0$                                                      | sensation but to be an inner flu                                                                                                     |
| cardona-qft-methods_F00883 | 47   | 1.000 | $\mathcal{D} \neq 0$                                                     | $\mathcal{D} \neq 0$                                                     | sensation but to be an inner flu                                                                                                     |
| cardona-qft-methods_F00884 | 47   | 0.728 | $(\mathcal{A}, \mathcal{D}, \mathcal{H})$                                | $(\mathcal{A}, \mathcal{D}, \mathcal{H})$                                | <b>Lemma 6.2.</b> Let $(\mathcal{A}, \mathcal{D}, \mathcal{H})$ be a spectral triple and $X \in \Psi(\mathcal{A})$ . Then            |
| cardona-qft-methods_F00885 | 47   | 0.728 | $X \in \Psi(\mathcal{A})$                                                | $X \in \Psi(\mathcal{A})$                                                | <b>Lemma 6.2.</b> Let $(\mathcal{A}, \mathcal{D}, \mathcal{H})$ be a spectral triple and $X \in \Psi(\mathcal{A})$ . Then            |
| cardona-qft-methods_F00886 | 47   | 0.592 | $f X^* = \overline{f X}$                                                 | $f X^* = \overline{f X}$                                                 | $f X^* = \overline{f X}$ . If the spectral triple is real, then, for $X \in \Psi(\mathcal{A})$ , $J X J^{-1} \in$                    |
| cardona-qft-methods_F00887 | 47   | 0.592 | $X \in \Psi(\mathcal{A}), J X J^{-1} \in$                                | $X \in \Psi(\mathcal{A}), J X J^{-1} \in$                                | $f X^* = \overline{f X}$ . If the spectral triple is real, then, for $X \in \Psi(\mathcal{A})$ , $J X J^{-1} \in$                    |
| cardona-qft-methods_F00888 | 47   | 0.997 | $\Psi(\mathcal{A})$                                                      | $\Psi(\mathcal{A})$                                                      | $\Psi(\mathcal{A})$ and $f J X J^{-1} = f X^* = \overline{f X}$ .<br>If $A = A^*$ then, for $k, l \in \mathbb{N}$                    |
| cardona-qft-methods_F00889 | 47   | 0.997 | $f J X J^{-1} = f X^* = \overline{f X}$                                  | $f J X J^{-1} = f X^* = \overline{f X}$                                  | $\Psi(\mathcal{A})$ and $f J X J^{-1} = f X^* = \overline{f X}$ .<br>If $A = A^*$ then, for $k, l \in \mathbb{N}$                    |
| cardona-qft-methods_F00890 | 47   | 0.999 | $A = A^*$                                                                | $A = A^*$                                                                | $f A^l  \mathcal{D} ^{-k}, f \chi A^l  \mathcal{D} ^{-k}$ and $f A^l \mathcal{D}  \mathcal{D} ^{-k}$ are real valued.                |
| cardona-qft-methods_F00891 | 47   | 0.999 | $k, l \in \mathbb{N}$                                                    | $k, l \in \mathbb{N}$                                                    | $f A^l  \mathcal{D} ^{-k}, f \chi A^l  \mathcal{D} ^{-k}$ and $f A^l \mathcal{D}  \mathcal{D} ^{-k}$ are real valued.                |
| cardona-qft-methods_F00892 | 47   | 0.999 | $f A^l  \mathcal{D} ^{-k}, f \left( A  \mathcal{D} ^{-1} \right)^k$      | $f A^l  \mathcal{D} ^{-k}, f \left( A \mathcal{D}^{-1} \right)^k$        | $f A^l  \mathcal{D} ^{-k}, f \chi A^l  \mathcal{D} ^{-k}$ and $f A^l \mathcal{D}  \mathcal{D} ^{-k}$ are real valued.                |
| cardona-qft-methods_F00893 | 47   | 0.919 | $f A^l  \mathcal{D} ^{-k}, f \chi A^l  \mathcal{D} ^{-k}$                | $f A^l  \mathcal{D} ^{-k}, f \chi A^l  \mathcal{D} ^{-k}$                | We remark that the fluctuations leave invariant the first term                                                                       |
| cardona-qft-methods_F00894 | 47   | 0.919 | $f A^l  \mathcal{D}   \mathcal{D} ^{-k}$                                 | $f A^l \mathcal{D}  \mathcal{D} ^{-k}$                                   | We remark that the fluctuations leave invariant the first term                                                                       |

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| cardona-qft-methods_F08895 | 47   | ■ 1.000 | $A+\epsilon J A J^{-1}$                                                                                                                                                                                  | $A+\epsilon J A J^{-1}$                                                                                 | $\zeta_{D_{\tilde{A}}}(0)=\zeta_D(0)+\sum_{q=1}^d\frac{(-1)^q}{q}f(\tilde{A}D^{-1})^q.$                                                            |
| cardona-qft-methods_F08896 | 47   | ■ 1.000 | $A\in\Omega_{\mathcal{D}}^1(\mathcal{A})$                                                                                                                                                                | $A\in\Omega_{\mathcal{D}}^1(\mathcal{A})$                                                               | notation                                                                                                                                           |
| cardona-qft-methods_F08897 | 47   | ■ 0.982 | $\zeta_{D_{\tilde{A}}}(\theta)=\zeta_D(\theta)+\sum_{q=1}^d\frac{(-1)^q}{q}f(\tilde{A}D^{-1})^q$                                                                                                         | $\zeta_{D_{\tilde{A}}}(0)=\zeta_D(0)+\sum_{q=1}^d\frac{(-1)^q}{q}f\left(\tilde{A}D^{-1}\right)^q$       | $\sigma_z(T)= D ^{2T} D ^{-z},\,z\in\mathbb{C},$                                                                                                   |
| cardona-qft-methods_F08898 | 47   | ■ 0.942 | $\nabla(T)=\left[\mathcal{D}^2,T\right]$                                                                                                                                                                 | $\nabla(T)=[\mathcal{D}^2,T]$                                                                           |                                                                                                                                                    |
| cardona-qft-methods_F08899 | 47   | ■ 1.000 | $\sigma_z$                                                                                                                                                                                               | $\sigma_z$                                                                                              |                                                                                                                                                    |
| cardona-qft-methods_F08900 | 47   | ■ 1.000 | $T\in OP^q$                                                                                                                                                                                              | $T\in OP^q$                                                                                             |                                                                                                                                                    |
| cardona-qft-methods_F08901 | 47   | ■ 1.000 | $g(z,\,r)=\frac{1}{r!}\left(\frac{z}{2}\right)\cdots\left(\frac{z}{2}-(r-1)\right)=\binom{z/2}{r}$                                                                                                       | $g(z,r)=\frac{1}{r!}\binom{z}{2}\cdots\left(\frac{z}{2}-(r-1)\right)=\binom{z/2}{r}$                    |                                                                                                                                                    |
| cardona-qft-methods_F08902 | 47   | ■ 1.000 | $g(z,\,\theta)=$                                                                                                                                                                                         | $g(z,0)=$                                                                                               |                                                                                                                                                    |
| cardona-qft-methods_F08903 | 48   | ■ 1.000 | $P_{\{A\}}$                                                                                                                                                                                              | $P_A$                                                                                                   | $\mathcal{D}_A=\mathcal{D}+A$ where $A=A+\varepsilon JAJ^{-1}$ , $\quad D_A=\mathcal{D}_A+P_A\quad (28)$                                           |
| cardona-qft-methods_F08904 | 48   | ■ 1.000 | $\widetilde{A}\in\mathcal{D}(\mathcal{A})\cap OP^0$                                                                                                                                                      | $\widetilde{A}\in\mathcal{D}(\mathcal{A})\cap OP^0$                                                     | $\mathcal{D}_A=\mathcal{D}+A$ where $A=A+\varepsilon JAJ^{-1}$ , $\quad D_A=\mathcal{D}_A+P_A\quad (28)$                                           |
| cardona-qft-methods_F08905 | 48   | ■ 1.000 | $\mathcal{D}_{\{A\}}\in\mathcal{D}(\mathcal{A})\cap OP^1$                                                                                                                                                | $\mathcal{D}_A\in\mathcal{D}(\mathcal{A})\cap OP^1$                                                     | where $P_A$ is<br>$P_A\in\mathcal{D}(\mathcal{A})$                                                                                                 |
| cardona-qft-methods_F08906 | 48   | ■ 1.000 | $V_{\{A\}}=P_{\{A\}}-P_0$                                                                                                                                                                                | $V_A=P_A-P_0$                                                                                           | We denote $V_A=P_A-P_0$ and, as the following lemma shows, $V_A$ is a                                                                              |
| cardona-qft-methods_F08907 | 48   | ■ 1.000 | $V_{\{A\}}$                                                                                                                                                                                              | $V_A$                                                                                                   | We denote $V_A=P_A-P_0$ and, as the following lemma shows, $V_A$ is a                                                                              |
| cardona-qft-methods_F08908 | 48   | ■ 1.000 | $\bigcap_{k\geq 1}\left(\mathcal{D}_A\right)^k\subseteq\bigcap_{k\geq 1} D ^k$<br>$\subseteq\bigcap_{k\geq 1} D ^k$                                                                                      | $\bigcap_{k\geq 1}(\mathcal{D}_A)^k\subseteq\bigcap_{k\geq 1} D ^k$                                     | (ii) $\mathcal{D}_A\subseteq\bigcap_{k\geq 1} D ^k$ .                                                                                              |
| cardona-qft-methods_F08909 | 48   | ■ 1.000 | $\mathcal{D}_{\{A\}}\subseteq\bigcap_{k\geq 1} \bar{D} ^k$                                                                                                                                               | $\mathcal{D}_A\subseteq\bigcap_{k\geq 1} \bar{D} ^k$                                                    | (iii) For any $\alpha$ ,<br>(iv) $P_A\in OP^{-\alpha}$                                                                                             |
| cardona-qft-methods_F08910 | 48   | ■ 0.808 | $\alpha,\beta\in\mathbb{R}, D ^{\beta}P_{\{A\}} D ^{\alpha}$                                                                                                                                             | $\alpha,\beta\in\mathbb{R}, D ^{\beta}P_A D ^{\alpha}$                                                  | Let us define $\mathbf{v}=\mathbf{v}^2-\mathbf{v}^2=\tilde{\mathbf{A}}\mathbf{v}+\mathbf{v}\tilde{\mathbf{A}}+\tilde{\mathbf{A}}^2-\mathbf{v}^2$ . |
| cardona-qft-methods_F08911 | 48   | ■ 1.000 | $P_{\{A\}}\in OP^{-\infty}$                                                                                                                                                                              | $P_A\in OP^{-\infty}$                                                                                   | $X\in\mathcal{D}_1(\mathcal{A})\cap$                                                                                                               |
| cardona-qft-methods_F08912 | 48   | ■ 1.000 | $X=\mathcal{D}_{\{A\}}^2-\mathcal{D}^2=\widetilde{A}\mathcal{D}+\mathcal{D}\widetilde{A}+\widetilde{A}^2,$<br>$\mathcal{D}_{\{A\}}+\mathcal{D}\widetilde{A}+\widetilde{A}^2,$<br>$\quad X_V=X+V_{\{A\}}$ | $X=\mathcal{D}_A^2-\mathcal{D}^2=\tilde{A}\mathcal{D}+\mathcal{D}\tilde{A}+\tilde{A}^2,\quad X_V=X+V_A$ | for any $A$ .                                                                                                                                      |
| cardona-qft-methods_F08913 | 48   | ■ 1.000 | $X\in\mathcal{D}_1(\mathcal{A})\cap OP^1$                                                                                                                                                                | $X\in\mathcal{D}_1(\mathcal{A})\cap OP^1$                                                               | By definition of $\Lambda_V$ , we get $Y=\log(L^s+\Lambda_V)-\log(L^s)$ .                                                                          |
| cardona-qft-methods_F08914 | 48   | ■ 1.000 | $X_V\sim X\bmod OP^{-\infty}$                                                                                                                                                                            | $X_V\sim X\bmod OP^{-\infty}$                                                                           | By definition of $\Lambda_V$ , we get $Y=\log(L^s+\Lambda_V)-\log(L^s)$ .                                                                          |
| cardona-qft-methods_F08915 | 48   | ■ 1.000 | $Y=\log\left(D_{\{A\}}^2\right)-\log\left(D^2\right)$                                                                                                                                                    | $Y=\log\left(D_A^2\right)-\log\left(D^2\right)$                                                         | <b>Lemma 6.4.</b> $Y$ is a pseudodifferential operator in $OP^{-1}$ with the expansion for any $N\in\mathbb{N}$                                    |
| cardona-qft-methods_F08916 | 48   | ■ 1.000 | $D_{\{A\}}^2=\mathcal{D}_{\{A\}}^2+P_{\{A\}}$                                                                                                                                                            | $D_A^2=\mathcal{D}_A^2+P_A$                                                                             | <b>Lemma 6.4.</b> $Y$ is a pseudodifferential operator in $OP^{-1}$ with the expansion for any $N\in\mathbb{N}$                                    |
| cardona-qft-methods_F08917 | 48   | ■ 1.000 | $X_{\{V\}}$                                                                                                                                                                                              | $X_V$                                                                                                   | $Y\sim\sum_{i=1}^N\sum_{p=0}^{\infty}\frac{(-1)^{k_1+pp+1}}{[k_1+p]}\nabla^{k_p}(X\nabla^{k_{p-1}}(\cdots X\nabla^{k_1}(X)\cdots))D^{-2(k$         |
| cardona-qft-methods_F08918 | 48   | ■ 1.000 | $Y=\log\left(D^2+X_{\{V\}}\right)-\log\left(D^2\right)$                                                                                                                                                  | $Y=\log\left(D^2+X_V\right)-\log\left(D^2\right)$                                                       | $Y\sim\sum_{i=1}^N\sum_{p=0}^{\infty}\frac{(-1)^{k_1+pp+1}}{[k_1+p]}\nabla^{k_p}(X\nabla^{k_{p-1}}(\cdots X\nabla^{k_1}(X)\cdots))D^{-2(k$         |
| cardona-qft-methods_F08919 | 48   | ■ 1.000 | $Y$                                                                                                                                                                                                      | $Y$                                                                                                     | mod $OP^{-N-1}$ . For any $N\in\mathbb{N}$ and $s\in\mathbb{C}$ ,                                                                                  |

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| cardona-qft-methods_F00920 | 48   | ■ 1.000 | $0 \in \mathbb{P}^{-1}$                                                                                    | $OP^{-1}$                                                                                                  | $\text{mod } OP^{-N-1}$ . For any $N \in \mathbb{N}$ and $s \in \mathbb{C}$ ,                                                   |
| cardona-qft-methods_F00921 | 48   | ■ 0.996 | $\bmod 0 \in \mathbb{P}^{-N-1}$                                                                            | $\text{mod } OP^{-N-1}$                                                                                    | <del>Proof of Theorem 9.1. See [1].</del> Since the operator (29) entails that $\zeta_{D_A}(0) - \zeta_D(0) = \text{Tr}(K_1(Y,$ |
| cardona-qft-methods_F00922 | 48   | ■ 0.996 | $s \in \mathbb{C}$                                                                                         | $s \in \mathbb{C}$                                                                                         | <del>Proof of Theorem 9.1. See [1].</del> Since the operator (29) entails that $\zeta_{D_A}(0) - \zeta_D(0) = \text{Tr}(K_1(Y,$ |
| cardona-qft-methods_F00923 | 48   | ■ 1.000 | $K_p(Y, s) \in \mathbb{P}^{-p}$                                                                            | $K_p(Y, s) \in OP^{-p}$                                                                                    |                                                                                                                                 |
| cardona-qft-methods_F00924 | 48   | ■ 1.000 | $p \in \mathbb{N}$                                                                                         | $p \in \mathbb{N}$                                                                                         |                                                                                                                                 |
| cardona-qft-methods_F00925 | 48   | ■ 1.000 | $r_1, \dots, r_p \in \mathbb{N}_0$                                                                         | $r_1, \dots, r_p \in \mathbb{N}_0$                                                                         |                                                                                                                                 |
| cardona-qft-methods_F00926 | 48   | ■ 0.977 | $\varepsilon^{r_1}(Y) \cdots \varepsilon^{r_p}(Y) \in \Psi(\mathcal{A})$                                   | $\varepsilon^{r_1}(Y) \cdots \varepsilon^{r_p}(Y) \in \Psi(\mathcal{A})$                                   |                                                                                                                                 |
| cardona-qft-methods_F00927 | 48   | ■ 1.000 | $\zeta_{D_A}(0) - \zeta_D(0) = \operatorname{Tr} \left( K_1(Y, s)  D ^{-s} \right)_{ s=0}$                 | $\zeta_{D_A}(0) - \zeta_D(0) = \text{Tr} \left( K_1(Y, s)  D ^{-s} \right)_{ s=0}$                         |                                                                                                                                 |
| cardona-qft-methods_F00928 | 48   | ■ 1.000 | $\zeta_{D_A}(0) - \zeta_D(0) = -\frac{1}{2} f Y$                                                           | $\zeta_{D_A}(0) - \zeta_D(0) = -\frac{1}{2} f Y$                                                           |                                                                                                                                 |
| cardona-qft-methods_F00929 | 48   | ■ 1.000 | $f \log((1+S)(1+T)) = f \log(1+S) +$                                                                       | $f \log((1+S)(1+T)) = f \log(1+S) +$                                                                       |                                                                                                                                 |
| cardona-qft-methods_F00930 | 48   | ■ 0.932 | $f \log(1+T)$                                                                                              | $f \log(1+T)$                                                                                              |                                                                                                                                 |
| cardona-qft-methods_F00931 | 48   | ■ 0.932 | $S, T \in \Psi(\mathcal{A}) \cap \mathbb{P}^{-1}$                                                          | $S, T \in \Psi(\mathcal{A}) \cap OP^{-1}$                                                                  |                                                                                                                                 |
| cardona-qft-methods_F00932 | 48   | ■ 0.932 | $\log(1+S) = \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} S^n$                                                 | $\log(1+S) = \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} S^n$                                                 |                                                                                                                                 |
| cardona-qft-methods_F00933 | 49   | ■ 0.999 | $S = D^{-1} A$                                                                                             | $S = D^{-1} A$                                                                                             |                                                                                                                                 |
| cardona-qft-methods_F00934 | 49   | ■ 0.999 | $T = A D^{-1}$                                                                                             | $T = A D^{-1}$                                                                                             |                                                                                                                                 |
| cardona-qft-methods_F00935 | 49   | ■ 0.999 | $f \log(1 + X D^{-2}) = 2 f \log(1 + A D^{-1})$                                                            | $f \log(1 + X D^{-2}) = 2 f \log(1 + A D^{-1})$                                                            |                                                                                                                                 |
| cardona-qft-methods_F00936 | 49   | ■ 1.000 | $-\frac{1}{2} f Y = \sum_{q=1}^d \frac{(-1)^q}{q} f \left( \tilde{A} D^{-1} \right)^q$                     | $-\frac{1}{2} f Y = \sum_{q=1}^d \frac{(-1)^q}{q} f \left( \tilde{A} D^{-1} \right)^q$                     | 38 B. Iochum                                                                                                                    |
| cardona-qft-methods_F00937 | 49   | ■ 1.000 | $k \in \mathbb{N}_0$                                                                                       | $k \in \mathbb{N}_0$                                                                                       | with $D = D^{-1} A$ and $A = A^*$ and $-\frac{1}{2} f Y = \sum_{q=1}^d \frac{(-1)^q}{q} f(\tilde{A} D^{-1})^q$                  |
| cardona-qft-methods_F00938 | 49   | ■ 0.953 | $\operatorname{Res}_{s=d-k} \zeta_{D_A}(s) = \operatorname{Res}_{s=d-k} \zeta_D(s) +$                      | $\operatorname{Res}_{s=d-k} \zeta_{D_A}(s) = \operatorname{Res}_{s=d-k} \zeta_D(s) +$                      | $\operatorname{Res}_{s=d-k} \zeta_{D_A}(s) = \operatorname{Res}_{s=d-k} \zeta_D(s) +$                                           |
| cardona-qft-methods_F00939 | 49   | ■ 1.000 | $h(s, r, p) = (-s/2)^p \int_{0 \leq t_1 \leq \dots \leq t_p \leq 1} g(-st_1, r_1) \cdots g(-st_p, r_p) dt$ | $h(s, r, p) = (-s/2)^p \int_{0 \leq t_1 \leq \dots \leq t_p \leq 1} g(-st_1, r_1) \cdots g(-st_p, r_p) dt$ | $h(s, r, p) = (-s/2)^p \int_{0 \leq t_1 \leq \dots \leq t_p \leq 1} g(-st_1, r_1) \cdots g(-st_p, r_p) dt$                      |
| cardona-qft-methods_F00940 | 49   | ■ 1.000 | $ D_A ^k$                                                                                                  | $ D_A ^k$                                                                                                  | $ D_A ^k \in \Psi^{-k}(\mathcal{A})$ .<br>The following result is quite important since it shows that one can use               |
| cardona-qft-methods_F00941 | 49   | ■ 1.000 | $ D_A ^k \in \Psi^k(\mathcal{A})$                                                                          | $ D_A ^k \in \Psi^k(\mathcal{A})$                                                                          | $f$ for $j$                                                                                                                     |
| cardona-qft-methods_F00942 | 49   | ■ 1.000 | $D_A$                                                                                                      | $D_A$                                                                                                      | pseud                                                                                                                           |
| cardona-qft-methods_F00943 | 49   | ■ 0.718 | $\operatorname{Res}_{s=0} \operatorname{Tr} \left( P  D_A ^{-s} \right) = f P$                             | $\operatorname{Res}_{s=0} \operatorname{Tr} \left( P  D_A ^{-s} \right) = f P$                             | $f  D_A ^{-d} = f  D ^{-d}$ .<br>(30)                                                                                           |

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| cardona-qft-methods_F00944 | 49   | ■ 0.972 | $k \in \mathbb{N} \setminus \{0\} \text{ f}\left D_A\right ^{\{-(d-k)\}}=$           | $k \in \mathbb{N}_0 f \left D_A\right ^{-(d-k)} =$                               |    |
| cardona-qft-methods_F00945 | 49   | ■ 0.999 | $\operatorname{Res}_{s=d-k} \zeta_{D_A}(s)$                                          | $\operatorname{Res}_{s=d-k} \zeta_{D_A}(s)$                                      |    |
| cardona-qft-methods_F00946 | 49   | ■ 0.945 | $\operatorname{Tad}_{\mathcal{D}+A}(k)$                                              | $\operatorname{Tad}_{\mathcal{D}+A}(k)$                                          |                                                                                       |
| cardona-qft-methods_F00947 | 49   | ■ 0.945 | $k \in$                                                                              | $k \in$                                                                          |                                                                                       |
| cardona-qft-methods_F00948 | 49   | ■ 1.000 | $\{d-l: l \in \mathbb{N}\}$                                                          | $\{d-l:l \in \mathbb{N}\}$                                                       |                                                                                       |
| cardona-qft-methods_F00949 | 49   | ■ 1.000 | $A=A^{\ast} \in \Omega_{\mathcal{D}}^1$                                              | $A=A^{\ast} \in \Omega_{\mathcal{D}}^1$                                          |                                                                                       |
| cardona-qft-methods_F00950 | 49   | ■ 1.000 | $\Lambda^k$                                                                          | $\Lambda^k$                                                                      |                                                                                       |
| cardona-qft-methods_F00951 | 49   | ■ 1.000 | $\mathcal{D} \rightarrow \mathcal{D}+A$                                              | $\mathcal{D} \rightarrow \mathcal{D}+A$                                          |                                                                                       |
| cardona-qft-methods_F00952 | 49   | ■ 0.655 | $\mathcal{A}, \mathcal{H}, \mathcal{D}, J$                                           | $\mathcal{A}, \mathcal{H}, \mathcal{D}, J$                                       |                                                                                       |
| cardona-qft-methods_F00953 | 49   | ■ 0.655 | $\operatorname{Tad}_{\mathcal{D}+\tilde{A}}(k)$                                      | $\operatorname{Tad}_{\mathcal{D}+\tilde{A}}(k)$                                  |                                                                                       |
| cardona-qft-methods_F00954 | 49   | ■ 1.000 | $\mathcal{D} \rightarrow \mathcal{D}+\widetilde{A}$                                  | $\mathcal{D} \rightarrow \mathcal{D}+\widetilde{A}$                              |                                                                                       |
| cardona-qft-methods_F00955 | 50   | ■ 1.000 | $\operatorname{Tad}_{\mathcal{D}+\widetilde{A}}=2\operatorname{Tad}_{\mathcal{D}+A}$ | $\operatorname{Tad}_{\mathcal{D}+\tilde{A}}=2\operatorname{Tad}_{\mathcal{D}+A}$ |    |
| cardona-qft-methods_F00956 | 50   | ■ 1.000 | $\widetilde{A}=0$                                                                    | $\tilde{A}=0$                                                                    |   |
| cardona-qft-methods_F00957 | 50   | ■ 1.000 | $\operatorname{Tad}_{\mathcal{D}+A}(k)=0$                                            | $\operatorname{Tad}_{\mathcal{D}+A}(k)=0$                                        |  |
| cardona-qft-methods_F00958 | 50   | ■ 1.000 | $k \in \mathbb{Z}, k \leq d$                                                         | $k \in \mathbb{Z}, k \leq d$                                                     |  |
| cardona-qft-methods_F00959 | 50   | ■ 0.968 | $\left(\operatorname{see}^{12}\right)$                                               | $\left(\operatorname{see}^{12}\right)$                                           |  |
| cardona-qft-methods_F00960 | 50   | ■ 1.000 | $\alpha(b):=\mathcal{D}b\,\mathcal{D}^{-1}$                                          | $\alpha(b):=\mathcal{D}b\mathcal{D}^{-1}$                                        |  |
| cardona-qft-methods_F00961 | 50   | ■ 1.000 | $A=0$                                                                                | $A=0$                                                                            |  |
| cardona-qft-methods_F00962 | 50   | ■ 0.999 | $\left(\mathcal{A}:=C^{\infty}(M),\mathcal{H}:=L^2(M,S),\not{D}\right)$              | $\left(\mathcal{A}:=C^{\infty}(M),\mathcal{H}:=L^2(M,S),\not{D}\right)$          |  |
| cardona-qft-methods_F00963 | 50   | ■ 1.000 | $J\,a\,J^{-1}=a^{\ast},\quad\forall a\in\mathcal{A}$                                 | $JaJ^{-1}=a^{\ast},\quad\forall a\in\mathcal{A}$                                 |  |
| cardona-qft-methods_F00964 | 50   | ■ 1.000 | $\chi:=(-i)^{d/2}\gamma^1\gamma^2\cdots\gamma^d$                                     | $\chi:=(-i)^{d/2}\gamma^1\gamma^2\cdots\gamma^d$                                 |  |
| cardona-qft-methods_F00965 | 50   | ■ 0.951 | $\mathcal{A}^{\circ}\simeq J\mathcal{A}J^{-1}\simeq\mathcal{A}$                      | $\mathcal{A}^{\circ}\simeq J\mathcal{A}J^{-1}\simeq\mathcal{A}$                  |                                                                                       |
| cardona-qft-methods_F00966 | 50   | ■ 0.951 | $JAJ^{-1}=$                                                                          | $JAJ^{-1}=$                                                                      |                                                                                       |
| cardona-qft-methods_F00967 | 50   | ■ 0.979 | $-\epsilon A^{\ast},\quad\forall A\in\Omega_{\mathcal{D}}^1(\mathcal{A})$            | $-\epsilon A^{\ast},\quad\forall A\in\Omega_{\mathcal{D}}^1(\mathcal{A})$        |                                                                                       |
| cardona-qft-methods_F00968 | 50   | ■ 0.999 | $f\,P$                                                                               | $fP$                                                                             |                                                                                       |
| cardona-qft-methods_F00969 | 50   | ■ 1.000 | $\log t$                                                                             | $\log t$                                                                         |                                                                                       |

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| cardona-qft-methods_F08976 | 50   | ■ 1.000 | $\operatorname{Tr}\left(P\,e^{-t\,\not{D}^2}\right)$                                                                                      | $\operatorname{Tr}\left(Pe^{-t\not{D}^2}\right)$                                                                                          |                                                                                                                                                                                                                                                         |
| cardona-qft-methods_F08971 | 50   | ■ 1.000 | $t\rightarrow 0$                                                                                                                          | $t\rightarrow 0$                                                                                                                          |                                                                                                                                                                                                                                                         |
| cardona-qft-methods_F08972 | 50   | ■ 1.000 | $\zeta$                                                                                                                                   | $\zeta$                                                                                                                                   |                                                                                                                                                                                                                                                         |
| cardona-qft-methods_F08973 | 51   | ■ 1.000 | $f\,P=2\,a_0'$                                                                                                                            | $fP=2a_0'$                                                                                                                                |                                                                                                                                                                                                                                                         |
| cardona-qft-methods_F08974 | 51   | ■ 1.000 | $\Psi\left(C^\infty(M)\right)$                                                                                                            | $\Psi\left(C^\infty(M)\right)$                                                                                                            |                                                                                                                                                                                                                                                         |
| cardona-qft-methods_F08975 | 51   | ■ 0.813 | $f\,P=c$                                                                                                                                  | $fP=c$                                                                                                                                    | 40 <i>B. Iochum</i>                                                                                                                                                                                                                                     |
| cardona-qft-methods_F08976 | 51   | ■ 0.813 | $\operatorname{Tr}\left(P\,\not{D}^{-2s}\right)=\frac{1}{\Gamma(s)}\int_0^\infty t^{s-1}\operatorname{Tr}\left(Pe^{-t\not{D}^2}\right)dt$ | $\operatorname{Tr}\left(P\,\not{D}^{-2s}\right)=\frac{1}{\Gamma(s)}\int_0^\infty t^{s-1}\operatorname{Tr}\left(Pe^{-t\not{D}^2}\right)dt$ | 40 <i>B. Iochum</i>                                                                                                                                                                                                                                     |
| cardona-qft-methods_F08977 | 51   | ■ 1.000 | $a_k^{\prime},k\neq 0$                                                                                                                    | $a_k',k\neq 0$                                                                                                                            | so $f\,P=2a_0'$ . Since $f$ <i>WRes</i> $P$ , because $\operatorname{Tr}(P\,\not{D}^{-2s})=\frac{1}{\Gamma(s)}\int_0^\infty t^{s-1}\operatorname{Tr}(Pe^{-t\not{D}^2})dt$ , the                                                                         |
| cardona-qft-methods_F08978 | 51   | ■ 1.000 | $\operatorname{Tr}\left(P\,\not{D}^{-2s}\right)$                                                                                          | $\operatorname{Tr}\left(P\,\not{D}^{-2s}\right)$                                                                                          | so $f\,P=2a_0'$ . Since $f$ <i>WRes</i> $P$ , because $\operatorname{Tr}(P\,\not{D}^{-2s})=\frac{1}{\Gamma(s)}\int_0^\infty t^{s-1}\operatorname{Tr}(Pe^{-t\not{D}^2})dt$ , the                                                                         |
| cardona-qft-methods_F08979 | 51   | ■ 1.000 | $k+2$                                                                                                                                     | $k+2$                                                                                                                                     | so $f\,P=2a_0'$ . Since $f$ <i>WRes</i> $P$ , because $\operatorname{Tr}(P\,\not{D}^{-2s})=\frac{1}{\Gamma(s)}\int_0^\infty t^{s-1}\operatorname{Tr}(Pe^{-t\not{D}^2})dt$ , the                                                                         |
| cardona-qft-methods_F08980 | 51   | ■ 0.998 | $\int_0^1 t^{s-1}\log(t)^kdt=\frac{(-1)^kk!}{s^{k+1}}$                                                                                    | $\int_0^1 t^{s-1}\log(t)^kdt=\frac{(-1)^kk!}{s^{k+1}}$                                                                                    | so $f\,P=2a_0'$ . Since $f$ , <i>WRes</i> are traces on $\Psi(C^\infty(M))$ , Corollary 2.2 gives $f\,P=c\,WRes\,P$ . Because $\operatorname{Tr}(P\,\not{D}^{-2s})=\frac{1}{\Gamma(s)}\int_0^\infty t^{s-1}\operatorname{Tr}(Pe^{-t\not{D}^2})dt$ , the |
| cardona-qft-methods_F08981 | 51   | ■ 0.998 | $\Gamma(s)=\frac{1}{s}+\gamma+sg(s)$                                                                                                      | $\Gamma(s)=\frac{1}{s}+\gamma+sg(s)$                                                                                                      | so $f\,P=2a_0'$ . Since $f$ , <i>WRes</i> are traces on $\Psi(C^\infty(M))$ , Corollary 2.2 gives $f\,P=c\,WRes\,P$ . Because $\operatorname{Tr}(P\,\not{D}^{-2s})=\frac{1}{\Gamma(s)}\int_0^\infty t^{s-1}\operatorname{Tr}(Pe^{-t\not{D}^2})dt$ , the |
| cardona-qft-methods_F08982 | 51   | ■ 0.982 | $\operatorname{Sp}(M)$                                                                                                                    | $\operatorname{Sp}(M)$                                                                                                                    | the Euler constant and the function $g$ is also holomorphic around zero.                                                                                                                                                                                |
| cardona-qft-methods_F08983 | 51   | ■ 0.988 | $\operatorname{Sp}(M)=\{d-k\mid k\in\mathbb{N}\}$                                                                                         | $\operatorname{Sp}(M)=\{d-k\mid k\in\mathbb{N}\}$                                                                                         | <b>Prop 6.4.</b> Let $Sp(M)$ be the dimension spectrum of a commutative geom-                                                                                                                                                                           |
| cardona-qft-methods_F08984 | 51   | ■ 1.000 | $\sigma_d^{ \mathcal{D}_A ^{-d}}=\sigma_d^{ \mathcal{D} ^{-d}}$                                                                           | $\sigma_d^{ \mathcal{D}_A ^{-d}}=\sigma_d^{ \mathcal{D} ^{-d}}$                                                                           | The equation (30) can be obtained via (8) as $\sigma_d^{ \mathcal{D}_A ^{-d}}=\sigma_d^{ \mathcal{D} ^{-d}}$ .<br>In dimension $d=4$ , the computation in (19) of coefficient $a_4(1,\mathcal{D}_A^2)$                                                  |
| cardona-qft-methods_F08985 | 51   | ■ 1.000 | $d=4$                                                                                                                                     | $d=4$                                                                                                                                     | gives                                                                                                                                                                                                                                                   |
| cardona-qft-methods_F08986 | 51   | ■ 1.000 | $a_4\left(1,\mathcal{D}_A^2\right)$                                                                                                       | $a_4\left(1,\mathcal{D}_A^2\right)$                                                                                                       | gives                                                                                                                                                                                                                                                   |
| cardona-qft-methods_F08987 | 51   | ■ 1.000 | $a_k\left(1,\mathcal{D}_A^2\right)$                                                                                                       | $a_k\left(1,\mathcal{D}_A^2\right)$                                                                                                       | According to Corollary 6.1, a commutative geometry has no tadpoles. <sup>58</sup>                                                                                                                                                                       |
| cardona-qft-methods_F08988 | 51   | ■ 1.000 | $\zeta_{\mathcal{D}_A}(0)$                                                                                                                | $\zeta_{\mathcal{D}_A}(0)$                                                                                                                | <b>6.7. Scalar curvature</b>                                                                                                                                                                                                                            |
| cardona-qft-methods_F08989 | 51   | ■ 1.000 | $ff(x)\,\not{D}^{-d+2}=$                                                                                                                  | $ff(x)\,\not{D}^{-d+2}=$                                                                                                                  | <b>Prop 6.5.</b> [30, Proposition 1.153] If $\mathcal{R}$ is a trace on $\mathcal{A}$ and the tadpoles $f\,A\mathcal{D}^{-d+1}$ are zero for all $A\in\Omega^1$ , $\mathcal{D}$ is invariant by inner fluctuations                                      |
| cardona-qft-methods_F08990 | 51   | ■ 1.000 | $\int_M f(x)s(x)d\,\mathrm{vol}(x)$                                                                                                       | $\int_M f(x)s(x)d\mathrm{vol}(x)$                                                                                                         | $f\,A\mathcal{D}^{-d+1}$ are zero for all $A\in\Omega^1$ , $\mathcal{D}$ is invariant by inner fluctuations<br>$\mathcal{D}\rightarrow\mathcal{D}+A$ .                                                                                                  |
| cardona-qft-methods_F08991 | 51   | ■ 1.000 | $\mathcal{R}:a\in\mathcal{A}\rightarrow\mathbb{C}$                                                                                        | $\mathcal{R}:a\in\mathcal{A}\rightarrow\mathbb{C}$                                                                                        | There is a natural notion of tensor for spectral triples which corresponds                                                                                                                                                                              |
| cardona-qft-methods_F08992 | 51   | ■ 1.000 | $\mathcal{R}(a)=fa\mathcal{D}^{-d+2}$                                                                                                     | $\mathcal{R}(a)=fa\mathcal{D}^{-d+2}$                                                                                                     | There is a natural notion of tensor for spectral triples which corresponds                                                                                                                                                                              |
| cardona-qft-methods_F08993 | 51   | ■ 1.000 | $\mathcal{R}$                                                                                                                             | $\mathcal{R}$                                                                                                                             |                                                                                                                                                                                                                                                         |
| cardona-qft-methods_F08994 | 51   | ■ 0.997 | $f\,A\,\mathcal{D}^{-d+1}$                                                                                                                | $fA\mathcal{D}^{-d+1}$                                                                                                                    |                                                                                                                                                                                                                                                         |

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| cardona-qft-methods_F00995 | 51   | ■ 0.997 | $A \in \Omega_{\mathcal{D}}^1, \mathcal{R}$                                                                                                                                                                                                             | $A \in \Omega_{\mathcal{D}}^1, \mathcal{R}$                                                                                                                                                                                                             |                                                                                                                                                                                                                                                                                                                                 |
| cardona-qft-methods_F00996 | 51   | ■ 0.932 | $(\mathcal{A}_i, \mathcal{D}_i, \mathcal{H}_i)$                                                                                                                                                                                                         | $(\mathcal{A}_i, \mathcal{D}_i, \mathcal{H}_i)$                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                 |
| cardona-qft-methods_F00997 | 52   | ■ 1.000 | $i=1,2$                                                                                                                                                                                                                                                 | $i = 1, 2$                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                                                 |
| cardona-qft-methods_F00998 | 52   | ■ 1.000 | $d_{\{i\}}$                                                                                                                                                                                                                                             | $d_i$                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                                 |
| cardona-qft-methods_F00999 | 52   | ■ 1.000 | $\chi_{\{1\}}$                                                                                                                                                                                                                                          | $\chi_1$                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                 |
| cardona-qft-methods_F01000 | 52   | ■ 1.000 | $\mathcal{D}^2 = \mathcal{D}_1^2 \otimes 1 + 1 \otimes \mathcal{D}_2^2$                                                                                                                                                                                 | $\mathcal{D}^2 = \mathcal{D}_1^2 \otimes 1 + 1 \otimes \mathcal{D}_2^2$                                                                                                                                                                                 | spectrum. Assume the first to be of even dimension, with grading $\chi_1$ . The spectral triple $(\mathcal{A}, \mathcal{D}, \mathcal{H})$ associated to the tensor product is defined by                                                                                                                                        |
| cardona-qft-methods_F01001 | 52   | ■ 1.000 | $\lim_{t \rightarrow 0} \text{Tr} \left( e^{-t \mathcal{D}_1^2} \right) a_1 t^{-d_1/2}$                                                                                                                                                                 | $\text{Tr} \left( e^{-t \mathcal{D}_1^2} \right) \sim_{t \rightarrow 0} a_1 t^{-d_1/2}$                                                                                                                                                                 | The interest of $\chi_1$ is to guarantee additivity: $\mathcal{D}^* = \mathcal{D}_1^* \otimes 1 + 1 \otimes \mathcal{D}_2^*$ .                                                                                                                                                                                                  |
| cardona-qft-methods_F01002 | 52   | ■ 1.000 | $\lim_{t \rightarrow 0} \text{Tr} \left( e^{-t \mathcal{D}_2^2} \right) = 0$                                                                                                                                                                            | $\text{Tr} \left( e^{-t \mathcal{D}_2^2} \right) \sim_{t \rightarrow 0} 0$                                                                                                                                                                              | The interest of $\chi_1$ is to guarantee additivity: $\mathcal{D}^* = \mathcal{D}_1^* \otimes 1 + 1 \otimes \mathcal{D}_2^*$ .                                                                                                                                                                                                  |
| cardona-qft-methods_F01003 | 52   | ■ 1.000 | $a_2 t^{-d_2/2}$                                                                                                                                                                                                                                        | $a_2 t^{-d_2/2}$                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                                 |
| cardona-qft-methods_F01004 | 52   | ■ 0.523 | $d = d_1 + d_2$                                                                                                                                                                                                                                         | $d = d_1 + d_2$                                                                                                                                                                                                                                         | $d_2 t^{-d_2/2}$ .<br>The triple $(\mathcal{A}, \mathcal{D}, \mathcal{H})$ has dimension $d = d_1 + d_2$ .                                                                                                                                                                                                                      |
| cardona-qft-methods_F01005 | 52   | ■ 1.000 | $\zeta_{\mathcal{D}}(s) = \text{Tr}( D ^{-s})$                                                                                                                                                                                                          | $\zeta_{\mathcal{D}}(s) = \text{Tr}( D ^{-s})$                                                                                                                                                                                                          | Moreover, the function $\zeta_{\mathcal{D}}(s) = \text{Tr}( D ^{-s})$ has a simple pole at $s = d_1 + d_2$                                                                                                                                                                                                                      |
| cardona-qft-methods_F01006 | 52   | ■ 1.000 | $s = d_1 + d_2$                                                                                                                                                                                                                                         | $s = d_1 + d_2$                                                                                                                                                                                                                                         | Moreover, the function $\zeta_{\mathcal{D}}(s) = \text{Tr}( D ^{-s})$ has a simple pole at $s = d_1 + d_2$                                                                                                                                                                                                                      |
| cardona-qft-methods_F01007 | 52   | ■ 1.000 | $\zeta_{\mathcal{D}}(2s) = \sum_{n=0}^{\infty} \mu_n \left( \mathcal{D}_1^2 \otimes 1 + 1 \otimes \mathcal{D}_2^2 \right)^{-s} = \sum_{n,m=0}^{\infty} \left( \mu_n \left( \mathcal{D}_1^2 \right) + \mu_m \left( \mathcal{D}_2^2 \right) \right)^{-s}$ | $\zeta_{\mathcal{D}}(2s) = \sum_{n=0}^{\infty} \mu_n \left( \mathcal{D}_1^2 \otimes 1 + 1 \otimes \mathcal{D}_2^2 \right)^{-s} = \sum_{n,m=0}^{\infty} \left( \mu_n \left( \mathcal{D}_1^2 \right) + \mu_m \left( \mathcal{D}_2^2 \right) \right)^{-s}$ | $\zeta_{\mathcal{D}}(c_1 + c_2) \leq \zeta_{\mathcal{D}_1}(c_1) \zeta_{\mathcal{D}_2}(c_2)$ if $c_i > d_i$ , so $d := \inf \{ c \in \mathbb{R}^+ : \zeta_{\mathcal{D}}(c) < \infty \} \leq d_1 + d_2$ . We claim that $d = d_1 + d_2$ : recall first that $a_i :=$                                                              |
| cardona-qft-methods_F01008 | 52   | ■ 1.000 | $\mu_m \left( \mathcal{D}_2^2 \right)^{-s}$                                                                                                                                                                                                             | $\mu_m \left( \mathcal{D}_2^2 \right)^{-s}$                                                                                                                                                                                                             | $\text{Res}_{s=d_i/2} \left( \Gamma(s) \zeta_{\mathcal{D}_i}(2s) \right) = \Gamma(d_i/2) \text{Res}_{s=d_i/2} \left( \zeta_{\mathcal{D}_i}(2s) \right) = \frac{1}{2} \Gamma(d_i/2) \text{Res}_{s=d_i} \left( \zeta_{\mathcal{D}_i}(s) \right)$ . If                                                                             |
| cardona-qft-methods_F01009 | 52   | ■ 1.000 | $\left( \mu_n \left( \mathcal{D}_1 \right)^2 + \mu_m \left( \mathcal{D}_2 \right)^2 \right)^{-(c_1 + c_2)/2} \leq \mu_n \left( \mathcal{D}_1 \right)^{-c_1} \mu_m \left( \mathcal{D}_2 \right)^{-c_2}$                                                  | $\left( \mu_n \left( \mathcal{D}_1 \right)^2 + \mu_m \left( \mathcal{D}_2 \right)^2 \right)^{-(c_1 + c_2)/2} \leq \mu_n \left( \mathcal{D}_1 \right)^{-c_1} \mu_m \left( \mathcal{D}_2 \right)^{-c_2}$                                                  | $\text{Res}_{s=d_i/2} \left( \Gamma(s) \zeta_{\mathcal{D}_i}(2s) \right) = \Gamma(d_i/2) \text{Res}_{s=d_i/2} \left( \zeta_{\mathcal{D}_i}(2s) \right) = \frac{1}{2} \Gamma(d_i/2) \text{Res}_{s=d_i} \left( \zeta_{\mathcal{D}_i}(s) \right)$ . If                                                                             |
| cardona-qft-methods_F01010 | 52   | ■ 0.979 | $\zeta_{\mathcal{D}}(c_1 + c_2) \leq \zeta_{\mathcal{D}_1}(c_1) \zeta_{\mathcal{D}_2}(c_2)$                                                                                                                                                             | $\zeta_{\mathcal{D}}(c_1 + c_2) \leq \zeta_{\mathcal{D}_1}(c_1) \zeta_{\mathcal{D}_2}(c_2)$                                                                                                                                                             | $f(s) := \Gamma(s) \zeta_{\mathcal{D}}(2s), \quad f(s) = \int_0^1 \text{Tr}(e^{-t \mathcal{D}^2}) t^{s-1} dt + g(s) =$                                                                                                                                                                                                          |
| cardona-qft-methods_F01011 | 52   | ■ 0.979 | $c_i > d_i$                                                                                                                                                                                                                                             | $c_i > d_i$                                                                                                                                                                                                                                             | $f(s) := \Gamma(s) \zeta_{\mathcal{D}}(2s), \quad f(s) = \int_0^1 \text{Tr}(e^{-t \mathcal{D}^2}) t^{s-1} dt + g(s) =$                                                                                                                                                                                                          |
| cardona-qft-methods_F01012 | 52   | ■ 0.979 | $d := \inf \{ c \in \mathbb{R}^+ : \zeta_{\mathcal{D}}(c) < \infty \} \leq$                                                                                                                                                                             | $d := \inf \{ c \in \mathbb{R}^+ : \zeta_{\mathcal{D}}(c) < \infty \} \leq$                                                                                                                                                                             | $f(s) := \Gamma(s) \zeta_{\mathcal{D}}(2s), \quad f(s) = \int_0^1 \text{Tr}(e^{-t \mathcal{D}^2}) t^{s-1} dt + g(s) =$                                                                                                                                                                                                          |
| cardona-qft-methods_F01013 | 52   | ■ 1.000 | $d_1 + d_2$                                                                                                                                                                                                                                             | $d_1 + d_2$                                                                                                                                                                                                                                             | $\int_0^1 \text{Tr}(e^{-t \mathcal{D}_1^2}) \text{Tr}(e^{-t \mathcal{D}_2^2}) t^{s-1} dt + g(s)$ where $g$ is holomorphic since the map                                                                                                                                                                                         |
| cardona-qft-methods_F01014 | 52   | ■ 1.000 | $a_i :=$                                                                                                                                                                                                                                                | $a_i :=$                                                                                                                                                                                                                                                | $\int_0^1 \text{Tr}(e^{-t \mathcal{D}_1^2}) \text{Tr}(e^{-t \mathcal{D}_2^2}) t^{s-1} dt + g(s)$ where $g$ is holomorphic since the map                                                                                                                                                                                         |
| cardona-qft-methods_F01015 | 52   | ■ 0.965 | $\text{Res}_{s=d_i/2} \left( \Gamma(s) \zeta_{\mathcal{D}_i}(2s) \right) = \Gamma(d_i/2) \text{Res}_{s=d_i/2} \left( \zeta_{\mathcal{D}_i}(2s) \right) = \frac{1}{2} \Gamma(d_i/2) \text{Res}_{s=d_i} \left( \zeta_{\mathcal{D}_i}(s) \right)$          | $\text{Res}_{s=d_i/2} \left( \Gamma(s) \zeta_{\mathcal{D}_i}(2s) \right) = \Gamma(d_i/2) \text{Res}_{s=d_i/2} \left( \zeta_{\mathcal{D}_i}(2s) \right) = \frac{1}{2} \Gamma(d_i/2) \text{Res}_{s=d_i} \left( \zeta_{\mathcal{D}_i}(s) \right)$          | $x \in \mathbb{R} \mapsto \int_1^e e^{-tx^2} dx$ is in Schwartz space.<br>Since $\text{Tr}(e^{-t \mathcal{D}_1^2}) \text{Tr}(e^{-t \mathcal{D}_2^2}) \sim_{t \rightarrow 0} a_1 a_2 t^{-(d_1 + d_2)/2}$ , the function $f(s)$ has a simple pole at $s = (d_1 + d_2)/2$ . We conclude that $\zeta_{\mathcal{D}}(s)$ has a simple |

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| cardona-qft-methods_F01016 | 52   | ■ 0.946 | $f(s) := \Gamma(s) \zeta_D(2s), \quad \text{quad } f(s) = \int_0^1 \operatorname{Tr} \left( e^{-t \operatorname{mathcal{D}}^2} \right) t^{s-1} dt + g(s) =$                                   | $f(s) := \Gamma(s) \zeta_D(2s), \quad f(s) = \int_0^1 \operatorname{Tr} \left( e^{-t \mathcal{D}^2} \right) t^{s-1} dt + g(s) =$                                  | a simple pole at $s = (d_1 + d_2)/2$ . We conclude that $\zeta_{\mathcal{D}}(s)$ has a simple pole at $s = d_1 + d_2$ . Moreover, using $a_i$ , $\frac{1}{2} \Gamma((d_1 + d_2)/2) \operatorname{Res}_{s=d}(\zeta_{\mathcal{D}}(s)) =$ |
| cardona-qft-methods_F01017 | 52   | ■ 0.999 | $\int_0^1 \operatorname{Tr} \left( e^{-t \operatorname{mathcal{D}}_1^2} \right) \operatorname{Tr} \left( e^{-t \operatorname{mathcal{D}}_2^2} \right) t^{s-1} dt + g(s)$                      | $\int_0^1 \operatorname{Tr} \left( e^{-t \mathcal{D}_1^2} \right) \operatorname{Tr} \left( e^{-t \mathcal{D}_2^2} \right) t^{s-1} dt + g(s)$                      | $\frac{1}{2} \Gamma(d_1/2) \operatorname{Res}_{s=d_1}(\zeta_{\mathcal{D}_1}(s)) \frac{1}{2} \Gamma(d_2/2) \operatorname{Res}_{s=d_2}(\zeta_{\mathcal{D}_2}(s)). \quad \square$                                                         |
| cardona-qft-methods_F01018 | 52   | ■ 1.000 | $x \in \mathbb{R} \rightarrow \int_1^\infty e^{-tx^2} t^{x-1} dt$                                                                                                                             | $x \in \mathbb{R} \rightarrow \int_1^\infty e^{-tx^2} t^{x-1} dt$                                                                                                 |                                                                                                                                                                                                                                        |
| cardona-qft-methods_F01019 | 52   | ■ 1.000 | $\operatorname{Tr} \left( e^{-t \operatorname{mathcal{D}}_1^2} \right) \operatorname{Tr} \left( e^{-t \operatorname{mathcal{D}}_2^2} \right) \sim_{t \rightarrow 0} a_1 a_2 t^{-(d_1+d_2)/2}$ | $\operatorname{Tr} \left( e^{-t \mathcal{D}_1^2} \right) \operatorname{Tr} \left( e^{-t \mathcal{D}_2^2} \right) \sim_{t \rightarrow 0} a_1 a_2 t^{-(d_1+d_2)/2}$ | 7. Spectral action                                                                                                                                                                                                                     |
| cardona-qft-methods_F01020 | 52   | ■ 1.000 | $f(s)$                                                                                                                                                                                        | $f(s)$                                                                                                                                                            | 7. Spectral action                                                                                                                                                                                                                     |
| cardona-qft-methods_F01021 | 52   | ■ 1.000 | $s = \left( d_1 + d_2 \right) / 2$                                                                                                                                                            | $s = (d_1 + d_2) / 2$                                                                                                                                             | 7.1. On the search for a good action functional                                                                                                                                                                                        |
| cardona-qft-methods_F01022 | 52   | ■ 1.000 | $\zeta_{\operatorname{mathcal{D}}}(s)$                                                                                                                                                        | $\zeta_{\mathcal{D}}(s)$                                                                                                                                          | 7.1. On the search for a good action functional                                                                                                                                                                                        |
| cardona-qft-methods_F01023 | 52   | ■ 0.996 | $a_i, \frac{1}{2} \Gamma \left( (d_1 + d_2) / 2 \right) \operatorname{Res}_{s=d} \left( \zeta_{\operatorname{mathcal{D}}}(s) \right) =$                                                       | $a_i, \frac{1}{2} \Gamma \left( (d_1 + d_2) / 2 \right) \operatorname{Res}_{s=d} \left( \zeta_{\mathcal{D}}(s) \right) =$                                         | We would like to obtain a good action for any spectral triple and for this it is useful to look at some examples in physics. In any physical theory based                                                                              |
| cardona-qft-methods_F01024 | 52   | ■ 0.998 | $\frac{1}{2} \Gamma(d_1/2) \operatorname{Res}_{s=d_1}(\zeta_{\operatorname{mathcal{D}}_1}(s)) \frac{1}{2} \Gamma(d_2/2) \operatorname{Res}_{s=d_2}(\zeta_{\operatorname{mathcal{D}}_2}(s))$   | $\frac{1}{2} \Gamma(d_1/2) \operatorname{Res}_{s=d_1}(\zeta_{\mathcal{D}_1}(s)) \frac{1}{2} \Gamma(d_2/2) \operatorname{Res}_{s=d_2}(\zeta_{\mathcal{D}_2}(s))$   | on geometry, the interest of an action function process, to exhibit a particular geometry, for inst                                                                                                                                    |
| cardona-qft-methods_F01025 | 53   | ■ 0.936 | $S_{\{E, H\}}(g) = - \int_M s_g(x) d \operatorname{vol}_g(x)$                                                                                                                                 | $S_{EH}(g) = - \int_M s_g(x) d \operatorname{vol}_g(x)$                                                                                                           | 42 B. Iochum                                                                                                                                                                                                                           |
| cardona-qft-methods_F01026 | 53   | ■ 1.000 | $f \not\sim^2$                                                                                                                                                                                | $f \not\sim^2$                                                                                                                                                    | 7.1.1 Einstein–Hilbert action                                                                                                                                                                                                          |
| cardona-qft-methods_F01027 | 53   | ■ 1.000 | $\operatorname{mathcal{M}}_1$                                                                                                                                                                 | $\mathcal{M}_1$                                                                                                                                                   | (chosen positive for the sphere). Up to a constant, this is $\frac{1}{2} \mathcal{D}^{-2}$ in dimension                                                                                                                                |
| cardona-qft-methods_F01028 | 53   | ■ 0.992 | $\int_M d \operatorname{vol}_g = 1$                                                                                                                                                           | $\int_M d \operatorname{vol}_g = 1$                                                                                                                               | 4 as quoted after (21).<br>This action is interesting for the following reason. Let $\mathcal{M}_1$ be the set                                                                                                                         |
| cardona-qft-methods_F01029 | 53   | ■ 1.000 | $\{g\} \in \operatorname{mathcal{M}}_1$                                                                                                                                                       | $g \in \mathcal{M}_1$                                                                                                                                             | This action is interesting for the following reason. Let $\mathcal{M}_1$ be the set of Riemannian metrics $g$ on $M$ such that $\int_M d \operatorname{vol}_g = 1$ . By a theorem of                                                   |
| cardona-qft-methods_F01030 | 53   | ■ 1.000 | $S_{\{E, H\}}(g)$                                                                                                                                                                             | $S_{EH}(g)$                                                                                                                                                       | This action is interesting for the following reason. Let $\mathcal{M}_1$ be the set of Riemannian metrics $g$ on $M$ such that $\int_M d \operatorname{vol}_g = 1$ . By a theorem of                                                   |
| cardona-qft-methods_F01031 | 53   | ■ 1.000 | $g: R = c \, g$                                                                                                                                                                               | $g: R = c g$                                                                                                                                                      | $(M, g)$ is an Einstein manifold (the Ricci curvature $R$ of $g$ is proportional by a constant to $g$ : $R = c g$ ). Taking the trace, this means that $s_g = c \dim(M)$                                                               |
| cardona-qft-methods_F01032 | 53   | ■ 1.000 | $s_g = c \operatorname{dim}(M)$                                                                                                                                                               | $s_g = c \dim(M)$                                                                                                                                                 | $(M, g)$ is an Einstein manifold (the Ricci curvature $R$ of $g$ is proportional by a constant to $g$ : $R = c g$ ). Taking the trace, this means that $s_g = c \dim(M)$                                                               |
| cardona-qft-methods_F01033 | 53   | ■ 0.945 | $\int_M f(s_g(x)) d \operatorname{vol}_g(x)$                                                                                                                                                  | $\int_M f(s_g(x)) d \operatorname{vol}_g(x)$                                                                                                                      | gravity in noncommutative geometry, the eigenvalues of the Dirac operator look as natural variables. <sup>69</sup> However we are looking for observables which                                                                        |
| cardona-qft-methods_F01034 | 53   | ■ 1.000 | $\operatorname{mathcal{D}}^{-1}$                                                                                                                                                              | $\mathcal{D}^{-1}$                                                                                                                                                | fluctuations via Morita equivalence given by a choice of a connection which is associated to a one form in $\Omega^1(M)$ . Thus, the guiding principle followed                                                                        |
| cardona-qft-methods_F01035 | 53   | ■ 1.000 | $d s$                                                                                                                                                                                         | $ds$                                                                                                                                                              | by Connes and Chamseddine is to use a theory which is pure gravity with                                                                                                                                                                |
| cardona-qft-methods_F01036 | 53   | ■ 1.000 | $\Omega_{\operatorname{mathcal{D}}}^1(\operatorname{mathcal{A}})$                                                                                                                             | $\Omega_{\mathcal{D}}^1(\mathcal{A})$                                                                                                                             | $f$ is any positive function (such that $f(\mathcal{D}^2/\Lambda^2)$ is a trace-class operator).                                                                                                                                       |

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| cardona-qft-methods_F01037 | 53   | 0.627 | $\mathcal{D}^{\cdot 10}$                                                                                      | $\mathcal{D}^{\cdot 10}$                                                                                      | For $f$ one can think of the characteristic                                                                                                                                                                                                       |
| cardona-qft-methods_F01038 | 53   | 0.995 | $\mathcal{S}(\mathcal{D}, f, \Lambda) := \text{Tr} \left( f \left( \mathcal{D}^2 / \Lambda^2 \right) \right)$ | $\mathcal{S}(\mathcal{D}, f, \Lambda) := \text{Tr} \left( f \left( \mathcal{D}^2 / \Lambda^2 \right) \right)$ | an effective theory. Naturally, $\mathcal{D}$ has to be replaced by $\mathcal{D}_A$ which is just a                                                                                                                                               |
| cardona-qft-methods_F01039 | 53   | 1.000 | $f \left( \mathcal{D}^2 / \Lambda^2 \right)$                                                                  | $f \left( \mathcal{D}^2 / \Lambda^2 \right)$                                                                  |                                                                                                                                                                                                                                                   |
| cardona-qft-methods_F01040 | 53   | 1.000 | $\mathcal{S}(\mathcal{D}, f, \Lambda) = \text{Tr}(f(\mathcal{D}/\Lambda))$                                    | $\mathcal{S}(\mathcal{D}, f, \Lambda) = \text{Tr}(f(\mathcal{D}/\Lambda))$                                    |                                                                                                                                                                                                                                                   |
| cardona-qft-methods_F01041 | 53   | 1.000 | $S(\mathcal{D}, g, \Lambda) = \text{Tr} \left( f \left( \mathcal{D}^2 / \Lambda^2 \right) \right)$            | $S(\mathcal{D}, g, \Lambda) = \text{Tr} \left( f \left( \mathcal{D}^2 / \Lambda^2 \right) \right)$            |                                                                                                                                                                                                                                                   |
| cardona-qft-methods_F01042 | 53   | 1.000 | $g(x) = f(x^2)$                                                                                               | $g(x) = f(x^2)$                                                                                               |                                                                                                                                                                                                                                                   |
| cardona-qft-methods_F01043 | 53   | 0.939 | $f(\mathcal{D} / \Lambda)$                                                                                    | $f(\mathcal{D} / \Lambda)$                                                                                    |                                                                                                                                                                                                                                                   |
| cardona-qft-methods_F01044 | 53   | 1.000 | $[-\Lambda, \Lambda]$                                                                                         | $[-\Lambda, \Lambda]$                                                                                         |                                                                                                                                                                                                                                                   |
| cardona-qft-methods_F01045 | 54   | 1.000 | $\frac{1}{2} \langle J \psi, \mathcal{D} \psi \rangle$                                                        | $\frac{1}{2} \langle J \psi, \mathcal{D} \psi \rangle$                                                        |                                                                                                                                                                                                                                                   |
| cardona-qft-methods_F01046 | 54   | 1.000 | $\psi \in \mathcal{H}$                                                                                        | $\psi \in \mathcal{H}$                                                                                        |                                                                                                                                                                                                                                                   |
| cardona-qft-methods_F01047 | 54   | 1.000 | $\langle \psi, \mathcal{D} \psi \rangle$                                                                      | $\langle \psi, \mathcal{D} \psi \rangle$                                                                      | for the coupling between gauge bosons and Higgs bosons with fermions.                                                                                                                                                                             |
| cardona-qft-methods_F01048 | 54   | 0.999 | $\psi = \chi \psi \in \mathcal{H}$                                                                            | $\psi = \chi \psi \in \mathcal{H}$                                                                            | Actually, this is of $KO$ -di                                                                                                                                                                                                                     |
| cardona-qft-methods_F01049 | 54   | 0.999 | $\mathfrak{g}$                                                                                                | $\mathfrak{g}$                                                                                                | situation, let $G$ be a compact Lie group with its Lie algebra $\mathfrak{g}$ and let $A \in \Omega^1(M, \mathfrak{a})$ be a connection. If $F = da + \frac{1}{2}[A, A] \in \Omega^2(M, \mathfrak{a})$ is the curvature                           |
| cardona-qft-methods_F01050 | 54   | 0.999 | $A \in$                                                                                                       | $A \in$                                                                                                       | situation, let $G$ be a compact Lie group with its Lie algebra $\mathfrak{g}$ and let $A \in \Omega^1(M, \mathfrak{a})$ be a connection. If $F = da + \frac{1}{2}[A, A] \in \Omega^2(M, \mathfrak{a})$ is the curvature                           |
| cardona-qft-methods_F01051 | 54   | 1.000 | $\Omega^1(M, \mathfrak{g})$                                                                                   | $\Omega^1(M, \mathfrak{g})$                                                                                   | (or field strength) of $A$ , then the Yang-Mills action is $S_{YM}(A) = \int_M \text{tr}(F \wedge$                                                                                                                                                |
| cardona-qft-methods_F01052 | 54   | 1.000 | $F = da + \frac{1}{2}[A, A] \in \Omega^2(M, \mathfrak{g})$                                                    | $F = da + \frac{1}{2}[A, A] \in \Omega^2(M, \mathfrak{g})$                                                    | (or field strength) of $A$ , then the Yang-Mills action is $S_{YM}(A) = \int_M \text{tr}(F \wedge$                                                                                                                                                |
| cardona-qft-methods_F01053 | 54   | 1.000 | $S_{YM}(A) = \int_M \text{tr}(F \wedge$                                                                       | $S_{YM}(A) = \int_M \text{tr}(F \wedge$                                                                       | $\star F) d\text{vol}_g$ . In the abelian case $G = U(1)$ , it is the Maxwell action and its quantum version is the quantum electrodynamics (QED) since the un-                                                                                   |
| cardona-qft-methods_F01054 | 54   | 0.706 | $\star F) d\text{vol}_g$                                                                                      | $\star F) d\text{vol}_g$                                                                                      | gauged $U(1)$ of electric charge conservation can be gauged and its gauging                                                                                                                                                                       |
| cardona-qft-methods_F01055 | 54   | 0.706 | $G = U(1)$                                                                                                    | $G = U(1)$                                                                                                    | gauged $U(1)$ of electric charge conservation can be gauged and its gauging                                                                                                                                                                       |
| cardona-qft-methods_F01056 | 54   | 1.000 | $U(1)$                                                                                                        | $U(1)$                                                                                                        | spectral triple $(\mathcal{A}, \mathcal{H}, \mathcal{D})$ of dimension $d$ . <sup>23,24</sup> let $A \in \Omega^1_{\mathcal{D}}(\mathcal{A})$ and curvature                                                                                       |
| cardona-qft-methods_F01057 | 54   | 1.000 | $\dim(M) = 4$                                                                                                 | $\dim(M) = 4$                                                                                                 | $\theta = dA + A^2$ ; then it is natural to consider $I(A) = \text{Tr}_{\text{Dix}}(\theta^2  \mathcal{D} ^{-d})$ since it coincides (up to a constant) with the quantum Yang-Mills action in the                                                 |
| cardona-qft-methods_F01058 | 54   | 0.843 | $d : ^{23,24}$                                                                                                | $d : ^{23,24}$                                                                                                | claim since for the principal symbol, $\text{tr}(\sigma^P(x, \xi)) = c \text{tr}(F \wedge \star F)(x)$ .                                                                                                                                          |
| cardona-qft-methods_F01059 | 54   | 1.000 | $\theta = dA + A^2$                                                                                           | $\theta = dA + A^2$                                                                                           | There is nevertheless a problem with the definition of $dA$ : if $A = \sum_i \pi(a_i)  \mathcal{D}, \pi(b_i) $ , then $dA = \sum_i  \mathcal{D}, \pi(a_i)   \mathcal{D}, \pi(b_i) $ can be non-zero                                               |
| cardona-qft-methods_F01060 | 54   | 1.000 | $I(A) = \text{Tr}_{\text{Dix}} \left( \theta^2  \mathcal{D} ^{-d} \right)$                                    | $I(A) = \text{Tr}_{\text{Dix}} \left( \theta^2  \mathcal{D} ^{-d} \right)$                                    | There is nevertheless a problem with the definition of $dA$ : if $A = \sum_i \pi(a_i)  \mathcal{D}, \pi(b_i) $ , then $dA = \sum_i  \mathcal{D}, \pi(a_i)   \mathcal{D}, \pi(b_i) $ can be non-zero                                               |
| cardona-qft-methods_F01061 | 54   | 1.000 | $P = \theta^2  \mathcal{D} ^{-d}$                                                                             | $P = \theta^2  \mathcal{D} ^{-d}$                                                                             | tial algebra $\Omega^2_{\mathcal{D}}(\mathcal{A})$ , one must divide by a junk, for instance $\Omega^2_{\mathcal{D}} \simeq \pi(\Omega^2 / \pi(\delta(Kor(\pi) \cap \Omega^1)))$ where $\Omega^k(\mathcal{A})$ is the set of universal $k$ -forms |
| cardona-qft-methods_F01062 | 54   | 1.000 | $\text{tr}(\sigma^P(x, \xi)) = c \text{tr}(F \wedge \star F)(x)$                                              | $\text{tr}(\sigma^P(x, \xi)) = c \text{tr}(F \wedge \star F)(x)$                                              | over $\mathcal{A}$ given by the set of $a_0 \delta a_1 \cdots \delta a_k$ (before representation on $\mathcal{H}$ ;                                                                                                                               |

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|----------------------------|------|---------|-------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| cardona-qft-methods_F01063 | 54   | ■ 1.000 | d A                                                                                                                     | $dA$                                                                                                                    | $\pi(a_0\partial a_1\cdots\partial a_k)=a_0[\mathcal{D},a_1]\cdots[\mathcal{D},a_k]$ .<br>That $\mathcal{D}$ is the Hestenes covariant of $\omega(\mathrm{Ker}(\mathcal{A}))$ with the scalar prod                                                                       |
| cardona-qft-methods_F01064 | 54   | ■ 1.000 | A=                                                                                                                      | $A=$                                                                                                                    | $\pi(a_0\partial a_1\cdots\partial a_k)=a_0[\mathcal{D},a_1]\cdots[\mathcal{D},a_k]$ .<br>That $\mathcal{D}$ is the Hestenes covariant of $\omega(\mathrm{Ker}(\mathcal{A}))$ with the scalar prod                                                                       |
| cardona-qft-methods_F01065 | 54   | ■ 1.000 | $\sum_j \pi\left(a_j\right)\left[\mathcal{D}, \pi\left(b_j\right)\right]$                                               | $\sum_j \pi\left(a_j\right)\left[\mathcal{D}, \pi\left(b_j\right)\right]$                                               | see eq. 3.6 the Hestenes covariant of $\omega(\mathrm{Ker}(\mathcal{A}))$ with the scalar prod<br>uct defined by $\langle A_1, A_2 \rangle_k = \mathrm{Tr}_{\mathrm{Dix}}\left(A_2^* A_1 \mathcal{D} ^{-d}\right)$ for $A_j \in \pi\left(\Omega^k(\mathcal{A})\right)$ . |
| cardona-qft-methods_F01066 | 54   | ■ 1.000 | $d A=\sum_j \pi\left(a_j\right)\left[\mathcal{D}, \pi\left(b_j\right)\right]$                                           | $d A=\sum_j \pi\left(a_j\right)\left[\mathcal{D}, \pi\left(b_j\right)\right]$                                           | see eq. 3.6 the Hestenes covariant of $\omega(\mathrm{Ker}(\mathcal{A}))$ with the scalar prod<br>uct defined by $\langle A_1, A_2 \rangle_k = \mathrm{Tr}_{\mathrm{Dix}}\left(A_2^* A_1 \mathcal{D} ^{-d}\right)$ for $A_j \in \pi\left(\Omega^k(\mathcal{A})\right)$ . |
| cardona-qft-methods_F01067 | 54   | ■ 1.000 | $\Omega_{\mathcal{D}}^*\left(\mathcal{A}\right)$                                                                        | $\Omega_{\mathcal{D}}^*\left(\mathcal{A}\right)$                                                                        | itive, quadratic and gauge invariant under $V \mapsto \pi(u)V\pi(u^*)+\pi(u)[\mathcal{D}, \pi(u^*)]$<br>when $u \in \mathcal{U}(\mathcal{A})$ . Moreover, $S_{YM}(V)=\inf \left\{I(\omega) \mid \omega \in \Omega^1(\mathcal{A}), \pi(\omega)=V\right\}$                 |
| cardona-qft-methods_F01068 | 54   | ■ 1.000 | $\Omega_{\mathcal{D}}^2 \simeq$                                                                                         | $\Omega_{\mathcal{D}}^2 \simeq$                                                                                         | itive, quadratic and gauge invariant under $V \mapsto \pi(u)V\pi(u^*)+\pi(u)[\mathcal{D}, \pi(u^*)]$<br>when $u \in \mathcal{U}(\mathcal{A})$ . Moreover, $S_{YM}(V)=\inf \left\{I(\omega) \mid \omega \in \Omega^1(\mathcal{A}), \pi(\omega)=V\right\}$                 |
| cardona-qft-methods_F01069 | 54   | ■ 1.000 | $\pi\left(\Omega^2 / \pi\left(\delta\left(\mathrm{Ker}(\pi) \cap \Omega^1\right)\right)\right)$                         | $\pi\left(\Omega^2 / \pi\left(\delta\left(\mathrm{Ker}(\pi) \cap \Omega^1\right)\right)\right)$                         | since the above ambiguity disappears when taking the infimum.<br>which $S_{YM}(V)$ can be extended to the infinitesimal of Yang-Mills                                                                                                                                    |
| cardona-qft-methods_F01070 | 54   | ■ 1.000 | $\Omega^k\left(\mathcal{A}\right)$                                                                                      | $\Omega^k\left(\mathcal{A}\right)$                                                                                      | since the above ambiguity disappears when taking the infimum.<br>which $S_{YM}(V)$ can be extended to the infinitesimal of Yang-Mills                                                                                                                                    |
| cardona-qft-methods_F01071 | 54   | ■ 1.000 | $a_{\{0\}} \delta a_{\{1\}} \cdots \delta a_{\{k\}}$                                                                    | $a_0 \delta a_1 \cdots \delta a_k$                                                                                      | This Yang-Mills action can be extended to the equivalent of Hermitian<br>vector bundles on $M$ , namely finitely projective modules over $\mathcal{A}$ .                                                                                                                 |
| cardona-qft-methods_F01072 | 54   | ■ 1.000 | $\pi\left(a_0 \delta a_1 \cdots \delta a_k\right)=a_0\left[\mathcal{D}, a_1\right] \cdots\left[\mathcal{D}, a_k\right]$ | $\pi\left(a_0 \delta a_1 \cdots \delta a_k\right)=a_0\left[\mathcal{D}, a_1\right] \cdots\left[\mathcal{D}, a_k\right]$ | The spectral action is more conceptu                                                                                                                                                                                                                                     |
| cardona-qft-methods_F01073 | 54   | ■ 1.000 | $\mathcal{H}_k$                                                                                                         | $\mathcal{H}_k$                                                                                                         |                                                                                                                                                                                                                                                                          |
| cardona-qft-methods_F01074 | 54   | ■ 1.000 | $\pi\left(\Omega^k\left(\mathcal{A}\right)\right)$                                                                      | $\pi\left(\Omega^k\left(\mathcal{A}\right)\right)$                                                                      |                                                                                                                                                                                                                                                                          |
| cardona-qft-methods_F01075 | 54   | ■ 1.000 | $\left\langle A_1, A_2\right\rangle_k=\mathrm{Tr}_{\mathrm{Dix}}\left(A_2^* A_1\left \mathcal{D}\right ^{-d}\right)$    | $\left\langle A_1, A_2\right\rangle_k=\mathrm{Tr}_{\mathrm{Dix}}\left(A_2^* A_1\left \mathcal{D}\right ^{-d}\right)$    |                                                                                                                                                                                                                                                                          |
| cardona-qft-methods_F01076 | 54   | ■ 1.000 | $A_j \in \pi\left(\Omega^k\left(\mathcal{A}\right)\right)$                                                              | $A_j \in \pi\left(\Omega^k\left(\mathcal{A}\right)\right)$                                                              |                                                                                                                                                                                                                                                                          |
| cardona-qft-methods_F01077 | 54   | ■ 1.000 | $\Omega^1\left(\mathcal{A}\right)$                                                                                      | $\Omega^1\left(\mathcal{A}\right)$                                                                                      |                                                                                                                                                                                                                                                                          |
| cardona-qft-methods_F01078 | 54   | ■ 1.000 | $S_{YM}(V)=\left\langle\delta V+V^2, \delta V+V^2\right\rangle$                                                         | $S_{YM}(V)=\left\langle\delta V+V^2, \delta V+V^2\right\rangle$                                                         |                                                                                                                                                                                                                                                                          |
| cardona-qft-methods_F01079 | 54   | ■ 1.000 | $V \rightarrow \pi(u) V \pi\left(u^*\right)+\pi(u)\left[\mathcal{D}, \pi\left(u^*\right)\right]$                        | $V \rightarrow \pi(u) V \pi\left(u^*\right)+\pi(u)\left[\mathcal{D}, \pi\left(u^*\right)\right]$                        |                                                                                                                                                                                                                                                                          |
| cardona-qft-methods_F01080 | 54   | ■ 1.000 | $u \in \mathcal{U}(\mathcal{A})$                                                                                        | $u \in \mathcal{U}(\mathcal{A})$                                                                                        |                                                                                                                                                                                                                                                                          |
| cardona-qft-methods_F01081 | 54   | ■ 1.000 | $S_{YM}(V)=\inf \left\{I(\omega) \mid \omega \in \Omega^1(\mathcal{A}), \pi(\omega)=V\right\}$                          | $S_{YM}(V)=\inf \left\{I(\omega) \mid \omega \in \Omega^1(\mathcal{A}), \pi(\omega)=V\right\}$                          |                                                                                                                                                                                                                                                                          |
| cardona-qft-methods_F01082 | 55   | ■ 1.000 | $\mathcal{D}_A=\mathcal{D}+A$                                                                                           | $\mathcal{D}_A=\mathcal{D}+A$                                                                                           |                                                                                                                                                                                                                                                                          |
| cardona-qft-methods_F01083 | 55   | ■ 1.000 | $\mathrm{Tr}\left(f\left(\mathcal{D} / \Lambda\right)\right) \geq 0$                                                    | $\mathrm{Tr}\left(f\left(\mathcal{D} / \Lambda\right)\right) \geq 0$                                                    | Moreover, when $f$ is positive (which is the case for a Euclidean Lagrangian),<br>the action $\mathrm{Tr}\left(f\left(\mathcal{D} / \Lambda\right)\right) \geq 0$ has the correct sign for a Euclidean action: the                                                       |
| cardona-qft-methods_F01084 | 55   | ■ 1.000 | $S(\mathcal{D}, f, \Lambda)$                                                                                            | $S(\mathcal{D}, f, \Lambda)$                                                                                            | $\mathrm{Tr}\left(f\left(\mathcal{D} / \Lambda\right)\right) \sim \sum_{\alpha \in S d} u_{\alpha} \quad \text { with } u_{\alpha} \neq 0.$                                                                                                                              |
| cardona-qft-methods_F01085 | 55   | ■ 1.000 | $\zeta_{\mathcal{D}}$                                                                                                   | $\zeta_{\mathcal{D}}$                                                                                                   | $\mathrm{Tr}\left(f\left(\mathcal{D} / \Lambda\right)\right) \sim \sum f_{\beta} \Lambda^{\beta} \int \mathcal{D} ^{\beta}+$                                                                                                                                             |
| cardona-qft-methods_F01086 | 55   | ■ 1.000 | $\alpha<0, \zeta_{\mathcal{D}}$                                                                                         | $\alpha<0, \zeta_{\mathcal{D}}$                                                                                         |                                                                                                                                                                                                                                                                          |
| cardona-qft-methods_F01087 | 55   | ■ 1.000 | $-2 \alpha$                                                                                                             | $-2 \alpha$                                                                                                             |                                                                                                                                                                                                                                                                          |



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| cardona-qft-methods_<br>F01112 | 56   | ■ 1.000 | $\mathcal{S}(\mathcal{D}, f, \Lambda) \underset{t \downarrow 0}{\sim} \sum_{k \in \{1, \dots, d\}} f_k \Lambda^k a_{d-k}(\mathcal{D}^2) + f(0) a_d(\mathcal{D}^2) + \dots$ |          | $\mathcal{D} + A$ as we saw in Theorem 6.2.                                                                                                                                                                            |
| cardona-qft-methods_<br>F01113 | 56   | ■ 0.840 | $f_k := \frac{1}{\Gamma(k/2)} \int_0^\infty f(s) s^{k/2-1} ds$                                                                                                             |          |                                                                                                                                                                                                                        |
| cardona-qft-methods_<br>F01114 | 56   | ■ 1.000 | $\Lambda^0$                                                                                                                                                                |          |                                                                                                                                                                                                                        |
| cardona-qft-methods_<br>F01115 | 56   | ■ 1.000 | $\mathcal{D} \rightarrow$                                                                                                                                                  |          |                                                                                                                                                                                                                        |
| cardona-qft-methods_<br>F01116 | 57   | ■ 0.813 | $S(\mathcal{D}, f, \Lambda) \underset{\Lambda \rightarrow +\infty}{\sim}$                                                                                                  |          | 46 B. Iochum                                                                                                                                                                                                           |
| cardona-qft-methods_<br>F01117 | 57   | ■ 1.000 | $\sum_{n=0}^\infty c_n \Lambda^{d-n} a_n(\mathcal{D}^2)$                                                                                                                   |          | 7.3. Remark on the use of Laplace transform                                                                                                                                                                            |
| cardona-qft-methods_<br>F01118 | 57   | ■ 1.000 | $f(x) = e^{-x}$                                                                                                                                                            |          | a Laplace transform for an arbitrary spectral triple (with simple c spectrum) satisfying (31). However, this hypothesis is too restrictive                                                                             |
| cardona-qft-methods_<br>F01119 | 57   | ■ 1.000 | $(-1)^m b_{2m+4}(\mathcal{D}^2) = \frac{(4\pi)^2}{m!} \mu_m(\mathcal{D}^2)$                                                                                                |          | The main point is that <i>this asymptotics makes sense in the Cesàro sense</i> (see <sup>37</sup> for definition) for $f$ in $\mathcal{K}'(\mathbb{R})$ , which is the dual of $\mathcal{K}(\mathbb{R})$ . This latter |
| cardona-qft-methods_<br>F01120 | 57   | ■ 1.000 | $\mathcal{K}'(\mathbb{R})$                                                                                                                                                 |          | Of course, the counting function is not smooth but is in $\mathcal{K}'(\mathbb{R})$ , so the                                                                                                                           |
| cardona-qft-methods_<br>F01121 | 57   | ■ 1.000 | $\mathcal{K}(\mathbb{R})$                                                                                                                                                  |          | Of course, the counting function is not smooth but is in $\mathcal{K}'(\mathbb{R})$ , so the                                                                                                                           |
| cardona-qft-methods_<br>F01122 | 57   | ■ 0.981 | $a \in \mathbb{R}, \phi^{(k)}(x) =$                                                                                                                                        |          | given asymptotic behavior is wrong beyond the first term, but is correct in the Cesàro sense. Actually there are more derivatives of $f$ at 0 as explained                                                             |
| cardona-qft-methods_<br>F01123 | 57   | ■ 1.000 | $\mathcal{O}( x ^{a-k})$                                                                                                                                                   |          | the Cesàro behavior (actually there are more derivatives of $f$ at 0 as explained on examples in [37, p. 243].                                                                                                         |
| cardona-qft-methods_<br>F01124 | 57   | ■ 1.000 | $ x  \rightarrow \infty$                                                                                                                                                   |          | the Cesàro behavior (actually there are more derivatives of $f$ at 0 as explained on examples in [37, p. 243].                                                                                                         |
| cardona-qft-methods_<br>F01125 | 57   | ■ 1.000 | $g(\Lambda^{-1})$                                                                                                                                                          |          |                                                                                                                                                                                                                        |
| cardona-qft-methods_<br>F01126 | 57   | ■ 1.000 | $M = \mathbb{T}^d$                                                                                                                                                         |          |                                                                                                                                                                                                                        |
| cardona-qft-methods_<br>F01127 | 57   | ■ 1.000 | $\Delta = \delta^{\mu\nu} \partial_\mu \partial_\nu, \text{Tr}(e^{t\Delta}) =$                                                                                             |          |                                                                                                                                                                                                                        |
| cardona-qft-methods_<br>F01128 | 57   | ■ 1.000 | $\frac{(4\pi)^{-d/2} \text{Vol}(\mathbb{T}^d)}{t^{d/2}} + \mathcal{O}(t^{-d/2} e^{-1/4t})$                                                                                 |          |                                                                                                                                                                                                                        |
| cardona-qft-methods_<br>F01129 | 57   | ■ 1.000 | $\text{Tr}(e^{t\Delta}) \simeq$                                                                                                                                            |          |                                                                                                                                                                                                                        |
| cardona-qft-methods_<br>F01130 | 57   | ■ 0.999 | $\frac{(4\pi)^{-d/2} \text{Vol}(\mathbb{T}^d)}{t^{d/2}}, t \rightarrow 0$                                                                                                  |          |                                                                                                                                                                                                                        |
| cardona-qft-methods_<br>F01131 | 57   | ■ 1.000 | $\mathbb{S}^4$                                                                                                                                                             |          |                                                                                                                                                                                                                        |
| cardona-qft-methods_<br>F01132 | 58   | ■ 0.530 | $(\text{see}^{19})$                                                                                                                                                        |          |                                                                                                                                                                                                                        |
| cardona-qft-methods_<br>F01133 | 58   | ■ 1.000 | $B_{2k}$                                                                                                                                                                   |          | with Bernoulli numbers $B_{2k}$ . Thus $t^2 \text{Tr}(e^{-t\mathcal{D}^2}) \simeq \frac{2}{3} + \frac{2}{3}t + \sum_{k=0}^\infty a_k t^{k+2}$                                                                          |

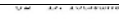
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| cardona-qft-methods_<br>F01134 | 58   | ■ 1.000 | $t^2 \operatorname{Tr} \left( e^{-t \not{D}^2} \right) \simeq \frac{2}{3} + \frac{2}{3} t + \sum_{k=0}^{\infty} a_k t^{k+2}$                                                                                                                                              |                                                                                                                                                                                                                                                                           | with Bernoulli numbers $B_{2k}$ . Thus $t^2 \operatorname{Tr}(e^{-t \not{D}^2}) \simeq \frac{2}{3} + \frac{2}{3} t + \sum_{k=0}^{\infty} a_k t^{k+2}$                                                                                                                                                          |
| cardona-qft-methods_<br>F01135 | 58   | ■ 1.000 | $a_{\{k\}} > \frac{4}{3k!} \frac{ B_{2k+4} }{2k+4} > 0$                                                                                                                                                                                                                   | $a_k > \frac{4}{3k!} \frac{ B_{2k+4} }{2k+4} > 0$                                                                                                                                                                                                                         | $a_k > \frac{4}{3k!} \frac{ B_{2k+4} }{2k+4} > 0$ and $ B_{2k+4}  = 2 \frac{(2k+4)!}{(2\pi)^{2k+4}} \zeta(2k+4) \simeq 4\sqrt{\pi(k+2)} \left(\frac{k+2}{2\pi}\right)^{2k+4} \rightarrow \infty$ if $k \rightarrow \infty$ .                                                                                   |
| cardona-qft-methods_<br>F01136 | 58   | ■ 1.000 | $ \operatorname{Tr} B_{2k+4}  = 2 \frac{(2k+4)!}{(2\pi)^{2k+4}} \zeta(2k+4) \simeq$                                                                                                                                                                                       | $ B_{2k+4}  = 2 \frac{(2k+4)!}{(2\pi)^{2k+4}} \zeta(2k+4) \simeq$                                                                                                                                                                                                         | $a_k > \frac{4}{3k!} \frac{ B_{2k+4} }{2k+4} > 0$ and $ B_{2k+4}  = 2 \frac{(2k+4)!}{(2\pi)^{2k+4}} \zeta(2k+4) \simeq 4\sqrt{\pi(k+2)} \left(\frac{k+2}{2\pi}\right)^{2k+4} \rightarrow \infty$ if $k \rightarrow \infty$ .                                                                                   |
| cardona-qft-methods_<br>F01137 | 58   | ■ 1.000 | $4 \sqrt{\pi(k+2)} \left(\frac{k+2}{\pi e}\right)^{2k+4} \rightarrow \infty$                                                                                                                                                                                              | $4\sqrt{\pi(k+2)} \left(\frac{k+2}{\pi e}\right)^{2k+4} \rightarrow \infty$                                                                                                                                                                                               | More generally, in the commutative case, the asymptotic expansion is only on a finite number of derivatives of a finite number of terms in the                                                                                                                                                                 |
| cardona-qft-methods_<br>F01138 | 58   | ■ 1.000 | $k \rightarrow \infty$                                                                                                                                                                                                                                                    | $k \rightarrow \infty$                                                                                                                                                                                                                                                    | More generally, in the commutative case, the asymptotic expansion is only on a finite number of derivatives of a finite number of terms in the                                                                                                                                                                 |
| cardona-qft-methods_<br>F01139 | 58   | ■ 1.000 | $2k > d$                                                                                                                                                                                                                                                                  | $2k > d$                                                                                                                                                                                                                                                                  | only on a finite number of derivatives of a finite number of terms in the                                                                                                                                                                                                                                      |
| cardona-qft-methods_<br>F01140 | 58   | ■ 1.000 | $2k \leq d$                                                                                                                                                                                                                                                               | $2k \leq d$                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                                                |
| cardona-qft-methods_<br>F01141 | 58   | ■ 1.000 | $e^{-t \sqrt{-\Delta}}$                                                                                                                                                                                                                                                   | $e^{-t \sqrt{-\Delta}}$                                                                                                                                                                                                                                                   | Thus we have an example where $t \rightarrow \infty$ cannot be used with the asymptotic series                                                                                                                                                                                                                 |
| cardona-qft-methods_<br>F01142 | 58   | ■ 1.000 | $\mathbb{T}^1$                                                                                                                                                                                                                                                            | $\mathbb{T}^1$                                                                                                                                                                                                                                                            | Thus we have an example where $t \rightarrow \infty$ cannot be used with the asymptotic series                                                                                                                                                                                                                 |
| cardona-qft-methods_<br>F01143 | 58   | ■ 1.000 | $\operatorname{Vol}(\mathbb{T}^1) = 2\pi$                                                                                                                                                                                                                                 | $\operatorname{Vol}(\mathbb{T}^1) = 2\pi$                                                                                                                                                                                                                                 | Thus we have an example where $t \rightarrow \infty$ cannot be used with the asymptotic series                                                                                                                                                                                                                 |
| cardona-qft-methods_<br>F01144 | 58   | ■ 1.000 | $\operatorname{Tr} \left( e^{-t \sqrt{-\Delta}} \right) = \frac{\sinh(t)}{\cosh(t)-1} = \coth\left(\frac{t}{2}\right) = \frac{2}{t} \sum_{k=0}^{\infty} \frac{B_{2k}}{(2k)!} t^{2k} = \frac{2}{t} \left[ 1 + \frac{t^2}{12} - \frac{t^4}{720} + \mathcal{O}(t^6) \right]$ | $\operatorname{Tr} \left( e^{-t \sqrt{-\Delta}} \right) = \frac{\sinh(t)}{\cosh(t)-1} = \coth\left(\frac{t}{2}\right) = \frac{2}{t} \sum_{k=0}^{\infty} \frac{B_{2k}}{(2k)!} t^{2k} = \frac{2}{t} \left[ 1 + \frac{t^2}{12} - \frac{t^4}{720} + \mathcal{O}(t^6) \right]$ | Thus the spectral action of Corollary 7.1 precisely encodes these local and nonlocal behavior which appear or not in its asymptotics for different $f$ . The coefficient of the action for the positive part (at least) of the dimension spectrum correspond to renormalized traces, namely the noncommutative |
| cardona-qft-methods_<br>F01145 | 58   | ■ 0.998 | $t < 2\pi$                                                                                                                                                                                                                                                                | $t < 2\pi$                                                                                                                                                                                                                                                                | integrals of (59). In conclusion, the asymptotics of spectral action may or may not have nonlocal coefficients.                                                                                                                                                                                                |
| cardona-qft-methods_<br>F01146 | 58   | ■ 0.998 | $\frac{B_{2k}}{(2k)!} = (-1)^{k+1} \frac{2\zeta(2k)}{(2\pi)^{2k}}$                                                                                                                                                                                                        | $\frac{B_{2k}}{(2k)!} = (-1)^{k+1} \frac{2\zeta(2k)}{(2\pi)^{2k}}$                                                                                                                                                                                                        | integrals of (59). In conclusion, the asymptotics of spectral action may or may not have nonlocal coefficients.                                                                                                                                                                                                |
| cardona-qft-methods_<br>F01147 | 58   | ■ 0.998 | $\frac{ B_{2k} }{(2k)!} \simeq$                                                                                                                                                                                                                                           | $\frac{ B_{2k} }{(2k)!} \simeq$                                                                                                                                                                                                                                           | integrals of (59). In conclusion, the asymptotics of spectral action may or may not have nonlocal coefficients.                                                                                                                                                                                                |
| cardona-qft-methods_<br>F01148 | 58   | ■ 1.000 | $\frac{2}{(2\pi)^{2k}}$                                                                                                                                                                                                                                                   | $\frac{2}{(2\pi)^{2k}}$                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                |
| cardona-qft-methods_<br>F01149 | 58   | ■ 1.000 | $t \rightarrow \infty$                                                                                                                                                                                                                                                    | $t \rightarrow \infty$                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                |
| cardona-qft-methods_<br>F01150 | 59   | ■ 1.000 | $\mathbb{T}^d$                                                                                                                                                                                                                                                            | $\mathbb{T}^d$                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                |
| cardona-qft-methods_<br>F01151 | 59   | ■ 1.000 | $\operatorname{Tr} \left( e^{t \Delta} \right)$                                                                                                                                                                                                                           | $\operatorname{Tr} \left( e^{t \Delta} \right)$                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                |
| cardona-qft-methods_<br>F01152 | 59   | ■ 1.000 | $N(\lambda)$                                                                                                                                                                                                                                                              | $N(\lambda)$                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                                |
| cardona-qft-methods_<br>F01153 | 59   | ■ 0.999 | $\lambda$                                                                                                                                                                                                                                                                 | $\lambda$                                                                                                                                                                                                                                                                 | For the flat torus $\mathbb{T}^d$ , the difference between $\operatorname{Tr}(e^{t \Delta})$ and its asymptotic expansion is                                                                                                                                                                                   |
| cardona-qft-methods_<br>F01154 | 59   | ■ 0.999 | $N(\lambda) =$                                                                                                                                                                                                                                                            | $N(\lambda) =$                                                                                                                                                                                                                                                            | For the flat torus $\mathbb{T}^d$ , the difference between $\operatorname{Tr}(e^{t \Delta})$ and its asymptotic expansion is                                                                                                                                                                                   |

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| cardona-qft-methods_F01155 | 59   | 0.892 | $\frac{(4\pi)^{-d/2} \text{Vol}(\mathbb{T}^d)}{\Gamma(d/2 + 1)} \lambda^{d/2} + o(\lambda^{d/2})$ |                                                          | otic series is an term which is related to periodic orbits of the geodesic flow on $\mathbb{T}^d$ . Similarly, the counting function $N(\lambda)$ (number of eigenval- |
| cardona-qft-methods_F01156 | 59   | 1.000 | $t$                                                                                               | $t$                                                      | the heat kernel and the formal expansion of the spectral measure is clear:                                                                                             |
| cardona-qft-methods_F01157 | 59   | 0.892 | $\lambda^{38}$                                                                                    | $\lambda^{38}$                                           | In <sup>16,76</sup> are examples of spectral actions on commutative                                                                                                    |
| cardona-qft-methods_F01158 | 59   | 1.000 | $\{ \}^{16,76}$                                                                                   | 16,76                                                    | dimension 4 whose asymptotics have only two terms. In the quantum group                                                                                                |
| cardona-qft-methods_F01159 | 59   | 1.000 | $SU_q(2)$                                                                                         | $SU_q(2)$                                                |                                                                                                                                                                        |
| cardona-qft-methods_F01160 | 59   | 0.958 | $A \in \Gamma^\infty(M, \text{End}(S))$                                                           | $A \in \Gamma^\infty(M, \text{End}(S))$                  | one is algebra is a sum of matrices has been deeply and intensively in-                                                                                                |
| cardona-qft-methods_F01161 | 59   | 1.000 | $\mathcal{D} = i\gamma^\mu (\partial_\mu + A_\mu)$                                                | $\mathcal{D} = i\gamma^\mu (\partial_\mu + A_\mu)$       | vestigated for the noncommutative approach to standard model of particle physics see 20,30 This approach offers a lot of interesting perspectives for                  |
| cardona-qft-methods_F01162 | 59   | 1.000 | $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu +$                                          | $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu +$ | vestigated for the noncommutative approach to standard model of particle physics see 20,30 This approach offers a lot of interesting perspectives for                  |
| cardona-qft-methods_F01163 | 59   | 1.000 | $[A_\mu, A_\nu]$                                                                                  | $[A_\mu, A_\nu]$                                         | instance, the possibility to compute the Higgs representations and mass (for                                                                                           |
| cardona-qft-methods_F01164 | 59   | 1.000 | $P = \mathcal{D}^2$                                                                               | $P = \mathcal{D}^2$                                      | instance, the possibility to compute the Higgs representations and mass (for                                                                                           |
| cardona-qft-methods_F01165 | 59   | 1.000 | $a_i(1, P)$                                                                                       | $a_i(1, P)$                                              | instance, the possibility to compute the Higgs representations and mass (for                                                                                           |
| cardona-qft-methods_F01166 | 59   | 1.000 | $i=0,2,4$                                                                                         | $i = 0, 2, 4$                                            | each noncommutative model) is particularly instructive, <sup>10,15,17,57,63,64,71,79</sup> Of course, since the first term in (29) is a geometrical term, one may be   |
| cardona-qft-methods_F01167 | 59   | 1.000 | $\{ \}^{20,30}$                                                                                   | 20,30                                                    |                                                                                                                                                                        |
| cardona-qft-methods_F01168 | 59   | 1.000 | $\{ \}^{10,15,17,57,63,64,71,79}$                                                                 | 10,15,17,57,63,64,71,79                                  |                                                                                                                                                                        |
| cardona-qft-methods_F01169 | 60   | 1.000 | $SU(2)$                                                                                           | $SU(2)$                                                  |                                                                                                                                                                        |
| cardona-qft-methods_F01170 | 60   | 1.000 | $\mathbb{S}^3 \times \mathbb{S}^1$                                                                | $\mathbb{S}^3 \times \mathbb{S}^1$                       | terms.<br>Due to the difficulties to deal with non-compact manifolds, the case of                                                                                      |
| cardona-qft-methods_F01171 | 60   | 1.000 | $\{ \}^{16,19}$                                                                                   | 16,19                                                    | terms.<br>Due to the difficulties to deal with non-compact manifolds, the case of                                                                                      |
| cardona-qft-methods_F01172 | 60   | 1.000 | $\{ \}^{66,76-78,81,95}$                                                                          | 66,76–78,81,95                                           | to inflation, see <sup>66,76–78</sup> ,<br>Constant inflation...                                                                                                       |
| cardona-qft-methods_F01173 | 60   | 1.000 | $\{ \}^{14,18,18,58,59,61}$                                                                       | 14,18,18,58,59,61                                        | conditions to the operator $\mathcal{D}$ to still get a selfadjoint operator and then, to                                                                              |
| cardona-qft-methods_F01174 | 60   | 1.000 | $\{ \}^{53,84,85}$                                                                                | 53,84,85                                                 | world.                                                                                                                                                                 |
| cardona-qft-methods_F01175 | 60   | 1.000 | $C^\infty(\mathbb{T}_\Theta^n)$                                                                   | $C^\infty(\mathbb{T}_\Theta^n)$                          |                                                                                                                                                                        |
| cardona-qft-methods_F01176 | 60   | 1.000 | $n$                                                                                               | $n$                                                      |                                                                                                                                                                        |
| cardona-qft-methods_F01177 | 60   | 1.000 | $\Theta \in M_n(\mathbb{R})$                                                                      | $\Theta \in M_n(\mathbb{R})$                             |                                                                                                                                                                        |
| cardona-qft-methods_F01178 | 60   | 1.000 | $\mathbb{T}^n$                                                                                    | $\mathbb{T}^n$                                           |                                                                                                                                                                        |
| cardona-qft-methods_F01179 | 60   | 1.000 | $u_i, i=1, \ldots, n$                                                                             | $u_i, i = 1, \dots, n$                                   |                                                                                                                                                                        |
| cardona-qft-methods_F01180 | 60   | 0.993 | $u_l u_j = e^{i\Theta_{lj}} u_j u_l$                                                              | $u_l u_j = e^{i\Theta_{lj}} u_j u_l$                     |                                                                                                                                                                        |

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| cardona-qft-methods_F01181 | 60   | ■ 0.999 | $a \in C^{\infty}(\mathbb{T}_{\Theta}^n)$                                                                                       | $a \in C^{\infty}(\mathbb{T}_{\Theta}^n)$                                                                                       |                                                                                                                                                                                                                                                           |
| cardona-qft-methods_F01182 | 60   | ■ 0.999 | $a=\sum_{k \in \mathbb{Z}^n} a_k U_k$                                                                                           | $a = \sum_{k \in \mathbb{Z}^n} a_k U_k$                                                                                         |                                                                                                                                                                                                                                                           |
| cardona-qft-methods_F01183 | 60   | ■ 0.999 | $\left\{a_k\right\} \in \mathcal{S}\left(\mathbb{Z}^n\right)$                                                                   | $\left\{a_k\right\} \in \mathcal{S}\left(\mathbb{Z}^n\right)$                                                                   |                                                                                                                                                                                                                                                           |
| cardona-qft-methods_F01184 | 60   | ■ 0.998 | $U_k=e^{-\frac{i}{2} k \cdot \chi} u_1^{k_1} \cdots u_n^{k_n}, k \in \mathbb{Z}^n$                                              | $U_k = e^{-\frac{i}{2} k \cdot \chi} u_1^{k_1} \cdots u_n^{k_n}, k \in \mathbb{Z}^n$                                            |                                                                                                                                                                                                                                                           |
| cardona-qft-methods_F01185 | 61   | ■ 1.000 | $\Theta$                                                                                                                        | $\Theta$                                                                                                                        |                                                                                                                                                                                                                                                           |
| cardona-qft-methods_F01186 | 61   | ■ 1.000 | $U_{-k}$                                                                                                                        | $U_k$                                                                                                                           | constraint relation reads                                                                                                                                                                                                                                 |
| cardona-qft-methods_F01187 | 61   | ■ 1.000 | $U_{-k}^*=U_{-k}$                                                                                                               | $U_k^*=U_{-k}$                                                                                                                  | constraint relation reads                                                                                                                                                                                                                                 |
| cardona-qft-methods_F01188 | 61   | ■ 1.000 | $\left[U_k, U_l\right]=-2 i \sin \left(\frac{1}{2} k \cdot \Theta l\right) U_{k+l}$                                             | $\left[U_k, U_l\right]=-2 i \sin \left(\frac{1}{2} k \cdot \Theta l\right) U_{k+l}$                                             | constraint relation reads                                                                                                                                                                                                                                 |
| cardona-qft-methods_F01189 | 61   | ■ 1.000 | $\tau\left(\sum_{k \in \mathbb{Z}^n} a_k U_k\right)=a_0$                                                                        | $\tau\left(\sum_{k \in \mathbb{Z}^n} a_k U_k\right)=a_0$                                                                        | $U_k U_q=e^{-i \chi(k, q)} U_{k+q}$ , and $U_k U_q=e^{-i \chi(k, q)} U_q U_k$<br>where $\chi$ is the matrix restriction of $\Theta$ to its upper triangular part. Thus                                                                                    |
| cardona-qft-methods_F01190 | 61   | ■ 1.000 | $\mathcal{H}_{\tau}$                                                                                                            | $\mathcal{H}_{\tau}$                                                                                                            | $U_k U_q=e^{-i \chi(k, q)} U_{k+q}$ , and $U_k U_q=e^{-i \chi(k, q)} U_q U_k$<br>where $\chi$ is the matrix restriction of $\Theta$ to its upper triangular part. Thus                                                                                    |
| cardona-qft-methods_F01191 | 61   | ■ 1.000 | $\langle a, b\rangle=\tau\left(a^* b\right)$                                                                                    | $\langle a, b\rangle=\tau\left(a^* b\right)$                                                                                    | Let $\tau$ be the trace on $C^{\infty}\left(\mathbb{T}_{\Theta}^n\right)$ defined by $\tau\left(\sum_{k \in \mathbb{Z}^n} a_k U_k\right)=a_0$ .<br>be the GNS Hilbert space obtained by completion of $C^{\infty}\left(\mathbb{T}_{\Theta}^n\right)$ with |
| cardona-qft-methods_F01192 | 61   | ■ 0.992 | $\mathcal{H}_{\tau}=\left\{\sum_{k \in \mathbb{Z}^n} a_k U_k \mid\left\{a_k\right\}_k \in l^2\left(\mathbb{Z}^n\right)\right\}$ | $\mathcal{H}_{\tau}=\left\{\sum_{k \in \mathbb{Z}^n} a_k U_k \mid\left\{a_k\right\}_k \in l^2\left(\mathbb{Z}^n\right)\right\}$ | of the norm induced by the scalar product $\langle a, b\rangle=\tau\left(a^* b\right)$ .                                                                                                                                                                  |
| cardona-qft-methods_F01193 | 61   | ■ 1.000 | $L(\cdot)$                                                                                                                      | $L(\cdot)$                                                                                                                      | denote respectively by $L(\cdot)$ and $R(\cdot)$ .                                                                                                                                                                                                        |
| cardona-qft-methods_F01194 | 61   | ■ 1.000 | $)$ and $R(\cdot)$                                                                                                              | $)andR(\cdot)$                                                                                                                  | denote respectively by $L(\cdot)$ and $R(\cdot)$ .                                                                                                                                                                                                        |
| cardona-qft-methods_F01195 | 61   | ■ 1.000 | $)$ .                                                                                                                           | $)$ .                                                                                                                           | denote respectively by $L(\cdot)$ and $R(\cdot)$ .                                                                                                                                                                                                        |
| cardona-qft-methods_F01196 | 61   | ■ 1.000 | $\delta_{\mu}, \mu \in\{1, \ldots, n\}$                                                                                         | $\delta_{\mu}, \mu \in\{1, \ldots, n\}$                                                                                         | Let also $\delta_{\mu}, \mu \in\{1, \ldots, n\}$ , be the $n$ (pairwise commuting) canonical derivations defined by $\delta_{\mu}(U_k)=i k_{\mu} U_k$ .                                                                                                   |
| cardona-qft-methods_F01197 | 61   | ■ 1.000 | $\delta_{\mu}\left(U_k\right)=i k_{\mu} U_k$                                                                                    | $\delta_{\mu}\left(U_k\right)=i k_{\mu} U_k$                                                                                    | We need to fix notations: let $\mathcal{A}_{\Theta}=\mathcal{C}$                                                                                                                                                                                          |
| cardona-qft-methods_F01198 | 61   | ■ 1.000 | $\mathcal{A}_{\Theta}=C^{\infty}\left(\mathbb{T}_{\Theta}^n\right)$                                                             | $\mathcal{A}_{\Theta}=C^{\infty}\left(\mathbb{T}_{\Theta}^n\right)$                                                             | with $n=2 m$ or $n=2 m+1$ (i.e., $m=\left[\frac{n}{2}\right]$ is the integer part of $\frac{n}{2}$ ),                                                                                                                                                     |
| cardona-qft-methods_F01199 | 61   | ■ 1.000 | $\mathcal{H}=\mathcal{H}_{\tau} \otimes \mathbb{C}^{2^m}$                                                                       | $\mathcal{H}=\mathcal{H}_{\tau} \otimes \mathbb{C}^{2^m}$                                                                       | with $n=2 m$ or $n=2 m+1$ (i.e., $m=\left[\frac{n}{2}\right]$ is the integer part of $\frac{n}{2}$ ),                                                                                                                                                     |
| cardona-qft-methods_F01200 | 61   | ■ 1.000 | $n=2 m$                                                                                                                         | $n=2 m$                                                                                                                         | the square integrable sections of the trivial spin bundle over $\mathbb{R}^n$ and each element of $\mathcal{A}_{\Theta}$ is represented on $\mathcal{H}$ as $L(a) \otimes 1_{2^m}$ . The Tomita conjugation                                               |
| cardona-qft-methods_F01201 | 61   | ■ 1.000 | $n=2 m+1$                                                                                                                       | $n=2 m+1$                                                                                                                       | the square integrable sections of the trivial spin bundle over $\mathbb{R}^n$ and each element of $\mathcal{A}_{\Theta}$ is represented on $\mathcal{H}$ as $L(a) \otimes 1_{2^m}$ . The Tomita conjugation                                               |
| cardona-qft-methods_F01202 | 61   | ■ 1.000 | $m=\left\lfloor\frac{n}{2}\right\rfloor$                                                                                        | $m=\left\lfloor\frac{n}{2}\right\rfloor$                                                                                        | the square integrable sections of the trivial spin bundle over $\mathbb{R}^n$ and each element of $\mathcal{A}_{\Theta}$ is represented on $\mathcal{H}$ as $L(a) \otimes 1_{2^m}$ . The Tomita conjugation                                               |
| cardona-qft-methods_F01203 | 61   | ■ 1.000 | $\frac{n}{2}$                                                                                                                   | $\frac{n}{2}$                                                                                                                   | the square integrable sections of the trivial spin bundle over $\mathbb{R}^n$ and each element of $\mathcal{A}_{\Theta}$ is represented on $\mathcal{H}$ as $L(a) \otimes 1_{2^m}$ . The Tomita conjugation                                               |
| cardona-qft-methods_F01204 | 61   | ■ 1.000 | $\mathcal{A}_{\Theta}$                                                                                                          | $\mathcal{A}_{\Theta}$                                                                                                          | operator on $\mathcal{C}^{\infty}$ . The Dirac-like operator is given by<br>$\mathcal{D}=-i \delta_{\mu} \otimes \gamma^{\mu}$ .                                                                                                                          |
| cardona-qft-methods_F01205 | 61   | ■ 1.000 | $L(a) \otimes 1_{2^m}$                                                                                                          | $L(a) \otimes 1_{2^m}$                                                                                                          | operator on $\mathcal{C}^{\infty}$ . The Dirac-like operator is given by<br>$\mathcal{D}=-i \delta_{\mu} \otimes \gamma^{\mu}$ .                                                                                                                          |

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| cardona-qft-methods_F01286 | 61   | ■ 1.000 | $J_{\{0\}}(a)=a^{[*]}$                                                                                                            | $J_0(a) = a^*$                                                                                      | where we use hermitian Dirac matrices $\gamma$ . It is defined and symmetric on                                                                                                                                                                                                        |
| cardona-qft-methods_F01287 | 61   | ■ 1.000 | $\left[J_{\{0\}}, \delta_{\mu}\right]=0$                                                                                          | $[J_0, \delta_{\mu}] = 0$                                                                           | where we use hermitian Dirac matrices $\gamma$ . It is defined and symmetric on                                                                                                                                                                                                        |
| cardona-qft-methods_F01288 | 61   | ■ 1.000 | $J=J_{\{0\}} \otimes C_{\{0\}}$                                                                                                   | $J = J_0 \otimes C_0$                                                                               | where we use hermitian Dirac matrices $\gamma$ . It is defined and symmetric on                                                                                                                                                                                                        |
| cardona-qft-methods_F01289 | 61   | ■ 1.000 | $C_{\{0\}}$                                                                                                                       | $C_0$                                                                                               | where we use hermitian Dirac matrices $\gamma$ . It is defined and symmetric on                                                                                                                                                                                                        |
| cardona-qft-methods_F01210 | 61   | ■ 1.000 | $\mathbb{C}^{2^m}$                                                                                                                | $\mathbb{C}^{2^m}$                                                                                  | the dense subset of $\mathcal{H}$ given by $C^\infty(\mathbb{T}_\Theta^n) \otimes \mathbb{C}^{2^m}$ and $\mathcal{D}$ denotes extension. Thus $C_0^\alpha = -\varepsilon^\alpha C_0$ and $\mathcal{D}(H_0 \otimes \sigma_0) = \mathcal{D}(H_0)$ .                                      |
| cardona-qft-methods_F01211 | 61   | ■ 1.000 | $C^{\{\infty\}}\left(\mathbb{T}_{\Theta}^n\right) \otimes \mathbb{C}^{2^m}$                                                       | $C^\infty(\mathbb{T}_\Theta^n) \otimes \mathbb{C}^{2^m}$                                            | this yields a spectral triple.                                                                                                                                                                                                                                                         |
| cardona-qft-methods_F01212 | 61   | ■ 1.000 | $C_{\{0\}} \gamma^\alpha=-\varepsilon \gamma^\alpha C_{\{0\}}$                                                                    | $C_0 \gamma^\alpha = -\varepsilon \gamma^\alpha C_0$                                                | <b>Theorem 8.1.</b> The $\varphi$ -tuple $(\mathcal{A}_\Theta, \mathcal{H}, \mathcal{D}, J, \chi)$ is a real regular spectral triple of dimension $n$ . It satisfies the finiteness and orientability conditions                                                                       |
| cardona-qft-methods_F01213 | 61   | ■ 1.000 | $\mathcal{D} U_k \otimes e_i=k_\mu U_k \otimes \gamma^\mu e_i$                                                                    | $\mathcal{D} U_k \otimes e_i = k_\mu U_k \otimes \gamma^\mu e_i$                                    | <b>Theorem 8.1.</b> The $\varphi$ -tuple $(\mathcal{A}_\Theta, \mathcal{H}, \mathcal{D}, J, \chi)$ is a real regular spectral triple of dimension $n$ . It satisfies the finiteness and orientability conditions                                                                       |
| cardona-qft-methods_F01214 | 61   | ■ 0.992 | $\left(e_i\right)$                                                                                                                | $(e_i)$                                                                                             | tuple of dimension $n$ is obtained by projecting the projection onto corresponding extension of Definition 6.2. It is $n$ -summable and its $KO$ -dimension is also $n$ .                                                                                                              |
| cardona-qft-methods_F01215 | 61   | ■ 0.992 | $C_{\{0\}}^2=\pm 1_{2^m}$                                                                                                         | $C_0^2 = \pm 1_{2^m}$                                                                               | tuple of dimension $n$ is obtained by projecting the projection onto corresponding extension of Definition 6.2. It is $n$ -summable and its $KO$ -dimension is also $n$ .                                                                                                              |
| cardona-qft-methods_F01216 | 61   | ■ 1.000 | $\chi=i d \otimes(-i)^m \gamma^1 \cdots \gamma^n$                                                                                 | $\chi = id \otimes (-i)^m \gamma^1 \cdots \gamma^n$                                                 | For every unitary $u \in \mathcal{A}$ , $uu^* = u^*u = U_0$ , the perturbed operator $V_u \mathcal{D} V_u^*$                                                                                                                                                                           |
| cardona-qft-methods_F01217 | 61   | ■ 0.710 | $\left(\mathcal{A}_{\Theta}, \mathcal{H}, \mathcal{D}, J, \chi\right)$                                                            | $(\mathcal{A}_\Theta, \mathcal{H}, \mathcal{D}, J, \chi)$                                           | operator $\mathcal{D}_A = \mathcal{D} + A + \epsilon J A J^{-1}$ since $V_u \mathcal{D} V_u^* = \mathcal{D}_{L(u) \otimes 1_{2^m}}[\mathcal{D}, L(u^*) \otimes 1_{2^m}]$ . Moreover, we get the gauge transformation $V_u \mathcal{D}_A V_u^* = \mathcal{D}_{\gamma_u(A)}$ , where the |
| cardona-qft-methods_F01218 | 61   | ■ 1.000 | $u \in \mathcal{A}, u u^*=u^* u=U_0$                                                                                              | $u \in \mathcal{A}, uu^* = u^*u = U_0$                                                              | $\gamma^\alpha$ , $A_\alpha = -A_\alpha^* \in \mathcal{A}_\Theta$ , thus $\mathcal{D}_A = -i \left( \delta_\alpha + L(A_\alpha) - R(A_\alpha) \right) \otimes \gamma^\alpha$ . Defining                                                                                                |
| cardona-qft-methods_F01219 | 61   | ■ 1.000 | $V_{\{u\}} \mathcal{D} V_{\{u\}}^*$                                                                                               | $V_u \mathcal{D} V_u^*$                                                                             | $\gamma^\alpha$ , $A_\alpha = -A_\alpha^* \in \mathcal{A}_\Theta$ , thus $\mathcal{D}_A = -i \left( \delta_\alpha + L(A_\alpha) - R(A_\alpha) \right) \otimes \gamma^\alpha$ . Defining                                                                                                |
| cardona-qft-methods_F01220 | 61   | ■ 1.000 | $V_{\{u\}}=\left(L(u) \otimes 1_{2^m}\right) J \left(L(u) \otimes 1_{2^m}\right) J^{-1}$                                          | $V_u = (L(u) \otimes 1_{2^m}) J (L(u) \otimes 1_{2^m}) J^{-1}$                                      |                                                                                                                                                                                                                                                                                        |
| cardona-qft-methods_F01221 | 61   | ■ 0.990 | $J \mathcal{D}=\epsilon \mathcal{D} J$                                                                                            | $J \mathcal{D} = \epsilon \mathcal{D} J$                                                            |                                                                                                                                                                                                                                                                                        |
| cardona-qft-methods_F01222 | 61   | ■ 0.552 | $\mathcal{D}_A=\mathcal{D}+A+\epsilon J A J^{-1}$                                                                                 | $\mathcal{D}_A = \mathcal{D} + A + \epsilon J A J^{-1}$                                             |                                                                                                                                                                                                                                                                                        |
| cardona-qft-methods_F01223 | 61   | ■ 0.552 | $V_{\{u\}} \mathcal{D} V_{\{u\}}^*=\mathcal{D}_{L(u) \otimes 1_{2^m}}\left[\mathcal{D}, L\left(u^*\right) \otimes 1_{2^m}\right]$ | $V_u \mathcal{D} V_u^* = \mathcal{D}_{L(u) \otimes 1_{2^m}}[\mathcal{D}, L(u^*) \otimes 1_{2^m}]$   |                                                                                                                                                                                                                                                                                        |
| cardona-qft-methods_F01224 | 61   | ■ 1.000 | $V_{\{u\}} \mathcal{D}_{\{A\}} V_{\{u\}}^*=\mathcal{D}_{\gamma_u(A)}$                                                             | $V_u \mathcal{D}_A V_u^* = \mathcal{D}_{\gamma_u(A)}$                                               |                                                                                                                                                                                                                                                                                        |
| cardona-qft-methods_F01225 | 61   | ■ 1.000 | $\gamma_u(A)=u\left[\mathcal{D}, u^*\right]+u A u^*$                                                                              | $\gamma_u(A) = u [\mathcal{D}, u^*] + u A u^*$                                                      |                                                                                                                                                                                                                                                                                        |
| cardona-qft-methods_F01226 | 61   | ■ 1.000 | $L(u) \otimes 1_{2^m} \longrightarrow u$                                                                                          | $L(u) \otimes 1_{2^m} \longrightarrow u$                                                            |                                                                                                                                                                                                                                                                                        |
| cardona-qft-methods_F01227 | 61   | ■ 1.000 | $\mathcal{S}\left(\mathcal{D}_A, f, \Lambda\right)=\mathcal{S}\left(\mathcal{D}_{\gamma_u(A)}, f, \Lambda\right)$                 | $\mathcal{S}(\mathcal{D}_A, f, \Lambda) = \mathcal{S}(\mathcal{D}_{\gamma_u(A)}, f, \Lambda)$       |                                                                                                                                                                                                                                                                                        |
| cardona-qft-methods_F01228 | 61   | ■ 1.000 | $A=L\left(-i A_{\alpha}\right) \otimes$                                                                                           | $A = L(-i A_\alpha) \otimes$                                                                        |                                                                                                                                                                                                                                                                                        |
| cardona-qft-methods_F01229 | 61   | ■ 0.981 | $\gamma^{\alpha}, A_{\alpha}=-A_{\alpha}^* \in \mathcal{A}_{\Theta}$                                                              | $\gamma^\alpha, A_\alpha = -A_\alpha^* \in \mathcal{A}_\Theta$                                      |                                                                                                                                                                                                                                                                                        |
| cardona-qft-methods_F01230 | 61   | ■ 0.981 | $\mathcal{D}_A=-i\left(\delta_{\alpha}+L\left(A_{\alpha}\right)-R\left(A_{\alpha}\right)\right) \otimes \gamma^{\alpha}$          | $\mathcal{D}_A = -i \left( \delta_\alpha + L(A_\alpha) - R(A_\alpha) \right) \otimes \gamma^\alpha$ |                                                                                                                                                                                                                                                                                        |

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| cardona-qft-methods_F01231 | 62   | ■ 1.000 | $\tilde{A}_{\alpha} = L(A_{\alpha}) - R(A_{\alpha})$                                                                                                                                                                                                   | $\tilde{A}_{\alpha} = L(A_{\alpha}) - R(A_{\alpha})$                                                                                                                                                                                                   |                                                                                                                                                                                                     |
| cardona-qft-methods_F01232 | 62   | ■ 1.000 | $\mathcal{D}_A^2 = -g^{\alpha_1 \alpha_2} \left( \delta_{\alpha_1} + \tilde{A}_{\alpha_1} \right) \left( \delta_{\alpha_2} + \tilde{A}_{\alpha_2} \right) \otimes 1_{2^m} - \frac{1}{2} \Omega_{\alpha_1 \alpha_2} \otimes \gamma^{\alpha_1 \alpha_2}$ | $\mathcal{D}_A^2 = -g^{\alpha_1 \alpha_2} \left( \delta_{\alpha_1} + \tilde{A}_{\alpha_1} \right) \left( \delta_{\alpha_2} + \tilde{A}_{\alpha_2} \right) \otimes 1_{2^m} - \frac{1}{2} \Omega_{\alpha_1 \alpha_2} \otimes \gamma^{\alpha_1 \alpha_2}$ | <i>Spectral norm</i> 51                                                                                                                                                                             |
| cardona-qft-methods_F01233 | 62   | ■ 1.000 | $\mathcal{D} = U_0 \otimes \mathbb{C}^{2^m}$                                                                                                                                                                                                           | $\mathcal{D} = U_0 \otimes \mathbb{C}^{2^m}$                                                                                                                                                                                                           | $\mathcal{D} \subseteq \mathcal{D}_A$ and or any unitary $u \in \mathcal{A}$ , $\mathcal{D}_{\gamma_u(A)} = V_u \mathcal{D}_A$ .                                                                    |
| cardona-qft-methods_F01234 | 62   | ■ 1.000 | $\operatorname{dim} \mathcal{D} = 2^m$                                                                                                                                                                                                                 | $\dim \mathcal{D} = 2^m$                                                                                                                                                                                                                               | $\mathcal{D} \subseteq \mathcal{D}_A$ and or any unitary $u \in \mathcal{A}$ , $\mathcal{D}_{\gamma_u(A)} = V_u \mathcal{D}_A$ .                                                                    |
| cardona-qft-methods_F01235 | 62   | ■ 1.000 | $\mathcal{D} \subseteq \mathcal{D}_A$                                                                                                                                                                                                                  | $\mathcal{D} \subseteq \mathcal{D}_A$                                                                                                                                                                                                                  | One shows that $\mathcal{D}_A = \mathcal{D}$ in the following cases:                                                                                                                                |
| cardona-qft-methods_F01236 | 62   | ■ 1.000 | $u \in \mathcal{A}, \mathcal{D}_{\gamma_u(A)} = V_u \mathcal{D}_A$                                                                                                                                                                                     | $u \in \mathcal{A}, \mathcal{D}_{\gamma_u(A)} = V_u \mathcal{D}_A$                                                                                                                                                                                     | One shows that $\mathcal{D}_A = \mathcal{D}$ in the following cases:                                                                                                                                |
| cardona-qft-methods_F01237 | 62   | ■ 1.000 | $\mathcal{D}_A = \mathcal{D}$                                                                                                                                                                                                                          | $\mathcal{D}_A = \mathcal{D}$                                                                                                                                                                                                                          | (ii) $\ A\  \leq \frac{1}{2}$ .<br>(iii) The matrix $\frac{1}{2} \Theta$ has only integral coeffici                                                                                                 |
| cardona-qft-methods_F01238 | 62   | ■ 1.000 | $A = A_u = L(u) \otimes 1_{2^m} [ \mathcal{D}, L(u^*) \otimes 1_{2^m} ]$                                                                                                                                                                               | $A = A_u = L(u) \otimes 1_{2^m} [ \mathcal{D}, L(u^*) \otimes 1_{2^m} ]$                                                                                                                                                                               | <i>Conjecture:</i> $\mathcal{D} = \mathcal{D}_A$ at least for generic $\Theta$ 's.                                                                                                                  |
| cardona-qft-methods_F01239 | 62   | ■ 1.000 | $u$                                                                                                                                                                                                                                                    | $u$                                                                                                                                                                                                                                                    | <i>Conjecture:</i> $\mathcal{D} = \mathcal{D}_A$ at least for generic $\Theta$ 's.                                                                                                                  |
| cardona-qft-methods_F01240 | 62   | ■ 0.995 | $\ A\  < \frac{1}{2}$                                                                                                                                                                                                                                  | $\ A\  < \frac{1}{2}$                                                                                                                                                                                                                                  | We will use<br>Since we w                                                                                                                                                                           |
| cardona-qft-methods_F01241 | 62   | ■ 1.000 | $\frac{1}{2\pi} \Theta$                                                                                                                                                                                                                                | $\frac{1}{2\pi} \Theta$                                                                                                                                                                                                                                | condition:                                                                                                                                                                                          |
| cardona-qft-methods_F01242 | 62   | ■ 1.000 | $\mathcal{D} = \mathcal{D}_A$                                                                                                                                                                                                                          | $\mathcal{D} = \mathcal{D}_A$                                                                                                                                                                                                                          | <b>Definition 8.1.</b> Let $\delta > 0$ . A vector $a \in \mathbb{R}^n$ is :                                                                                                                        |
| cardona-qft-methods_F01243 | 62   | ■ 1.000 | $\delta > 0$                                                                                                                                                                                                                                           | $\delta > 0$                                                                                                                                                                                                                                           | approximable if there exists $u \in \mathcal{Z}$ such that $\Theta(u)$ is a badly approximable vector of $\mathbb{R}^n$                                                                             |
| cardona-qft-methods_F01244 | 62   | ■ 1.000 | $a \in \mathbb{R}^n$                                                                                                                                                                                                                                   | $a \in \mathbb{R}^n$                                                                                                                                                                                                                                   | approximable if there exists $u \in \mathcal{Z}$ such that $\Theta(u)$ is a badly approximable vector of $\mathbb{R}^n$                                                                             |
| cardona-qft-methods_F01245 | 62   | ■ 1.000 | $\delta$                                                                                                                                                                                                                                               | $\delta$                                                                                                                                                                                                                                               | approximable if there exists $u \in \mathcal{Z}$ such that $\Theta(u)$ is a badly approximable vector of $\mathbb{R}^n$                                                                             |
| cardona-qft-methods_F01246 | 62   | ■ 0.947 | $c > 0$                                                                                                                                                                                                                                                | $c > 0$                                                                                                                                                                                                                                                |                                                                                                                                                                                                     |
| cardona-qft-methods_F01247 | 62   | ■ 0.947 | $ q \cdot a - m  \geq c  q ^{-\delta}, \forall q \in \mathbb{Z}^n \setminus \{0\}$                                                                                                                                                                     | $ q \cdot a - m  \geq c  q ^{-\delta}, \forall q \in \mathbb{Z}^n \setminus \{0\}$                                                                                                                                                                     |                                                                                                                                                                                                     |
| cardona-qft-methods_F01248 | 62   | ■ 1.000 | $\forall m \in \mathbb{Z}$                                                                                                                                                                                                                             | $\forall m \in \mathbb{Z}$                                                                                                                                                                                                                             |                                                                                                                                                                                                     |
| cardona-qft-methods_F01249 | 62   | ■ 0.996 | $\mathcal{BV}(\delta)$                                                                                                                                                                                                                                 | $\mathcal{BV}(\delta)$                                                                                                                                                                                                                                 | is $\delta$ -badly approximable.)                                                                                                                                                                   |
| cardona-qft-methods_F01250 | 62   | ■ 0.996 | $\mathcal{BV} :=$                                                                                                                                                                                                                                      | $\mathcal{BV} :=$                                                                                                                                                                                                                                      | is $\delta$ -badly approximable.)                                                                                                                                                                   |
| cardona-qft-methods_F01251 | 62   | ■ 1.000 | $\cup_{\delta > 0} \mathcal{BV}(\delta)$                                                                                                                                                                                                               | $\cup_{\delta > 0} \mathcal{BV}(\delta)$                                                                                                                                                                                                               | Let $\Theta \in \mathcal{M}_n(\mathbb{R})$ . If its row of index $i$ is a badly ap of $\mathbb{R}^n$ (i.e. if $I_{\omega} \in \mathcal{BV}$ ) then $\Theta(e_i) \in \mathcal{BV}$ and thus $\Theta$ |
| cardona-qft-methods_F01252 | 62   | ■ 1.000 | $\Theta \in \mathcal{M}_n(\mathbb{R})$                                                                                                                                                                                                                 | $\Theta \in \mathcal{M}_n(\mathbb{R})$                                                                                                                                                                                                                 | imable matrix. It follows that almost any matrix of $\mathcal{M}_n(\mathbb{R}) \approx \mathbb{R}^{n^2}$ is badly                                                                                   |
| cardona-qft-methods_F01253 | 62   | ■ 1.000 | $n \times n$                                                                                                                                                                                                                                           | $n \times n$                                                                                                                                                                                                                                           | imable matrix. It follows that almost any matrix of $\mathcal{M}_n(\mathbb{R}) \approx \mathbb{R}^{n^2}$ is badly                                                                                   |
| cardona-qft-methods_F01254 | 62   | ■ 1.000 | $u \in \mathbb{Z}^n$                                                                                                                                                                                                                                   | $u \in \mathbb{Z}^n$                                                                                                                                                                                                                                   | approximable.                                                                                                                                                                                       |
| cardona-qft-methods_F01255 | 62   | ■ 1.000 | ${}^t \Theta(u)$                                                                                                                                                                                                                                       | ${}^t \Theta(u)$                                                                                                                                                                                                                                       | approximable.                                                                                                                                                                                       |

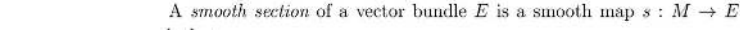
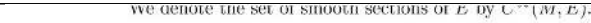
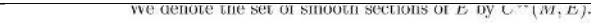
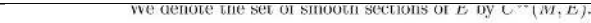
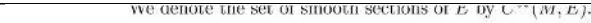
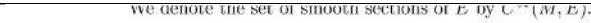
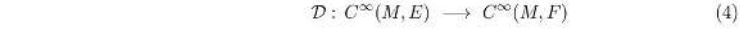
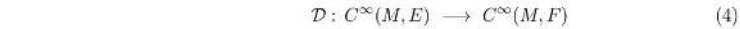
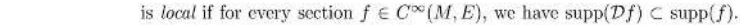
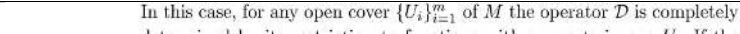
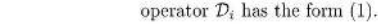
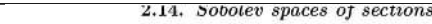
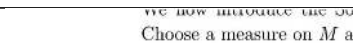
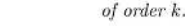
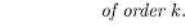
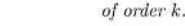
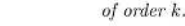
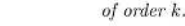
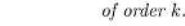
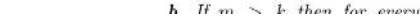
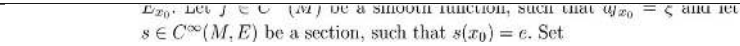
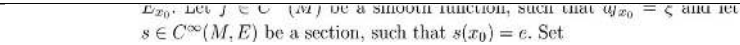
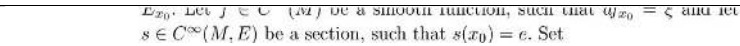
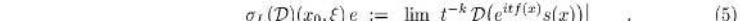
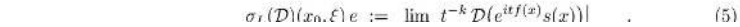
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| cardona-qft-methods_F01256 | 62   | ■ 1.000 | $\mathbb{R}^n$                                                                                                                                       | $\mathbb{R}^n$                                                                                                                                       |                                                                                                                                                                                                                |
| cardona-qft-methods_F01257 | 62   | ■ 1.000 | $\delta > n$                                                                                                                                         | $\delta > n$                                                                                                                                         |                                                                                                                                                                                                                |
| cardona-qft-methods_F01258 | 62   | ■ 0.975 | $\mathbb{R}^n \setminus \mathcal{BV}(\delta)$                                                                                                        | $\mathbb{R}^n \setminus \mathcal{BV}(\delta)$                                                                                                        |                                                                                                                                                                                                                |
| cardona-qft-methods_F01259 | 62   | ■ 1.000 | $i$                                                                                                                                                  | $i$                                                                                                                                                  |                                                                                                                                                                                                                |
| cardona-qft-methods_F01260 | 62   | ■ 0.666 | $L_i \in \mathcal{BV}$                                                                                                                               | $L_i \in \mathcal{BV}$                                                                                                                               |                                                                                                                                                                                                                |
| cardona-qft-methods_F01261 | 62   | ■ 0.666 | ${}^t\Theta(e_i) \in \mathcal{BV}$                                                                                                                   | ${}^t\Theta(e_i) \in \mathcal{BV}$                                                                                                                   |                                                                                                                                                                                                                |
| cardona-qft-methods_F01262 | 62   | ■ 1.000 | $\mathcal{M}_n(\mathbb{R}) \approx \mathbb{R}^{n^2}$                                                                                                 | $\mathcal{M}_n(\mathbb{R}) \approx \mathbb{R}^{n^2}$                                                                                                 |                                                                                                                                                                                                                |
| cardona-qft-methods_F01263 | 63   | ■ 1.000 | $(C^\infty(\mathbb{T}_\Theta^n), \mathcal{H}, \mathcal{D})$                                                                                          | $(C^\infty(\mathbb{T}_\Theta^n), \mathcal{H}, \mathcal{D})$                                                                                          |                                                                                                                             |
| cardona-qft-methods_F01264 | 63   | ■ 1.000 | $\{n-k : k \in \mathbb{N}_0\}$                                                                                                                       | $\{n-k : k \in \mathbb{N}_0\}$                                                                                                                       |                                                                                                                             |
| cardona-qft-methods_F01265 | 63   | ■ 1.000 | $\zeta_D(0)=0$                                                                                                                                       | $\zeta_D(0)=0$                                                                                                                                       |                                                                                                                             |
| cardona-qft-methods_F01266 | 63   | ■ 1.000 | $\zeta_D(0)$                                                                                                                                         | $\zeta_D(0)$                                                                                                                                         | the sufficient Diophantine condition.                                                                                                                                                                          |
| cardona-qft-methods_F01267 | 63   | ■ 1.000 | $\zeta_{D_A}(0)$                                                                                                                                     | $\zeta_{D_A}(0)$                                                                                                                                     | We consider                                                                                                                                                                                                    |
| cardona-qft-methods_F01268 | 63   | ■ 0.813 | $C^\infty(\mathbb{T}_\Theta^n), \mathcal{H}, \mathcal{D}$                                                                                            | $C^\infty(\mathbb{T}_\Theta^n), \mathcal{H}, \mathcal{D}$                                                                                            | (iii) More generally, in $\mathcal{S}(\mathcal{D}_A, f, \Lambda) = \sum_{k=0}^\infty f_{n-k} c_{n-k}(A) \Lambda^{n-k} + \mathcal{O}(\Lambda^{-1})$ $c_{n-k}(A) = 0$ for $k$ odd. In particular, $c_0(A) = 0$ . |
| cardona-qft-methods_F01269 | 63   | ■ 1.000 | $A=L\left(-iA_\alpha\right)\otimes\gamma^\alpha$                                                                                                     | $A=L\left(-iA_\alpha\right)\otimes\gamma^\alpha$                                                                                                     | This result (for $n=4$ ) has also been obtained in <sup>41</sup> using the heat kernel                                                                                                                         |
| cardona-qft-methods_F01270 | 63   | ■ 1.000 | $n=2, \mathcal{S}(\mathcal{D}_A, f, \Lambda) = 4\pi f_2 \Lambda^2 + \mathcal{O}(\Lambda^{-2})$                                                       | $n=2, \mathcal{S}(\mathcal{D}_A, f, \Lambda) = 4\pi f_2 \Lambda^2 + \mathcal{O}(\Lambda^{-2})$                                                       | or (59) since it shows how this formula is obtained.<br><b>Remark 8.1.</b> Note that all terms must be gauge invariant.                                                                                        |
| cardona-qft-methods_F01271 | 63   | ■ 1.000 | $n=4, \mathcal{S}(\mathcal{D}_A, f, \Lambda) = 8\pi^2 f_4 \Lambda^4 - \frac{4\pi^2}{3} f(0) \tau(F_{\mu\nu} F^{\mu\nu}) + \mathcal{O}(\Lambda^{-2})$ | $n=4, \mathcal{S}(\mathcal{D}_A, f, \Lambda) = 8\pi^2 f_4 \Lambda^4 - \frac{4\pi^2}{3} f(0) \tau(F_{\mu\nu} F^{\mu\nu}) + \mathcal{O}(\Lambda^{-2})$ | <b>Remark 8.1.</b> Note that all terms must be gauge invariant, namely invariant by $A_\alpha \rightarrow \gamma_u(A_\alpha) = u A_\alpha u^* + u \delta_\alpha(u^*)$ . A particular case is $u = U_k$ .       |
| cardona-qft-methods_F01272 | 63   | ■ 1.000 | $\mathcal{S}(\mathcal{D}_A, f, \Lambda) = \sum_{k=0}^n f_{n-k} c_{n-k}(A) \Lambda^{n-k} +$                                                           | $\mathcal{S}(\mathcal{D}_A, f, \Lambda) = \sum_{k=0}^n f_{n-k} c_{n-k}(A) \Lambda^{n-k} +$                                                           | where $U_k \delta_\alpha(U_k^*) = -i k_\alpha U_0$ .                                                                                                                                                           |
| cardona-qft-methods_F01273 | 63   | ■ 1.000 | $\mathcal{O}(\Lambda^{-1}), c_{n-2}(A) = 0, c_{n-k}(A) = 0$                                                                                          | $\mathcal{O}(\Lambda^{-1}), c_{n-2}(A) = 0, c_{n-k}(A) = 0$                                                                                          |                                                                                                                                                                                                                |
| cardona-qft-methods_F01274 | 63   | ■ 1.000 | $c_0(A)=0$                                                                                                                                           | $c_0(A)=0$                                                                                                                                           |                                                                                                                                                                                                                |
| cardona-qft-methods_F01275 | 63   | ■ 1.000 | $n=4$                                                                                                                                                | $n=4$                                                                                                                                                |                                                                                                                                                                                                                |
| cardona-qft-methods_F01276 | 63   | ■ 1.000 | $A_\alpha \rightarrow \gamma_u(A_\alpha) = u A_\alpha u^* + u \delta_\alpha(u^*)$                                                                    | $A_\alpha \rightarrow \gamma_u(A_\alpha) = u A_\alpha u^* + u \delta_\alpha(u^*)$                                                                    |                                                                                                                                                                                                                |
| cardona-qft-methods_F01277 | 63   | ■ 1.000 | $u=U_k$                                                                                                                                              | $u=U_k$                                                                                                                                              |                                                                                                                                                                                                                |
| cardona-qft-methods_F01278 | 63   | ■ 1.000 | $U_k \delta_\alpha(U_k^*) = -i k_\alpha U_0$                                                                                                         | $U_k \delta_\alpha(U_k^*) = -i k_\alpha U_0$                                                                                                         |                                                                                                                                                                                                                |
| cardona-qft-methods_F01279 | 64   | ■ 1.000 | $A, \mathcal{D}_A = \mathcal{D}$                                                                                                                     | $A, \mathcal{D}_A = \mathcal{D}$                                                                                                                     |                                                                                                                                                                                                                |

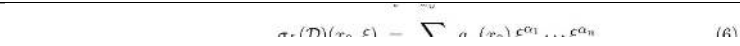
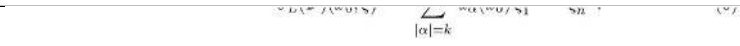
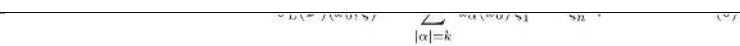
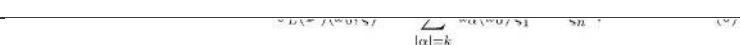
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| cardona-qft-methods_F01280 | 64   | 1.000 | $\Theta=\theta\left(\begin{array}{cc}\theta&1\\-1&0\end{array}\right)$                               | $\Theta = \theta \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$              | $\Theta = \nu \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}$ with $\nu \in \mathbb{R}$ . In fact, the Hochschild cohomology $HH^j(\mathcal{A}_\Theta, \mathcal{A}_\Theta^*)$ satisfies $\dim H^j(\mathcal{A}_\Theta, \mathcal{A}_\Theta^*) = 2$ (or 1) for $j = 1$ (or $j = 2$ ) if and only if the |
| cardona-qft-methods_F01281 | 64   | 1.000 | $\theta \in \mathbb{R}$                                                                              | $\theta \in \mathbb{R}$                                                      | $\Theta = \nu \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}$ with $\nu \in \mathbb{R}$ . In fact, the Hochschild cohomology $HH^j(\mathcal{A}_\Theta, \mathcal{A}_\Theta^*)$ satisfies $\dim H^j(\mathcal{A}_\Theta, \mathcal{A}_\Theta^*) = 2$ (or 1) for $j = 1$ (or $j = 2$ ) if and only if the |
| cardona-qft-methods_F01282 | 64   | 1.000 | $H\left(\mathrm{mathcal{A}}_{\Theta}, \mathrm{mathcal{A}}_{\Theta}^*\right)^{\ast}$                  | $H(\mathcal{A}_\Theta, \mathcal{A}_\Theta^*)$                                | $\Theta = \nu \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}$ with $\nu \in \mathbb{R}$ . In fact, the Hochschild cohomology $HH^j(\mathcal{A}_\Theta, \mathcal{A}_\Theta^*)$ satisfies $\dim H^j(\mathcal{A}_\Theta, \mathcal{A}_\Theta^*) = 2$ (or 1) for $j = 1$ (or $j = 2$ ) if and only if the |
| cardona-qft-methods_F01283 | 64   | 0.996 | $\operatorname{dim} H^j\left(\mathrm{mathcal{A}}_{\Theta}, \mathrm{mathcal{A}}_{\Theta}^*\right)=2$  | $\dim H^j(\mathcal{A}_\Theta, \mathcal{A}_\Theta^*) = 2$                     | irrational number $\theta$ satisfies a Diophantine condition like $ 1 - e^{i2\pi n\theta} ^{-1} =$                                                                                                                                                                                                  |
| cardona-qft-methods_F01284 | 64   | 0.996 | $j=1$                                                                                                | $j = 1$                                                                      | irrational number $\theta$ satisfies a Diophantine condition like $ 1 - e^{i2\pi n\theta} ^{-1} =$                                                                                                                                                                                                  |
| cardona-qft-methods_F01285 | 64   | 0.996 | $j=2$                                                                                                | $j = 2$                                                                      | irrational number $\theta$ satisfies a Diophantine condition like $ 1 - e^{i2\pi n\theta} ^{-1} =$                                                                                                                                                                                                  |
| cardona-qft-methods_F01286 | 64   | 1.000 | $\theta$                                                                                             | $\theta$                                                                     | $\mathcal{O}(n^j)$ for some $k$ .<br>The result of Theorem 8.2 without this Diophantine condition is un-                                                                                                                                                                                            |
| cardona-qft-methods_F01287 | 64   | 1.000 | $ 1-e^{i2\pi n\theta} ^{-1}=$                                                                        | $ 1 - e^{i2\pi n\theta} ^{-1} =$                                             | $\mathcal{O}(n^j)$ for some $k$ .<br>The result of Theorem 8.2 without this Diophantine condition is un-                                                                                                                                                                                            |
| cardona-qft-methods_F01288 | 64   | 1.000 | $\mathrm{mathcal{O}}\left(n^k\right)$                                                                | $\mathcal{O}(n^k)$                                                           | known.                                                                                                                                                                                                                                                                                              |
| cardona-qft-methods_F01289 | 64   | 1.000 | $\mathbb{Z}^n$                                                                                       | $\mathbb{Z}^n$                                                               |                                                                                                                                                                                                                                                                                                     |
| cardona-qft-methods_F01290 | 71   | 1.000 | $\boldsymbol{G}$                                                                                     | $\boldsymbol{G}$                                                             |                                                                                                                                                                                                                                                                                                     |
| cardona-qft-methods_F01291 | 71   | 0.998 | $\{ \}^{\{3-5,13,17\}}$                                                                              | 3−5,13,17                                                                    | <i>Remarks:</i> Index theorem: non-compact manifolds                                                                                                                                                                                                                                                |
| cardona-qft-methods_F01292 | 71   | 0.997 | $\{ \}^{\{11,25\}}$                                                                                  | 11,25                                                                        | finite-dimensional vector<br>let us start with the finite                                                                                                                                                                                                                                           |
| cardona-qft-methods_F01293 | 72   | 1.000 | $A: \mathrm{mathcal{H}}_1 \rightarrow \mathrm{mathcal{H}}_2$                                         | $A: \mathcal{H}_1 \rightarrow \mathcal{H}_2$                                 | <i>Index theory for non-compact G-manifolds</i> 61                                                                                                                                                                                                                                                  |
| cardona-qft-methods_F01294 | 72   | 0.997 | $\mathrm{mathcal{H}}_1$                                                                              | $\mathcal{H}_1$                                                              | $\mathcal{H}_1 = \mathrm{mathcal{H}}_1 \oplus \mathrm{mathcal{H}}_2 \oplus \mathrm{mathcal{H}}_3 \oplus \dots$                                                                                                                                                                                      |
| cardona-qft-methods_F01295 | 72   | 0.997 | $\mathrm{mathcal{H}}_2$                                                                              | $\mathcal{H}_2$                                                              | $\mathcal{H}_2 = \mathrm{mathcal{H}}_1 \oplus \mathrm{mathcal{H}}_2 \oplus \mathrm{mathcal{H}}_3 \oplus \dots$                                                                                                                                                                                      |
| cardona-qft-methods_F01296 | 72   | 0.840 | $\operatorname{Coker}(A):=\mathrm{mathcal{H}}_2/\operatorname{Im}(A)$                                | $\operatorname{Coker}(A) := \mathcal{H}_2/\operatorname{Im}(A)$              | $\operatorname{Ind}(A) := \dim \operatorname{Ker}(A) - \dim \operatorname{Coker}(A) \in \mathbb{Z}, \quad (1)$                                                                                                                                                                                      |
| cardona-qft-methods_F01297 | 72   | 0.840 | $\operatorname{dim} \operatorname{Ker}(A)$                                                           | $\dim \operatorname{Ker}(A)$                                                 | $\operatorname{Ind}(A) := \dim \operatorname{Ker}(A) - \dim \operatorname{Coker}(A) \in \mathbb{Z}, \quad (1)$                                                                                                                                                                                      |
| cardona-qft-methods_F01298 | 72   | 0.974 | $\operatorname{dim} \operatorname{Coker}(A)$                                                         | $\dim \operatorname{Coker}(A)$                                               | where $\operatorname{Coker}(A) := \mathcal{H}_2/\operatorname{Im}(A)$ . Noti                                                                                                                                                                                                                        |
| cardona-qft-methods_F01299 | 72   | 0.999 | $\operatorname{Ker}(A)$                                                                              | $\operatorname{Ker}(A)$                                                      | Exercise 1.3. Show that if the dimension of $\operatorname{Coker}(A)$ is finite, then $\operatorname{Im}(A)$                                                                                                                                                                                        |
| cardona-qft-methods_F01300 | 72   | 0.999 | $\operatorname{Coker}(A)$                                                                            | $\operatorname{Coker}(A)$                                                    | Exercise 1.3. Show that if the dimension of $\operatorname{Coker}(A)$ is finite, then $\operatorname{Im}(A)$                                                                                                                                                                                        |
| cardona-qft-methods_F01301 | 72   | 0.994 | $\operatorname{Im}(A)$                                                                               | $\operatorname{Im}(A)$                                                       | be a bounded linear operator. We say that $A$ is a <i>Fredholm operator</i> iff                                                                                                                                                                                                                     |
| cardona-qft-methods_F01302 | 72   | 0.985 | $\mathrm{mathcal{H}}_1, \mathrm{mathcal{H}}_2$                                                       | $\mathcal{H}_1, \mathcal{H}_2$                                               | defined by (1). The notion of index of a Fredholm operator was introduced by Fritz Neuberger <sup>20a</sup> , who also established the following “stability” property:                                                                                                                              |
| cardona-qft-methods_F01303 | 72   | 0.506 | $\operatorname{dim} \operatorname{Ker}(A)<\infty, \operatorname{dim} \operatorname{Coker}(A)<\infty$ | $\dim \operatorname{Ker}(A) < \infty, \dim \operatorname{Coker}(A) < \infty$ | <b>Theorem 1.5.</b> <b>a.</b> If $A(t)$ is a Fredh                                                                                                                                                                                                                                                  |
| cardona-qft-methods_F01304 | 72   | 1.000 | $\operatorname{Ind}(A)$                                                                              | $\operatorname{Ind}(A)$                                                      | <b>b.</b> If $K$ is compact and $A$ Fredholm, then $\operatorname{Ind}(A+K) = \operatorname{Ind}(A)$ .                                                                                                                                                                                              |
| cardona-qft-methods_F01305 | 72   | 1.000 | $\{ \}^{\{20 \mathrm{a}\}}$                                                                          | 20a                                                                          |                                                                                                                                                                                                                                                                                                     |
| cardona-qft-methods_F01306 | 72   | 1.000 | $A(t)$                                                                                               | $A(t)$                                                                       |                                                                                                                                                                                                                                                                                                     |

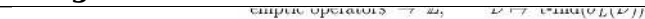
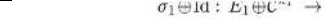
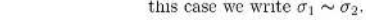
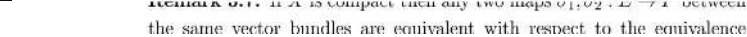
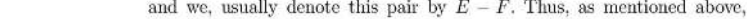
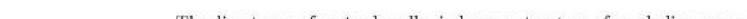
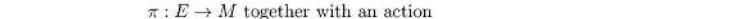
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| cardona-qft-methods_F01307 | 72   | ■ 1.000 | $\operatorname{Ind}(A(t))$                          | $\operatorname{Ind}(A(t))$                          |                                                                                                                                                                                |
| cardona-qft-methods_F01308 | 72   | ■ 1.000 | $\operatorname{Ind}(A+K)=\operatorname{Ind}(A)$     | $\operatorname{Ind}(A+K)=\operatorname{Ind}(A)$     |                                                                                                                                                                                |
| cardona-qft-methods_F01309 | 72   | ■ 0.992 | $\{ \ }^{\mathrm{a}}$                               | a                                                   |                                                                                                                                                                                |
| cardona-qft-methods_F01310 | 72   | ■ 1.000 | $\{ \ }^{21,22}$                                    | 21,22                                               |                                                                                                                                                                                |
| cardona-qft-methods_F01311 | 73   | ■ 1.000 | $T: \mathcal{H} \rightarrow \mathcal{H}$            | $T: \mathcal{H} \rightarrow \mathcal{H}$            | $\mathcal{H} = \ell_2 := \{ (x_1, x_2, \dots) \mid x_j \in \mathbb{C}, \sum_{j=1}^{\infty}  x_j ^2 < \infty \},$                                                               |
| cardona-qft-methods_F01312 | 73   | ■ 1.000 | $\operatorname{dim} \operatorname{Ker}(T)=2$        | $\dim \operatorname{Ker}(T) = 2$                    | Since $\operatorname{Im}(T) = \mathbb{R}$ , since the range of $T$ is one dimensional, we conclude that $\operatorname{Coker}(T) = \{0\}$ . Thus $\operatorname{Ind}(T) = 2$ . |
| cardona-qft-methods_F01313 | 73   | ■ 1.000 | $\operatorname{Coker}(T)=\{0\}$                     | $\operatorname{Coker}(T) = \{0\}$                   | <b>Remark 1.7.</b> If $\mathcal{H}$ is a finite dimensional vector space                                                                                                       |
| cardona-qft-methods_F01314 | 73   | ■ 1.000 | $\operatorname{Ind}(T)=2$                           | $\operatorname{Ind}(T) = 2$                         | <b>Remark 1.7.</b> If $\mathcal{H}$ is a finite dimensional vector space                                                                                                       |
| cardona-qft-methods_F01315 | 73   | ■ 0.999 | $\operatorname{Ind}(A)=0$                           | $\operatorname{Ind}(A) = 0$                         | Exercise 1.8. Suppose $A: \mathcal{H}_1 \rightarrow \mathcal{H}_2$ and $B: \mathcal{H}_2 \rightarrow \mathcal{H}_3$ are Fredholm                                               |
| cardona-qft-methods_F01316 | 73   | ■ 0.999 | $A: \mathcal{H} \rightarrow \mathcal{H}$            | $A: \mathcal{H} \rightarrow \mathcal{H}$            | Exercise 1.8. Suppose $A: \mathcal{H}_1 \rightarrow \mathcal{H}_2$ and $B: \mathcal{H}_2 \rightarrow \mathcal{H}_3$ are Fredholm                                               |
| cardona-qft-methods_F01317 | 73   | ■ 1.000 | $\operatorname{dim} \mathcal{H}=\infty$             | $\dim \mathcal{H} = \infty$                         | operators. Show that the operator $BA: \mathcal{H}_1 \rightarrow \mathcal{H}_3$ is                                                                                             |
| cardona-qft-methods_F01318 | 73   | ■ 1.000 | $B: \mathcal{H}_2 \rightarrow \mathcal{H}_3$        | $B: \mathcal{H}_2 \rightarrow \mathcal{H}_3$        | Exercise 1.9. Let $A: \mathcal{H}_1 \rightarrow \mathcal{H}_2$ be a Fredholm operator. Show that the                                                                           |
| cardona-qft-methods_F01319 | 73   | ■ 1.000 | $BA: \mathcal{H}_1 \rightarrow \mathcal{H}_3$       | $BA: \mathcal{H}_1 \rightarrow \mathcal{H}_3$       | adjoint operator $A^*: \mathcal{H}_2 \rightarrow \mathcal{H}_1$ is Fredholm and that                                                                                           |
| cardona-qft-methods_F01320 | 73   | ■ 0.983 | $A^*: \mathcal{H}_2 \rightarrow \mathcal{H}_1$      | $A^*: \mathcal{H}_2 \rightarrow \mathcal{H}_1$      | One of the most common applications of the index is based on. Suppose for example that $\operatorname{Ind}(A) > 0$ . This means that $\operatorname{Ker}($                     |
| cardona-qft-methods_F01321 | 73   | ■ 0.991 | $\operatorname{Ind}(A)>0$                           | $\operatorname{Ind}(A) > 0$                         | we conclude that equation (5) has a non-trivial solution, even though we                                                                                                       |
| cardona-qft-methods_F01322 | 73   | ■ 0.991 | $\operatorname{Ker}(A) \neq \{0\}$                  | $\operatorname{Ker}(A) \neq \{0\}$                  | we conclude that equation (5) has a non-trivial solution, even though we                                                                                                       |
| cardona-qft-methods_F01323 | 73   | ■ 1.000 | $Ax=0$                                              | $Ax = 0$                                            | cannot find this solution explicitly.                                                                                                                                          |
| cardona-qft-methods_F01324 | 73   | ■ 1.000 | $\operatorname{Ind}(A+K)=\operatorname{Ind}(K)>0$   | $\operatorname{Ind}(A+K)=\operatorname{Ind}(K)>0$   |                                                                                                                                                                                |
| cardona-qft-methods_F01325 | 73   | ■ 1.000 | $B$                                                 | $B$                                                 |                                                                                                                                                                                |
| cardona-qft-methods_F01326 | 73   | ■ 1.000 | $B=A+K$                                             | $B = A + K$                                         |                                                                                                                                                                                |
| cardona-qft-methods_F01327 | 74   | ■ 1.000 | $\operatorname{Fred}(\mathcal{H}_1, \mathcal{H}_2)$ | $\operatorname{Fred}(\mathcal{H}_1, \mathcal{H}_2)$ | <b>1.11 Connected components of the set of Fredholm</b>                                                                                                                        |
| cardona-qft-methods_F01328 | 74   | ■ 1.000 | $d(A, B):=\ A-B\ $                                  | $d(A, B) := \ A - B\ $                              | <i>operators</i>                                                                                                                                                               |
| cardona-qft-methods_F01329 | 74   | ■ 1.000 | $\operatorname{Ind}(A)=\operatorname{Ind}(B)$       | $\operatorname{Ind}(A) = \operatorname{Ind}(B)$     | is a metric space with the distance defined by $d(A, B) := \ A - B\ $ . Theorem 1.5 implies that if operators $A$ and $B$ belong to the same connected                         |
| cardona-qft-methods_F01330 | 74   | ■ 1.000 | $\operatorname{Irr}(G)$                             | $\operatorname{Irr}(G)$                             | <b>Theorem 1.13</b> Any finite dimensional representation $U$ of $G$ has a unique                                                                                              |
| cardona-qft-methods_F01331 | 74   | ■ 1.000 | $m_V V$                                             | $m_V V$                                             | commutes with the action of the group, i.e.,                                                                                                                                   |
| cardona-qft-methods_F01332 | 74   | ■ 1.000 | $V \oplus V \oplus \cdots \oplus V$                 | $V \oplus V \oplus \cdots \oplus V$                 | commutes with the action of the group, i.e.,                                                                                                                                   |
| cardona-qft-methods_F01333 | 74   | ■ 1.000 | $m_V$                                               | $m_V$                                               | commutes with the action of the group, i.e.,                                                                                                                                   |

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| cardona-qft-methods_F01334 | 74   | ■ 1.000 | $m_{\{V\}} \in \mathbb{N}$                                                                                        | $m_V \in \mathbb{N}$                                        | Exercise 1.15. Let $G$ be a compact group acting on Hilbert spaces $\mathcal{H}_1$ and                                                                                                                  |
| cardona-qft-methods_F01335 | 74   | ■ 1.000 | $x \in \operatorname{Ker}(A), g \in G$                                                                            | $x \in \operatorname{Ker}(A), g \in G$                      |                                                                                                                                                                                                         |
| cardona-qft-methods_F01336 | 74   | ■ 1.000 | $g \, x \in \operatorname{Ker}(A)$                                                                                | $gx \in \operatorname{Ker}(A)$                              |                                                                                                                                                                                                         |
| cardona-qft-methods_F01337 | 75   | ■ 1.000 | $m_{\{+\}}, m_{\{-}}$                                                                                             | $m_+, m_-$                                                  | <b>Remark 1.16.</b> Note the though the set $\operatorname{irr}(G)$ is infinite both sums above are actually finite, since only finitely many numbers in $\operatorname{irr}(G)$ can be non-            |
| cardona-qft-methods_F01338 | 75   | ■ 0.886 | $R(G)$                                                                                                            | $R(G)$                                                      | <b>Definition 1.18.</b> Consider the set of formal differences, $V_1 - V_2$ , of finite                                                                                                                 |
| cardona-qft-methods_F01339 | 75   | ■ 0.886 | $V-U$                                                                                                             | $V - U$                                                     | <b>Definition 1.18.</b> Consider the set of formal differences, $V_1 - V_2$ , of finite                                                                                                                 |
| cardona-qft-methods_F01340 | 75   | ■ 1.000 | $V_{\{1\}}-V_{\{2\}}$                                                                                             | $V_1 - V_2$                                                 | $V - U \sim A - B \iff V \oplus B \simeq A \oplus U,$                                                                                                                                                   |
| cardona-qft-methods_F01341 | 75   | ■ 1.000 | $V_{\{1\}}$                                                                                                       | $V_1$                                                       | is an abelian group denoted by $R(G)$ and called the <i>group of characters of</i>                                                                                                                      |
| cardona-qft-methods_F01342 | 75   | ■ 1.000 | $V_{\{2\}}$                                                                                                       | $V_2$                                                       | is an abelian group denoted by $R(G)$ and called the <i>group of characters of</i>                                                                                                                      |
| cardona-qft-methods_F01343 | 75   | ■ 1.000 | $\{ \}^{\{\text{b} \}}$                                                                                           | $\mathfrak{b}$                                              | $G$ . Formally,                                                                                                                                                                                         |
| cardona-qft-methods_F01344 | 75   | ■ 0.997 | $R(G):=\{(V, U) \mid V, U$                                                                                        | $R(G) := \{(V, U) \mid V, U$                                | tion of $\operatorname{irr}(G)$ ,<br>is the set of all $(V, U) \in \operatorname{irr}(G) \times \operatorname{irr}(G)$ such that                                                                        |
| cardona-qft-methods_F01345 | 75   | ■ 0.997 | $G\} / \sim$                                                                                                      | $G\} / \sim$                                                | tion of $\operatorname{irr}(G)$ ,<br>is the set of all $(V, U) \in \operatorname{irr}(G) \times \operatorname{irr}(G)$ such that                                                                        |
| cardona-qft-methods_F01346 | 75   | ■ 1.000 | $\{ \}^{\{\mathrm{b}\}}$                                                                                          | $\mathfrak{b}$                                              |                                                                                                                                                                                                         |
| cardona-qft-methods_F01347 | 76   | ■ 1.000 | $\operatorname{Ind}_{\{G\}}(A)$                                                                                   | $\operatorname{Ind}_G(A)$                                   | $V \in \operatorname{Irr}(G)$                                                                                                                                                                           |
| cardona-qft-methods_F01348 | 76   | ■ 1.000 | $V_{\{0\}}$                                                                                                       | $V_0$                                                       | To avoid confusion we denote by $q$ the non-trivial element of $G$ (thus $q =$                                                                                                                          |
| cardona-qft-methods_F01349 | 76   | ■ 1.000 | $m_{\{V_{\{0\}}\}^{\{+\}}}-m_{\{V_{\{0\}}\}^{\{-}}}>0$                                                            | $m_{V_0}^+ - m_{V_0}^- > 0$                                 | As the datum on section of $G$ are $\pm \varrho_2$<br>$\sigma(x_1,x_2,\ldots)=(x_2,x_1,x_4,x_3,\ldots).$                                                                                                |
| cardona-qft-methods_F01350 | 76   | ■ 1.000 | $q$                                                                                                               | $q$                                                         |                                                                                                                                                                                                         |
| cardona-qft-methods_F01351 | 76   | ■ 1.000 | $q=$                                                                                                              | $q =$                                                       |                                                                                                                                                                                                         |
| cardona-qft-methods_F01352 | 76   | ■ 0.582 | $l^{\{2\}}$                                                                                                       | $l^2$                                                       |                                                                                                                                                                                                         |
| cardona-qft-methods_F01353 | 76   | ■ 1.000 | $V_{\{0\}}=\mathbb{C}$                                                                                            | $V_0 = \mathbb{C}$                                          |                                                                                                                                                                                                         |
| cardona-qft-methods_F01354 | 76   | ■ 1.000 | $V_{\{1\}}=\mathbb{C}$                                                                                            | $V_1 = \mathbb{C}$                                          |                                                                                                                                                                                                         |
| cardona-qft-methods_F01355 | 76   | ■ 1.000 | $q \, z=-z$                                                                                                       | $qz = -z$                                                   |                                                                                                                                                                                                         |
| cardona-qft-methods_F01356 | 76   | ■ 0.977 | $\operatorname{Ker}(T)=V_{\{0\}} \oplus V_{\{1\}}$                                                                | $\operatorname{Ker}(T) = V_0 \oplus V_1$                    |                                                                                                                                                                                                         |
| cardona-qft-methods_F01357 | 76   | ■ 1.000 | $\operatorname{Ind}_{\{G\}}(T)$                                                                                   | $\operatorname{Ind}_G(T)$                                   |                                                                                                                                                                                                         |
| cardona-qft-methods_F01358 | 77   | ■ 1.000 | $\mathcal{D}_{\{j\}}$                                                                                             | $D_j$                                                       | We recall the definition of linear differential operators in $\mathbb{R}^n$ . Let $D_j :$<br>$C^\infty(\mathbb{R}^n) \rightarrow C^\infty(\mathbb{R}^n)$ denote the operator                            |
| cardona-qft-methods_F01359 | 77   | ■ 1.000 | $C^{\{\infty\}}\!\left(\mathbb{R}^{\{n\}}\right) \rightarrowtail C^{\{\infty\}}\!\left(\mathbb{R}^{\{n\}}\right)$ | $C^\infty(\mathbb{R}^n) \rightarrow C^\infty(\mathbb{R}^n)$ | $C^{\{\infty\}}(\mathbb{R}^{\{n\}}) \subset C^{\{\infty\}}(\mathbb{R}^{\{n\}})$ denote the operators<br>$\partial_1 = \frac{\partial}{\partial x_1}, \dots, \partial_n = \frac{\partial}{\partial x_n}$ |
| cardona-qft-methods_F01360 | 77   | ■ 0.999 | $\alpha=\left(\alpha_{\{1\}}, \, \ldots, \, \alpha_{\{n\}}\right) \in$                                            | $\alpha = (\alpha_1, \dots, \alpha_n) \in$                  | $ \alpha  := \alpha_1 + \dots + \alpha_n$                                                                                                                                                               |

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| cardona-qft-methods_F01361 | 77   | ■ 1.000 | $\mathbb{Z}_{\geq 0}^n$                                                                                           | $\mathbb{Z}_{\geq 0}^n$                                                                                           | and                                                                                                                                                                                     |
| cardona-qft-methods_F01362 | 77   | ■ 1.000 | $a_{\alpha}(x) \in C^{\infty}(\mathbb{R}^n)$                                                                      | $a_{\alpha}(x) \in C^{\infty}(\mathbb{R}^n)$                                                                      | formula                                                                                                                                                                                 |
| cardona-qft-methods_F01363 | 77   | ■ 1.000 | $C_c^{\infty}(\mathbb{R}^n)$                                                                                      | $C_c^{\infty}(\mathbb{R}^n)$                                                                                      | $\sum_{ \alpha =k}$                                                                                                                                                                     |
| cardona-qft-methods_F01364 | 77   | ■ 1.000 | $\langle \cdot, \cdot \rangle$                                                                                    | $\langle \cdot, \cdot \rangle$                                                                                    | The corresponding norm                                                                                                                                                                  |
| cardona-qft-methods_F01365 | 78   | ■ 1.000 | $H^k(\mathbb{R}^n) \subset L^2(\mathbb{R}^n)$                                                                     | $H^k(\mathbb{R}^n) \subset L^2(\mathbb{R}^n)$                                                                     | Let $\mathcal{D}(\mathbb{R}^n)$ denote the space of smooth functions with compact support                                                                                               |
| cardona-qft-methods_F01366 | 78   | ■ 1.000 | $H^k(\mathbb{R}^n)$                                                                                               | $H^k(\mathbb{R}^n)$                                                                                               | $C_c^{\infty}(\mathbb{R}^n)$ with respect to the                                                                                                                                        |
| cardona-qft-methods_F01367 | 78   | ■ 1.000 | $L^2(\mathbb{R}^n)$                                                                                               | $L^2(\mathbb{R}^n)$                                                                                               | integrable function $L^2(\mathbb{R}^n)$ . More exp                                                                                                                                      |
| cardona-qft-methods_F01368 | 78   | ■ 1.000 | $\Omega \subset \mathbb{R}^n$                                                                                     | $\Omega \subset \mathbb{R}^n$                                                                                     | $H^k(\Omega)$ of functions on $\Omega$ . An analogue of Theorem 2.4 still holds if the                                                                                                  |
| cardona-qft-methods_F01369 | 78   | ■ 1.000 | $H^k(\Omega)$                                                                                                     | $H^k(\Omega)$                                                                                                     | boundary of $\Omega$ is sufficiently nice, see [1, Theorem 3.18].                                                                                                                       |
| cardona-qft-methods_F01370 | 78   | ■ 1.000 | $K \subset M$                                                                                                     | $K \subset M$                                                                                                     | For every $m \geq k$ a differential operator $\mathcal{D} : C_c^{\infty}(\mathbb{R}^n) \rightarrow C_c^{\infty}(\mathbb{R}^n)$                                                          |
| cardona-qft-methods_F01371 | 78   | ■ 0.997 | $m \geq k$                                                                                                        | $m \geq k$                                                                                                        | With a bit more work one can show the following result, cf. [1, Theorem 0.2].                                                                                                           |
| cardona-qft-methods_F01372 | 78   | ■ 0.997 | $\mathcal{D} : C_c^{\infty}(\mathbb{R}^n) \rightarrow C_c^{\infty}(\mathbb{R}^n)$                                 | $\mathcal{D} : C_c^{\infty}(\mathbb{R}^n) \rightarrow C_c^{\infty}(\mathbb{R}^n)$                                 | With a bit more work one can show the following result, cf. [1, Theorem 0.2].                                                                                                           |
| cardona-qft-methods_F01373 | 78   | ■ 1.000 | $\mathcal{D} : H_K^m(\mathbb{R}^n) \rightarrow$                                                                   | $\mathcal{D} : H_K^m(\mathbb{R}^n) \rightarrow$                                                                   | <b>Theorem 2.7. (Rellich’s lemma)</b> <i>If <math>K \subset \mathbb{R}^n</math> is a compact set, then the</i>                                                                          |
| cardona-qft-methods_F01374 | 78   | ■ 1.000 | $K \subset \mathbb{R}^n$                                                                                          | $K \subset \mathbb{R}^n$                                                                                          | <b>2.9. Matrix-valued differential operators</b>                                                                                                                                        |
| cardona-qft-methods_F01375 | 78   | ■ 1.000 | $H_K^m(\mathbb{R}^n) \hookrightarrow H^k(\mathbb{R}^n)$                                                           | $H_K^m(\mathbb{R}^n) \hookrightarrow H^k(\mathbb{R}^n)$                                                           | We extend Definition 2.2 to operators acting on vector valued functions on                                                                                                              |
| cardona-qft-methods_F01376 | 78   | ■ 1.000 | $m > k$                                                                                                           | $m > k$                                                                                                           | We extend Definition 2.2 to operators acting on vector valued functions on                                                                                                              |
| cardona-qft-methods_F01377 | 78   | ■ 1.000 | $k < m$                                                                                                           | $k < m$                                                                                                           | Using the scalar product of $\mathbb{R}^N$ we can generalize the constructions of                                                                                                       |
| cardona-qft-methods_F01378 | 78   | ■ 1.000 | $H_K^m(\mathbb{R}^n)$                                                                                             | $H_K^m(\mathbb{R}^n)$                                                                                             | Using the scalar product of $\mathbb{R}^N$ we can generalize the constructions of                                                                                                       |
| cardona-qft-methods_F01379 | 78   | ■ 1.000 | $H_K^m(\mathbb{R}^n) \rightarrow L^2(\mathbb{R}^n)$                                                               | $H_K^m(\mathbb{R}^n) \rightarrow L^2(\mathbb{R}^n)$                                                               | the previous                                                                                                                                                                            |
| cardona-qft-methods_F01380 | 78   | ■ 1.000 | $C^{\infty}(\mathbb{R}^n, \mathbb{R}^N)$                                                                          | $C^{\infty}(\mathbb{R}^n, \mathbb{R}^N)$                                                                          |                                                                                                                                                                                         |
| cardona-qft-methods_F01381 | 78   | ■ 1.000 | $\mathbb{R}^N$                                                                                                    | $\mathbb{R}^N$                                                                                                    |                                                                                                                                                                                         |
| cardona-qft-methods_F01382 | 78   | ■ 1.000 | $L^2(\mathbb{R}^n, \mathbb{R}^N), H^k(\mathbb{R}^n, \mathbb{R}^N)$                                                | $L^2(\mathbb{R}^n, \mathbb{R}^N), H^k(\mathbb{R}^n, \mathbb{R}^N)$                                                |                                                                                                                                                                                         |
| cardona-qft-methods_F01383 | 78   | ■ 1.000 | $H_K^k(\mathbb{R}^n, \mathbb{R}^N)$                                                                               | $H_K^k(\mathbb{R}^n, \mathbb{R}^N)$                                                                               |                                                                                                                                                                                         |
| cardona-qft-methods_F01384 | 79   | ■ 0.998 | $\mathcal{D} : C^{\infty}(\mathbb{R}^n, \mathbb{R}^{N_1}) \rightarrow C^{\infty}(\mathbb{R}^n, \mathbb{R}^{N_2})$ | $\mathcal{D} : C^{\infty}(\mathbb{R}^n, \mathbb{R}^{N_1}) \rightarrow C^{\infty}(\mathbb{R}^n, \mathbb{R}^{N_2})$ |                                                                                                                                                                                         |
| cardona-qft-methods_F01385 | 79   | ■ 1.000 | $\pi : E \rightarrow M$                                                                                           | $\pi : E \rightarrow M$                                                                                           |                                                                                                                                                                                         |
| cardona-qft-methods_F01386 | 79   | ■ 1.000 | $\pi^{-1}(x)$                                                                                                     | $\pi^{-1}(x)$                                                                                                     | <b>Example 2.11.</b> a. The trivial bundle $E = M \times \mathbb{C}$ . Here $\pi : E \rightarrow M$ is the projection on the first factor. In this case each fiber $E_x$ is canonically |

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| cardona-qft-methods_F01387 | 79   | ■ 1.000 | $E=M \times \mathcal{C}^{\mathrm{N}}$                               | $E = M \times \mathcal{C}^N$                                        |    |
| cardona-qft-methods_F01388 | 79   | ■ 1.000 | $\mathcal{C}^{\mathrm{N}}$                                          | $\mathcal{C}^N$                                                     |    |
| cardona-qft-methods_F01389 | 79   | ■ 1.000 | $s\colon M \rightarrow E$                                           | $s : M \rightarrow E$                                               |    |
| cardona-qft-methods_F01390 | 79   | ■ 1.000 | $s(x) \in E_x$                                                      | $s(x) \in E_x$                                                      |    |
| cardona-qft-methods_F01391 | 79   | ■ 1.000 | $M \rightarrow \mathcal{C}^{\mathrm{N}}$                            | $M \rightarrow \mathcal{C}^N$                                       |    |
| cardona-qft-methods_F01392 | 79   | ■ 1.000 | $C^{\infty}(M, E)$                                                  | $C^{\infty}(M, E)$                                                  |    |
| cardona-qft-methods_F01393 | 80   | ■ 1.000 | $f \in C^{\infty}(M, E)$                                            | $f \in C^{\infty}(M, E)$                                            |    |
| cardona-qft-methods_F01394 | 80   | ■ 1.000 | $\operatorname{supp}(\mathcal{D} f) \subset \operatorname{supp}(f)$ | $\operatorname{supp}(\mathcal{D} f) \subset \operatorname{supp}(f)$ |    |
| cardona-qft-methods_F01395 | 80   | ■ 1.000 | $\left\{U_i\right\}_{i=1}^m$                                        | $\{U_i\}_{i=1}^m$                                                   |    |
| cardona-qft-methods_F01396 | 80   | ■ 1.000 | $U_i$                                                               | $U_i$                                                               |    |
| cardona-qft-methods_F01397 | 80   | ■ 0.881 | $C^{\infty}(M, E) \rightarrow C^{\infty}(M, F)$                     | $C^{\infty}(M, E) \rightarrow C^{\infty}(M, F)$                     |    |
| cardona-qft-methods_F01398 | 80   | ■ 1.000 | $i=1, \ldots, m$                                                    | $i = 1, \dots, m$                                                   |    |
| cardona-qft-methods_F01399 | 80   | ■ 1.000 | $\mathcal{D}_i$                                                     | $\mathcal{D}_i$                                                     |    |
| cardona-qft-methods_F01400 | 80   | ■ 1.000 | $H^k(M, E)$                                                         | $H^k(M, E)$                                                         |  |
| cardona-qft-methods_F01401 | 80   | ■ 1.000 | $\langle \cdot, \cdot \rangle_k$                                    | $\langle \cdot, \cdot \rangle_k$                                    |  |
| cardona-qft-methods_F01402 | 80   | ■ 1.000 | $C_k^{\infty}(M, E)$                                                | $C_k^{\infty}(M, E)$                                                |  |
| cardona-qft-methods_F01403 | 80   | ■ 1.000 | $C_c^{\infty}(M, E)$                                                | $C_c^{\infty}(M, E)$                                                |  |
| cardona-qft-methods_F01404 | 80   | ■ 1.000 | $M=\mathbb{R}^n$                                                    | $M = \mathbb{R}^n$                                                  |  |
| cardona-qft-methods_F01405 | 80   | ■ 1.000 | $\mathcal{D}\colon C^{\infty}(M, E) \rightarrow C^{\infty}(M, F)$   | $\mathcal{D} : C^{\infty}(M, E) \rightarrow C^{\infty}(M, F)$       |  |
| cardona-qft-methods_F01406 | 81   | ■ 1.000 | $H_K^m(M, E) \rightarrow L^2(M, F)$                                 | $H_K^m(M, E) \rightarrow L^2(M, F)$                                 |  |
| cardona-qft-methods_F01407 | 81   | ■ 1.000 | $x_{\emptyset} \in M$                                               | $x_0 \in M$                                                         |  |
| cardona-qft-methods_F01408 | 81   | ■ 1.000 | $\xi \in T_{x_{\emptyset}}^* M$                                     | $\xi \in T_{x_0}^* M$                                               |  |
| cardona-qft-methods_F01409 | 81   | ■ 1.000 | $e \in$                                                             | $e \in$                                                             |  |
| cardona-qft-methods_F01410 | 81   | ■ 1.000 | $E_{x_{\emptyset}}$                                                 | $E_{x_0}$                                                           |  |
| cardona-qft-methods_F01411 | 81   | ■ 1.000 | $df_{x_{\emptyset}}=\xi$                                            | $df_{x_0} = \xi$                                                    |  |
| cardona-qft-methods_F01412 | 81   | ■ 1.000 | $s \in C^{\infty}(M, E)$                                            | $s \in C^{\infty}(M, E)$                                            |  |
| cardona-qft-methods_F01413 | 81   | ■ 1.000 | $s\left(x_{\emptyset}\right)=e$                                     | $s\left(x_0\right)=e$                                               |  |

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| cardona-qft-methods_F01414 | 81   | ■ 1.000 | $\sigma_L(\mathcal{D})(\mathcal{D})\left(x_0,\xi\right) e$                                      | $\sigma_L(\mathcal{D})(x_0,\xi) e$                                    |                                 |
| cardona-qft-methods_F01415 | 81   | ■ 1.000 | $e$                                                                                             | $e$                                                                   |                                 |
| cardona-qft-methods_F01416 | 81   | ■ 1.000 | $\sigma_L(\mathcal{D})(\mathcal{D})\left(x_0,\xi\right)$                                        | $\sigma_L(\mathcal{D})(x_0,\xi)$                                      |                                 |
| cardona-qft-methods_F01417 | 81   | ■ 1.000 | $E_{\{x_0\}} \rightarrow F_{\{x_0\}}$                                                           | $E_{x_0} \rightarrow F_{x_0}$                                         |                                 |
| cardona-qft-methods_F01418 | 81   | ■ 1.000 | $T_{\{x_0\}} M$                                                                                 | $T_{x_0} M$                                                           | hand side of (6) is independent of the choice of a coordinate system.                                              |
| cardona-qft-methods_F01419 | 81   | ■ 0.999 | $\sigma_L(\mathcal{D})(\mathcal{D})\left(x_0,\xi\right)\left(x_0\in M,\xi\in T_{x_0}^*M\right)$ | $\sigma_L(\mathcal{D})(x_0,\xi)(x_0\in M,\xi\in T_{x_0}^*M)$          | of the local coordinates.                                                                                          |
| cardona-qft-methods_F01420 | 81   | ■ 1.000 | $\operatorname{Hom}\left(E_x,F_x\right)$                                                        | $\operatorname{Hom}(E_x,F_x)$                                         |                                                                                                                    |
| cardona-qft-methods_F01421 | 82   | ■ 1.000 | $\pi\colon T^*M\rightarrow M$                                                                   | $\pi:T^*M\rightarrow M$                                               | Index theory for non-compact $G$ -manifolds 71                                                                     |
| cardona-qft-methods_F01422 | 82   | ■ 1.000 | $\pi^*E$                                                                                        | $\pi^*E$                                                              | 3.20. The bundle associated to a section of the pullback bundle.                                                   |
| cardona-qft-methods_F01423 | 82   | ■ 1.000 | $(x,\xi)\in T^*M$                                                                               | $(x,\xi)\in T^*M$                                                     | 3.20. The bundle associated to a section of the pullback bundle.                                                   |
| cardona-qft-methods_F01424 | 82   | ■ 1.000 | $\sigma_L(\mathcal{D})(x,\xi)$                                                                  | $\sigma_L(\mathcal{D})(x,\xi)$                                        | Let $\mathcal{D}:C^\infty(M,E)\rightarrow C^\infty(M,F)$ be a differential operator. Then f                        |
| cardona-qft-methods_F01425 | 82   | ■ 1.000 | $\sigma_L(\mathcal{D})$                                                                         | $\sigma_L(\mathcal{D})$                                               | Hence, we can view $\sigma_L(\mathcal{D})$ as a section of the vector bundle $\operatorname{Hom}(\pi^*E,\pi^*F)$ . |
| cardona-qft-methods_F01426 | 82   | ■ 1.000 | $\operatorname{Hom}\left(\pi^*E,\pi^*F\right)$                                                  | $\operatorname{Hom}(\pi^*E,\pi^*F)$                                   | Hence, we can view $\sigma_L(\mathcal{D})$ as a section of the vector bundle $\operatorname{Hom}(\pi^*E,\pi^*F)$ . |
| cardona-qft-methods_F01427 | 82   | ■ 1.000 | $\xi\neq 0$                                                                                     | $\xi\neq 0$                                                           | Example 3.20. Consider the Laplace operator                                                                        |
| cardona-qft-methods_F01428 | 82   | ■ 0.999 | $\sigma_L(\Delta)$                                                                              | $\sigma_L(\Delta)$                                                    | Elliptic operators play a very important role in analysis and                                                      |
| cardona-qft-methods_F01429 | 82   | ■ 0.999 | $\xi=\left(\xi_1,\ldots,\xi_n\right)\neq 0$                                                     | $\xi=(\xi_1,\dots,\xi_n)\neq 0$                                       | ferential operator.                                                                                                |
| cardona-qft-methods_F01430 | 82   | ■ 0.999 | $\mathcal{D} f=u$                                                                               | $\mathcal{D} f=u$                                                     |                                                                                                                    |
| cardona-qft-methods_F01431 | 82   | ■ 0.999 | $u\in C^\infty(M,F)$                                                                            | $u\in C^\infty(M,F)$                                                  |                                                                                                                    |
| cardona-qft-methods_F01432 | 82   | ■ 0.999 | $\operatorname{Ind}(\mathcal{D})$                                                               | $\operatorname{Ind}(\mathcal{D})$                                     |                                                                                                                    |
| cardona-qft-methods_F01433 | 82   | ■ 1.000 | $\sigma_L(\mathcal{D}_1)=\sigma_L(\mathcal{D}_2)$                                               | $\sigma_L(\mathcal{D}_1)=\sigma_L(\mathcal{D}_2)$                     |                                                                                                                    |
| cardona-qft-methods_F01434 | 82   | ■ 1.000 | $\operatorname{Ind}(\mathcal{D}_1)=\operatorname{Ind}(\mathcal{D}_2)$                           | $\operatorname{Ind}(\mathcal{D}_1)=\operatorname{Ind}(\mathcal{D}_2)$ |                                                                                                                    |
| cardona-qft-methods_F01435 | 83   | ■ 1.000 | $M=S^1$                                                                                         | $M=S^1$                                                               |                                                                                                                    |
| cardona-qft-methods_F01436 | 83   | ■ 1.000 | $\operatorname{SEl}(E,F)$                                                                       | $\operatorname{SEl}(E,F)$                                             | Theorem 3.2. Let $E$ and $F$ be vector bundles over a compact manifold $M$ .                                       |
| cardona-qft-methods_F01437 | 83   | ■ 1.000 | $\sigma(x,\xi)\in \operatorname{Hom}\left(\pi^*E,\pi^*F\right)$                                 | $\sigma(x,\xi)\in \operatorname{Hom}(\pi^*E,\pi^*F)$                  | Theorem 3.2. Let $E$ and $F$ be vector bundles over a compact manifold $M$ .                                       |
| cardona-qft-methods_F01438 | 83   | ■ 1.000 | $\{ \}^{14,15}$                                                                                 | 14,15                                                                 | The composition of the arrows in the diagram above leads to a map                                                  |
| cardona-qft-methods_F01439 | 83   | ■ 1.000 | $\{ \}^{6,7}$                                                                                   | 6,7                                                                   | Empic operators $\sigma_L(\mathcal{D})=\sigma_L(\mathcal{D}_1)=\sigma_L(\mathcal{D}_2)$                            |

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| cardona-qft-methods_F01440 | 83   | ■ 1.000 | $\{ \} ^{9,23}$                                                                                                                                                | 9,23                                                                                       |    |
| cardona-qft-methods_F01441 | 83   | ■ 0.942 | $\mathrm{t}\text{-}\operatorname{Ind}(\operatorname{\mathcal{D}})$                                                                                             | $\mathbf{t} - \operatorname{Ind}(\mathcal{D})$                                             |    |
| cardona-qft-methods_F01442 | 84   | ■ 1.000 | $\operatorname{Ind}(\operatorname{\mathcal{D}})\!=\!\mathrm{t}\text{-}\operatorname{Ind}(\operatorname{\sigma}_L(\operatorname{\mathcal{D}})\text{\rm right})$ | $\operatorname{Ind}(\mathcal{D}) = \mathbf{t} - \operatorname{Ind}(\sigma_L(\mathcal{D}))$ |    |
| cardona-qft-methods_F01443 | 84   | ■ 1.000 | $\operatorname{\mathcal{D}} \ f\!=\!0$                                                                                                                         | $\mathcal{D}f = 0$                                                                         |    |
| cardona-qft-methods_F01444 | 84   | ■ 1.000 | $\operatorname{\mathcal{D}}^* \ u\!=\!0$                                                                                                                       | $\mathcal{D}^*u = 0$                                                                       |    |
| cardona-qft-methods_F01445 | 84   | ■ 1.000 | $\sigma(x)\colon E_x \rightarrow F_x$                                                                                                                          | $\sigma(x) : E_x \rightarrow F_x$                                                          |    |
| cardona-qft-methods_F01446 | 84   | ■ 1.000 | $K \subset X$                                                                                                                                                  | $K \subset X$                                                                              |    |
| cardona-qft-methods_F01447 | 84   | ■ 1.000 | $\sigma_1\colon E_1 \rightarrow F_1$                                                                                                                           | $\sigma_1 : E_1 \rightarrow F_1$                                                           |    |
| cardona-qft-methods_F01448 | 84   | ■ 0.999 | $\sigma_2\colon E_2 \rightarrow F_2$                                                                                                                           | $\sigma_2 : E_2 \rightarrow F_2$                                                           |    |
| cardona-qft-methods_F01449 | 84   | ■ 0.999 | $k_1, \ k_2 \geq 0$                                                                                                                                            | $k_1, k_2 \geq 0$                                                                          |    |
| cardona-qft-methods_F01450 | 85   | ■ 1.000 | $\sigma_1 \sim \sigma_2$                                                                                                                                       | $\sigma_1 \sim \sigma_2$                                                                   |    |
| cardona-qft-methods_F01451 | 85   | ■ 1.000 | $K(X)$                                                                                                                                                         | $K(X)$                                                                                     |    |
| cardona-qft-methods_F01452 | 85   | ■ 1.000 | $\sigma_1, \sigma_2\colon E \rightarrow F$                                                                                                                     | $\sigma_1, \sigma_2 : E \rightarrow F$                                                     |    |
| cardona-qft-methods_F01453 | 85   | ■ 1.000 | $(E, \ F)$                                                                                                                                                     | $(E, F)$                                                                                   |  |
| cardona-qft-methods_F01454 | 85   | ■ 1.000 | $E\text{-}F$                                                                                                                                                   | $E - F$                                                                                    |  |
| cardona-qft-methods_F01455 | 85   | ■ 1.000 | $X\!=\!\{p \ t\}$                                                                                                                                              | $X = \{pt\}$                                                                               |  |
| cardona-qft-methods_F01456 | 85   | ■ 1.000 | $E_x \rightarrow F_x$                                                                                                                                          | $E_x \rightarrow F_x$                                                                      |  |
| cardona-qft-methods_F01457 | 85   | ■ 1.000 | $\sigma(\operatorname{\mathcal{D}})$                                                                                                                           | $\sigma(\mathcal{D})$                                                                      |  |
| cardona-qft-methods_F01458 | 86   | ■ 1.000 | $X\!=\!T^{\wedge\,*} \ M$                                                                                                                                      | $X = T^*M$                                                                                 |  |
| cardona-qft-methods_F01459 | 86   | ■ 1.000 | $K\!\left(T^{\wedge\,*} \ M\right)$                                                                                                                            | $K\left(T^*M\right)$                                                                       |  |
| cardona-qft-methods_F01460 | 86   | ■ 1.000 | $j\colon Y \hookrightarrow X$                                                                                                                                  | $j : Y \hookrightarrow X$                                                                  |  |
| cardona-qft-methods_F01461 | 86   | ■ 1.000 | $i\colon\{p \ t\} \hookrightarrow \operatorname{\mathcal{C}}^N$                                                                                                | $i : \{pt\} \hookrightarrow \mathcal{C}^N$                                                 |  |
| cardona-qft-methods_F01462 | 86   | ■ 1.000 | $M \hookrightarrow \operatorname{\mathbb{R}}^N$                                                                                                                | $M \hookrightarrow \mathbb{R}^N$                                                           |  |
| cardona-qft-methods_F01463 | 87   | ■ 0.971 | $i_!$                                                                                                                                                          | $i_!$                                                                                      |  |
| cardona-qft-methods_F01464 | 87   | ■ 1.000 | $j\colon M \hookrightarrow \operatorname{\mathbb{R}}^N$                                                                                                        | $j : M \hookrightarrow \mathbb{R}^N$                                                       |  |
| cardona-qft-methods_F01465 | 87   | ■ 1.000 | $g \in G$                                                                                                                                                      | $g \in G$                                                                                  |  |
| cardona-qft-methods_F01466 | 87   | ■ 1.000 | $\phi(g)\colon M \rightarrow M$                                                                                                                                | $\phi(g) : M \rightarrow M$                                                                |  |

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| cardona-qft-methods_F01467 | 87   | ■ 1.000 | $\phi\!\left(g_1\right)\circ\phi\!\left(g_2\right)=\phi\!\left(g_1\,g_2\right)$                                       | $\phi\left(g_1\right)\circ\phi\left(g_2\right)=\phi\left(g_1g_2\right)$                     | $\pi:E\rightarrow M$ together with an action                                                                                                                                                                                                                |
| cardona-qft-methods_F01468 | 87   | ■ 1.000 | $g\cdot x$                                                                                                            | $g\cdot x$                                                                                  | $g\mapsto l_g:C^\infty(M,E)\longrightarrow C^\infty(M,E),\quad l_g(f)(x):=g\cdot f(g^{-1}\cdot x). \quad (5)$                                                                                                                                               |
| cardona-qft-methods_F01469 | 87   | ■ 1.000 | $g\cdot e$                                                                                                            | $g\cdot e$                                                                                  | $g\mapsto l_g:C^\infty(M,E)\longrightarrow C^\infty(M,E),\quad l_g(f)(x):=g\cdot f(g^{-1}\cdot x). \quad (5)$                                                                                                                                               |
| cardona-qft-methods_F01470 | 87   | ■ 1.000 | $\phi(g)(x)$                                                                                                          | $\phi(g)(x)$                                                                                | Similarly, we define the action                                                                                                                                                                                                                             |
| cardona-qft-methods_F01471 | 87   | ■ 1.000 | $\psi(g)(e)$                                                                                                          | $\psi(g)(e)$                                                                                | Similarly, we define the action                                                                                                                                                                                                                             |
| cardona-qft-methods_F01472 | 87   | ■ 1.000 | $l_g\colon L^2(M,E)\rightarrow L^2(M,E)$                                                                              | $l_g:L^2(M,E)\rightarrow L^2(M,E)$                                                          |                                                                                                                                                                                                                                                             |
| cardona-qft-methods_F01473 | 87   | ■ 1.000 | $L^2(G)$                                                                                                              | $L^2(G)$                                                                                    |                                                                                                                                                                                                                                                             |
| cardona-qft-methods_F01474 | 87   | ■ 1.000 | $g\cdot\mathcal{D}=\mathcal{D}\cdot g$                                                                                | $g\cdot\mathcal{D}=\mathcal{D}\cdot g$                                                      |                                                                                                                                                                                                                                                             |
| cardona-qft-methods_F01475 | 88   | ■ 1.000 | $K_G(X)$                                                                                                              | $K_G(X)$                                                                                    | <b>2.17. Equivalent <math>K</math> theory</b>                                                                                                                                                                                                               |
| cardona-qft-methods_F01476 | 88   | ■ 1.000 | $M\subset T^*M$                                                                                                       | $M\subset T^*M$                                                                             | Suppose now that the manifold $M$ is not compact. The topological index                                                                                                                                                                                     |
| cardona-qft-methods_F01477 | 88   | ■ 1.000 | $\sigma_L\in$                                                                                                         | $\sigma_L\in$                                                                               | Suppose now that the manifold $M$ is not compact. The topological index                                                                                                                                                                                     |
| cardona-qft-methods_F01478 | 88   | ■ 0.612 | $K_G\!\left(T^*M\right)$                                                                                              | $K_G\left(T^*M\right)$                                                                      | $\mathrm{map}\left(\star\right)$ is surjective, however the symbol of an empty operator $\mathcal{D}$ does not define an element of $K(T^*M)$ , since it is not invertible on the non-                                                                      |
| cardona-qft-methods_F01479 | 88   | ■ 0.612 | $\mathrm{t}-\mathrm{Ind}\!\left(\sigma_L\!\left(\mathcal{D}\right)\right)$                                            | $\mathfrak{t}-\mathrm{Ind}\left(\sigma_L(\mathcal{D})\right)$                               | $\mathrm{map}\left(\star\right)$ is surjective, however the symbol of an empty operator $\mathcal{D}$ does not define an element of $K(T^*M)$ , since it is not invertible on the non-                                                                      |
| cardona-qft-methods_F01480 | 88   | ■ 1.000 | $\mathrm{Ind}_G\!\left(\mathcal{D}\right)=\mathrm{t}-\mathrm{Ind}_G\!\left(\sigma_L\!\left(\mathcal{D}\right)\right)$ | $\mathrm{Ind}_G(\mathcal{D})=\mathfrak{t}-\mathrm{Ind}_G\left(\sigma_L(\mathcal{D})\right)$ |                                                                                                                                                                                                                                                             |
| cardona-qft-methods_F01481 | 89   | ■ 1.000 | $\sigma\colon\pi^*E\rightarrow\pi^*F$                                                                                 | $\sigma:\pi^*E\rightarrow\pi^*F$                                                            | <b>Question 1</b>                                                                                                                                                                                                                                           |
| cardona-qft-methods_F01482 | 89   | ■ 0.920 | $C^{\infty}(N)\rightarrow C^{\infty}(N)$                                                                              | $C^\infty(N)\rightarrow C^\infty(N)$                                                        | $g_1\cdot(x,g_2):=(x,g_1g_2),\quad x\in N,\;g_1,g_2\in G.$                                                                                                                                                                                                  |
| cardona-qft-methods_F01483 | 89   | ■ 1.000 | $\left(x_1,\ldots,x_n\right)$                                                                                         | $(x_1,\ldots,x_n)$                                                                          | $\{x_1,\dots,x_n\}$                                                                                                                                                                                                                                         |
| cardona-qft-methods_F01484 | 89   | ■ 0.558 | $\left(y_1,\cdots,y_r\right)$                                                                                         | $(y_1,\cdots,y_r)$                                                                          |                                                                                                                                                                                                                                                             |
| cardona-qft-methods_F01485 | 89   | ■ 0.558 | $\left(x_1,\ldots,x_n,y_1,\ldots,y_r\right)$                                                                          | $(x_1,\ldots,x_n,y_1,\ldots,y_r)$                                                           |                                                                                                                                                                                                                                                             |
| cardona-qft-methods_F01486 | 89   | ■ 1.000 | $\tilde{\mathcal{D}}$                                                                                                 | $\tilde{\mathcal{D}}$                                                                       |                                                                                                                                                                                                                                                             |
| cardona-qft-methods_F01487 | 90   | ■ 1.000 | $\mathrm{Ker}\!\left(\tilde{\mathcal{D}}\right)$                                                                      | $\mathrm{Ker}(\tilde{\mathcal{D}})$                                                         | Notice that $\pi_{\mathfrak{m}}(\tilde{\mathcal{D}})=\pi_{\mathfrak{m}}(\mathcal{D})\circ\tau_2(\mathcal{D})=\mathcal{O}_{1,\mathfrak{m}}(\tilde{\mathcal{D}})=\mathcal{O}_{1,\mathfrak{m}}(\mathcal{D})\circ\tau_2(\mathcal{D})=\mathcal{O}(\mathfrak{m})$ |
| cardona-qft-methods_F01488 | 90   | ■ 1.000 | $\mathrm{Coker}\!\left(\tilde{\mathcal{D}}\right)$                                                                    | $\mathrm{Coker}(\tilde{\mathcal{D}})$                                                       | Notice that $\pi_{\mathfrak{m}}(\tilde{\mathcal{D}})=\pi_{\mathfrak{m}}(\mathcal{D})\circ\tau_2(\mathcal{D})=\mathcal{O}_{1,\mathfrak{m}}(\tilde{\mathcal{D}})=\mathcal{O}_{1,\mathfrak{m}}(\mathcal{D})\circ\tau_2(\mathcal{D})=\mathcal{O}(\mathfrak{m})$ |
| cardona-qft-methods_F01489 | 90   | ■ 1.000 | $\dim V$                                                                                                              | $\dim V$                                                                                    | Equations (1) i                                                                                                                                                                                                                                             |
| cardona-qft-methods_F01490 | 90   | ■ 1.000 | $\mathrm{Ind}\!\left(\tilde{\mathcal{D}}\right)$                                                                      | $\mathrm{Ind}(\tilde{\mathcal{D}})$                                                         | or $\mathcal{D}$ is that although this operator is not empty on $\mathfrak{m}$ , it is empty on the quotient $N=M/G$ . The purpose of this section is to define the analogue                                                                                |
| cardona-qft-methods_F01491 | 90   | ■ 1.000 | $\widehat{R}(G)$                                                                                                      | $\widehat{R}(G)$                                                                            | or $\mathcal{D}$ is that although this operator is not empty on $\mathfrak{m}$ , it is empty on the quotient $N=M/G$ . The purpose of this section is to define the analogue                                                                                |
| cardona-qft-methods_F01492 | 90   | ■ 1.000 | $N=M\,/\,G$                                                                                                           | $N=M/G$                                                                                     | Suppose a compact group $G$ acts on a smooth manifold $M$ . Recall that an                                                                                                                                                                                  |

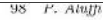
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| cardona-qft-methods_F01493 | 90   | ■ 1.000 | $\xi \in T^*M$                                       | $\xi \in T^*M$                                       |                                                                                                                                            |
| cardona-qft-methods_F01494 | 90   | ■ 0.994 | $v \in T\mathcal{O}(x)$                              | $v \in T\mathcal{O}(x)$                              |                                                                                                                                            |
| cardona-qft-methods_F01495 | 90   | ■ 0.994 | $\xi(v)=0$                                           | $\xi(v)=0$                                           |                                                                                                                                            |
| cardona-qft-methods_F01496 | 90   | ■ 1.000 | $T_{G^*}^*M$                                         | $T_G^*M$                                             |                                                                                                                                            |
| cardona-qft-methods_F01497 | 90   | ■ 1.000 | $T_{\{G, x\}^*}^*M:=T_{G^*}^*M \cap T_x^*M$          | $T_{G,x}^*M:=T_G^*M \cap T_x^*M$                     |                                                                                                                                            |
| cardona-qft-methods_F01498 | 91   | ■ 1.000 | $0 \neq \xi \in T_G^*M$                              | $0 \neq \xi \in T_G^*M$                              | for every non-zero $0 \neq \xi \in T_G^*M$                                                                                                 |
| cardona-qft-methods_F01499 | 91   | ■ 0.999 | $\operatorname{Ker}(\mathcal{D})$                    | $\operatorname{Ker}(\mathcal{D})$                    |                                                         |
| cardona-qft-methods_F01500 | 91   | ■ 0.999 | $\operatorname{Coker}(\mathcal{D})$                  | $\operatorname{Coker}(\mathcal{D})$                  |                                                         |
| cardona-qft-methods_F01501 | 91   | ■ 1.000 | $m_V^{\pm} \in$                                      | $m_V^{\pm} \in$                                      | $\operatorname{Ind}_G(\mathcal{D}) := \bigoplus_{V \in \operatorname{Irr}(G)} (m_V^+ - m_V^-) V \in R(G), \quad (6)$                       |
| cardona-qft-methods_F01502 | 91   | ■ 1.000 | $\mathbb{Z}_{\geq 0}$                                | $\mathbb{Z}_{\geq 0}$                                | where                                                                                                                                      |
| cardona-qft-methods_F01503 | 91   | ■ 1.000 | $m_V^{\pm} \in \mathbb{Z}_{\geq 0}$                  | $m_V^{\pm} \in \mathbb{Z}_{\geq 0}$                  | numbers $\operatorname{Ind}_{G,x}^{\pm} M$ might depend on $x$ . But it is a topological space, and one can                                |
| cardona-qft-methods_F01504 | 91   | ■ 1.000 | $T_{\{G, x\}^*}^*M$                                  | $T_{G,x}^*M$                                         |                                                                                                                                            |
| cardona-qft-methods_F01505 | 92   | ■ 1.000 | $K_G(T_G^*M)$                                        | $K_G(T_G^*M)$                                        |                                                                                                                                            |
| cardona-qft-methods_F01506 | 92   | ■ 1.000 | $\sigma \in K_G(T_G^*M)$                             | $\sigma \in K_G(T_G^*M)$                             | is still defined, but the image lies not in the ring of characters and the defi-                                                           |
| cardona-qft-methods_F01507 | 92   | ■ 1.000 | $\sigma_L(D)=\sigma$                                 | $\sigma_L(D)=\sigma$                                 | nition is more involved. In fact, for compact manifold $M$ , for $\sigma \in K_G(T_G^*M)$                                                  |
| cardona-qft-methods_F01508 | 93   | ■ 1.000 | $\mathbb{R}$                                         | $\mathbb{R}$                                         | §2 <i>M. Braverman and L. Cono</i>                                                                                                         |
| cardona-qft-methods_F01509 | 93   | ■ 1.000 | $W$                                                  | $W$                                                  | Let $V$ be a finite dimensional vector space over $\mathbb{K}$ endowed with a scalar product $\langle \cdot, \cdot \rangle$ .              |
| cardona-qft-methods_F01510 | 93   | ■ 1.000 | $v \in V$                                            | $v \in V$                                            | such that for any $v \in V$ we have                                                                                                        |
| cardona-qft-methods_F01511 | 93   | ■ 0.994 | $c: V \rightarrow \operatorname{End}(W)$             | $c: V \rightarrow \operatorname{End}(W)$             | if and only if                                                                                                                             |
| cardona-qft-methods_F01512 | 93   | ■ 1.000 | $\mathbb{R}^3$                                       | $\mathbb{R}^3$                                       | be the exterior algebra of $V$ . For $v$ exterior multiplication by $v$                                                                    |
| cardona-qft-methods_F01513 | 93   | ■ 1.000 | $\mathcal{C}^2$                                      | $\mathcal{C}^2$                                      | be the exterior algebra of $V$ . For $v$ exterior multiplication by $v$                                                                    |
| cardona-qft-methods_F01514 | 93   | ■ 1.000 | $V^{\mathcal{C}}=V \otimes_{\mathbb{R}} \mathcal{C}$ | $V^{\mathcal{C}}=V \otimes_{\mathbb{R}} \mathcal{C}$ | Using the scalar product on $V$ , we identify $v$ with an element $V^*$ and denote by $\iota: W \rightarrow W$ the interior multi-         |
| cardona-qft-methods_F01515 | 93   | ■ 1.000 | $V^{\mathcal{C}}$                                    | $V^{\mathcal{C}}$                                    | show that the map $\mathcal{C} \ni c \mapsto \mathcal{C}(c)$ defines a $\mathbb{C}$ -linear action of $\mathcal{C}$ on $V^{\mathcal{C}}$ . |
| cardona-qft-methods_F01516 | 93   | ■ 1.000 | $\varepsilon(v): W \rightarrow W$                    | $\varepsilon(v): W \rightarrow W$                    | show that the map $\mathcal{C} \ni c \mapsto \mathcal{C}(c)$ defines a $\mathbb{C}$ -linear action of $\mathcal{C}$ on $V^{\mathcal{C}}$ . |
| cardona-qft-methods_F01517 | 93   | ■ 1.000 | $v$                                                  | $v$                                                  |                                                                                                                                            |
| cardona-qft-methods_F01518 | 93   | ■ 1.000 | $V^*$                                                | $V^*$                                                |                                                                                                                                            |

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| cardona-qft-methods_ F01519 | 93   | ■ 1.000 | $\iota_v: W \rightarrow W$                          | $\iota_v: W \rightarrow W$                          |                                                                                                                                                                                       |
| cardona-qft-methods_ F01520 | 93   | ■ 1.000 | $c: v \mapsto c(v)$                                 | $c: v \mapsto c(v)$                                 |                                                                                                                                                                                       |
| cardona-qft-methods_ F01521 | 94   | ■ 1.000 | $E \rightarrow$                                     | $E \rightarrow$                                     | $T_x^*M$ are endowed with a scalar product.                                                                                                                                           |
| cardona-qft-methods_ F01522 | 94   | ■ 0.999 | $v \in T^*M$                                        | $v \in T^*M$                                        | such that for any $v \in T^*M$ we have $c(v)^2 = - v ^2 \text{Id}$ .                                                                                                                  |
| cardona-qft-methods_ F01523 | 94   | ■ 0.999 | $c(v)^2 = - v ^2 \mathrm{Id}$                       | $c(v)^2 = - v ^2 \text{Id}$                         | such that for any $v \in T^*M$ we have $c(v)^2 = - v ^2 \text{Id}$ .                                                                                                                  |
| cardona-qft-methods_ F01524 | 94   | ■ 0.999 | $M = \mathbb{R}^3$                                  | $M = \mathbb{R}^3$                                  | <b>Example 5.9.</b> Let $M$ be a Riemannian manifold and let $E = \Lambda^*(T^*M \otimes \mathcal{C})$ .                                                                              |
| cardona-qft-methods_ F01525 | 94   | ■ 0.999 | $E = \mathbb{R}^3 \times \mathcal{C}^2$             | $E = \mathbb{R}^3 \times \mathcal{C}^2$             | <b>Example 5.9.</b> Let $M$ be a Riemannian manifold and let $E = \Lambda^*(T^*M \otimes \mathcal{C})$ .                                                                              |
| cardona-qft-methods_ F01526 | 94   | ■ 1.000 | $E = \Lambda^*(T^*M \otimes \mathcal{C})$           | $E = \Lambda^*(T^*M \otimes \mathcal{C})$           |                                                                                                                                                                                       |
| cardona-qft-methods_ F01527 | 94   | ■ 1.000 | $\Omega^*(M)$                                       | $\Omega^*(M)$                                       | <b>5.10. A Clifford connection</b>                                                                                                                                                    |
| cardona-qft-methods_ F01528 | 94   | ■ 1.000 | $\omega \in \Omega^*(M)$                            | $\omega \in \Omega^*(M)$                            | <i>Levi-Civita connection and denoted by <math>\nabla</math>, i.e.</i>                                                                                                                |
| cardona-qft-methods_ F01529 | 94   | ■ 0.993 | $\nabla: C^\infty(M, E) \rightarrow \Omega^1(M, E)$ | $\nabla: C^\infty(M, E) \rightarrow \Omega^1(M, E)$ |                                                                                                                                                                                       |
| cardona-qft-methods_ F01530 | 94   | ■ 1.000 | $u \in TM, v \in T^*M, s \in C^\infty(M, E)$        | $u \in TM, v \in T^*M, s \in C^\infty(M, E)$        |                                                                                                                                                                                       |
| cardona-qft-methods_ F01531 | 95   | ■ 0.949 | $e_1, \dots, e_n$                                   | $e_1, \dots, e_n$                                   | where $e_1, \dots, e_n$ is an orthonormal basis of $TM$ . <sup>c</sup>                                                                                                                |
| cardona-qft-methods_ F01532 | 95   | ■ 0.949 | $TM^c$                                              | $TM^c$                                              | where $e_1, \dots, e_n$ is an orthonormal basis of $TM$ . <sup>c</sup>                                                                                                                |
| cardona-qft-methods_ F01533 | 95   | ■ 1.000 | $d: \Omega^*(M, E) \rightarrow$                     | $d: \Omega^*(M, E) \rightarrow$                     |                                                                                                                                                                                       |
| cardona-qft-methods_ F01534 | 95   | ■ 1.000 | $\Omega^{*+1}(M, E)$                                | $\Omega^{*+1}(M, E)$                                |                                                                                                                                                                                       |
| cardona-qft-methods_ F01535 | 95   | ■ 0.997 | $\Omega^*(M, E)$                                    | $\Omega^*(M, E)$                                    |                                                                                                                                                                                       |
| cardona-qft-methods_ F01536 | 95   | ■ 1.000 | $\{ \}^{\mathrm{c}}$                                | $^c$                                                |                                                                                                                                                                                       |
| cardona-qft-methods_ F01537 | 97   | ■ 1.000 | $C^\infty(M, E^+)$                                  | $C^\infty(M, E^+)$                                  |                                                                                                                                                                                       |
| cardona-qft-methods_ F01538 | 97   | ■ 0.950 | $C^\infty(M, E^-)$                                  | $C^\infty(M, E^-)$                                  |                                                                                                                                                                                       |
| cardona-qft-methods_ F01539 | 97   | ■ 0.950 | $\mathcal{D}^+$                                     | $\mathcal{D}^+$                                     |                                                                                                                                                                                       |
| cardona-qft-methods_ F01540 | 97   | ■ 0.950 | $\mathcal{D}^-$                                     | $\mathcal{D}^-$                                     |                                                                                                                                                                                       |
| cardona-qft-methods_ F01541 | 97   | ■ 1.000 | $\mathcal{D} = \mathcal{D}^+ \oplus \mathcal{D}^-$  | $\mathcal{D} = \mathcal{D}^+ \oplus \mathcal{D}^-$  |                                                                                                                                                                                       |
| cardona-qft-methods_ F01542 | 97   | ■ 1.000 | $\mathcal{D}^+: C^\infty(M, E^+) \rightarrow$       | $\mathcal{D}^+: C^\infty(M, E^+) \rightarrow$       | is presented in the form of the so-called <i>graded Dirac operator</i> : in such theory one usually considers the index of the operator $\mathcal{D}^+: C^\infty(M, E^+) \rightarrow$ |
| cardona-qft-methods_ F01543 | 97   | ■ 0.827 | $C^\infty(M, E^-)$                                  | $C^\infty(M, E^-)$                                  | $C^c$                                                                                                                                                                                 |
| cardona-qft-methods_ F01544 | 97   | ■ 1.000 | $E = \Lambda^*(T^*M \otimes$                        | $E = \Lambda^*(T^*M \otimes$                        | $\mathcal{C})$ . Set                                                                                                                                                                  |
| cardona-qft-methods_ F01545 | 97   | ■ 0.997 | $\mathcal{C}$                                       | $\mathcal{C}$                                       |                                                                                                                                                                                       |

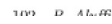
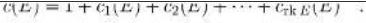
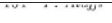
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| cardona-qft-methods_F01546 | 97   | ■ 0.998 | $\operatorname{Ind}\!\left(\mathcal{D}^+\right)$                    | $\operatorname{Ind}\left(\mathcal{D}^+\right)$    | <b>Example 5.24.</b> Suppose the dimension of $M$ is even, $\dim M = n = 2l$ .                                                                                                                                |
| cardona-qft-methods_F01547 | 97   | ■ 1.000 | $\operatorname{dim} M=n=2\,\mathfrak{l}$                            | $\dim M = n = 2l$                                 | $\{1,-1\}$ . Let $E^+$ (respectively, $E^-$ ) denote the eigenspace of $\Gamma$ correspond-                                                                                                                   |
| cardona-qft-methods_F01548 | 97   | ■ 1.000 | $\ast\colon E\rightarrowtail E$                                     | $\ast:E\rightarrow E$                             | Then $E = E^+ \oplus E^-$ and the Levi-Civita connection preserves this grad-                                                                                                                                 |
| cardona-qft-methods_F01549 | 97   | ■ 1.000 | $\Gamma\colon E\rightarrowtail E$                                   | $\Gamma:E\rightarrow E$                           | ing, hence, $(10)$ becomes a gr                                                                                                                                                                               |
| cardona-qft-methods_F01550 | 97   | ■ 0.998 | $\Gamma^2=1$                                                        | $\Gamma^2 = 1$                                    |                                                                                                                                                                                                               |
| cardona-qft-methods_F01551 | 97   | ■ 1.000 | $\{1,-1\}$                                                          | $\{1,-1\}$                                        |                                                                                                                                                                                                               |
| cardona-qft-methods_F01552 | 97   | ■ 1.000 | $E^{\{+\}}$                                                         | $E^+$                                             |                                                                                                                                                                                                               |
| cardona-qft-methods_F01553 | 97   | ■ 1.000 | $E^{\{-}}$                                                          | $E^-$                                             |                                                                                                                                                                                                               |
| cardona-qft-methods_F01554 | 98   | ■ 0.925 | $\mathcal{D}^{\pm}$                                                 | $\mathcal{D}^\pm$                                 |                                                                                                                                                                                                               |
| cardona-qft-methods_F01555 | 99   | ■ 0.963 | $\mathfrak{g}=\operatorname{Lie} S^1$                               | $\mathfrak{g} = \operatorname{Lie} S^1$           | $G = S^1 = \{z \in \mathcal{C} :  z  = 1\}$                                                                                                                                                                   |
| cardona-qft-methods_F01556 | 99   | ■ 1.000 | $\mathbf{v}\colon M\rightarrowtail$                                 | $\mathbf{v}:M\rightarrow$                         | be the circle group. Then the Lie algebra $\mathfrak{g} = \operatorname{Lie} S^1$ is naturally isomorphic                                                                                                     |
| cardona-qft-methods_F01557 | 99   | ■ 0.998 | $\mathbf{v}(x)\equivtail 1$                                         | $\mathbf{v}(x) \equiv 1$                          | to as, suppose $G$ acts on a complex Riemannian manifold $(M, g)$ . Let $\mathbf{v} : M \rightarrow \mathfrak{g}$ be the constant map $\mathbf{v}(x) \equiv 1$ . The corresponding vector field $v$ is called |
| cardona-qft-methods_F01558 | 99   | ■ 1.000 | $G=S^1$                                                             | $G = S^1$                                         | the <i>generating vector field</i> for the action of $G = S^1$ , since it completely                                                                                                                          |
| cardona-qft-methods_F01559 | 99   | ■ 1.000 | $M=\mathcal{C}$                                                     | $M = \mathcal{C}$                                 | Compute the generating vector field $v(x)$ .                                                                                                                                                                  |
| cardona-qft-methods_F01560 | 99   | ■ 1.000 | $\left(e^{\mathfrak{i}\,t},z\right)\mapsto e^{\mathfrak{i}\,t}z$    | $\left(e^{it},z\right)\mapsto e^{it}z$            | Compute the generating vector field $v(x)$ .                                                                                                                                                                  |
| cardona-qft-methods_F01561 | 99   | ■ 1.000 | $v(\mathfrak{x})$                                                   | $v(x)$                                            | <b>6.4. Tamed <math>G</math>-manifolds</b>                                                                                                                                                                    |
| cardona-qft-methods_F01562 | 99   | ■ 0.885 | $(M,\mathbf{v})$                                                    | $(M,\mathbf{v})$                                  | transversal cotangent bundle to $M$ , cf. Definition 4.4, and let $\pi:T_G^*M\rightarrowtail M$                                                                                                               |
| cardona-qft-methods_F01563 | 99   | ■ 0.885 | $\mathbf{v}$                                                        | $\mathbf{v}$                                      | transversal cotangent bundle to $M$ , cf. Definition 4.4, and let $\pi:T_G^*M\rightarrowtail M$                                                                                                               |
| cardona-qft-methods_F01564 | 99   | ■ 1.000 | $\pi\colon T_G^*M\rightarrowtail M$                                 | $\pi:T_G^*M\rightarrow M$                         |                                                                                                                                                                                                               |
| cardona-qft-methods_F01565 | 99   | ■ 1.000 | $\pi^*E=\pi^*E^+\oplus\pi^*E^-$                                     | $\pi^*E = \pi^*E^+ \oplus \pi^*E^-$               |                                                                                                                                                                                                               |
| cardona-qft-methods_F01566 | 99   | ■ 1.000 | $x\in M,\,\xi\in T^*M$                                              | $x\in M,\xi\in T^*M$                              |                                                                                                                                                                                                               |
| cardona-qft-methods_F01567 | 100  | ■ 1.000 | $(x,\xi)\in T_G^*M$                                                 | $(x,\xi)\in T_G^*M$                               |                                                                                                                                                                                                               |
| cardona-qft-methods_F01568 | 100  | ■ 1.000 | $c(\xi+v(x))$                                                       | $c(\xi+v(x))$                                     | The collection of these maps for all $(x,\xi)\in T_G^*M$ defines a bundle map                                                                                                                                 |
| cardona-qft-methods_F01569 | 100  | ■ 1.000 | $(x,\xi)\in$                                                        | $(x,\xi)\in$                                      | The collection of these maps for all $(x,\xi)\in T_G^*M$ defines a bundle map                                                                                                                                 |
| cardona-qft-methods_F01570 | 100  | ■ 1.000 | $v(x)\neq 0,\,\xi\neq 0$                                            | $v(x)\neq 0,\xi\neq 0$                            | $c(\xi+v):\pi^*E^+\rightarrowtail\pi^*E^-$ .                                                                                                                                                                  |
| cardona-qft-methods_F01571 | 100  | ■ 0.572 | $(E,\mathbf{v})$                                                    | $(E,\mathbf{v})$                                  | over a tamed $G$ -manifold $(M,\mathbf{v})$ , we refer to the pair $(E,\mathbf{v})$ as a <i>tamed Clifford bundle</i> over $M$ . The topological index $\mathfrak{t}\text{-Ind}_G(E,\mathbf{v})$ of a tamed   |
| cardona-qft-methods_F01572 | 100  | ■ 1.000 | $\mathrm{t}\text{-}\operatorname{Ind}_{\mathfrak{G}}(E,\mathbf{v})$ | $\mathfrak{t}-\operatorname{Ind}_G(E,\mathbf{v})$ | Clifford bundle is defined by                                                                                                                                                                                 |

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| cardona-qft-methods_F01573 | 100  | ■ 1.000 | $\{ \}^{24,25}$                              | $_{24,25}$                                   | 6.9. The Dirac operator                                                                                                             |
| cardona-qft-methods_F01574 | 100  | ■ 1.000 | $c(\xi)$                                     | $c(\xi)$                                     | positive $G$ -invariant function. Roughly speaking, we demand that $f(x)v(x)$                                                       |
| cardona-qft-methods_F01575 | 100  | ■ 0.991 | $\boldsymbol{v}$                             | $\boldsymbol{v}$                             | about to define turn                                                                                                                |
| cardona-qft-methods_F01576 | 100  | ■ 1.000 | $f(x) \ v(x)$                                | $f(x)v(x)$                                   |                                                                                                                                     |
| cardona-qft-methods_F01577 | 100  | ■ 1.000 | $f\colon M \rightarrow [0, \infty)$          | $f : M \rightarrow [0, \infty)$              |                                                                                                                                     |
| cardona-qft-methods_F01578 | 100  | ■ 1.000 | $M, \ E$                                     | $M, E$                                       |                                                                                                                                     |
| cardona-qft-methods_F01579 | 100  | ■ 1.000 | $\nabla$                                     | $\nabla$                                     |                                                                                                                                     |
| cardona-qft-methods_F01580 | 100  | ■ 0.817 | $M, \ E, \ \nabla, \ \mathbf{v}$             | $M, E, \nabla, \mathbf{v}$                   |                                                                                                                                     |
| cardona-qft-methods_F01581 | 101  | ■ 0.994 | $(M, \ E, \ \nabla, \ \mathbf{v})$           | $(M, E, \nabla, \mathbf{v})$                 | 90 M. Braverman and L. Cano                                                                                                         |
| cardona-qft-methods_F01582 | 101  | ■ 1.000 | $\mathcal{D}_{fv}^+$                         | $\mathcal{D}_{fv}^+$                         | $\mathcal{D}_{f\vartheta} = \mathcal{D} + ic(fv), \tag{6}$                                                                          |
| cardona-qft-methods_F01583 | 101  | ■ 1.000 | $\mathcal{D}_{fv}$                           | $\mathcal{D}_{fv}$                           | $\mathcal{D}_{f\vartheta} = \mathcal{D} + ic(fv), \tag{6}$                                                                          |
| cardona-qft-methods_F01584 | 101  | ■ 1.000 | $\operatorname{Ker}(\mathcal{D}_{fv}^+)$     | $\operatorname{Ker}(\mathcal{D}_{fv}^+)$     | $G$ acts on $\operatorname{Ker}(\mathcal{D}_{f\vartheta})$ and $\operatorname{Coker}(\mathcal{D}_{f\vartheta})$ .                   |
| cardona-qft-methods_F01585 | 101  | ■ 1.000 | $\operatorname{Coker}(\mathcal{D}_{fv}^+)$   | $\operatorname{Coker}(\mathcal{D}_{fv}^+)$   | $G$ acts on $\operatorname{Ker}(\mathcal{D}_{f\vartheta})$ and $\operatorname{Coker}(\mathcal{D}_{f\vartheta})$ .                   |
| cardona-qft-methods_F01586 | 101  | ■ 1.000 | $\mathcal{D}_{fv}^+$                         | $\mathcal{D}_{fv}^+$                         | where $\mathcal{D}_{fv}^+$ is the restriction of $\mathcal{D}_{fv}$ to $\mathcal{D}_{fv}^+$ .                                       |
| cardona-qft-methods_F01587 | 101  | ■ 1.000 | $m_V^{\pm}$                                  | $m_V^{\pm}$                                  | $\operatorname{Ind}_G \mathcal{D}^+ = \sum_V m_V^+ V, \operatorname{Coker} \mathcal{D}^+ = \sum_V m_V^- V. \tag{7}$                 |
| cardona-qft-methods_F01588 | 101  | ■ 1.000 | $\operatorname{Ker} \mathcal{D}_{fv}^{\pm}$  | $\operatorname{Ker} \mathcal{D}_{fv}^{\pm}$  | $\operatorname{Ind}_G(\mathcal{D}_{fv}^+) = \sum_{V \in \operatorname{Irr}(G)} (m_V^+ - m_V^-) \cdot V \in \widehat{R}(G). \tag{8}$ |
| cardona-qft-methods_F01589 | 101  | ■ 1.000 | $V \in \operatorname{Irr}(G)$                | $V \in \operatorname{Irr}(G)$                | $\operatorname{Ind}_G(\mathcal{D}, \mathbf{v}) := \operatorname{Ind}_G(\mathcal{D}_{fv}^+), \tag{9}$                                |
| cardona-qft-methods_F01590 | 101  | ■ 1.000 | $m_V^+ - m_V^-(V \in \operatorname{Irr}(G))$ | $m_V^+ - m_V^-(V \in \operatorname{Irr}(G))$ | where $f$ is an admissible function for $(M, \mathbf{v})$ .                                                                         |
| cardona-qft-methods_F01591 | 101  | ■ 0.611 | $(\mathcal{D}, \mathbf{v})$                  | $(\mathcal{D}, \mathbf{v})$                  |                                                                                                                                     |
| cardona-qft-methods_F01592 | 102  | ■ 0.666 | $M, \ \mathbf{v}$                            | $M, \mathbf{v}$                              |                                                                                                                                     |
| cardona-qft-methods_F01593 | 102  | ■ 1.000 | $\sigma_1$                                   | $\sigma_1$                                   | where $\sigma_1$ and $\sigma_2$ are the first two Pauli matrices, cf. (2).                                                          |
| cardona-qft-methods_F01594 | 102  | ■ 1.000 | $\sigma_2$                                   | $\sigma_2$                                   | where $\sigma_1$ and $\sigma_2$ are the first two Pauli matrices, cf. (2).                                                          |
| cardona-qft-methods_F01595 | 102  | ■ 1.000 | $E^{\pm}$                                    | $E^{\pm}$                                    | The $G$ action on $M$ lifts to $E^{\pm}$ so that                                                                                    |
| cardona-qft-methods_F01596 | 102  | ■ 1.000 | $f \equiv 1$                                 | $f \equiv 1$                                 | d. Compute $\operatorname{Ind}_G(\mathcal{D}, \mathbf{v})$ .                                                                        |
| cardona-qft-methods_F01597 | 102  | ■ 1.000 | $\mathcal{D}_v$                              | $\mathcal{D}_v$                              |                                                                                                                                     |
| cardona-qft-methods_F01598 | 102  | ■ 1.000 | $\mathcal{D}_v^2$                            | $\mathcal{D}_v^2$                            |                                                                                                                                     |

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| cardona-qft-methods_F01599 | 102  | 1.000 | $\operatorname{Ind}_{\mathcal{G}}(\mathcal{D}, \mathbf{v})$                                        | $\operatorname{Ind}_G(\mathcal{D}, \mathbf{v})$ | We are now ready to for<br>inday theorem for tamed                                                                                                                                                                                                               |
| cardona-qft-methods_F01600 | 103  | 0.854 | $\Sigma \subset M$                                                                                 | $\Sigma \subset M$                              | Let $(M, \mathbf{v})$ be a tamed $G$ -manifold. Suppose $\Sigma \subset M$ is a smooth $G$ -invariant                                                                                                                                                            |
| cardona-qft-methods_F01601 | 103  | 1.000 | $M \backslash \Sigma$                                                                              | $M \backslash \Sigma$                           | appearance in the fact that $\Sigma$ is a disjoint union of two open manifolds<br>$M_1$ and $M_2$ :                                                                                                                                                              |
| cardona-qft-methods_F01602 | 103  | 1.000 | $M_{\{1\}}$                                                                                        | $M_1$                                           |                                                                                                                                                                                                                                                                  |
| cardona-qft-methods_F01603 | 103  | 1.000 | $M_{\{2\}}$                                                                                        | $M_2$                                           |                                                                                                                                                                                                                                                                  |
| cardona-qft-methods_F01604 | 103  | 1.000 | $\Sigma$                                                                                           | $\Sigma$                                        | complete riemannian metric on $M_j$ ( $j = 1, 2$ ). Let $\mathbf{v}_j$ denote the restriction<br>of $\mathbf{v}$ to $M_j$ . Then $(M_i, \mathbf{v}_i)$ ( $i = 1, 2$ ) are tamed $G$ -manifolds                                                                   |
| cardona-qft-methods_F01605 | 103  | 1.000 | $M_{\{j\}}(j=1,2)$                                                                                 | $M_j(j = 1, 2)$                                 | over $M$ . Denote by $E_j$ the restriction of $E$ to $M_j$ ( $j = 1, 2$ ). Let $\mathcal{D}_j$ ( $j = 1, 2$ )<br>denote the restriction of $\mathcal{D}$ to $M_j$ .                                                                                              |
| cardona-qft-methods_F01606 | 103  | 1.000 | $\mathbf{v}_{\{j\}}$                                                                               | $\mathbf{v}_j$                                  | over $M$ . Denote by $E_j$ the restriction of $E$ to $M_j$ ( $j = 1, 2$ ). Let $\mathcal{D}_j$ ( $j = 1, 2$ )<br>denote the restriction of $\mathcal{D}$ to $M_j$ .                                                                                              |
| cardona-qft-methods_F01607 | 103  | 0.783 | $M_{\{j\}}$                                                                                        | $M_j$                                           | denote the restriction of $\mathcal{D}$ to $M_j$ ;                                                                                                                                                                                                               |
| cardona-qft-methods_F01608 | 103  | 0.783 | $\left(M_{\{j\}}, \mathbf{v}_{\{j\}}\right)(j=1,2)$                                                | $(M_j, \mathbf{v}_j)(j = 1, 2)$                 | denote the restriction of $\mathcal{D}$ to $M_j$ ;                                                                                                                                                                                                               |
| cardona-qft-methods_F01609 | 103  | 1.000 | $E_{\{j\}}$                                                                                        | $E_j$                                           | $\operatorname{Ind}_G(\mathcal{D}, \mathbf{v}) = \operatorname{Ind}_G(\mathcal{D}_1, \mathbf{v}_1) + \operatorname{Ind}_G(\mathcal{D}_2, \mathbf{v}_2). \quad (12)$                                                                                              |
| cardona-qft-methods_F01610 | 103  | 1.000 | $\mathcal{D}_{\{j\}}(j=1,2)$                                                                       | $\mathcal{D}_j(j = 1, 2)$                       | $\operatorname{Ind}_G(\mathcal{D}, \mathbf{v}) = \operatorname{Ind}_G(\mathcal{D}_1, \mathbf{v}_1) + \operatorname{Ind}_G(\mathcal{D}_2, \mathbf{v}_2). \quad (12)$                                                                                              |
| cardona-qft-methods_F01611 | 104  | 0.663 | $K \, K$                                                                                           | $KK$                                            |                                                                                                                                                                                                                                                                  |
| cardona-qft-methods_F01612 | 107  | 0.374 | $\{ \}^{\text{\texttt{\text{AM11a}}}, \text{\texttt{\text{AM11b}}}, \text{\texttt{\text{AM11c}}}}$ | AM11a , AM11b , AM11c                           | with infinite $\operatorname{Aut}(\mathcal{G})$ (especially $\operatorname{Aut}(\mathcal{G}) = \mathbb{Z}$ ), we define a notion<br>of <i>algebra-geometric Feynman rules</i> (Definition 4), i.e., graph invariants                                             |
| cardona-qft-methods_F01613 | 107  | 1.000 | $C_{\mathcal{G}}(t)$                                                                               | $C_G(t)$                                        | ent ways: for example, the coefficient of $t$ in $C_G(t)$ equals the Euler charac-                                                                                                                                                                               |
| cardona-qft-methods_F01614 | 107  | 0.987 | $\{ \}^{\text{\texttt{\text{Mar10}}}}$                                                             | Mar10                                           | together this in                                                                                                                                                                                                                                                 |
| cardona-qft-methods_F01615 | 108  | 1.000 | $\mathbb{C}\!\left[x_1,\ldots,x_n\right]$                                                          | $\mathbb{C}\left[x_1,\ldots,x_n\right]$         | —An <i>affine algebraic set</i> is the set of zeros of a collection of polynomials<br>in $\mathbb{C}[x_1,\ldots,x_n]$ , viewed as a subset of affine space $A^n(=\mathbb{C}^n)$ .                                                                                |
| cardona-qft-methods_F01616 | 108  | 1.000 | $\mathbb{A}_{\mathbb{C}}\!\left[\mathbb{C}\right]^n\!\left(=\mathbb{C}\right)^n$                   | $A_{\mathbb{C}}^n(=\mathbb{C}^n)$               | —An <i>affine algebraic set</i> is the set of zeros of a collection of polynomials<br>in $\mathbb{C}[x_1,\ldots,x_n]$ , viewed as a subset of affine space $A^n(=\mathbb{C}^n)$ .                                                                                |
| cardona-qft-methods_F01617 | 108  | 1.000 | $\mathbb{C}\!\left[z_0,\ldots,z_n\right]$                                                          | $\mathbb{C}\left[z_0,\ldots,z_n\right]$         | homogeneous polynomials in $\mathbb{C}[z_0,\ldots,z_n]$ , viewed as a subset of projective<br>space $\mathbb{P}^n=\mathbb{P}^n$ .                                                                                                                                |
| cardona-qft-methods_F01618 | 108  | 1.000 | $\mathbb{P}^n\!=\!\mathbb{P}_{\mathbb{C}}\!\left[\mathbb{C}\right]^n$                              | $\mathbb{P}^n=\mathbb{P}_{\mathbb{C}}^n$        | A                                                                                                                                                                                                                                                                |
| cardona-qft-methods_F01619 | 108  | 1.000 | $Y \backslash Z$                                                                                   | $Y \backslash Z$                                | normal functions on members of an affine cover. A <i>variational isomorphism</i><br>$X \dashrightarrow Y$ is the datum of an isomorphism between dense open subsets of                                                                                           |
| cardona-qft-methods_F01620 | 108  | 1.000 | $Z$                                                                                                | $Z$                                             | normal functions on members of an affine cover. A <i>variational isomorphism</i><br>$X \dashrightarrow Y$ is the datum of an isomorphism between dense open subsets of                                                                                           |
| cardona-qft-methods_F01621 | 108  | 0.947 | $X \rightarrowtail Y$                                                                              | $X \rightarrow Y$                               | to $\mathbb{R}$ (or $\mathbb{C}$ ). If $\nabla_p f=0$ , then $f$ is called a <i>critical point</i> of $f$ at $p$ . If $\nabla_p f \neq 0$ , then $f$ is called<br>zero; the linear form defined by $\nabla_p f$ is the <i>tangent space</i> to $X$ at $p$ . If a |
| cardona-qft-methods_F01622 | 108  | 1.000 | $Y \, ; \, X$                                                                                      | $Y;X$                                           | variety $X$ is nonsingular at all of its points, the collection of the tangent<br>spaces forms the tangent bundle $TX$ of the variety. If $Z$ is a subvariety                                                                                                    |
| cardona-qft-methods_F01623 | 108  | 1.000 | $\mathbb{A}^n$                                                                                     | $A^n$                                           | $Z$ by an <i>exceptional divisor</i> , which is a copy of the projectivized normal                                                                                                                                                                               |
| cardona-qft-methods_F01624 | 108  | 1.000 | $f\!\left(x_1,\ldots,x_n\right)\!=\!0$                                                             | $f\left(x_1,\ldots,x_n\right)=0$                |                                                                                                                                                                                                                                                                  |
| cardona-qft-methods_F01625 | 108  | 1.000 | $\nabla_{\mathfrak{p}} f$                                                                          | $\nabla_p f$                                    |                                                                                                                                                                                                                                                                  |

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| cardona-qft-methods_F01626 | 108  | ■ 1.000 | $\tau X$                                                                                         | $TX$                                                                                  |                                                                                                                                                                                                                           |
| cardona-qft-methods_F01627 | 108  | ■ 1.000 | $N_Z X$                                                                                          | $N_Z X$                                                                               |                                                                                                                                                                                                                           |
| cardona-qft-methods_F01628 | 108  | ■ 1.000 | $ \left. \tau X \right _Z  / \tau Z$                                                             | $TX _Z / TZ$                                                                          |                                                                                                                                                                                                                           |
| cardona-qft-methods_F01629 | 108  | ■ 1.000 | $X \backslash Z$                                                                                 | $X \backslash Z$                                                                      |                                                                                                                                                                                                                           |
| cardona-qft-methods_F01630 | 109  | ■ 1.000 | $\{\}^{\text{\texttt{Hir64}}}$                                                                   | Hir64                                                                                 |                                                                                                                                        |
| cardona-qft-methods_F01631 | 109  | ■ 0.988 | $\operatorname{sum} \chi(X) := \sum_i (-1)^i \operatorname{dim} H_c^i(X, \mathbb{Q})$            | $\operatorname{sum} \chi(X) := \sum_i (-1)^i \operatorname{dim} H_c^i(X, \mathbb{Q})$ | <b>Definition 1.</b> The (topological) <i>Eule</i>                                                                                                                                                                        |
| cardona-qft-methods_F01632 | 109  | ■ 1.000 | $H_{\{c\}}$                                                                                      | $H_c$                                                                                 | Here $H_{\bullet}$ denotes cohomology with compact support. As it happens, ev-                                                                                                                                            |
| cardona-qft-methods_F01633 | 109  | ■ 1.000 | $\{\}^{\text{\texttt{Ful93}}}$                                                                   | Ful93                                                                                 | As every projective variety $X$ may be triangulated ( <sup>Hir75</sup> ), we can also define the Euler characteristic as                                                                                                  |
| cardona-qft-methods_F01634 | 109  | ■ 1.000 | $\{\}^{\text{\texttt{Hir75}}}$                                                                   | Hir75                                                                                 | $\chi(X) = e_0 - e_1 + e_2 - \dots$                                                                                                                                                                                       |
| cardona-qft-methods_F01635 | 109  | ■ 1.000 | $s_{\{i\}}$                                                                                      | $s_i$                                                                                 | The following properties are easy consequences of the definition:                                                                                                                                                         |
| cardona-qft-methods_F01636 | 109  | ■ 1.000 | $X=Y \backslash Z$                                                                               | $X = Y \backslash Z$                                                                  | (i) If $X$ and $X'$ are isomorphic, then $\chi(X) = \chi(X')$ ;<br>(ii) If $Z \subset X$ is a closed subvariety then $\chi(X) = \chi(Z) + \chi(X \backslash Z)$ .                                                         |
| cardona-qft-methods_F01637 | 109  | ■ 0.999 | $\chi(X)=\chi(Y)-\chi(Z)$                                                                        | $\chi(X) = \chi(Y) - \chi(Z)$                                                         | (iii) $\chi(X \times Y) = \chi(X) \cdot \chi(Y)$ .                                                                                                                                                                        |
| cardona-qft-methods_F01638 | 109  | ■ 0.999 | $X^{\{\prime\}}$                                                                                 | $X'$                                                                                  | tained from homomorphisms from the Grothendieck group/ring of v.                                                                                                                                                          |
| cardona-qft-methods_F01639 | 109  | ■ 0.999 | $\chi(X)=\chi(\left(X^{\{\prime\}}\right))$                                                      | $\chi(X) = \chi(X')$                                                                  | tained from homomorphisms from the Grothendieck group/ring of v.                                                                                                                                                          |
| cardona-qft-methods_F01640 | 109  | ■ 1.000 | $Z \subseteq X$                                                                                  | $Z \subseteq X$                                                                       | and the Chern-Schwartz-MacPherson class.<br>Note that a formal consequence of (ii) is an inclusion-exclusion principle                                                                                                    |
| cardona-qft-methods_F01641 | 109  | ■ 1.000 | $\chi(X)=\chi(Z)+\chi(X \backslash Z)$                                                           | $\chi(X) = \chi(Z) + \chi(X \backslash Z)$                                            | and the Chern-Schwartz-MacPherson class.<br>Note that a formal consequence of (ii) is an inclusion-exclusion principle                                                                                                    |
| cardona-qft-methods_F01642 | 109  | ■ 1.000 | $\chi(X \times Y)=\chi(X) \cdot \chi(Y)$                                                         | $\chi(X \times Y) = \chi(X) \cdot \chi(Y)$                                            | for the Euler characteristic: if $Y$ :                                                                                                                                                                                    |
| cardona-qft-methods_F01643 | 109  | ■ 1.000 | $\chi(X)=\chi(Y) \cdot \chi(F)$                                                                  | $\chi(X) = \chi(Y) \cdot \chi(F)$                                                     |                                                                                                                                                                                                                           |
| cardona-qft-methods_F01644 | 109  | ■ 0.744 | $\boldsymbol{K}_{\{\mathbf{0}\}}$                                                                | $\boldsymbol{K}_0$                                                                    |                                                                                                                                                                                                                           |
| cardona-qft-methods_F01645 | 110  | ■ 1.000 | $\mathbb{Q}, \mathbb{F}_q$                                                                       | $\mathbb{Q}, \mathbb{F}_q$                                                            |                                                                                                                                                                                                                           |
| cardona-qft-methods_F01646 | 110  | ■ 0.792 | $K_{\{0\}}($                                                                                     | $K_0($                                                                                | to consider other possibilities, such as $\mathbb{Q}, \mathbb{F}_q$ (the finite field with a prime                                                                                                                        |
| cardona-qft-methods_F01647 | 110  | ■ 0.792 | $)$                                                                                              | $)$                                                                                   | to consider other possibilities, such as $\mathbb{Q}, \mathbb{F}_q$ (the finite field with a prime                                                                                                                        |
| cardona-qft-methods_F01648 | 110  | ■ 1.000 | $[M]=\left[M^{\{\prime\}}\right]+\left[M^{\{\prime\prime}\prime}\right]$                         | $[M] = [M'] + [M'']$                                                                  | defined as the quotient of the free abelian group on isomorphism classes                                                                                                                                                  |
| cardona-qft-methods_F01649 | 110  | ■ 1.000 | $0 \rightarrow M^{\{\prime\}} \rightarrow M \rightarrow M^{\{\prime\prime}\prime} \rightarrow 0$ | $0 \rightarrow M' \rightarrow M \rightarrow M'' \rightarrow 0$                        | of $R$ -modules by a relation $[M] = [M'] + [M'']$ for every exact sequence $0 \rightarrow M' \rightarrow M \rightarrow M'' \rightarrow 0$ . More generally, the same concept may be                                      |
| cardona-qft-methods_F01650 | 110  | ■ 1.000 | $\{\}^{\text{\texttt{Lev08}}}$                                                                   | Lev08                                                                                 | ‘Chow’ motives are based on an equivalence that is especially germane to algebraic geometry.                                                                                                                              |
| cardona-qft-methods_F01651 | 110  | ■ 0.590 | $2 \operatorname{in}^{\text{\texttt{Mar10}}}$                                                    | $2\operatorname{in}^{\text{\texttt{Mar10}}}$                                          | algebraic geometry.<br>In any case, in this category there ought to be a distinguished trian-                                                                                                                             |
| cardona-qft-methods_F01652 | 110  | ■ 1.000 | $Z, X$                                                                                           | $Z, X$                                                                                | the free abelian group $\mathbb{Q}_k$ is the tensor product $\mathbb{Q} \otimes_{\mathbb{Z}} \mathbb{Z}_k$ of $\mathbb{Q}$ with $\mathbb{Z}_k$ .<br>operation defined on generators by $[X] \cdot [Y] = [X \times_k Y]$ . |

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| cardona-qft-methods_F01653 | 110  | ■ 1.000 | $K_{\{0\}}\left(\operatorname{Var}_{\{k\}}\right):=\frac{\text{Free abelian group on isom. classes of quasi-proj. } k \text{ -varieties }}{\langle [X]=[Z]+[X\backslash Z] \text{ for every closed embedding } Z\subseteq X \rangle}$ | $K_0(\operatorname{Var}_k) := \frac{\text{Free abelian group on isom. classes of quasi-proj. } k\text{-varieties}}{\langle [X] = [Z] + [X \setminus Z] \text{ for every closed embedding } Z \subseteq X \rangle}$ |                                                                                                                                                                                                                                              |
| cardona-qft-methods_F01654 | 110  | ■ 1.000 | $[X] \cdot [Y] = [X \times_k Y]$                                                                                                                                                                                                      | $[X] \cdot [Y] = [X \times_k Y]$                                                                                                                                                                                   |                                                                                                                                                                                                                                              |
| cardona-qft-methods_F01655 | 110  | ■ 0.995 | $K_{\{0\}}\left(\operatorname{Var}_{\{k\}}\right)$                                                                                                                                                                                    | $K_0(\operatorname{Var}_k)$                                                                                                                                                                                        |                                                                                                                                                                                                                                              |
| cardona-qft-methods_F01656 | 110  | ■ 1.000 | $k=\mathbb{C}$                                                                                                                                                                                                                        | $k = \mathbb{C}$                                                                                                                                                                                                   |                                                                                                                                                                                                                                              |
| cardona-qft-methods_F01657 | 111  | ■ 1.000 | $[X]$                                                                                                                                                                                                                                 | $[X]$                                                                                                                                                                                                              | at least as much information as the Grothendieck group obtained from the                                                                                                                                                                     |
| cardona-qft-methods_F01658 | 111  | ■ 0.916 | $K_{\{0\}}(\operatorname{Var})\left(\text{Bit04}\right)$                                                                                                                                                                              | $K_0(\operatorname{Var})\left(\text{Bit04}\right)$                                                                                                                                                                 | V. Navarro Aznar ( <sup>GrNA02</sup> ), and recovered by F. Bittner from a very useful alternative description of $K_0(\operatorname{Var})$ (Bit04) which will be recalled below                                                             |
| cardona-qft-methods_F01659 | 111  | ■ 1.000 | $e(p)=1$                                                                                                                                                                                                                              | $e(p) = 1$                                                                                                                                                                                                         | <ul style="list-style-type: none"> <li>If <math>X</math> and <math>X'</math> are isomorphic, <math>e(X) = e(X')</math>.</li> <li>If <math>Z \subset X</math> is a closed subvariety, <math>e(X) = e(Z) + e(X \setminus Z)</math>.</li> </ul> |
| cardona-qft-methods_F01660 | 111  | ■ 1.000 | $e(X)=e\left(X^{\prime}\right)$                                                                                                                                                                                                       | $e(X) = e(X')$                                                                                                                                                                                                     | <ul style="list-style-type: none"> <li><math>e(X \times Y) = e(X)e(Y)</math>.</li> </ul>                                                                                                                                                     |
| cardona-qft-methods_F01661 | 111  | ■ 1.000 | $e(X)=e(Z)+e(X\backslash Z)$                                                                                                                                                                                                          | $e(X) = e(Z) + e(X \setminus Z)$                                                                                                                                                                                   | Then $e$ is induced by a unique ring homomorphism $K_0(\operatorname{Var}) \rightarrow R$ .                                                                                                                                                  |
| cardona-qft-methods_F01662 | 111  | ■ 1.000 | $e(X \times Y)=e(X) \cdot e(Y)$                                                                                                                                                                                                       | $e(X \times Y) = e(X)e(Y)$                                                                                                                                                                                         | Theorem 2.1 is essenti                                                                                                                                                                                                                       |
| cardona-qft-methods_F01663 | 111  | ■ 0.968 | $K_{\{0\}}(\operatorname{Var}) \rightarrow R$                                                                                                                                                                                         | $K_0(\operatorname{Var}) \rightarrow R$                                                                                                                                                                            | jects satisfying a universal requirement, and Theorem 2.1 does nothing but                                                                                                                                                                   |
| cardona-qft-methods_F01664 | 111  | ■ 0.970 | $\rightarrow \mathbb{Z}$                                                                                                                                                                                                              | $\rightarrow \mathbb{Z}$                                                                                                                                                                                           | $K_0(\operatorname{Var})$ : as mentioned above, there is a homomorphism from $K_0(\operatorname{Var})$ to                                                                                                                                    |
| cardona-qft-methods_F01665 | 111  | ■ 0.621 | $[X] \in K_{\{0\}}(\operatorname{Var})$                                                                                                                                                                                               | $[X] \in K_0(\operatorname{Var})$                                                                                                                                                                                  | and every prime power $q$ we can define homomorphisms $K_0(\operatorname{Var}_{\mathbb{F}_q}) \rightarrow \mathbb{Z}$ by sending $[X]$ to the number of points of $X$ over the finite field $\mathbb{F}_q$ (i.e.,                            |
| cardona-qft-methods_F01666 | 111  | ■ 0.623 | $K_{\{0\}}(\operatorname{Var})$                                                                                                                                                                                                       | $K_0(\operatorname{Var})$                                                                                                                                                                                          |                                                                                                                                                                                                                                              |
| cardona-qft-methods_F01667 | 111  | ■ 1.000 | $\mathbb{Z}[u, v]$                                                                                                                                                                                                                    | $\mathbb{Z}[u, v]$                                                                                                                                                                                                 |                                                                                                                                                                                                                                              |
| cardona-qft-methods_F01668 | 111  | ■ 1.000 | $K\left(\operatorname{Var}_{\mathbb{Z}}\right) \rightarrow \mathbb{Z}$                                                                                                                                                                | $K(\operatorname{Var}_{\mathbb{Z}}) \rightarrow \mathbb{Z}$                                                                                                                                                        |                                                                                                                                                                                                                                              |
| cardona-qft-methods_F01669 | 111  | ■ 1.000 | $\mathbb{F}_q$                                                                                                                                                                                                                        | $\mathbb{F}_q$                                                                                                                                                                                                     |                                                                                                                                                                                                                                              |
| cardona-qft-methods_F01670 | 112  | ■ 1.000 | $\left(\text{Bit04}\right)$                                                                                                                                                                                                           | Bit04                                                                                                                                                                                                              | Generalized Euler characteristics, graph hypersurfaces, and Feynman periods 101                                                                                                                                                              |
| cardona-qft-methods_F01671 | 112  | ■ 1.000 | $Z \subseteq X$                                                                                                                                                                                                                       | $Z \subset X$                                                                                                                                                                                                      | versal Euler characteristic. In this paper, F. Bittner gives a very useful                                                                                                                                                                   |
| cardona-qft-methods_F01672 | 112  | ■ 1.000 | $\widetilde{X} \rightarrow X$                                                                                                                                                                                                         | $\widetilde{X} \rightarrow X$                                                                                                                                                                                      | alternative presentation for $K_0(\operatorname{Var})$ . I have recalled that if $X$ is nonsingular and $Z \subset X$ is a nonsingular subvariety, then there exists a ‘blow-up’                                                             |
| cardona-qft-methods_F01673 | 112  | ■ 1.000 | $[X]=[Z]+[X\backslash Z]$                                                                                                                                                                                                             | $[X] = [Z] + [X \setminus Z]$                                                                                                                                                                                      | $[X]$ of the finite field. Elements of the subgroup $\mathbb{Z}[X] \subseteq K_0(\operatorname{Var})$ Tate (naive) motives.                                                                                                                  |
| cardona-qft-methods_F01674 | 112  | ■ 1.000 | $\mathbb{L}:=$                                                                                                                                                                                                                        | $\mathbb{L} :=$                                                                                                                                                                                                    | The (not very standard) terminology introduced here reflects the fact that classes in $\mathbb{Z}[\mathbb{L}]$ correspond to mixed Tate motives in the category of                                                                           |
| cardona-qft-methods_F01675 | 112  | ■ 0.994 | $\left[\mathbb{A}^1\right]$                                                                                                                                                                                                           | $[\mathbb{A}^1]$                                                                                                                                                                                                   | motives. (For the sake of simplicity of exposition I am neglecting a local-                                                                                                                                                                  |
| cardona-qft-methods_F01676 | 112  | ■ 0.994 | $\mathbb{Z}[\mathbb{L}] \subseteq K_{\{0\}}(\mathbb{L})$                                                                                                                                                                              | $\mathbb{Z}[\mathbb{L}] \subseteq K_0(\mathbb{L})$                                                                                                                                                                 | motives. (For the sake of simplicity of exposition I am neglecting a local-                                                                                                                                                                  |
| cardona-qft-methods_F01677 | 112  | ■ 1.000 | $\mathbb{Z}[\mathbb{L}]$                                                                                                                                                                                                              | $\mathbb{Z}[\mathbb{L}]$                                                                                                                                                                                           | for example,                                                                                                                                                                                                                                 |

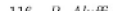
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| cardona-qft-methods_F01678 | 112  | 1.000 | $\widetilde{X}$                                                           | $\widetilde{X}$                                                           |                                                                                                                                                                                                                 |
| cardona-qft-methods_F01679 | 112  | 1.000 | $[\widetilde{X}]$                                                         | $[\widetilde{X}]$                                                         |                                                                                                                                                                                                                 |
| cardona-qft-methods_F01680 | 112  | 1.000 | $[Z]$                                                                     | $[Z]$                                                                     |                                                                                                                                                                                                                 |
| cardona-qft-methods_F01681 | 113  | 1.000 | $[X] \in \mathbb{Z}[\mathbb{L}] \subseteq K_0(\mathbf{Var}_{\mathbb{Z}})$ | $[X] \in \mathbb{Z}[\mathbb{L}] \subseteq K_0(\mathbf{Var}_{\mathbb{Z}})$ |                                                                                                                              |
| cardona-qft-methods_F01682 | 113  | 0.644 | $\chi(X)=\sum_i a_i$                                                      | $\chi(X)=\sum_i a_i$                                                      | value obtained by formally replacing $q$ by 1 in (3). The Euler characteristic                                                                                                                                  |
| cardona-qft-methods_F01683 | 113  | 0.984 | $\mathbb{F}_1$                                                            | $\mathbb{F}_1$                                                            |                                                                                                                              |
| cardona-qft-methods_F01684 | 113  | 0.999 | $c_{\mathrm{SM}}$                                                         | $c_{SM}$                                                                  | monomorphisms'. Ind                                                                                                                                                                                             |
| cardona-qft-methods_F01685 | 113  | 1.000 | $c_{\mathrm{SM}}(X)$                                                      | $c_{\mathrm{SM}}(X)$                                                      | ment of intersection theory is given in; the hurried reader will benefit from consulting Ful84b). To the case of motivic spaces $A \in \mathbb{D}^{\mathrm{b}}_{\mathrm{ct}}(\mathbf{Spc})$ with                |
| cardona-qft-methods_F01686 | 113  | 1.000 | $A_*X$                                                                    | $A_*X$                                                                    | homology, and equals the free abelian group on generators $[\mathbb{P}^0], \dots, [\mathbb{P}^n]$ . It                                                                                                          |
| cardona-qft-methods_F01687 | 113  | 1.000 | $\mathrm{Ful84b}$                                                         | $\mathrm{Ful84b}$                                                         |                                                                                                                                                                                                                 |
| cardona-qft-methods_F01688 | 113  | 1.000 | $A_*\mathbb{P}^n$                                                         | $A_*\mathbb{P}^n$                                                         |                                                                                                                                                                                                                 |
| cardona-qft-methods_F01689 | 113  | 1.000 | $[\mathbb{P}^0], \dots, [\mathbb{P}^n]$                                   | $[\mathbb{P}^0], \dots, [\mathbb{P}^n]$                                   |                                                                                                                                                                                                                 |
| cardona-qft-methods_F01690 | 113  | 0.950 | $r=\dim X$                                                                | $r=\dim X$                                                                |                                                                                                                                                                                                                 |
| cardona-qft-methods_F01691 | 114  | 1.000 | $c_i(E) \cap [X]$                                                         | $c_i(E) \cap [X]$                                                         |                                                                                                                            |
| cardona-qft-methods_F01692 | 114  | 0.994 | $\mathrm{rk} E - i + 1$                                                   | $\mathrm{rk} E - i + 1$                                                   | For our purposes, we will rely on the intuition that the $i$ -th Chern class of $\mathbb{P}^n$ defined as $c_i(\mathbb{P}^n) \in \mathbb{Z}[\mathbb{P}^1]$ is a class in codimension $i$ according to intuition |
| cardona-qft-methods_F01693 | 114  | 1.000 | $\mathrm{Ful84a}$                                                         | $\mathrm{Ful84a}$                                                         | example, $E$ may have no nontrivial sections whatsoever; here we are simply capturing the gist of Examples 14.4.1 and 14.4.2 in. <sup>Ful84a</sup> According to this                                            |
| cardona-qft-methods_F01694 | 114  | 0.988 | $c_{\dim X}(TX) \cap [X]$                                                 | $c_{\dim X}(TX) \cap [X]$                                                 | rough interpretation, the ‘top’ Chern class $c_{\dim X}(TX) \cap [X]$ of the tangent                                                                                                                            |
| cardona-qft-methods_F01695 | 114  | 0.981 | $\dim X$                                                                  | $\dim X$                                                                  | bundle of $X$ is a class in codimension $\dim X$ (i.e., in dimension 0), recording the locus where a general section of $TX$ (i.e., a general vector field) is                                                  |
| cardona-qft-methods_F01696 | 114  | 1.000 | $\int$                                                                    | $\int$                                                                    | $\chi(X) < 0$ .) A similar statement may be given relating all Chern classes of $TX$ and loci at which there arise obstructions to the construction of frames                                                   |
| cardona-qft-methods_F01697 | 114  | 1.000 | $\chi(X) < 0$                                                             | $\chi(X) < 0$                                                             | meaning.<br>In the 1960s, Marie-Hélène Schwartz answered this question by introduc-                                                                                                                             |
| cardona-qft-methods_F01698 | 114  | 0.885 | $\mathrm{Mac74}$                                                          | $\mathrm{Mac74}$                                                          |                                                                                                                                                                                                                 |
| cardona-qft-methods_F01699 | 115  | 0.594 | $(\mathrm{BS81AB08})$                                                     | $(\mathrm{BS81AB08})$                                                     |                                                                                                                                                                                                                 |
| cardona-qft-methods_F01700 | 115  | 0.987 | $c_{\mathrm{SM}}$                                                         | $c_{\mathrm{SM}}$                                                         |                                                                                                                            |
| cardona-qft-methods_F01701 | 115  | 0.987 | $c_{\mathrm{SM}}(\varphi)$                                                | $c_{\mathrm{SM}}(\varphi)$                                                |                                                                                                                            |
| cardona-qft-methods_F01702 | 115  | 1.000 | $\varphi = \sum m_i \mathbb{1}_{Z_i}$                                     | $\varphi = \sum m_i \mathbb{1}_{Z_i}$                                     | Chow group; cf., e.g., Example 19.1.1.) for every <i>constructible function</i> on a variety $X$ . A constructible function is a linear combination of indicator                                                |
| cardona-qft-methods_F01703 | 115  | 1.000 | $C(X)$                                                                    | $C(X)$                                                                    | functions of subvarieties: $\varphi = \sum m_i \mathbb{1}_{Z_i}$ . Constructible functions on $X$ form                                                                                                          |
| cardona-qft-methods_F01704 | 115  | 1.000 | $C$                                                                       | $C$                                                                       | functions of subvarieties: $\varphi = \sum m_i \mathbb{1}_{Z_i}$ . Constructible functions on $X$ form                                                                                                          |

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| cardona-qft-methods_F01785 | 115  | ■ 1.000 | f: X \rightarrow Y                                              | $f : X \rightarrow Y$                                                              | functions of subvarieties: $\varphi = \sum m_i \mathbb{1}_{Z_i}$ . Constructible functions on $X$ form                                                                                           |
| cardona-qft-methods_F01786 | 115  | ■ 1.000 | C(X) \rightarrow C(Y)                                           | $C(X) \rightarrow C(Y)$                                                            | an abelian group $C(X)$ , and in fact $C$ is a <i>covariant functor</i> : if $f : X \rightarrow Y$ is a regular map, we can define a homomorphism $C(X) \rightarrow C(Y)$ by setting             |
| cardona-qft-methods_F01787 | 115  | ■ 1.000 | p \in Y                                                         | $p \in Y$                                                                          | linearity. MacPherson's definition has the following properties:                                                                                                                                 |
| cardona-qft-methods_F01788 | 115  | ■ 1.000 | X, c_{\mathrm{SM}}: C(X) \rightarrow A_* X                      | $X, c_{\mathrm{SM}} : C(X) \rightarrow A_* X$                                      | <ul style="list-style-type: none"> <li>If <math>f : X \rightarrow Y</math> is a proper morphism and <math>\varphi \in C(X)</math>, th</li> </ul>                                                 |
| cardona-qft-methods_F01789 | 115  | ■ 1.000 | c_{\mathrm{SM}}(\mathbb{1}_X) = c(TX) \cap [X]                  | $c_{\mathrm{SM}}(\mathbb{1}_X) = c(TX) \cap [X]$                                   | $c_{\mathrm{SM}}(f_* \varphi) = f_* c_{\mathrm{SM}}(\varphi)$ .                                                                                                                                  |
| cardona-qft-methods_F01710 | 115  | ■ 1.000 | \varphi \in C(X)                                                | $\varphi \in C(X)$                                                                 | in the third property, $f$ is required to be proper in order to have a good notion of push-forward of classes in the Chow group. Various extensions of the                                       |
| cardona-qft-methods_F01711 | 115  | ■ 1.000 | c_{\mathrm{SM}}(f_* \varphi) = f_* c_{\mathrm{SM}}(\varphi)     | $c_{\mathrm{SM}}(f_* \varphi) = f_* c_{\mathrm{SM}}(\varphi)$                      | theory, to more general field                                                                                                                                                                    |
| cardona-qft-methods_F01712 | 115  | ■ 0.549 | { }^{\text {Ken90Alu06 }}                                       | Ken90Alu06                                                                         | The class $c_{\mathrm{SM}}(\mathbb{1}_X)$ is usually denoted $c_{\mathrm{SM}}(X)$ , and called the <i>Schwartz-MacPherson</i> class of $X$ . Abusing language, we denote by $c_{\mathrm{SM}}(Y)$ |
| cardona-qft-methods_F01713 | 115  | ■ 1.000 | c_{\mathrm{SM}}(\mathbb{1}_X)                                   | $c_{\mathrm{SM}}(\mathbb{1}_X)$                                                    | group,                                                                                                                                                                                           |
| cardona-qft-methods_F01714 | 115  | ■ 1.000 | c_{\mathrm{SM}}(Y)                                              | $c_{\mathrm{SM}}(Y)$                                                               | $\kappa_* : A_* X \rightarrow A_* \{\mathrm{pt}\} = \mathbb{Z}$ ,                                                                                                                                |
| cardona-qft-methods_F01715 | 115  | ■ 0.998 | c_{\mathrm{SM}}(\mathbb{1}_Y)                                   | $c_{\mathrm{SM}}(\mathbb{1}_Y)$                                                    | is nothing but the ‘integral’ $\int$ mentioned above. According to the above                                                                                                                     |
| cardona-qft-methods_F01716 | 115  | ■ 0.821 | \kappa: X \rightarrow \{\mathrm{pt}\}                           | $\kappa : X \rightarrow \{\mathrm{pt}\}$                                           |                                                                                                                                                                                                  |
| cardona-qft-methods_F01717 | 116  | ■ 0.736 | { }^{\text {Kwi92 }}                                            | Kwi92                                                                              | of course $c_{\mathrm{SM}}(X)$ and $c_{\mathrm{SM}}(X')$ agree up to the identification $A_* X \cong A_* X'$ . However, if $V$ and $V'$ are viewed as subvarieties of (for example) projective   |
| cardona-qft-methods_F01718 | 116  | ■ 0.736 | { }^{\text {Alu06 }}                                            | Alu06                                                                              | of course $c_{\mathrm{SM}}(X)$ and $c_{\mathrm{SM}}(X')$ agree up to the identification $A_* X \cong A_* X'$ . However, if $V$ and $V'$ are viewed as subvarieties of (for example) projective   |
| cardona-qft-methods_F01719 | 116  | ■ 1.000 | c_{\mathrm{SM}}(X')                                             | $c_{\mathrm{SM}}(X')$                                                              | hence may differ. Thus, the notation $c_{\mathrm{SM}}(X)$ is ambiguous. This is harmless, unless the inclusion of $V$ is taken into                                                              |
| cardona-qft-methods_F01720 | 116  | ■ 1.000 | A_* X \cong A_* X'                                              | $A_* X \cong A_* X'$                                                               | hence may differ. Thus, the notation $c_{\mathrm{SM}}(X)$ is ambiguous. This is harmless, unless the inclusion of $V$ is taken into                                                              |
| cardona-qft-methods_F01721 | 116  | ■ 1.000 | c_{\mathrm{SM}}(\mathbb{1}_{X'})                                | $c_{\mathrm{SM}}(\mathbb{1}_{X'})$                                                 | <b>Example 2.2.</b> —Viewing $\mathbb{P}^1$ as a subvariety of itself, $c_{\mathrm{SM}}(\mathbb{P}^1) = [\mathbb{P}^1] + 0[\mathrm{pt}]$ ; indeed, since $\mathbb{P}^1$ is nonempty we have      |
| cardona-qft-methods_F01722 | 116  | ■ 0.999 | \mathbb{P}^1                                                    | $\mathbb{P}^1$                                                                     | $c_{\mathrm{SM}}(\mathbb{P}^n) = [\mathbb{P}^n] + \binom{n+1}{1} [\mathbb{P}^{n-1}] + \binom{n+1}{2} [\mathbb{P}^{n-2}] + \cdots + \binom{n+1}{n} [\mathbb{P}^0]$ . (6)                          |
| cardona-qft-methods_F01723 | 116  | ■ 0.999 | c_{\mathrm{SM}}(\mathbb{P}^1) = [\mathbb{P}^1] +                | $c_{\mathrm{SM}}(\mathbb{P}^1) = [\mathbb{P}^1] +$                                 | $c_{\mathrm{SM}}(\mathbb{P}^n) = [\mathbb{P}^n] + \binom{n+1}{1} [\mathbb{P}^{n-1}] + \binom{n+1}{2} [\mathbb{P}^{n-2}] + \cdots + \binom{n+1}{n} [\mathbb{P}^0]$ . (6)                          |
| cardona-qft-methods_F01724 | 116  | ■ 1.000 | 2 [\mathbb{P}^0]                                                | $2 [\mathbb{P}^0]$                                                                 | This again follows from the normalization prope                                                                                                                                                  |
| cardona-qft-methods_F01725 | 116  | ■ 0.974 | T \mathbb{P}^n                                                  | $T\mathbb{P}^n$                                                                    |                                                                                                                                                                                                  |
| cardona-qft-methods_F01726 | 116  | ■ 1.000 | \int c_{\mathrm{SM}}(\mathbb{P}^n) = n + 1 = \chi(\mathbb{P}^n) | $\int c_{\mathrm{SM}}(\mathbb{P}^n) = \binom{n+1}{n} = n + 1 = \chi(\mathbb{P}^n)$ |                                                                                                                                                                                                  |
| cardona-qft-methods_F01727 | 116  | ■ 1.000 | H                                                               | $H$                                                                                |                                                                                                                                                                                                  |
| cardona-qft-methods_F01728 | 116  | ■ 1.000 | H^i                                                             | $H^i$                                                                              |                                                                                                                                                                                                  |
| cardona-qft-methods_F01729 | 116  | ■ 1.000 | [\mathbb{P}^{n-i}]                                              | $[\mathbb{P}^{n-i}]$                                                               |                                                                                                                                                                                                  |
| cardona-qft-methods_F01730 | 116  | ■ 1.000 | H^{n+1} = 0                                                     | $H^{n+1} = 0$                                                                      |                                                                                                                                                                                                  |

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| cardona-qft-methods_F01731 | 117  | ■ 1.000 | $\mathbb{P}^2$                                         | $\mathbb{P}^2$                                         |                                                                                                                                                         |
| cardona-qft-methods_F01732 | 117  | ■ 1.000 | $c_{\mathrm{SM}}(\mathbb{P}^1)$                        | $c_{\mathrm{SM}}(\mathbb{P}^1)$                        |                                                                                                                                                         |
| cardona-qft-methods_F01733 | 117  | ■ 0.999 | $L_1, L_2$                                             | $L_1, L_2$                                             | forward to $\mathbb{P}^2$ differs from the class $c_{\mathrm{SM}}(\mathbb{P}^1)$ recorded above.                                                        |
| cardona-qft-methods_F01734 | 117  | ■ 1.000 | $c_{\mathrm{SM}}(X) = c_{\mathrm{SM}}(L_1 \cup L_2)$   | $c_{\mathrm{SM}}(X) = c_{\mathrm{SM}}(L_1 \cup L_2)$   | —A union $X$ of two distinct lines $L_1, L_2$ in $\mathbb{P}^2$ is a <i>singular</i>                                                                    |
| cardona-qft-methods_F01735 | 117  | ■ 0.988 | $c_{\mathrm{SM}}$                                      | $c_{\mathrm{SM}}$                                      | to express the <i>Milnor number</i> of a singularity, and more general <i>Milnor</i>                                                                    |
| cardona-qft-methods_F01736 | 117  | ■ 1.000 | $\mathrm{PP01}$                                        | PP01                                                   | —Exercise for the reader: For subsets of projective space obtained as                                                                                   |
| cardona-qft-methods_F01737 | 117  | ■ 1.000 | $\mathrm{AM09}$                                        | AM09                                                   | classes. It is not hard to see that this information is the <i>characteristic</i>                                                                       |
| cardona-qft-methods_F01738 | 117  | ■ 0.693 | $\mathrm{Alu12}$                                       | Alu12                                                  | terms of a relative Grothendieck group ( <sup>BSY10</sup> ). For projective sets, there                                                                 |
| cardona-qft-methods_F01739 | 117  | ■ 0.998 | $X \subseteq \mathbb{P}^n$                             | $X \subseteq \mathbb{P}^n$                             | interesting, and they will play a role behind the scenes later in these notes                                                                           |
| cardona-qft-methods_F01740 | 117  | ■ 0.996 | $\mathrm{Alub}$                                        | Alub                                                   |                                                                                                                                                         |
| cardona-qft-methods_F01741 | 117  | ■ 0.980 | $\left(\mathrm{BSY10}\right)$                          | $\left(\mathrm{BSY10}\right)$                          |                                                                                                                                                         |
| cardona-qft-methods_F01742 | 117  | ■ 1.000 | $\mathbb{P}^\infty$                                    | $\mathbb{P}^\infty$                                    |                                                                                                                                                         |
| cardona-qft-methods_F01743 | 117  | ■ 1.000 | $\mathbb{Z}[T]$                                        | $\mathbb{Z}[T]$                                        |                                                                                                                                                         |
| cardona-qft-methods_F01744 | 117  | ■ 0.753 | $\left(\mathrm{Alub}\right)$                           | $\left(\mathrm{Alub}\right)$                           |                                                                                                                                                         |
| cardona-qft-methods_F01745 | 118  | ■ 1.000 | $L(\phi)$                                              | $L(\phi)$                                              | be studied by means of a perturbative expansion, which takes the form of                                                                                |
| cardona-qft-methods_F01746 | 118  | ■ 1.000 | $\langle G \rangle$                                    | $\langle G \rangle$                                    | causes $G$ to become disconnected (i.e., without ‘bridges’); also see §4.1. The reader is addressed to any text in quantum field theory (such as Zee10) |
| cardona-qft-methods_F01747 | 118  | ■ 1.000 | $\frac{G(\phi)}{\#\mathrm{Aut}(G)}$                    | $\frac{G(\phi)}{\#\mathrm{Aut}(G)}$                    | causes $G$ to become disconnected (i.e., without ‘bridges’); also see §4.1. The reader is addressed to any text in quantum field theory (such as Zee10) |
| cardona-qft-methods_F01748 | 118  | ■ 1.000 | $G(\phi)$                                              | $G(\phi)$                                              | The reader is addressed to any text in quantum field theory (such as Mar10) for information on what all of this really means, and to Mar10 for a more   |
| cardona-qft-methods_F01749 | 118  | ■ 0.970 | $\mathrm{Zee10}$                                       | Zee10                                                  | Here $N$ is the number of external edges attached to $G$ , and $n_i$ denotes a                                                                          |
| cardona-qft-methods_F01750 | 118  | ■ 1.000 | $p_i$                                                  | $p_i$                                                  |                                                                                                                                                         |
| cardona-qft-methods_F01751 | 118  | ■ 1.000 | $\sum_i p_i = 0$                                       | $\sum_i p_i = 0$                                       |                                                                                                                                                         |
| cardona-qft-methods_F01752 | 118  | ■ 1.000 | $\hat{\phi}$                                           | $\hat{\phi}$                                           |                                                                                                                                                         |
| cardona-qft-methods_F01753 | 118  | ■ 0.874 | $\mathrm{CM}$                                          | CM                                                     |                                                                                                                                                         |
| cardona-qft-methods_F01754 | 119  | ■ 1.000 | $I_G$                                                  | $I_G$                                                  |                                                                                                                                                         |
| cardona-qft-methods_F01755 | 119  | ■ 1.000 | $U(G)$                                                 | $U(G)$                                                 | Clever use of various reductions (keywords: ‘Schwinger parameters’,                                                                                     |
| cardona-qft-methods_F01756 | 119  | ■ 1.000 | $\sigma_n = \left\{ \sum_i t_i = 1 \right\}; \omega_n$ | $\sigma_n = \left\{ \sum_i t_i = 1 \right\}; \omega_n$ | certain polynomials determined by the graph $G$ . The reader should take                                                                                |

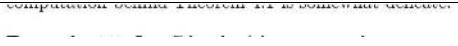
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| cardona-qft-methods_F01757 | 119  | ■ 1.000 | $P_{\{G\}}, \Psi_{\{G\}}$                    | $P_G, \Psi_G$                                | certain polynomials determined by the graph $G$ . The reader should take                                                                                                                             |
| cardona-qft-methods_F01758 | 119  | ■ 1.000 | $\Psi_{\{G\}}$                               | $\Psi_G$                                     | in. The polynomial $P_G(\underline{t}, p)$ is given by                                                                                                                                               |
| cardona-qft-methods_F01759 | 119  | ■ 0.957 | $P_{\{G\}}(\underline{\text{t}}, p)$         | $P_G(\underline{t}, p)$                      | where the sum ranges over the <i>cut sets</i> $C$ of $G$ ,                                                                                                                                           |
| cardona-qft-methods_F01760 | 119  | ■ 1.000 | $s_{\{C\}}$                                  | $s_C$                                        | for the ‘2-banana’ graph                                                                                                                                                                             |
| cardona-qft-methods_F01761 | 119  | ■ 1.000 | $D \setminus \ell \mid 2 - n \geq 0$         | $D\ell/2 - n \geq 0$                         |                                                                                                                                                                                                      |
| cardona-qft-methods_F01762 | 119  | ■ 1.000 | $P_{\{G\}}$                                  | $P_G$                                        |                                                                                                                                                                                                      |
| cardona-qft-methods_F01763 | 120  | ■ 1.000 | $\Gamma(n - D \setminus \ell \mid 2)$        | $\Gamma(n - D\ell/2)$                        |                                                                                                                                                                                                      |
| cardona-qft-methods_F01764 | 120  | ■ 1.000 | $M \geq 0$                                   | $M \geq 0$                                   | $\int_{\text{cycle } \sigma} \overline{\Psi_G(t)}^M$                                                                                                                                                 |
| cardona-qft-methods_F01765 | 120  | ■ 1.000 | $\hat{X}_{\{G\}}$                            | $\hat{X}_G$                                  | with $M \geq 0$ , an integral of an algebraic differential form over the <i>comple-</i>                                                                                                              |
| cardona-qft-methods_F01766 | 120  | ■ 1.000 | $\Psi_{\{G\}} = 0$                           | $\Psi_G = 0$                                 | with $M \geq 0$ , an integral of an algebraic differential form over the <i>comple-</i>                                                                                                              |
| cardona-qft-methods_F01767 | 120  | ■ 1.000 | $\overline{\mathbb{Q}}$                      | $\overline{\mathbb{Q}}$                      | view article by M. Kontsevich and D. Zagier, <sup>KZ01</sup> ‘The set of periods forms a ring (often enlarged by including $\sqrt{-1}$ for technical reasons) and is count-                          |
| cardona-qft-methods_F01768 | 120  | ■ 1.000 | $\{ \}^{\text{KZ01}}$                        | KZ01                                         | surprisingly difficult to prove that a given number is <i>not</i> a period, although                                                                                                                 |
| cardona-qft-methods_F01769 | 120  | ■ 1.000 | $\frac{1}{2\pi i}$                           | $\frac{1}{2\pi i}$                           | commonly believed that these numbers are not. (For example, $e$ is not be-<br>lieved to be a period, but this is not known.) A period is an expression of                                            |
| cardona-qft-methods_F01770 | 120  | ■ 1.000 | $\{ \}^{\text{Bro12}}$                       | Bro12                                        | for the (motive corresponding to) the complement $Y_G$ of the <i>graph hyper-</i><br><i>surface</i> $\hat{Y}_G \subset \mathbb{A}^n$ defined by $\Psi_G(t) = 0$ . Again, let me point out that there |
| cardona-qft-methods_F01771 | 120  | ■ 1.000 | $n_i \geq 1, n_r \geq 2$                     | $n_i \geq 1, n_r \geq 2$                     |                                                                                                                                                                                                      |
| cardona-qft-methods_F01772 | 120  | ■ 1.000 | $\mathbb{Q} \left[ \frac{1}{2\pi i} \right]$ | $\mathbb{Q} \left[ \frac{1}{2\pi i} \right]$ |                                                                                                                                                                                                      |
| cardona-qft-methods_F01773 | 120  | ■ 1.000 | $\hat{Y}_{\{G\}}$                            | $\hat{Y}_G$                                  |                                                                                                                                                                                                      |
| cardona-qft-methods_F01774 | 120  | ■ 1.000 | $\hat{X}_{\{G\}} \subseteq \mathbb{A}^n$     | $\hat{X}_G \subseteq \mathbb{A}^n$           |                                                                                                                                                                                                      |
| cardona-qft-methods_F01775 | 120  | ■ 1.000 | $\Psi_{\{G\}}(\underline{\text{t}}) = 0$     | $\Psi_G(\underline{t}) = 0$                  |                                                                                                                                                                                                      |
| cardona-qft-methods_F01776 | 121  | ■ 1.000 | $\{ \}^{\text{BK97}}$                        | BK97                                         | very extensive computations is that <i>Feynman periods appear to be (combi-</i>                                                                                                                      |
| cardona-qft-methods_F01777 | 121  | ■ 1.000 | $t_{\{e\}}$                                  | $t_e$                                        |                                                                                                                                                                                                      |
| cardona-qft-methods_F01778 | 122  | ■ 1.000 | $n > 0$                                      | $n > 0$                                      |                                                                                                                                                                                                      |
| cardona-qft-methods_F01779 | 123  | ■ 1.000 | $\Psi_{\{G\}}(\underline{\text{t}})$         | $\Psi_G(\underline{t})$                      | <i>Symanzik</i> polynomial. The number of edges in a maximal spanning forest                                                                                                                         |
| cardona-qft-methods_F01780 | 123  | ■ 0.943 | $X_{\{G\}}$                                  | $X_G$                                        | for $G$ is the same for all such forests (exercise!), therefore $\Psi_G(\underline{t})$ is homoge-<br>neous. Thus, its vanishing defines a hypersurface $X_G$ in $\mathbb{P}^{n-1}$ ( $n$ =number    |
| cardona-qft-methods_F01781 | 123  | ■ 0.943 | $\mathbb{P}^{n-1}$                           | $\mathbb{P}^{n-1}$                           | for $G$ is the same for all such forests (exercise!), therefore $\Psi_G(\underline{t})$ is homoge-<br>neous. Thus, its vanishing defines a hypersurface $X_G$ in $\mathbb{P}^{n-1}$ ( $n$ =number    |
| cardona-qft-methods_F01782 | 123  | ■ 0.943 | $n =$                                        | $n =$                                        | for $G$ is the same for all such forests (exercise!), therefore $\Psi_G(\underline{t})$ is homoge-<br>neous. Thus, its vanishing defines a hypersurface $X_G$ in $\mathbb{P}^{n-1}$ ( $n$ =number    |

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| cardona-qft-methods_F01783 | 123  | ■ 1.000 | $Y_{\{G\}}$                                                           | $Y_G$                                       | corresponding to the points of $\Lambda_G$ ). We will denote by $Y_G$ the complement of $\bar{Y}_G$ in $\mathbb{P}^{n-1}$ and by $\bar{Y}_G$ the complement of $\bar{Y}_G$ in $\mathbb{A}^n$ . Of course each of |
| cardona-qft-methods_F01784 | 123  | ■ 1.000 | $b_{\{1\}}(G)$                                                        | $b_1(G)$                                    | For very small graphs, $\chi_G$ may be nonnegative. In the example of a 3-banana given above, the corresponding $X_G$ is a nonsingular conic in $\mathbb{P}^2$ .                                                 |
| cardona-qft-methods_F01785 | 123  | ■ 1.000 | $G \backslash e$                                                      | $G \backslash e$                            | irreducible; and it is known (see for example, <sup>SMJ077</sup> Proposition 5.2) that                                                                                                                           |
| cardona-qft-methods_F01786 | 123  | ■ 0.980 | $\{ \}^{\{\text{\texttt{Pat10}}\}}$                                   | Pat10                                       | $\Psi_G$ is reducible if and only if $G$ can be ‘separated’ as a union of disjoint subgraphs or graphs joined at a vertex. Also see MA §5 for a discussion of                                                    |
| cardona-qft-methods_F01787 | 123  | ■ 0.974 | $\{ \}^{\{\text{\texttt{SMJO}}\} \, 77}$                              | SMJO 77                                     | $X_G$ should determine a mixed Tate motive (over $\mathbb{A}^1$ ). As a consequence, the Grothendieck class $[\bar{Y}_G]$ should belong to the subring $\mathbb{Z}[\mathbb{L}]$ of the                           |
| cardona-qft-methods_F01788 | 123  | ■ 0.998 | $\{ \}^{\{\text{\texttt{MA}}\}}$                                      | MA                                          | countable (cf. Definition 5, Example 2.1). That is, as it is defined over $\mathbb{Z}$ , $\Psi_G$ defines a polynomial in $\mathbb{F}_q[t_1, \dots, t_n]$ for all finite fields $\mathbb{F}_q$ (here $q$ is      |
| cardona-qft-methods_F01789 | 123  | ■ 1.000 | $\left[\hat{Y}_{\{G\}}\right]$                                        | $\left[\hat{Y}_G\right]$                    | In this sense $G$ is ‘polynomially countable’. The conjecture on the mixed-                                                                                                                                      |
| cardona-qft-methods_F01790 | 123  | ■ 1.000 | $K_{\{0\}}(\operatorname{Var}_{\{\mathbb{Z}\}})$                      | $K_0(\operatorname{Var}_{\mathbb{Z}})$      | ‘finiteness’ of $Y_G$ can therefore be tested by verifying whether $G$ is polynomially countable. The physicists’ evidence mentioned in the previous section                                                     |
| cardona-qft-methods_F01791 | 123  | ■ 1.000 | $\mathbb{F}_{\{q\}}\left[t_{\{1\}}, \ldots, t_{\{n\}}\right]$         | $\mathbb{F}_q[t_1, \dots, t_n]$             |                                                                                                                                                                                                                  |
| cardona-qft-methods_F01792 | 123  | ■ 1.000 | $N_{\{q\}}(G)$                                                        | $N_q(G)$                                    |                                                                                                                                                                                                                  |
| cardona-qft-methods_F01793 | 123  | ■ 1.000 | $\left(t_{\{1\}}, \ldots, t_{\{n\}}\right)$                           | $(t_1, \dots, t_n)$                         |                                                                                                                                                                                                                  |
| cardona-qft-methods_F01794 | 123  | ■ 1.000 | $\mathbb{F}_{\{q\}}^{\{n\}}$                                          | $\mathbb{F}_q^n$                            |                                                                                                                                                                                                                  |
| cardona-qft-methods_F01795 | 123  | ■ 1.000 | $\Psi_{\{G\}}(\bar{t}) \neq 0$                                        | $\Psi_G(\bar{t}) \neq 0$                    |                                                                                                                                                                                                                  |
| cardona-qft-methods_F01796 | 124  | ■ 0.977 | $\{ \}^{\{\text{\texttt{Sta98}}\}}$                                   | Sta98                                       | motivated the conjecture (attributed to Kontsevich) that all graphs would                                                                                                                                        |
| cardona-qft-methods_F01797 | 124  | ■ 0.977 | $\{ \}^{\{\text{\texttt{Ste98}}\}}$                                   | Ste98                                       | motivated the conjecture (attributed to Kontsevich) that all graphs would                                                                                                                                        |
| cardona-qft-methods_F01798 | 124  | ■ 0.997 | $\{ \}^{\{\text{\texttt{BB03}}\}}$                                    | BB03                                        | to provide very substantial evidence for the conjecture: all graphs with 12 or fewer edges were found to satisfy it. But the conjecture was soon <i>dis-</i>                                                     |
| cardona-qft-methods_F01799 | 124  | ■ 0.999 | $\{ \}^{\{\text{\texttt{Dor11Sch11}}\}}$                              | Dor11Sch11                                  | proved. Subsequently, it has been shown that all graphs with 10 or fewer edges satisfy the conjecture, and specific graphs $G$ with 14 and 16 edges                                                              |
| cardona-qft-methods_F01800 | 124  | ■ 1.000 | $\mathbb{P}^{\{3\}}$                                                  | $\mathbb{P}^3$                              | The polynomial $\Psi_G$ may be written as a determinant of a matrix easily concocted from the graph and recording which edges belong to a fixed basis                                                            |
| cardona-qft-methods_F01801 | 124  | ■ 1.000 | $\{ \}^{\{\text{\texttt{BS12}}\}}$                                    | BS12                                        | for $H^1(G)$ . Thus, graph hypersurfaces may be viewed as specie                                                                                                                                                 |
| cardona-qft-methods_F01802 | 124  | ■ 1.000 | $H^{\{1\}}(G)$                                                        | $H^1(G)$                                    | $Y_G$ is not mixed-Tate, it may be that the part of its cohomology that is responsible for the period is mixed-Tate. Realizing the period as a mixed Tate                                                        |
| cardona-qft-methods_F01803 | 124  | ■ 1.000 | $\{ \}^{\{\text{\texttt{AM10}}\}}$                                    | AM10                                        | thing $\langle G \rangle = \chi_G(\text{pt})$ may have the interpretation of being the ‘naïve’ version of something to say about the other ‘generalized Euler characteristic’ considered                         |
| cardona-qft-methods_F01804 | 124  | ■ 1.000 | $\{ \}^{\{\text{\texttt{Vak06}}\}}$                                   | Vak06                                       |                                                                                                                                                                                                                  |
| cardona-qft-methods_F01805 | 124  | ■ 0.948 | $K_{\{0\}}(\operatorname{Var}_{\{\}})$                                | $K_0(\operatorname{Var})$                   |                                                                                                                                                                                                                  |
| cardona-qft-methods_F01806 | 125  | ■ 0.990 | $c_{\{\mathrm{SM}\}}\left(X_{\{G\}}\right)$                           | $c_{\mathrm{SM}}(X_G)$                      |                                                                                                                                                                                                                  |
| cardona-qft-methods_F01807 | 125  | ■ 0.749 | $\sum_{\{\text{\texttt{graphs}}\}}\{ \}_{\{G\}}\lrcorner G \lrcorner$ | $\sum_{\text{graphs } G} \langle G \rangle$ | a symmetry factor. If $G_i$ are the connected components of $G$ , then                                                                                                                                           |
| cardona-qft-methods_F01808 | 125  | ■ 0.999 | $G_{\{i\}}$                                                           | $G_i$                                       | $\langle G \rangle = \prod \langle G_i \rangle / \text{extra symmetries}$ .                                                                                                                                      |

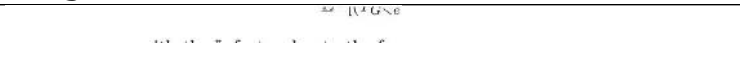
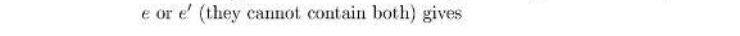
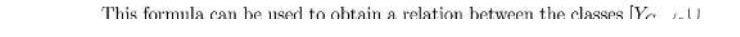
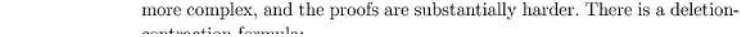
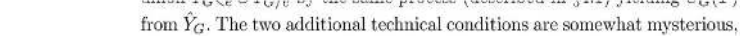
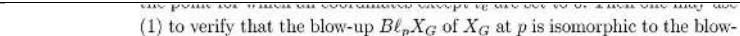
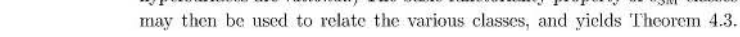
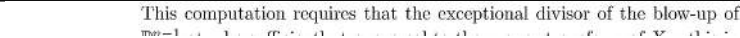
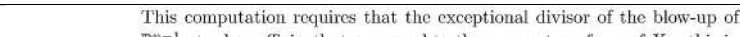
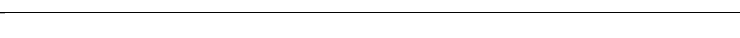
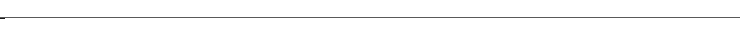
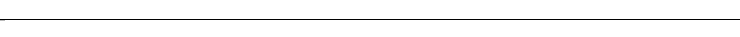
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| cardona-qft-methods_F01889 | 125  | 1.000 | $G_{\{1\}}$                                                                                         | $G_1$                                                    | to the other across the bridge joining the two graphs.<br>These reductions are parts of the lore of ‘Earmann rules’ – a set of maxims                                                     |
| cardona-qft-methods_F01810 | 125  | 1.000 | $G_{\{2\}}$                                                                                         | $G_2$                                                    | to the other across the bridge joining the two graphs.<br>These reductions are parts of the lore of ‘Earmann rules’ – a set of maxims                                                     |
| cardona-qft-methods_F01811 | 126  | 1.000 | $\{ \}^{\text{\texttt{\text{AM11a}}}}$                                                              | AM11a                                                    | directly some information carried by the combinatorics of the graph.<br>This material is addressed in more detail in AM11a (The reader should be                                          |
| cardona-qft-methods_F01812 | 126  | 1.000 | $G_{\{1\}}, G_{\{2\}}$                                                                              | $G_1, G_2$                                               | by bridges:                                                                                            |
| cardona-qft-methods_F01813 | 126  | 1.000 | $\Psi_G = \left( \Psi_{G_{\{1\}}} \right) \left( \Psi_{G_{\{2\}}} \right)$                          | $\Psi_G = (\Psi_{G_1}) (\Psi_{G_2})$                     | <b>Lemma 4.</b>                                                                                                                                                                           |
| cardona-qft-methods_F01814 | 127  | 1.000 | $t_{\{4\}}$                                                                                         | $t_4$                                                    |                                                                                                        |
| cardona-qft-methods_F01815 | 127  | 0.988 | $\left( t_{\{1\}} + t_{\{2\}} + t_{\{3\}} \right) \left( t_{\{5\}} + t_{\{6\}} + t_{\{7\}} \right)$ | $(t_1 + t_2 + t_3) (t_5 + t_6 + t_7)$                    |                                                                                                        |
| cardona-qft-methods_F01816 | 127  | 1.000 | $\Psi_G \left( t_{\{1\}}, \ldots, t_{\{n\}} \right) = 0$                                            | $\Psi_G(t_1, \dots, t_n) = 0$                            | <b>Prop. 4.1.</b> Let $G_1, G_2$ be graphs                                                                                                                                                |
| cardona-qft-methods_F01817 | 127  | 1.000 | $\mathbb{A}^n \backslash \hat{X}_G$                                                                 | $\mathbb{A}^n \backslash \hat{X}_G$                      | <br>                |
| cardona-qft-methods_F01818 | 127  | 1.000 | $G = G_{\{1\}} \amalg G_{\{2\}}$                                                                    | $G = G_1 \amalg G_2$                                     | <b>Proof.</b> Let $t_1, \dots, t_{n_1}$ be the variables correspon                                                                                                                        |
| cardona-qft-methods_F01819 | 127  | 1.000 | $\hat{Y}_G \cong \hat{Y}_{G_{\{1\}}} \times \hat{Y}_{G_{\{2\}}}$                                    | $\hat{Y}_G \cong \hat{Y}_{G_1} \times \hat{Y}_{G_2}$     | <b>Proof.</b> Let $t_1, \dots, t_{n_1}$ be the variables correspon                                                                                                                        |
| cardona-qft-methods_F01820 | 127  | 1.000 | $\hat{Y}_G \cong$                                                                                   | $\hat{Y}_G \cong$                                        | $G_1$ , and let $u_1, \dots, u_{n_2}$ be the variables corresponding to the edges of $G_2$ . If $G = G_1 \amalg G_2$ , then $\Psi_G = \Psi_{G_1} \Psi_{G_2}$ , as observed above. A point |
| cardona-qft-methods_F01821 | 127  | 1.000 | $\hat{Y}_{G_{\{1\}}} \times \hat{Y}_{G_{\{2\}}} \times \mathbb{A}^1$                                | $\hat{Y}_{G_1} \times \hat{Y}_{G_2} \times \mathbb{A}^1$ | $(t_1, \dots, t_{n_1}, u_1, \dots,$                                                                                                                                                       |
| cardona-qft-methods_F01822 | 127  | 1.000 | $t_{\{1\}}, \ldots, t_{\{n_{\{1\}}\}}$                                                              | $t_1, \dots, t_{n_1}$                                    | if and only if                                                                                                                                                                            |
| cardona-qft-methods_F01823 | 127  | 1.000 | $u_{\{1\}}, \ldots, u_{\{n_{\{2\}}\}}$                                                              | $u_1, \dots, u_{n_2}$                                    | $\Psi_{G_1}(t_1, \dots, t_{n_1}) \Psi_{G_2}(u_1, \dots, u_{n_2}) \neq 0$ ,                                                                                                                |
| cardona-qft-methods_F01824 | 127  | 1.000 | $\Psi_G = \Psi_{G_{\{1\}}} \Psi_{G_{\{2\}}}$                                                        | $\Psi_G = \Psi_{G_1} \Psi_{G_2}$                         | if and only if                                                                                                                                                                            |
| cardona-qft-methods_F01825 | 127  | 1.000 | $\left( t_{\{1\}}, \ldots, t_{\{n_{\{1\}}\}}, u_{\{1\}}, \ldots, u_{\{n_{\{2\}}\}} \right)$         | $(t_1, \dots, t_{n_1}, u_1, \dots, u_{n_2})$             | $\Psi_{G_1}(t_1, \dots, t_{n_1}) \neq 0$ and $\Psi_{G_2}(u_1, \dots, u_{n_2}) \neq 0$ .                                                                                                   |
| cardona-qft-methods_F01826 | 127  | 1.000 | $\mathbb{A}^{n_1+n_2}$                                                                              | $\mathbb{A}^{n_1+n_2}$                                   | $\Psi_{G_1}(t_1, \dots, t_{n_1}) \neq 0$ and $\Psi_{G_2}(u_1, \dots, u_{n_2}) \neq 0$ .                                                                                                   |
| cardona-qft-methods_F01827 | 127  | 1.000 | $\left( t_{\{1\}}, \ldots, t_{\{n_{\{1\}}\}} \right) \in \hat{Y}_{G_{\{1\}}}$                       | $(t_1, \dots, t_{n_1}) \in \hat{Y}_{G_1}$                |                                                                                                                                                                                           |
| cardona-qft-methods_F01828 | 127  | 1.000 | $\left( u_{\{1\}} \ldots, u_{\{n_{\{2\}}\}} \right) \in \hat{Y}_{G_{\{2\}}}$                        | $(u_1 \ldots, u_{n_2}) \in \hat{Y}_2$                    |                                                                                                                                                                                           |
| cardona-qft-methods_F01829 | 128  | 1.000 | $\mathbb{A}^1$                                                                                      | $\mathbb{A}^1$                                           |                                                                                                                                                                                           |
| cardona-qft-methods_F01830 | 128  | 1.000 | $G \mapsto U(G)$                                                                                    | $G \mapsto U(G)$                                         | us (and for lack of a better term), ‘abstract Feynman rules’ con                                                                                                                          |
| cardona-qft-methods_F01831 | 128  | 1.000 | $U(G) = U \left( G_{\{1\}} \right) U \left( G_{\{2\}} \right)$                                      | $U(G) = U(G_1) U(G_2)$                                   | assignment from graphs to rings $G \mapsto U(G)$ such that                                                                                                                                |
| cardona-qft-methods_F01832 | 128  | 1.000 | $U(G) = U \left( G_{\{1\}} \right) U \left( G_{\{2\}} \right) U(e)$                                 | $U(G) = U(G_1) U(G_2) U(e)$                              | vertex, then $U(G) = U(G_1) U(G_2)$ .                                                                                                                                                     |
| cardona-qft-methods_F01833 | 128  | 1.000 | $U(e)$                                                                                              | $U(e)$                                                   | by joining $G_1$ and $G_2$ by a bridge $e$ . This identifies $U(e)$ as the (inverse) propagator .<br>Abstract Feynman rules are of course a dime for a dozen: for example,                |
| cardona-qft-methods_F01834 | 128  | 1.000 | $\mathcal{SB}(G) \in \mathbb{Z}$                                                                    | $\mathcal{SB}(G) \in \mathbb{Z}$                         | and independently defined for all graphs, and verify that these happen                                                                                                                    |

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| cardona-qft-methods_F01835 | 128  | ■ 1.000 | $(-1)^{\{n\}}$                             | $(-1)^n$                                   | to satisfy the multiplicativity prescribed by the rules. We are interested in those invariants that arise from algebraic/combinatorial properties of graphs.                             |
| cardona-qft-methods_F01836 | 128  | ■ 1.000 | $F \rightarrowtail R$                      | $F \rightarrow R$                          |                                                                                                                                                                                          |
| cardona-qft-methods_F01837 | 129  | ■ 1.000 | $\mathcal{SB}(G)$                          | $SB(G)$                                    | Two examples are discussed below; there are others, and they await the reader.                                                                                                           |
| cardona-qft-methods_F01838 | 129  | ■ 1.000 | $\chi(Y_G)$                                | $\chi(Y_G)$                                | Also, it may be useful to point out that meaningful algebro-geometric invariants need not be Feynman rules. My favorite example is the Euler                                             |
| cardona-qft-methods_F01839 | 129  | ■ 0.960 | $[Y_G]$                                    | $[Y_G]$                                    | projective space: if $G$ is the disjoint union of $G_1$ and $G_2$ , and neither $G_1$ nor $G_2$ is a forest, then the class $[Y_G]$ is a multiple of $(T+1)$ (overruled) so              |
| cardona-qft-methods_F01840 | 129  | ■ 0.960 | $(\mathbb{L}-1)$                           | $(\mathbb{L}-1)$                           | projective space: if $G$ is the disjoint union of $G_1$ and $G_2$ , and neither $G_1$ nor $G_2$ is a forest, then the class $[Y_G]$ is a multiple of $(T+1)$ (overruled) so              |
| cardona-qft-methods_F01841 | 129  | ■ 1.000 | $\chi(Y_G)=0$                              | $\chi(Y_G)=0$                              | $\chi(Y_G)=0$ no matter what $\chi(Y_{G_1})$ and $\chi(Y_{G_2})$ may be. By itself, this in-                                                                                             |
| cardona-qft-methods_F01842 | 129  | ■ 1.000 | $\chi(Y_{G_1})$                            | $\chi(Y_{G_1})$                            | $\chi(Y_G)=0$ no matter what $\chi(Y_{G_1})$ and $\chi(Y_{G_2})$ may be. By itself, this in-                                                                                             |
| cardona-qft-methods_F01843 | 129  | ■ 1.000 | $\chi(Y_{G_2})$                            | $\chi(Y_{G_2})$                            | $\chi(Y_G)=0$ no matter what $\chi(Y_{G_1})$ and $\chi(Y_{G_2})$ may be. By itself, this in-                                                                                             |
| cardona-qft-methods_F01844 | 129  | ■ 1.000 | $\mathbb{U}(G):=\left[\hat{Y}_G\right]$    | $\mathbb{U}(G):=\left[\hat{Y}_G\right]$    | this assignment gives ‘abstract Feynman rules’ in the sense specified above. The ‘motivic propagator’ is the class $\mathbb{L}=(T+1)$ of $\mathbb{A}^1$ . It is a good                   |
| cardona-qft-methods_F01845 | 129  | ■ 0.999 | $\mathbb{L}=(T+1)$                         | $\mathbb{L}=(T+1)$                         | While this example is very straightforward, the reader should note that                                                                                                                  |
| cardona-qft-methods_F01846 | 129  | ■ 1.000 | $\mathbb{U}(G)$                            | $\mathbb{U}(G)$                            | variations on this theme—such as $[\Lambda G]$ , $[\Lambda G]$ , or the class $[X_G]$ of the hypersurface complement in <i>projective</i> space, would not work. The class $[\hat{Y}_G]$ |
| cardona-qft-methods_F01847 | 129  | ■ 0.516 | $(T+1)$                                    | $(T+1)$                                    | of the affine complement is just right.                                                                                                                                                  |
| cardona-qft-methods_F01848 | 129  | ■ 1.000 | $\mathbb{T}$                               | $\mathbb{T}$                               | —Polynomial algebro-geometric Feynman rules:                                                                                                                                             |
| cardona-qft-methods_F01849 | 129  | ■ 1.000 | $\left[\hat{X}_G\right],[X_G]$             | $\left[\hat{X}_G\right],[X_G]$             | corresponding Feynman rules were defined in Fig. 1.00 they may be interpreted as (truncated) polynomials in $\mathbb{Z}[T]$ .                                                            |
| cardona-qft-methods_F01850 | 129  | ■ 1.000 | $\hat{X}$                                  | $\hat{X}$                                  |                                                                                                                                                                                          |
| cardona-qft-methods_F01851 | 129  | ■ 0.996 | $\mathbb{P}^n$                             | $\mathbb{P}^n$                             |                                                                                                                                                                                          |
| cardona-qft-methods_F01852 | 129  | ■ 1.000 | $F_{\hat{X}}(T):=a_0+a_1 T+\cdots+a_n T^n$ | $F_{\hat{X}}(T):=a_0+a_1 T+\cdots+a_n T^n$ |                                                                                                                                                                                          |
| cardona-qft-methods_F01853 | 129  | ■ 1.000 | $F_{\mathbb{A}^n}(T)$                      | $F_{\mathbb{A}^n}(T)$                      |                                                                                                                                                                                          |
| cardona-qft-methods_F01854 | 130  | ■ 1.000 | $F_{\mathbb{A}^n}(T)=(1+T)^n$              | $F_{\mathbb{A}^n}(T)=(1+T)^n$              |                                                                                                                                                                                          |
| cardona-qft-methods_F01855 | 130  | ■ 1.000 | $F \rightarrowtail \mathbb{Z}[T]$          | $F \rightarrow \mathbb{Z}[T]$              | <b>Prop 4.2 (Easy).</b> $F$ defines a group homomorphism $F \rightarrow \mathbb{Z}[T]$ .                                                                                                 |
| cardona-qft-methods_F01856 | 130  | ■ 1.000 | $F \rightarrowtail$                        | $F \rightarrow$                            |                                                                                                                                                                                          |
| cardona-qft-methods_F01857 | 130  | ■ 1.000 | $\left[\mathbb{P}^i\right]$                | $\left[\mathbb{P}^i\right]$                | Once Theorem 4.1 is established, we can define ‘polynomial’, or ‘csm’ Feynman rules by associating with a graph $G$ the polynomial $C_G(T):=$                                            |
| cardona-qft-methods_F01858 | 130  | ■ 1.000 | $\left[\mathbb{P}^j\right]$                | $\left[\mathbb{P}^j\right]$                | $F_{Y_G}(T) \in \mathbb{Z}[T]$ . The csm propagator is the polynomial $C_{A^1}(T)=1+T$ .                                                                                                 |
| cardona-qft-methods_F01859 | 130  | ■ 1.000 | $\left[\mathbb{P}^{i+j}\right]$            | $\left[\mathbb{P}^{i+j}\right]$            | $F_{Y_G}(T) \in \mathbb{Z}[T]$ . The csm propagator is the polynomial $C_{A^1}(T)=1+T$ .                                                                                                 |
| cardona-qft-methods_F01860 | 130  | ■ 1.000 | $i+j$                                      | $i+j$                                      | $F_{Y_G}(T) \in \mathbb{Z}[T]$ . The csm propagator is the polynomial $C_{A^1}(T)=1+T$ .                                                                                                 |

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| cardona-qft-methods_F01861 | 130  | ■ 1.000 | $C_{\{G\}}(T) :=$                                                           | $C_G(T) :=$                            | <p>to verify this, note that the graph polynomials for the <math>n</math>-sided polygon <math>G</math> is simply <math>t_1 + \cdots + t_n</math>. Therefore, <math>\hat{Y}_G</math> consists of the complement of a</p> |
| cardona-qft-methods_F01862 | 130  | ■ 0.997 | $F_{\{\hat{Y}_{\{G\}}(T) \in \mathbb{Z}[T]$                                 | $F_{\hat{Y}_G}(T) \in \mathbb{Z}[T]$   | <p>hyperplane <math>\mathbb{A}^{n-1}</math> in <math>\mathbb{A}^n</math>. As verified in Example 4.3, <math>F_{A^r}(T) = (1 + T)^r</math>,</p>                                                                          |
| cardona-qft-methods_F01863 | 130  | ■ 0.997 | $C_{\{\mathbb{A}^{\{1\}}(T)=1+T$                                            | $C_{\mathbb{A}^1}(T) = 1 + T$          | <p>hyperplane <math>\mathbb{A}^{n-1}</math> in <math>\mathbb{A}^n</math>. As verified in Example 4.3, <math>F_{A^r}(T) = (1 + T)^r</math>,</p>                                                                          |
| cardona-qft-methods_F01864 | 130  | ■ 1.000 | $C_{\{G\}}(T)=T(1+T)^{n-1}$                                                 | $C_G(T) = T(1 + T)^{n-1}$              | <p>as claimed. <math>\square</math></p>                                                                                                                                                                                 |
| cardona-qft-methods_F01865 | 130  | ■ 1.000 | $t_1+\cdots+t_n$                                                            | $t_1 + \cdots + t_n$                   |                                                                                                                                                                                                                         |
| cardona-qft-methods_F01866 | 130  | ■ 1.000 | $\mathbb{A}^{n-1}$                                                          | $\mathbb{A}^{n-1}$                     |                                                                                                                                                                                                                         |
| cardona-qft-methods_F01867 | 130  | ■ 1.000 | $F_{\{\mathbb{A}^{\{r\}}(T)=(1+T)^{\{r\}}$                                  | $F_{\mathbb{A}^r}(T) = (1 + T)^r$      |                                                                                                                                                                                                                         |
| cardona-qft-methods_F01868 | 131  | ■ 1.000 | $C_{\{G\}}(T)$                                                              | $C_G(T)$                               |                                                                                                                                                                                                                         |
| cardona-qft-methods_F01869 | 131  | ■ 1.000 | $T^{n-1}$                                                                   | $T^{n-1}$                              | <p>(i) <math>C_G(T)</math> is a monic polynomial, of degree equal to the number of <math>G</math>;</p>                                                                                                                  |
| cardona-qft-methods_F01870 | 131  | ■ 1.000 | $n-b_{\{1\}}(G)$                                                            | $n - b_1(G)$                           | <p>(i) <math>C_G(T)</math> is a monic polynomial, of degree equal to the number of <math>G</math>;</p>                                                                                                                  |
| cardona-qft-methods_F01871 | 131  | ■ 0.975 | $C_{\{G\}}(T)=(1+T)^{\{n\}}$                                                | $C_G(T) = (1 + T)^n$                   | <p>(ii) The coefficient of <math>T^{n-1}</math> in <math>C_G(T)</math> equals <math>n - b_1(G)</math>;</p>                                                                                                              |
| cardona-qft-methods_F01872 | 131  | ■ 1.000 | $C_{\{G\}}(\emptyset)=\emptyset$                                            | $C_G(0) = 0$                           | <p>(iii) If <math>G</math> is a forest with <math>n</math> edges, then <math>C_G</math><br/>(iv) If <math>G</math> is not a forest, then <math>C_G(0) = 0</math>;</p>                                                   |
| cardona-qft-methods_F01873 | 131  | ■ 1.000 | $C_{\{G\}}^{\{\prime\}}(\emptyset)$                                         | $C_G'(0)$                              | <p>(v) <math>C_G'(0)</math> equals the Euler characteristic of <math>Y_G = \mathbb{P}^{n-1} \setminus X_G</math>;</p>                                                                                                   |
| cardona-qft-methods_F01874 | 131  | ■ 1.000 | $Y_{\{G\}}=\mathbb{P}^{\{n-1\}} \backslash X_{\{G\}}$                       | $Y_G = \mathbb{P}^{n-1} \setminus X_G$ | <p>(v) <math>C_G'(0)</math> equals the Euler characteristic of <math>Y_G = \mathbb{P}^{n-1} \setminus X_G</math>;</p>                                                                                                   |
| cardona-qft-methods_F01875 | 131  | ■ 1.000 | $G^{\{\prime\}}$                                                            | $G'$                                   | <p>(vi) If <math>G'</math> is obtained from <math>G</math> by attaching an edge or splitting an edge, then <math>C_{G'}(T) = (1 + T)C_G(T)</math>;</p>                                                                  |
| cardona-qft-methods_F01876 | 131  | ■ 0.829 | $C_{\{G^{\{\prime\}}\}}(T)=(1+T) C_{\{G\}}(T)$                              | $C_{G'}(T) = (1 + T)C_G(T)$            | <p>(vii) If <math>G'</math> is obtained from</p>                                                                                                                                                                        |
| cardona-qft-methods_F01877 | 131  | ■ 1.000 | $C_{\{G^{\{\prime\}}\}}(T)=$                                                | $C_{G'}(T) =$                          | <p><math>TC_G(T)</math>;<br/>(viii) If <math>G</math> is not 1-particle irreducible, then <math>C_G(-1) = 0</math></p>                                                                                                  |
| cardona-qft-methods_F01878 | 131  | ■ 1.000 | $T C_{\{G\}}(T)$                                                            | $TC_G(T)$                              | <p>1</p>                                                                                                                                                                                                                |
| cardona-qft-methods_F01879 | 131  | ■ 0.999 | $C_{\{G\}}(-1)=\emptyset$                                                   | $C_G(-1) = 0$                          | <p>(again) <math>C_G(T)</math> for an <math>n</math>-sided polygon, start from a single vertex</p>                                                                                                                      |
| cardona-qft-methods_F01880 | 131  | ■ 1.000 | $n-1$                                                                       | $n - 1$                                | <p><math>C_{G_2}(T) = \chi(Y_{G_2})T + \text{h.o.t.}</math> By Theorem 4.1,</p>                                                                                                                                         |
| cardona-qft-methods_F01881 | 131  | ■ 1.000 | $T(1+T)^{n-1}$                                                              | $T(1 + T)^{n-1}$                       | <p><math>C_{G_1 \amalg G_2}(T) = \chi(Y_{G_1})\chi(Y_{G_2})T^2 + \text{h.o.t.}</math> <math>\square</math></p>                                                                                                          |
| cardona-qft-methods_F01882 | 131  | ■ 1.000 | $C_{\{G_{\{1\}}\}}(T)=\chi\!\left(Y_{\{G_{\{1\}}\}}\right) T+$              | $C_{G_1}(T) = \chi(Y_{G_1})T +$        | <p>of <math>T^2</math> in the corresponding <math>c_{\text{SM}}</math> Feynman rules.</p>                                                                                                                               |
| cardona-qft-methods_F01883 | 131  | ■ 1.000 | $C_{\{G_{\{2\}}\}}(T)=\chi\!\left(Y_{\{G_{\{2\}}\}}\right) T+$              | $C_{G_2}(T) = \chi(Y_{G_2})T +$        | <p>Incidentally: In all examples I know, <math>\chi(Y_G)</math> is number of computations for small graphs are odd</p>                                                                                                  |
| cardona-qft-methods_F01884 | 131  | ■ 1.000 | $\chi\!\left(Y_{\{G_{\{1\}}\}} \amalg G_{\{2\}}\right)\!=\!0$               | $\chi(Y_{G_1 \amalg G_2}) = 0$         | <p>nonzero coefficient of <math>U_G(T)</math> should always be <math>\pm 1</math>.<br/>If <math>G</math> is not a forest, then one may verify that <math>C_G(T)</math> is the polynomial</p>                            |
| cardona-qft-methods_F01885 | 131  | ■ 1.000 | $\chi\!\left(Y_{\{G_{\{1\}}\}}\right) \chi\!\left(Y_{\{G_{\{2\}}\}}\right)$ | $\chi(Y_{G_1})\chi(Y_{G_2})$           | <p>obtained from <math>c_{\text{SM}}(\mathbb{1}_{\mathbb{P}^{n-1} \setminus X_G})</math> by replacing <math>[\mathbb{P}^k]</math> with <math>T^{k+1}</math>. Naive varia-</p>                                           |
| cardona-qft-methods_F01886 | 131  | ■ 1.000 | $T^{\{2\}}$                                                                 | $T^2$                                  | <p>tions on this recipe, e.g., using <math>c_{\text{SM}}(\mathbb{1}_{X_G})</math> instead, since</p>                                                                                                                    |

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| cardona-qft-methods_F01887 | 131  | 0.803 | <code>\pm 1</code>                                                                                                                                              | $\pm 1$                                                                                                                                              |                                                                                                                       |
| cardona-qft-methods_F01888 | 131  | 1.000 | <code>{ }^{\text {Str11 }}</code>                                                                                                                               | Str11                                                                                                                                                | Example 4.5. Let $G$ be the 3-banana graph:                                                                                                                                                              |
| cardona-qft-methods_F01889 | 131  | 1.000 | <code>c_{\mathrm{SM}}\left(\mathrm{1}_{\mathrm{P}^{n-1} \backslash X_G}\right)</code>                                                                           | $c_{\mathrm{SM}}\left(\mathbb{1}_{\mathbb{P}^{n-1} \backslash X_G}\right)$                                                                           |                                                                                                                                                                                                          |
| cardona-qft-methods_F01890 | 131  | 1.000 | <code>\left[\mathrm{P}^k\right]</code>                                                                                                                          | $\left[\mathbb{P}^k\right]$                                                                                                                          |                                                                                                                                                                                                          |
| cardona-qft-methods_F01891 | 131  | 1.000 | <code>T^{k+1}</code>                                                                                                                                            | $T^{k+1}$                                                                                                                                            |                                                                                                                                                                                                          |
| cardona-qft-methods_F01892 | 131  | 0.997 | <code>c_{\mathrm{SM}}\left(\mathrm{1}_{X_G}\right)</code>                                                                                                       | $c_{\mathrm{SM}}\left(\mathbb{1}_{X_G}\right)$                                                                                                       |                                                                                                                                                                                                          |
| cardona-qft-methods_F01893 | 132  | 1.000 | <code>t_{1} t_2+t_{1} t_{3}+t_{2} t_3</code>                                                                                                                    | $t_1 t_2+t_1 t_3+t_2 t_3$                                                                                                                            |                                                                                                                       |
| cardona-qft-methods_F01894 | 132  | 1.000 | <code>c_{\mathrm{SM}}\left(\mathrm{1}_{X_G}\right)=</code>                                                                                                      | $c_{\mathrm{SM}}\left(\mathbb{1}_{X_G}\right)=$                                                                                                      |                                                                                                                                                                                                          |
| cardona-qft-methods_F01895 | 132  | 1.000 | <code>2\left[\mathrm{P}^0\right]+2\left[\mathrm{P}^1\right]</code>                                                                                              | $2\left[\mathbb{P}^0\right]+2\left[\mathbb{P}^1\right]$                                                                                              | The corresponding graph polynomial is $t_1 t_2+t_1 t_3+t_2 t_3$ ; therefore, the graph hypersurface $X_G$ is a nonsingular conic in $\mathbb{P}^2$ hence $c_{\mathrm{SM}}\left(\mathbb{1}_{X_G}\right)=$ |
| cardona-qft-methods_F01896 | 132  | 1.000 | <code>c_{\mathrm{SM}}\left(\mathrm{1}_{\mathrm{P}^2 \backslash X_G}\right)=\left[\mathrm{P}^0\right]+\left[\mathrm{P}^1\right]+\left[\mathrm{P}^2\right]</code> | $c_{\mathrm{SM}}\left(\mathbb{1}_{\mathbb{P}^2 \backslash X_G}\right)=\left[\mathbb{P}^0\right]+\left[\mathbb{P}^1\right]+\left[\mathbb{P}^2\right]$ | The corresponding graph polynomial is $t_1 t_2+t_1 t_3+t_2 t_3$ ; therefore, the graph hypersurface $X_G$ is a nonsingular conic in $\mathbb{P}^2$ hence $c_{\mathrm{SM}}\left(\mathbb{1}_{X_G}\right)=$ |
| cardona-qft-methods_F01897 | 132  | 0.651 | <code>C_{\{G\}}(T)=T\left(1+T+T^2\right)</code>                                                                                                                 | $C_G(T)=T\left(1+T+T^2\right)$                                                                                                                       | $2\left[\mathbb{P}^0\right]+2\left[\mathbb{P}^1\right]$ . It folk                                                                                                                                        |
| cardona-qft-methods_F01898 | 132  | 1.000 | <code>{ }^{\text {Web12 }}</code>                                                                                                                               | Web12                                                                                                                                                |                                                                                                                                                                                                          |
| cardona-qft-methods_F01899 | 132  | 1.000 | <code>{ }^{\text {Bol98 }}</code>                                                                                                                               | Bol98                                                                                                                                                | and joining the two resulting vertices by an edge. More precisely: the Tutte polynomial is $T_G(x, y)=x^i y^j$ if $G$ contains $i$ bridges, $j$ looping edges, and                                       |
| cardona-qft-methods_F01900 | 132  | 1.000 | <code>T_{\{G\}}(x, y)=x^{i} y^j</code>                                                                                                                          | $T_G(x, y)=x^i y^j$                                                                                                                                  | $G/e$ , $G \setminus e$ denote respectively the operations of contracting an edge $e$ and                                                                                                                |
| cardona-qft-methods_F01901 | 132  | 0.999 | <code>G / e, G \backslash e</code>                                                                                                                              | $G/e, G \setminus e$                                                                                                                                 | <ul style="list-style-type: none"> <li><math>\tau(G)=y \tau(G/e)</math> if <math>e</math> is a looping edge;</li> </ul>                                                                                  |
| cardona-qft-methods_F01902 | 132  | 1.000 | <code>R[\alpha, \beta, \gamma, x, y]</code>                                                                                                                     | $R[\alpha, \beta, \gamma, x, y]$                                                                                                                     | function of the Ising model; and many more. Up to a change of variables                                                                                                                                  |
| cardona-qft-methods_F01903 | 132  | 1.000 | <code>\tau(G)=\gamma^{\# \text { vertices }}</code>                                                                                                             | $\tau(G)=\gamma^{\# \text { vertices }}$                                                                                                             |                                                                                                                                                                                                          |
| cardona-qft-methods_F01904 | 132  | 1.000 | <code>\tau(G)=x \tau(G \backslash e)</code>                                                                                                                     | $\tau(G)=x \tau(G \setminus e)$                                                                                                                      |                                                                                                                                                                                                          |
| cardona-qft-methods_F01905 | 132  | 1.000 | <code>\tau(G)=y \tau(G / e)</code>                                                                                                                              | $\tau(G)=y \tau(G/e)$                                                                                                                                |                                                                                                                                                                                                          |
| cardona-qft-methods_F01906 | 132  | 0.999 | <code>\tau(G)=\alpha \tau(G / e)+\beta \tau(G \backslash e)</code>                                                                                              | $\tau(G)=\alpha \tau(G/e)+\beta \tau(G \setminus e)$                                                                                                 |                                                                                                                                                                                                          |
| cardona-qft-methods_F01907 | 133  | 1.000 | <code>\alpha=\beta=\gamma=1</code>                                                                                                                              | $\alpha=\beta=\gamma=1$                                                                                                                              |                                                                                                                     |
| cardona-qft-methods_F01908 | 133  | 1.000 | <code>b_{\{0\}}(G)</code>                                                                                                                                       | $b_0(G)$                                                                                                                                             | satisfies the recursion spelled out above, where $b_0(G)$ is the number of                                                                                                                               |
| cardona-qft-methods_F01909 | 134  | 1.000 | <code>x=1-\lambda, y=0</code>                                                                                                                                   | $x=1-\lambda, y=0$                                                                                                                                   |                                                                                                                     |
| cardona-qft-methods_F01910 | 134  | 1.000 | <code>(1-\lambda)^2+(1-\lambda)=(1-\lambda)(2-\lambda)</code>                                                                                                   | $(1-\lambda)^2+(1-\lambda)=(1-\lambda)(2-\lambda)$                                                                                                   | result mentioned above, the chromatic polynomial of a graph may be re-                                                                                                                                   |
| cardona-qft-methods_F01911 | 134  | 1.000 | <code>T_{\{G\}}(x, y)</code>                                                                                                                                    | $T_G(x, y)$                                                                                                                                          | factor is $x$ . Also, the recursive definition of $T_G(x, y)$ implies a formula for deciding a single edge in a graph: if $e$ is an edge of $G$ , denote by $G_e$ the                                    |
| cardona-qft-methods_F01912 | 134  | 1.000 | <code>G_{2 e}</code>                                                                                                                                            | $G_{2 e}$                                                                                                                                            | obtaining a single edge in a graph: if $e$ is an edge of $G$ , denote by $G_e$ the graph obtained by inserting an edge parallel to $e$ in $G$ ; then                                                     |

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| cardona-qft-methods_F01913 | 134  | ■ 0.998 | $x^2+x+y$                                                                                                    | $x^2+x+y$                                                                                                                                         | $\sum_{m>0} \frac{I_{G_{me}}(x,y)}{m!} = e^x \left( I_{G \setminus e}(x,y) + \frac{1}{y-1} I_{G/e}(x,y) \right) \quad ,$                                                |
| cardona-qft-methods_F01914 | 134  | ■ 0.998 | $x+y$                                                                                                        | $x+y$                                                                                                                                             | $\sum_{m>0} \frac{I_{G_{me}}(x,y)}{m!} = e^x \left( I_{G \setminus e}(x,y) + \frac{1}{y-1} I_{G/e}(x,y) \right) \quad ,$                                                |
| cardona-qft-methods_F01915 | 135  | ■ 0.997 | $\{ \}^{\text{\texttt{AM11b}}}$                                                                              | AM11b                                                                                                                                             | establish dublin                                                                                                                                                        |
| cardona-qft-methods_F01916 | 135  | ■ 0.997 | $\{ \}^{\text{\texttt{Alua}}}$                                                                               | Alua                                                                                                                                              | establish dublin                                                                                                                                                        |
| cardona-qft-methods_F01917 | 135  | ■ 0.453 | $\mathbb{U}(G)=\mathbb{U}(G \setminus e) + \mathbb{U}(G/e)$                                                  | $\mathbb{U}(G) = \mathbb{U}(G \setminus e) + \mathbb{U}(G/e)$                                                                                     | <ul style="list-style-type: none"> <li><math>\mathbb{U}(G_{2e}) = (\mathbb{T} - 1) \cdot \mathbb{U}(G) + \mathbb{T} \cdot \mathbb{U}(G \setminus e) +</math></li> </ul> |
| cardona-qft-methods_F01918 | 135  | ■ 1.000 | $\mathbb{U}(G \setminus e) = \mathbb{U}(G) - \mathbb{U}(G/e)$                                                | $\mathbb{U}(G_{2e}) = (\mathbb{T} - 1) \cdot \mathbb{U}(G) + \mathbb{T} \cdot \mathbb{U}(G \setminus e) + (\mathbb{T} + 1) \cdot \mathbb{U}(G/e)$ | The first formula is a version of a deletion-contraction relation for mo-                                                                                               |
| cardona-qft-methods_F01919 | 135  | ■ 1.000 | $\mathbb{U}(G \setminus e) + \mathbb{U}(G/e)$                                                                | $\hat{Y}_{G \setminus e} \cup \hat{Y}_{G/e}$                                                                                                      | motivic Feynman rules for graphs obtained from the given one by combina-<br>torial operations. The formulae are the analogues of $\mathbb{U}(G)$ for the moduli         |
| cardona-qft-methods_F01920 | 135  | ■ 1.000 | $\mathbb{L}^3 - \mathbb{L}^2 = \mathbb{T}(\mathbb{T} + 1)^2, (\mathbb{T} + 1)^2, \mathbb{T}(\mathbb{T} + 1)$ | $\mathbb{L}^3 - \mathbb{L}^2 = \mathbb{T}(\mathbb{T} + 1)^2, (\mathbb{T} + 1)^2, \mathbb{T}(\mathbb{T} + 1)$                                      | Fig. 1. $G, G \setminus e, G/e$ .                                                                                                                                       |
| cardona-qft-methods_F01921 | 135  | ■ 0.997 | $G, G \setminus e, G/e$                                                                                      | $G, G \setminus e, G/e$                                                                                                                           |                                                                                                                                                                         |
| cardona-qft-methods_F01922 | 135  | ■ 0.999 | $\mathbb{T}(\mathbb{T} + 1)^2$                                                                               | $\mathbb{T}(\mathbb{T} + 1)^2$                                                                                                                    |                                                                                                                                                                         |
| cardona-qft-methods_F01923 | 136  | ■ 1.000 | $G \setminus e, G/e$                                                                                         | $G \setminus e, G/e$                                                                                                                              | $\chi(Y_{G \setminus e}) = (-1)^{m-1}(\chi(Y_G) - \chi(Y_{G/e})) \quad .$                                                                                               |
| cardona-qft-methods_F01924 | 136  | ■ 1.000 | $G_{me}$                                                                                                     | $G_{me}$                                                                                                                                          | $\chi(Y_{G \setminus e}) = (-1)^{m-1}(\chi(Y_G) - \chi(Y_{G/e})) \quad .$                                                                                               |
| cardona-qft-methods_F01925 | 136  | ■ 1.000 | $T_y$                                                                                                        | $T_y$                                                                                                                                             | edge in $G$ , and further that $G \setminus e$ is not a forest. As mentioned in §4.1, $\chi(Y_G)$                                                                       |
| cardona-qft-methods_F01926 | 136  | ■ 1.000 | $\{ \}^{\text{\texttt{Hir95}}}$                                                                              | Hir95                                                                                                                                             |                                                                                                                                                                         |
| cardona-qft-methods_F01927 | 136  | ■ 1.000 | $\mathbb{U}(G_{me})$                                                                                         | $\mathbb{U}(G_{me})$                                                                                                                              | cannot be both nonzero and opposite in sign under these assumpt                                                                                                         |
| cardona-qft-methods_F01928 | 136  | ■ 1.000 | $m \geq 2$                                                                                                   | $m \geq 2$                                                                                                                                        | in $\mathcal{A}^{(m)}$ and, $\mathcal{A}^{(m)}$                                                                                                                         |
| cardona-qft-methods_F01929 | 136  | ■ 1.000 | $\chi(Y_{G \setminus e})$                                                                                    | $\chi(Y_{G/e})$                                                                                                                                   |                                                                                                                                                                         |
| cardona-qft-methods_F01930 | 136  | ■ 0.590 | $G = \mathbf{a}^2$                                                                                           | $G = \mathbf{a}^2$                                                                                                                                |                                                                                                                                                                         |
| cardona-qft-methods_F01931 | 136  | ■ 0.590 | $\mathbb{U}(G) = \mathbb{U}(G \setminus e) + \mathbb{U}(G/e)$                                                | $\mathbb{U}(G) = \mathbb{U}(G \setminus e) + \mathbb{U}(G/e)$                                                                                     |                                                                                                                                                                         |
| cardona-qft-methods_F01932 | 136  | ■ 0.998 | $\mathbb{U}(G/e) = \mathbb{U}(G) - \mathbb{U}(G \setminus e)$                                                | $\mathbb{U}(G/e) = \mathbb{U}(G) - \mathbb{U}(G \setminus e)$                                                                                     |                                                                                                                                                                         |
| cardona-qft-methods_F01933 | 136  | ■ 0.998 | $G_{me} =$                                                                                                   | $G_{me} =$                                                                                                                                        |                                                                                                                                                                         |
| cardona-qft-methods_F01934 | 136  | ■ 1.000 | $(m+1)$                                                                                                      | $(m+1)$                                                                                                                                           |                                                                                                                                                                         |
| cardona-qft-methods_F01935 | 137  | ■ 1.000 | $G \setminus e$                                                                                              | $G/e$                                                                                                                                             | of $G$ not containing $e$ and the maximal forests of $G \setminus e$ , and a one-to-one                                                                                 |
| cardona-qft-methods_F01936 | 137  | ■ 0.998 | $\underline{t} \in \mathbb{A}^n, n =$                                                                        | $\underline{t} \in \mathbb{A}^n, n =$                                                                                                             | Now let $\underline{t} \in \mathbb{A}^n$ , $n$ =number of edges of $G$ . We have                                                                                        |
| cardona-qft-methods_F01937 | 137  | ■ 1.000 | $\underline{t}'$                                                                                             | $\underline{t}'$                                                                                                                                  | the first formula in Theorem 4.2. For example, the case in which $\underline{t}' \notin Y_{G \setminus e}$                                                              |
| cardona-qft-methods_F01938 | 137  | ■ 1.000 | $\underline{t}' \notin Y_{G \setminus e}$                                                                    | $\underline{t}' \notin Y_{G \setminus e}$                                                                                                         | $\mathbb{U}(G) = \mathbb{U}(G \setminus e) + \mathbb{U}(G/e)$                                                                                                           |

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| cardona-qft-methods_F01939 | 137  | ■ 1.000 | $\underline{t}^{\prime} \in Y_{G/e}$           | $\underline{t}' \in Y_{G/e}$                   |    |
| cardona-qft-methods_F01940 | 137  | ■ 1.000 | $\mathbb{L}$                                   | $\mathbb{L}$                                   |    |
| cardona-qft-methods_F01941 | 137  | ■ 1.000 | $e^{\prime}$                                   | $e'$                                           |    |
| cardona-qft-methods_F01942 | 137  | ■ 0.999 | $\left[Y_{G_{2e}/e} \cup \right.$              | $\left[Y_{G_{2e}/e} \cup \right.$              |    |
| cardona-qft-methods_F01943 | 137  | ■ 0.950 | $\left.Y_{G_{2e}} \backslash e\right]$         | $\left.Y_{G_{2e}} \backslash e\right]$         |    |
| cardona-qft-methods_F01944 | 137  | ■ 0.950 | $\left[Y_{G/e} \cup Y_{G \backslash e}\right]$ | $\left[Y_{G/e} \cup Y_{G \backslash e}\right]$ |    |
| cardona-qft-methods_F01945 | 138  | ■ 1.000 | $C_{G \backslash e, G/e}(T)$                   | $C_{G \backslash e, G/e}(T)$                   |    |
| cardona-qft-methods_F01946 | 138  | ■ 1.000 | $\Psi_{G/e}(T)$                                | $\Psi_{G/e}(T)$                                |    |
| cardona-qft-methods_F01947 | 138  | ■ 1.000 | $\Psi_{G \backslash e}$                        | $\Psi_{G \backslash e}$                        |    |
| cardona-qft-methods_F01948 | 138  | ■ 0.573 | $C_{3-\text{banana}}(T)=$                      | $C_{3-\text{banana}}(T)=$                      |    |
| cardona-qft-methods_F01949 | 138  | ■ 1.000 | $T\left(1+T+T^2\right)$                        | $T\left(1+T+T^2\right)$                        |    |
| cardona-qft-methods_F01950 | 139  | ■ 1.000 | $m \geq 1$                                     | $m \geq 1$                                     |    |
| cardona-qft-methods_F01951 | 139  | ■ 1.000 | $p \in \mathbb{P}^{n-1}$                       | $p \in \mathbb{P}^{n-1}$                       |    |
| cardona-qft-methods_F01952 | 139  | ■ 1.000 | $B_{\ell, p} X_G$                              | $B_{\ell, p} X_G$                              |  |
| cardona-qft-methods_F01953 | 139  | ■ 1.000 | $\mathbb{P}^{n-2}$                             | $\mathbb{P}^{n-2}$                             |  |
| cardona-qft-methods_F01954 | 139  | ■ 1.000 | $X_{G \backslash e} \cap X_{G/e}$              | $X_{G \backslash e} \cap X_{G/e}$              |  |
| cardona-qft-methods_F01955 | 139  | ■ 0.998 | $X \times \mathbb{P}^{\ell}$                   | $X \times \mathbb{P}^{\ell}$                   |  |
| cardona-qft-methods_F01956 | 139  | ■ 0.998 | $Y \times \mathbb{P}^m$                        | $Y \times \mathbb{P}^m$                        |  |
| cardona-qft-methods_F01957 | 139  | ■ 1.000 | $\ell, m$                                      | $\ell, m$                                      |  |
| cardona-qft-methods_F01958 | 140  | ■ 1.000 | $X_1$                                          | $X_1$                                          |  |
| cardona-qft-methods_F01959 | 140  | ■ 1.000 | $X_2$                                          | $X_2$                                          |  |
| cardona-qft-methods_F01960 | 140  | ■ 1.000 | $Y_1$                                          | $Y_1$                                          |  |
| cardona-qft-methods_F01961 | 140  | ■ 1.000 | $Y_2$                                          | $Y_2$                                          |  |
| cardona-qft-methods_F01962 | 140  | ■ 1.000 | $X_1 \times Y_1$                               | $X_1 \times Y_1$                               |  |
| cardona-qft-methods_F01963 | 140  | ■ 1.000 | $X_2 \times Y_2$                               | $X_2 \times Y_2$                               |  |
| cardona-qft-methods_F01964 | 140  | ■ 1.000 | $\geq 2$                                       | $\geq 2$                                       |  |
| cardona-qft-methods_F01965 | 140  | ■ 1.000 | $\{ \}^{\text{AKMW02}}$                        | AKMW02                                         |  |

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| cardona-qft-methods_F01966 | 140  | 0.873 | $\left(\text{\texttt{LL03}}\right)$                                                     | $\left(\text{\texttt{LL03}}\right)$                                           | <ul style="list-style-type: none"> <li>Then let <math>\mathbb{Z}[SB]</math> be the corresponding monoid ring.</li> </ul>                                                                                          |
| cardona-qft-methods_F01967 | 140  | 1.000 | $S \times B$                                                                            | $SB$                                                                          | be written as formal integer linear combinations of elements $[X]_{SB}$ ; if $X$ is                                                                                                                               |
| cardona-qft-methods_F01968 | 140  | 1.000 | $X \times Y$                                                                            | $X \times Y$                                                                  | smooth irreducible compact <i>and</i> rational, then $[X]_{SB} = 1$ .                                                                                                                                             |
| cardona-qft-methods_F01969 | 140  | 1.000 | $\mathbb{Z}[S \times B]$                                                                | $\mathbb{Z}[SB]$                                                              | class mentioned above. Instead, one can determine the class $[X]_{SB}$ by (for example) obtaining $X$ from smooth irreducible com                                                                                 |
| cardona-qft-methods_F01970 | 140  | 1.000 | $[X]_{S \times B}$                                                                      | $[X]_{SB}$                                                                    | interpreting disjoint union as addition. Of course one must show that this                                                                                                                                        |
| cardona-qft-methods_F01971 | 140  | 1.000 | $[X]_{S \times B}=1$                                                                    | $[X]_{SB} = 1$                                                                |                                                                                                                                                                                                                   |
| cardona-qft-methods_F01972 | 141  | 1.000 | $\mathbb{P}^1 \setminus \mathbb{P}^0$                                                   | $\mathbb{P}^1 \setminus \mathbb{P}^0$                                         | leads to a well-defined class; see Theorem 4.6 below.                                                                                                                                                             |
| cardona-qft-methods_F01973 | 141  | 1.000 | $\left[\mathbb{P}^1\right]_{S \times B} - \left[\mathbb{P}^0\right]_{S \times B}=1-1=0$ | $\left[\mathbb{P}^1\right]_{SB} - \left[\mathbb{P}^0\right]_{SB} = 1 - 1 = 0$ | leads to a well-defined class; see Theorem 4.6 below.                                                                                                                                                             |
| cardona-qft-methods_F01974 | 141  | 0.922 | $\left[\mathbb{P}^1\right]_{S \times B} -$                                              | $\left[\mathbb{P}^1\right]_{SB} -$                                            | $\mathbb{P}^1 \setminus \mathbb{P}^0$ , hence map it to $\left[\mathbb{P}^1\right]_{SB} - \left[\mathbb{P}^0\right]_{SB} = 1 - 1 = 0$ .<br>The class of a nodal plane cubic is also 0; the complement of the node |
| cardona-qft-methods_F01975 | 141  | 1.000 | $2\left[\mathbb{P}^0\right]_{S \times B}=-1$                                            | $2\left[\mathbb{P}^0\right]_{SB} = -1$                                        | The class of a nodal plane cubic is also 0; the complement of the node<br>may be realized as the complement of 2 points in                                                                                        |
| cardona-qft-methods_F01976 | 141  | 0.999 | $\text{\texttt{AM11c}}$                                                                 | AM11c                                                                         | rem 4.2 and from the following key description of the ring $\mathbb{Z}[SB]$ :                                                                                                                                     |
| cardona-qft-methods_F01977 | 141  | 0.999 | $\text{\texttt{LL03}}$                                                                  | LL03                                                                          | rem 4.2 and from the following key description of the ring $\mathbb{Z}[SB]$ :                                                                                                                                     |
| cardona-qft-methods_F01978 | 141  | 0.998 | $X \mapsto [X]_{S \times B}$                                                            | $X \mapsto [X]_{SB}$                                                          |                                                                                                                                                                                                                   |
| cardona-qft-methods_F01979 | 141  | 0.996 | $K_0(\text{\texttt{Var}}) \rightarrow \mathbb{Z}[S \times B]$                           | $K_0(\text{\texttt{Var}}) \rightarrow \mathbb{Z}[SB]$                         |                                                                                                                                                                                                                   |
| cardona-qft-methods_F01980 | 141  | 1.000 | $(\mathbb{L})$                                                                          | $(\mathbb{L})$                                                                |                                                                                                                                                                                                                   |
| cardona-qft-methods_F01981 | 142  | 1.000 | $B \times \mathbb{P}^r$                                                                 | $B \times \mathbb{P}^r$                                                       | because if $B$ is a nonsingular compact variety and $E$ is the total space of a projective bundle over $B$ , then $B$ and $E$ are stably birationally equivalent:                                                 |
| cardona-qft-methods_F01982 | 142  | 1.000 | $B \subset X$                                                                           | $B \subset X$                                                                 | indeed, $E$ is birationally equivalent to $B \times \mathbb{P}^r$ for some $r$ . It follows that if $X$                                                                                                           |
| cardona-qft-methods_F01983 | 142  | 1.000 | $B \ell_B X$                                                                            | $B \ell_B X$                                                                  | This shows that there is a homomorphism $K_0(\text{\texttt{Var}}) \rightarrow \mathbb{Z}[SB]$ as claimed,                                                                                                         |
| cardona-qft-methods_F01984 | 142  | 0.903 | $) \rightarrow \mathbb{Z}[S \times B]$                                                  | $) \rightarrow \mathbb{Z}[SB]$                                                | and this homomorphism is onto since classes of nonsingular compact irreducible varieties generate $\mathbb{Z}[SB]$ . It is clear that $(1)$ is in the kernel of this                                              |
| cardona-qft-methods_F01985 | 142  | 0.998 | $\left[B \ell_B X\right] \equiv [X] \text{ modulo } (\mathbb{L})$                       | $[B \ell_B X] \equiv [X] \text{ modulo } (\mathbb{L})$                        | the ring of universal algebro-geometric Feynman rules to the Larsen-Lunts                                                                                                                                         |
| cardona-qft-methods_F01986 | 142  | 1.000 | $F \rightarrow \mathbb{Z}[S \times B]$                                                  | $F \rightarrow \mathbb{Z}[SB]$                                                | <b>Proof of Theorem 4.5.</b> By Theorem 4.6, it suffices to compute the class                                                                                                                                     |
| cardona-qft-methods_F01987 | 142  | 1.000 | $\left[\hat{Y}_G\right] = \mathbb{U}(G)$                                                | $\left[\hat{Y}_G\right] = \mathbb{U}(G)$                                      | and if $e$ is a bridge, then                                                                                                                                                                                      |
| cardona-qft-methods_F01988 | 143  | 1.000 | $\mathbb{U}(G) \bmod$                                                                   | $\mathbb{U}(G) \bmod$                                                         |                                                                                                                                                                                                                   |
| cardona-qft-methods_F01989 | 143  | 1.000 | $\mathbb{Z} \rightarrow \mathbb{Z}[S \times B]$                                         | $\mathbb{Z} \rightarrow \mathbb{Z}[SB]$                                       | This is of course nothing new, but it is useful to see that $\mathbb{Z}$ is in the kernel of this by the inclusion $\mathbb{Z} \rightarrow \mathbb{Z}[\xi]$                                                       |
| cardona-qft-methods_F01990 | 143  | 0.939 | $15 \text{\texttt{in}}^{\text{\texttt{BS12}}}$                                          | $15 \text{\texttt{in}}^{\text{\texttt{BS12}}}$                                | An interesting consequence of Theorem 4.5 is that the classes of graph                                                                                                                                            |
| cardona-qft-methods_F01991 | 143  | 1.000 | $\mathbb{L}^2$                                                                          | $\mathbb{L}^2$                                                                | hypersurfaces in $\mathbb{Z}[SB]$ span $\mathbb{Z}$ , and this is a rather small<br>of stable birational equivalence classes. By Theorem 4.6,                                                                     |

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| cardona-qft-methods_F01992 | 143  | ■ 0.475 | $c+\mathbb{L}\cdot M^{\prime}$                                     | $c+\mathbb{L}\cdot M'$                      | be written as a combination of classes of smooth complete varieties that                                                                                                                             |
| cardona-qft-methods_F01993 | 143  | ■ 0.475 | $c\in\mathbb{Z}$                                                   | $c\in\mathbb{Z}$                            | be written as a combination of classes of smooth complete varieties that                                                                                                                             |
| cardona-qft-methods_F01994 | 143  | ■ 0.475 | $M^{\prime}\in K_0(\mathrm{Var})$                                  | $M'\in K_0(\mathrm{Var})$                   | be written as a combination of classes of smooth complete varieties that                                                                                                                             |
| cardona-qft-methods_F01995 | 143  | ■ 1.000 | $h^{\{p,\,q\}}$                                                    | $h^{p,q}$                                   | are <i>unquely</i> determined up to stable rational equivalence. On the basis of such results, one might suspect that if the class of a variety is a com-                                            |
| cardona-qft-methods_F01996 | 143  | ■ 1.000 | $p=0,\,q>0$                                                        | $p=0,q>0$                                   | bination of classes of $\mathfrak{g}$                                                                                                                                                                |
| cardona-qft-methods_F01997 | 143  | ■ 1.000 | $p>0,\,q=0$                                                        | $p>0,q=0$                                   | bination of classes of $\mathfrak{g}$                                                                                                                                                                |
| cardona-qft-methods_F01998 | 144  | ■ 1.000 | $\mathbb{L}^n-1\,(n>0)$                                            | $\mathbb{L}^n-1\,(n>0)$                     |                                                                                                                                                                                                      |
| cardona-qft-methods_F01999 | 144  | ■ 1.000 | $\mathbb{L}^n-1$                                                   | $\mathbb{L}^n-1$                            | THEOREM 4.5 THAT THE LOCALIZATION AT $\mathbb{L}$ IS NECESSARY FOR THIS RESULT TO HOLD: classes of graph hypersurfaces do <i>not</i> generate the localization of $K_0(\mathrm{Var}_{\mathbb{A}^1})$ |
| cardona-qft-methods_F02000 | 144  | ■ 1.000 | $K\left(\operatorname{Var}_{\mathbb{Z}}\right)_{\mathbb{L}}$       | $K(\mathrm{Var}_{\mathbb{Z}})_{\mathbb{L}}$ | <i>Question:</i> Do graph hypersurfaces generate $K(\mathrm{Var}_{\mathbb{Z}})_{\mathbb{L}}$ over $\mathbb{Z}[\mathbb{L}]_{\mathbb{L}}$ ?                                                            |
| cardona-qft-methods_F02001 | 144  | ■ 1.000 | $\mathbb{Z}[\mathbb{L}]_{\mathbb{L}}$                              | $\mathbb{Z}[\mathbb{L}]_{\mathbb{L}}$       | <i>Question:</i> Do graph hypersurfaces generate $K(\mathrm{Var}_{\mathbb{Z}})_{\mathbb{L}}$ over $\mathbb{Z}[\mathbb{L}]_{\mathbb{L}}$ ?                                                            |
| cardona-qft-methods_F02002 | 145  | ■ 1.000 | $\phi^4$                                                           | $\phi^4$                                    | in configuration spaces. <small>GLAY (2013) 3094.</small><br>Dor11.     Dzmityr Doryn. On one example and one counterexample in counting                                                             |
| cardona-qft-methods_F02003 | 149  | ■ 1.000 | $\{ \}^{\{2,3\}}$                                                  | $2,3$                                       | <i>right and it and its generalization are the subject of this paper.</i><br>In the forty years since their discovery, CS forms have opened new areas                                                |
| cardona-qft-methods_F02004 | 149  | ■ 1.000 | $\{ \}^{\{4-6\}}$                                                  | $4-6$                                       | applications of CS gravities, like. <sup>4-6</sup> Here, the emphasis is on the construction of the action localization and in the geometric features that make CS                                   |
| cardona-qft-methods_F02005 | 149  | ■ 1.000 | $SO(D,\,1)$                                                        | $SO(D,1)$                                   | Section 6 contains a summary and a few comments on future projections.<br><b>Note to the readers:</b> I found the wonderful quote of V.I.Arnol'd in, <sup>8</sup>                                    |
| cardona-qft-methods_F02006 | 149  | ■ 1.000 | $SO(D-1,2)$                                                        | $SO(D-1,2)$                                 | Section 6 contains a summary and a few comments on future projections.<br><b>Note to the readers:</b> I found the wonderful quote of V.I.Arnol'd in, <sup>8</sup>                                    |
| cardona-qft-methods_F02007 | 149  | ■ 0.926 | $ISO(D-1,1)$                                                       | $ISO(D-1,1)$                                | that illustrates the attitude of most physicists, including myself, when con-                                                                                                                        |
| cardona-qft-methods_F02008 | 149  | ■ 1.000 | $2\,p$                                                             | $2p$                                        | $t_1\rightarrow Petya(t_1)$ belongs to the set of dirty nanas, and a $t_2$ , $t_1< t_2\leq 0$ , such that the image of $t_2$ under the above-mentioned manninas belongs to                           |
| cardona-qft-methods_F02009 | 149  | ■ 1.000 | $t_{\{1\}<0}$                                                      | $t_1<0$                                     |                                                                                                                                                                                                      |
| cardona-qft-methods_F02010 | 149  | ■ 1.000 | $t_{\{1\}}$                                                        | $t_1$                                       |                                                                                                                                                                                                      |
| cardona-qft-methods_F02011 | 149  | ■ 0.985 | $t_{\{1\}}\rightarrow\operatorname{Petya}\!\left(t_{\{1\}}\right)$ | $t_1\rightarrow\mathrm{Petya}(t_1)$         |                                                                                                                                                                                                      |
| cardona-qft-methods_F02012 | 149  | ■ 0.985 | $t_{\{2\}},\,t_{\{1\}<t_{\{2\}}\,\leq\,0$                          | $t_2,t_1<t_2\leq 0$                         |                                                                                                                                                                                                      |
| cardona-qft-methods_F02013 | 149  | ■ 1.000 | $t_{\{2\}}$                                                        | $t_2$                                       |                                                                                                                                                                                                      |
| cardona-qft-methods_F02014 | 150  | ■ 1.000 | $\mathbb{G}$                                                       | $\mathbb{G}$                                | theory i.e., the structure group $\mathbb{G}$ of the principal fibre bundle over the                                                                                                                 |
| cardona-qft-methods_F02015 | 150  | ■ 1.000 | $\forall x\in M,\,g(x)\in\mathbb{G}$                               | $\forall x\in M,g(x)\in\mathbb{G}$          | constructed with the connection $\mathbf{A}$ transforms as                                                                                                                                           |
| cardona-qft-methods_F02016 | 150  | ■ 0.999 | $\mathcal{C}(\mathbf{A})$                                          | $\mathcal{C}(\mathbf{A})$                   | tion,                                                                                                                                                                                                |
| cardona-qft-methods_F02017 | 150  | ■ 0.999 | $(2\,n+1)$                                                         | $(2n+1)$                                    | tion,                                                                                                                                                                                                |
| cardona-qft-methods_F02018 | 150  | ■ 0.999 | $n\,\geq\,0$                                                       | $n\geq 0$                                   | tion,                                                                                                                                                                                                |

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| cardona-qft-methods_F02019 | 150  | ■ 1.000 | $A_{\mu}$                                                                                                                                                                                                 | $A_{\mu}$                                                                                                                                                                                                 | the Hamiltonian flow in phase space, like the energy-momentum or the angular momentum of an isolated system. In the second case, nontrivial |
| cardona-qft-methods_F02020 | 151  | ■ 1.000 | $\partial_{\mu} j^{\mu} = 0$                                                                                                                                                                              | $\partial_{\mu} j^{\mu} = 0$                                                                                                                                                                              | external current leads to the conserved Noether current made up of point electric charges (electric charge conservation).                   |
| cardona-qft-methods_F02021 | 151  | ■ 1.000 | $\partial$                                                                                                                                                                                                | $\partial$                                                                                                                                                                                                | external current leads to the conserved Noether current made up of point electric charges (electric charge conservation).                   |
| cardona-qft-methods_F02022 | 151  | ■ 0.686 | $(\mathbb{G})$                                                                                                                                                                                            | $(\mathbb{G})$                                                                                                                                                                                            | a gauge invariant theory. This idea has been explored in theoretical physics                                                                |
| cardona-qft-methods_F02023 | 151  | ■ 0.686 | $d \Omega$                                                                                                                                                                                                | $d\Omega$                                                                                                                                                                                                 | a gauge invariant theory. This idea has been explored in theoretical physics                                                                |
| cardona-qft-methods_F02024 | 151  | ■ 1.000 | $(2 \ n)$                                                                                                                                                                                                 | $(2n)$                                                                                                                                                                                                    | over the past thirty years, beginning with the pioneering works by Deser, Jackiw and Templeton. <sup>9</sup>                                |
| cardona-qft-methods_F02025 | 151  | ■ 1.000 | $j^{\mu}$                                                                                                                                                                                                 | $j^{\mu}$                                                                                                                                                                                                 |                                                          |
| cardona-qft-methods_F02026 | 151  | ■ 1.000 | $\partial M$                                                                                                                                                                                              | $\partial M$                                                                                                                                                                                              |                                                                                                                                             |
| cardona-qft-methods_F02027 | 151  | ■ 1.000 | $d \star j = 0$                                                                                                                                                                                           | $d \star j = 0$                                                                                                                                                                                           |                                                                                                                                             |
| cardona-qft-methods_F02028 | 151  | ■ 0.989 | $\{ \ }^{\text{\texttt{a}}}$                                                                                                                                                                              | $\mathbf{a}$                                                                                                                                                                                              |                                                                                                                                             |
| cardona-qft-methods_F02029 | 151  | ■ 0.989 | $\alpha=(p!)^{-1} \alpha_{\mu_1 \mu_2 \cdots \mu_p} d x^{\mu_1} \wedge d x^{\mu_2} \wedge \cdots \wedge d x^{\mu_p}$<br>$d x^{\mu_1} \wedge d x^{\mu_2} \wedge \cdots \wedge d x^{\mu_p}$                 | $\alpha=(p!)^{-1} \alpha_{\mu_1 \mu_2 \cdots \mu_p} d x^{\mu_1} \wedge d x^{\mu_2} \wedge \cdots \wedge d x^{\mu_p}$                                                                                      |                                                                                                                                             |
| cardona-qft-methods_F02030 | 151  | ■ 0.989 | $(D-p)$                                                                                                                                                                                                   | $(D-p)$                                                                                                                                                                                                   |                                                                                                                                             |
| cardona-qft-methods_F02031 | 151  | ■ 0.694 | $\star \alpha=[p!(D-p)!]^{-1} \epsilon_{\nu_1 \nu_2 \cdots \nu_{D-p}}{}^{\mu_1 \mu_2 \cdots \mu_p} \alpha_{\mu_1 \mu_2 \cdots \mu_p} d x^{\nu_1} \wedge d x^{\nu_2} \wedge \cdots \wedge d x^{\nu_{D-p}}$ | $\star \alpha=[p!(D-p)!]^{-1} \epsilon_{\nu_1 \nu_2 \cdots \nu_{D-p}}{}^{\mu_1 \mu_2 \cdots \mu_p} \alpha_{\mu_1 \mu_2 \cdots \mu_p} d x^{\nu_1} \wedge d x^{\nu_2} \wedge \cdots \wedge d x^{\nu_{D-p}}$ |                                                                                                                                             |
| cardona-qft-methods_F02032 | 152  | ■ 1.000 | $\star j$                                                                                                                                                                                                 | $\star j$                                                                                                                                                                                                 |                                                                                                                                             |
| cardona-qft-methods_F02033 | 152  | ■ 0.998 | $(D-2 \ n-1)$                                                                                                                                                                                             | $(D-2n-1)$                                                                                                                                                                                                |                                                                                                                                             |
| cardona-qft-methods_F02034 | 152  | ■ 1.000 | $j =$                                                                                                                                                                                                     | $j =$                                                                                                                                                                                                     | $\mathcal{C}$ couples to a conserved gauge-invariant source. If the spacetime $M$ is $D$ -                                                  |
| cardona-qft-methods_F02035 | 152  | ■ 1.000 | $q \delta(\Gamma) d z$                                                                                                                                                                                    | $q \delta(\Gamma) d z$                                                                                                                                                                                    | dimensional, a $(2n+1)$ -form $\mathcal{C}$ couples to a source $j$ , whose dual $\star j$ is a $(D-2n-1)$ form.                            |
| cardona-qft-methods_F02036 | 152  | ■ 1.000 | $\delta(\Gamma)$                                                                                                                                                                                          | $\delta(\Gamma)$                                                                                                                                                                                          | dimensional, a $(2n+1)$ -form $\mathcal{C}$ couples to a source $j$ , whose dual $\star j$ is a $(D-2n-1)$ form.                            |
| cardona-qft-methods_F02037 | 152  | ■ 1.000 | $2 \ n$                                                                                                                                                                                                   | $2n$                                                                                                                                                                                                      | Analogously, the $(2n+1)$ -form $j$ in (6) can also be viewed as describing the                                                             |
| cardona-qft-methods_F02038 | 152  | ■ 1.000 | $\mathbf{K}_a(a=1,2,\ldots,N)$                                                                                                                                                                            | $\mathbf{K}_a(a=1,2,\ldots,N)$                                                                                                                                                                            | representation of the gauge group. If $g(x)=\exp[\alpha^a(x)\mathbf{K}_a]$ is an element of the group, it acts on the connection as (1)     |
| cardona-qft-methods_F02039 | 152  | ■ 1.000 | $\mathbf{K}_a$                                                                                                                                                                                            | $\mathbf{K}_a$                                                                                                                                                                                            |                                                        |
| cardona-qft-methods_F02040 | 152  | ■ 1.000 | $g(x)=\exp \left[\alpha^a(x) \mathbf{K}_a\right]$                                                                                                                                                         | $g(x)=\exp \left[\alpha^a(x) \mathbf{K}_a\right]$                                                                                                                                                         | This transformation law is prompted by the need to set up a covariant                                                                       |
| cardona-qft-methods_F02041 | 152  | ■ 1.000 | $D=d-\mathbf{A}$                                                                                                                                                                                          | $D=d-\mathbf{A}$                                                                                                                                                                                          | The curvature like $\vec{E}$ and $\vec{B}$ has directly observable local features. For                                                      |
| cardona-qft-methods_F02042 | 152  | ■ 1.000 | $\partial_{\mu}-i A_{\mu}$                                                                                                                                                                                | $\partial_{\mu}-i A_{\mu}$                                                                                                                                                                                | The curvature like $\vec{E}$ and $\vec{B}$ has directly observable local features. For                                                      |
| cardona-qft-methods_F02043 | 152  | ■ 1.000 | $\nabla_{\mu}=\partial_{\mu}+\boldsymbol{\Gamma}_{\mu}$                                                                                                                                                   | $\nabla_{\mu}=\partial_{\mu}+\boldsymbol{\Gamma}_{\mu}$                                                                                                                                                   | example, the $2k$ -form                                                                                                                     |
| cardona-qft-methods_F02044 | 152  | ■ 1.000 | $\mathbf{F}=d \mathbf{A}+\mathbf{A} \wedge \mathbf{A}$                                                                                                                                                    | $\mathbf{F}=d \mathbf{A}+\mathbf{A} \wedge \mathbf{A}$                                                                                                                                                    | We follow the notation and conventions of. <sup>7</sup>                                                                                     |

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| cardona-qft-methods_F02045 | 152  | ■ 1.000 | $\vec{E}$                                                       | $\vec{E}$                                                 |                                                                                                                                                                                       |
| cardona-qft-methods_F02046 | 152  | ■ 1.000 | $\vec{B}$                                                       | $\vec{B}$                                                 |                                                                                                                                                                                       |
| cardona-qft-methods_F02047 | 152  | ■ 1.000 | $2\,k$                                                          | $2k$                                                      |                                                                                                                                                                                       |
| cardona-qft-methods_F02048 | 153  | ■ 0.993 | $\mathbf{A}: M \mapsto \mathbb{L}$                              | $\mathbf{A}: M \mapsto \mathbb{L}$                        | or this kind (or more generally, the trace of any polynomial in $\mathbf{F}$ ), like the Euler or the Pontrvagin forms, known as <b>characteristic classes</b> , capture              |
| cardona-qft-methods_F02049 | 153  | ■ 1.000 | $P_{2\,k}(\mathbf{F})=\left\langle \mathbf{F}^k\right\rangle$   | $P_{2k}(\mathbf{F})=\langle \mathbf{F}^k \rangle$         | A <b>CS form</b> is defined in association with a characteristic class. Let's                                                                                                         |
| cardona-qft-methods_F02050 | 153  | ■ 1.000 | $\langle \cdots \rangle$                                        | $\langle \cdots \rangle$                                  | A <b>CS form</b> is defined in association with a characteristic class. Let's                                                                                                         |
| cardona-qft-methods_F02051 | 153  | ■ 0.973 | $Tr$                                                            | $Tr$                                                      | is a homogeneous polynomial in the curvature $\mathbf{F}$ associated to a gauge                                                                                                       |
| cardona-qft-methods_F02052 | 153  | ■ 0.973 | $P_{2\,k}$                                                      | $P_{2k}$                                                  | is a homogeneous polynomial in the curvature $\mathbf{F}$ associated to a gauge                                                                                                       |
| cardona-qft-methods_F02053 | 153  | ■ 1.000 | $\delta_{\text{gauge}}\,P_{2\,k}=0$                             | $\delta_{\text{gauge}}\,P_{2k}=0$                         | iii. Its co-                                                                                                                                                                          |
| cardona-qft-methods_F02054 | 153  | ■ 1.000 | $d\,P_{2\,k}=0$                                                 | $dP_{2k}=0$                                               | iv. Its integral over                                                                                                                                                                 |
| cardona-qft-methods_F02055 | 153  | ■ 1.000 | $(2\,k-1)$                                                      | $(2k-1)$                                                  | boundary, is a topological invariant, $\int_M P_{2k}=c_{2k}(M)\in\mathbf{Z}$ .<br>Condition (i) is satisfied by virtue of the cyclic property of the product of                       |
| cardona-qft-methods_F02056 | 153  | ■ 1.000 | $P_{2\,k}=$                                                     | $P_{2k}=$                                                 | boundary, is a topological invariant, $\int_M P_{2k}=c_{2k}(M)\in\mathbf{Z}$ .<br>Condition (i) is satisfied by virtue of the cyclic property of the product of                       |
| cardona-qft-methods_F02057 | 153  | ■ 1.000 | $d\,\mathcal{C}_{2\,k-1}$                                       | $d\mathcal{C}_{2k-1}$                                     |                                                                                                                                                                                       |
| cardona-qft-methods_F02058 | 153  | ■ 1.000 | $\int_M P_{2\,k}=c_{2\,k}(M)\in\mathbf{Z}$                      | $\int_M P_{2k}=c_{2k}(M)\in\mathbf{Z}$                    | follows from (ii). By Poincaré lemma: If $d\phi=0$ then, locally $\phi$ is                                                                                                            |
| cardona-qft-methods_F02059 | 153  | ■ 0.988 | $\mathbf{F}$                                                    | $\mathbf{F}$                                              | that in $\mathbf{A}$ and $d\mathbf{A}$ that cannot be written as a local function involving only the curvature $\mathbf{F}$ . This makes the CS forms rather cumbersome to write, but |
| cardona-qft-methods_F02060 | 153  | ■ 0.988 | $D\,\mathbf{F}=d\,\mathbf{F}+[\mathbf{A},\,\mathbf{F}]\equiv 0$ | $D\mathbf{F}=d\mathbf{F}+[\mathbf{A},\mathbf{F}]\equiv 0$ | that in $\mathbf{A}$ and $d\mathbf{A}$ that cannot be written as a local function involving only the curvature $\mathbf{F}$ . This makes the CS forms rather cumbersome to write, but |
| cardona-qft-methods_F02061 | 153  | ■ 1.000 | $d\,\phi=0$                                                     | $d\phi=0$                                                 | its exact expression is not needed in order to establish its most important                                                                                                           |
| cardona-qft-methods_F02062 | 153  | ■ 0.996 | $\phi=d$                                                        | $\phi=d$                                                  | property, as stated in the<br><b>Lemma:</b> Under a gauge                                                                                                                             |
| cardona-qft-methods_F02063 | 153  | ■ 1.000 | $\mathcal{C}_{2\,n-1}$                                          | $\mathcal{C}_{2n-1}$                                      | $Tr[\mathbf{F}^k]$ , due to the cyclic property of the trace). Performing a gauge trans-                                                                                              |
| cardona-qft-methods_F02064 | 153  | ■ 0.991 | $d\,\mathbf{A}$                                                 | $d\mathbf{A}$                                             |                                                                                                                                                                                       |
| cardona-qft-methods_F02065 | 153  | ■ 0.993 | $Tr\left[\mathbf{F}^k\right]$                                   | $Tr\left[\mathbf{F}^k\right]$                             |                                                                                                                                                                                       |
| cardona-qft-methods_F02066 | 154  | ■ 1.000 | $\mathcal{C}_{2\,k-1}$                                          | $\mathcal{C}_{2k-1}$                                      | By Poincaré's lemma, this last equation implies that $\mathcal{C}_{2k-1}$ can be written locally as an exact form                                                                     |
| cardona-qft-methods_F02067 | 154  | ■ 1.000 | $(2\,n-1)$                                                      | $(2n-1)$                                                  | defines gauge-invariant action in a $(2n-1)$ -dimensional manifold,                                                                                                                   |
| cardona-qft-methods_F02068 | 154  | ■ 1.000 | $M^{2\,n-1}$                                                    | $M^{2n-1}$                                                | and the inertial mass must be regarded as phenomenological constructs of classical or semi-classical nature: an emerging phenomenon.                                                  |
| cardona-qft-methods_F02069 | 155  | ■ 1.000 | $\xi^\mu$                                                       | $\xi^\mu$                                                 | which is a different point on the manifold. One could try to write (10) in a form similar to a gauge transformation                                                                   |
| cardona-qft-methods_F02070 | 155  | ■ 0.999 | $v^{\prime\,\mu}$                                               | $v'^\mu$                                                  |                                                                                                                                                                                       |
| cardona-qft-methods_F02071 | 155  | ■ 0.999 | $x^{\prime}$                                                    | $x'$                                                      |                                                                                                                                                                                       |

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| cardona-qft-methods_F02672 | 156  | ■ 1.000 | $\mathcal{H}_{\mu}$                                                             | $\mathcal{H}_{\mu}$                                                     | along the spatial direction $x^i$ ; $\delta(x, y)$ is the Dirac delta with support on                                                                       |
| cardona-qft-methods_F02673 | 156  | ■ 1.000 | $g^{ij}(x)$                                                                     | $g^{ij}(x)$                                                             |                                                                                                                                                             |
| cardona-qft-methods_F02674 | 156  | ■ 1.000 | $x, \mathcal{H}_{\perp}$                                                        | $x, \mathcal{H}_{\perp}$                                                |                                                                                                                                                             |
| cardona-qft-methods_F02675 | 156  | ■ 1.000 | $x^{\perp}, \mathcal{H}_i$                                                      | $x^{\perp}, \mathcal{H}_i$                                              |                                                                                                                                                             |
| cardona-qft-methods_F02676 | 156  | ■ 1.000 | $x^{\perp}, \mathcal{H}_i$                                                      | $x^{\perp}, \mathcal{H}_i$                                              |                                                                                                                                                             |
| cardona-qft-methods_F02676 | 156  | ■ 1.000 | $x^i; \delta(x, y)$                                                             | $x^i; \delta(x, y)$                                                     |                                                                                                                                                             |
| cardona-qft-methods_F02677 | 156  | ■ 0.965 | $x=y$                                                                           | $x = y$                                                                 |                                                                                                                                                             |
| cardona-qft-methods_F02678 | 156  | ■ 0.965 | $\delta(x, y), \{ \}_{i} = \left( \partial / \partial y^i \right) \delta(x, y)$ | $\delta(x, y), i = \left( \partial / \partial y^i \right) \delta(x, y)$ |                                                                                                                                                             |
| cardona-qft-methods_F02679 | 157  | ■ 0.999 | $S O(3,1)$                                                                      | $SO(3,1)$                                                               |                                                                                                                                                             |
| cardona-qft-methods_F02680 | 158  | ■ 0.833 | $\mathbb{G} \supseteq S O(3,1)$                                                 | $\mathbb{G} \supseteq SO(3,1)$                                          |                                                                                                                                                             |
| cardona-qft-methods_F02681 | 158  | ■ 0.833 | $(S O(4,1))$                                                                    | $(SO(4,1))$                                                             |                                                                                                                                                             |
| cardona-qft-methods_F02682 | 158  | ■ 0.855 | $(S O(3,2))$                                                                    | $(SO(3,2))$                                                             | <i>Gravitation theory and Chern-Simons forms</i> 141                                                                                                        |
| cardona-qft-methods_F02683 | 158  | ■ 0.855 | $(S O(4,2))$                                                                    | $(SO(4,2))$                                                             | <i>Gravitation theory and Chern-Simons forms</i> 141                                                                                                        |
| cardona-qft-methods_F02684 | 158  | ■ 0.855 | $I S O(3,1)$                                                                    | $ISO(3,1)$                                                              | <i>Gravitation theory and Chern-Simons forms</i> 141                                                                                                        |
| cardona-qft-methods_F02685 | 158  | ■ 1.000 | $S O(D-1,1)$                                                                    | $SO(D-1,1)$                                                             | that correspond to some nontrivial representation of $SO(D-1,1)$ , where $D$ is the spacetime dimension. (This is most effectively done if the matrix and   |
| cardona-qft-methods_F02686 | 158  | ■ 1.000 | $d^2 \equiv 0$                                                                  | $d^2 \equiv 0$                                                          | with the standard tensor calculus in coordinate bases. <sup>16</sup> Nevertheless, it is                                                                    |
| cardona-qft-methods_F02687 | 158  | ■ 1.000 | $e_{\mu}^a(x)$                                                                  | $e_{\mu}^a(x)$                                                          |                                                                                                                                                             |
| cardona-qft-methods_F02688 | 158  | ■ 0.615 | $\omega^a_{\ b\mu}(x)$                                                          | $\omega^a_{\ b\mu}(x)$                                                  |                                                                                                                                                             |
| cardona-qft-methods_F02689 | 159  | ■ 1.000 | $(-1,1,1, \cdots, 1)$                                                           | $(-1,1,1,\cdots,1)$                                                     | <b>2.1 The vielbein</b>                                                                                                                                     |
| cardona-qft-methods_F02690 | 159  | ■ 1.000 | $T_{\{x\}}$                                                                     | $T_x$                                                                   | Spacetime is a smooth $D$ -dimensional manifold $M$ of Lorentzian signature                                                                                 |
| cardona-qft-methods_F02691 | 159  | ■ 1.000 | $\left\{ T_{\{x\}}, x \in M \right\}$                                           | $\{T_x, x \in M\}$                                                      | $M$ , labelled by the coordinates $x^{\mu}$ is the spacetime where we live, and the collection of tangent spaces over it is where the symmetries occur (see |
| cardona-qft-methods_F02692 | 159  | ■ 1.000 | $dx^{\mu}$                                                                      | $dx^{\mu}$                                                              | For this reason the vielbein is also viewed as a local orthonormal frame.                                                                                   |
| cardona-qft-methods_F02693 | 159  | ■ 1.000 | $z^a$                                                                           | $z^a$                                                                   | $= g_{\mu\nu}(x) dx^{\mu} dx^{\nu}, \quad (2)$                                                                                                              |
| cardona-qft-methods_F02694 | 159  | ■ 1.000 | $\eta_{ab}$                                                                     | $\eta_{ab}$                                                             | $g_{\mu\nu}(x) \equiv \eta_{ab} e_{\mu}^a(x) e_{\nu}^b(x), \quad (3)$                                                                                       |
| cardona-qft-methods_F02695 | 159  | ■ 1.000 | $e_{\mu}^a$                                                                     | $e_{\mu}^a$                                                             | is the metric on $M$ , induced by the vielbein $e_{\mu}^a(x)$ and the tangent s                                                                             |
| cardona-qft-methods_F02696 | 160  | ■ 1.000 | $T_{\{x\}}, S O(D-1,1)$                                                         | $T_x, SO(D-1,1)$                                                        | By the definition (1) the viel                                                                                                                              |
| cardona-qft-methods_F02697 | 160  | ■ 1.000 | $\boldsymbol{\Lambda}(x)$                                                       | $\boldsymbol{\Lambda}(x)$                                               | where the matrix $\boldsymbol{\Lambda}(x)$ leaves the metric in the tangent space unchanged.                                                                |
| cardona-qft-methods_F02698 | 160  | ■ 1.000 | $D(D+1)/2$                                                                      | $D(D+1)/2$                                                              | In this way, the <b>vielbein one-forms</b> $e^a_{\mu}(x) = e^a_{\mu}(x) dx^{\mu}$ , define the metric structure of the manifold                             |

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| cardona-qft-methods_F02099 | 160  | ■ 1.000 | $D(D-1) / 2$                                                                   | $D(D-1)/2$                                                                     |                                                                                                                                                                                                                                                                          |
| cardona-qft-methods_F02100 | 160  | ■ 1.000 | $e^a(x)=e_{\mu}^a(x) \, d x^{\mu}$                                             | $e^a(x) = e_{\mu}^a(x)dx^{\mu}$                                                | 3.2. The Lorentz Connection                                                                                                                                                                                                                                              |
| cardona-qft-methods_F02101 | 160  | ■ 1.000 | $\Lambda(x)$                                                                   | $\Lambda(x)$                                                                   | tions defined on the tangent $T_x$ . The parallel-transported field from $x+dx$ to $x$ is defined as                                                                                                                                                                     |
| cardona-qft-methods_F02102 | 160  | ■ 0.955 | $\{\mathrm{d}\}$                                                               | $d$                                                                            | $d\phi^a(x) = d\phi^a(x) + \omega^a_b(x)\phi^b(x) = D\phi^a(x)$                                                                                                                                                                                                          |
| cardona-qft-methods_F02103 | 160  | ■ 1.000 | $T_{x+d \, x}$                                                                 | $T_{x+dx}$                                                                     | $\phi _{x+dx}$                                                                                                                                                                                                                                                           |
| cardona-qft-methods_F02104 | 160  | ■ 1.000 | $\phi^a(x)$                                                                    | $\phi^a(x)$                                                                    | that it arises naturally in the discussion of spinors, which carry a special representation of the group of rotations in the tangent space. For a more extended discussion, there are                                                                                    |
| cardona-qft-methods_F02105 | 160  | ■ 1.000 | $x+d \, x$                                                                     | $x + dx$                                                                       | several texts such as those of Refs. <sup>2,7,16</sup> and <sup>17</sup>                                                                                                                                                                                                 |
| cardona-qft-methods_F02106 | 160  | ■ 1.000 | $\{ \}^{2,7,16}$                                                               | $^{2,7,16}$                                                                    |                                                                                                                                                                                                                                                                          |
| cardona-qft-methods_F02107 | 161  | ■ 0.979 | $\phi^a$                                                                       | $\phi^a$                                                                       |                                                                                                                                                                                                                                                                          |
| cardona-qft-methods_F02108 | 161  | ■ 0.979 | $\phi_{  }^a$                                                                  | $\phi_{  }^a$                                                                  |                                                                                                                                                                                                                                                                          |
| cardona-qft-methods_F02109 | 161  | ■ 0.993 | $\omega_{b\mu}^a$                                                              | $\omega_{b\mu}^a$                                                              | $\phi_{  }^a(x) - \phi^a(x) = d\phi^a(x) + \omega^a_b(x)\phi^b(x) = D\phi^a(x)$                                                                                                                                                                                          |
| cardona-qft-methods_F02110 | 161  | ■ 0.993 | $\Gamma_{\lambda\rho}^{\mu}$                                                   | $\Gamma_{\lambda\rho}^{\mu}$                                                   | $\phi_{  }^a(x) - \phi^a(x) = d\phi^a(x) + \omega^a_b(x)\phi^b(x) = D\phi^a(x)$                                                                                                                                                                                          |
| cardona-qft-methods_F02111 | 161  | ■ 1.000 | $\omega^a{}_b=\omega^a_{b\mu}dx^{\mu}$                                         | $\omega^a{}_b = \omega^a_{b\mu}dx^{\mu}$                                       | tensor, $\epsilon_{a_1a_2\cdots a_D}$ . Because they are the same in every tangent space, they are constant ( $\partial_{\mu}\eta_{ab} = 0$ , $\partial_{\mu}\epsilon_{a_1a_2\cdots a_D} = 0$ ) and since they are                                                       |
| cardona-qft-methods_F02112 | 161  | ■ 1.000 | $\epsilon_{a_1a_2\cdots a_D}$                                                  | $\epsilon_{a_1a_2\cdots a_D}$                                                  | $\epsilon_{b_1a_2\cdots a_D}\omega^{b_1}_{a_1} + \epsilon_{a_1b_2\cdots a_D}\omega^{b_2}_{a_2} + \cdots + \epsilon_{a_1a_2\cdots b_D}\omega^{b_D}_{a_D} = 0. \quad (13)$                                                                                                 |
| cardona-qft-methods_F02113 | 161  | ■ 1.000 | $(\partial_{\mu}\eta_{ab} = 0, \partial_{\mu}\epsilon_{a_1a_2\cdots a_D} = 0)$ | $(\partial_{\mu}\eta_{ab} = 0, \partial_{\mu}\epsilon_{a_1a_2\cdots a_D} = 0)$ |                                                                                                                                                                                                                                                                          |
| cardona-qft-methods_F02114 | 162  | ■ 1.000 | $\omega^{ab}=-\omega^{ba}$                                                     | $\omega^{ab} = -\omega^{ba}$                                                   | Gravitation theory and Chern-Simons forms 161                                                                                                                                                                                                                            |
| cardona-qft-methods_F02115 | 162  | ■ 0.977 | $D^2(D-1) / 2$                                                                 | $D^2(D-1)/2$                                                                   | patible with the metric structure of the tangent space (12), implies anti-symmetry of the connection $\omega^{ab} = -\omega^{ba}$ . The second relation (13) does                                                                                                        |
| cardona-qft-methods_F02116 | 162  | ■ 0.968 | $(D^2(D+1) / 2)$                                                               | $(D^2(D+1)/2)$                                                                 | tion. Then, the number of independent components of $\omega^{ab}$ is $D(D-1)/2$ , which is less than                                                                                                                                                                     |
| cardona-qft-methods_F02117 | 162  | ■ 1.000 | $dx^{\mu}\partial_{\mu}\wedge$                                                 | $dx^{\mu}\partial_{\mu}\wedge$                                                 | The 1-form exterior derivative operator, $dx^{\mu}\partial_{\mu}\wedge$ is such that acting on a 1-form $\omega = \omega_a dx^a$ it yields a $(p+1)$ -form $d\omega = d\omega_a dx^a + \omega_a dx^a \wedge \partial_{\mu} dx^{\mu}$ . One of the fundamental properties |
| cardona-qft-methods_F02118 | 162  | ■ 0.468 | $\alpha_p$                                                                     | $\alpha_p$                                                                     | $p$ -form, $\alpha_p$ is yields a $(p+1)$ -form $d\alpha_p$ . One of the fundamental properties of exterior calculus is that the second exterior derivative of a differential                                                                                            |
| cardona-qft-methods_F02119 | 162  | ■ 0.468 | $d \, \alpha_p$                                                                | $d\alpha_p$                                                                    | $p$ -form, $\alpha_p$ is yields a $(p+1)$ -form $d\alpha_p$ . One of the fundamental properties of exterior calculus is that the second exterior derivative of a differential                                                                                            |
| cardona-qft-methods_F02120 | 162  | ■ 1.000 | $\{ \}^{16,17}$                                                                | $^{16,17}$                                                                     | $A^a_{b\mu}dx^{\mu}$ , are both 1-forms and have similar properties is not an accident                                                                                                                                                                                   |
| cardona-qft-methods_F02121 | 162  | ■ 0.924 | $R^a{}_b$                                                                      | $R^a{}_b$                                                                      | since they are both tensors, they must have the same form                                                                                                                                                                                                                |
| cardona-qft-methods_F02122 | 162  | ■ 0.788 | $\omega^a{}_b(x)$                                                              | $\omega^a{}_b(x)$                                                              |                                                                                                                                                                                                                                                                          |
| cardona-qft-methods_F02123 | 162  | ■ 0.788 | $A^a{}_b=$                                                                     | $A^a{}_b =$                                                                    |                                                                                                                                                                                                                                                                          |
| cardona-qft-methods_F02124 | 162  | ■ 0.955 | $A^a_{b\mu}dx^{\mu}$                                                           | $A^a_{b\mu}dx^{\mu}$                                                           |                                                                                                                                                                                                                                                                          |

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| cardona-qft-methods_F02125 | 162  | ■ 0.953 | $R_{\mathrm{b}}^{\mathrm{a}}$                                                                                                                                                                                                                                               | $R_b^a$                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| cardona-qft-methods_F02126 | 162  | ■ 1.000 | $R^{\{\alpha\ \beta\}}_{\{\mu\ \nu\}}$                                                                                                                                                                                                                                      | $R^{\alpha\beta}_{\mu\nu}$                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| cardona-qft-methods_F02127 | 163  | ■ 1.000 | $e^{\mathrm{a}}$                                                                                                                                                                                                                                                            | $e^a$                                                                 | to construct a tensor, the covariant derivative, and the curvature tensor, in contrast with the curvature which is uniquely defined by the connection.                                                                                                                                                                                                                                                                                            |
| cardona-qft-methods_F02128 | 163  | ■ 1.000 | $T^{\mathrm{a}}$                                                                                                                                                                                                                                                            | $T^a$                                                                 | into a torsion-free part $\omega$ , and the so-called contorsion $\kappa$ ,                                                                                                                                                                                                                                                                                                                                                                       |
| cardona-qft-methods_F02129 | 163  | ■ 0.999 | $\bar{\omega}$                                                                                                                                                                                                                                                              | $\bar{\omega}$                                                        | $R_b^a = \bar{R}_b^a + \bar{D}\kappa_b^a + \kappa_c^a \kappa_b^c,$                                                                                                                                                                                                                                                                                                                                                                                |
| cardona-qft-methods_F02130 | 163  | ■ 0.999 | $\kappa$                                                                                                                                                                                                                                                                    | $\kappa$                                                              | $R_b^a = \bar{R}_b^a + \bar{D}\kappa_b^a + \kappa_c^a \kappa_b^c,$                                                                                                                                                                                                                                                                                                                                                                                |
| cardona-qft-methods_F02131 | 163  | ■ 1.000 | $\bar{R}_{\mathrm{b}}^{\mathrm{a}}$                                                                                                                                                                                                                                         | $\bar{R}_b^a$                                                         | to multiplying by the curvature 2-form. A consequence of this is an important property known as Bianchi identity,                                                                                                                                                                                                                                                                                                                                 |
| cardona-qft-methods_F02132 | 163  | ■ 1.000 | $\bar{D}$                                                                                                                                                                                                                                                                   | $\bar{D}$                                                             | to multiplying by the curvature 2-form. A consequence of this is an important property known as Bianchi identity,                                                                                                                                                                                                                                                                                                                                 |
| cardona-qft-methods_F02133 | 164  | ■ 0.896 | $\omega^{\mathrm{a}}_{\mathrm{b}\ \mu}$                                                                                                                                                                                                                                     | $\omega^a_{b\mu}$                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| cardona-qft-methods_F02134 | 164  | ■ 1.000 | $R^{\mathrm{a}\ \mathrm{b}}$                                                                                                                                                                                                                                                | $R^{ab}$                                                              | <i>Gravitation theory and Chern-Simons forms</i> 133                                                                                                                                                                                                                                                                                                                                                                                              |
| cardona-qft-methods_F02135 | 164  | ■ 1.000 | $e^{\mathrm{a}},\ \omega^{\mathrm{a}\ \mathrm{b}}$                                                                                                                                                                                                                          | $e^a, \omega^{ab}$                                                    | An important direct consequence of this identity is that by taking suc-                                                                                                                                                                                                                                                                                                                                                                           |
| cardona-qft-methods_F02136 | 164  | ■ 0.793 | $\omega_{\mathrm{b}}^{\mathrm{a}}$                                                                                                                                                                                                                                          | $\omega_b^a$                                                          | put together an action. We are interested in objects that transform in a controlled way under Lorentz rotations (scalars, vectors, tensors of various                                                                                                                                                                                                                                                                                             |
| cardona-qft-methods_F02137 | 164  | ■ 0.793 | $R_{\mathrm{b}}^{\mathrm{a}},\ T^{\mathrm{a}}$                                                                                                                                                                                                                              | $R_b^a, T^a$                                                          | put together an action. We are interested in objects that transform in a controlled way under Lorentz rotations (scalars, vectors, tensors of various                                                                                                                                                                                                                                                                                             |
| cardona-qft-methods_F02138 | 164  | ■ 0.508 | $I[e,\ \omega]$                                                                                                                                                                                                                                                             | $I[e, \omega]$                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| cardona-qft-methods_F02139 | 164  | ■ 0.508 | $\omega^{\mathrm{a}}_{\mathrm{b}}$                                                                                                                                                                                                                                          | $\omega^a_b$                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| cardona-qft-methods_F02140 | 164  | ■ 0.852 | $\mu,\ \nu$                                                                                                                                                                                                                                                                 | $\mu, \nu$                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| cardona-qft-methods_F02141 | 165  | ■ 0.666 | $e^{\mathrm{a}},\ \omega^{\mathrm{a}}_{\mathrm{b}}$                                                                                                                                                                                                                         | $e^a, \omega^a_b$                                                     | 4. Covariant actions                                                                                                                                                                                                                                                                                                                                                                                                                              |
| cardona-qft-methods_F02142 | 165  | ■ 1.000 | $\epsilon_{a_1\cdots a_D}$                                                                                                                                                                                                                                                  | $\epsilon_{a_1\cdots a_D}$                                            | above, we expect it to be a local functional of the one-forms $e^a, \omega^a_b$ and their                                                                                                                                                                                                                                                                                                                                                         |
| cardona-qft-methods_F02143 | 165  | ■ 1.000 | $\{ \mathrm{e} \}_{\mu\ \nu} = E_{\mathrm{a}}^{\{\lambda\}} \eta_{\mathrm{b}\ \mathrm{c}} \epsilon_{\mu}^{\mathrm{c}} R_{\lambda\nu}^{\mathrm{a}\ \mathrm{b}}$<br>$\{ \mathrm{e} \}_{\lambda\ \nu}^{\{\mu\}} R_{\lambda\ \nu}^{\{\mu\}} \epsilon_{\mathrm{a}}^{\mathrm{b}}$ | ${}^e R_{\mu\nu} = E_a^\lambda \eta_{bc} e_\mu^c R_{\lambda\nu}^{ab}$ | cludes tensors like the Ricci tensor <sup>a</sup> $R_{\mu\nu} = E_a^\lambda \eta_{bc} e_\mu^c R_{\lambda\nu}^{ab}$ , or $R_{\alpha\beta} R_{\mu\nu} R^{\alpha\mu\beta\nu}$ ,                                                                                                                                                                                                                                                                      |
| cardona-qft-methods_F02144 | 165  | ■ 1.000 | $R_{\{\alpha\ \beta\}} R_{\{\mu\ \nu\}} R^{\{\alpha\ \mu\ \beta\ \nu\}}$                                                                                                                                                                                                    | $R_{\alpha\beta} R_{\mu\nu} R^{\alpha\mu\beta\nu}$                    | cludes tensors like the Ricci tensor <sup>a</sup> $R_{\mu\nu} = E_a^\lambda \eta_{bc} e_\mu^c R_{\lambda\nu}^{ab}$ , or $R_{\alpha\beta} R_{\mu\nu} R^{\alpha\mu\beta\nu}$ ,                                                                                                                                                                                                                                                                      |
| cardona-qft-methods_F02145 | 165  | ■ 0.988 | $e^{\mathrm{a}},\ R_{\mathrm{b}}^{\mathrm{a}},\ T^{\mathrm{a}}$                                                                                                                                                                                                             | $e^a, R_b^a, T^a$                                                     | $\zeta_k =: e_{a_1} R^{\alpha_1}_{\phantom{\alpha_1}a_2} R^{\alpha_2}_{\phantom{\alpha_2}a_3} \cdots R^{\alpha_k}_{\phantom{\alpha_k}b} L^{\alpha_k}_{\phantom{\alpha_k}b} \quad (4)$<br>$\mathbf{P} = \left( \begin{array}{cccc} D^{0102} & D^{0104} & D^{0204} & D^{0206} \\ D^{0102} & D^{0104} & D^{0204} & D^{0206} \\ D^{0102} & D^{0104} & D^{0204} & D^{0206} \\ D^{0102} & D^{0104} & D^{0204} & D^{0206} \end{array} \right) \quad (5)$ |
| cardona-qft-methods_F02146 | 165  | ■ 1.000 | $\mathbf{P}_{2\ k}$                                                                                                                                                                                                                                                         | $\mathbf{P}_{2k}$                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| cardona-qft-methods_F02147 | 165  | ■ 1.000 | $\mathbf{E}_{2\ n}$                                                                                                                                                                                                                                                         | $\mathbf{E}_{2n}$                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| cardona-qft-methods_F02148 | 165  | ■ 1.000 | $4\ k$                                                                                                                                                                                                                                                                      | $4k$                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| cardona-qft-methods_F02149 | 165  | ■ 0.774 | $\{ \mathrm{e} \}$                                                                                                                                                                                                                                                          | $e$                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| cardona-qft-methods_F02150 | 165  | ■ 0.774 | $E_{\mathrm{a}}^{\{\lambda\}}$                                                                                                                                                                                                                                              | $E_a^\lambda$                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                   |

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| cardona-qft-methods_F02151 | 165  | 0.774 | $E_{\{a\}^{\{\lambda\}} e^{\{b\}}\}_{\lambda}=\delta_{\{b\}}^{\{a\}}$   | $E_a^\lambda e^b_{\lambda} = \delta^a_b$           |                                                                                                                                                                       |
| cardona-qft-methods_F02152 | 166  | 1.000 | $\Omega_n$                                                              | $\Omega_n$                                         | $\Omega_n \int_{\dots} \mathbf{E}_{2n} \in \mathbb{Z}, \quad \tilde{\Omega}_k \int_{\dots} \mathbf{P}_{4k} \in \mathbb{Z}, \quad (7)$                                 |
| cardona-qft-methods_F02153 | 166  | 1.000 | $\tilde{\Omega}_k$                                                      | $\tilde{\Omega}_k$                                 | $\Omega_n \int_{\dots} \mathbf{E}_{2n} \in \mathbb{Z}, \quad \tilde{\Omega}_k \int_{\dots} \mathbf{P}_{4k} \in \mathbb{Z}, \quad (7)$                                 |
| cardona-qft-methods_F02154 | 166  | 1.000 | $D=2\ n$                                                                | $D=2n$                                             | $D=2n$                                                                                                                                                                |
| cardona-qft-methods_F02155 | 166  | 1.000 | $4\left(k_1+k_2+\cdots+k_r\right)=D$                                    | $4\left(k_1+k_2+\cdots+k_r\right)=D$               | where $4(k_1+k_2+\cdots+k_r)=D$ , so the number of 1-oriental involutions grows with the number of different ways to express the number $D/4$ as a                    |
| cardona-qft-methods_F02156 | 166  | 1.000 | $D\ /\ 4$                                                               | $D/4$                                              | sum of integers. <sup>f</sup>                                                                                                                                         |
| cardona-qft-methods_F02157 | 166  | 0.994 | $\{ \ }^{\{\mathrm{f}\}}$                                               | $\mathfrak{f}$                                     |                                                                                                                                                                       |
| cardona-qft-methods_F02158 | 166  | 1.000 | $\tau_{\{k\}},\ \zeta_{\{k\}}$                                          | $\tau_k,\zeta_k$                                   | <b>Theorem</b> [Lovelock, 1970 <sup>19</sup> and Zumino, 1986 <sup>12</sup> ]: In the absence of                                                                      |
| cardona-qft-methods_F02159 | 166  | 1.000 | $v_{\{k\}}$                                                             | $v_k$                                              | <b>Theorem</b> [Lovelock, 1970 <sup>19</sup> and Zumino, 1986 <sup>12</sup> ]: In the absence of                                                                      |
| cardona-qft-methods_F02160 | 166  | 1.000 | $R^{\{a\}\{ \}_{\{b\}} e^{\{b\}}$                                       | $R^a_b e^b$                                        | torsion, the most general action for gravity $I[e,\omega]$ invariant, under Lorentz transformations that does not explicitly involve the metric, is of the form       |
| cardona-qft-methods_F02161 | 166  | 1.000 | $a_{\{p\}}$                                                             | $a_p$                                              |                                                                                                                                                                       |
| cardona-qft-methods_F02162 | 166  | 1.000 | $L_{\{p\}}^{\{D\}}$                                                     | $L_p^D$                                            |                                                                                                                                                                       |
| cardona-qft-methods_F02163 | 166  | 1.000 | $L_{\{D\ /\ 2\}}^{\{D\}}=\mathrm{mathcal{E}}_{\{D\}}$                   | $L_{D/2}^D=\mathcal{E}_D$                          |                                                                                                                                                                       |
| cardona-qft-methods_F02164 | 166  | 1.000 | $p(D\ /\ 4)\ \sim\ \frac{1}{\sqrt{3D}}\ \exp\left[\pi\sqrt{D/6}\right]$ | $p(D/4)\sim\frac{1}{\sqrt{3D}}\exp[\pi\sqrt{D/6}]$ |                                                                                                                                                                       |
| cardona-qft-methods_F02165 | 167  | 0.994 | $D=2$                                                                   | $D=2$                                              |                                                                                                                                                                       |
| cardona-qft-methods_F02166 | 167  | 1.000 | $V_{\{2\}}=0$                                                           | $V_2=0$                                            | This action has as a local extremum for $V_0=0$ , which reflects the fact that,                                                                                       |
| cardona-qft-methods_F02167 | 167  | 1.000 | $I_{\{2\}}$                                                             | $I_2$                                              | unless matter is included, $I_2$ does not produce a very interesting dynamical                                                                                        |
| cardona-qft-methods_F02168 | 167  | 0.998 | $D=3$                                                                   | $D=3$                                              | Euler invariant $\mathbf{E}_4$ ,                                                                                                                                      |
| cardona-qft-methods_F02169 | 167  | 0.998 | $D=4$                                                                   | $D=4$                                              | Euler invariant $\mathbf{E}_4$ ,                                                                                                                                      |
| cardona-qft-methods_F02170 | 167  | 1.000 | $\mathbf{E}_{\{4\}}$                                                    | $\mathbf{E}_4$                                     | $\frac{JM}{f}$                                                                                                                                                        |
| cardona-qft-methods_F02171 | 167  | 1.000 | $D=5$                                                                   | $D=5$                                              | five dimensions seems to have been known for many years, and is commonly attributed to Lovelock. <sup>21</sup>                                                        |
| cardona-qft-methods_F02172 | 167  | 1.000 | $D(D-3)\ /\ 2$                                                          | $D(D-3)/2$                                         |                                                                                                                                                                       |
| cardona-qft-methods_F02173 | 168  | 1.000 | $\{ \ }^{\{\mathrm{g}\}}$                                               | $\mathfrak{g}$                                     |                                                                                                                                                                       |
| cardona-qft-methods_F02174 | 168  | 1.000 | $\{ \ }^{\{12,23\}}$                                                    | $12,23$                                            |                                                                                                                                                                       |
| cardona-qft-methods_F02175 | 168  | 1.000 | $k^{\{-2\}}$                                                            | $k^{-2}$                                           | string theory at low energy. <sup>12,13,22</sup> The unexpected and nontrivial absence of ghosts seems to reflect the fact that, in the absence of torsion, the Love- |
| cardona-qft-methods_F02176 | 168  | 1.000 | $k^{\{-2\}}+k^{\{-4\}}$                                                 | $k^{-2}+k^{-4}$                                    | string theory at low energy. <sup>12,13,22</sup> The unexpected and nontrivial absence of ghosts seems to reflect the fact that, in the absence of torsion, the Love- |
| cardona-qft-methods_F02177 | 168  | 1.000 | $\omega^{\{a\ b\}}$                                                     | $\omega^{ab}$                                      | and higher derivatives.                                                                                                                                               |

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| cardona-qft-methods_F02178 | 168  | ■ 1.000 | $I_{\{D\}}$                                                     | $I_D$                                        | modulo surface terms. The condition for $I_D$ to have an extremum to first order under arbitrary infinitesimal variations is that $\mathcal{E}_\mu$ and $\mathcal{E}_{\mu\nu}$ vanish.                                                                                                              |
| cardona-qft-methods_F02179 | 168  | ■ 1.000 | $\mathcal{E}_{\{a\}}$                                           | $\mathcal{E}_a$                              | $[\frac{\partial-1}{\partial}]$                                                                                                                                                                                                                                                                     |
| cardona-qft-methods_F02180 | 168  | ■ 1.000 | $\mathcal{E}_{\{a\} \{b\}}$                                     | $\mathcal{E}_{ab}$                           | $[\frac{\partial-1}{\partial}]$                                                                                                                                                                                                                                                                     |
| cardona-qft-methods_F02181 | 168  | ■ 1.000 | $d^{\{2\}}=0$                                                   | $d^2 = 0$                                    | derivatives,                                                                                                                                                                                                                                                                                        |
| cardona-qft-methods_F02182 | 168  | ■ 0.826 | $\left(E_{\{a\}}^{\{\mu\}}\right)$                              | $(E_a^\mu)$                                  |                                                                                                                                                                                                                                                                                                     |
| cardona-qft-methods_F02183 | 168  | ■ 1.000 | $\Gamma_{\{\mu\} \{\nu\}}^{\{\lambda\}}$                        | $\Gamma_{\mu\nu}^\lambda$                    |                                                                                                                                                                                                                                                                                                     |
| cardona-qft-methods_F02184 | 168  | ■ 1.000 | $\mu\ \nu$                                                      | $\mu\nu$                                     |                                                                                                                                                                                                                                                                                                     |
| cardona-qft-methods_F02185 | 169  | ■ 1.000 | $g_{\{\mu\} \{\nu\}}$                                           | $g_{\mu\nu}$                                 | from one side determines the full metric tensor of $g_{\mu\nu}$ where the other equations for the metric are second order and not fourth order, in spite of                                                                                                                                         |
| cardona-qft-methods_F02186 | 169  | ■ 1.000 | $f(R)$                                                          | $f(R)$                                       | to vanish identically. In $f(R)$ theories, for example, this first condition is not respected and they generically give rise to higher order field equations. In                                                                                                                                    |
| cardona-qft-methods_F02187 | 169  | ■ 1.000 | $D \leq 4$                                                      | $D \leq 4$                                   | components of the curvature and torsion tensors completely undetermined. For example, Eq (16) has the form of a polynomial in $R^{ab}$ times $T^a$ and it                                                                                                                                           |
| cardona-qft-methods_F02188 | 169  | ■ 1.000 | $D>4$                                                           | $D > 4$                                      | components of the curvature and torsion tensors completely undetermined. For example, Eq (16) has the form of a polynomial in $R^{ab}$ times $T^a$ and it                                                                                                                                           |
| cardona-qft-methods_F02189 | 169  | ■ 1.000 | $(15, 16)$                                                      | $(15, 16)$                                   | is possible that the polynomial vanishes identically, imposing no conditions                                                                                                                                                                                                                        |
| cardona-qft-methods_F02190 | 170  | ■ 1.000 | $I[\omega, e]=I[\omega(e, \partial e), e]$                      | $I[\omega, e] = I[\omega(e, \partial e), e]$ | in general, (16), (19) does                                                                                                                                                                                                                                                                         |
| cardona-qft-methods_F02191 | 170  | ■ 1.000 | $\omega(e, \partial e)$                                         | $\omega(e, \partial e)$                      | action in which torsion <i>is not</i> assumed to vanish identically, adding all                                                                                                                                                                                                                     |
| cardona-qft-methods_F02192 | 170  | ■ 0.857 | $T^{\{a\}}=0 \quad .^{\{18\}}$                                  | $T^a = 0$ . <sup>18</sup>                    | For $D = 0$ , there is one torsion term in addition to the Lorentz symmetry, $\mathcal{L} = -\frac{1}{2} \partial_\mu T^\mu$ (19)                                                                                                                                                                   |
| cardona-qft-methods_F02193 | 170  | ■ 1.000 | $\mathbf{P}_{\{4\}} \ v_{\{1\}} \ \tau_{\{0\}} \ \zeta_{\{0\}}$ | $\mathbf{P}_4 v_1 \tau_0 \zeta_0$            | $\mathcal{L}_1 = \frac{1}{2} \omega_{\mu\nu} \tau^\mu \tau^\nu + \frac{1}{2} \omega_{\mu\nu} \tau^\mu \tau^\nu + \frac{1}{2} \omega_{\mu\nu} \tau^\mu \tau^\nu + \frac{1}{2} \omega_{\mu\nu} \tau^\mu \tau^\nu$<br>The integral of the last term in (99) is a topological invariant (the Pontryagin |
| cardona-qft-methods_F02194 | 170  | ■ 0.964 | $S\ 0(5)$                                                       | $SO(5)$                                      |                                                                                                                                                                                                                                                                                                     |
| cardona-qft-methods_F02195 | 170  | ■ 0.964 | $S\ 0(4)$                                                       | $SO(4)$                                      |                                                                                                                                                                                                                                                                                                     |
| cardona-qft-methods_F02196 | 170  | ■ 0.948 | $S\ 0(n, \ 5-n)$                                                | $SO(n, 5 - n$                                |                                                                                                                                                                                                                                                                                                     |
| cardona-qft-methods_F02197 | 170  | ■ 0.948 | $S\ 0(m, \ 4-m)) \quad .^{\{28\}}$                              | $SO(m, 4 - m)).$ <sup>28</sup>               |                                                                                                                                                                                                                                                                                                     |
| cardona-qft-methods_F02198 | 171  | ■ 1.000 | $\mathbf{P}_{\{4\}}$                                            | $\mathbf{P}_4$                               | Both these terms are invariant under $SO(2, 1)$ (Lorentz invariant in three                                                                                                                                                                                                                         |
| cardona-qft-methods_F02199 | 171  | ■ 1.000 | $\mathbf{N}_{\{4\}}$                                            | $\mathbf{N}_4$                               | Both these terms are invariant under $SO(2, 1)$ (Lorentz invariant in three                                                                                                                                                                                                                         |
| cardona-qft-methods_F02200 | 171  | ■ 0.982 | $S\ 0(2, 1)$                                                    | $SO(2, 1)$                                   | since in a 4d space-time there is a common vector and tensor in 3 + 1 dimensions. For example, in $D = 7$ , the Lorentz CS form is                                                                                                                                                                  |
| cardona-qft-methods_F02201 | 171  | ■ 1.000 | $4\ k-1$                                                        | $4k - 1$                                     |                                                                                                                                                                                                                                                                                                     |
| cardona-qft-methods_F02202 | 171  | ■ 1.000 | $D=7$                                                           | $D = 7$                                      |                                                                                                                                                                                                                                                                                                     |
| cardona-qft-methods_F02203 | 172  | ■ 1.000 | $v, \ \tau$                                                     | $v, \tau$                                    | terms appear in a given dimension and the number of elementary terms of                                                                                                                                                                                                                             |
| cardona-qft-methods_F02204 | 172  | ■ 0.999 | $\{ \}^{\{-2\}}$                                                | $-2$                                         | in a quantum version of the theory. Dimensionful parameters in the action are potentially dangerous because they are likely to remain uncontrolled                                                                                                                                                  |

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| cardona-qft-methods_F02285 | 173  | ■ 1.000 | (\kappa)                                                                                                                                          | $(\kappa)$                                                 | seen to depend on a unique (dimensionless) coefficient ( $\kappa$ ), analogous to Newton's constant. This coefficient can be shown to be quantized, follow |
| cardona-qft-methods_F02286 | 173  | ■ 1.000 | l^{D-2 p}                                                                                                                                         | $l^{D-2p}$                                                 |                                                                                                                                                            |
| cardona-qft-methods_F02287 | 173  | ■ 1.000 | \left[e^{\mathrm{a}}\right]=l                                                                                                                     | $[e^a] = l$                                                |                                                                                                                                                            |
| cardona-qft-methods_F02288 | 173  | ■ 1.000 | \left[\omega^{\mathrm{a}\,\mathrm{b}}\right]=l^0                                                                                                  | $[\omega^{ab}] = l^0$                                      | 5.1.1. <i>Poincaré group</i>                                                                                                                               |
| cardona-qft-methods_F02289 | 173  | ■ 1.000 | \hat{\omega}^{\mathrm{a}\,\mathrm{b}}=\omega^{\mathrm{a}\,\mathrm{b}}+\xi\,\epsilon^{\mathrm{a}\,\mathrm{b}\,\mathrm{c}}\,\mathrm{e}_{\mathrm{c}} | $\hat{\omega}^{ab} = \omega^{ab} + \xi \epsilon^{abc} e_c$ |                                                                                                                                                            |
| cardona-qft-methods_F02210 | 173  | ■ 1.000 | \lambda^{\mathrm{a}}_{\,\,\mathrm{b}}                                                                                                             | $\lambda^a_b$                                              |                                                                                                                                                            |
| cardona-qft-methods_F02211 | 174  | ■ 1.000 | e^{\mathrm{c}},\,\,R^{\mathrm{a}\,\mathrm{b}}                                                                                                     | $e^c, R^{ab}$                                              | $\delta\omega^a_b = d\lambda^a_b + \omega^a_c \lambda^c_b - \omega^c_b \lambda^a_c$                                                                        |
| cardona-qft-methods_F02212 | 174  | ■ 1.000 | \epsilon_{\mathrm{a}\,\mathrm{b}\,\mathrm{c}}                                                                                                     | $\epsilon_{abc}$                                           | $\delta\omega^a_b = d\lambda^a_b + \omega^a_c \lambda^c_b - \omega^c_b \lambda^a_c$                                                                        |
| cardona-qft-methods_F02213 | 174  | ■ 1.000 | L_{\mathrm{3}}                                                                                                                                    | $L_3$                                                      | Combining these relations, the Lorentz invariance of $L_3$ can be explicitly                                                                               |
| cardona-qft-methods_F02214 | 174  | ■ 0.531 | {\mathrm{h}}                                                                                                                                      | $\mathrm{h}$                                               | $X^a{}_a = iY^a \perp_{\perp \dots a} Y^b$                                                                                                                 |
| cardona-qft-methods_F02215 | 174  | ■ 1.000 | \lambda^{\mathrm{a}}                                                                                                                              | $\lambda^a$                                                | $= D_\omega \lambda^a$                                                                                                                                     |
| cardona-qft-methods_F02216 | 174  | ■ 1.000 | \delta\,\mathrm{e}                                                                                                                                | $\delta e$                                                 | of the appropriate tangent space is not the local Poincaré symmetry, but                                                                                   |
| cardona-qft-methods_F02217 | 174  | ■ 1.000 | \delta\,\omega                                                                                                                                    | $\delta\omega$                                             | of the appropriate tangent space is not the local Poincaré symmetry, but                                                                                   |
| cardona-qft-methods_F02218 | 174  | ■ 0.439 | 2+1                                                                                                                                               | $2+1$                                                      |                                                                                                                                                            |
| cardona-qft-methods_F02219 | 174  | ■ 1.000 | \Lambda=\mp\,l^{-2}                                                                                                                               | $\Lambda = \mp l^{-2}$                                     |                                                                                                                                                            |
| cardona-qft-methods_F02220 | 175  | ■ 1.000 | \Lambda\neq 0                                                                                                                                     | $\Lambda \neq 0$                                           | corresponds to the covering space of a vacuum solution of the Einstein                                                                                     |
| cardona-qft-methods_F02221 | 175  | ■ 1.000 | \Lambda>0                                                                                                                                         | $\Lambda > 0$                                              | longer flat spacetime, but the de Sitter or anti-de Sitter space,                                                                                          |
| cardona-qft-methods_F02222 | 175  | ■ 1.000 | \Lambda<0                                                                                                                                         | $\Lambda < 0$                                              | longer flat spacetime, but the de Sitter or anti-de Sitter space,                                                                                          |
| cardona-qft-methods_F02223 | 175  | ■ 0.995 | W^{\mathrm{A}\,\mathrm{B}}                                                                                                                        | $W^{AB}$                                                   | formation parameters, but for which group? Since $\Lambda^{AB} = -\Lambda^{BA}$ , this indi-                                                               |
| cardona-qft-methods_F02224 | 175  | ■ 0.995 | \Lambda^{\mathrm{A}\,\mathrm{B}}                                                                                                                  | $\Lambda^{AB}$                                             | formation parameters, but for which group? Since $\Lambda^{AB} = -\Lambda^{BA}$ , this indi-                                                               |
| cardona-qft-methods_F02225 | 175  | ■ 0.999 | a,\,\mathrm{b},\,\ldots=1,2,\,\ldots,D                                                                                                            | $a,b,..=1,2,..D$                                           |                                                                                                                                                            |
| cardona-qft-methods_F02226 | 175  | ■ 0.999 | A,\,\mathrm{B},\,\ldots=1,2,\,\ldots,D+1                                                                                                          | $A,B,...=1,2,..,D+1$                                       |                                                                                                                                                            |
| cardona-qft-methods_F02227 | 175  | ■ 1.000 | \Lambda^{\mathrm{A}\,\mathrm{B}}=-\Lambda^{\mathrm{B}\,\mathrm{A}}                                                                                | $\Lambda^{AB} = -\Lambda^{BA}$                             |                                                                                                                                                            |
| cardona-qft-methods_F02228 | 175  | ■ 0.957 | S\,O(r,\,s)                                                                                                                                       | $SO(r,s)$                                                  |                                                                                                                                                            |
| cardona-qft-methods_F02229 | 176  | ■ 1.000 | \Upsilon^{\mathrm{A}\,\mathrm{B}}                                                                                                                 | $\Upsilon^{AB}$                                            |                                                                                                                                                            |
| cardona-qft-methods_F02230 | 176  | ■ 1.000 | W^{\mathrm{A}\,\mathrm{B}}=\Upsilon^{\mathrm{B}\,\mathrm{C}}\,W_{\mathrm{C}}^{\mathrm{A}}                                                         | $W^{AB} = \Upsilon^{BC} W^A_C$                             | $\Upsilon^{AB} = \left  \begin{array}{cc} \eta^{ab} & 0 \\ 0 & \pm 1 \end{array} \right , \quad (12)$                                                      |
| cardona-qft-methods_F02231 | 176  | ■ 1.000 | W^{\mathrm{A}\,\mathrm{B}}+W^{\mathrm{B}\,\mathrm{A}}=0                                                                                           | $W^{AB} + W^{BA} = 0$                                      | Since $\Upsilon^{AB}$ is constant, this last expression implies $W^{AB} + W^{BA} = 0$ , in                                                                 |



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| cardona-qft-methods_F02259                                                        | 179                                                           | ■ 1.000                                | $2\ n-1$                                                                                                                                                                | $2n-1$                                                                                  | <ul style="list-style-type: none"><li><math>L_{2n-1}</math> can be used as a Lagrangian in <math>(2n-1)</math> dimensions.</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                      |                        |        |                                                               |                                                               |                                        |                                                                                  |                                     |           |                    |                                    |           |                  |                          |        |                                                                                   |                                                 |         |
| cardona-qft-methods_F02260                                                        | 179                                                           | ■ 1.000                                | $\{ \ }^{\mathrm{j}}$                                                                                                                                                   | j                                                                                       | <sup>†</sup> This identifica                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                      |                        |        |                                                               |                                                               |                                        |                                                                                  |                                     |           |                    |                                    |           |                  |                          |        |                                                                                   |                                                 |         |
| cardona-qft-methods_F02261                                                        | 179                                                           | ■ 1.000                                | $L_{2\ n-1}$                                                                                                                                                            | $L_{2n-1}$                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                      |                        |        |                                                               |                                                               |                                        |                                                                                  |                                     |           |                    |                                    |           |                  |                          |        |                                                                                   |                                                 |         |
| cardona-qft-methods_F02262                                                        | 179                                                           | ■ 1.000                                | $\{ \ }^{\{36,37\}}$                                                                                                                                                    | 36,37                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                      |                        |        |                                                               |                                                               |                                        |                                                                                  |                                     |           |                    |                                    |           |                  |                          |        |                                                                                   |                                                 |         |
| cardona-qft-methods_F02263                                                        | 179                                                           | ■ 1.000                                | $\{ \ }^{\{38,39\}}$                                                                                                                                                    | 38,39                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                      |                        |        |                                                               |                                                               |                                        |                                                                                  |                                     |           |                    |                                    |           |                  |                          |        |                                                                                   |                                                 |         |
| cardona-qft-methods_F02264                                                        | 180                                                           | ■ 1.000                                | $\bar{\alpha}_{\{p\}}$                                                                                                                                                  | $\bar{\alpha}_p$                                                                        | where $L_n^D$ is given by (10) of section 4. This is a particular case of a Lovelock                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                      |                        |        |                                                               |                                                               |                                        |                                                                                  |                                     |           |                    |                                    |           |                  |                          |        |                                                                                   |                                                 |         |
| cardona-qft-methods_F02265                                                        | 180                                                           | ■ 1.000                                | $R_{\{t\}^{\{a\ b\}}:=R^{\{a\ b\}}+\left(t^2 / l^2\right) e^{\{a\} } e^{\{b\} }$                                                                                        | $R_t^{ab}:=R^{ab}+\left(t^2/l^2\right)e^ae^b$                                           | $L_5^{(5)}=-\frac{1}{l}\epsilon_{abcde}\left[e^{\cdot}R^{\cdot\cdot}R^{\cdot\cdot\cdot}\pm\frac{e^{\cdot}e^{\cdot}e^{\cdot}R^{\cdot\cdot\cdot}}{3l^2}+\frac{\cdot}{5l^4}\right]$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                      |                        |        |                                                               |                                                               |                                        |                                                                                  |                                     |           |                    |                                    |           |                  |                          |        |                                                                                   |                                                 |         |
| cardona-qft-methods_F02266                                                        | 180                                                           | ■ 1.000                                | $a_{\{3\}}$                                                                                                                                                             | $a_3$                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                      |                        |        |                                                               |                                                               |                                        |                                                                                  |                                     |           |                    |                                    |           |                  |                          |        |                                                                                   |                                                 |         |
| cardona-qft-methods_F02267                                                        | 180                                                           | ■ 1.000                                | $\cdots a_{2\ n-1}$                                                                                                                                                     | $\cdots a_{2n-1}$                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                      |                        |        |                                                               |                                                               |                                        |                                                                                  |                                     |           |                    |                                    |           |                  |                          |        |                                                                                   |                                                 |         |
| cardona-qft-methods_F02268                                                        | 180                                                           | ■ 1.000                                | $d\ L_{2\ n-1}^{\{(A\ d\ S\}}=\mathrm{d}\mathbf{E}_{\{2\ n\}}$                                                                                                          | $dL_{2n-1}^{(A)dS}=\mathbf{E}_{2n}$                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                      |                        |        |                                                               |                                                               |                                        |                                                                                  |                                     |           |                    |                                    |           |                  |                          |        |                                                                                   |                                                 |         |
| cardona-qft-methods_F02269                                                        | 181                                                           | ■ 1.000                                | $\{ \ }^{\{2,41\}}$                                                                                                                                                     | 2,41                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                      |                        |        |                                                               |                                                               |                                        |                                                                                  |                                     |           |                    |                                    |           |                  |                          |        |                                                                                   |                                                 |         |
| cardona-qft-methods_F02270                                                        | 181                                                           | —                                      | $L_{\{3\}^{\{(A\ d\ S\}}=\epsilon_{\{a\ b\ c\}}\left(R^{\{a\ b\}}\pm\frac{\epsilon^{\{a\ b\}}}{3l^2}\right)\frac{e^{\{a\}}e^{\{b\}}}{3l^2}\right)\frac{e^{\{c\}}}{l^2}$ | $L_3^{(A)dS}=\epsilon_{abc}\left(R^{ab}\pm\frac{e^ae^b}{3l^2}\right)e^c$                | —                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                      |                        |        |                                                               |                                                               |                                        |                                                                                  |                                     |           |                    |                                    |           |                  |                          |        |                                                                                   |                                                 |         |
| cardona-qft-methods_F02271                                                        | 181                                                           | ■ 0.546                                | $\mathbf{E}_{\{4\}}=\epsilon_{\{a\ b\ c\}}\left(R^{\{a\ b\}}\pm\frac{\epsilon^{\{a\ b\}}}{l^2}\right)\frac{e^{\{a\}}e^{\{b\}}}{l^2}\right)\frac{e^{\{c\}}}{l^2}$        | $\mathbf{E}_4=\epsilon_{abc}\left(R^{ab}\pm\frac{e^ae^b}{l^2}\right)T^c$                | topological invariants,<br><table><tr><th><math>D=3</math> CS Lagrangians</th><th>Characteristic classes</th><th>Groups</th></tr><tr><td><math>L_3^{(A)dS}=\epsilon_{abc}(R^{ab}\pm\frac{e^ae^b}{3l^2})e^c</math></td><td><math>\mathbf{E}_4=\epsilon_{abc}(R^{ab}\pm\frac{e^ae^b}{l^2})T^c</math></td><td><math>SO(4)</math>,<br/><math>SO(3,1)</math><br/>or <math>SO(2,2)</math></td></tr><tr><td><math>L_3^{Lor}=\omega^a{}_bd\omega^b{}_a+\frac{2}{3}\omega^a{}_b\omega^b{}_ce^c{}_a</math></td><td><math>\mathbf{P}_4^{Lor}=R^a{}_bR^b{}_a</math></td><td><math>SO(2,1)</math></td></tr><tr><td><math>L_3^{Tor}=e^aT_a</math></td><td><math>\mathbf{N}_4=T^aT_a-e^ae^bR_{ab}</math></td><td><math>SO(2,1)</math></td></tr><tr><td><math>L_3^{U(1)}=AdA</math></td><td><math>\mathbf{P}_4^{U(1)}=FF</math></td><td><math>U(1)</math></td></tr><tr><td><math>L_3^{SU(N)}=Tr[\mathbf{A}d\mathbf{A}+\frac{2}{3}\mathbf{A}\mathbf{A}\mathbf{A}]</math></td><td><math>\mathbf{P}_4^{SU(4)}=Tr[\mathbf{F}\mathbf{F}]</math></td><td><math>SU(N)</math></td></tr></table> <b>Table 1:</b> Three-dimensional gravitational CS Lagrangians, their related | $D=3$ CS Lagrangians | Characteristic classes | Groups | $L_3^{(A)dS}=\epsilon_{abc}(R^{ab}\pm\frac{e^ae^b}{3l^2})e^c$ | $\mathbf{E}_4=\epsilon_{abc}(R^{ab}\pm\frac{e^ae^b}{l^2})T^c$ | $SO(4)$ ,<br>$SO(3,1)$<br>or $SO(2,2)$ | $L_3^{Lor}=\omega^a{}_bd\omega^b{}_a+\frac{2}{3}\omega^a{}_b\omega^b{}_ce^c{}_a$ | $\mathbf{P}_4^{Lor}=R^a{}_bR^b{}_a$ | $SO(2,1)$ | $L_3^{Tor}=e^aT_a$ | $\mathbf{N}_4=T^aT_a-e^ae^bR_{ab}$ | $SO(2,1)$ | $L_3^{U(1)}=AdA$ | $\mathbf{P}_4^{U(1)}=FF$ | $U(1)$ | $L_3^{SU(N)}=Tr[\mathbf{A}d\mathbf{A}+\frac{2}{3}\mathbf{A}\mathbf{A}\mathbf{A}]$ | $\mathbf{P}_4^{SU(4)}=Tr[\mathbf{F}\mathbf{F}]$ | $SU(N)$ |
| $D=3$ CS Lagrangians                                                              | Characteristic classes                                        | Groups                                 |                                                                                                                                                                         |                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                      |                        |        |                                                               |                                                               |                                        |                                                                                  |                                     |           |                    |                                    |           |                  |                          |        |                                                                                   |                                                 |         |
| $L_3^{(A)dS}=\epsilon_{abc}(R^{ab}\pm\frac{e^ae^b}{3l^2})e^c$                     | $\mathbf{E}_4=\epsilon_{abc}(R^{ab}\pm\frac{e^ae^b}{l^2})T^c$ | $SO(4)$ ,<br>$SO(3,1)$<br>or $SO(2,2)$ |                                                                                                                                                                         |                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                      |                        |        |                                                               |                                                               |                                        |                                                                                  |                                     |           |                    |                                    |           |                  |                          |        |                                                                                   |                                                 |         |
| $L_3^{Lor}=\omega^a{}_bd\omega^b{}_a+\frac{2}{3}\omega^a{}_b\omega^b{}_ce^c{}_a$  | $\mathbf{P}_4^{Lor}=R^a{}_bR^b{}_a$                           | $SO(2,1)$                              |                                                                                                                                                                         |                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                      |                        |        |                                                               |                                                               |                                        |                                                                                  |                                     |           |                    |                                    |           |                  |                          |        |                                                                                   |                                                 |         |
| $L_3^{Tor}=e^aT_a$                                                                | $\mathbf{N}_4=T^aT_a-e^ae^bR_{ab}$                            | $SO(2,1)$                              |                                                                                                                                                                         |                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                      |                        |        |                                                               |                                                               |                                        |                                                                                  |                                     |           |                    |                                    |           |                  |                          |        |                                                                                   |                                                 |         |
| $L_3^{U(1)}=AdA$                                                                  | $\mathbf{P}_4^{U(1)}=FF$                                      | $U(1)$                                 |                                                                                                                                                                         |                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                      |                        |        |                                                               |                                                               |                                        |                                                                                  |                                     |           |                    |                                    |           |                  |                          |        |                                                                                   |                                                 |         |
| $L_3^{SU(N)}=Tr[\mathbf{A}d\mathbf{A}+\frac{2}{3}\mathbf{A}\mathbf{A}\mathbf{A}]$ | $\mathbf{P}_4^{SU(4)}=Tr[\mathbf{F}\mathbf{F}]$               | $SU(N)$                                |                                                                                                                                                                         |                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                      |                        |        |                                                               |                                                               |                                        |                                                                                  |                                     |           |                    |                                    |           |                  |                          |        |                                                                                   |                                                 |         |
| cardona-qft-methods_F02272                                                        | 181                                                           | ■ 0.546                                | $S\ O(2,2)$                                                                                                                                                             | $SO(2,2)$                                                                               | topological invariants,<br><table><tr><th><math>D=3</math> CS Lagrangians</th><th>Characteristic classes</th><th>Groups</th></tr><tr><td><math>L_3^{(A)dS}=\epsilon_{abc}(R^{ab}\pm\frac{e^ae^b}{3l^2})e^c</math></td><td><math>\mathbf{E}_4=\epsilon_{abc}(R^{ab}\pm\frac{e^ae^b}{l^2})T^c</math></td><td><math>SO(4)</math>,<br/><math>SO(3,1)</math><br/>or <math>SO(2,2)</math></td></tr><tr><td><math>L_3^{Lor}=\omega^a{}_bd\omega^b{}_a+\frac{2}{3}\omega^a{}_b\omega^b{}_ce^c{}_a</math></td><td><math>\mathbf{P}_4^{Lor}=R^a{}_bR^b{}_a</math></td><td><math>SO(2,1)</math></td></tr><tr><td><math>L_3^{Tor}=e^aT_a</math></td><td><math>\mathbf{N}_4=T^aT_a-e^ae^bR_{ab}</math></td><td><math>SO(2,1)</math></td></tr><tr><td><math>L_3^{U(1)}=AdA</math></td><td><math>\mathbf{P}_4^{U(1)}=FF</math></td><td><math>U(1)</math></td></tr><tr><td><math>L_3^{SU(N)}=Tr[\mathbf{A}d\mathbf{A}+\frac{2}{3}\mathbf{A}\mathbf{A}\mathbf{A}]</math></td><td><math>\mathbf{P}_4^{SU(4)}=Tr[\mathbf{F}\mathbf{F}]</math></td><td><math>SU(N)</math></td></tr></table> <b>Table 1:</b> Three-dimensional gravitational CS Lagrangians, their related | $D=3$ CS Lagrangians | Characteristic classes | Groups | $L_3^{(A)dS}=\epsilon_{abc}(R^{ab}\pm\frac{e^ae^b}{3l^2})e^c$ | $\mathbf{E}_4=\epsilon_{abc}(R^{ab}\pm\frac{e^ae^b}{l^2})T^c$ | $SO(4)$ ,<br>$SO(3,1)$<br>or $SO(2,2)$ | $L_3^{Lor}=\omega^a{}_bd\omega^b{}_a+\frac{2}{3}\omega^a{}_b\omega^b{}_ce^c{}_a$ | $\mathbf{P}_4^{Lor}=R^a{}_bR^b{}_a$ | $SO(2,1)$ | $L_3^{Tor}=e^aT_a$ | $\mathbf{N}_4=T^aT_a-e^ae^bR_{ab}$ | $SO(2,1)$ | $L_3^{U(1)}=AdA$ | $\mathbf{P}_4^{U(1)}=FF$ | $U(1)$ | $L_3^{SU(N)}=Tr[\mathbf{A}d\mathbf{A}+\frac{2}{3}\mathbf{A}\mathbf{A}\mathbf{A}]$ | $\mathbf{P}_4^{SU(4)}=Tr[\mathbf{F}\mathbf{F}]$ | $SU(N)$ |
| $D=3$ CS Lagrangians                                                              | Characteristic classes                                        | Groups                                 |                                                                                                                                                                         |                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                      |                        |        |                                                               |                                                               |                                        |                                                                                  |                                     |           |                    |                                    |           |                  |                          |        |                                                                                   |                                                 |         |
| $L_3^{(A)dS}=\epsilon_{abc}(R^{ab}\pm\frac{e^ae^b}{3l^2})e^c$                     | $\mathbf{E}_4=\epsilon_{abc}(R^{ab}\pm\frac{e^ae^b}{l^2})T^c$ | $SO(4)$ ,<br>$SO(3,1)$<br>or $SO(2,2)$ |                                                                                                                                                                         |                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                      |                        |        |                                                               |                                                               |                                        |                                                                                  |                                     |           |                    |                                    |           |                  |                          |        |                                                                                   |                                                 |         |
| $L_3^{Lor}=\omega^a{}_bd\omega^b{}_a+\frac{2}{3}\omega^a{}_b\omega^b{}_ce^c{}_a$  | $\mathbf{P}_4^{Lor}=R^a{}_bR^b{}_a$                           | $SO(2,1)$                              |                                                                                                                                                                         |                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                      |                        |        |                                                               |                                                               |                                        |                                                                                  |                                     |           |                    |                                    |           |                  |                          |        |                                                                                   |                                                 |         |
| $L_3^{Tor}=e^aT_a$                                                                | $\mathbf{N}_4=T^aT_a-e^ae^bR_{ab}$                            | $SO(2,1)$                              |                                                                                                                                                                         |                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                      |                        |        |                                                               |                                                               |                                        |                                                                                  |                                     |           |                    |                                    |           |                  |                          |        |                                                                                   |                                                 |         |
| $L_3^{U(1)}=AdA$                                                                  | $\mathbf{P}_4^{U(1)}=FF$                                      | $U(1)$                                 |                                                                                                                                                                         |                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                      |                        |        |                                                               |                                                               |                                        |                                                                                  |                                     |           |                    |                                    |           |                  |                          |        |                                                                                   |                                                 |         |
| $L_3^{SU(N)}=Tr[\mathbf{A}d\mathbf{A}+\frac{2}{3}\mathbf{A}\mathbf{A}\mathbf{A}]$ | $\mathbf{P}_4^{SU(4)}=Tr[\mathbf{F}\mathbf{F}]$               | $SU(N)$                                |                                                                                                                                                                         |                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                      |                        |        |                                                               |                                                               |                                        |                                                                                  |                                     |           |                    |                                    |           |                  |                          |        |                                                                                   |                                                 |         |
| cardona-qft-methods_F02273                                                        | 181                                                           | ■ 0.546                                | $L_{\{3\}^{\{\text{Lor}\}}}=\omega^{\{a\}\{ \}_b}d\ \omega^{\{b\}\{ \}_a}+\frac{2}{3}\ \omega^{\{a\}\{ \}_b}\ \omega^{\{b\}\{ \}_c}\ \omega^{\{c\}\{ \}_a}$             | $L_3^{\text{Lor}}=\omega^a{}_bd\omega^b{}_a+\frac{2}{3}\omega^a{}_b\omega^b{}_ce^c{}_a$ | topological invariants,<br><table><tr><th><math>D=3</math> CS Lagrangians</th><th>Characteristic classes</th><th>Groups</th></tr><tr><td><math>L_3^{(A)dS}=\epsilon_{abc}(R^{ab}\pm\frac{e^ae^b}{3l^2})e^c</math></td><td><math>\mathbf{E}_4=\epsilon_{abc}(R^{ab}\pm\frac{e^ae^b}{l^2})T^c</math></td><td><math>SO(4)</math>,<br/><math>SO(3,1)</math><br/>or <math>SO(2,2)</math></td></tr><tr><td><math>L_3^{Lor}=\omega^a{}_bd\omega^b{}_a+\frac{2}{3}\omega^a{}_b\omega^b{}_ce^c{}_a</math></td><td><math>\mathbf{P}_4^{Lor}=R^a{}_bR^b{}_a</math></td><td><math>SO(2,1)</math></td></tr><tr><td><math>L_3^{Tor}=e^aT_a</math></td><td><math>\mathbf{N}_4=T^aT_a-e^ae^bR_{ab}</math></td><td><math>SO(2,1)</math></td></tr><tr><td><math>L_3^{U(1)}=AdA</math></td><td><math>\mathbf{P}_4^{U(1)}=FF</math></td><td><math>U(1)</math></td></tr><tr><td><math>L_3^{SU(N)}=Tr[\mathbf{A}d\mathbf{A}+\frac{2}{3}\mathbf{A}\mathbf{A}\mathbf{A}]</math></td><td><math>\mathbf{P}_4^{SU(4)}=Tr[\mathbf{F}\mathbf{F}]</math></td><td><math>SU(N)</math></td></tr></table> <b>Table 1:</b> Three-dimensional gravitational CS Lagrangians, their related | $D=3$ CS Lagrangians | Characteristic classes | Groups | $L_3^{(A)dS}=\epsilon_{abc}(R^{ab}\pm\frac{e^ae^b}{3l^2})e^c$ | $\mathbf{E}_4=\epsilon_{abc}(R^{ab}\pm\frac{e^ae^b}{l^2})T^c$ | $SO(4)$ ,<br>$SO(3,1)$<br>or $SO(2,2)$ | $L_3^{Lor}=\omega^a{}_bd\omega^b{}_a+\frac{2}{3}\omega^a{}_b\omega^b{}_ce^c{}_a$ | $\mathbf{P}_4^{Lor}=R^a{}_bR^b{}_a$ | $SO(2,1)$ | $L_3^{Tor}=e^aT_a$ | $\mathbf{N}_4=T^aT_a-e^ae^bR_{ab}$ | $SO(2,1)$ | $L_3^{U(1)}=AdA$ | $\mathbf{P}_4^{U(1)}=FF$ | $U(1)$ | $L_3^{SU(N)}=Tr[\mathbf{A}d\mathbf{A}+\frac{2}{3}\mathbf{A}\mathbf{A}\mathbf{A}]$ | $\mathbf{P}_4^{SU(4)}=Tr[\mathbf{F}\mathbf{F}]$ | $SU(N)$ |
| $D=3$ CS Lagrangians                                                              | Characteristic classes                                        | Groups                                 |                                                                                                                                                                         |                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                      |                        |        |                                                               |                                                               |                                        |                                                                                  |                                     |           |                    |                                    |           |                  |                          |        |                                                                                   |                                                 |         |
| $L_3^{(A)dS}=\epsilon_{abc}(R^{ab}\pm\frac{e^ae^b}{3l^2})e^c$                     | $\mathbf{E}_4=\epsilon_{abc}(R^{ab}\pm\frac{e^ae^b}{l^2})T^c$ | $SO(4)$ ,<br>$SO(3,1)$<br>or $SO(2,2)$ |                                                                                                                                                                         |                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                      |                        |        |                                                               |                                                               |                                        |                                                                                  |                                     |           |                    |                                    |           |                  |                          |        |                                                                                   |                                                 |         |
| $L_3^{Lor}=\omega^a{}_bd\omega^b{}_a+\frac{2}{3}\omega^a{}_b\omega^b{}_ce^c{}_a$  | $\mathbf{P}_4^{Lor}=R^a{}_bR^b{}_a$                           | $SO(2,1)$                              |                                                                                                                                                                         |                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                      |                        |        |                                                               |                                                               |                                        |                                                                                  |                                     |           |                    |                                    |           |                  |                          |        |                                                                                   |                                                 |         |
| $L_3^{Tor}=e^aT_a$                                                                | $\mathbf{N}_4=T^aT_a-e^ae^bR_{ab}$                            | $SO(2,1)$                              |                                                                                                                                                                         |                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                      |                        |        |                                                               |                                                               |                                        |                                                                                  |                                     |           |                    |                                    |           |                  |                          |        |                                                                                   |                                                 |         |
| $L_3^{U(1)}=AdA$                                                                  | $\mathbf{P}_4^{U(1)}=FF$                                      | $U(1)$                                 |                                                                                                                                                                         |                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                      |                        |        |                                                               |                                                               |                                        |                                                                                  |                                     |           |                    |                                    |           |                  |                          |        |                                                                                   |                                                 |         |
| $L_3^{SU(N)}=Tr[\mathbf{A}d\mathbf{A}+\frac{2}{3}\mathbf{A}\mathbf{A}\mathbf{A}]$ | $\mathbf{P}_4^{SU(4)}=Tr[\mathbf{F}\mathbf{F}]$               | $SU(N)$                                |                                                                                                                                                                         |                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                      |                        |        |                                                               |                                                               |                                        |                                                                                  |                                     |           |                    |                                    |           |                  |                          |        |                                                                                   |                                                 |         |

| Identifier                                                                                   | Page                                                               | Conf.                                   | LaTeX source                                  | Rendered                                      | Image                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                        |                        |        |                                                                    |                                                                    |                                         |                                                                                              |                                        |            |                       |                                           |            |                    |                            |        |                                                            |                                          |         |
|----------------------------------------------------------------------------------------------|--------------------------------------------------------------------|-----------------------------------------|-----------------------------------------------|-----------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|------------------------|--------|--------------------------------------------------------------------|--------------------------------------------------------------------|-----------------------------------------|----------------------------------------------------------------------------------------------|----------------------------------------|------------|-----------------------|-------------------------------------------|------------|--------------------|----------------------------|--------|------------------------------------------------------------|------------------------------------------|---------|
| cardona-qft-methods_F02274                                                                   | 181                                                                | 0.546                                   | $\mathbf{P}_4^{\text{Lor}} = R^a{}_b R^b{}_a$ | $\mathbf{P}_4^{\text{Lor}} = R^a{}_b R^b{}_a$ | <div>topological invariants,</div> <table><tr><th><math>D = 3</math> CS Lagrangians</th><th>Characteristic classes</th><th>Groups</th></tr><tr><td><math>L_3^{(A)dS} = \epsilon_{abc}(R^{ab} \pm \frac{e^a e^b}{3l^2})e^c</math></td><td><math>\mathbf{E}_4 = \epsilon_{abc}(R^{ab} \pm \frac{e^a e^b}{l^2})T^c</math></td><td><math>SO(4),</math><br/><math>SO(3, 1)</math><br/>or <math>SO(2, 2)</math></td></tr><tr><td><math>L_3^{Lor} = \omega^a{}_b d\omega^b{}_a + \frac{2}{3}\omega^a{}_b \omega^b{}_c \omega^c{}_a</math></td><td><math>\mathbf{P}_4^{Lor} = R^a{}_b R^b{}_a</math></td><td><math>SO(2, 1)</math></td></tr><tr><td><math>L_3^{Tor} = e^a T_a</math></td><td><math>\mathbf{N}_4 = T^a T_a - e^a e^b R_{ab}</math></td><td><math>SO(2, 1)</math></td></tr><tr><td><math>L_3^{U(1)} = AdA</math></td><td><math>\mathbf{P}_4^{U(1)} = FF</math></td><td><math>U(1)</math></td></tr><tr><td><math>L_3^{SU(N)} = Tr[\mathbf{AdA} + \frac{2}{3}\mathbf{AAA}]</math></td><td><math>\mathbf{P}_4^{SU(4)} = Tr[\mathbf{FF}]</math></td><td><math>SU(N)</math></td></tr></table> <div>Table 1: Three-dimensional gravitational CS Lagrangians, their related</div> | $D = 3$ CS Lagrangians | Characteristic classes | Groups | $L_3^{(A)dS} = \epsilon_{abc}(R^{ab} \pm \frac{e^a e^b}{3l^2})e^c$ | $\mathbf{E}_4 = \epsilon_{abc}(R^{ab} \pm \frac{e^a e^b}{l^2})T^c$ | $SO(4),$<br>$SO(3, 1)$<br>or $SO(2, 2)$ | $L_3^{Lor} = \omega^a{}_b d\omega^b{}_a + \frac{2}{3}\omega^a{}_b \omega^b{}_c \omega^c{}_a$ | $\mathbf{P}_4^{Lor} = R^a{}_b R^b{}_a$ | $SO(2, 1)$ | $L_3^{Tor} = e^a T_a$ | $\mathbf{N}_4 = T^a T_a - e^a e^b R_{ab}$ | $SO(2, 1)$ | $L_3^{U(1)} = AdA$ | $\mathbf{P}_4^{U(1)} = FF$ | $U(1)$ | $L_3^{SU(N)} = Tr[\mathbf{AdA} + \frac{2}{3}\mathbf{AAA}]$ | $\mathbf{P}_4^{SU(4)} = Tr[\mathbf{FF}]$ | $SU(N)$ |
| $D = 3$ CS Lagrangians                                                                       | Characteristic classes                                             | Groups                                  |                                               |                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                        |                        |        |                                                                    |                                                                    |                                         |                                                                                              |                                        |            |                       |                                           |            |                    |                            |        |                                                            |                                          |         |
| $L_3^{(A)dS} = \epsilon_{abc}(R^{ab} \pm \frac{e^a e^b}{3l^2})e^c$                           | $\mathbf{E}_4 = \epsilon_{abc}(R^{ab} \pm \frac{e^a e^b}{l^2})T^c$ | $SO(4),$<br>$SO(3, 1)$<br>or $SO(2, 2)$ |                                               |                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                        |                        |        |                                                                    |                                                                    |                                         |                                                                                              |                                        |            |                       |                                           |            |                    |                            |        |                                                            |                                          |         |
| $L_3^{Lor} = \omega^a{}_b d\omega^b{}_a + \frac{2}{3}\omega^a{}_b \omega^b{}_c \omega^c{}_a$ | $\mathbf{P}_4^{Lor} = R^a{}_b R^b{}_a$                             | $SO(2, 1)$                              |                                               |                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                        |                        |        |                                                                    |                                                                    |                                         |                                                                                              |                                        |            |                       |                                           |            |                    |                            |        |                                                            |                                          |         |
| $L_3^{Tor} = e^a T_a$                                                                        | $\mathbf{N}_4 = T^a T_a - e^a e^b R_{ab}$                          | $SO(2, 1)$                              |                                               |                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                        |                        |        |                                                                    |                                                                    |                                         |                                                                                              |                                        |            |                       |                                           |            |                    |                            |        |                                                            |                                          |         |
| $L_3^{U(1)} = AdA$                                                                           | $\mathbf{P}_4^{U(1)} = FF$                                         | $U(1)$                                  |                                               |                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                        |                        |        |                                                                    |                                                                    |                                         |                                                                                              |                                        |            |                       |                                           |            |                    |                            |        |                                                            |                                          |         |
| $L_3^{SU(N)} = Tr[\mathbf{AdA} + \frac{2}{3}\mathbf{AAA}]$                                   | $\mathbf{P}_4^{SU(4)} = Tr[\mathbf{FF}]$                           | $SU(N)$                                 |                                               |                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                        |                        |        |                                                                    |                                                                    |                                         |                                                                                              |                                        |            |                       |                                           |            |                    |                            |        |                                                            |                                          |         |
| cardona-qft-methods_F02275                                                                   | 181                                                                | 0.546                                   | $L_3^{\text{Tor}} = e^a T_a$                  | $L_3^{\text{Tor}} = e^a T_a$                  | <div>topological invariants,</div> <table><tr><th><math>D = 3</math> CS Lagrangians</th><th>Characteristic classes</th><th>Groups</th></tr><tr><td><math>L_3^{(A)dS} = \epsilon_{abc}(R^{ab} \pm \frac{e^a e^b}{3l^2})e^c</math></td><td><math>\mathbf{E}_4 = \epsilon_{abc}(R^{ab} \pm \frac{e^a e^b}{l^2})T^c</math></td><td><math>SO(4),</math><br/><math>SO(3, 1)</math><br/>or <math>SO(2, 2)</math></td></tr><tr><td><math>L_3^{Lor} = \omega^a{}_b d\omega^b{}_a + \frac{2}{3}\omega^a{}_b \omega^b{}_c \omega^c{}_a</math></td><td><math>\mathbf{P}_4^{Lor} = R^a{}_b R^b{}_a</math></td><td><math>SO(2, 1)</math></td></tr><tr><td><math>L_3^{Tor} = e^a T_a</math></td><td><math>\mathbf{N}_4 = T^a T_a - e^a e^b R_{ab}</math></td><td><math>SO(2, 1)</math></td></tr><tr><td><math>L_3^{U(1)} = AdA</math></td><td><math>\mathbf{P}_4^{U(1)} = FF</math></td><td><math>U(1)</math></td></tr><tr><td><math>L_3^{SU(N)} = Tr[\mathbf{AdA} + \frac{2}{3}\mathbf{AAA}]</math></td><td><math>\mathbf{P}_4^{SU(4)} = Tr[\mathbf{FF}]</math></td><td><math>SU(N)</math></td></tr></table> <div>Table 1: Three-dimensional gravitational CS Lagrangians, their related</div> | $D = 3$ CS Lagrangians | Characteristic classes | Groups | $L_3^{(A)dS} = \epsilon_{abc}(R^{ab} \pm \frac{e^a e^b}{3l^2})e^c$ | $\mathbf{E}_4 = \epsilon_{abc}(R^{ab} \pm \frac{e^a e^b}{l^2})T^c$ | $SO(4),$<br>$SO(3, 1)$<br>or $SO(2, 2)$ | $L_3^{Lor} = \omega^a{}_b d\omega^b{}_a + \frac{2}{3}\omega^a{}_b \omega^b{}_c \omega^c{}_a$ | $\mathbf{P}_4^{Lor} = R^a{}_b R^b{}_a$ | $SO(2, 1)$ | $L_3^{Tor} = e^a T_a$ | $\mathbf{N}_4 = T^a T_a - e^a e^b R_{ab}$ | $SO(2, 1)$ | $L_3^{U(1)} = AdA$ | $\mathbf{P}_4^{U(1)} = FF$ | $U(1)$ | $L_3^{SU(N)} = Tr[\mathbf{AdA} + \frac{2}{3}\mathbf{AAA}]$ | $\mathbf{P}_4^{SU(4)} = Tr[\mathbf{FF}]$ | $SU(N)$ |
| $D = 3$ CS Lagrangians                                                                       | Characteristic classes                                             | Groups                                  |                                               |                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                        |                        |        |                                                                    |                                                                    |                                         |                                                                                              |                                        |            |                       |                                           |            |                    |                            |        |                                                            |                                          |         |
| $L_3^{(A)dS} = \epsilon_{abc}(R^{ab} \pm \frac{e^a e^b}{3l^2})e^c$                           | $\mathbf{E}_4 = \epsilon_{abc}(R^{ab} \pm \frac{e^a e^b}{l^2})T^c$ | $SO(4),$<br>$SO(3, 1)$<br>or $SO(2, 2)$ |                                               |                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                        |                        |        |                                                                    |                                                                    |                                         |                                                                                              |                                        |            |                       |                                           |            |                    |                            |        |                                                            |                                          |         |
| $L_3^{Lor} = \omega^a{}_b d\omega^b{}_a + \frac{2}{3}\omega^a{}_b \omega^b{}_c \omega^c{}_a$ | $\mathbf{P}_4^{Lor} = R^a{}_b R^b{}_a$                             | $SO(2, 1)$                              |                                               |                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                        |                        |        |                                                                    |                                                                    |                                         |                                                                                              |                                        |            |                       |                                           |            |                    |                            |        |                                                            |                                          |         |
| $L_3^{Tor} = e^a T_a$                                                                        | $\mathbf{N}_4 = T^a T_a - e^a e^b R_{ab}$                          | $SO(2, 1)$                              |                                               |                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                        |                        |        |                                                                    |                                                                    |                                         |                                                                                              |                                        |            |                       |                                           |            |                    |                            |        |                                                            |                                          |         |
| $L_3^{U(1)} = AdA$                                                                           | $\mathbf{P}_4^{U(1)} = FF$                                         | $U(1)$                                  |                                               |                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                        |                        |        |                                                                    |                                                                    |                                         |                                                                                              |                                        |            |                       |                                           |            |                    |                            |        |                                                            |                                          |         |
| $L_3^{SU(N)} = Tr[\mathbf{AdA} + \frac{2}{3}\mathbf{AAA}]$                                   | $\mathbf{P}_4^{SU(4)} = Tr[\mathbf{FF}]$                           | $SU(N)$                                 |                                               |                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                        |                        |        |                                                                    |                                                                    |                                         |                                                                                              |                                        |            |                       |                                           |            |                    |                            |        |                                                            |                                          |         |
| cardona-qft-methods_F02276                                                                   | 181                                                                | 0.546                                   | $\mathbf{N}_4 = T^a T_a - e^a e^b R_{ab}$     | $\mathbf{N}_4 = T^a T_a - e^a e^b R_{ab}$     | <div>topological invariants,</div> <table><tr><th><math>D = 3</math> CS Lagrangians</th><th>Characteristic classes</th><th>Groups</th></tr><tr><td><math>L_3^{(A)dS} = \epsilon_{abc}(R^{ab} \pm \frac{e^a e^b}{3l^2})e^c</math></td><td><math>\mathbf{E}_4 = \epsilon_{abc}(R^{ab} \pm \frac{e^a e^b}{l^2})T^c</math></td><td><math>SO(4),</math><br/><math>SO(3, 1)</math><br/>or <math>SO(2, 2)</math></td></tr><tr><td><math>L_3^{Lor} = \omega^a{}_b d\omega^b{}_a + \frac{2}{3}\omega^a{}_b \omega^b{}_c \omega^c{}_a</math></td><td><math>\mathbf{P}_4^{Lor} = R^a{}_b R^b{}_a</math></td><td><math>SO(2, 1)</math></td></tr><tr><td><math>L_3^{Tor} = e^a T_a</math></td><td><math>\mathbf{N}_4 = T^a T_a - e^a e^b R_{ab}</math></td><td><math>SO(2, 1)</math></td></tr><tr><td><math>L_3^{U(1)} = AdA</math></td><td><math>\mathbf{P}_4^{U(1)} = FF</math></td><td><math>U(1)</math></td></tr><tr><td><math>L_3^{SU(N)} = Tr[\mathbf{AdA} + \frac{2}{3}\mathbf{AAA}]</math></td><td><math>\mathbf{P}_4^{SU(4)} = Tr[\mathbf{FF}]</math></td><td><math>SU(N)</math></td></tr></table> <div>Table 1: Three-dimensional gravitational CS Lagrangians, their related</div> | $D = 3$ CS Lagrangians | Characteristic classes | Groups | $L_3^{(A)dS} = \epsilon_{abc}(R^{ab} \pm \frac{e^a e^b}{3l^2})e^c$ | $\mathbf{E}_4 = \epsilon_{abc}(R^{ab} \pm \frac{e^a e^b}{l^2})T^c$ | $SO(4),$<br>$SO(3, 1)$<br>or $SO(2, 2)$ | $L_3^{Lor} = \omega^a{}_b d\omega^b{}_a + \frac{2}{3}\omega^a{}_b \omega^b{}_c \omega^c{}_a$ | $\mathbf{P}_4^{Lor} = R^a{}_b R^b{}_a$ | $SO(2, 1)$ | $L_3^{Tor} = e^a T_a$ | $\mathbf{N}_4 = T^a T_a - e^a e^b R_{ab}$ | $SO(2, 1)$ | $L_3^{U(1)} = AdA$ | $\mathbf{P}_4^{U(1)} = FF$ | $U(1)$ | $L_3^{SU(N)} = Tr[\mathbf{AdA} + \frac{2}{3}\mathbf{AAA}]$ | $\mathbf{P}_4^{SU(4)} = Tr[\mathbf{FF}]$ | $SU(N)$ |
| $D = 3$ CS Lagrangians                                                                       | Characteristic classes                                             | Groups                                  |                                               |                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                        |                        |        |                                                                    |                                                                    |                                         |                                                                                              |                                        |            |                       |                                           |            |                    |                            |        |                                                            |                                          |         |
| $L_3^{(A)dS} = \epsilon_{abc}(R^{ab} \pm \frac{e^a e^b}{3l^2})e^c$                           | $\mathbf{E}_4 = \epsilon_{abc}(R^{ab} \pm \frac{e^a e^b}{l^2})T^c$ | $SO(4),$<br>$SO(3, 1)$<br>or $SO(2, 2)$ |                                               |                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                        |                        |        |                                                                    |                                                                    |                                         |                                                                                              |                                        |            |                       |                                           |            |                    |                            |        |                                                            |                                          |         |
| $L_3^{Lor} = \omega^a{}_b d\omega^b{}_a + \frac{2}{3}\omega^a{}_b \omega^b{}_c \omega^c{}_a$ | $\mathbf{P}_4^{Lor} = R^a{}_b R^b{}_a$                             | $SO(2, 1)$                              |                                               |                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                        |                        |        |                                                                    |                                                                    |                                         |                                                                                              |                                        |            |                       |                                           |            |                    |                            |        |                                                            |                                          |         |
| $L_3^{Tor} = e^a T_a$                                                                        | $\mathbf{N}_4 = T^a T_a - e^a e^b R_{ab}$                          | $SO(2, 1)$                              |                                               |                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                        |                        |        |                                                                    |                                                                    |                                         |                                                                                              |                                        |            |                       |                                           |            |                    |                            |        |                                                            |                                          |         |
| $L_3^{U(1)} = AdA$                                                                           | $\mathbf{P}_4^{U(1)} = FF$                                         | $U(1)$                                  |                                               |                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                        |                        |        |                                                                    |                                                                    |                                         |                                                                                              |                                        |            |                       |                                           |            |                    |                            |        |                                                            |                                          |         |
| $L_3^{SU(N)} = Tr[\mathbf{AdA} + \frac{2}{3}\mathbf{AAA}]$                                   | $\mathbf{P}_4^{SU(4)} = Tr[\mathbf{FF}]$                           | $SU(N)$                                 |                                               |                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                        |                        |        |                                                                    |                                                                    |                                         |                                                                                              |                                        |            |                       |                                           |            |                    |                            |        |                                                            |                                          |         |
| cardona-qft-methods_F02277                                                                   | 181                                                                | 0.546                                   | $L_3^{U(1)} = AdA$                            | $L_3^{U(1)} = AdA$                            | <div>topological invariants,</div> <table><tr><th><math>D = 3</math> CS Lagrangians</th><th>Characteristic classes</th><th>Groups</th></tr><tr><td><math>L_3^{(A)dS} = \epsilon_{abc}(R^{ab} \pm \frac{e^a e^b}{3l^2})e^c</math></td><td><math>\mathbf{E}_4 = \epsilon_{abc}(R^{ab} \pm \frac{e^a e^b}{l^2})T^c</math></td><td><math>SO(4),</math><br/><math>SO(3, 1)</math><br/>or <math>SO(2, 2)</math></td></tr><tr><td><math>L_3^{Lor} = \omega^a{}_b d\omega^b{}_a + \frac{2}{3}\omega^a{}_b \omega^b{}_c \omega^c{}_a</math></td><td><math>\mathbf{P}_4^{Lor} = R^a{}_b R^b{}_a</math></td><td><math>SO(2, 1)</math></td></tr><tr><td><math>L_3^{Tor} = e^a T_a</math></td><td><math>\mathbf{N}_4 = T^a T_a - e^a e^b R_{ab}</math></td><td><math>SO(2, 1)</math></td></tr><tr><td><math>L_3^{U(1)} = AdA</math></td><td><math>\mathbf{P}_4^{U(1)} = FF</math></td><td><math>U(1)</math></td></tr><tr><td><math>L_3^{SU(N)} = Tr[\mathbf{AdA} + \frac{2}{3}\mathbf{AAA}]</math></td><td><math>\mathbf{P}_4^{SU(4)} = Tr[\mathbf{FF}]</math></td><td><math>SU(N)</math></td></tr></table> <div>Table 1: Three-dimensional gravitational CS Lagrangians, their related</div> | $D = 3$ CS Lagrangians | Characteristic classes | Groups | $L_3^{(A)dS} = \epsilon_{abc}(R^{ab} \pm \frac{e^a e^b}{3l^2})e^c$ | $\mathbf{E}_4 = \epsilon_{abc}(R^{ab} \pm \frac{e^a e^b}{l^2})T^c$ | $SO(4),$<br>$SO(3, 1)$<br>or $SO(2, 2)$ | $L_3^{Lor} = \omega^a{}_b d\omega^b{}_a + \frac{2}{3}\omega^a{}_b \omega^b{}_c \omega^c{}_a$ | $\mathbf{P}_4^{Lor} = R^a{}_b R^b{}_a$ | $SO(2, 1)$ | $L_3^{Tor} = e^a T_a$ | $\mathbf{N}_4 = T^a T_a - e^a e^b R_{ab}$ | $SO(2, 1)$ | $L_3^{U(1)} = AdA$ | $\mathbf{P}_4^{U(1)} = FF$ | $U(1)$ | $L_3^{SU(N)} = Tr[\mathbf{AdA} + \frac{2}{3}\mathbf{AAA}]$ | $\mathbf{P}_4^{SU(4)} = Tr[\mathbf{FF}]$ | $SU(N)$ |
| $D = 3$ CS Lagrangians                                                                       | Characteristic classes                                             | Groups                                  |                                               |                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                        |                        |        |                                                                    |                                                                    |                                         |                                                                                              |                                        |            |                       |                                           |            |                    |                            |        |                                                            |                                          |         |
| $L_3^{(A)dS} = \epsilon_{abc}(R^{ab} \pm \frac{e^a e^b}{3l^2})e^c$                           | $\mathbf{E}_4 = \epsilon_{abc}(R^{ab} \pm \frac{e^a e^b}{l^2})T^c$ | $SO(4),$<br>$SO(3, 1)$<br>or $SO(2, 2)$ |                                               |                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                        |                        |        |                                                                    |                                                                    |                                         |                                                                                              |                                        |            |                       |                                           |            |                    |                            |        |                                                            |                                          |         |
| $L_3^{Lor} = \omega^a{}_b d\omega^b{}_a + \frac{2}{3}\omega^a{}_b \omega^b{}_c \omega^c{}_a$ | $\mathbf{P}_4^{Lor} = R^a{}_b R^b{}_a$                             | $SO(2, 1)$                              |                                               |                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                        |                        |        |                                                                    |                                                                    |                                         |                                                                                              |                                        |            |                       |                                           |            |                    |                            |        |                                                            |                                          |         |
| $L_3^{Tor} = e^a T_a$                                                                        | $\mathbf{N}_4 = T^a T_a - e^a e^b R_{ab}$                          | $SO(2, 1)$                              |                                               |                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                        |                        |        |                                                                    |                                                                    |                                         |                                                                                              |                                        |            |                       |                                           |            |                    |                            |        |                                                            |                                          |         |
| $L_3^{U(1)} = AdA$                                                                           | $\mathbf{P}_4^{U(1)} = FF$                                         | $U(1)$                                  |                                               |                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                        |                        |        |                                                                    |                                                                    |                                         |                                                                                              |                                        |            |                       |                                           |            |                    |                            |        |                                                            |                                          |         |
| $L_3^{SU(N)} = Tr[\mathbf{AdA} + \frac{2}{3}\mathbf{AAA}]$                                   | $\mathbf{P}_4^{SU(4)} = Tr[\mathbf{FF}]$                           | $SU(N)$                                 |                                               |                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                        |                        |        |                                                                    |                                                                    |                                         |                                                                                              |                                        |            |                       |                                           |            |                    |                            |        |                                                            |                                          |         |
| cardona-qft-methods_F02278                                                                   | 181                                                                | 0.546                                   | $\mathbf{P}_4^{U(1)} = FF$                    | $\mathbf{P}_4^{U(1)} = FF$                    | <div>topological invariants,</div> <table><tr><th><math>D = 3</math> CS Lagrangians</th><th>Characteristic classes</th><th>Groups</th></tr><tr><td><math>L_3^{(A)dS} = \epsilon_{abc}(R^{ab} \pm \frac{e^a e^b}{3l^2})e^c</math></td><td><math>\mathbf{E}_4 = \epsilon_{abc}(R^{ab} \pm \frac{e^a e^b}{l^2})T^c</math></td><td><math>SO(4),</math><br/><math>SO(3, 1)</math><br/>or <math>SO(2, 2)</math></td></tr><tr><td><math>L_3^{Lor} = \omega^a{}_b d\omega^b{}_a + \frac{2}{3}\omega^a{}_b \omega^b{}_c \omega^c{}_a</math></td><td><math>\mathbf{P}_4^{Lor} = R^a{}_b R^b{}_a</math></td><td><math>SO(2, 1)</math></td></tr><tr><td><math>L_3^{Tor} = e^a T_a</math></td><td><math>\mathbf{N}_4 = T^a T_a - e^a e^b R_{ab}</math></td><td><math>SO(2, 1)</math></td></tr><tr><td><math>L_3^{U(1)} = AdA</math></td><td><math>\mathbf{P}_4^{U(1)} = FF</math></td><td><math>U(1)</math></td></tr><tr><td><math>L_3^{SU(N)} = Tr[\mathbf{AdA} + \frac{2}{3}\mathbf{AAA}]</math></td><td><math>\mathbf{P}_4^{SU(4)} = Tr[\mathbf{FF}]</math></td><td><math>SU(N)</math></td></tr></table> <div>Table 1: Three-dimensional gravitational CS Lagrangians, their related</div> | $D = 3$ CS Lagrangians | Characteristic classes | Groups | $L_3^{(A)dS} = \epsilon_{abc}(R^{ab} \pm \frac{e^a e^b}{3l^2})e^c$ | $\mathbf{E}_4 = \epsilon_{abc}(R^{ab} \pm \frac{e^a e^b}{l^2})T^c$ | $SO(4),$<br>$SO(3, 1)$<br>or $SO(2, 2)$ | $L_3^{Lor} = \omega^a{}_b d\omega^b{}_a + \frac{2}{3}\omega^a{}_b \omega^b{}_c \omega^c{}_a$ | $\mathbf{P}_4^{Lor} = R^a{}_b R^b{}_a$ | $SO(2, 1)$ | $L_3^{Tor} = e^a T_a$ | $\mathbf{N}_4 = T^a T_a - e^a e^b R_{ab}$ | $SO(2, 1)$ | $L_3^{U(1)} = AdA$ | $\mathbf{P}_4^{U(1)} = FF$ | $U(1)$ | $L_3^{SU(N)} = Tr[\mathbf{AdA} + \frac{2}{3}\mathbf{AAA}]$ | $\mathbf{P}_4^{SU(4)} = Tr[\mathbf{FF}]$ | $SU(N)$ |
| $D = 3$ CS Lagrangians                                                                       | Characteristic classes                                             | Groups                                  |                                               |                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                        |                        |        |                                                                    |                                                                    |                                         |                                                                                              |                                        |            |                       |                                           |            |                    |                            |        |                                                            |                                          |         |
| $L_3^{(A)dS} = \epsilon_{abc}(R^{ab} \pm \frac{e^a e^b}{3l^2})e^c$                           | $\mathbf{E}_4 = \epsilon_{abc}(R^{ab} \pm \frac{e^a e^b}{l^2})T^c$ | $SO(4),$<br>$SO(3, 1)$<br>or $SO(2, 2)$ |                                               |                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                        |                        |        |                                                                    |                                                                    |                                         |                                                                                              |                                        |            |                       |                                           |            |                    |                            |        |                                                            |                                          |         |
| $L_3^{Lor} = \omega^a{}_b d\omega^b{}_a + \frac{2}{3}\omega^a{}_b \omega^b{}_c \omega^c{}_a$ | $\mathbf{P}_4^{Lor} = R^a{}_b R^b{}_a$                             | $SO(2, 1)$                              |                                               |                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                        |                        |        |                                                                    |                                                                    |                                         |                                                                                              |                                        |            |                       |                                           |            |                    |                            |        |                                                            |                                          |         |
| $L_3^{Tor} = e^a T_a$                                                                        | $\mathbf{N}_4 = T^a T_a - e^a e^b R_{ab}$                          | $SO(2, 1)$                              |                                               |                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                        |                        |        |                                                                    |                                                                    |                                         |                                                                                              |                                        |            |                       |                                           |            |                    |                            |        |                                                            |                                          |         |
| $L_3^{U(1)} = AdA$                                                                           | $\mathbf{P}_4^{U(1)} = FF$                                         | $U(1)$                                  |                                               |                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                        |                        |        |                                                                    |                                                                    |                                         |                                                                                              |                                        |            |                       |                                           |            |                    |                            |        |                                                            |                                          |         |
| $L_3^{SU(N)} = Tr[\mathbf{AdA} + \frac{2}{3}\mathbf{AAA}]$                                   | $\mathbf{P}_4^{SU(4)} = Tr[\mathbf{FF}]$                           | $SU(N)$                                 |                                               |                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                        |                        |        |                                                                    |                                                                    |                                         |                                                                                              |                                        |            |                       |                                           |            |                    |                            |        |                                                            |                                          |         |

| Identifier                                                                                        | Page                                                                 | Conf.                                  | LaTeX source                                                                                                                    | Rendered                                                                                                    | Image                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                      |                        |                                                                                         |                                                               |                                                                                                   |                                             |                                                                                       |                                                                      |           |                       |                                    |           |                  |                          |        |                                                                                   |                                                 |         |
|---------------------------------------------------------------------------------------------------|----------------------------------------------------------------------|----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|------------------------|-----------------------------------------------------------------------------------------|---------------------------------------------------------------|---------------------------------------------------------------------------------------------------|---------------------------------------------|---------------------------------------------------------------------------------------|----------------------------------------------------------------------|-----------|-----------------------|------------------------------------|-----------|------------------|--------------------------|--------|-----------------------------------------------------------------------------------|-------------------------------------------------|---------|
| cardona-qft-methods_ F02279                                                                       | 181                                                                  | 0.546                                  | $L_{\{3\}^{\{S\ U(N)\}}}=\operatorname{Tr}\left[\mathbf{A}\ d\ \mathbf{A}+\frac{2}{3}\ \mathbf{A}\mathbf{A}\ \mathbf{A}\right]$ | $L_3^{SU(N)}=\operatorname{Tr}\left[\mathbf{A}d\mathbf{A}+\frac{2}{3}\mathbf{A}\mathbf{A}\mathbf{A}\right]$ | <p>topological invariants,</p> <table><tr><th><math>D=3</math> CS Lagrangians</th><th>Characteristic classes</th><th>Groups</th></tr><tr><td><math>L_3^{(A)dS}=\epsilon_{abc}(R^{ab}\pm\frac{e^ae^b}{3l^2})e^c</math></td><td><math>\mathbf{E}_4=\epsilon_{abc}(R^{ab}\pm\frac{e^ae^b}{l^2})T^c</math></td><td><math>SO(4)</math>,<br/><math>SO(3,1)</math><br/>or <math>SO(2,2)</math></td></tr><tr><td><math>L_3^{Lor}=\omega^a{}_bd\omega^b{}_a+\frac{2}{3}\omega^a{}_b\omega^b{}_c\omega^c{}_a</math></td><td><math>\mathbf{P}_4^{Lor}=R^a{}_bR^b{}_a</math></td><td><math>SO(2,1)</math></td></tr><tr><td><math>L_3^{T_{or}}=e^aT_a</math></td><td><math>\mathbf{N}_4=T^aT_a-e^ae^bR_{ab}</math></td><td><math>SO(2,1)</math></td></tr><tr><td><math>L_3^{U(1)}=AdA</math></td><td><math>\mathbf{P}_4^{U(1)}=FF</math></td><td><math>U(1)</math></td></tr><tr><td><math>L_3^{SU(N)}=Tr[\mathbf{A}d\mathbf{A}+\frac{2}{3}\mathbf{A}\mathbf{A}\mathbf{A}]</math></td><td><math>\mathbf{P}_4^{SU(4)}=Tr[\mathbf{F}\mathbf{F}]</math></td><td><math>SU(N)</math></td></tr></table> <p><b>Table 1:</b> Three-dimensional gravitational CS Lagrangians, their related</p> | $D=3$ CS Lagrangians | Characteristic classes | Groups                                                                                  | $L_3^{(A)dS}=\epsilon_{abc}(R^{ab}\pm\frac{e^ae^b}{3l^2})e^c$ | $\mathbf{E}_4=\epsilon_{abc}(R^{ab}\pm\frac{e^ae^b}{l^2})T^c$                                     | $SO(4)$ ,<br>$SO(3,1)$<br>or $SO(2,2)$      | $L_3^{Lor}=\omega^a{}_bd\omega^b{}_a+\frac{2}{3}\omega^a{}_b\omega^b{}_c\omega^c{}_a$ | $\mathbf{P}_4^{Lor}=R^a{}_bR^b{}_a$                                  | $SO(2,1)$ | $L_3^{T_{or}}=e^aT_a$ | $\mathbf{N}_4=T^aT_a-e^ae^bR_{ab}$ | $SO(2,1)$ | $L_3^{U(1)}=AdA$ | $\mathbf{P}_4^{U(1)}=FF$ | $U(1)$ | $L_3^{SU(N)}=Tr[\mathbf{A}d\mathbf{A}+\frac{2}{3}\mathbf{A}\mathbf{A}\mathbf{A}]$ | $\mathbf{P}_4^{SU(4)}=Tr[\mathbf{F}\mathbf{F}]$ | $SU(N)$ |
| $D=3$ CS Lagrangians                                                                              | Characteristic classes                                               | Groups                                 |                                                                                                                                 |                                                                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                      |                        |                                                                                         |                                                               |                                                                                                   |                                             |                                                                                       |                                                                      |           |                       |                                    |           |                  |                          |        |                                                                                   |                                                 |         |
| $L_3^{(A)dS}=\epsilon_{abc}(R^{ab}\pm\frac{e^ae^b}{3l^2})e^c$                                     | $\mathbf{E}_4=\epsilon_{abc}(R^{ab}\pm\frac{e^ae^b}{l^2})T^c$        | $SO(4)$ ,<br>$SO(3,1)$<br>or $SO(2,2)$ |                                                                                                                                 |                                                                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                      |                        |                                                                                         |                                                               |                                                                                                   |                                             |                                                                                       |                                                                      |           |                       |                                    |           |                  |                          |        |                                                                                   |                                                 |         |
| $L_3^{Lor}=\omega^a{}_bd\omega^b{}_a+\frac{2}{3}\omega^a{}_b\omega^b{}_c\omega^c{}_a$             | $\mathbf{P}_4^{Lor}=R^a{}_bR^b{}_a$                                  | $SO(2,1)$                              |                                                                                                                                 |                                                                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                      |                        |                                                                                         |                                                               |                                                                                                   |                                             |                                                                                       |                                                                      |           |                       |                                    |           |                  |                          |        |                                                                                   |                                                 |         |
| $L_3^{T_{or}}=e^aT_a$                                                                             | $\mathbf{N}_4=T^aT_a-e^ae^bR_{ab}$                                   | $SO(2,1)$                              |                                                                                                                                 |                                                                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                      |                        |                                                                                         |                                                               |                                                                                                   |                                             |                                                                                       |                                                                      |           |                       |                                    |           |                  |                          |        |                                                                                   |                                                 |         |
| $L_3^{U(1)}=AdA$                                                                                  | $\mathbf{P}_4^{U(1)}=FF$                                             | $U(1)$                                 |                                                                                                                                 |                                                                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                      |                        |                                                                                         |                                                               |                                                                                                   |                                             |                                                                                       |                                                                      |           |                       |                                    |           |                  |                          |        |                                                                                   |                                                 |         |
| $L_3^{SU(N)}=Tr[\mathbf{A}d\mathbf{A}+\frac{2}{3}\mathbf{A}\mathbf{A}\mathbf{A}]$                 | $\mathbf{P}_4^{SU(4)}=Tr[\mathbf{F}\mathbf{F}]$                      | $SU(N)$                                |                                                                                                                                 |                                                                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                      |                        |                                                                                         |                                                               |                                                                                                   |                                             |                                                                                       |                                                                      |           |                       |                                    |           |                  |                          |        |                                                                                   |                                                 |         |
| cardona-qft-methods_ F02280                                                                       | 181                                                                  | 0.546                                  | $\mathbf{P}_{\{4\}^{\{S\ U(4)\}}}=\operatorname{Tr}[\mathbf{F}\ F]$                                                             | $\mathbf{P}_4^{SU(4)}=\operatorname{Tr}[\mathbf{F}\mathbf{F}]$                                              | <p>topological invariants,</p> <table><tr><th><math>D=3</math> CS Lagrangians</th><th>Characteristic classes</th><th>Groups</th></tr><tr><td><math>L_3^{(A)dS}=\epsilon_{abc}(R^{ab}\pm\frac{e^ae^b}{3l^2})e^c</math></td><td><math>\mathbf{E}_4=\epsilon_{abc}(R^{ab}\pm\frac{e^ae^b}{l^2})T^c</math></td><td><math>SO(4)</math>,<br/><math>SO(3,1)</math><br/>or <math>SO(2,2)</math></td></tr><tr><td><math>L_3^{Lor}=\omega^a{}_bd\omega^b{}_a+\frac{2}{3}\omega^a{}_b\omega^b{}_c\omega^c{}_a</math></td><td><math>\mathbf{P}_4^{Lor}=R^a{}_bR^b{}_a</math></td><td><math>SO(2,1)</math></td></tr><tr><td><math>L_3^{T_{or}}=e^aT_a</math></td><td><math>\mathbf{N}_4=T^aT_a-e^ae^bR_{ab}</math></td><td><math>SO(2,1)</math></td></tr><tr><td><math>L_3^{U(1)}=AdA</math></td><td><math>\mathbf{P}_4^{U(1)}=FF</math></td><td><math>U(1)</math></td></tr><tr><td><math>L_3^{SU(N)}=Tr[\mathbf{A}d\mathbf{A}+\frac{2}{3}\mathbf{A}\mathbf{A}\mathbf{A}]</math></td><td><math>\mathbf{P}_4^{SU(4)}=Tr[\mathbf{F}\mathbf{F}]</math></td><td><math>SU(N)</math></td></tr></table> <p><b>Table 1:</b> Three-dimensional gravitational CS Lagrangians, their related</p> | $D=3$ CS Lagrangians | Characteristic classes | Groups                                                                                  | $L_3^{(A)dS}=\epsilon_{abc}(R^{ab}\pm\frac{e^ae^b}{3l^2})e^c$ | $\mathbf{E}_4=\epsilon_{abc}(R^{ab}\pm\frac{e^ae^b}{l^2})T^c$                                     | $SO(4)$ ,<br>$SO(3,1)$<br>or $SO(2,2)$      | $L_3^{Lor}=\omega^a{}_bd\omega^b{}_a+\frac{2}{3}\omega^a{}_b\omega^b{}_c\omega^c{}_a$ | $\mathbf{P}_4^{Lor}=R^a{}_bR^b{}_a$                                  | $SO(2,1)$ | $L_3^{T_{or}}=e^aT_a$ | $\mathbf{N}_4=T^aT_a-e^ae^bR_{ab}$ | $SO(2,1)$ | $L_3^{U(1)}=AdA$ | $\mathbf{P}_4^{U(1)}=FF$ | $U(1)$ | $L_3^{SU(N)}=Tr[\mathbf{A}d\mathbf{A}+\frac{2}{3}\mathbf{A}\mathbf{A}\mathbf{A}]$ | $\mathbf{P}_4^{SU(4)}=Tr[\mathbf{F}\mathbf{F}]$ | $SU(N)$ |
| $D=3$ CS Lagrangians                                                                              | Characteristic classes                                               | Groups                                 |                                                                                                                                 |                                                                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                      |                        |                                                                                         |                                                               |                                                                                                   |                                             |                                                                                       |                                                                      |           |                       |                                    |           |                  |                          |        |                                                                                   |                                                 |         |
| $L_3^{(A)dS}=\epsilon_{abc}(R^{ab}\pm\frac{e^ae^b}{3l^2})e^c$                                     | $\mathbf{E}_4=\epsilon_{abc}(R^{ab}\pm\frac{e^ae^b}{l^2})T^c$        | $SO(4)$ ,<br>$SO(3,1)$<br>or $SO(2,2)$ |                                                                                                                                 |                                                                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                      |                        |                                                                                         |                                                               |                                                                                                   |                                             |                                                                                       |                                                                      |           |                       |                                    |           |                  |                          |        |                                                                                   |                                                 |         |
| $L_3^{Lor}=\omega^a{}_bd\omega^b{}_a+\frac{2}{3}\omega^a{}_b\omega^b{}_c\omega^c{}_a$             | $\mathbf{P}_4^{Lor}=R^a{}_bR^b{}_a$                                  | $SO(2,1)$                              |                                                                                                                                 |                                                                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                      |                        |                                                                                         |                                                               |                                                                                                   |                                             |                                                                                       |                                                                      |           |                       |                                    |           |                  |                          |        |                                                                                   |                                                 |         |
| $L_3^{T_{or}}=e^aT_a$                                                                             | $\mathbf{N}_4=T^aT_a-e^ae^bR_{ab}$                                   | $SO(2,1)$                              |                                                                                                                                 |                                                                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                      |                        |                                                                                         |                                                               |                                                                                                   |                                             |                                                                                       |                                                                      |           |                       |                                    |           |                  |                          |        |                                                                                   |                                                 |         |
| $L_3^{U(1)}=AdA$                                                                                  | $\mathbf{P}_4^{U(1)}=FF$                                             | $U(1)$                                 |                                                                                                                                 |                                                                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                      |                        |                                                                                         |                                                               |                                                                                                   |                                             |                                                                                       |                                                                      |           |                       |                                    |           |                  |                          |        |                                                                                   |                                                 |         |
| $L_3^{SU(N)}=Tr[\mathbf{A}d\mathbf{A}+\frac{2}{3}\mathbf{A}\mathbf{A}\mathbf{A}]$                 | $\mathbf{P}_4^{SU(4)}=Tr[\mathbf{F}\mathbf{F}]$                      | $SU(N)$                                |                                                                                                                                 |                                                                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                      |                        |                                                                                         |                                                               |                                                                                                   |                                             |                                                                                       |                                                                      |           |                       |                                    |           |                  |                          |        |                                                                                   |                                                 |         |
| cardona-qft-methods_ F02281                                                                       | 181                                                                  | 0.546                                  | $S\ U(N)$                                                                                                                       | $SU(N)$                                                                                                     | <p>topological invariants,</p> <table><tr><th><math>D=3</math> CS Lagrangians</th><th>Characteristic classes</th><th>Groups</th></tr><tr><td><math>L_3^{(A)dS}=\epsilon_{abc}(R^{ab}\pm\frac{e^ae^b}{3l^2})e^c</math></td><td><math>\mathbf{E}_4=\epsilon_{abc}(R^{ab}\pm\frac{e^ae^b}{l^2})T^c</math></td><td><math>SO(4)</math>,<br/><math>SO(3,1)</math><br/>or <math>SO(2,2)</math></td></tr><tr><td><math>L_3^{Lor}=\omega^a{}_bd\omega^b{}_a+\frac{2}{3}\omega^a{}_b\omega^b{}_c\omega^c{}_a</math></td><td><math>\mathbf{P}_4^{Lor}=R^a{}_bR^b{}_a</math></td><td><math>SO(2,1)</math></td></tr><tr><td><math>L_3^{T_{or}}=e^aT_a</math></td><td><math>\mathbf{N}_4=T^aT_a-e^ae^bR_{ab}</math></td><td><math>SO(2,1)</math></td></tr><tr><td><math>L_3^{U(1)}=AdA</math></td><td><math>\mathbf{P}_4^{U(1)}=FF</math></td><td><math>U(1)</math></td></tr><tr><td><math>L_3^{SU(N)}=Tr[\mathbf{A}d\mathbf{A}+\frac{2}{3}\mathbf{A}\mathbf{A}\mathbf{A}]</math></td><td><math>\mathbf{P}_4^{SU(4)}=Tr[\mathbf{F}\mathbf{F}]</math></td><td><math>SU(N)</math></td></tr></table> <p><b>Table 1:</b> Three-dimensional gravitational CS Lagrangians, their related</p> | $D=3$ CS Lagrangians | Characteristic classes | Groups                                                                                  | $L_3^{(A)dS}=\epsilon_{abc}(R^{ab}\pm\frac{e^ae^b}{3l^2})e^c$ | $\mathbf{E}_4=\epsilon_{abc}(R^{ab}\pm\frac{e^ae^b}{l^2})T^c$                                     | $SO(4)$ ,<br>$SO(3,1)$<br>or $SO(2,2)$      | $L_3^{Lor}=\omega^a{}_bd\omega^b{}_a+\frac{2}{3}\omega^a{}_b\omega^b{}_c\omega^c{}_a$ | $\mathbf{P}_4^{Lor}=R^a{}_bR^b{}_a$                                  | $SO(2,1)$ | $L_3^{T_{or}}=e^aT_a$ | $\mathbf{N}_4=T^aT_a-e^ae^bR_{ab}$ | $SO(2,1)$ | $L_3^{U(1)}=AdA$ | $\mathbf{P}_4^{U(1)}=FF$ | $U(1)$ | $L_3^{SU(N)}=Tr[\mathbf{A}d\mathbf{A}+\frac{2}{3}\mathbf{A}\mathbf{A}\mathbf{A}]$ | $\mathbf{P}_4^{SU(4)}=Tr[\mathbf{F}\mathbf{F}]$ | $SU(N)$ |
| $D=3$ CS Lagrangians                                                                              | Characteristic classes                                               | Groups                                 |                                                                                                                                 |                                                                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                      |                        |                                                                                         |                                                               |                                                                                                   |                                             |                                                                                       |                                                                      |           |                       |                                    |           |                  |                          |        |                                                                                   |                                                 |         |
| $L_3^{(A)dS}=\epsilon_{abc}(R^{ab}\pm\frac{e^ae^b}{3l^2})e^c$                                     | $\mathbf{E}_4=\epsilon_{abc}(R^{ab}\pm\frac{e^ae^b}{l^2})T^c$        | $SO(4)$ ,<br>$SO(3,1)$<br>or $SO(2,2)$ |                                                                                                                                 |                                                                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                      |                        |                                                                                         |                                                               |                                                                                                   |                                             |                                                                                       |                                                                      |           |                       |                                    |           |                  |                          |        |                                                                                   |                                                 |         |
| $L_3^{Lor}=\omega^a{}_bd\omega^b{}_a+\frac{2}{3}\omega^a{}_b\omega^b{}_c\omega^c{}_a$             | $\mathbf{P}_4^{Lor}=R^a{}_bR^b{}_a$                                  | $SO(2,1)$                              |                                                                                                                                 |                                                                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                      |                        |                                                                                         |                                                               |                                                                                                   |                                             |                                                                                       |                                                                      |           |                       |                                    |           |                  |                          |        |                                                                                   |                                                 |         |
| $L_3^{T_{or}}=e^aT_a$                                                                             | $\mathbf{N}_4=T^aT_a-e^ae^bR_{ab}$                                   | $SO(2,1)$                              |                                                                                                                                 |                                                                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                      |                        |                                                                                         |                                                               |                                                                                                   |                                             |                                                                                       |                                                                      |           |                       |                                    |           |                  |                          |        |                                                                                   |                                                 |         |
| $L_3^{U(1)}=AdA$                                                                                  | $\mathbf{P}_4^{U(1)}=FF$                                             | $U(1)$                                 |                                                                                                                                 |                                                                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                      |                        |                                                                                         |                                                               |                                                                                                   |                                             |                                                                                       |                                                                      |           |                       |                                    |           |                  |                          |        |                                                                                   |                                                 |         |
| $L_3^{SU(N)}=Tr[\mathbf{A}d\mathbf{A}+\frac{2}{3}\mathbf{A}\mathbf{A}\mathbf{A}]$                 | $\mathbf{P}_4^{SU(4)}=Tr[\mathbf{F}\mathbf{F}]$                      | $SU(N)$                                |                                                                                                                                 |                                                                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                      |                        |                                                                                         |                                                               |                                                                                                   |                                             |                                                                                       |                                                                      |           |                       |                                    |           |                  |                          |        |                                                                                   |                                                 |         |
| cardona-qft-methods_ F02282                                                                       | 181                                                                  | 0.951                                  | $R,\ F$                                                                                                                         | $R,F$                                                                                                       | Pontryagin densities for the Lorentz group, <sup>1b</sup> and $\mathbf{N}_4$ is the Nieh-Yan in-variant <sup>27</sup>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                      |                        |                                                                                         |                                                               |                                                                                                   |                                             |                                                                                       |                                                                      |           |                       |                                    |           |                  |                          |        |                                                                                   |                                                 |         |
| cardona-qft-methods_ F02283                                                                       | 181                                                                  | 0.951                                  | $\omega_b^a,\ A$                                                                                                                | $\omega_b^a,A$                                                                                              | Pontryagin densities for the Lorentz group, <sup>1b</sup> and $\mathbf{N}_4$ is the Nieh-Yan in-variant <sup>27</sup>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                      |                        |                                                                                         |                                                               |                                                                                                   |                                             |                                                                                       |                                                                      |           |                       |                                    |           |                  |                          |        |                                                                                   |                                                 |         |
| cardona-qft-methods_ F02284                                                                       | 181                                                                  | 0.951                                  | $\mathbf{A}$                                                                                                                    | $\mathbf{A}$                                                                                                | Pontryagin densities for the Lorentz group, <sup>1b</sup> and $\mathbf{N}_4$ is the Nieh-Yan in-variant <sup>27</sup>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                      |                        |                                                                                         |                                                               |                                                                                                   |                                             |                                                                                       |                                                                      |           |                       |                                    |           |                  |                          |        |                                                                                   |                                                 |         |
| cardona-qft-methods_ F02285                                                                       | 181                                                                  | 1.000                                  | $d\ L_{\{3\}^{\{A\ d\ S\}}}$                                                                                                    | $dL_3^{AdS}$                                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                      |                        |                                                                                         |                                                               |                                                                                                   |                                             |                                                                                       |                                                                      |           |                       |                                    |           |                  |                          |        |                                                                                   |                                                 |         |
| cardona-qft-methods_ F02286                                                                       | 182                                                                  | 1.000                                  | $L_{\{3\}^{\{\text{Exotic}\}}}$                                                                                                 | $L_3^{\text{Exotic}}$                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                      |                        |                                                                                         |                                                               |                                                                                                   |                                             |                                                                                       |                                                                      |           |                       |                                    |           |                  |                          |        |                                                                                   |                                                 |         |
| cardona-qft-methods_ F02287                                                                       | 182                                                                  | 1.000                                  | $R^{\{a\ b\}}\ T_{\{a\}}\ e_{\{b\}}$                                                                                            | $R^{ab}T_ae_b$                                                                                              | with respect to $e^a$ gives the same equation as varying $L_3^{\text{Exotic}}$ with respect to $e^a$ .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                      |                        |                                                                                         |                                                               |                                                                                                   |                                             |                                                                                       |                                                                      |           |                       |                                    |           |                  |                          |        |                                                                                   |                                                 |         |
| cardona-qft-methods_ F02288                                                                       | 182                                                                  | 0.562                                  | $D=7\ \mathrm{CS}$                                                                                                              | $D=7\text{CS}$                                                                                              | <p>two Lorentz-Pontryagin, and one torsional CS forms,</p> <table><tr><th><math>D=7</math> CS Lagrangians</th><th>Characteristic classes</th></tr><tr><td><math>L_7^{Lor}=\omega(d\omega)^3+\frac{8}{3}\omega^3(d\omega)^2+\cdots+\frac{1}{7}\omega^7</math></td><td><math>\mathbf{P}_8^{Lor}=R^a{}_bR^b{}_cR^c{}_dR^d{}_a</math></td></tr><tr><td><math>L_7^T=(\omega^a{}_bd\omega^b{}_a+\frac{2}{3}\omega^a{}_b\omega^b{}_c\omega^c{}_a)R^a{}_bR^b{}_a</math></td><td><math>(\mathbf{P}_4^{Lor})^2=(R^a{}_bR^b{}_a)^2</math></td></tr><tr><td><math>L_7^{T^T}=(e^aT_a)(R^a{}_bR^b{}_a)</math></td><td><math>\mathbf{N}_4\mathbf{P}_4^{Lor}=(T^aT_a-e^ae^bR_{ab})R^c{}_dR^d{}_c</math></td></tr></table> <p><b>Table 2:</b> Seven dimensional gravitational CS Lagrangians, their related</p>                                                                                                                                                                                                                                                                                                                                                                           | $D=7$ CS Lagrangians | Characteristic classes | $L_7^{Lor}=\omega(d\omega)^3+\frac{8}{3}\omega^3(d\omega)^2+\cdots+\frac{1}{7}\omega^7$ | $\mathbf{P}_8^{Lor}=R^a{}_bR^b{}_cR^c{}_dR^d{}_a$             | $L_7^T=(\omega^a{}_bd\omega^b{}_a+\frac{2}{3}\omega^a{}_b\omega^b{}_c\omega^c{}_a)R^a{}_bR^b{}_a$ | $(\mathbf{P}_4^{Lor})^2=(R^a{}_bR^b{}_a)^2$ | $L_7^{T^T}=(e^aT_a)(R^a{}_bR^b{}_a)$                                                  | $\mathbf{N}_4\mathbf{P}_4^{Lor}=(T^aT_a-e^ae^bR_{ab})R^c{}_dR^d{}_c$ |           |                       |                                    |           |                  |                          |        |                                                                                   |                                                 |         |
| $D=7$ CS Lagrangians                                                                              | Characteristic classes                                               |                                        |                                                                                                                                 |                                                                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                      |                        |                                                                                         |                                                               |                                                                                                   |                                             |                                                                                       |                                                                      |           |                       |                                    |           |                  |                          |        |                                                                                   |                                                 |         |
| $L_7^{Lor}=\omega(d\omega)^3+\frac{8}{3}\omega^3(d\omega)^2+\cdots+\frac{1}{7}\omega^7$           | $\mathbf{P}_8^{Lor}=R^a{}_bR^b{}_cR^c{}_dR^d{}_a$                    |                                        |                                                                                                                                 |                                                                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                      |                        |                                                                                         |                                                               |                                                                                                   |                                             |                                                                                       |                                                                      |           |                       |                                    |           |                  |                          |        |                                                                                   |                                                 |         |
| $L_7^T=(\omega^a{}_bd\omega^b{}_a+\frac{2}{3}\omega^a{}_b\omega^b{}_c\omega^c{}_a)R^a{}_bR^b{}_a$ | $(\mathbf{P}_4^{Lor})^2=(R^a{}_bR^b{}_a)^2$                          |                                        |                                                                                                                                 |                                                                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                      |                        |                                                                                         |                                                               |                                                                                                   |                                             |                                                                                       |                                                                      |           |                       |                                    |           |                  |                          |        |                                                                                   |                                                 |         |
| $L_7^{T^T}=(e^aT_a)(R^a{}_bR^b{}_a)$                                                              | $\mathbf{N}_4\mathbf{P}_4^{Lor}=(T^aT_a-e^ae^bR_{ab})R^c{}_dR^d{}_c$ |                                        |                                                                                                                                 |                                                                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                      |                        |                                                                                         |                                                               |                                                                                                   |                                             |                                                                                       |                                                                      |           |                       |                                    |           |                  |                          |        |                                                                                   |                                                 |         |

| Identifier                                                                                        | Page                                                                 | Conf. | LaTeX source                                                                                                                              | Rendered                                                                                                     | Image                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                      |                        |                                                                                         |                                                   |                                                                                                   |                                             |                                     |                                                                      |
|---------------------------------------------------------------------------------------------------|----------------------------------------------------------------------|-------|-------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|------------------------|-----------------------------------------------------------------------------------------|---------------------------------------------------|---------------------------------------------------------------------------------------------------|---------------------------------------------|-------------------------------------|----------------------------------------------------------------------|
| cardona-qft-methods_F02289                                                                        | 182                                                                  | 0.562 | $\text{L}_{7}^{\text{\text{Lor}}}=\omega(d\,\omega)^3+\frac{8}{5}\,\omega^3(d\,\omega)^2+\cdots+\frac{4}{7}\,\omega^7$                    | $L_7^{\text{Lor}}=\omega(d\omega)^3+\frac{8}{5}\omega^3(d\omega)^2+\cdots+\frac{4}{7}\omega^7$               | <div>two Lorentz-Pontryagin, and one torsional CS forms,</div> <table><tr><th><math>D=7</math> CS Lagrangians</th><th>Characteristic classes</th></tr><tr><td><math>L_7^{Lor}=\omega(d\omega)^3+\frac{8}{5}\omega^3(d\omega)^2+\cdots+\frac{4}{7}\omega^7</math></td><td><math>\mathbf{P}_8^{Lor}=R^a{}_bR^b{}_cR^c{}_dR^d{}_a</math></td></tr><tr><td><math>L_7^I=(\omega^a{}_bd\omega^b{}_a+\frac{2}{3}\omega^a{}_b\omega^b{}_c\omega^c{}_a)R^a{}_bR^b{}_a</math></td><td><math>(\mathbf{P}_4^{Lor})^2=(R^a{}_bR^b{}_a)^2</math></td></tr><tr><td><math>L_7^{II}=(e^aT_a)(R^a{}_bR^b{}_a)</math></td><td><math>\mathbf{N}_4\mathbf{P}_4^{Lor}=(T^aT_a-e^ae^bR_{ab})R^c{}_dR^d{}_c</math></td></tr></table> <div>Table 9: Seven dimensional gravitational CS Lagrangians, their related</div> | $D=7$ CS Lagrangians | Characteristic classes | $L_7^{Lor}=\omega(d\omega)^3+\frac{8}{5}\omega^3(d\omega)^2+\cdots+\frac{4}{7}\omega^7$ | $\mathbf{P}_8^{Lor}=R^a{}_bR^b{}_cR^c{}_dR^d{}_a$ | $L_7^I=(\omega^a{}_bd\omega^b{}_a+\frac{2}{3}\omega^a{}_b\omega^b{}_c\omega^c{}_a)R^a{}_bR^b{}_a$ | $(\mathbf{P}_4^{Lor})^2=(R^a{}_bR^b{}_a)^2$ | $L_7^{II}=(e^aT_a)(R^a{}_bR^b{}_a)$ | $\mathbf{N}_4\mathbf{P}_4^{Lor}=(T^aT_a-e^ae^bR_{ab})R^c{}_dR^d{}_c$ |
| $D=7$ CS Lagrangians                                                                              | Characteristic classes                                               |       |                                                                                                                                           |                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                      |                        |                                                                                         |                                                   |                                                                                                   |                                             |                                     |                                                                      |
| $L_7^{Lor}=\omega(d\omega)^3+\frac{8}{5}\omega^3(d\omega)^2+\cdots+\frac{4}{7}\omega^7$           | $\mathbf{P}_8^{Lor}=R^a{}_bR^b{}_cR^c{}_dR^d{}_a$                    |       |                                                                                                                                           |                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                      |                        |                                                                                         |                                                   |                                                                                                   |                                             |                                     |                                                                      |
| $L_7^I=(\omega^a{}_bd\omega^b{}_a+\frac{2}{3}\omega^a{}_b\omega^b{}_c\omega^c{}_a)R^a{}_bR^b{}_a$ | $(\mathbf{P}_4^{Lor})^2=(R^a{}_bR^b{}_a)^2$                          |       |                                                                                                                                           |                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                      |                        |                                                                                         |                                                   |                                                                                                   |                                             |                                     |                                                                      |
| $L_7^{II}=(e^aT_a)(R^a{}_bR^b{}_a)$                                                               | $\mathbf{N}_4\mathbf{P}_4^{Lor}=(T^aT_a-e^ae^bR_{ab})R^c{}_dR^d{}_c$ |       |                                                                                                                                           |                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                      |                        |                                                                                         |                                                   |                                                                                                   |                                             |                                     |                                                                      |
| cardona-qft-methods_F02290                                                                        | 182                                                                  | 0.562 | $\mathbf{P}_8^{\text{\text{Lor}}}=R^a{}_bR^b{}_cR^c{}_dR^d{}_a$                                                                           | $\mathbf{P}_8^{\text{Lor}}=R^a{}_bR^b{}_cR^c{}_dR^d{}_a$                                                     | <div>two Lorentz-Pontryagin, and one torsional CS forms,</div> <table><tr><th><math>D=7</math> CS Lagrangians</th><th>Characteristic classes</th></tr><tr><td><math>L_7^{Lor}=\omega(d\omega)^3+\frac{8}{5}\omega^3(d\omega)^2+\cdots+\frac{4}{7}\omega^7</math></td><td><math>\mathbf{P}_8^{Lor}=R^a{}_bR^b{}_cR^c{}_dR^d{}_a</math></td></tr><tr><td><math>L_7^I=(\omega^a{}_bd\omega^b{}_a+\frac{2}{3}\omega^a{}_b\omega^b{}_c\omega^c{}_a)R^a{}_bR^b{}_a</math></td><td><math>(\mathbf{P}_4^{Lor})^2=(R^a{}_bR^b{}_a)^2</math></td></tr><tr><td><math>L_7^{II}=(e^aT_a)(R^a{}_bR^b{}_a)</math></td><td><math>\mathbf{N}_4\mathbf{P}_4^{Lor}=(T^aT_a-e^ae^bR_{ab})R^c{}_dR^d{}_c</math></td></tr></table> <div>Table 9: Seven dimensional gravitational CS Lagrangians, their related</div> | $D=7$ CS Lagrangians | Characteristic classes | $L_7^{Lor}=\omega(d\omega)^3+\frac{8}{5}\omega^3(d\omega)^2+\cdots+\frac{4}{7}\omega^7$ | $\mathbf{P}_8^{Lor}=R^a{}_bR^b{}_cR^c{}_dR^d{}_a$ | $L_7^I=(\omega^a{}_bd\omega^b{}_a+\frac{2}{3}\omega^a{}_b\omega^b{}_c\omega^c{}_a)R^a{}_bR^b{}_a$ | $(\mathbf{P}_4^{Lor})^2=(R^a{}_bR^b{}_a)^2$ | $L_7^{II}=(e^aT_a)(R^a{}_bR^b{}_a)$ | $\mathbf{N}_4\mathbf{P}_4^{Lor}=(T^aT_a-e^ae^bR_{ab})R^c{}_dR^d{}_c$ |
| $D=7$ CS Lagrangians                                                                              | Characteristic classes                                               |       |                                                                                                                                           |                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                      |                        |                                                                                         |                                                   |                                                                                                   |                                             |                                     |                                                                      |
| $L_7^{Lor}=\omega(d\omega)^3+\frac{8}{5}\omega^3(d\omega)^2+\cdots+\frac{4}{7}\omega^7$           | $\mathbf{P}_8^{Lor}=R^a{}_bR^b{}_cR^c{}_dR^d{}_a$                    |       |                                                                                                                                           |                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                      |                        |                                                                                         |                                                   |                                                                                                   |                                             |                                     |                                                                      |
| $L_7^I=(\omega^a{}_bd\omega^b{}_a+\frac{2}{3}\omega^a{}_b\omega^b{}_c\omega^c{}_a)R^a{}_bR^b{}_a$ | $(\mathbf{P}_4^{Lor})^2=(R^a{}_bR^b{}_a)^2$                          |       |                                                                                                                                           |                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                      |                        |                                                                                         |                                                   |                                                                                                   |                                             |                                     |                                                                      |
| $L_7^{II}=(e^aT_a)(R^a{}_bR^b{}_a)$                                                               | $\mathbf{N}_4\mathbf{P}_4^{Lor}=(T^aT_a-e^ae^bR_{ab})R^c{}_dR^d{}_c$ |       |                                                                                                                                           |                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                      |                        |                                                                                         |                                                   |                                                                                                   |                                             |                                     |                                                                      |
| cardona-qft-methods_F02291                                                                        | 182                                                                  | 0.562 | $\text{L}_{7}^{\text{\text{I}}}=\left(\omega^a{}_bd\,\omega^b{}_a+\frac{2}{3}\,\omega^a{}_b\omega^b{}_c\omega^c{}_a\right)R^a{}_bR^b{}_a$ | $L_7^I=\left(\omega^a{}_bd\omega^b{}_a+\frac{2}{3}\omega^a{}_b\omega^b{}_c\omega^c{}_a\right)R^a{}_bR^b{}_a$ | <div>two Lorentz-Pontryagin, and one torsional CS forms,</div> <table><tr><th><math>D=7</math> CS Lagrangians</th><th>Characteristic classes</th></tr><tr><td><math>L_7^{Lor}=\omega(d\omega)^3+\frac{8}{5}\omega^3(d\omega)^2+\cdots+\frac{4}{7}\omega^7</math></td><td><math>\mathbf{P}_8^{Lor}=R^a{}_bR^b{}_cR^c{}_dR^d{}_a</math></td></tr><tr><td><math>L_7^I=(\omega^a{}_bd\omega^b{}_a+\frac{2}{3}\omega^a{}_b\omega^b{}_c\omega^c{}_a)R^a{}_bR^b{}_a</math></td><td><math>(\mathbf{P}_4^{Lor})^2=(R^a{}_bR^b{}_a)^2</math></td></tr><tr><td><math>L_7^{II}=(e^aT_a)(R^a{}_bR^b{}_a)</math></td><td><math>\mathbf{N}_4\mathbf{P}_4^{Lor}=(T^aT_a-e^ae^bR_{ab})R^c{}_dR^d{}_c</math></td></tr></table> <div>Table 9: Seven dimensional gravitational CS Lagrangians, their related</div> | $D=7$ CS Lagrangians | Characteristic classes | $L_7^{Lor}=\omega(d\omega)^3+\frac{8}{5}\omega^3(d\omega)^2+\cdots+\frac{4}{7}\omega^7$ | $\mathbf{P}_8^{Lor}=R^a{}_bR^b{}_cR^c{}_dR^d{}_a$ | $L_7^I=(\omega^a{}_bd\omega^b{}_a+\frac{2}{3}\omega^a{}_b\omega^b{}_c\omega^c{}_a)R^a{}_bR^b{}_a$ | $(\mathbf{P}_4^{Lor})^2=(R^a{}_bR^b{}_a)^2$ | $L_7^{II}=(e^aT_a)(R^a{}_bR^b{}_a)$ | $\mathbf{N}_4\mathbf{P}_4^{Lor}=(T^aT_a-e^ae^bR_{ab})R^c{}_dR^d{}_c$ |
| $D=7$ CS Lagrangians                                                                              | Characteristic classes                                               |       |                                                                                                                                           |                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                      |                        |                                                                                         |                                                   |                                                                                                   |                                             |                                     |                                                                      |
| $L_7^{Lor}=\omega(d\omega)^3+\frac{8}{5}\omega^3(d\omega)^2+\cdots+\frac{4}{7}\omega^7$           | $\mathbf{P}_8^{Lor}=R^a{}_bR^b{}_cR^c{}_dR^d{}_a$                    |       |                                                                                                                                           |                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                      |                        |                                                                                         |                                                   |                                                                                                   |                                             |                                     |                                                                      |
| $L_7^I=(\omega^a{}_bd\omega^b{}_a+\frac{2}{3}\omega^a{}_b\omega^b{}_c\omega^c{}_a)R^a{}_bR^b{}_a$ | $(\mathbf{P}_4^{Lor})^2=(R^a{}_bR^b{}_a)^2$                          |       |                                                                                                                                           |                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                      |                        |                                                                                         |                                                   |                                                                                                   |                                             |                                     |                                                                      |
| $L_7^{II}=(e^aT_a)(R^a{}_bR^b{}_a)$                                                               | $\mathbf{N}_4\mathbf{P}_4^{Lor}=(T^aT_a-e^ae^bR_{ab})R^c{}_dR^d{}_c$ |       |                                                                                                                                           |                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                      |                        |                                                                                         |                                                   |                                                                                                   |                                             |                                     |                                                                      |
| cardona-qft-methods_F02292                                                                        | 182                                                                  | 0.562 | $\left(\mathbf{P}_4^{\text{\text{Lor}}}\right)^2=\left(R^a{}_bR^b{}_a\right)^2$                                                           | $\left(\mathbf{P}_4^{\text{Lor}}\right)^2=\left(R^a{}_bR^b{}_a\right)^2$                                     | <div>two Lorentz-Pontryagin, and one torsional CS forms,</div> <table><tr><th><math>D=7</math> CS Lagrangians</th><th>Characteristic classes</th></tr><tr><td><math>L_7^{Lor}=\omega(d\omega)^3+\frac{8}{5}\omega^3(d\omega)^2+\cdots+\frac{4}{7}\omega^7</math></td><td><math>\mathbf{P}_8^{Lor}=R^a{}_bR^b{}_cR^c{}_dR^d{}_a</math></td></tr><tr><td><math>L_7^I=(\omega^a{}_bd\omega^b{}_a+\frac{2}{3}\omega^a{}_b\omega^b{}_c\omega^c{}_a)R^a{}_bR^b{}_a</math></td><td><math>(\mathbf{P}_4^{Lor})^2=(R^a{}_bR^b{}_a)^2</math></td></tr><tr><td><math>L_7^{II}=(e^aT_a)(R^a{}_bR^b{}_a)</math></td><td><math>\mathbf{N}_4\mathbf{P}_4^{Lor}=(T^aT_a-e^ae^bR_{ab})R^c{}_dR^d{}_c</math></td></tr></table> <div>Table 9: Seven dimensional gravitational CS Lagrangians, their related</div> | $D=7$ CS Lagrangians | Characteristic classes | $L_7^{Lor}=\omega(d\omega)^3+\frac{8}{5}\omega^3(d\omega)^2+\cdots+\frac{4}{7}\omega^7$ | $\mathbf{P}_8^{Lor}=R^a{}_bR^b{}_cR^c{}_dR^d{}_a$ | $L_7^I=(\omega^a{}_bd\omega^b{}_a+\frac{2}{3}\omega^a{}_b\omega^b{}_c\omega^c{}_a)R^a{}_bR^b{}_a$ | $(\mathbf{P}_4^{Lor})^2=(R^a{}_bR^b{}_a)^2$ | $L_7^{II}=(e^aT_a)(R^a{}_bR^b{}_a)$ | $\mathbf{N}_4\mathbf{P}_4^{Lor}=(T^aT_a-e^ae^bR_{ab})R^c{}_dR^d{}_c$ |
| $D=7$ CS Lagrangians                                                                              | Characteristic classes                                               |       |                                                                                                                                           |                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                      |                        |                                                                                         |                                                   |                                                                                                   |                                             |                                     |                                                                      |
| $L_7^{Lor}=\omega(d\omega)^3+\frac{8}{5}\omega^3(d\omega)^2+\cdots+\frac{4}{7}\omega^7$           | $\mathbf{P}_8^{Lor}=R^a{}_bR^b{}_cR^c{}_dR^d{}_a$                    |       |                                                                                                                                           |                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                      |                        |                                                                                         |                                                   |                                                                                                   |                                             |                                     |                                                                      |
| $L_7^I=(\omega^a{}_bd\omega^b{}_a+\frac{2}{3}\omega^a{}_b\omega^b{}_c\omega^c{}_a)R^a{}_bR^b{}_a$ | $(\mathbf{P}_4^{Lor})^2=(R^a{}_bR^b{}_a)^2$                          |       |                                                                                                                                           |                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                      |                        |                                                                                         |                                                   |                                                                                                   |                                             |                                     |                                                                      |
| $L_7^{II}=(e^aT_a)(R^a{}_bR^b{}_a)$                                                               | $\mathbf{N}_4\mathbf{P}_4^{Lor}=(T^aT_a-e^ae^bR_{ab})R^c{}_dR^d{}_c$ |       |                                                                                                                                           |                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                      |                        |                                                                                         |                                                   |                                                                                                   |                                             |                                     |                                                                      |
| cardona-qft-methods_F02293                                                                        | 182                                                                  | 0.562 | $\text{L}_{7}^{\text{\text{I I}}}=\left(e^a\,\text{T}_{\text{a}}\right)\left(R^a{}_bR^b{}_a\right)$                                       | $L_7^{II}=(e^aT_a)\left(R^a{}_bR^b{}_a\right)$                                                               | <div>two Lorentz-Pontryagin, and one torsional CS forms,</div> <table><tr><th><math>D=7</math> CS Lagrangians</th><th>Characteristic classes</th></tr><tr><td><math>L_7^{Lor}=\omega(d\omega)^3+\frac{8}{5}\omega^3(d\omega)^2+\cdots+\frac{4}{7}\omega^7</math></td><td><math>\mathbf{P}_8^{Lor}=R^a{}_bR^b{}_cR^c{}_dR^d{}_a</math></td></tr><tr><td><math>L_7^I=(\omega^a{}_bd\omega^b{}_a+\frac{2}{3}\omega^a{}_b\omega^b{}_c\omega^c{}_a)R^a{}_bR^b{}_a</math></td><td><math>(\mathbf{P}_4^{Lor})^2=(R^a{}_bR^b{}_a)^2</math></td></tr><tr><td><math>L_7^{II}=(e^aT_a)(R^a{}_bR^b{}_a)</math></td><td><math>\mathbf{N}_4\mathbf{P}_4^{Lor}=(T^aT_a-e^ae^bR_{ab})R^c{}_dR^d{}_c</math></td></tr></table> <div>Table 9: Seven dimensional gravitational CS Lagrangians, their related</div> | $D=7$ CS Lagrangians | Characteristic classes | $L_7^{Lor}=\omega(d\omega)^3+\frac{8}{5}\omega^3(d\omega)^2+\cdots+\frac{4}{7}\omega^7$ | $\mathbf{P}_8^{Lor}=R^a{}_bR^b{}_cR^c{}_dR^d{}_a$ | $L_7^I=(\omega^a{}_bd\omega^b{}_a+\frac{2}{3}\omega^a{}_b\omega^b{}_c\omega^c{}_a)R^a{}_bR^b{}_a$ | $(\mathbf{P}_4^{Lor})^2=(R^a{}_bR^b{}_a)^2$ | $L_7^{II}=(e^aT_a)(R^a{}_bR^b{}_a)$ | $\mathbf{N}_4\mathbf{P}_4^{Lor}=(T^aT_a-e^ae^bR_{ab})R^c{}_dR^d{}_c$ |
| $D=7$ CS Lagrangians                                                                              | Characteristic classes                                               |       |                                                                                                                                           |                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                      |                        |                                                                                         |                                                   |                                                                                                   |                                             |                                     |                                                                      |
| $L_7^{Lor}=\omega(d\omega)^3+\frac{8}{5}\omega^3(d\omega)^2+\cdots+\frac{4}{7}\omega^7$           | $\mathbf{P}_8^{Lor}=R^a{}_bR^b{}_cR^c{}_dR^d{}_a$                    |       |                                                                                                                                           |                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                      |                        |                                                                                         |                                                   |                                                                                                   |                                             |                                     |                                                                      |
| $L_7^I=(\omega^a{}_bd\omega^b{}_a+\frac{2}{3}\omega^a{}_b\omega^b{}_c\omega^c{}_a)R^a{}_bR^b{}_a$ | $(\mathbf{P}_4^{Lor})^2=(R^a{}_bR^b{}_a)^2$                          |       |                                                                                                                                           |                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                      |                        |                                                                                         |                                                   |                                                                                                   |                                             |                                     |                                                                      |
| $L_7^{II}=(e^aT_a)(R^a{}_bR^b{}_a)$                                                               | $\mathbf{N}_4\mathbf{P}_4^{Lor}=(T^aT_a-e^ae^bR_{ab})R^c{}_dR^d{}_c$ |       |                                                                                                                                           |                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                      |                        |                                                                                         |                                                   |                                                                                                   |                                             |                                     |                                                                      |
| cardona-qft-methods_F02294                                                                        | 182                                                                  | —     | $\begin{aligned} & \mathbf{N}_4\mathbf{P}_4^{\text{\text{Lor}}} = \\ & \left(T^aT_a-e^ae^bR_{ab}\right)R^c{}_dR^d{}_c \end{aligned}$      | $\mathbf{N}_4\mathbf{P}_4^{\text{Lor}} = \left(T^aT_a-e^ae^bR_{ab}\right)R^c{}_dR^d{}_c$                     | —                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                      |                        |                                                                                         |                                                   |                                                                                                   |                                             |                                     |                                                                      |
| cardona-qft-methods_F02295                                                                        | 182                                                                  | 1.000 | $O(D-n,\,n)$                                                                                                                              | $O(D-n,n)$                                                                                                   | ions for gravity for even $D$ , invariant under the (anti-) de Sitter or Poincaré groups. In this light, it is fairly obvious that although ordinary Einstein-                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                      |                        |                                                                                         |                                                   |                                                                                                   |                                             |                                     |                                                                      |
| cardona-qft-methods_F02296                                                                        | 182                                                                  | 1.000 | $(g)$                                                                                                                                     | $(g)$                                                                                                        | Hilbert gravity can be given a fibre bundle structure for the Lorentz group,                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                      |                        |                                                                                         |                                                   |                                                                                                   |                                             |                                     |                                                                      |
| cardona-qft-methods_F02297                                                                        | 182                                                                  | 1.000 | $\chi=2-2\,g$                                                                                                                             | $\chi=2-2g$                                                                                                  | Hilbert gravity can be given a fibre bundle structure for the Lorentz group,                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                      |                        |                                                                                         |                                                   |                                                                                                   |                                             |                                     |                                                                      |
| cardona-qft-methods_F02298                                                                        | 182                                                                  | 1.000 | $S\,O(D+1)$                                                                                                                               | $SO(D+1)$                                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                      |                        |                                                                                         |                                                   |                                                                                                   |                                             |                                     |                                                                      |
| cardona-qft-methods_F02299                                                                        | 182                                                                  | 1.000 | $S\,O(D)$                                                                                                                                 | $SO(D)$                                                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                      |                        |                                                                                         |                                                   |                                                                                                   |                                             |                                     |                                                                      |
| cardona-qft-methods_F02300                                                                        | 183                                                                  | 1.000 | $\Omega^{\prime}$                                                                                                                         | $\Omega'$                                                                                                    | leave the path integral unchanged, a minimal requirement would be                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                      |                        |                                                                                         |                                                   |                                                                                                   |                                             |                                     |                                                                      |
| cardona-qft-methods_F02301                                                                        | 183                                                                  | 1.000 | $\Omega^{\prime}\rightarrow\widetilde{\Omega}^{\prime}=-\Omega^{\prime}$                                                                  | $\Omega'\rightarrow\widetilde{\Omega}'=-\Omega'$                                                             | With this orientation, the quantity in brackets –with the appropriate                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                      |                        |                                                                                         |                                                   |                                                                                                   |                                             |                                     |                                                                      |
| cardona-qft-methods_F02302                                                                        | 183                                                                  | 1.000 | $\int_{\Omega^{\prime}}\rightarrow-\int_{\widetilde{\Omega}^{\prime}}$                                                                    | $\int_{\Omega'}\rightarrow-\int_{\widetilde{\Omega}'}$                                                       | With this orientation, the quantity in brackets –with the appropriate                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                      |                        |                                                                                         |                                                   |                                                                                                   |                                             |                                     |                                                                      |

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| cardona-qft-methods_F02303 | 183  | ■ 1.000 | $\widetilde{\Omega}^{\prime}$                                                   | $\widetilde{\Omega}'$                                      | 5.6. CS and Born-Infeld degenerate vacua                                                                                                                                 |
| cardona-qft-methods_F02304 | 183  | ■ 0.955 | $\chi\left[\Omega\cup\widetilde{\Omega}'\right]$                                | $\chi\left[\Omega\cup\widetilde{\Omega}'\right]$           | If the Lovelock coefficients $\alpha_p$ are chosen according to the CS prescription                                                                                      |
| cardona-qft-methods_F02305 | 183  | ■ 1.000 | $D=2\ n+1$                                                                      | $D=2n+1$                                                   |                                                                                                                                                                          |
| cardona-qft-methods_F02306 | 184  | ■ 0.979 | $R^{\{a\ b\}}=\frac{1}{l^2}\ e^{\{a\}}\ e^{\{b\}}$                              | $R^{ab}=\frac{1}{l^2}e^ae^b$                               | $R^{ab}=\frac{1}{l^2}e^ae^b\qquad\qquad\qquad(36)$                                                                                                                       |
| cardona-qft-methods_F02307 | 184  | ■ 0.840 | $\{ \ }^{\{ \}}\!\!\text{\rm k} \}$                                             | $\mathbf{k}$                                               | one has a n-fold degenerate root $R^{\mu\nu}=\frac{1}{l^2}e^\mu e^\nu$ , which corresponds to a locally constant negative curvature geometry (locally AdS). The globally |
| cardona-qft-methods_F02308 | 184  | ■ 1.000 | $A\ d\ S\ /\ K$                                                                 | $AdS/K$                                                    | Although as already mentioned, in even dimensions there are no CS theories, there exists a choice of Lovelock coefficients that yields equations                         |
| cardona-qft-methods_F02309 | 184  | ■ 0.793 | $\alpha_p\ \mathrm{S}$                                                          | $\alpha_p\ \mathbf{S}$                                     | to a CS theory in even dimensions. <sup>xxxxxx</sup> The name of fact that the Pfaffian of an antisymmetric matrix $M^a$                                                 |
| cardona-qft-methods_F02310 | 184  | ■ 1.000 | $\{ \ }^{\{ \}}\!\!\{38,39,42\}$                                                | 38,39,42                                                   |                                                                                                                                                                          |
| cardona-qft-methods_F02311 | 184  | ■ 1.000 | $M^{\{a\ b\}}$                                                                  | $M^{ab}$                                                   |                                                                                                                                                                          |
| cardona-qft-methods_F02312 | 184  | ■ 0.995 | $\{ \ }^{\{ \}}\!\!\{\mathrm{k}\}$                                              | $\mathbf{k}$                                               |                                                                                                                                                                          |
| cardona-qft-methods_F02313 | 184  | ■ 0.995 | $S\ 0(D,\ 2)\ /\ S\ 0(D-1,1)$                                                   | $SO(D,2)/SO(D-1,1)$                                        |                                                                                                                                                                          |
| cardona-qft-methods_F02314 | 184  | ■ 1.000 | $S\ 0(D,\ 2)$                                                                   | $SO(D,2)$                                                  |                                                                                                                                                                          |
| cardona-qft-methods_F02315 | 185  | ■ 1.000 | $L_{2n}^{BI}$                                                                   | $L_{2n}^{BI}$                                              | determinant as<br>$\mathrm{Pfaff}[M]=e_{a_1\ldots a_n}\ldots e_{a_n}M^{a_1b_1}\ldots M^{a_nb_n}=\sqrt{\det[M]}$                                                          |
| cardona-qft-methods_F02316 | 185  | ■ 1.000 | $\sqrt{\det\left[R^{\{a\ b\}}+\frac{1}{l^2}\ e^{\{a\}}\ e^{\{b\}}\right]}$      | $\sqrt{\det\left[R^{ab}+\frac{1}{l^2}e^ae^b\right]}$       | determinant as<br>$\mathrm{Pfaff}[M]=e_{a_1\ldots a_n}\ldots e_{a_n}M^{a_1b_1}\ldots M^{a_nb_n}=\sqrt{\det[M]}$                                                          |
| cardona-qft-methods_F02317 | 185  | ■ 1.000 | $L_{EM}^{BI}=\sqrt{\det[F_{\mu\nu}-\lambda\eta_{\mu\nu}-\lambda\eta_{\mu\nu}]}$ | $L_{EM}^{BI}=\sqrt{\det[F_{\mu\nu}-\lambda\eta_{\mu\nu}]}$ | $L_{EM}^{BI}=\sqrt{\det[Dab+\lambda\eta_{ab}]}$ which is invariant                                                                                                       |
| cardona-qft-methods_F02318 | 185  | ■ 1.000 | $\{ \ }^{\{ \}}\!\!\{45,46\}$                                                   | 45,46                                                      | from a generic configuration with maximum number of degrees of freedom, and its evolution takes it into a degenerate surface. It can be shown that in                    |
| cardona-qft-methods_F02319 | 185  | ■ 1.000 | $R^{\{a\ b\}}+\frac{1}{l^2}\ e^{\{a\}}\ e^{\{b\}}=0$                            | $R^{ab}+\frac{1}{l^2}e^ae^b=0$                             | ary conditions could be those that keep the values of the fields fixed at                                                                                                |
| cardona-qft-methods_F02320 | 186  | ■ 1.000 | $I\left[\phi^r\right]$                                                          | $I\left[\phi^r\right]$                                     | more relevant to fix gauge invariant properties at the boundary –like the curvature. These boundary conditions are not of the Dirichlet type, but                        |
| cardona-qft-methods_F02321 | 186  | ■ 1.000 | $\Theta=0$                                                                      | $\Theta=0$                                                 | but this is not always physically acceptable. The boundary conditions under                                                                                              |
| cardona-qft-methods_F02322 | 186  | ■ 1.000 | $B\left(\left.\phi^r\right _{\partial M}\right)$                                | $B\left(\phi^r _{\partial M}\right)$                       | cause the Euler-Lagrange equations are not changed, but it is a delicate op-                                                                                             |
| cardona-qft-methods_F02323 | 187  | ■ 0.901 | $\left(\epsilon_{abc\dots}\rightarrow\epsilon\right)$                           | $(\epsilon_{abc\dots}\rightarrow\epsilon)$                 | attains an extremum for boundary conditions that fix the extrinsic curva- ture of the boundary. In <sup>40</sup> it is also shown that this action principle yields      |
| cardona-qft-methods_F02324 | 187  | ■ 0.901 | $\widetilde{R}$                                                                 | $\widetilde{R}$                                            | attains an extremum for boundary conditions that fix the extrinsic curva- ture of the boundary. In <sup>40</sup> it is also shown that this action principle yields      |
| cardona-qft-methods_F02325 | 188  | ■ 1.000 | $\bar{A}$                                                                       | $\bar{A}$                                                  | explicit expression for the transgression form in terms of the Chern-Simons forms for $A$ and $\bar{A}$ ,                                                                |
| cardona-qft-methods_F02326 | 188  | ■ 1.000 | $\mathcal{B}_{2n}$                                                              | $\mathcal{B}_{2n}$                                         | one accepts the existence of this unwelcome guest, an explanation would be                                                                                               |
| cardona-qft-methods_F02327 | 188  | ■ 0.892 | $(\bar{A})$                                                                     | $(\bar{A})$                                                |                                                                                                                                                                          |
| cardona-qft-methods_F02328 | 189  | ■ 1.000 | $\mathcal{B}_{2n}(A,\ \bar{A})$                                                 | $\mathcal{B}_{2n}(A,\ \bar{A})$                            | ary, where condition (45) makes sense; they are independent but <i>cobordant</i> manifolds                                                                               |

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| cardona-qft-methods_F02329 | 189  | ■ 1.000 | $\mathcal{C}(\bar{A})$                                                             | $\mathcal{C}(\bar{A})$                                                             | $I(A, \bar{A}) = \int \mathcal{L}(A) + \int \mathcal{L}(\bar{A}) + \int B(A, \bar{A}), \quad (46)$                                                                                                                                 |
| cardona-qft-methods_F02330 | 189  | ■ 1.000 | $\bar{M}$                                                                          | $\bar{M}$                                                                          | and matching gauge symmetry. This second term $A$ extends onto a compact manifold $\bar{M}$ , to which we have no direct access except through the                                                                                 |
| cardona-qft-methods_F02331 | 189  | ■ 1.000 | $\partial M = -\partial \bar{M}$                                                   | $\partial M = -\partial \bar{M}$                                                   | interaction of $\bar{A}$ with our $A$ . <sup>49</sup> If the spacetime we live in is asymptotically                                                                                                                                |
| cardona-qft-methods_F02332 | 189  | ■ 0.998 | $(M)$                                                                              | $(M)$                                                                              |                                                                                                                                                                                                                                    |
| cardona-qft-methods_F02333 | 190  | ■ 1.000 | $\{ \}^{9,52}$                                                                     | 9,52                                                                               | <b>9. CS as brane couplings</b><br>Although CS forms appeared in high-energy physics more than 20 years ago,                                                                                                                       |
| cardona-qft-methods_F02334 | 190  | ■ 1.000 | $\{ \}^{53-55}$                                                                    | 53–55                                                                              | some gauge colc                                                                                                                                                                                                                    |
| cardona-qft-methods_F02335 | 190  | ■ 1.000 | $A_{\mu_1 \mu_2 \cdots \mu_p} j^{\mu_1 \mu_2 \cdots \mu_p}$                        | $A_{\mu_1 \mu_2 \cdots \mu_p} j^{\mu_1 \mu_2 \cdots \mu_p}$                        | only for an abelian potential. <sup>39</sup>                                                                                                                                                                                       |
| cardona-qft-methods_F02336 | 190  | ■ 1.000 | $I_{\text{Int}}$                                                                   | $I_{\text{Int}}$                                                                   | the generator of the symmetry that implies its conservation.                                                                                                                                                                       |
| cardona-qft-methods_F02337 | 190  | ■ 1.000 | $j^{\prime \mu} = j^{\mu}$                                                         | $j^{\prime \mu} = j^{\mu}$                                                         | <sup>1</sup> This discussion is based on. <sup>57</sup>                                                                                                                                                                            |
| cardona-qft-methods_F02338 | 191  | ■ 0.999 | $\sqrt{ g }$                                                                       | $\sqrt{ g }$                                                                       | The metric independence of the minimal coupling has a more subtle                                                                                                                                                                  |
| cardona-qft-methods_F02339 | 191  | ■ 1.000 | $j = q * \delta(\Gamma)$                                                           | $j = q * \delta(\Gamma)$                                                           | the parameter $j = q * \delta(\Gamma)$ where $\delta$ is the Dirac delta function, $j \mu j$ , $\delta x = q A \wedge \delta(\Gamma)$ , and (1) reads                                                                              |
| cardona-qft-methods_F02340 | 191  | ■ 1.000 | $A_{\mu} j^{\mu} d^4 x =$                                                          | $A_{\mu} j^{\mu} d^4 x =$                                                          | the parameter $j = q * \delta(\Gamma)$ where $\delta$ is the Dirac delta function, $j \mu j$ , $\delta x = q A \wedge \delta(\Gamma)$ , and (1) reads                                                                              |
| cardona-qft-methods_F02341 | 191  | ■ 1.000 | $q A \wedge \delta(\Gamma)$                                                        | $q A \wedge \delta(\Gamma)$                                                        |                                                                                                                                                                                                                                    |
| cardona-qft-methods_F02342 | 191  | ■ 1.000 | $z^{\mu}$                                                                          | $z^{\mu}$                                                                          | Dirac delta, it yields the value of the test function at the support, which is                                                                                                                                                     |
| cardona-qft-methods_F02343 | 191  | ■ 0.484 | $(2p+1)$                                                                           | $(2p+1)$                                                                           |                                                                                                                                                                                                                                    |
| cardona-qft-methods_F02344 | 192  | ■ 0.881 | $\mathfrak{C}_{2p+1}$                                                              | $\mathfrak{C}_{2p+1}$                                                              | $I(\mathbf{A}, \mathbf{j}) = \int_{\Gamma} \mathbf{A} \wedge \mathbf{j} = I(\mathbf{A}) \quad (9)$                                                                                                                                 |
| cardona-qft-methods_F02345 | 192  | ■ 1.000 | $\mathcal{C}_{2p+1}(\mathbf{A}) = \langle \mathfrak{C}_{2p+1}(\mathbf{A}) \rangle$ | $\mathcal{C}_{2p+1}(\mathbf{A}) = \langle \mathfrak{C}_{2p+1}(\mathbf{A}) \rangle$ | $\mathfrak{A}(\mathbf{A}, \mathbf{j}) = \int \mathfrak{A} d p \wedge \mathfrak{C}_{2p+1}(\mathbf{A}, \mathbf{j}) \quad (9)$                                                                                                        |
| cardona-qft-methods_F02346 | 192  | ■ 0.852 | $D-2p-1$                                                                           | $D-2p-1$                                                                           | where $\mathfrak{C}_{2p+1}$ is the algebra-valued form whose trace is the $(2p+1)$ -CS form living on the brane history $\mathcal{C}_{2p+1}(\mathbf{A}) = \langle \mathfrak{C}_{2p+1}(\mathbf{A}) \rangle$ . The current generated |
| cardona-qft-methods_F02347 | 192  | ■ 0.852 | $\mathbf{j}$                                                                       | $\mathbf{j}$                                                                       | where $\mathfrak{C}_{2p+1}$ is the algebra-valued form whose trace is the $(2p+1)$ -CS form living on the brane history $\mathcal{C}_{2p+1}(\mathbf{A}) = \langle \mathfrak{C}_{2p+1}(\mathbf{A}) \rangle$ . The current generated |
| cardona-qft-methods_F02348 | 192  | ■ 1.000 | $dx^{\alpha_i}$                                                                    | $dx^{\alpha_i}$                                                                    | where $dx^{\alpha_i}$ are transverse directions to $\Gamma$ . The integration over the $D-2p-1$                                                                                                                                    |
| cardona-qft-methods_F02349 | 192  | ■ 1.000 | $I[\mathbf{A}; \mathbf{j}]$                                                        | $I[\mathbf{A}; \mathbf{j}]$                                                        | form, provided the current is covariantly conserved,                                                                                                                                                                               |
| cardona-qft-methods_F02350 | 192  | ■ 1.000 | $\mathbf{j}_{2p}$                                                                  | $\mathbf{j}_{2p}$                                                                  | teed by consistency. On the worldvolume, however, the gauge symmetry is reduced to the subgroup that commutes with $i\theta_1 \theta_2^{-1} a_s \mathbf{K} = \mathbf{K} = \cdots \mathbf{K}$                                       |
| cardona-qft-methods_F02351 | 192  | ■ 0.999 | $j^{a_1 a_2 \cdots a_s} \mathbf{K}_{a_1} \mathbf{K}_{a_2} \cdots \mathbf{K}_{a_s}$ | $j^{a_1 a_2 \cdots a_s} \mathbf{K}_{a_1} \mathbf{K}_{a_2} \cdots \mathbf{K}_{a_s}$ | with a point particle; <sup>59</sup> the singularity is the worldline of the particle where                                                                                                                                        |
| cardona-qft-methods_F02352 | 192  | ■ 0.737 | $\mathrm{AdS}_3$                                                                   | $\mathrm{AdS}_3$                                                                   | sign in the mass parameter of the solution. <sup>61</sup> A similar situation arises also                                                                                                                                          |
| cardona-qft-methods_F02353 | 193  | ■ 1.000 | $\{ \}^{62,63}$                                                                    | 62,63                                                                              | 182 J. Zanelli                                                                                                                                                                                                                     |
| cardona-qft-methods_F02354 | 193  | ■ 0.531 | $(2+1)$                                                                            | $(2+1)$                                                                            | dimensional spacetime whose excitations propagate on two-dimensional lay-                                                                                                                                                          |
| cardona-qft-methods_F02355 | 193  | ■ 1.000 | $\theta_1$                                                                         | $\theta_1$                                                                         | $\Theta$ takes a constant value $\theta_1$ in the region $M \subset M$ and $\theta_2$ elsewhere, then this term can also be written as a coupling between the Chern-Simons and                                                     |

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| cardona-qft-methods_F02356 | 193  | ■ 1.000 | $\tilde{M} \subset M$                                                                                                                                                                  | $\tilde{M} \subset M$                                          | $\Theta$ takes a constant value $\theta_1$ in the region $M \subset M$ and $\theta_2$ elsewhere, then this term can also be written as a coupling between the Chern-Simons and                                                               |
| cardona-qft-methods_F02357 | 193  | ■ 1.000 | $\theta_2$                                                                                                                                                                             | $\theta_2$                                                     | $\Theta$ takes a constant value $\theta_1$ in the region $M \subset M$ and $\theta_2$ elsewhere, then this term can also be written as a coupling between the Chern-Simons and                                                               |
| cardona-qft-methods_F02358 | 193  | ■ 1.000 | $j=d \backslash \Theta=\backslash \left(\backslash \theta_{\{1\}}-\backslash \theta_{\{2\}}\backslash \right) \backslash \delta(\Sigma) d z$                                           | $j=d \Theta=\left(\theta_1-\theta_2\right) \delta(\Sigma) d z$ | are the same as in vacuum. However, this term modifies the behavior of the                                                                                                                                                                   |
| cardona-qft-methods_F02359 | 193  | ■ 1.000 | $\tilde{M}, \backslash \Sigma=\backslash \partial \tilde{M}$                                                                                                                           | $\tilde{M}, \Sigma=\partial \tilde{M}$                         | field at the surface $\Sigma$ .<br>This coupling between the CS term and the spacetime boundary of the                                                                                                                                       |
| cardona-qft-methods_F02360 | 193  | ■ 1.000 | $\tilde{M}$                                                                                                                                                                            | $\tilde{M}$                                                    | This coupling between the CS term and the spacetime boundary of the region $\tilde{M}$ has physical consequences even in the simple case of the abelian                                                                                      |
| cardona-qft-methods_F02361 | 193  | ■ 0.934 | $\theta \neq 0 .\{ \} ^{65}$                                                                                                                                                           | $\theta \neq 0 .^{65}$                                         | One of the great achievements of physics in the last century was to establish                                                                                                                                                                |
| cardona-qft-methods_F02362 | 200  | ■ 1.000 | $\{ \} ^{12,33,35,36,46}$                                                                                                                                                              | 12,33,35,36,46                                                 | what extent, geometry can be used to model physics. It is well known that Riemannian geometry was extremely successful in providing the mathemat.                                                                                            |
| cardona-qft-methods_F02363 | 201  | ■ 0.996 | $S \mathrm{~U}(3)$                                                                                                                                                                     | $SU(3)$                                                        | Functional spectral moments, where the gauge potentials are again described by connections and the fermions they interact with are described                                                                                                 |
| cardona-qft-methods_F02364 | 202  | ■ 1.000 | $S=S_{\{E \ H\}}+S_{\{S \ M\}}$                                                                                                                                                        | $S=S_{EH}+S_{SM}$                                              | there is a term in the action functional involving the weyl curvature tensor                                                                                                                                                                 |
| cardona-qft-methods_F02365 | 202  | ■ 1.000 | $S_{\{S \ M\}}$                                                                                                                                                                        | $S_{SM}$                                                       | there is a term in the action functional involving the weyl curvature tensor                                                                                                                                                                 |
| cardona-qft-methods_F02366 | 202  | ■ 1.000 | $f\backslash \left(R, R^{\{\mu \ \nu\}}, C_{\backslash \lambda \mu \nu \kappa}\backslash \right)$                                                                                      | $f\left(R, R^{\mu \nu}, C_{\lambda \mu \nu \kappa}\right)$     |                                                                                                                                                                                                                                              |
| cardona-qft-methods_F02367 | 203  | ■ 1.000 | $C^{\{\infty\}}(X)$                                                                                                                                                                    | $C^{\infty}(X)$                                                |                                                                                                                                                                                                                                              |
| cardona-qft-methods_F02368 | 204  | ■ 1.000 | $(\backslash \mathrm{mathcal}\{A\}, \backslash \mathrm{mathcal}\{H\}, D)$                                                                                                              | $(\mathcal{A}, \mathcal{H}, D)$                                | <ul style="list-style-type: none"> <li>a representation <math>\pi: \mathcal{A} \rightarrow \mathcal{L}(\mathcal{H})</math> in the algebra <math>\mathcal{L}(\mathcal{H})</math> of bounded operators on a separable Hilbert space</li> </ul> |
| cardona-qft-methods_F02369 | 204  | ■ 1.000 | $\backslash \pi: \backslash \mathrm{mathcal}\{A\} \rightarrow \backslash \mathrm{mathcal}\{L\}(\backslash \mathrm{mathcal}\{H\})$                                                      | $\pi: \mathcal{A} \rightarrow \mathcal{L}(\mathcal{H})$        | $\mathcal{K}$ of compact operators;                                                                                                                                                                                                          |
| cardona-qft-methods_F02370 | 204  | ■ 1.000 | $\backslash \mathrm{mathcal}\{L\}(\backslash \mathrm{mathcal}\{H\})$                                                                                                                   | $\mathcal{L}(\mathcal{H})$                                     | $\mathcal{K}$ of compact operators;                                                                                                                                                                                                          |
| cardona-qft-methods_F02371 | 204  | ■ 1.000 | $D=D^{\{*\}}$                                                                                                                                                                          | $D=D^*$                                                        | <ul style="list-style-type: none"> <li>The spectral triples is <i>even</i> if there is also a <math>\mathbb{Z}/2</math>-grading <math>\gamma</math> on <math>\mathcal{H}</math> such</li> </ul>                                              |
| cardona-qft-methods_F02372 | 204  | ■ 1.000 | $\backslash \left(1+D^{\{2\}}\backslash \right)^{\{-1 \ / \ 2\}}$                                                                                                                      | $\left(1+D^2\right)^{-1/2}$                                    | that                                                                                                                                                                                                                                         |
| cardona-qft-methods_F02373 | 204  | ■ 1.000 | $\backslash \mathrm{mathcal}\{K\}$                                                                                                                                                     | $\mathcal{K}$                                                  | $[\gamma, a]$                                                                                                                                                                                                                                |
| cardona-qft-methods_F02374 | 204  | ■ 0.999 | $[a, \ D]$                                                                                                                                                                             | $[a, D]$                                                       | The main example of a classical space that gives rise to data of a spectral triple is a compact smooth spin Riemannian manifold $M$ for which                                                                                                |
| cardona-qft-methods_F02375 | 204  | ■ 1.000 | $\backslash \mathrm{mathbb}\{Z\} \ / \ 2$                                                                                                                                              | $\mathbb{Z}/2$                                                 | $S$ the spinor bundle and $L^2(M, S)$ the Hilbert space of square integrable                                                                                                                                                                 |
| cardona-qft-methods_F02376 | 204  | ■ 0.895 | $\backslash \left(C^{\{\infty\}}(M), \ L^{\{2\}}(M, \ S), \ \neq_{\backslash M}\backslash \right)$                                                                                     | $\left(C^{\infty}(M), L^2(M, S), \neq_M\right)$                |                                                                                                                                                                                                                                              |
| cardona-qft-methods_F02377 | 204  | ■ 1.000 | $L^{\{2\}}(M, \ S)$                                                                                                                                                                    | $L^2(M, S)$                                                    |                                                                                                                                                                                                                                              |
| cardona-qft-methods_F02378 | 204  | ■ 1.000 | $D=\backslash \partial \backslash _{\backslash M}$                                                                                                                                     | $D=\partial_M$                                                 |                                                                                                                                                                                                                                              |
| cardona-qft-methods_F02379 | 204  | ■ 0.997 | $\gamma_5$                                                                                                                                                                             | $\gamma_5$                                                     |                                                                                                                                                                                                                                              |
| cardona-qft-methods_F02380 | 204  | ■ 1.000 | $\{ \} ^{22,31,49}$                                                                                                                                                                    | 22,31,49                                                       |                                                                                                                                                                                                                                              |
| cardona-qft-methods_F02381 | 205  | ■ 1.000 | $n \ \backslash \mathrm{in} \ \backslash \mathrm{mathbb}\{Z\} \ / \ 8 \ \backslash \mathrm{mathbb}\{Z\}$                                                                               | $n \in \mathbb{Z}/8\mathbb{Z}$                                 | where $\epsilon, \epsilon', \epsilon'' \in \{\pm 1\}$ are sign                                                                                                                                                                               |
| cardona-qft-methods_F02382 | 205  | ■ 1.000 | $\backslash \epsilon, \backslash \epsilon^{\{\prime\}}, \backslash \epsilon^{\{\prime\prime \ \prime\prime\}} \ \backslash \mathrm{in} \ \backslash \{ \ \backslash \mathrm{pm} \ 1\}$ | $\epsilon, \epsilon', \epsilon'' \in \{\pm 1\}$                | The antilinear involution $J$ is also required to satisfy the commutation                                                                                                                                                                    |

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| cardona-qft-methods_ F02383 | 205  | ■ 0.993 | $\varepsilon$                                                                                         | $\varepsilon$                                                               | $[[D, a], b^0] = 0 \quad \forall a, b \in \mathcal{A}.$ <p>table of signs listed above is modeled on the case of ordinary where it reflects the properties of the mod eight periodicity of gy and real <math>K</math>-theory. However, imposing that real structure</p> |
| cardona-qft-methods_ F02384 | 205  | ■ 0.993 | $\varepsilon'$                                                                                        | $\varepsilon'$                                                              | $[[D, a], b^0] = 0 \quad \forall a, b \in \mathcal{A}.$ <p>table of signs listed above is modeled on the case of ordinary where it reflects the properties of the mod eight periodicity of gy and real <math>K</math>-theory. However, imposing that real structure</p> |
| cardona-qft-methods_ F02385 | 205  | ■ 0.993 | $\varepsilon''$                                                                                       | $\varepsilon''$                                                             | $[[D, a], b^0] = 0 \quad \forall a, b \in \mathcal{A}.$ <p>table of signs listed above is modeled on the case of ordinary where it reflects the properties of the mod eight periodicity of gy and real <math>K</math>-theory. However, imposing that real structure</p> |
| cardona-qft-methods_ F02386 | 205  | ■ 0.999 | $\left[ a, b^{\{0\}} \right] = 0$                                                                     | $[a, b^0] = 0$                                                              |                                                                                                                                                                                                                                                                         |
| cardona-qft-methods_ F02387 | 205  | ■ 0.999 | $a, b \in \mathcal{A}$                                                                                | $a, b \in \mathcal{A}$                                                      |                                                                                                                                                                                                                                                                         |
| cardona-qft-methods_ F02388 | 205  | ■ 0.999 | $b^{\{0\}}$                                                                                           | $b^0$                                                                       |                                                                                                                                                                                                                                                                         |
| cardona-qft-methods_ F02389 | 205  | ■ 0.999 | $b^{\{0\}} =$                                                                                         | $b^0 =$                                                                     |                                                                                                                                                                                                                                                                         |
| cardona-qft-methods_ F02390 | 205  | ■ 1.000 | $J b^{\{*\}} J^{-1}$                                                                                  | $J b^* J^{-1}$                                                              |                                                                                                                                                                                                                                                                         |
| cardona-qft-methods_ F02391 | 205  | ■ 1.000 | $b \in \mathcal{A}$                                                                                   | $b \in \mathcal{A}$                                                         |                                                                                                                                                                                                                                                                         |
| cardona-qft-methods_ F02392 | 206  | ■ 1.000 | $\{ \}^{\{5,9\}}$                                                                                     | 5,9                                                                         | <p>inclusion of noncommutative geometry, there can be a finite <math>KO</math>-dimension.</p>                                                                                                                                                                           |
| cardona-qft-methods_ F02393 | 206  | ■ 1.000 | $F =$                                                                                                 | $F =$                                                                       | <p>The data of <math>F = (\mathcal{A}_F, \mathcal{H}_F, D_F)</math> are given by</p>                                                                                                                                                                                    |
| cardona-qft-methods_ F02394 | 206  | ■ 0.986 | $\left( \mathcal{A}_F, \mathcal{H}_F, D_F \right)$                                                    | $(\mathcal{A}_F, \mathcal{H}_F, D_F)$                                       | <ul style="list-style-type: none"> <li>• a finite dimensional (real) <math>C^*</math>-algebra <math>\mathcal{A}</math>, which by Wedderburn theorem</li> </ul>                                                                                                          |
| cardona-qft-methods_ F02395 | 206  | ■ 0.986 | $\mathcal{A}_F$                                                                                       | $\mathcal{A}_F$                                                             | <ul style="list-style-type: none"> <li>• a finite dimensional (real) <math>C^*</math>-algebra <math>\mathcal{A}</math>, which by Wedderburn theorem</li> </ul>                                                                                                          |
| cardona-qft-methods_ F02396 | 206  | ■ 1.000 | $F = \left( \mathcal{A}_F, \mathcal{H}_F, D_F \right)$                                                | $F = (\mathcal{A}_F, \mathcal{H}_F, D_F)$                                   | <ul style="list-style-type: none"> <li>• an antilinear involution <math>J</math> on <math>\mathcal{H}_F</math> with signs <math>\epsilon, \epsilon, \epsilon</math> (the real structure):</li> </ul>                                                                    |
| cardona-qft-methods_ F02397 | 206  | ■ 1.000 | $\mathbb{K}_i = \mathbb{R}$                                                                           | $\mathbb{K}_i = \mathbb{R}$                                                 |                                                                                                                                                                                                                                                                         |
| cardona-qft-methods_ F02398 | 206  | ■ 1.000 | $\mathbb{H}$                                                                                          | $\mathbb{H}$                                                                |                                                                                                                                                                                                                                                                         |
| cardona-qft-methods_ F02399 | 206  | ■ 1.000 | $\mathcal{H}_F$                                                                                       | $\mathcal{H}_F$                                                             |                                                                                                                                                                                                                                                                         |
| cardona-qft-methods_ F02400 | 207  | ■ 1.000 | $D^{\{*\}} = D$                                                                                       | $D^* = D$                                                                   |                                                                                                                                                                                                                                                                         |
| cardona-qft-methods_ F02401 | 207  | ■ 0.957 | $\left[ [D, a], b^{\{0\}} \right] = 0$                                                                | $[[D, a], b^0] = 0$                                                         | <ul style="list-style-type: none"> <li>• a bimodule</li> </ul>                                                                                                                                                                                                          |
| cardona-qft-methods_ F02402 | 207  | ■ 0.992 | $\left( \mathcal{A}_F, \mathcal{H}_F \right)$                                                         | $(\mathcal{A}_F, \mathcal{H}_F)$                                            | <p>One can consider the moduli spaces of all linear operators as up to unitary equivalence on a given <math>(\mathcal{A}_F, \mathcal{H}_F)</math>. These moduli spaces will provide</p>                                                                                 |
| cardona-qft-methods_ F02403 | 207  | ■ 1.000 | $\left( \lambda, q_L, q_R, m \right) \mapsto \left( \bar{\lambda}, \bar{q}_L, \bar{q}_R, m^* \right)$ | $(\lambda, q_L, q_R, m) \mapsto (\bar{\lambda}, \bar{q}_L, \bar{q}_R, m^*)$ |                                                                                                                                                                                                                                                                         |
| cardona-qft-methods_ F02404 | 207  | ■ 1.000 | $\mathbb{C} \oplus M_3(\mathbb{C})$                                                                   | $\mathbb{C} \oplus M_3(\mathbb{C})$                                         |                                                                                                                                                                                                                                                                         |



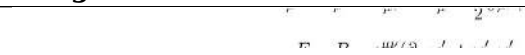
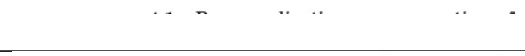


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| cardona-qft-methods_F02457 | 210  | ■ 1.000 | $\mathcal{H}_f$                                                                                            | $\mathcal{H}_f$                                                                                            |                                                                                                                                                                                                   |
| cardona-qft-methods_F02458 | 210  | ■ 1.000 | $\mathcal{H}_{\bar{f}}$                                                                                    | $\mathcal{H}_{\bar{f}}$                                                                                    |                                                                                                                                                                                                   |
| cardona-qft-methods_F02459 | 210  | ■ 1.000 | $\left(\mathcal{A}_F, D_F\right)$                                                                          | $(\mathcal{A}_F, D_F)$                                                                                     | <i>noncommutative geometry models for particle physics</i> 139                                                                                                                                    |
| cardona-qft-methods_F02460 | 210  | ■ 0.973 | $\mathcal{A}_F \subset \mathcal{A}_{LR}$                                                                   | $\mathcal{A}_F \subset \mathcal{A}_{LR}$                                                                   | mixes $\mathcal{H}_f$ and $\mathcal{H}_{\bar{f}}$ would not satisfy the order one condition for $\mathcal{A}_{LR}$ . Thus,                                                                        |
| cardona-qft-methods_F02461 | 210  | ■ 0.973 | $D_F$                                                                                                      | $D_F$                                                                                                      | mixes $\mathcal{H}_f$ and $\mathcal{H}_{\bar{f}}$ would not satisfy the order one condition for $\mathcal{A}_{LR}$ . Thus,                                                                        |
| cardona-qft-methods_F02462 | 210  | ■ 0.996 | $\mathcal{A}_F$                                                                                            | $\mathcal{A}_F$                                                                                            | the question of mixing a finite spectral triple $(\mathcal{A}_F, \pi_F, D_F)$ can be phrased as a coupled problem for the pair $(\mathcal{A}_F, D_F)$ , for fixed $\mathcal{H}_F$ as above, where |
| cardona-qft-methods_F02463 | 210  | ■ 1.000 | $\operatorname{Aut}\left(\mathcal{A}_{LR}\right)$                                                          | $\operatorname{Aut}(\mathcal{A}_{LR})$                                                                     | unique up to $\operatorname{Aut}(\mathcal{A}_{LR})$ , which represents a spontaneous breaking of the left-right symmetry of $\mathcal{A}_{LR}$ imposed by the order one condition of the Dirac    |
| cardona-qft-methods_F02464 | 210  | ■ 1.000 | $\lambda \in \mathrm{U}(1)$                                                                                | $\lambda \in \mathrm{U}(1)$                                                                                | ;                                                                                                                                                                                                 |
| cardona-qft-methods_F02465 | 210  | ■ 1.000 | $\left(\mathcal{A}_F, \mathcal{H}_F, \gamma_F, J_F\right)$                                                 | $(\mathcal{A}_F, \mathcal{H}_F, \gamma_F, J_F)$                                                            |                                                                                                                                                                                                   |
| cardona-qft-methods_F02466 | 210  | ■ 1.000 | $J_F$                                                                                                      | $J_F$                                                                                                      |                                                                                                                                                                                                   |
| cardona-qft-methods_F02467 | 210  | ■ 1.000 | $\gamma_F$                                                                                                 | $\gamma_F$                                                                                                 |                                                                                                                                                                                                   |
| cardona-qft-methods_F02468 | 210  | ■ 1.000 | $a, b \in \mathcal{A}_F$                                                                                   | $a, b \in \mathcal{A}_F$                                                                                   |                                                                                                                                                                                                   |
| cardona-qft-methods_F02469 | 211  | ■ 1.000 | $S_3$                                                                                                      | $S_3$                                                                                                      |                                                                                                                                                                                                   |
| cardona-qft-methods_F02470 | 211  | ■ 1.000 | $Y_{(\downarrow 1)}, Y_{(\uparrow 1)}, Y_{(\downarrow 3)}, Y_{(\uparrow 3)} \in \mathrm{GL}_3(\mathbb{C})$ | $Y_{(\downarrow 1)}, Y_{(\uparrow 1)}, Y_{(\downarrow 3)}, Y_{(\uparrow 3)} \in \mathrm{GL}_3(\mathbb{C})$ | moduli space $\mathcal{C}_3 \times \mathcal{C}_1$ , where $\mathcal{C}_3$ is the set of pairs $(I_{(\downarrow 3)}, I_{(\uparrow 3)})$ equivalence                                                |
| cardona-qft-methods_F02471 | 211  | ■ 1.000 | $Y_R$                                                                                                      | $Y_R$                                                                                                      | moduli space $\mathcal{C}_3 \times \mathcal{C}_1$ , where $\mathcal{C}_3$ is the set of pairs $(I_{(\downarrow 3)}, I_{(\uparrow 3)})$ equivalence                                                |
| cardona-qft-methods_F02472 | 211  | ■ 1.000 | $\mathcal{C}_3 \times \mathcal{C}_1$                                                                       | $\mathcal{C}_3 \times \mathcal{C}_1$                                                                       | with $G = \mathrm{GL}_3(\mathbb{C})$ and $K = U(3)$ . The real dimension is $\dim_{\mathbb{R}} \mathcal{C}_3 = 10 =$                                                                              |
| cardona-qft-methods_F02473 | 211  | ■ 1.000 | $\mathcal{C}_3$                                                                                            | $\mathcal{C}_3$                                                                                            | with $G = \mathrm{GL}_3(\mathbb{C})$ and $K = U(3)$ . The real dimension is $\dim_{\mathbb{R}} \mathcal{C}_3 = 10 =$                                                                              |
| cardona-qft-methods_F02474 | 211  | ■ 1.000 | $\left(Y_{(\downarrow 3)}, Y_{(\uparrow 3)}\right)$                                                        | $(Y_{(\downarrow 3)}, Y_{(\uparrow 3)})$                                                                   | with $G = \mathrm{GL}_3(\mathbb{C})$ and $K = U(3)$ . The real dimension is $\dim_{\mathbb{R}} \mathcal{C}_3 = 10 =$                                                                              |
| cardona-qft-methods_F02475 | 211  | ■ 1.000 | $W_j$                                                                                                      | $W_j$                                                                                                      | $A_{(\downarrow 1)} = \sqrt{1} A_{(\downarrow 1)} \otimes 3, \quad A_{(\uparrow 1)} = \sqrt{2} A_{(\uparrow 1)} \otimes$<br>the following relations from the definition                           |
| cardona-qft-methods_F02476 | 211  | ■ 1.000 | $G=\mathrm{GL}_3(\mathbb{C})$                                                                              | $G = \mathrm{GL}_3(\mathbb{C})$                                                                            | $12-1=11,$                                                                                                                                                                                        |
| cardona-qft-methods_F02477 | 211  | ■ 1.000 | $K=U(3)$                                                                                                   | $K = U(3)$                                                                                                 | $12-1=11,$                                                                                                                                                                                        |
| cardona-qft-methods_F02478 | 211  | ■ 1.000 | $\operatorname{dim}_{\mathbb{R}} \mathcal{C}_3=10=$                                                        | $\dim_{\mathbb{R}} \mathcal{C}_3 = 10 =$                                                                   | $12-1=11,$                                                                                                                                                                                        |
| cardona-qft-methods_F02479 | 211  | ■ 0.978 | $3+3+4$                                                                                                    | $3+3+4$                                                                                                    |                                                                                                                                                                                                   |
| cardona-qft-methods_F02480 | 211  | ■ 0.978 | $3+3$                                                                                                      | $3+3$                                                                                                      |                                                                                                                                                                                                   |
| cardona-qft-methods_F02481 | 211  | ■ 1.000 | $\mathcal{C}_1$                                                                                            | $\mathcal{C}_1$                                                                                            |                                                                                                                                                                                                   |
| cardona-qft-methods_F02482 | 211  | ■ 1.000 | $\left(Y_{(\downarrow 1)}, Y_{(\uparrow 1)}, Y_R\right)$                                                   | $(Y_{(\downarrow 1)}, Y_{(\uparrow 1)}, Y_R)$                                                              |                                                                                                                                                                                                   |
| cardona-qft-methods_F02483 | 211  | ■ 1.000 | $\pi: \mathcal{C}_1 \rightarrow \mathcal{C}_3$                                                             | $\pi: \mathcal{C}_1 \rightarrow \mathcal{C}_3$                                                             |                                                                                                                                                                                                   |

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| cardona-qft-methods_F02484 | 211  | ■ 1.000 | $Y_{\{R\}} \mapsto \lambda^2 Y_{\{R\}}$                                                                                 | $Y_R \mapsto \lambda^2 Y_R$                                                                       |                                                                                                                                                                                                    |
| cardona-qft-methods_F02485 | 211  | ■ 1.000 | $\operatorname{dim}_{\mathbb{R}}(\mathcal{C}_3 \times \mathcal{C}_1) = 31$                                              | $\dim_{\mathbb{R}}(\mathcal{C}_3 \times \mathcal{C}_1) = 31$                                      |                                                                                                                                                                                                    |
| cardona-qft-methods_F02486 | 211  | ■ 1.000 | $12-1=11$                                                                                                               | $12-1=11$                                                                                         |                                                                                                                                                                                                    |
| cardona-qft-methods_F02487 | 212  | ■ 1.000 | $\delta_{\uparrow}, \delta_{\downarrow}$                                                                                | $\delta_{\uparrow}, \delta_{\downarrow}$                                                          | $\mathcal{I}_{(\uparrow 1)} = U_{PMNS} \theta_{(\uparrow 1)} U_{PMNS}^{\dagger} \quad \mathcal{I}_{(\downarrow 1)} = \theta_{(\downarrow 1)}$                                                      |
| cardona-qft-methods_F02488 | 212  | ■ 1.000 | $c_{\{i\}}=\cos \theta_{\{i\}}, \; s_{\{i\}}=\sin \theta_{\{i\}}, \; e_{\{\delta\}}=\exp \left(i \delta_{\{i\}}\right)$ | $c_i = \cos \theta_i, s_i = \sin \theta_i, e_{\delta} = \exp(i \delta)$                           | For the quark and lepton sector ( $\mathcal{C}_3$ and $\mathcal{C}_1$ , respectively) these are the                                                                                                |
| cardona-qft-methods_F02489 | 212  | ■ 1.000 | $U_{\{C \; K \; M\}}$                                                                                                   | $U_{CKM}$                                                                                         | the presence in the model of the latter matrix, this model can account for                                                                                                                         |
| cardona-qft-methods_F02490 | 212  | ■ 1.000 | $U_{\{P \; M \; N \; S\}}$                                                                                              | $U_{PMNS}$                                                                                        | the experimental phenomenon of neutrino mixing. The matrix $U_{\mathcal{R}}$ contains the Majorana mass terms for the right-handed neutrinos. The CKM matrix                                       |
| cardona-qft-methods_F02491 | 212  | ■ 0.969 | $V^{*} V$                                                                                                               | $V^* V$                                                                                           | subvarieties in the moduli space; the moduli spaces are described by sym-                                                                                                                          |
| cardona-qft-methods_F02492 | 212  | ■ 1.000 | $(K \times K) \backslash (G \times G) / K$                                                                              | $(K \times K) \backslash (G \times G) / K$                                                        | <ul style="list-style-type: none"> <li>3 coupling constants</li> </ul>                                                                                                                             |
| cardona-qft-methods_F02493 | 212  | ■ 1.000 | $\nu$                                                                                                                   | $\nu$                                                                                             |                                                                                                                                                                                                    |
| cardona-qft-methods_F02494 | 213  | ■ 0.699 | $\left(\mathcal{A}_{\{i\}}, \mathcal{H}_{\{i\}}, D_{\{i\}}, \gamma_{\{i\}}, J_{\{i\}}\right)$                           | $(\mathcal{A}_i, \mathcal{H}_i, D_i, \gamma_i, J_i)$                                              |                                                                                                                                                                                                    |
| cardona-qft-methods_F02495 | 214  | ■ 1.000 | $J, \gamma$                                                                                                             | $J, \gamma$                                                                                       | the dimension of the space associated to the spectral triple.                                                                                                                                      |
| cardona-qft-methods_F02496 | 214  | ■ 1.000 | $\zeta_{a,D}(s)=$                                                                                                       | $\zeta_{a,D}(s) =$                                                                                | In the case of ordinary manifolds the first two notions agree (the $KO$ -                                                                                                                          |
| cardona-qft-methods_F02497 | 214  | ■ 1.000 | $\operatorname{Tr}\left(a D ^{-s}\right)$                                                                               | $\operatorname{Tr}\left(a D ^{-s}\right)$                                                         | dimension is the metric dimension modulo eight) a                                                                                                                                                  |
| cardona-qft-methods_F02498 | 214  | ■ 1.000 | $X=M \times F$                                                                                                          | $X=M \times F$                                                                                    | where                                                                                                                                                                                              |
| cardona-qft-methods_F02499 | 214  | ■ 0.974 | $10=4+6$                                                                                                                | $10=4+6$                                                                                          | <ul style="list-style-type: none"> <li><math>M</math> is a smooth manifold. <math>\mathcal{E}</math> is an algebra bundle with fiber <math>\mathcal{E}_x</math>, a finite</li> </ul>               |
| cardona-qft-methods_F02500 | 214  | ■ 0.974 | $2 \bmod 8$                                                                                                             | $2 \bmod 8$                                                                                       | <ul style="list-style-type: none"> <li><math>M</math> is a smooth manifold. <math>\mathcal{E}</math> is an algebra bundle with fiber <math>\mathcal{E}_x</math>, a finite</li> </ul>               |
| cardona-qft-methods_F02501 | 214  | ■ 0.983 | $\left\{k \in \mathbb{Z}_{\geq 0}, k \leq 4\right\}$                                                                    | $\{k \in \mathbb{Z}_{\geq 0}, k \leq 4\}$                                                         | dimensional algebra $\mathcal{A}_F$                                                                                                                                                                |
| cardona-qft-methods_F02502 | 214  | ■ 1.000 | $\mathcal{E}$                                                                                                           | $\mathcal{E}$                                                                                     |                                                                                                                                                                                                    |
| cardona-qft-methods_F02503 | 214  | ■ 1.000 | $\mathcal{E}_{\{x\}}$                                                                                                   | $\mathcal{E}_x$                                                                                   |                                                                                                                                                                                                    |
| cardona-qft-methods_F02504 | 214  | ■ 1.000 | $C^{\infty}(M, \mathcal{E})$                                                                                            | $C^{\infty}(M, \mathcal{E})$                                                                      |                                                                                                                                                                                                    |
| cardona-qft-methods_F02505 | 214  | ■ 1.000 | $\mathcal{D}_{\mathcal{E}}=c \circ\left(\nabla^{\mathcal{E}} \otimes 1+1 \otimes \nabla^S\right)$                       | $\mathcal{D}_{\mathcal{E}}=c \circ\left(\nabla^{\mathcal{E}} \otimes 1+1 \otimes \nabla^S\right)$ |                                                                                                                                                                                                    |
| cardona-qft-methods_F02506 | 215  | ■ 0.998 | $\mathrm{U}(\mathcal{A})=\left\{u \in \mathcal{A} \mid u u^*=\right.$                                                   | $\mathrm{U}(\mathcal{A})=\{u \in \mathcal{A} \mid uu^*=$                                          |                                                                                                                                                                                                    |
| cardona-qft-methods_F02507 | 215  | ■ 0.999 | $\left.u^*\right\}$                                                                                                     | $u^*u=1\}$                                                                                        |                                                                                                                                                                                                    |
| cardona-qft-methods_F02508 | 216  | ■ 1.000 | $\operatorname{Ad}(u)$                                                                                                  | $\operatorname{Ad}(u)$                                                                            | the unimodular subgroup of $\mathcal{U}(\mathcal{A})$ in the adjoint representation $\operatorname{Ad}(u)$ on $\mathcal{H}$ is the gauge group of the standard model. The unimodular inner fluctu- |
| cardona-qft-methods_F02509 | 216  | ■ 0.991 | $U(1), \; S \; U(2)$                                                                                                    | $U(1), SU(2)$                                                                                     | provide the Higgs field of the particle physics model.                                                                                                                                             |
| cardona-qft-methods_F02510 | 216  | ■ 1.000 | $A^{\{1,0\}}=\sum_{\{i\}} a_{\{i\}}\left[\not{\partial}_M \otimes\right.$                                               | $A^{(1,0)}=\sum_i a_i \left[\not{\partial}_M \otimes\right.$                                      | <ul style="list-style-type: none"> <li>a <math>U(1)</math> gauge field <math>A=\sum_i \lambda_i d\lambda_i'=\sum_i \lambda_i [\partial_M \otimes 1, \lambda_i']</math>;</li> </ul>                 |

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| cardona-qft-methods_F02511 | 216  | ■ 0.749 | $\left.1, a_i^{\prime}\right]$                                                                                                                                                                                                                    | $1, a_i']$                                                                                                                                                              | <ul style="list-style-type: none"> <li>an <math>SU(2)</math> gauge field <math>Q = \sum_i q_i dq_i'</math>, with <math>q = f_0 + \sum_{\alpha} i f_{\alpha} \sigma^{\alpha}</math> and <math>Q =</math></li> </ul> |
| cardona-qft-methods_F02512 | 216  | ■ 0.749 | $a_i=\left(\lambda _i, q_i, m_i\right)$                                                                                                                                                                                                           | $a_i = (\lambda_i, q_i, m_i)$                                                                                                                                           | <ul style="list-style-type: none"> <li>an <math>SU(2)</math> gauge field <math>Q = \sum_i q_i dq_i'</math>, with <math>q = f_0 + \sum_{\alpha} i f_{\alpha} \sigma^{\alpha}</math> and <math>Q =</math></li> </ul> |
| cardona-qft-methods_F02513 | 216  | ■ 0.749 | $a_i^{\prime}=\left(\lambda _i^{\prime}, q_i^{\prime}, m_i^{\prime}\right)$                                                                                                                                                                       | $a_i' = (\lambda_i', q_i', m_i')$                                                                                                                                       | <ul style="list-style-type: none"> <li>an <math>SU(2)</math> gauge field <math>Q = \sum_i q_i dq_i'</math>, with <math>q = f_0 + \sum_{\alpha} i f_{\alpha} \sigma^{\alpha}</math> and <math>Q =</math></li> </ul> |
| cardona-qft-methods_F02514 | 216  | ■ 0.749 | $\mathcal{A}=\mathcal{C}^{\infty}\left(M, \mathcal{A}_F\right)$                                                                                                                                                                                   | $\mathcal{A} = C^{\infty}(M, \mathcal{A}_F)$                                                                                                                            | <ul style="list-style-type: none"> <li>an <math>SU(2)</math> gauge field <math>Q = \sum_i q_i dq_i'</math>, with <math>q = f_0 + \sum_{\alpha} i f_{\alpha} \sigma^{\alpha}</math> and <math>Q =</math></li> </ul> |
| cardona-qft-methods_F02515 | 216  | ■ 1.000 | $\Lambda=\sum_i \lambda _i d \lambda _i^{\prime}=\sum_i \lambda _i\left[\partial _M \otimes 1, \lambda _i^{\prime}\right]$                                                                                                                        | $\Lambda = \sum_i \lambda_i d\lambda_i' = \sum_i \lambda_i [\partial_M \otimes 1, \lambda_i']$                                                                          | <ul style="list-style-type: none"> <li>an <math>SU(2)</math> gauge field <math>Q = \sum_i q_i dq_i'</math>, with <math>q = f_0 + \sum_{\alpha} i f_{\alpha} \sigma^{\alpha}</math> and <math>Q =</math></li> </ul> |
| cardona-qft-methods_F02516 | 216  | ■ 0.988 | $Q=\sum_i q_i d q_i^{\prime}$                                                                                                                                                                                                                     | $Q = \sum_i q_i dq_i'$                                                                                                                                                  | gets an $SU(3)$ -gauge field                                                                                                                                                                                       |
| cardona-qft-methods_F02517 | 216  | ■ 0.988 | $q=f_0+\sum_{\alpha} i f_{\alpha} \sigma^{\alpha}$                                                                                                                                                                                                | $q = f_0 + \sum_{\alpha} i f_{\alpha} \sigma^{\alpha}$                                                                                                                  | gets an $SU(3)$ -gauge field                                                                                                                                                                                       |
| cardona-qft-methods_F02518 | 216  | ■ 0.988 | $Q=$                                                                                                                                                                                                                                              | $Q =$                                                                                                                                                                   | gets an $SU(3)$ -gauge field                                                                                                                                                                                       |
| cardona-qft-methods_F02519 | 216  | ■ 0.966 | $\sum_{\alpha} f_{\alpha}\left[\partial _M \otimes 1, i f_{\alpha}^{\prime} \sigma^{\alpha}\right] ;$                                                                                                                                             | $\sum_{\alpha} f_{\alpha} [\partial_M \otimes 1, i f_{\alpha}' \sigma^{\alpha}];$                                                                                       |                                                                                                                                                                                                                    |
| cardona-qft-methods_F02520 | 216  | ■ 1.000 | $U(3)$                                                                                                                                                                                                                                            | $U(3)$                                                                                                                                                                  |                                                                                                                                                                                                                    |
| cardona-qft-methods_F02521 | 216  | ■ 1.000 | $V^{\prime}=\sum_i m_i d m_i^{\prime}=\sum_i m_i\left[\partial _M \otimes 1, m_i^{\prime}\right]$                                                                                                                                                 | $V' = \sum_i m_i dm_i' = \sum_i m_i [\partial_M \otimes 1, m_i']$                                                                                                       |                                                                                                                                                                                                                    |
| cardona-qft-methods_F02522 | 216  | ■ 0.806 | $V^{\prime}$                                                                                                                                                                                                                                      | $V'$                                                                                                                                                                    | As in the case of the fermions, the identification with the physical parti-                                                                                                                                        |
| cardona-qft-methods_F02523 | 216  | ■ 1.000 | $\operatorname{SU}\left(\mathcal{A}_F\right)$                                                                                                                                                                                                     | $SU(\mathcal{A}_F)$                                                                                                                                                     | cles is justified by the fact that they have the correct hypercharges for gauge                                                                                                                                    |
| cardona-qft-methods_F02524 | 216  | ■ 1.000 | $\operatorname{Tr}(A)=0$                                                                                                                                                                                                                          | $\operatorname{Tr}(A) = 0$                                                                                                                                              | cles is justified by the fact that they have the correct hypercharges for gauge                                                                                                                                    |
| cardona-qft-methods_F02525 | 216  | ■ 1.000 | $(1,0)$                                                                                                                                                                                                                                           | $(1, 0)$                                                                                                                                                                |                                                                                                                                                                                                                    |
| cardona-qft-methods_F02526 | 216  | ■ 1.000 | $A+J A J^{-1}$                                                                                                                                                                                                                                    | $A + J A J^{-1}$                                                                                                                                                        |                                                                                                                                                                                                                    |
| cardona-qft-methods_F02527 | 217  | ■ 1.000 | $A^{\{0,1\}}$                                                                                                                                                                                                                                     | $A^{(0,1)}$                                                                                                                                                             | The Higgs boson corresponds instead to the inner fluctuation from $A^{(0,1)}$ in the F-term equation. These are given by                                                                                           |
| cardona-qft-methods_F02528 | 217  | ■ 1.000 | $\varphi_1=\sum \lambda _i\left(\alpha _i^{\prime}-\lambda _i^{\prime}\right), \varphi_2=\sum \lambda _i \beta _i^{\prime} \varphi_1^{\prime}=\sum \alpha _i\left(\lambda _i^{\prime}-\alpha _i^{\prime}\right)+\beta _i \bar{\beta }_i^{\prime}$ | $\varphi_1 = \sum \lambda_i (\alpha_i' - \lambda_i'), \varphi_2 = \sum \lambda_i \beta_i' \varphi_1' = \sum \alpha_i (\lambda_i' - \alpha_i') + \beta_i \bar{\beta}_i'$ | Thus, this discrete part of the inner fluctuations gives rise to a quaternion                                                                                                                                      |
| cardona-qft-methods_F02529 | 217  | ■ 1.000 | $\varphi_2^{\prime}=\sum\left(-\alpha _i \beta _i^{\prime}+\beta _i\left(\bar{\lambda }_i^{\prime}-\bar{\alpha }_i^{\prime}\right)\right)$                                                                                                        | $\varphi_2' = \sum (-\alpha_i \beta_i' + \beta_i (\bar{\lambda}_i' - \bar{\alpha}_i'))$                                                                                 | valued function $H = \varphi_1 + \varphi_2 j$ or $\varphi = (\varphi_1, \varphi_2)$ , which is the Higgs field of                                                                                                  |
| cardona-qft-methods_F02530 | 217  | ■ 1.000 | $a_i(x)=\left(\lambda _i, q_i, m_i\right)$                                                                                                                                                                                                        | $a_i(x) = (\lambda_i, q_i, m_i)$                                                                                                                                        | valued function $H = \varphi_1 + \varphi_2 j$ or $\varphi = (\varphi_1, \varphi_2)$ , which is the Higgs field of                                                                                                  |
| cardona-qft-methods_F02531 | 217  | ■ 1.000 | $a_i^{\prime}(x)=\left(\lambda _i^{\prime}, q_i^{\prime}, m_i^{\prime}\right)$                                                                                                                                                                    | $a_i'(x) = (\lambda_i', q_i', m_i')$                                                                                                                                    | valued function $H = \varphi_1 + \varphi_2 j$ or $\varphi = (\varphi_1, \varphi_2)$ , which is the Higgs field of                                                                                                  |
| cardona-qft-methods_F02532 | 217  | ■ 1.000 | $H=\varphi_1+\varphi_2 j$                                                                                                                                                                                                                         | $H = \varphi_1 + \varphi_2 j$                                                                                                                                           | and                                                                                                                                                                                                                |
| cardona-qft-methods_F02533 | 217  | ■ 1.000 | $\varphi=\left(\varphi_1, \varphi_2\right)$                                                                                                                                                                                                       | $\varphi = (\varphi_1, \varphi_2)$                                                                                                                                      | and                                                                                                                                                                                                                |
| cardona-qft-methods_F02534 | 217  | ■ 0.998 | $D_A^2=\nabla^* \nabla-E$                                                                                                                                                                                                                         | $D_A^2 = \nabla^* \nabla - E$                                                                                                                                           |                                                                                                                                                                                                                    |

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| cardona-qft-methods_F02535 | 217  | ■ 0.998 | $\nabla^{*}\nabla$                                                        | $\nabla^{*}\nabla$                                                        |                                                                                                                                                                                 |
| cardona-qft-methods_F02536 | 217  | ■ 0.998 | $\nabla=\nabla^{s}+\mathbb{A}$                                            | $\nabla=\nabla^s+\mathbb{A}$                                              |                                                                                                                                                                                 |
| cardona-qft-methods_F02537 | 218  | ■ 0.996 | $s=-R$                                                                    | $s=-R$                                                                    |                                                                                                                                                                                 |
| cardona-qft-methods_F02538 | 218  | ■ 0.996 | $\mathbb{F}_{\mu\nu}$                                                     | $\mathbb{F}_{\mu\nu}$                                                     |                                                                                                                                                                                 |
| cardona-qft-methods_F02539 | 218  | ■ 0.996 | $\mathbb{A}$                                                              | $\mathbb{A}$                                                              |                                                                                                                                                                                 |
| cardona-qft-methods_F02540 | 218  | ■ 1.000 | $D\,/\,\Lambda$                                                           | $D/\Lambda$                                                               | usually taken to be a smooth approximation of a cutoff function.<br>In the case where the spectral triple has simple dimension spectrum,                                        |
| cardona-qft-methods_F02541 | 218  | ■ 1.000 | $f>0$                                                                     | $f>0$                                                                     | usually taken to be a smooth approximation of a cutoff function.<br>In the case where the spectral triple has simple dimension spectrum,                                        |
| cardona-qft-methods_F02542 | 218  | ■ 1.000 | $f_{\mathrm{k}}=\int_0^{\infty}f(v)v^{k-1}\,\mathrm{d}\,v$                | $f_k=\int_0^\infty f(v)v^{k-1}dv$                                         | $\mathrm{Tr}(e^{-t\Delta})\sim\sum a_\alpha t^\alpha\quad(t\rightarrow 0)$                                                                                                      |
| cardona-qft-methods_F02543 | 218  | ■ 0.999 | $\operatorname{DimSp}(\mathcal{A},\mathcal{H},D)$                         | $\operatorname{DimSp}(\mathcal{A},\mathcal{H},D)$                         | and its relation to the the $\zeta$ function                                                                                                                                    |
| cardona-qft-methods_F02544 | 218  | ■ 1.000 | $\zeta_{b,D}(s)=\operatorname{Tr}\big(b D ^{-s}\big)$                     | $\zeta_{b,D}(s)=\operatorname{Tr}\big(b D ^{-s}\big)$                     | $\zeta_{\mathfrak{A}}(s)=\operatorname{Tr}(\Lambda^{-s/2})$                                                                                                                     |
| cardona-qft-methods_F02545 | 218  | ■ 1.000 | $a_{\alpha}$                                                              | $a_\alpha$                                                                |                                                                                                                                                                                 |
| cardona-qft-methods_F02546 | 218  | ■ 1.000 | $\alpha<0$                                                                | $\alpha<0$                                                                |                                                                                                                                                                                 |
| cardona-qft-methods_F02547 | 218  | ■ 1.000 | $\zeta_D$                                                                 | $\zeta_D$                                                                 |                                                                                                                                                                                 |
| cardona-qft-methods_F02548 | 219  | ■ 1.000 | $\int_0^1t^{\alpha+s/2-1}dt=(\alpha+s/2)^{-1}$                            | $\int_0^1t^{\alpha+s/2-1}dt=(\alpha+s/2)^{-1}$                            |                                                                                                                                                                                 |
| cardona-qft-methods_F02549 | 219  | ■ 1.000 | $\zeta_D(0)=a_0$                                                          | $\zeta_D(0)=a_0$                                                          | 208 M. Marzulli                                                                                                                                                                 |
| cardona-qft-methods_F02550 | 219  | ■ 1.000 | $a_0\int_0^1t^{s/2-1}dt=a_0\frac{2}{s}$                                   | $a_0\int_0^1t^{s/2-1}dt=a_0\frac{2}{s}$                                   | is given by $a_0\int_0^1t^{s/2-1}dt=a_0\frac{2}{s}$                                                                                                                             |
| cardona-qft-methods_F02551 | 219  | ■ 1.000 | $\mathcal{H}_{cl}^+$                                                      | $\mathcal{H}_{cl}^+$                                                      | is an antisymmetric form $\mathfrak{A}(\tilde{\xi})$ on                                                                                                                         |
| cardona-qft-methods_F02552 | 219  | ■ 0.999 | $\mathfrak{A}(\tilde{\xi})$                                               | $\mathfrak{A}(\tilde{\xi})$                                               | symmetric $\langle \xi,D_A\xi\rangle$ for a finite n                                                                                                                            |
| cardona-qft-methods_F02553 | 219  | ■ 1.000 | $\langle \xi,D_A\xi\rangle$                                               | $\langle \xi,D_A\xi\rangle$                                               |                                                                                                                                                                                 |
| cardona-qft-methods_F02554 | 220  | ■ 1.000 | $f_0,f_2,f_4$                                                             | $f_0,f_2,f_4$                                                             |                                                                                                                                                                                 |
| cardona-qft-methods_F02555 | 220  | ■ 1.000 | $f_0=f(0)$                                                                | $f_0=f(0)$                                                                |                                                                                                                                                                                 |
| cardona-qft-methods_F02556 | 220  | ■ 1.000 | $k>0$                                                                     | $k>0$                                                                     |                                                                                                                                                                                 |
| cardona-qft-methods_F02557 | 220  | ■ 0.997 | $\mathfrak{a},\mathfrak{b},\mathfrak{c},\mathfrak{d},\mathfrak{e}$        | $\mathfrak{a},\mathfrak{b},\mathfrak{c},\mathfrak{d},\mathfrak{e}$        |                                                                                                                                                                                 |
| cardona-qft-methods_F02558 | 221  | ■ 1.000 | $P=-\left(g^{\mu\nu}I\partial_\mu\partial_\nu+A^\mu\partial_\mu+B\right)$ | $P=-\left(g^{\mu\nu}I\partial_\mu\partial_\nu+A^\mu\partial_\mu+B\right)$ | theorem for a differential operator $P=-(g^{\mu\nu}I\partial_\mu\partial_\nu+A^\mu\partial_\mu+B)$ with $A,B$ bundle endomorphisms and $m=\dim M$ (see a detailed account in §4 |
| cardona-qft-methods_F02559 | 221  | ■ 1.000 | $m=\operatorname{dim} M$                                                  | $m=\dim M$                                                                | $D$ bundle endomorphisms and $m=\dim M$ (see a detailed account in §4 of <sup>24</sup> ). One has                                                                               |
| cardona-qft-methods_F02560 | 221  | ■ 1.000 | $P=\nabla^{*}\nabla-E$                                                    | $P=\nabla^{*}\nabla-E$                                                    | $\mathfrak{P}=\mathfrak{D}+\mathfrak{A}\mathfrak{D}+\mathfrak{D}\mathfrak{A}+\mathfrak{B}+\mathfrak{C}+\mathfrak{D}+\mathfrak{E}$                                               |

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| cardona-qft-methods_F02561 | 221  | ■ 1.000 | $E_{;\mu}{}^\mu := \nabla_{\mu} \nabla^{\mu} E$                                                                                                  | $E_{;\mu}{}^\mu := \nabla_{\mu} \nabla^{\mu} E$                                                                                          |                                                                                                                                                  |
| cardona-qft-methods_F02562 | 221  | ■ 0.712 | $H=\frac{\sqrt{a}f_0}{\pi}\varphi$                                                                                                               | $H = \frac{\sqrt{a}f_0}{\pi}\varphi$                                                                                                     |                                                                                                                                                  |
| cardona-qft-methods_F02563 | 222  | ■ 1.000 | $R^{[*]}R^{[*]}=\frac{1}{4}\,\epsilon^{\mu\nu\rho\sigma}\epsilon_{\alpha\beta\gamma\delta}R_{\mu\nu}^{\alpha\beta}R_{\rho\sigma}^{\gamma\delta}$ | $R^*R^* = \frac{1}{4}\epsilon^{\mu\nu\rho\sigma}\epsilon_{\alpha\beta\gamma\delta}R_{\mu\nu}^{\alpha\beta}R_{\rho\sigma}^{\gamma\delta}$ | $\chi(M)$ and $C^{\mu\nu\rho\sigma}$ is the Weyl curvature tensor. The coefficients are given by:                                                                                                                                   |
| cardona-qft-methods_F02564 | 222  | ■ 1.000 | $\chi(M)$                                                                                                                                        | $\chi(M)$                                                                                                                                |                                                                                                                                                  |
| cardona-qft-methods_F02565 | 222  | ■ 1.000 | $C^{\mu\nu\rho\sigma}$                                                                                                                           | $C^{\mu\nu\rho\sigma}$                                                                                                                   |                                                                                                                                                  |
| cardona-qft-methods_F02566 | 222  | ■ 1.000 | $10^{15}$                                                                                                                                        | $10^{15}$                                                                                                                                | on parameters $a, b, c, d, e$ , which in turn are functions of the Yukawa parameters. The latter run with the renormalization group flow according to                                                                               |
| cardona-qft-methods_F02567 | 222  | ■ 1.000 | $10^{17}\,\mathrm{GeV}$                                                                                                                          | $10^{17}\mathrm{GeV}$                                                                                                                    | on parameters $a, b, c, d, e$ , which in turn are functions of the Yukawa parameters. The latter run with the renormalization group flow according to                                                                               |
| cardona-qft-methods_F02568 | 222  | ■ 1.000 | $f_{\scriptscriptstyle 0}$                                                                                                                       | $f_0$                                                                                                                                    | consistency also serves as a guide for the renormalization group flow, according to the renormalization group equations (RGE). Moreover, the fact that the                                                                          |
| cardona-qft-methods_F02569 | 223  | ■ 1.000 | $\{ \ }^{\scriptscriptstyle 17,29,30,33}$                                                                                                        | 17,29,30,33                                                                                                                              | conditions for the renormalization group flow should be compatible with observed physics at low energies.                                                                                                                           |
| cardona-qft-methods_F02570 | 223  | ■ 0.983 | $\nu\,M\,S\,M$                                                                                                                                   | $\nu MSM$                                                                                                                                |                                                                                                                                                  |
| cardona-qft-methods_F02571 | 223  | ■ 0.999 | $\Lambda\frac{df}{d\Lambda}=\beta_f(\Lambda)$                                                                                                    | $\Lambda\frac{df}{d\Lambda} = \beta_f(\Lambda)$                                                                                          | $16\pi^2\,\beta_{Y_u}=Y_u(\frac{3}{2}Y_u^\dagger Y_u-\frac{3}{2}Y_d^\dagger Y_d+a-\frac{17}{22}g_1^2-\frac{9}{2}g_2^2-8g_3^2)$                                                                                                      |
| cardona-qft-methods_F02572 | 224  | ■ 0.978 | $\Lambda_{\text{unif}}$                                                                                                                          | $\Lambda_{\text{unif}}$                                                                                                                  | consists of the following steps:                                                                                                                                                                                                    |
| cardona-qft-methods_F02573 | 224  | ■ 0.822 | $Y_{\nu}^{\scriptscriptstyle (3)}$                                                                                                               | $Y_{\nu}^{(3)}$                                                                                                                          | <ul style="list-style-type: none"> <li>• Iteration ends the comparison with the eigenvalue of the matrix.</li> <li>• Pass to an effective field theory with <math>Y_{\nu}^{(3)}</math> obtained by removing the last</li> </ul>     |
| cardona-qft-methods_F02574 | 224  | ■ 1.000 | $Y_{\nu}$                                                                                                                                        | $Y_{\nu}$                                                                                                                                | row of $Y_{\nu}$ in a basis in which $M$ is diagonal and with $M^{(3)}$ obtained by removing the last row and columns                                                                                                               |
| cardona-qft-methods_F02575 | 224  | ■ 1.000 | $M^{\scriptscriptstyle (3)}$                                                                                                                     | $M^{(3)}$                                                                                                                                | row of $Y_{\nu}$ in a basis in which $M$ is diagonal and with $M^{(3)}$ obtained by removing the last row and columns                                                                                                               |
| cardona-qft-methods_F02576 | 224  | ■ 0.997 | $Y_{\nu}^{\scriptscriptstyle (2)}$                                                                                                               | $Y_{\nu}^{(2)}$                                                                                                                          | <ul style="list-style-type: none"> <li>• Run the induced RGE flow down to first see-saw scale.</li> <li>• Introduce a new effective field theory with <math>Y_{\nu}^{(1)}</math> and <math>M^{(1)}</math> again obtained</li> </ul> |
| cardona-qft-methods_F02577 | 224  | ■ 0.997 | $M^{\scriptscriptstyle (2)}$                                                                                                                     | $M^{(2)}$                                                                                                                                | <ul style="list-style-type: none"> <li>• Run the induced RGE flow down to first see-saw scale.</li> <li>• Introduce a new effective field theory with <math>Y_{\nu}^{(1)}</math> and <math>M^{(1)}</math> again obtained</li> </ul> |
| cardona-qft-methods_F02578 | 224  | ■ 1.000 | $Y_{\nu}^{\scriptscriptstyle (1)}$                                                                                                               | $Y_{\nu}^{(1)}$                                                                                                                          | Using the effective field theories $Y_{\nu}^{(N)}$ and $M^{(N)}$ between see-saw scales                                                                                                                                             |
| cardona-qft-methods_F02579 | 224  | ■ 1.000 | $M^{\scriptscriptstyle (1)}$                                                                                                                     | $M^{(1)}$                                                                                                                                | Using the effective field theories $Y_{\nu}^{(N)}$ and $M^{(N)}$ between see-saw scales                                                                                                                                             |
| cardona-qft-methods_F02580 | 224  | ■ 1.000 | $\Lambda_{ew}$                                                                                                                                   | $\Lambda_{ew}$                                                                                                                           | see Section 2.4.2. Since we are going to discuss the renormalization of $a, b, c, d, e$ according to this flow was analyzed in, <sup>33</sup> in terms of possible                                                                  |
| cardona-qft-methods_F02581 | 224  | ■ 1.000 | $Y_{\nu}^{\scriptscriptstyle (N)}$                                                                                                               | $Y_{\nu}^{(N)}$                                                                                                                          | ncation and the electroweak epoch in the cosmological timeline. We will discuss here a simplified model, where one only keeps some dominant terms                                                                                   |
| cardona-qft-methods_F02582 | 224  | ■ 1.000 | $M^{\scriptscriptstyle (N)}$                                                                                                                     | $M^{(N)}$                                                                                                                                | ncation and the electroweak epoch in the cosmological timeline. We will discuss here a simplified model, where one only keeps some dominant terms                                                                                   |
| cardona-qft-methods_F02583 | 225  | ■ 1.000 | $t=\log\left(\Lambda/M_Z\right)$                                                                                                                 | $t = \log\left(\Lambda/M_Z\right)$                                                                                                       | of the form                                                                                                                                                                                                                         |
| cardona-qft-methods_F02584 | 225  | ■ 1.000 | $\Lambda=M_Z$                                                                                                                                    | $\Lambda = M_Z$                                                                                                                          | These equations can be solved exactly. The values of the coupling constants at $\Lambda = M_Z$ , determined in agreement with experiments, are                                                                                      |
| cardona-qft-methods_F02585 | 225  | ■ 1.000 | $g_{\scriptscriptstyle 3}^2=g_{\scriptscriptstyle 2}^2=5\,g_{\scriptscriptstyle 1}^2\,/\,3$                                                      | $g_3^2 = g_2^2 = 5g_1^2/3$                                                                                                               | the existence of new physics beyond the Standard Model. It is known, for instance, that in supersymmetric extensions of the Standard Model the                                                                                      |

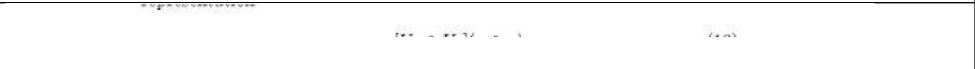
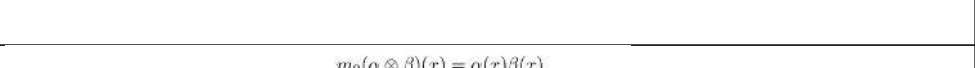
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| cardona-qft-methods_F02586 | 226  | ■ 0.994 | $\mathfrak{c}$                                                                                                                                             | $\mathfrak{c}$                                                                            | <ul style="list-style-type: none"> <li>Mass relation at unification</li> </ul>                                                                                                                                               |
| cardona-qft-methods_F02587 | 227  | ■ 1.000 | $g_{\{1\}}, g_{\{2\}}, g_{\{3\}}$                                                                                                                          | $g_1, g_2, g_3$                                                                           |                                                                                                                                                                                                                              |
| cardona-qft-methods_F02588 | 228  | ■ 1.000 | $\tilde{\lambda}$                                                                                                                                          | $\tilde{\lambda}$                                                                         | and uses the relation at unification, where $\tilde{\lambda}$ is the $ \mathbf{H} ^4$ coupling,                                                                                                                              |
| cardona-qft-methods_F02589 | 228  | ■ 1.000 | $ \mathbf{H} ^4$                                                                                                                                           | $ \mathbf{H} ^4$                                                                          | and uses the relation at unification, where $\tilde{\lambda}$ is the $ \mathbf{H} ^4$ coupling,                                                                                                                              |
| cardona-qft-methods_F02590 | 228  | ■ 0.585 | $\lambda\left(M_Z\right) \sim 0.241$                                                                                                                       | $\lambda\left(M_Z\right) \sim 0.241$                                                      | needed, in order to obtain compatibility with the experimental Higgs mass.<br>(Please use at present two different methods of computing the NLO model)                                                                       |
| cardona-qft-methods_F02591 | 228  | ■ 0.585 | $\sim 170 \mathrm{GeV}$                                                                                                                                    | $\sim 170 \mathrm{GeV}$                                                                   | needed, in order to obtain compatibility with the experimental Higgs mass.<br>(Please use at present two different methods of computing the NLO model)                                                                       |
| cardona-qft-methods_F02592 | 229  | ■ 1.000 | $\kappa_0, \gamma_0, \alpha_0, \tau_0, \mu_0, \xi_0, \lambda_0, \xi_{\mathrm{i}_0}, \lambda_{\mathrm{b}_0}$                                                | $\kappa_0, \gamma_0, \alpha_0, \tau_0, \mu_0, \xi_0, \lambda_0$                           | otic expansion of the spectral action.<br>The method followed in [17] assumes that the dimension of the space                                                                                                                |
| cardona-qft-methods_F02593 | 229  | ■ 0.609 | $\{ \}^{3,18,27}$                                                                                                                                          | 3,18,27                                                                                   | $J \setminus 2\eta \rightarrow \eta \rightarrow 3\eta \rightarrow \eta \rightarrow J$ and the commutative relations of the form                                                                                              |
| cardona-qft-methods_F02594 | 230  | ■ 1.000 | $10^{-12}$                                                                                                                                                 | $10^{-12}$                                                                                | <i>Noncommutative geometry models for particle physics</i> 219                                                                                                                                                               |
| cardona-qft-methods_F02595 | 230  | ■ 1.000 | $\lambda_{\mathrm{b}_0}=\lambda \cdot\left(\pi^2 \mathfrak{b}\right) /\left(f_0 \mathfrak{a}^2\right) \wedge \left(f_{\mathrm{b}_0} \mathfrak{a}^2\right)$ | $\lambda_0=\lambda \cdot\left(\pi^2 \mathfrak{b}\right) /\left(f_0 \mathfrak{a}^2\right)$ | these couplings have changed with respect to the tree couplings of [10] since instead of applying the relation $\lambda_0=\lambda \cdot\left(\pi^2 \mathfrak{b}\right) /\left(f_0 \mathfrak{a}^2\right)$ only at the initial |
| cardona-qft-methods_F02596 | 230  | ■ 0.992 | $x_{\{j\}}$                                                                                                                                                | $x_j$                                                                                     | $M_P^2(\Lambda)=M_P^2+2 \rho_0 \Lambda^2, \quad (7)$                                                                                                                                                                         |
| cardona-qft-methods_F02597 | 230  | ■ 0.992 | $\beta_{\mathrm{j}}^{\mathrm{SM}}$                                                                                                                         | $\beta_j^{\mathrm{SM}}$                                                                   | $M_P^2(\Lambda)=M_P^2+2 \rho_0 \Lambda^2, \quad (7)$                                                                                                                                                                         |
| cardona-qft-methods_F02598 | 230  | ■ 1.000 | $\beta_{\mathrm{j}}^{\mathrm{grav}}$                                                                                                                       | $\beta_j^{\mathrm{grav}}$                                                                 | with $\rho_0 \sim 0.024$ . The anomalous dimensions are usually taken to be $a_1=a_2=a_3=a_{\mathrm{a}}$ , with $ a_{\mathrm{a}}  \sim 1$ and a negative sign for the coupling constants                                     |
| cardona-qft-methods_F02599 | 230  | ■ 1.000 | $\rho_0 \sim 0.024$                                                                                                                                        | $\rho_0 \sim 0.024$                                                                       |                                                                                                                                                                                                                              |
| cardona-qft-methods_F02600 | 230  | ■ 1.000 | $\mathfrak{a}_{\{1\}}=$                                                                                                                                    | $a_1=$                                                                                    |                                                                                                                                                                                                                              |
| cardona-qft-methods_F02601 | 230  | ■ 0.993 | $\mathfrak{a}_{\{2\}}=\mathfrak{a}_{\{3\}}=\mathfrak{a}_{\mathrm{g}}$                                                                                      | $a_2=a_3=a_g$                                                                             |                                                                                                                                                                                                                              |
| cardona-qft-methods_F02602 | 230  | ■ 0.993 | $\left \mathfrak{a}_{\mathrm{g}}\right  \sim 1$                                                                                                            | $ a_g  \sim 1$                                                                            |                                                                                                                                                                                                                              |
| cardona-qft-methods_F02603 | 230  | ■ 0.473 | $\mathfrak{a}_{\mathrm{g}} \leq \mathfrak{a}_{\mathrm{y}} < 0$                                                                                             | $a_g \leq a_y < 0$                                                                        |                                                                                                                                                                                                                              |
| cardona-qft-methods_F02604 | 230  | ■ 0.473 | $\mathfrak{a}_{\lambda}$                                                                                                                                   | $a_\lambda$                                                                               |                                                                                                                                                                                                                              |
| cardona-qft-methods_F02605 | 230  | ■ 0.473 | $\left(\right.$                                                                                                                                            | $($                                                                                       |                                                                                                                                                                                                                              |
| cardona-qft-methods_F02606 | 230  | ■ 0.473 | $\left.^{\mathrm{45}}\right)$                                                                                                                              | $\left.^{45}\right)$                                                                      |                                                                                                                                                                                                                              |
| cardona-qft-methods_F02607 | 231  | ■ 1.000 | $m_{\mathrm{H}} \sim 125.4$                                                                                                                                | $m_H \sim 125.4$                                                                          | that it admits solutions that give a realistic Higgs mass $m_H \sim 125.4$ , com-                                                                                                                                            |
| cardona-qft-methods_F02608 | 231  | ■ 0.999 | $m_{\mathrm{t}} \sim 171.3 \mathrm{GeV}$                                                                                                                   | $m_t \sim 171.3 \mathrm{GeV}$                                                             | patible with a realistic top quark mass of $m_t \sim 171.3 \mathrm{GeV}$ .                                                                                                                                                   |
| cardona-qft-methods_F02609 | 231  | ■ 1.000 | $\{ \}^{42-44}$                                                                                                                                            | 42–44                                                                                     | As a final consideration, while the construction of particle physics models via noncommutative geometry has had so far as main drawback reducing                                                                             |
| cardona-qft-methods_F02610 | 236  | ■ 1.000 | $10^{-32}$                                                                                                                                                 | $10^{-32}$                                                                                | <i>Noncommutative spacetimes and quantum physics</i> 225                                                                                                                                                                     |
| cardona-qft-methods_F02611 | 236  | ■ 1.000 | $10^9$                                                                                                                                                     | $10^9$                                                                                    | <i>Noncommutative spacetimes and quantum physics</i> 225                                                                                                                                                                     |

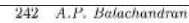
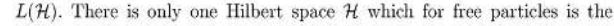
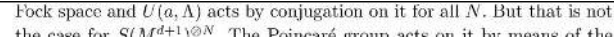
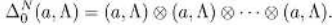
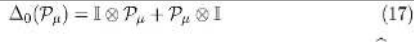
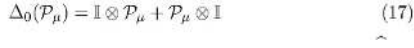
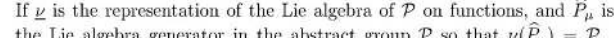
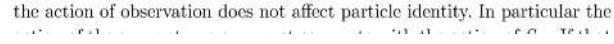
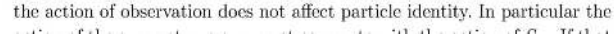
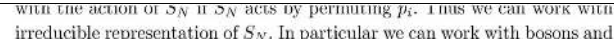
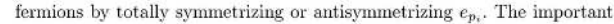
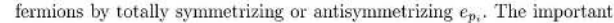
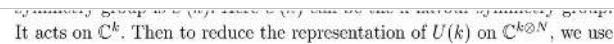
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| cardona-qft-methods_F02612 | 236  | ■ 1.000 | $L \simeq 10^{-32} \mathrm{~cm}$                                                                                | $L \simeq 10^{-32} \mathrm{~cm}$                                                 | must therefore fulfill                                                                                                                                                           |
| cardona-qft-methods_F02613 | 236  | ■ 1.000 | $\lambda_C$                                                                                                     | $\lambda_C$                                                                      | must therefore fulfill                                                                                                                                                           |
| cardona-qft-methods_F02614 | 236  | ■ 1.000 | $L^3 \simeq \left(10^{-32} \mathrm{~cm}\right)^3$                                                               | $L^3 \simeq \left(10^{-32} \mathrm{~cm}\right)^3$                                |                                                                                                                                                                                  |
| cardona-qft-methods_F02615 | 236  | ■ 1.000 | $\Delta x$                                                                                                      | $\Delta x$                                                                       | where $\theta$ is a new Planck-like fundamental constant with dimensions of                                                                                                      |
| cardona-qft-methods_F02616 | 236  | ■ 1.000 | $\Delta p$                                                                                                      | $\Delta p$                                                                       | where $\theta$ is a new Planck-like fundamental constant with dimensions of                                                                                                      |
| cardona-qft-methods_F02617 | 236  | ■ 0.994 | $x \, p - p \, x = i \hbar$                                                                                     | $xp - px = i\hbar$                                                               | Now, from (3) we obtain                                                                                                                                                          |
| cardona-qft-methods_F02618 | 236  | ■ 1.000 | $\Delta y$                                                                                                      | $\Delta y$                                                                       |                                                                                                                                                                                  |
| cardona-qft-methods_F02619 | 237  | ■ 0.640 | $G-M$                                                                                                           | $G - M$                                                                          | 226 A.P. Balachandran                                                                                                                                                            |
| cardona-qft-methods_F02620 | 237  | ■ 0.640 | $\mathcal{A}_\theta$                                                                                            | $\mathcal{A}_\theta$                                                             | 226 A.P. Balachandran                                                                                                                                                            |
| cardona-qft-methods_F02621 | 237  | ■ 0.640 | $\mathbb{R}^{d+1}$                                                                                              | $\mathbb{R}^{d+1}$                                                               | 226 A.P. Balachandran                                                                                                                                                            |
| cardona-qft-methods_F02622 | 238  | ■ 1.000 | $\mathbb{C}G$                                                                                                   | $\mathbb{C}G$                                                                    | principle. We then formulate them in terms of $\mathbb{C}G$ , the group algebra of $G$ . Now $\mathbb{C}G$ is a Hopf algebra of a special kind. Besides the multiplication $\mu$ |
| cardona-qft-methods_F02623 | 238  | ■ 1.000 | $U(g) \, \Psi$                                                                                                  | $U(g)\Psi$                                                                       | with this role in fundamental theories of nature. Although gauge groups                                                                                                          |
| cardona-qft-methods_F02624 | 239  | ■ 1.000 | $\mathcal{H}^{(1)}$                                                                                             | $\mathcal{H}^{(1)}$                                                              | consider appropriate symmetrisation postulates. The symmetry group $G$                                                                                                           |
| cardona-qft-methods_F02625 | 239  | ■ 1.000 | $ \theta\rangle$                                                                                                | $ 0\rangle$                                                                      | $U(a) = \bigoplus U^{(N)}(a). \quad (1)$                                                                                                                                         |
| cardona-qft-methods_F02626 | 239  | ■ 1.000 | $\mathcal{H}^{(N)} = \mathcal{H}^{(1)} \otimes \cdots \otimes \mathcal{H}^{(1)}$<br>$\otimes \mathcal{H}^{(1)}$ | $\mathcal{H}^{(N)} = \mathcal{H}^{(1)} \otimes \cdots \otimes \mathcal{H}^{(1)}$ |                                                                                                                                                                                  |
| cardona-qft-methods_F02627 | 239  | ■ 1.000 | $\mathcal{F} = \bigoplus_N \mathcal{H}^{(N)}$                                                                   | $\mathcal{F} = \bigoplus_N \mathcal{H}^{(N)}$                                    | that is,                                                                                                                                                                         |
| cardona-qft-methods_F02628 | 239  | ■ 1.000 | $U^{(N)}$                                                                                                       | $U^{(N)}$                                                                        |                                                                                                                                                                                  |
| cardona-qft-methods_F02629 | 239  | ■ 1.000 | $\mathcal{H}^{(N)}$                                                                                             | $\mathcal{H}^{(N)}$                                                              |                                                                                                                                                                                  |
| cardona-qft-methods_F02630 | 239  | ■ 1.000 | $U^{(1)}$                                                                                                       | $U^{(1)}$                                                                        |                                                                                                                                                                                  |
| cardona-qft-methods_F02631 | 240  | ■ 1.000 | $\alpha^*$                                                                                                      | $\alpha^*$                                                                       | with $\alpha$ a function on $G$ . The product of two elements $\alpha^*$ and $\beta^*$ is defined                                                                                |
| cardona-qft-methods_F02632 | 240  | ■ 1.000 | $\beta^*$                                                                                                       | $\beta^*$                                                                        | with $\alpha$ a function on $G$ . The product of two elements $\alpha^*$ and $\beta^*$ is defined                                                                                |
| cardona-qft-methods_F02633 | 240  | ■ 1.000 | $\alpha * \beta$                                                                                                | $\alpha * \beta$                                                                 | $(\alpha * \beta)(g) = \sum_{h, h^{-1}} \alpha(gh^{-1})\beta(h)$                                                                                                                 |
| cardona-qft-methods_F02634 | 240  | ■ 1.000 | $\{ \}^{7,8}$                                                                                                   | $7,8$                                                                            | $U(\alpha^*) = \int d\mu(g) \alpha(g) U(g). \quad (7)$                                                                                                                           |
| cardona-qft-methods_F02635 | 240  | ■ 1.000 | $\mathcal{F}$                                                                                                   | $\mathcal{F}$                                                                    | only if we have a homomorphism $\mathcal{L}$                                                                                                                                     |
| cardona-qft-methods_F02636 | 240  | ■ 1.000 | $U^{(N)}(N \geq 2)$                                                                                             | $U^{(N)}(N \geq 2)$                                                              |                                                                                                                                                                                  |
| cardona-qft-methods_F02637 | 240  | ■ 1.000 | $\mathbb{C}G \otimes \mathbb{C}G$                                                                               | $\mathbb{C}G \otimes \mathbb{C}G$                                                |                                                                                                                                                                                  |
| cardona-qft-methods_F02638 | 240  | ■ 1.000 | $U^{(0)}$                                                                                                       | $U^{(0)}$                                                                        |                                                                                                                                                                                  |

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| cardona-qft-methods_F02639 | 240  | ■ 1.000 | $U^{\{(\emptyset)\}}(g) \emptyset\rangle= \emptyset\rangle$                        | $U^{(0)}(g) 0\rangle= 0\rangle$                                                    |                                                                                                                                                                                        |
| cardona-qft-methods_F02640 | 241  | ■ 1.000 | $D(g)$                                                                             | $D(g)$                                                                             | We can write it as                                                                                                                                                                     |
| cardona-qft-methods_F02641 | 241  | ■ 1.000 | $U^{\{(1)\}}(g)$                                                                   | $U^{(1)}(g)$                                                                       | We can write it as                                                                                                                                                                     |
| cardona-qft-methods_F02642 | 241  | ■ 1.000 | $U^{\{(2)\}}$                                                                      | $U^{(2)}$                                                                          | $U^{(2)}(g) \rho\rangle= \sigma\rangle U_{\sigma}$                                                                                                                                     |
| cardona-qft-methods_F02643 | 241  | ■ 1.000 | $\Delta: G \rightarrow \mathbb{C}G \otimes \mathbb{C}G$                            | $\Delta: G \rightarrow \mathbb{C}G \otimes \mathbb{C}G$                            | (iii) The two-particle representation $U^{(2)}$ :<br>Group theory does not fix the action of $G$ on two-particle states. We                                                            |
| cardona-qft-methods_F02644 | 241  | ■ 1.000 | $U^{\{(2)\}}(g)=U^{\{(1)\}} \otimes U^{\{(1)\}} \Delta(g)$                         | $U^{(2)}(g) = U^{(1)} \otimes U^{(1)} \Delta(g)$                                   | need a homomorphism $\Delta: G \rightarrow \mathbb{C}G \otimes \mathbb{C}G$ to fix $U^{(2)}$ . Given $\Delta$ , we                                                                     |
| cardona-qft-methods_F02645 | 241  | ■ 1.000 | $\Delta: \mathbb{C}G \rightarrow \mathbb{C}G \otimes \mathbb{C}G$                  | $\Delta: \mathbb{C}G \rightarrow \mathbb{C}G \otimes \mathbb{C}G$                  | (iv) <i>*-particle representation</i> $U^{(2)}$ :<br>Once again we are given a homomorphism $\Delta$ . We have seen how $U^{(2)}$                                                      |
| cardona-qft-methods_F02646 | 241  | ■ 1.000 | $\Delta(g)=g \otimes g$                                                            | $\Delta(g) = g \otimes g$                                                          | The importance of this property is that it provides a way to induct<br>obtain a unique homomorphism                                                                                    |
| cardona-qft-methods_F02647 | 242  | ■ 1.000 | $\mathcal{E}_3$                                                                    | $\mathcal{E}_3$                                                                    | independently of the splitting we choose.                                                                                                                                              |
| cardona-qft-methods_F02648 | 242  | ■ 1.000 | $g \in \mathcal{E}_3$                                                              | $g \in \mathcal{E}_3$                                                              | <b>Example 2.1.</b> Consider the underlying $\mathfrak{g}$                                                                                                                             |
| cardona-qft-methods_F02649 | 242  | ■ 1.000 | $\theta_{\mu\nu}$                                                                  | $\theta_{\mu\nu}$                                                                  | $\theta_{\mu\nu}$ being a real antisymmetric matrix and the $T_\nu$ being the translation<br>generators of $\mathcal{E}_3$ . Notice that the right hand side of (15) is not an element |
| cardona-qft-methods_F02650 | 242  | ■ 1.000 | $P_\nu$                                                                            | $P_\nu$                                                                            | $\theta_{\mu\nu}$ being a real antisymmetric matrix and the $T_\nu$ being the translation<br>generators of $\mathcal{E}_3$ . Notice that the right hand side of (15) is not an element |
| cardona-qft-methods_F02651 | 242  | ■ 1.000 | $G \otimes G$                                                                      | $G \otimes G$                                                                      | eralized if we replace the Euclidean group by the Poincare group $\mathcal{P}_+$ , thus<br>considering as examples of particular interest in quantum field theory its rep-             |
| cardona-qft-methods_F02652 | 242  | ■ 1.000 | $\mathcal{P}_+^\uparrow$                                                           | $\mathcal{P}_+^\uparrow$                                                           | providing us examples of particular interest in quantum field theory as par-<br>ticular for investigations of quantum physics at the Planck scale, where the                           |
| cardona-qft-methods_F02653 | 242  | ■ 1.000 | $\Delta_\theta$                                                                    | $\Delta_\theta$                                                                    | The group algebra $\mathbb{C}G$ has further important properties besides the copro-<br>duct. They are significant for physics. Let us now discuss them.                                |
| cardona-qft-methods_F02654 | 243  | ■ 1.000 | $g^{-1}$                                                                           | $g^{-1}$                                                                           |                                                                                                                                                                                        |
| cardona-qft-methods_F02655 | 243  | ■ 1.000 | $g \rightarrow g^{-1}$                                                             | $g \rightarrow g^{-1}$                                                             | 999 A D Dabachandani                                                                                                                                                                   |
| cardona-qft-methods_F02656 | 243  | ■ 1.000 | $(S(g\,h)=S(h)\,S(g))$                                                             | $(S(gh) = S(h)S(g))$                                                               | 999 A D Dabachandani                                                                                                                                                                   |
| cardona-qft-methods_F02657 | 243  | ■ 0.954 | $U\left(g^{-1}\right)=U(g)^{*}$                                                    | $U\left(g^{-1}\right)=U(g)^{*}$                                                    | The adjoint representation of the representation $U$ is characterized by the<br>condition $U(\hat{\alpha}^*)=U(\hat{\alpha})^*$ . $U$ is thus a “*-representation” of $\mathbb{C}G$ .  |
| cardona-qft-methods_F02658 | 243  | ■ 0.970 | $U\left(\hat{\alpha}^*\right)=U\left(\hat{\alpha}\right)^{*} . U$                  | $U\left(\hat{\alpha}^*\right)=U\left(\hat{\alpha}\right)^{*}.U$                    | <b>Definition 2.1.</b> A *- Hopf algebra $H$ is an associative, *-algebra with unit<br>element $e$ . This means that                                                                   |
| cardona-qft-methods_F02659 | 243  | ■ 1.000 | $(\alpha\,\beta)\,\gamma=\alpha(\beta\,\gamma)$                                    | $(\alpha\beta)\gamma = \alpha(\beta\gamma)$                                        |                                                                                                                                                                                        |
| cardona-qft-methods_F02660 | 243  | ■ 1.000 | $\alpha,\beta,\gamma\in H$                                                         | $\alpha,\beta,\gamma \in H$                                                        |                                                                                                                                                                                        |
| cardona-qft-methods_F02661 | 244  | ■ 1.000 | $*:\alpha\in H\rightarrow\alpha^*\in H$                                            | $*:\alpha \in H \rightarrow \alpha^* \in H$                                        |                                                                                                                                                                                        |
| cardona-qft-methods_F02662 | 244  | ■ 0.953 | $(\alpha\,\beta)^{*}=\beta^{*}\alpha^{*},\alpha,\beta\in H$                        | $(\alpha\beta)^{*}=\beta^{*}\alpha^{*},\alpha,\beta \in H$                         |                                                                                                                                                                                        |
| cardona-qft-methods_F02663 | 244  | ■ 0.992 | $(\lambda\alpha)^{*}=\bar{\lambda}\alpha^{*}\quad\lambda\in\mathbb{C},\alpha\in H$ | $(\lambda\alpha)^{*}=\bar{\lambda}\alpha^{*}\quad\lambda\in\mathbb{C},\alpha\in H$ |                                                                                                                                                                                        |
| cardona-qft-methods_F02664 | 244  | ■ 0.719 | $e^{*}=e,\quad\alpha e=e\alpha=\alpha,\forall\alpha\in H$                          | $e^{*}=e,\quad\alpha e=e\alpha=\alpha,\forall\alpha\in H$                          | $(\lambda\alpha)^{*}=\bar{\lambda}\alpha^{*}\quad\lambda\in\mathbb{C},\alpha\in H,$                                                                                                    |

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| cardona-qft-methods_F02665 | 244  | 0.672 | $\backslash epsilon: H \rightarrow \mathbb{C}$                                                                                                                                          | $\epsilon : H \rightarrow \mathbb{C}$                                                                   | $(\text{Id} \otimes \epsilon) \Delta = \text{Id} = (\epsilon \otimes \text{Id}) \Delta. \quad (17)$                                                                                                   |
| cardona-qft-methods_F02666 | 244  | 0.683 | $S: H \rightarrow H$                                                                                                                                                                    | $S : H \rightarrow H$                                                                                   | $m_r(\xi \otimes \eta \otimes \rho) = \eta \xi \otimes \rho$                                                                                                                                          |
| cardona-qft-methods_F02667 | 244  | 0.683 | $S(\backslash alpha \backslash beta)=S(\backslash beta \backslash alpha)$                                                                                                               | $S(\alpha\beta) = S(\beta\alpha)$                                                                       | $m_r(\xi \otimes \eta \otimes \rho) = \eta \xi \otimes \rho$                                                                                                                                          |
| cardona-qft-methods_F02668 | 244  | 1.000 | $m_{\{13\}} \otimes m_{\{24\}}$                                                                                                                                                         | $m_{13} \otimes m_{24}$                                                                                 |                                                                                                                                                                                                       |
| cardona-qft-methods_F02669 | 244  | 1.000 | $H \otimes H \otimes H \otimes H$                                                                                                                                                       | $H \otimes H \otimes H \otimes H$                                                                       |                                                                                                                                                                                                       |
| cardona-qft-methods_F02670 | 245  | 1.000 | $b$                                                                                                                                                                                     | $b$                                                                                                     |                                                                                                                                                                                                       |
| cardona-qft-methods_F02671 | 245  | 1.000 | $b^{\{n\}}=0$                                                                                                                                                                           | $b^n = 0$                                                                                               |                                                                                                                                                                                                       |
| cardona-qft-methods_F02672 | 245  | 1.000 | $a \ b=r \ b \ a$                                                                                                                                                                       | $ab = rba$                                                                                              |                                                                                                                                                                                                       |
| cardona-qft-methods_F02673 | 245  | 0.999 | $\backslash Delta(a)=a \otimes a, \backslash Delta(b)=a \otimes b+b \otimes a^{\{-1\}}, \backslash epsilon(a)=1, \backslash epsilon(b)=0$                                               | $\Delta(a) = a \otimes a, \Delta(b) = a \otimes b + b \otimes a^{-1}, \epsilon(a) = 1, \epsilon(b) = 0$ | <b>PROBLEM 2.2.</b> Let $\mathcal{A}$ be the algebra generated by an invertible element $a$ and an element $b$ such that $b^p = 0$ and $ab = rba$ , where $r$ is a primitive                          |
| cardona-qft-methods_F02674 | 245  | 0.998 | $H \otimes H$                                                                                                                                                                           | $H \otimes H$                                                                                           | and determine the antipode $S$ .                                                                                                                                                                      |
| cardona-qft-methods_F02675 | 245  | 1.000 | $\backslash Delta_{\{F\}}$                                                                                                                                                              | $\Delta_F$                                                                                              | Show that $\Delta_F$ defined by $\Delta_F(h) = F^{-1} \Delta(h) F$ satisfies the coassociativity condition $(\Lambda_v \otimes \text{Id}) \Delta_v = (\text{Id} \otimes \Lambda_v) \Delta_v$ .        |
| cardona-qft-methods_F02676 | 245  | 1.000 | $\backslash Delta_{\{F\}}(h)=F^{\{-1\}} \backslash Delta(h) \ F$                                                                                                                        | $\Delta_F(h) = F^{-1} \Delta(h) F$                                                                      | Show that $\Delta_F$ defined by $\Delta_F(h) = F^{-1} \Delta(h) F$ satisfies the coassociativity condition $(\Lambda_v \otimes \text{Id}) \Delta_v = (\text{Id} \otimes \Lambda_v) \Delta_v$ .        |
| cardona-qft-methods_F02677 | 245  | 1.000 | $\backslash left(\backslash Delta_{\{F\}} \otimes \text{Id}\right) \backslash Delta_{\{F\}}=\backslash left(\text{Id} \otimes \backslash Delta_{\{F\}}\right) \backslash Delta_{\{F\}}$ | $(\Delta_F \otimes \text{Id}) \Delta_F = (\text{Id} \otimes \Delta_F) \Delta_F$                         | There are several known examples of Hopf                                                                                                                                                              |
| cardona-qft-methods_F02678 | 245  | 1.000 | $U_{\{q\}}(S \ L(S, \mathbb{C}))$                                                                                                                                                       | $U_q(SL(S, \mathbb{C}))$                                                                                | example as the twisted coproduct of the Poincar group element involves functions of momenta which cannot be expressed in terms of group ele                                                           |
| cardona-qft-methods_F02679 | 245  | 1.000 | $\{ \}^{\{6,7,9-12\}}$                                                                                                                                                                  | $6,7,9-12$                                                                                              | <b>2.5. Identical Particles</b>                                                                                                                                                                       |
| cardona-qft-methods_F02680 | 245  | 1.000 | $V \otimes V$                                                                                                                                                                           | $V \otimes V$                                                                                           | $\mathcal{H} = \mathcal{H}_1 \oplus \mathcal{H}_2 \oplus \mathcal{H}_3 \oplus \mathcal{H}_4$<br>$\mathcal{H}_1 = \mathcal{H}_2 = \mathcal{H}_3 = \mathcal{H}_4 = \mathcal{H}$                         |
| cardona-qft-methods_F02681 | 245  | 1.000 | $\backslash xi \otimes \backslash eta \in V \otimes V$                                                                                                                                  | $\xi \otimes \eta \in V \otimes V$                                                                      |                                                                                                                                                                                                       |
| cardona-qft-methods_F02682 | 245  | 0.744 | $(\backslash rho \otimes \backslash rho) \backslash Delta(g)$                                                                                                                           | $(\rho \otimes \rho) \Delta(g)$                                                                         |                                                                                                                                                                                                       |
| cardona-qft-methods_F02683 | 245  | 1.000 | $V \otimes_{\{S\}} V$                                                                                                                                                                   | $V \otimes_S V$                                                                                         |                                                                                                                                                                                                       |
| cardona-qft-methods_F02684 | 246  | 0.985 | $V \otimes_{\{A\}} V$                                                                                                                                                                   | $V \otimes_A V$                                                                                         |                                                                                                                                                                                                       |
| cardona-qft-methods_F02685 | 246  | 1.000 | $V \otimes V \otimes \cdots \otimes V$                                                                                                                                                  | $V \otimes V \otimes \cdots \otimes V$                                                                  |                                                                                                                                                                                                       |
| cardona-qft-methods_F02686 | 246  | 1.000 | $S_{\{N\}}$                                                                                                                                                                             | $S_N$                                                                                                   | They generate the full permutation group $S_N$ and commute with the ac-                                                                                                                               |
| cardona-qft-methods_F02687 | 246  | 1.000 | $\backslash tau_{\{i, \ i+1\}} \rightarrow \pm 1$                                                                                                                                       | $\tau_{i,i+1} \rightarrow \pm 1$                                                                        | $V \otimes_{S,A} V \otimes_{S,A} \cdots \otimes_{S,A} V \equiv V^{\otimes_S}, V^{\otimes_A}.$                                                                                                         |
| cardona-qft-methods_F02688 | 246  | 1.000 | $V \otimes_{\{S, \ A\}} V \otimes_{\{S, \ A\}} \cdots \otimes_{\{S, \ A\}} V \equiv V^{\{\otimes_S\}}, V^{\{\otimes_A\}}$                                                               | $V \otimes_{S,A} V \otimes_{S,A} \cdots \otimes_{S,A} V \equiv V^{\otimes_S}, V^{\otimes_A}$            | subspaces are invariant under the full obs                                                                                                                                                            |
| cardona-qft-methods_F02689 | 246  | 0.998 | $\backslash Delta^{\{\mathrm{op}\}}$                                                                                                                                                    | $\Delta^{\text{op}}$                                                                                    | triangular". From these properties it follows that, if $\rho$ is a representation of $H$ then $(\rho \otimes \rho) \Delta(\rho)$ commutes with $\tau(\rho \otimes \rho) R$ . This means that at least |
| cardona-qft-methods_F02690 | 246  | 1.000 | $R \in H \otimes H$                                                                                                                                                                     | $R \in H \otimes H$                                                                                     |                                                                                                                                                                                                       |
| cardona-qft-methods_F02691 | 246  | 1.000 | $\backslash Delta^{\{\mathrm{op}\}}(\backslash eta)=$                                                                                                                                   | $\Delta^{\text{op}}(\eta) =$                                                                            |                                                                                                                                                                                                       |

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| cardona-qft-methods_F02692 | 246  | 1.000 | $R \Delta(\eta) R^{-1}$                                                | $R\Delta(\eta)R^{-1}$                                                  |                                                                                                                                                                                                                            |
| cardona-qft-methods_F02693 | 246  | 0.658 | $\rho$                                                                 | $\rho$                                                                 |                                                                                                                                                                                                                            |
| cardona-qft-methods_F02694 | 246  | 1.000 | $(\rho \otimes \rho) \Delta(\alpha)$                                   | $(\rho \otimes \rho) \Delta(\alpha)$                                   |                                                                                                                                                                                                                            |
| cardona-qft-methods_F02695 | 246  | 1.000 | $\tau(\rho \otimes \rho) R$                                            | $\tau(\rho \otimes \rho) R$                                            |                                                                                                                                                                                                                            |
| cardona-qft-methods_F02696 | 246  | 1.000 | $\tau R$                                                               | $\tau R$                                                               |                                                                                                                                                                                                                            |
| cardona-qft-methods_F02697 | 247  | 1.000 | $\tau_{i, i+1}$                                                        | $\tau_{i, i+1}$                                                        |                                                                                                                                                                                                                            |
| cardona-qft-methods_F02698 | 247  | 1.000 | $\mathcal{B}_N$                                                        | $\mathcal{B}_N$                                                        | generates $\tau_{i,i+1}$ of $S_N$ on $\tau \otimes \tau \otimes \dots \otimes \tau$ . The group it generates is precisely $S_N$ (it is the braid group $\mathcal{B}_N$ ). Notice that the square of $\tau R$ is not neces- |
| cardona-qft-methods_F02699 | 247  | 1.000 | $R \Delta R^{-1}$                                                      | $R\Delta R^{-1}$                                                       | CONJECTURE ON $\tau$ LEADS TO PROBABLY CORRECT RESULTS FOR LARGE QUANTUM ENTANGLEMENT                                                                                                                                      |
| cardona-qft-methods_F02700 | 247  | 0.974 | $\Delta_{\mathrm{op}}$                                                 | $\Delta_{\mathrm{op}}$                                                 | <b>Example 2.2.</b> As a simple example, we consider the twisted coproduct                                                                                                                                                 |
| cardona-qft-methods_F02701 | 247  | 1.000 | $(\rho \otimes \rho) \Delta_0(g)$                                      | $(\rho \otimes \rho) \Delta_0(g)$                                      | $\tau_\theta = \tau(\rho \otimes \rho) F_\theta^2,$                                                                                                                                                                        |
| cardona-qft-methods_F02702 | 247  | 1.000 | $\Delta_\theta(g)$                                                     | $\Delta_\theta(g)$                                                     | from which we obtain, for the $R$ -matrix,                                                                                                                                                                                 |
| cardona-qft-methods_F02703 | 247  | 1.000 | $(\rho \otimes \rho) \Delta_\theta(g)$                                 | $(\rho \otimes \rho) \Delta_\theta(g)$                                 | $\Delta_\theta^{\mathrm{op}}(g) = F_\theta(g \otimes g) F_\theta^{-1}$ . Clearly, $R_\theta \Delta_\theta(g) = \Delta_\theta^{\mathrm{op}}(g) R_\theta$ , as we s                                                          |
| cardona-qft-methods_F02704 | 247  | 1.000 | $\tau_\theta$                                                          | $\tau_\theta$                                                          | $\Delta_\theta^{\mathrm{op}}(g) = F_\theta(g \otimes g) F_\theta^{-1}$ . Clearly, $R_\theta \Delta_\theta(g) = \Delta_\theta^{\mathrm{op}}(g) R_\theta$ , as we s                                                          |
| cardona-qft-methods_F02705 | 247  | 0.666 | $\Delta_\theta^{\mathrm{op}}$                                          | $\Delta_\theta^{\mathrm{op}}$                                          |                                                                                                                                                                                                                            |
| cardona-qft-methods_F02706 | 247  | 1.000 | $\Delta_\theta^{\mathrm{op}}(g) = F_\theta(g \otimes g) F_\theta^{-1}$ | $\Delta_\theta^{\mathrm{op}}(g) = F_\theta(g \otimes g) F_\theta^{-1}$ |                                                                                                                                                                                                                            |
| cardona-qft-methods_F02707 | 247  | 1.000 | $R_\theta \Delta_\theta(g) = \Delta_\theta^{\mathrm{op}}(g) R_\theta$  | $R_\theta \Delta_\theta(g) = \Delta_\theta^{\mathrm{op}}(g) R_\theta$  |                                                                                                                                                                                                                            |
| cardona-qft-methods_F02708 | 247  | 1.000 | $\tau_\theta^2 = \mathrm{Id}$                                          | $\tau_\theta^2 = \mathrm{Id}$                                          |                                                                                                                                                                                                                            |
| cardona-qft-methods_F02709 | 248  | 1.000 | $R(t)$                                                                 | $R(t)$                                                                 | respond before the time at which it is disturbed. Hence we must have:                                                                                                                                                      |
| cardona-qft-methods_F02710 | 248  | 1.000 | $t < 0$                                                                | $t < 0$                                                                | $R(t) = 0 \quad t < 0 \tag{1}$                                                                                                                                                                                             |
| cardona-qft-methods_F02711 | 248  | 1.000 | $A > B$                                                                | $A > B$                                                                | Local observables $\rho(x)$ , $\eta(y)$ commute if $x$ and $y$ are spacelike separated:                                                                                                                                    |
| cardona-qft-methods_F02712 | 248  | 1.000 | $O$                                                                    | $O$                                                                    | $[\rho(x), \eta(y)] = 0 \quad \text{if} \tag{4}$                                                                                                                                                                           |
| cardona-qft-methods_F02713 | 248  | 1.000 | $C > 0$                                                                | $C > O$                                                                | $(x^0 - y^0)^2 - (\vec{x} - \vec{y})^2 < 0 \tag{5}$                                                                                                                                                                        |
| cardona-qft-methods_F02714 | 248  | 1.000 | $O > A$                                                                | $O > A$                                                                |                                                                                                                                                                                                                            |
| cardona-qft-methods_F02715 | 248  | 0.817 | $>$                                                                    | $>$                                                                    |                                                                                                                                                                                                                            |
| cardona-qft-methods_F02716 | 248  | 1.000 | $\rho(x), \eta(y)$                                                     | $\rho(x), \eta(y)$                                                     |                                                                                                                                                                                                                            |
| cardona-qft-methods_F02717 | 249  | 1.000 | $x_\mu$                                                                | $x_\mu$                                                                | $G$ with elements $g$ acting on single particle Hilbert spaces $\mathcal{H}_i$ by unitary                                                                                                                                  |
| cardona-qft-methods_F02718 | 249  | 1.000 | $\widehat{x}_\mu$                                                      | $\widehat{x}_\mu$                                                      | $G$ with elements $g$ acting on single particle Hilbert spaces $\mathcal{H}_i$ by unitary                                                                                                                                  |

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| cardona-qft-methods_F02719 | 249  | ■ 1.000 | G M                                                                                     | $GM$                                                                                    |                                                                                            |
| cardona-qft-methods_F02720 | 249  | ■ 1.000 | $\mathcal{H}_i$                                                                         | $\mathcal{H}_i$                                                                         |                                                                                            |
| cardona-qft-methods_F02721 | 249  | ■ 1.000 | $g \rightarrow U_i(g)$                                                                  | $g \rightarrow U_i(g)$                                                                  |                                                                                            |
| cardona-qft-methods_F02722 | 249  | ■ 1.000 | $\mathcal{H}_1 \otimes \mathcal{H}_2$                                                   | $\mathcal{H}_1 \otimes \mathcal{H}_2$                                                   |                                                                                            |
| cardona-qft-methods_F02723 | 250  | ■ 1.000 | $m_{\{0\}}$                                                                             | $m_0$                                                                                   | $m_0(\alpha \otimes \beta)(x) = \alpha(x)\beta(x)$                                                                                                                            |
| cardona-qft-methods_F02724 | 250  | ■ 1.000 | $\mathcal{A}_{\{0\}}$                                                                   | $\mathcal{A}_0$                                                                         | $m_0(\alpha \otimes \beta)(x) = \alpha(x)\beta(x)$                                                                                                                            |
| cardona-qft-methods_F02725 | 250  | ■ 1.000 | $x \in \mathbb{R}^N$                                                                    | $x \in \mathbb{R}^N$                                                                    | It acts on functions on $\mathbb{R}^N$ by pull-back:                                                                                                                          |
| cardona-qft-methods_F02726 | 250  | ■ 1.000 | $\mathcal{A}_{\theta}(\mathbb{R}^N)$                                                    | $\mathcal{A}_{\theta}(\mathbb{R}^N)$                                                    | In order to understand how this comes about, let us consider a quantum mechanical two-particle system.                                                                        |
| cardona-qft-methods_F02727 | 250  | ■ 1.000 | $m_{\{\theta\}}$                                                                        | $m_{\theta}$                                                                            | In order to understand how this comes about, let us consider a quantum mechanical two-particle system.                                                                        |
| cardona-qft-methods_F02728 | 250  | ■ 1.000 | $\theta^{\mu\nu} = 0$                                                                   | $\theta^{\mu\nu} = 0$                                                                   | $\phi \otimes_{S,A} \chi \equiv \frac{1}{2}(\phi \otimes \chi \pm \chi \otimes \phi)$ (16)                                                                                    |
| cardona-qft-methods_F02729 | 250  | ■ 1.000 | $\mathcal{A}_{\{0\}} \otimes \mathcal{A}_{\{0\}}$                                       | $\mathcal{A}_0 \otimes \mathcal{A}_0$                                                   |                                                                                                                                                                               |
| cardona-qft-methods_F02730 | 250  | ■ 1.000 | $\Delta_{\{0\}}$                                                                        | $\Delta_0$                                                                              |                                                                                                                                                                               |
| cardona-qft-methods_F02731 | 250  | ■ 1.000 | $\mathcal{A}_{\theta} \otimes \mathcal{A}_{\theta}$                                     | $\mathcal{A}_{\theta} \otimes \mathcal{A}_{\theta}$                                     |                                                                                                                                                                               |
| cardona-qft-methods_F02732 | 250  | ■ 1.000 | $\theta_{\mu\nu} = 0$                                                                   | $\theta_{\mu\nu} = 0$                                                                   |                                                                                                                                                                               |
| cardona-qft-methods_F02733 | 251  | ■ 1.000 | $\tau_{\{0\}}$                                                                          | $\tau_0$                                                                                | of reference. But one easily sees that the twisted coproduct action of the Lorentz group is not compatible with the usual symmetriza-                                         |
| cardona-qft-methods_F02734 | 251  | ■ 1.000 | $\Delta_0(\Lambda) = \Lambda \otimes \Lambda$                                           | $\Delta_0(\Lambda) = \Lambda \otimes \Lambda$                                           | Lorentz group action, $\Delta_0(\Lambda) = \Lambda \otimes \Lambda$ , must and does commute with the                                                                          |
| cardona-qft-methods_F02735 | 251  | ■ 1.000 | $\phi \otimes \chi$                                                                     | $\phi \otimes \chi$                                                                     | $(1 \pm \tau_0) \chi \otimes \chi$                                                                                                                                            |
| cardona-qft-methods_F02736 | 252  | ■ 1.000 | $\mathcal{P}$                                                                           | $\mathcal{P}$                                                                           | 4. Covariant quantum fields on noncommutative spacetimes                                                                                                                      |
| cardona-qft-methods_F02737 | 252  | ■ 1.000 | $M^{d+1}$                                                                               | $M^{d+1}$                                                                               | 4.1. Poincaré group                                                                                                                                                           |
| cardona-qft-methods_F02738 | 252  | ■ 1.000 | $\mathcal{A}_{\{0\}}(M^{d+1})$                                                          | $\mathcal{A}_0(M^{d+1})$                                                                | $(a, \Lambda) \in P: (a, \Lambda)x = \Lambda x + a.$ (1)                                                                                                                      |
| cardona-qft-methods_F02739 | 252  | ■ 1.000 | $\varphi$                                                                               | $\varphi$                                                                               | The commutative algebra of functions on $M^{d+1}$ is denoted by $\mathcal{A}_0(M^{d+1})$ . If $\phi$ is a quantum relativistic scalar field on $M^{d+1}$ we assume that there |
| cardona-qft-methods_F02740 | 252  | ■ 1.000 | $\varphi \in L(\mathcal{H}) \otimes S(M^{d+1})$                                         | $\varphi \in L(\mathcal{H}) \otimes S(M^{d+1})$                                         | and the action on $\mathcal{Q}(M^{d+1})$ is given by                                                                                                                          |
| cardona-qft-methods_F02741 | 252  | ■ 1.000 | $L(\mathcal{H})$                                                                        | $L(\mathcal{H})$                                                                        | and its action on $\mathcal{Q}(M^{d+1})$ is given by                                                                                                                          |
| cardona-qft-methods_F02742 | 252  | ■ 1.000 | $S(M^{d+1})$                                                                            | $S(M^{d+1})$                                                                            | and its action on $\mathcal{Q}(M^{d+1})$ is given by                                                                                                                          |
| cardona-qft-methods_F02743 | 253  | ■ 1.000 | $\varphi \otimes \varphi \otimes \dots \otimes \varphi$                                 | $\varphi \otimes \varphi \otimes \dots \otimes \varphi$                                 |                                                                                                                                                                               |
| cardona-qft-methods_F02744 | 253  | ■ 1.000 | $L(\mathcal{H}) \otimes S(M^{d+1}) \otimes S(M^{d+1}) \otimes \dots \otimes S(M^{d+1})$ | $L(\mathcal{H}) \otimes S(M^{d+1}) \otimes S(M^{d+1}) \otimes \dots \otimes S(M^{d+1})$ |                                                                                                                                                                               |

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| cardona-qft-methods_F02745 | 253  | ■ 1.000 | $x_{\{1\}}, x_{\{2\}}, \ldots x_{\{N\}}$                                                                                                                      | $x_1, x_2, \ldots x_N$                                                                                                                                        |    |
| cardona-qft-methods_F02746 | 253  | ■ 1.000 | $\varphi^{\otimes N} = \varphi \otimes \varphi \otimes \ldots \otimes \varphi \in L(\mathcal{H}) \otimes \left( S \left( M^{d+1} \right)^{\otimes N} \right)$ | $\varphi^{\otimes N} = \varphi \otimes \varphi \otimes \ldots \otimes \varphi \in L(\mathcal{H}) \otimes \left( S \left( M^{d+1} \right)^{\otimes N} \right)$ |    |
| cardona-qft-methods_F02747 | 253  | ■ 1.000 | $U(a, \Lambda)$                                                                                                                                               | $U(a, \Lambda)$                                                                                                                                               |    |
| cardona-qft-methods_F02748 | 253  | ■ 1.000 | $S \left( M^{d+1} \right)^{\otimes N}$                                                                                                                        | $S \left( M^{d+1} \right)^{\otimes N}$                                                                                                                        |    |
| cardona-qft-methods_F02749 | 253  | ■ 1.000 | $(a, \Lambda)$                                                                                                                                                | $(a, \Lambda)$                                                                                                                                                |    |
| cardona-qft-methods_F02750 | 253  | ■ 1.000 | $\Delta_0^N$                                                                                                                                                  | $\Delta_0^N$                                                                                                                                                  |    |
| cardona-qft-methods_F02751 | 253  | ■ 1.000 | $(a, \Lambda) \in \mathcal{P}$                                                                                                                                | $(a, \Lambda) \in \mathcal{P}$                                                                                                                                |    |
| cardona-qft-methods_F02752 | 253  | ■ 1.000 | $c_{\{p\}}, c_{\{p\}}^{\dagger}$                                                                                                                              | $c_p, c_p^{\dagger}$                                                                                                                                          |    |
| cardona-qft-methods_F02753 | 253  | ■ 1.000 | $\varphi^{\{ \pm \}}$                                                                                                                                         | $\varphi^{(\pm)}$                                                                                                                                             |    |
| cardona-qft-methods_F02754 | 253  | ■ 1.000 | $\varphi^{(-)}$                                                                                                                                               | $\varphi^{(-)}$                                                                                                                                               |                                                                                       |
| cardona-qft-methods_F02755 | 253  | ■ 1.000 | $P_{\mu}$                                                                                                                                                     | $P_{\mu}$                                                                                                                                                     |                                                                                       |
| cardona-qft-methods_F02756 | 254  | ■ 0.999 | $\mathcal{P}_{\mu} = -i \partial_{\mu}$                                                                                                                       | $\mathcal{P}_{\mu} = -i \partial_{\mu}$                                                                                                                       |                                                                                       |
| cardona-qft-methods_F02757 | 254  | ■ 1.000 | $\mathcal{P}_{\mu}$                                                                                                                                           | $\mathcal{P}_{\mu}$                                                                                                                                           |  |
| cardona-qft-methods_F02758 | 254  | ■ 0.999 | $\underline{\nu}$                                                                                                                                             | $\underline{\nu}$                                                                                                                                             |  |
| cardona-qft-methods_F02759 | 254  | ■ 0.999 | $\widehat{P}_{\mu}$                                                                                                                                           | $\widehat{P}_{\mu}$                                                                                                                                           |  |
| cardona-qft-methods_F02760 | 254  | ■ 1.000 | $\underline{\nu} \left( \widehat{P}_{\mu} \right) = \mathcal{P}_{\mu}$                                                                                        | $\underline{\nu} \left( \widehat{P}_{\mu} \right) = \mathcal{P}_{\mu}$                                                                                        |  |
| cardona-qft-methods_F02761 | 254  | ■ 1.000 | $e_{\{p\}}$                                                                                                                                                   | $e_p$                                                                                                                                                         |  |
| cardona-qft-methods_F02762 | 254  | ■ 1.000 | $\Lambda e_p = e_{\Lambda p}$                                                                                                                                 | $\Lambda e_p = e_{\Lambda p}$                                                                                                                                 |  |
| cardona-qft-methods_F02763 | 254  | ■ 1.000 | $d \geq 3$                                                                                                                                                    | $d \geq 3$                                                                                                                                                    |                                                                                       |
| cardona-qft-methods_F02764 | 255  | ■ 0.964 | $(a, \Delta)$                                                                                                                                                 | $(a, \Delta)$                                                                                                                                                 |  |
| cardona-qft-methods_F02765 | 255  | ■ 0.964 | $e_{\{p_{\{1\}}\}} \otimes e_{\{p_{\{2\}}\}} \otimes \ldots \otimes e_{\{p_{\{N\}}\}}$                                                                        | $e_{p_1} \otimes e_{p_2} \otimes \ldots \otimes e_{p_N}$                                                                                                      |  |
| cardona-qft-methods_F02766 | 255  | ■ 1.000 | $e_{\{p_{\{i\}}\}}$                                                                                                                                           | $e_{p_i}$                                                                                                                                                     |  |
| cardona-qft-methods_F02767 | 255  | ■ 1.000 | $\mathbb{C} \mathcal{P}$                                                                                                                                      | $\mathbb{C} \mathcal{P}$                                                                                                                                      |  |
| cardona-qft-methods_F02768 | 255  | ■ 1.000 | $\mathbb{C} S_N$                                                                                                                                              | $\mathbb{C} S_N$                                                                                                                                              |  |
| cardona-qft-methods_F02769 | 255  | ■ 0.833 | $U(k)$                                                                                                                                                        | $U(k)$                                                                                                                                                        |  |
| cardona-qft-methods_F02770 | 255  | ■ 1.000 | $\mathbb{C}^k$                                                                                                                                                | $\mathbb{C}^k$                                                                                                                                                |  |

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| cardona-qft-methods_F02771 | 255  | 1.000 | $\mathbb{C}^k \otimes N$                      | $\mathbb{C}^{k \otimes N}$                    | the fact that $\mathbb{C}S_N$ commutes with $\mathbb{C}U(k)$ . That lets us use Young tableaux                                                                            |
| cardona-qft-methods_F02772 | 255  | 1.000 | $\mathbb{C} \cup (k)$                         | $\mathbb{C}U(k)$                              | methods. It is to take the case where $\mathbb{C}U(k)$ and $\mathbb{C}S_N$ commute. The commutators of each other. This result and the Young tableaux methods are part of |
| cardona-qft-methods_F02773 | 255  | 1.000 | $\tau_{ij}$                                   | $\tau_{ij}$                                   |                                                                                                                                                                           |
| cardona-qft-methods_F02774 | 255  | 0.999 | $\alpha: p \rightarrow \alpha p, p \in M$     | $\alpha: p \rightarrow \alpha p, p \in M$     |                                                                                                                                                                           |
| cardona-qft-methods_F02775 | 256  | 1.000 | $\mathbb{C}^{\infty}(M)$                      | $\mathbb{C}^{\infty}(M)$                      |                                                                                                                                                                           |
| cardona-qft-methods_F02776 | 256  | 1.000 | $f_1, f_2 \in C^{\infty}(M)$                  | $f_1, f_2 \in C^{\infty}(M)$                  |                                                                                                                                                                           |
| cardona-qft-methods_F02777 | 256  | 1.000 | $C^0(M)$                                      | $C^0(M)$                                      | pletion of this algebra under the supremum norm gives the commutative algebra of $C^0(M)$ a commutative $C^*$ -algebra. By the Gelfand-Naimark                            |
| cardona-qft-methods_F02778 | 256  | 1.000 | $C^{\star}$                                   | $C^*$                                         | pletion of this algebra under the supremum norm gives the commutative algebra of $C^0(M)$ a commutative $C^*$ -algebra. By the Gelfand-Naimark                            |
| cardona-qft-methods_F02779 | 256  | 1.000 | $\Psi^{\dagger}$                              | $\Psi^{\dagger}$                              | postulates) compatible with $m$ - and $\star$ -covariance.                                                                                                                |
| cardona-qft-methods_F02780 | 256  | 0.996 | $U(a, \Delta)$                                | $U(a, \Delta)$                                |                                                                                                                                                                           |
| cardona-qft-methods_F02781 | 256  | 1.000 | $\mathcal{A}_{\theta}(M^{d+1})$               | $\mathcal{A}_{\theta}(M^{d+1})$               |                                                                                                                                                                           |
| cardona-qft-methods_F02782 | 256  | 1.000 | $M^{(d+1)}$                                   | $M^{(d+1)}$                                   |                                                                                                                                                                           |
| cardona-qft-methods_F02783 | 257  | 0.978 | $\mathcal{A}_{\theta}(M^{(d+1)})$             | $\mathcal{A}_{\theta}(M^{(d+1)})$             | $\mathcal{P} \ni (a, \Delta): \alpha \rightarrow (a, \Delta)\alpha, \quad ((a, \Delta)\alpha)(x) = \alpha(((a, \Delta)^{-1}x). \quad (31)$                                |
| cardona-qft-methods_F02784 | 257  | 1.000 | $F_{\theta}$                                  | $F_{\theta}$                                  | tor in $\mathcal{P}$ with representatives $\mathcal{P}_{\mu} = -i\partial_{\mu}$ and $P_{\mu}$ on functions and $L(\mathcal{H})$                                          |
| cardona-qft-methods_F02785 | 257  | 1.000 | $\hat{P}_{\mu}$                               | $\hat{P}_{\mu}$                               | tor in $\mathcal{P}$ with representatives $\mathcal{P}_{\mu} = -i\partial_{\mu}$ and $P_{\mu}$ on functions and $L(\mathcal{H})$                                          |
| cardona-qft-methods_F02786 | 257  | 1.000 | $\varphi_{\theta}$                            | $\varphi_{\theta}$                            | $U_{\theta}(a, \Lambda)\varphi_{\theta}((a, \Lambda)^{-1}x)U_{\theta}(a, \Lambda)^{-1} = \varphi_{\theta}(x) \quad (33)$                                                  |
| cardona-qft-methods_F02787 | 257  | 1.000 | $U_{\theta}$                                  | $U_{\theta}$                                  | $U_{\theta}(a, \Lambda)\varphi_{\theta}((a, \Lambda)^{-1}x)U_{\theta}(a, \Lambda)^{-1} = \varphi_{\theta}(x) \quad (33)$                                                  |
| cardona-qft-methods_F02788 | 257  | 1.000 | $\varphi_{\theta}^{\dagger}$                  | $\varphi_{\theta}^{\dagger}$                  | $\varphi = \int d\mu(p)[a_p^{\dagger}e_p + a_pe_{-p}] = \varphi_{\theta}^{(-)} + \varphi_{\theta}^{(+)}, \quad d\mu(p) = \frac{d^dp}{\alpha_{1-1}}. \quad (34)$           |
| cardona-qft-methods_F02789 | 257  | 1.000 | $U_{\theta}(a, \Lambda) 0\rangle =  0\rangle$ | $U_{\theta}(a, \Lambda) 0\rangle =  0\rangle$ |                                                                                                                                                                           |
| cardona-qft-methods_F02790 | 258  | 1.000 | $a_p, a_p^{\dagger}$                          | $a_p, a_p^{\dagger}$                          |                                                                                                                                                                           |
| cardona-qft-methods_F02791 | 258  | 1.000 | $e_{p_1} \otimes e_{p_2}$                     | $e_{p_1} \otimes e_{p_2}$                     | $\int \prod d\mu(p_i)a_{p_1}^{\dagger}a_{p_2}^{\dagger} 0\rangle e_{p_1} \otimes e_{p_2}$                                                                                 |
| cardona-qft-methods_F02792 | 258  | 1.000 | $a_{p_1}^{\dagger}$                           | $a_{p_1}^{\dagger}$                           | Translational covariance requires the standard transformation of $a_{p_1}^{\dagger}$ :                                                                                    |
| cardona-qft-methods_F02793 | 258  | 1.000 | $P_{\mu}^{\theta}$                            | $P_{\mu}^{\theta}$                            | tions are more interesting. We have that                                                                                                                                  |
| cardona-qft-methods_F02794 | 258  | 1.000 | $a_p^{\dagger}$                               | $a_p^{\dagger}$                               | $U_{\theta}(a, \Lambda) = U_0(a, \Lambda) = U(a, \Lambda). \quad (43)$                                                                                                    |
| cardona-qft-methods_F02795 | 258  | 1.000 | $c_p^{\dagger}$                               | $c_p^{\dagger}$                               | $U_{\theta}(a, \Lambda) = U_0(a, \Lambda) = U(a, \Lambda). \quad (43)$                                                                                                    |
| cardona-qft-methods_F02796 | 258  | 0.996 | $c_p$                                         | $c_p$                                         |                                                                                                                                                                           |

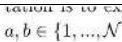
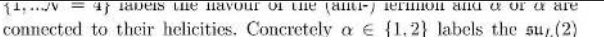
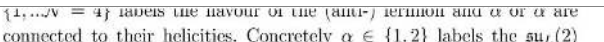
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| cardona-qft-methods_F02797 | 258  | ■ 0.996 | $\backslash\mathrm{varphi}_{\{0\}} \backslash\mathrm{equiv} \backslash\mathrm{varphi}_{\{}$                                                 | $\varphi_0 \equiv \varphi :$                      |                                                                                                                                                   |
| cardona-qft-methods_F02798 | 258  | ■ 1.000 | $\backslash\mathrm{varphi}_{\{0\}} \backslash\mathrm{rightarrow} \backslash\mathrm{varphi}_{\{\mathrm{theta}\}}$                            | $\varphi_0 \rightarrow \varphi_\theta$            |                                                                                                                                                   |
| cardona-qft-methods_F02799 | 258  | ■ 1.000 | $\backslash\mathrm{varphi}_{\{0\}}$                                                                                                         | $\varphi_0$                                       |                                                                                                                                                   |
| cardona-qft-methods_F02800 | 259  | ■ 0.999 | $\backslash\mathrm{rangle} \ 0 $                                                                                                            | $\langle 0 $                                      | $U(a,\Lambda)\varphi_\theta(x_1)\varphi_\theta(x_2)\dots\varphi_\theta(x_N)U(a,\Lambda)^{-1} 0\rangle=$                                           |
| cardona-qft-methods_F02801 | 259  | ■ 1.000 | $\backslash\mathrm{alpha}, \backslash\mathrm{beta}$                                                                                         | $\alpha, \beta$                                   | But soon we will generalise them to N-particles to get $S_N$ . Thus twisted                                                                       |
| cardona-qft-methods_F02802 | 259  | ■ 1.000 | $S_{\{2\}}$                                                                                                                                 | $S_2$                                             | bosons (fermions) have the two-particle plane wave stat                                                                                           |
| cardona-qft-methods_F02803 | 259  | ■ 1.000 | $S_{\{\mathrm{theta}\}}$                                                                                                                    | $S_\theta$                                        | the group                                                                                                                                         |
| cardona-qft-methods_F02804 | 260  | ■ 0.932 | $\mathrm{a}^{\{\mathrm{dagger}\}}$                                                                                                          | $a^\dagger$                                       | We can extend this demonstration regarding the consistency of the twist to                                                                        |
| cardona-qft-methods_F02805 | 260  | ■ 0.932 | $\backslash\mathrm{mathcal}\{F\}_{\{\mathrm{theta}\}}^{\{i\ j\}}$                                                                           | $\mathcal{F}_\theta^{ij}$                         | We can extend this demonstration regarding the consistency of the twist to                                                                        |
| cardona-qft-methods_F02806 | 260  | ■ 0.939 | $\backslash\mathrm{partial}_{\{\mu\}} \backslash\mathrm{otimes} \backslash\mathrm{partial}_{\{\nu\}}, \backslash\mathrm{partial}_{\{\mu\}}$ | $\partial_\mu \otimes \partial_\nu, \partial_\mu$ | multinomials in $a^\dagger$ 's and $a$ s. In the $N$ -particle sector, call $\mathcal{F}_\theta^{ij}$ the Drinfeld                                |
| cardona-qft-methods_F02807 | 260  | ■ 0.939 | $\mathrm{i}^{\{\mathrm{text} \ \mathrm{th} \ \}}$                                                                                           | $i^{\mathrm{th}}$                                 | multinomials in $a^\dagger$ 's and $a$ s. In the $N$ -particle sector, call $\mathcal{F}_\theta^{ij}$ the Drinfeld                                |
| cardona-qft-methods_F02808 | 260  | ■ 0.939 | $\backslash\mathrm{partial}_{\{\nu\}}$                                                                                                      | $\partial_\nu$                                    | multinomials in $a^\dagger$ 's and $a$ s. In the $N$ -particle sector, call $\mathcal{F}_\theta^{ij}$ the Drinfeld                                |
| cardona-qft-methods_F02809 | 260  | ■ 0.939 | $\mathrm{j}^{\{\mathrm{text} \ \mathrm{th} \ \}}$                                                                                           | $j^{\mathrm{th}}$                                 | multinomials in $a^\dagger$ 's and $a$ s. In the $N$ -particle sector, call $\mathcal{F}_\theta^{ij}$ the Drinfeld                                |
| cardona-qft-methods_F02810 | 260  | ■ 0.976 | $\backslash\mathrm{tau}_{\{0\}}^{\{i\ j\}}$                                                                                                 | $\tau_0^{ij}$                                     | and $j^{\mathrm{th}}$ entries. Then                                                                                                               |
| cardona-qft-methods_F02811 | 260  | ■ 0.976 | $\mathrm{i}^{\{\mathrm{t} \ \mathrm{h}\}}$                                                                                                  | $i^{\mathrm{th}}$                                 | and $j^{\mathrm{th}}$ entries. Then                                                                                                               |
| cardona-qft-methods_F02812 | 260  | ■ 0.999 | $\backslash\mathrm{tau}_{\{\mathrm{theta}\}}^{\{i, \ i+1\}} \backslash\mathrm{mathrm}\{\sim\mathrm{s}\}$                                    | $\tau_\theta^{i,i+1} \mathbf{s}$                  | field. We now require it to be compatible with the $\star$ -operation. That is if                                                                 |
| cardona-qft-methods_F02813 | 260  | ■ 1.000 | $\backslash\mathrm{varphi}_{\{0\}}^{\{*\}}=\backslash\mathrm{varphi}_{\{0\}}$                                                               | $\varphi_0^* = \varphi_0$                         | $\varphi_\theta = \varphi_\theta \psi^*$                                                                                                          |
| cardona-qft-methods_F02814 | 260  | ■ 1.000 | $\backslash\mathrm{varphi}_{\{\mathrm{theta}\}}^{\{*\}}=\backslash\mathrm{varphi}_{\{\mathrm{theta}\}}$                                     | $\varphi_\theta^* = \varphi_\theta$               | $\varphi_\theta = \varphi_\theta \psi^*$                                                                                                          |
| cardona-qft-methods_F02815 | 260  | ■ 1.000 | $\backslash\mathrm{partial}_{\{\mu\}}$                                                                                                      | $\partial_\mu$                                    | is the conceptual basis behind the dressing transformation. This means that symmetry transformations on spacetime and associated structures like  |
| cardona-qft-methods_F02816 | 260  | ■ 1.000 | $\backslash\mathrm{varphi}_{\{0\}}, \ \mathrm{P}_{\{\nu\}}$                                                                                 | $\varphi_0, P_\nu$                                | is the conceptual basis behind the dressing transformation. This means that symmetry transformations on spacetime and associated structures like  |
| cardona-qft-methods_F02817 | 260  | ■ 1.000 | $\backslash\mathrm{varphi}_{\{0\}}-\backslash\mathrm{partial}_{\{\nu\}} \backslash\mathrm{varphi}_{\{0\}}$                                  | $\varphi_0 - \partial_\nu \varphi_0$              | suitable symmetrization postulates of particle wave fun                                                                                           |
| cardona-qft-methods_F02818 | 260  | ■ 1.000 | $\backslash\mathrm{partial} \ \backslash\mathrm{wedge} \ \backslash\mathrm{partial}=0$                                                      | $\partial \wedge \partial = 0$                    | suitable symmetrization postulates of particle wave fun                                                                                           |
| cardona-qft-methods_F02819 | 261  | ■ 1.000 | $\{ \ \}^{\{14-17\}}$                                                                                                                       | 14–17                                             | In this last section we (very briefly) mention some of the most striking phenomenological implications of spacetime noncommutativity. They follow |
| cardona-qft-methods_F02820 | 261  | ■ 0.997 | $\mathrm{V}_{\{\mathrm{S} \ \mathrm{T} \ \mathrm{A} \ \mathrm{T}\}}$                                                                        | $V_{STAT}$                                        | STAT and a large $p$ -multiples<br>In an interesting paper <sup>18</sup> the statistical potential $V_{\mathrm{stat}}$ is between fermions        |
| cardona-qft-methods_F02821 | 261  | ■ 1.000 | $\backslash\mathrm{left}\{x_{\{1\}}, \ x_{\{2\}}\}\mathrm{right}\rangle\mathrm{rangle}$                                                     | $ x_1, x_2\rangle$                                | $\langle 0 \psi(x)\psi(x')=0$<br>It is explicitly shown not to have an infi                                                                       |
| cardona-qft-methods_F02822 | 261  | ■ 1.000 | $0^{\{16\}}\backslash\mathrm{left}(\mathrm{C}^{\{12\}}\mathrm{right})$                                                                      | $O^{16} (C^{12})$                                 | models are yet to be studied. A more detailed discussion of this issues can                                                                       |

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| cardona-qft-methods_F02823 | 261  | ■ 1.000 | $\tilde{0}^{16}\left(\tilde{C}^{12}\right)$                                             | $\tilde{O}^{16}\left(\tilde{C}^{12}\right)$ | models are yet to be studied. A more detailed discussion of this issues can                                                                                                                                                                                                                              |
| cardona-qft-methods_F02824 | 261  | ■ 1.000 | $1\ S_{1\ /\ 2}$                                                                        | $1S_{1/2}$                                  | of the form                                                                                                                                                                                                                                                                                              |
| cardona-qft-methods_F02825 | 261  | ■ 1.000 | $10^{27}$                                                                               | $10^{27}$                                   | <b>5.2. Causality</b>                                                                                                                                                                                                                                                                                    |
| cardona-qft-methods_F02826 | 261  | ■ 1.000 | $10^{-25}$                                                                              | $10^{-25}$                                  | $\phi_\alpha(x)\phi_\alpha(y)=e^{-\frac{1}{2}\frac{\partial}{\partial x^\mu}\theta^{\mu\nu}\frac{\partial}{\partial y^\nu}\phi_\alpha(x)\phi_\alpha(y)}e^{\frac{1}{2}\left(\frac{\partial}{\partial x^\mu}+\frac{\partial}{\partial y^\mu}\right)\theta^{\mu\nu}P_\nu}\neq\phi_\alpha(x)\phi_\alpha(y).$ |
| cardona-qft-methods_F02827 | 261  | ■ 1.000 | $\phi_{\theta}$                                                                         | $\phi_\theta$                               |                                                                                                                                                                                                                                                                                                          |
| cardona-qft-methods_F02828 | 262  | ■ 1.000 | $\rho_{\theta},\eta_{\theta}$                                                           | $\rho_\theta,\eta_\theta$                   |                                                                                                                                                                                                                                                                                                          |
| cardona-qft-methods_F02829 | 262  | ■ 1.000 | $\omega_{\beta}$                                                                        | $\omega_\beta$                              | of the form                                                                                                                                                                                                                                                                                              |
| cardona-qft-methods_F02830 | 262  | ■ 1.000 | $H_{\mathrm{I}}$                                                                        | $H_I$                                       | Bogoliubov and Shirkov and then Weinberg long ago deduced from causality                                                                                                                                                                                                                                 |
| cardona-qft-methods_F02831 | 262  | ■ 1.000 | $x\sim y$                                                                               | $x\sim y$                                   |                                                                                                                                                                                                                                                                                                          |
| cardona-qft-methods_F02832 | 263  | ■ 1.000 | $S^{\{2\}}$                                                                             | $S^{(2)}$                                   | $U(\Lambda)S^{(2)}U$                                                                                                                                                                                                                                                                                     |
| cardona-qft-methods_F02833 | 263  | ■ 0.998 | $\vec{P}_{\text{inc}}$                                                                  | $\vec{P}_{\text{inc}}$                      | $\theta_{0i}\left(\vec{P}_{\text{inc}}\right)_i\quad (17)$                                                                                                                                                                                                                                               |
| cardona-qft-methods_F02834 | 263  | ■ 1.000 | $\theta_{0\ i}$                                                                         | $\theta_{0i}$                               | if $j$ is odd.                                                                                                                                                                                                                                                                                           |
| cardona-qft-methods_F02835 | 263  | ■ 0.866 | $\left\{ \right\}_{0\ i}\!\!\!\left(\vec{P}_{\text{inc}}\right)\!\!\!\right)_{\!\!\!i}$ | $_{0i}\left(\vec{P}_{\text{inc}}\right)_i$  | <b>5.3. Cosmic microwave ba</b>                                                                                                                                                                                                                                                                          |
| cardona-qft-methods_F02836 | 263  | ■ 1.000 | $Z^{\{0\}}\rightarrow 2\ \gamma$                                                        | $Z^0\rightarrow 2\gamma$                    | which led to the conclusion that the early universe was not smooth; there                                                                                                                                                                                                                                |
| cardona-qft-methods_F02837 | 264  | ■ 1.000 | $a_{lm}$                                                                                | $a_{lm}$                                    | $T=\sum_{lm}a_{lm}e^{im\varphi}e^{ikz}$                                                                                                                                                                                                                                                                  |
| cardona-qft-methods_F02838 | 264  | ■ 1.000 | $\Phi$                                                                                  | $\Phi$                                      |                                                                                                                                                                                                                                                                                                          |
| cardona-qft-methods_F02839 | 264  | ■ 1.000 | $\Delta_l^T(k)$                                                                         | $\Delta_l^T(k)$                             | where $\Delta_l^T(k)$ are called the transfer functions. We find                                                                                                                                                                                                                                         |
| cardona-qft-methods_F02840 | 264  | ■ 1.000 | $\vec{\theta}^0=\theta(0,0,1)\cdot P_{\Phi}$                                            | $\vec{\theta}^0=\theta(0,0,1).P_\Phi$       | with $H$ the Hubble parameter and $\varepsilon$ the slow-roll parameter                                                                                                                                                                                                                                  |
| cardona-qft-methods_F02841 | 264  | ■ 1.000 | $\left\langle a_{lm}a_{l'\prime m'}^*\right\rangle _\theta$                             | $\langle a_{lm}a_{l'm'}^*\rangle_\theta$    | $\varphi(\alpha,x)=\left(\frac{\alpha}{\pi}\right)^{\frac{3}{2}}\int d^3x'\varphi_\theta(x)\exp^{-\alpha(x'-x)^2},\quad (22)$                                                                                                                                                                            |
| cardona-qft-methods_F02842 | 264  | ■ 1.000 | $C\ M\ B$                                                                               | $CMB$                                       | $\Delta\varphi_\theta(\alpha,x_1)\Delta\varphi_\theta(\alpha,x_2)\geq\langle 0 \varphi_\theta(\alpha,x_1),\varphi_\theta(\alpha,x_2) 0\rangle\quad (23)$                                                                                                                                                 |
| cardona-qft-methods_F02843 | 264  | ■ 1.000 | $\sqrt{\theta}\leq 10^{-32}$                                                            | $\sqrt{\theta}\leq 10^{-32}$                | by uncertainty principle, and it is non-zero by causality violati                                                                                                                                                                                                                                        |
| cardona-qft-methods_F02844 | 264  | ■ 1.000 | $\geq 10^{16}\mathrm{GeV}$                                                              | $\geq 10^{16}\mathrm{GeV}$                  | by uncertainty principle, and it is non-zero by causality violati                                                                                                                                                                                                                                        |
| cardona-qft-methods_F02845 | 264  | ■ 1.000 | $x_{\{1\}}\sim$                                                                         | $x_1\sim$                                   |                                                                                                                                                                                                                                                                                                          |
| cardona-qft-methods_F02846 | 264  | ■ 1.000 | $x_{\{2\}}$                                                                             | $x_2$                                       |                                                                                                                                                                                                                                                                                                          |
| cardona-qft-methods_F02847 | 267  | ■ 1.000 | $5+5=10$                                                                                | $5+5=10$                                    | ding space is the maximally supersymmetric completion of the $5+5=10$                                                                                                                                                                                                                                    |
| cardona-qft-methods_F02848 | 267  | ■ 0.998 | $A\ d\ S_5\ \times\ S^5$                                                                | $AdS_5\times S^5$                           | dimensional product spaces $AdS_5\times S^5$ , the 5-dimensional Anti-de-Sitter space and the 5-sphere. The fermionic coordinates can be intuitively con                                                                                                                                                 |

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| cardona-qft-methods_F02849 | 267  | ■ 1.000 | A d S_{5}                                   | $AdS_5$                                         | reason is that all supercharges carry one index referring to the $AdS$ space                                                                                                                                                             |
| cardona-qft-methods_F02850 | 267  | ■ 1.000 | S^{5}                                       | $S^5$                                           | reason is that all supercharges carry one index referring to the $AdS$ space                                                                                                                                                             |
| cardona-qft-methods_F02851 | 267  | ■ 1.000 | A d S                                       | $AdS$                                           | and a second index referring to the sphere $S^5$ .<br>In 1997 Juan Maldacena <sup>2</sup> conjectured an exact correspondence between                                                                                                    |
| cardona-qft-methods_F02852 | 267  | ■ 1.000 | C F T_{4}                                   | $CFT_4$                                         | More colloquially: For every string observable at the boundary of $AdS_5$                                                                                                                                                                |
| cardona-qft-methods_F02853 | 268  | ■ 1.000 | X X X                                       | $XXX$                                           | chain followed by the Heisenberg spin chain, $XXX$ model. After that we                                                                                                                                                                  |
| cardona-qft-methods_F02854 | 268  | ■ 1.000 | I I B                                       | $IIB$                                           | the interest of the reader. Many crucial topics are not discussed here, such as                                                                                                                                                          |
| cardona-qft-methods_F02855 | 268  | ■ 1.000 | \frac{1}{2}                                 | $\frac{1}{2}$                                   | $\sigma^1 = \begin{pmatrix} 0 & +1 \\ +1 & 0 \end{pmatrix}, \quad \sigma^2 = \begin{pmatrix} 0 & -i \\ +i & 0 \end{pmatrix}, \quad \sigma^3 = \begin{pmatrix} +1 & 0 \\ 0 & -1 \end{pmatrix}. \quad (2)$                                 |
| cardona-qft-methods_F02856 | 268  | ■ 1.000 | \mathbb{C}^2                                | $\mathbb{C}^2$                                  | These matrices generate the simplest non-trivial (fundamental) representa-                                                                                                                                                               |
| cardona-qft-methods_F02857 | 268  | ■ 1.000 | \vec{S}                                     | $\vec{S}$                                       | tion of the Lie algebra $\mathfrak{su}(2)$ defined by the commutation relations                                                                                                                                                          |
| cardona-qft-methods_F02858 | 268  | ■ 1.000 | \mathfrak{su}(2)                            | $\mathfrak{su}(2)$                              |                                                                                                                                                                                                                                          |
| cardona-qft-methods_F02859 | 269  | ■ 1.000 | s=\frac{1}{2}                               | $s = \frac{1}{2}$                               | For this basis one has                                                                                                                                                                                                                   |
| cardona-qft-methods_F02860 | 269  | ■ 1.000 | \sigma^3                                    | $\sigma^3$                                      | $S^+ \downarrow\rangle = 0, \quad S^+ \uparrow\rangle =  \uparrow\rangle, \\ S^+ \downarrow\rangle =  \uparrow\rangle, \quad S^- \uparrow\rangle = 0.$                                                                                   |
| cardona-qft-methods_F02861 | 270  | ■ 1.000 | S^{+}                                       | $S^+$                                           | From these relations the name ladder operator becomes obvious, $S^+$ raises                                                                                                                                                              |
| cardona-qft-methods_F02862 | 270  | ■ 1.000 | S^{-}                                       | $S^-$                                           | the spin and $S^-$ lowers it. Usually $ \uparrow\rangle$ is called highest weight state and                                                                                                                                              |
| cardona-qft-methods_F02863 | 270  | ■ 1.000 | 2^L                                         | $2^L$                                           | $ \uparrow\uparrow\cdots\downarrow\rangle = \begin{pmatrix} 1 \\ n \end{pmatrix} \otimes \begin{pmatrix} 1 \\ n \end{pmatrix} \otimes \cdots \otimes \begin{pmatrix} 0 \\ 1 \end{pmatrix} \otimes \begin{pmatrix} 1 \\ n \end{pmatrix}.$ |
| cardona-qft-methods_F02864 | 270  | ■ 1.000 | L=2                                         | $L = 2$                                         | On this basis                                                                                                                                                                                                                            |
| cardona-qft-methods_F02865 | 270  | ■ 1.000 | \mathbb{C}^2 \otimes \mathbb{C}^2           | $\mathbb{C}^2 \otimes \mathbb{C}^2$             | On this basis                                                                                                                                                                                                                            |
| cardona-qft-methods_F02866 | 271  | ■ 1.000 | L(L \sim 10)                                | $L(L \sim 10)$                                  | $\hat{H} \psi\rangle = E \psi\rangle, \quad  \psi\rangle \in (\mathbb{C}^2)^{\otimes L}, \quad (9)$                                                                                                                                      |
| cardona-qft-methods_F02867 | 271  | ■ 1.000 | 2^L \times 2^L                              | $2^L \times 2^L$                                | trivial. For larger                                                                                                                                                                                                                      |
| cardona-qft-methods_F02868 | 271  | ■ 0.873 | l^{\text{th}}                               | $l^{\text{th}}$                                 | $\begin{pmatrix} 0 & 0 & 0 & 0 \end{pmatrix}$                                                                                                                                                                                            |
| cardona-qft-methods_F02869 | 271  | ■ 0.873 | l+1^{\text{th}}                             | $l+1^{\text{th}}$                               | $\begin{pmatrix} 0 & 0 & 0 & 0 \end{pmatrix}$                                                                                                                                                                                            |
| cardona-qft-methods_F02870 | 271  | ■ 0.953 | \vec{S}_{\text{tot}}                        | $\vec{S}_{\text{tot}}$                          |                                                                                                                                                                                                                                          |
| cardona-qft-methods_F02871 | 271  | ■ 0.953 | \vec{S}_{\text{tot}}=\sum_{l=1}^L \vec{S}_l | $\vec{S}_{\text{tot}} = \sum_{l=1}^L \vec{S}_l$ |                                                                                                                                                                                                                                          |
| cardona-qft-methods_F02872 | 271  | ■ 0.953 | \vec{S}_l                                   | $\vec{S}_l$                                     |                                                                                                                                                                                                                                          |

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| cardona-qft-methods_F02873 | 272  | ■ 0.985 | L+1                                                  | $L+1$                                                |                                                                                                                                                                                                                                                                                                                                                              |
| cardona-qft-methods_F02874 | 272  | ■ 0.985 | $\binom{L}{M} \times \binom{L}{M}$                   | $\binom{L}{M} \times \binom{L}{M}$                   |                                                                                                                                                                                                                                                                                                                                                              |
| cardona-qft-methods_F02875 | 272  | ■ 1.000 | $[\hat{H}, \hat{U}] = 0$                             | $[\hat{H}, \hat{U}] = 0$                             | When it acts on any spin chain state, its name becomes obvious <sup>a</sup> . In the                                                                                                                                                                                                                                                                         |
| cardona-qft-methods_F02876 | 272  | ■ 1.000 | $\hat{U}$                                            | $\hat{U}$                                            | case of the closed chain, the shift operator satisfies $[\hat{H}, \hat{U}] = 0$ . If one acts                                                                                                                                                                                                                                                                |
| cardona-qft-methods_F02877 | 272  | ■ 1.000 | $n=0,1,\ldots,L-1$                                   | $n=0,1,\ldots,L-1$                                   | operator was beautif                                                                                                                                                                                                                                                                                                                                         |
| cardona-qft-methods_F02878 | 272  | ■ 1.000 | $u_{\{k\}}$                                          | $u_k$                                                | $E=2\sum_{k=1}^L\frac{1}{u_k^2+\frac{1}{4}},\quad U=\prod_{k=1}^L\frac{u_k-\frac{1}{2}}{u_k+\frac{1}{2}}.$                                                                                                                                                                                                                                                   |
| cardona-qft-methods_F02879 | 273  | ■ 0.883 | $l_{\{i\}}$                                          | $l_i$                                                |                                                                                                                                                                                                                                                                                                                                                              |
| cardona-qft-methods_F02880 | 273  | ■ 0.883 | $ 0\rangle=\lvert\uparrow\cdots\uparrow\rangle$      | $ 0\rangle=\lvert\uparrow\cdots\uparrow\rangle$      |                                                                                                                                                                                                                                                                                                                                                              |
| cardona-qft-methods_F02881 | 273  | ■ 1.000 | $S_{\{n\}}$                                          | $S_n$                                                | $\psi(l_1,\dots,l_M)=\sum_{\sigma\in S_n}A(\sigma)e^{ip_{\sigma_1}l_1+\dots+ip_{\sigma_M}l_M},\quad(2)$                                                                                                                                                                                                                                                      |
| cardona-qft-methods_F02882 | 273  | ■ 1.000 | $A(\tau)$                                            | $A(\tau)$                                            | where $S_n$ is the permutation group of a set containing $n$ elements. The                                                                                                                                                                                                                                                                                   |
| cardona-qft-methods_F02883 | 273  | ■ 1.000 | $\operatorname{sgn}(\tau)$                           | $\operatorname{sgn}(\tau)$                           | energy eigenvalue is determined to be                                                                                                                                                                                                                                                                                                                        |
| cardona-qft-methods_F02884 | 274  | ■ 1.000 | $\mathbb{C}^2\otimes\mathbb{C}^2\otimes\mathbb{C}^2$ | $\mathbb{C}^2\otimes\mathbb{C}^2\otimes\mathbb{C}^2$ | $\mathbf{D}=\begin{pmatrix}1&0&0&0\\0&1&0&0\\0&0&1&0\\0&0&0&1\end{pmatrix}$                                                                                                                                                                                                                                                                                  |
| cardona-qft-methods_F02885 | 274  | ■ 0.999 | $L_{\{a,\,l\}}(u)=R_{a,\,l}\big(u-\frac{i}{2}\big)$  | $L_{a,l}(u)=R_{a,l}\left(u-\frac{i}{2}\right)$       |                                                                                                                                                                                                                                                                                                                                                              |
| cardona-qft-methods_F02886 | 274  | ■ 0.876 | $8\times8$                                           | $8\times8$                                           |                                                                                                                                                                                                                                                                                                                                                              |
| cardona-qft-methods_F02887 | 275  | ■ 1.000 | $\left(\mathbb{C}^2\right)^{\otimes L}$              | $(\mathbb{C}^2)^{\otimes L}$                         | $D^{-1}(a_1\cdots a^L)M^{-1}(a)MM^{-1}(a^L)=M^{-1}(a^L)M^{-1}(a)D^{-1}(a_1\cdots a^L)$                                                                                                                                                                                                                                                                       |
| cardona-qft-methods_F02888 | 275  | ■ 1.000 | $1,\ldots,L$                                         | $1,\ldots,L$                                         | $\frac{\partial}{\partial u_j} \left( \frac{1}{\det M} \right) = - \frac{1}{\det M} \operatorname{tr} \left( M^{-1} \frac{\partial M}{\partial u_j} \right) = - \frac{1}{\det M} \sum_{i=1}^L \frac{\partial}{\partial u_j} \left( \frac{1}{\det M} \right) = - \frac{1}{\det M} \sum_{i=1}^L \frac{\partial}{\partial u_j} \left( \frac{1}{\det M} \right)$ |
| cardona-qft-methods_F02889 | 275  | ■ 1.000 | $1,2,\ldots,L$                                       | $1,2,\ldots,L$                                       |                                                                                                                                                                                                                                                                                                                                                              |
| cardona-qft-methods_F02890 | 276  | ■ 1.000 | $R_{\{a,\,b\}}(u-u')$                                | $R_{a,b}(u-u')$                                      |                                                                                                                                                                                                                                                                                                                                                              |
| cardona-qft-methods_F02891 | 276  | ■ 1.000 | $u=\frac{i}{2}$                                      | $u=\frac{i}{2}$                                      | that at the special point $u=\frac{i}{2}$ of the spectral parameter the Lax operator reduces essentially to a permutation: $T^{-1}(i/2)=iD^{-1}$ . Using this observation                                                                                                                                                                                    |
| cardona-qft-methods_F02892 | 276  | ■ 1.000 | $L_{\{a,\,l\}}(\frac{i}{2})=iP_{a,\,l}$              | $L_{a,l}\left(\frac{i}{2}\right)=iP_{a,l}$           | can also be obtained by a straightforward calculation. Using also the definition and properties of the permutation operator we obtain                                                                                                                                                                                                                        |
| cardona-qft-methods_F02893 | 276  | ■ 1.000 | $\hat{P}$                                            | $\hat{P}$                                            | $\epsilon^{\pm}(u)=\left.\frac{d}{du}\epsilon^{\pm}(u)\right _{u=0}$                                                                                                                                                                                                                                                                                         |
| cardona-qft-methods_F02894 | 276  | ■ 1.000 | $\left[\hat{T}(u),\hat{T}(u')\right]=0$              | $\left[\hat{T}(u),\hat{T}(u')\right]=0$              |                                                                                                                                                                                                                                                                                                                                                              |
| cardona-qft-methods_F02895 | 276  | ■ 1.000 | $\left\{\hat{Q}_n\right\}$                           | $\left\{\hat{Q}_n\right\}$                           |                                                                                                                                                                                                                                                                                                                                                              |
| cardona-qft-methods_F02896 | 277  | ■ 1.000 | $\hat{H} \psi\rangle=E \psi\rangle$                  | $\hat{H} \psi\rangle=E \psi\rangle$                  | leads from the Yang-Baxter equation to a <i>quantum integrable system</i> . However, a substitute for the coordinate Bethe ansatz is still missing.                                                                                                                                                                                                          |
| cardona-qft-methods_F02897 | 277  | ■ 1.000 | $ \psi\rangle$                                       | $ \psi\rangle$                                       | matrix. We write the monodromy as a matrix in the auxiliary space $a$ ,                                                                                                                                                                                                                                                                                      |
| cardona-qft-methods_F02898 | 277  | ■ 1.000 | $\hat{A}(u),\hat{B}(u),\hat{C}(u)$                   | $\hat{A}(u),\hat{B}(u),\hat{C}(u)$                   | show that the ansatz                                                                                                                                                                                                                                                                                                                                         |

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| cardona-qft-methods_F02899 | 277  | ■ 1.000 | $\hat{D}(u)$                                                             | $\hat{D}(u)$                                                             | show that the ansatz                                                                                                                                                                                                                                |
| cardona-qft-methods_F02900 | 277  | ■ 1.000 | $\hat{C}(u) 0\rangle=0$                                                  | $\hat{C}(u) 0\rangle=0$                                                  | yields eigenvectors of the transfer matrix, if the <i>Bethe roots</i> $\{u_k\}$ satisfy                                                                                                                                                             |
| cardona-qft-methods_F02901 | 277  | ■ 0.960 | $\left\{u_k\right\}$                                                     | $\{u_k\}$                                                                | The corresponding eigenvalue of the transfer matrix $\hat{T}(u)$ is                                                                                                                                                                                 |
| cardona-qft-methods_F02902 | 277  | ■ 1.000 | $\hat{T}(u)$                                                             | $\hat{T}(u)$                                                             | $u = u_k$ have to vanish. Demanding $\text{res}_{u=u_k}T(u) = 0$ yields again the B                                                                                                                                                                 |
| cardona-qft-methods_F02903 | 277  | ■ 0.894 | $T(u)$                                                                   | $T(u)$                                                                   |                                                                                                                                                                                                                                                     |
| cardona-qft-methods_F02904 | 277  | ■ 1.000 | $u=u_k$                                                                  | $u = u_k$                                                                |                                                                                                                                                                                                                                                     |
| cardona-qft-methods_F02905 | 277  | ■ 1.000 | $\operatorname{res}_{u=u_k}T(u)=0$                                       | $\text{res}_{u=u_k}T(u)=0$                                               |                                                                                                                                                                                                                                                     |
| cardona-qft-methods_F02906 | 278  | ■ 0.706 | $TQ$                                                                     | $TQ$                                                                     | $Q(u)=\prod (u-u_j).$                                                                                                                                                                                                                               |
| cardona-qft-methods_F02907 | 278  | ■ 1.000 | $\operatorname{SU}(N)$                                                   | $\text{SU}(N)$                                                           | language of quantum field theory, this means that its $\beta$ -function vanishes to all orders in perturbation theory. For a more detailed introduction to this                                                                                     |
| cardona-qft-methods_F02908 | 278  | ■ 1.000 | $\beta$                                                                  | $\beta$                                                                  | be expanded as $Y_\mu=\sum_{a=1}^{(N^2-1)}Y_\mu^aT_a$ , where $\mu\in\{0,1,2,3\}$ (if present) is a                                                                                                                                                 |
| cardona-qft-methods_F02909 | 278  | ■ 1.000 | $Y_{\mu}=\sum_{a=1}^{\left(N^2-1\right)}Y_{\mu}^aT_a$                    | $Y_\mu=\sum_{a=1}^{(N^2-1)}Y_\mu^aT_a$                                   |                                                                                                                                                                                                                                                     |
| cardona-qft-methods_F02910 | 278  | ■ 1.000 | $\mu\in\{0,1,2,3\}$                                                      | $\mu\in\{0,1,2,3\}$                                                      |                                                                                                                                                                                                                                                     |
| cardona-qft-methods_F02911 | 278  | ■ 1.000 | $T_a$                                                                    | $T_a$                                                                    |                                                                                                                                                                                                                                                     |
| cardona-qft-methods_F02912 | 278  | ■ 1.000 | $N\times N$                                                              | $N\times N$                                                              |                                                                                                                                                                                                                                                     |
| cardona-qft-methods_F02913 | 278  | ■ 1.000 | $\left[T_a,T_b\right]=if_{abc}T_c$                                       | $[T_a,T_b]=if_{abc}T_c$                                                  |                                                                                                                                                                                                                                                     |
| cardona-qft-methods_F02914 | 278  | ■ 0.954 | $Y\rightarrow UYU^{-1}$                                                  | $Y\rightarrow UYU^{-1}$                                                  |                                                                                                                                                                                                                                                     |
| cardona-qft-methods_F02915 | 278  | ■ 0.954 | $U\equiv U(x)\in\operatorname{SU}(N)$                                    | $U\equiv U(x)\in\text{SU}(N)$                                            |                                                                                                                                                                                                                                                     |
| cardona-qft-methods_F02916 | 279  | ■ 1.000 | $x\in\mathbb{R}^{1,3}$                                                   | $x\in\mathbb{R}^{1,3}$                                                   |                                                                                                                                                                                                                                                     |
| cardona-qft-methods_F02917 | 279  | ■ 1.000 | $A_{\mu}\rightarrow UA_{\mu}U^{-1}-i\left(\partial_{\mu}U\right)$        | $A_\mu\rightarrow UA_\mu U^{-1}-i(\partial_\mu U)$                       |                                                                                                                                                                                                                                                     |
| cardona-qft-methods_F02918 | 279  | ■ 1.000 | $U^{-1}$                                                                 | $U^{-1}$                                                                 |                                                                                                                                                                                                                                                     |
| cardona-qft-methods_F02919 | 279  | ■ 1.000 | $\varphi_m$                                                              | $\varphi_m$                                                              | $U^{-1}$ . This transformation ensures the gauge invariance of the theory. The                                                                                                                                                                      |
| cardona-qft-methods_F02920 | 279  | ■ 1.000 | $m\in\{1,\ldots,6\}$                                                     | $m\in\{1,\dots,6\}$                                                      | $U^{-1}$ . This transformation ensures the gauge invariance of the theory. The                                                                                                                                                                      |
| cardona-qft-methods_F02921 | 279  | ■ 1.000 | $\Delta_0=1$                                                             | $\Delta_0=1$                                                             | field content of SYM consists of:                                                                                                                                                                                                                   |
| cardona-qft-methods_F02922 | 279  | ■ 1.000 | $X=\varphi_1+i\varphi_2, Y=\varphi_3+i\varphi_4, Z=\varphi_5+i\varphi_6$ | $X=\varphi_1+i\varphi_2, Y=\varphi_3+i\varphi_4, Z=\varphi_5+i\varphi_6$ | <ul style="list-style-type: none"> <li>6 massless, real scalars, <math>\varphi_m</math> with <math>m\in\{1,\dots,6\}</math> with classical mass-dimension <math>\Delta_\varphi=1</math>. Equivalently, these may be represented by 3 com</li> </ul> |
| cardona-qft-methods_F02923 | 279  | ■ 1.000 | $\bar{X},\bar{Y},\bar{Z}$                                                | $\bar{X},\bar{Y},\bar{Z}$                                                | dimension $\Delta_\varphi=1$ . Equivalently, these may be represented by 3 complex scalars: $X=\varphi_1+i\varphi_2, Y=\varphi_3+i\varphi_4, Z=\varphi_5+i\varphi_6$ and their                                                                      |
| cardona-qft-methods_F02924 | 279  | ■ 1.000 | $\varphi_{ab}$                                                           | $\varphi_{ab}$                                                           | complex conjugates, denoted by $\bar{X}, \bar{Y}, \bar{Z}$ . Yet another, important, notation is to express the scalars as adjoint-state bilinears $\varphi_{ab}=\varphi_a\varphi_b$                                                                |

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| cardona-qft-methods_F02925 | 279  | ■ 1.000 | $a, b \in \{1, \ldots, \mathcal{N}=4\}$                                                                          | $a, b \in \{1, \ldots, \mathcal{N} = 4\}$                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| cardona-qft-methods_F02926 | 279  | ■ 1.000 | $\psi_{a\alpha}$                                                                                                 | $\psi_{a\alpha}$                                                                                                 | <ul style="list-style-type: none"> <li>4 massless Weyl-fermions, <math>\psi_{a\alpha}</math>, and their 4 conjugates <math>\bar{\psi}_{\dot{\alpha}}^a</math>. Here <math>a \in \{1, \ldots, \mathcal{N} = 4\}</math> labels the flavour of the (anti-) fermion and <math>\alpha</math> or <math>\dot{\alpha}</math> are connected to their helicities. Concretely <math>\alpha \in \{1, 2\}</math> labels the <math>\mathfrak{su}_L(2)</math> algebra and <math>\dot{\alpha} \in \{\dot{1}, \dot{2}\}</math> the <math>\mathfrak{su}_R(2)</math> in the Lie algebra given by the isomorphism with the complexification of the Lorentz algebra, <math>\mathfrak{so}(1, 3) \simeq \mathfrak{su}_L(2) \oplus \mathfrak{su}_R(2)</math>. The fermions have classical mass-dimension <math>\Lambda_0 = 3/2</math>.</li> </ul> |
| cardona-qft-methods_F02927 | 279  | ■ 1.000 | $\bar{\psi}_{\dot{\alpha}}^a$                                                                                    | $\bar{\psi}_{\dot{\alpha}}^a$                                                                                    | <ul style="list-style-type: none"> <li>4 massless Weyl-fermions, <math>\psi_{a\alpha}</math>, and their 4 conjugates <math>\bar{\psi}_{\dot{\alpha}}^a</math>. Here <math>a \in \{1, \ldots, \mathcal{N} = 4\}</math> labels the flavour of the (anti-) fermion and <math>\alpha</math> or <math>\dot{\alpha}</math> are connected to their helicities. Concretely <math>\alpha \in \{1, 2\}</math> labels the <math>\mathfrak{su}_L(2)</math> algebra and <math>\dot{\alpha} \in \{\dot{1}, \dot{2}\}</math> the <math>\mathfrak{su}_R(2)</math> in the Lie algebra given by the isomorphism with the complexification of the Lorentz algebra, <math>\mathfrak{so}(1, 3) \simeq \mathfrak{su}_L(2) \oplus \mathfrak{su}_R(2)</math>. The fermions have classical mass-dimension <math>\Lambda_0 = 3/2</math>.</li> </ul> |
| cardona-qft-methods_F02928 | 279  | ■ 1.000 | $a \in$                                                                                                          | $a \in$                                                                                                          | <ul style="list-style-type: none"> <li>4 massless Weyl-fermions, <math>\psi_{a\alpha}</math>, and their 4 conjugates <math>\bar{\psi}_{\dot{\alpha}}^a</math>. Here <math>a \in \{1, \ldots, \mathcal{N} = 4\}</math> labels the flavour of the (anti-) fermion and <math>\alpha</math> or <math>\dot{\alpha}</math> are connected to their helicities. Concretely <math>\alpha \in \{1, 2\}</math> labels the <math>\mathfrak{su}_L(2)</math> algebra and <math>\dot{\alpha} \in \{\dot{1}, \dot{2}\}</math> the <math>\mathfrak{su}_R(2)</math> in the Lie algebra given by the isomorphism with the complexification of the Lorentz algebra, <math>\mathfrak{so}(1, 3) \simeq \mathfrak{su}_L(2) \oplus \mathfrak{su}_R(2)</math>. The fermions have classical mass-dimension <math>\Lambda_0 = 3/2</math>.</li> </ul> |
| cardona-qft-methods_F02929 | 279  | ■ 1.000 | $\{1, \ldots \mathcal{N}=4\}$                                                                                    | $\{1, \ldots \mathcal{N} = 4\}$                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| cardona-qft-methods_F02930 | 279  | ■ 1.000 | $\dot{\alpha}$                                                                                                   | $\dot{\alpha}$                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| cardona-qft-methods_F02931 | 279  | ■ 0.993 | $\alpha \in \{1, 2\}$                                                                                            | $\alpha \in \{1, 2\}$                                                                                            | algebra and $\dot{\alpha} \in \{\dot{1}, \dot{2}\}$ the $\mathfrak{su}_R(2)$ in the Lie algebra given by the isomorphism with the complexification of the Lorentz algebra, $\mathfrak{so}(1, 3) \simeq \mathfrak{su}_L(2) \oplus \mathfrak{su}_R(2)$ . The fermions have classical mass-dimension $\Lambda_0 = 3/2$ .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| cardona-qft-methods_F02932 | 279  | ■ 0.993 | $\mathfrak{su}_L(2)$                                                                                             | $\mathfrak{su}_L(2)$                                                                                             | algebra and $\dot{\alpha} \in \{\dot{1}, \dot{2}\}$ the $\mathfrak{su}_R(2)$ in the Lie algebra given by the isomorphism with the complexification of the Lorentz algebra, $\mathfrak{so}(1, 3) \simeq \mathfrak{su}_L(2) \oplus \mathfrak{su}_R(2)$ . The fermions have classical mass-dimension $\Lambda_0 = 3/2$ .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| cardona-qft-methods_F02933 | 279  | ■ 1.000 | $\dot{\alpha} \in \{\dot{1}, \dot{2}\}$                                                                          | $\dot{\alpha} \in \{\dot{1}, \dot{2}\}$                                                                          | morphism with the complexification of the Lorentz algebra, $\mathfrak{so}(1, 3) \simeq \mathfrak{su}_L(2) \oplus \mathfrak{su}_R(2)$ . The fermions have classical mass-dimension $\Lambda_0 = 3/2$ .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| cardona-qft-methods_F02934 | 279  | ■ 1.000 | $\mathfrak{su}_R(2)$                                                                                             | $\mathfrak{su}_R(2)$                                                                                             | morphism with the complexification of the Lorentz algebra, $\mathfrak{so}(1, 3) \simeq \mathfrak{su}_L(2) \oplus \mathfrak{su}_R(2)$ . The fermions have classical mass-dimension $\Lambda_0 = 3/2$ .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| cardona-qft-methods_F02935 | 279  | ■ 1.000 | $\mathfrak{so}(1, 3) \simeq$                                                                                     | $\mathfrak{so}(1, 3) \simeq$                                                                                     | <ul style="list-style-type: none"> <li>1 field strength tensor, <math>F_{\mu\nu} = i[D_\mu, D_\nu] = \partial_\mu A_\nu - \partial_\nu A_\mu - i[A_\mu, A_\nu]</math>,</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| cardona-qft-methods_F02936 | 279  | ■ 1.000 | $\mathfrak{su}_L(2) \oplus \mathfrak{su}_R(2)$                                                                   | $\mathfrak{su}_L(2) \oplus \mathfrak{su}_R(2)$                                                                   | with $D_\mu = \partial_\mu - iA_\mu$ . The field strength has classical mass-dimensions $\Lambda_0 = 2$ .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| cardona-qft-methods_F02937 | 279  | ■ 1.000 | $\Delta_0 = 3/2$                                                                                                 | $\Delta_0 = 3/2$                                                                                                 | with $D_\mu = \partial_\mu - iA_\mu$ . The field strength has classical mass-dimensions $\Lambda_0 = 2$ .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| cardona-qft-methods_F02938 | 279  | ■ 1.000 | $F_{\mu\nu} = i\left[D_\mu, D_\nu\right] = \partial_\mu A_\nu - \partial_\nu A_\mu - i\left[A_\mu, A_\nu\right]$ | $F_{\mu\nu} = i\left[D_\mu, D_\nu\right] = \partial_\mu A_\nu - \partial_\nu A_\mu - i\left[A_\mu, A_\nu\right]$ | <ul style="list-style-type: none"> <li>In addition to these fundamental fields infinitely many local fields can</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| cardona-qft-methods_F02939 | 279  | ■ 1.000 | $D_\mu = \partial_\mu - iA_\mu$                                                                                  | $D_\mu = \partial_\mu - iA_\mu$                                                                                  | be generated by acting with covariant derivatives on all the fields just discussed. For this note that a covariant derivative acts on a field in                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| cardona-qft-methods_F02940 | 279  | ■ 1.000 | $\Delta_0 = 2$                                                                                                   | $\Delta_0 = 2$                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| cardona-qft-methods_F02941 | 279  | ■ 1.000 | $D_\mu \star = \partial_\mu \star - i\left[A_\mu, \star\right]$                                                  | $D_\mu \star = \partial_\mu \star - i\left[A_\mu, \star\right]$                                                  | $S = \frac{i\star}{\lambda} \int \frac{\star}{(2\pi)^4} \text{Tr} \left( \frac{\star}{4} F^{\mu\nu} F_{\mu\nu} + \frac{\star}{2} (D^\mu)^\star \varphi_m D_\mu \varphi_m + \bar{\psi}_\alpha^a (\sigma_\mu)^{\dot{\alpha}\beta} D^\mu \psi_{a\dot{\beta}} \right)$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| cardona-qft-methods_F02942 | 279  | ■ 1.000 | $\lambda = Ng_{YM}^2$                                                                                            | $\lambda = Ng_{YM}^2$                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| cardona-qft-methods_F02943 | 279  | ■ 1.000 | $(\Sigma_m)^{ab}$                                                                                                | $(\Sigma_m)^{ab}$                                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| cardona-qft-methods_F02944 | 279  | ■ 0.523 | $(\sigma_\mu)^{\alpha\dot{\beta}}$                                                                               | $(\sigma_\mu)^{\alpha\dot{\beta}}$                                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| cardona-qft-methods_F02945 | 279  | ■ 1.000 | $N^{-1}$                                                                                                         | $N^{-1}$                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| cardona-qft-methods_F02946 | 280  | ■ 1.000 | $N \rightarrow \infty, g_{YM} \rightarrow 0$                                                                     | $N \rightarrow \infty, g_{YM} \rightarrow 0$                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| cardona-qft-methods_F02947 | 280  | ■ 1.000 | $Y_\mu^{ij} = Y_\mu^a (T_a)^{ij}$                                                                                | $Y_\mu^{ij} = Y_\mu^a (T_a)^{ij}$                                                                                | $T_\mu^{\pm} = T_\mu (T_a)^{\pm}, \text{ with } i, j \in \{1, 2, \dots, 2N\}.$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| cardona-qft-methods_F02948 | 280  | ■ 1.000 | $i, j \in \{1, 2, \ldots, N\}$                                                                                   | $i, j \in \{1, 2, \ldots, N\}$                                                                                   | $T_\mu^{\pm} = T_\mu (T_a)^{\pm}, \text{ with } i, j \in \{1, 2, \dots, 2N\}.$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| cardona-qft-methods_F02949 | 280  | ■ 0.878 | $\mathfrak{sl}(2, \mathbb{C}) \simeq \mathfrak{so}(1, 3)$                                                        | $\mathfrak{sl}(2, \mathbb{C}) \simeq \mathfrak{so}(1, 3)$                                                        | metric $\eta_{\mu\nu} = \text{diag}(1, -1, -1, -1)$ it is given by the following symmetry algebra                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| cardona-qft-methods_F02950 | 280  | ■ 1.000 | $M_{\mu\nu}$                                                                                                     | $M_{\mu\nu}$                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |

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| cardona-qft-methods_F02951 | 280       | 1.000     | $\partial_{\mu} \rightarrow D_{\mu}$           | $\partial_{\mu} \rightarrow D_{\mu}$           | $[M_{\mu\nu}, P_{\rho}] = i(\eta_{\rho\nu} P_{\mu} - \eta_{\mu\rho} P_{\nu}),$ (3)                                                                                                                                                                                                                                                  |     |     |     |     |           |           |     |           |     |
| cardona-qft-methods_F02952 | 280       | 1.000     | $P_{\mu} = i D_{\mu}$                          | $P_{\mu} = i D_{\mu}$                          | $[M_{\mu\nu}, P_{\rho}] = i(\eta_{\rho\nu} P_{\mu} - \eta_{\mu\rho} P_{\nu}),$ (3)                                                                                                                                                                                                                                                  |     |     |     |     |           |           |     |           |     |
| cardona-qft-methods_F02953 | 280       | 1.000     | $(M_{\mu\nu}, P_{\rho})$                       | $(M_{\mu\nu}, P_{\rho})$                       | $[P_{\mu}, P_{\nu}] = 0,$ (4)                                                                                                                                                                                                                                                                                                       |     |     |     |     |           |           |     |           |     |
| cardona-qft-methods_F02954 | 280       | 1.000     | $\eta_{\mu\nu} = \text{diag}(1, -1, -1, -1)$   | $\eta_{\mu\nu} = \text{diag}(1, -1, -1, -1)$   | <ul style="list-style-type: none"><li>Another 4 symmetry operations are the special conformal transforma-</li></ul>                                                                                                                                                                                                                 |     |     |     |     |           |           |     |           |     |
| cardona-qft-methods_F02955 | 280       | 1.000     | $K_{\mu}$                                      | $K_{\mu}$                                      |                                                                                                                                                                                                                                                                                                                                     |     |     |     |     |           |           |     |           |     |
| cardona-qft-methods_F02956 | 280       | 1.000     | $I: x_{\mu} \rightarrow x_{\mu} / x^2$         | $I: x_{\mu} \rightarrow x_{\mu} / x^2$         |                                                                                                                                                                                                                                                                                                                                     |     |     |     |     |           |           |     |           |     |
| cardona-qft-methods_F02957 | 280       | 1.000     | $K_{\mu} = I P_{\mu} I$                        | $K_{\mu} = I P_{\mu} I$                        |                                                                                                                                                                                                                                                                                                                                     |     |     |     |     |           |           |     |           |     |
| cardona-qft-methods_F02958 | 281       | 1.000     | $6+4+4+1=15$                                   | $6 + 4 + 4 + 1 = 15$                           | remaining two commutation relations are                                                                                                                                                                                                                                                                                             |     |     |     |     |           |           |     |           |     |
| cardona-qft-methods_F02959 | 281       | 1.000     | $\mathfrak{so}(2,4) \simeq \mathfrak{su}(2,2)$ | $\mathfrak{so}(2,4) \simeq \mathfrak{su}(2,2)$ | $[\mathcal{D}, P_{\mu}] = i P_{\mu} \quad , \quad [\mathcal{D}, K_{\mu}] = -i K_{\mu} \quad (7)$                                                                                                                                                                                                                                    |     |     |     |     |           |           |     |           |     |
| cardona-qft-methods_F02960 | 281       | 1.000     | $\mathfrak{so}(6)$                             | $\mathfrak{so}(6)$                             | <ul style="list-style-type: none"><li>the <math>\mathfrak{so}(6)</math> symmetry transforming the 6 scalars into each other,</li><li>the <math>\mathfrak{su}(4)</math> symmetry acting on the 4 fermionic flavours</li></ul>                                                                                                        |     |     |     |     |           |           |     |           |     |
| cardona-qft-methods_F02961 | 281       | 0.998     | $\mathfrak{su}(4)$                             | $\mathfrak{su}(4)$                             | In fact the $\mathfrak{so}(6)$ and the $\mathfrak{su}(4)$ symmetry are the same in the sense th-                                                                                                                                                                                                                                    |     |     |     |     |           |           |     |           |     |
| cardona-qft-methods_F02962 | 281       | 0.907     | $2 \times 2$                                   | $2 \times 2$                                   | Now by gluing together a $\mathfrak{su}(2)$ and a $\mathfrak{su}(2)$ one obtains the $4 \times 4$ table                                                                                                                                                                                                                             |     |     |     |     |           |           |     |           |     |
| cardona-qft-methods_F02963 | 281       | 0.971     | $\overline{\mathfrak{su}(2)}$                  | $\overline{\mathfrak{su}(2)}$                  | incorporates the generators of a $\mathfrak{su}(2,2)$ .                                                                                                                                                                                                                                                                             |     |     |     |     |           |           |     |           |     |
| cardona-qft-methods_F02964 | 281       | 0.574     | $p \, p \, p \, p$                             | $pppp$                                         | gives the $\mathfrak{su}(4)$ algebra, expressed in<br><table><tr><td><math>L</math></td><td><math>P</math></td><td><math>Q</math></td></tr><tr><td><math>K</math></td><td><math>\bar{L}</math></td><td><math>\bar{S}</math></td></tr><tr><td><math>S</math></td><td><math>\bar{Q}</math></td><td><math>R</math></td></tr></table> . | $L$ | $P$ | $Q$ | $K$ | $\bar{L}$ | $\bar{S}$ | $S$ | $\bar{Q}$ | $R$ |
| $L$                        | $P$       | $Q$       |                                                |                                                |                                                                                                                                                                                                                                                                                                                                     |     |     |     |     |           |           |     |           |     |
| $K$                        | $\bar{L}$ | $\bar{S}$ |                                                |                                                |                                                                                                                                                                                                                                                                                                                                     |     |     |     |     |           |           |     |           |     |
| $S$                        | $\bar{Q}$ | $R$       |                                                |                                                |                                                                                                                                                                                                                                                                                                                                     |     |     |     |     |           |           |     |           |     |
| cardona-qft-methods_F02965 | 281       | 0.574     | $k \, k$                                       | $kk$                                           | gives the $\mathfrak{su}(4)$ algebra, expressed in<br><table><tr><td><math>L</math></td><td><math>P</math></td><td><math>Q</math></td></tr><tr><td><math>K</math></td><td><math>\bar{L}</math></td><td><math>\bar{S}</math></td></tr><tr><td><math>S</math></td><td><math>\bar{Q}</math></td><td><math>R</math></td></tr></table> . | $L$ | $P$ | $Q$ | $K$ | $\bar{L}$ | $\bar{S}$ | $S$ | $\bar{Q}$ | $R$ |
| $L$                        | $P$       | $Q$       |                                                |                                                |                                                                                                                                                                                                                                                                                                                                     |     |     |     |     |           |           |     |           |     |
| $K$                        | $\bar{L}$ | $\bar{S}$ |                                                |                                                |                                                                                                                                                                                                                                                                                                                                     |     |     |     |     |           |           |     |           |     |
| $S$                        | $\bar{Q}$ | $R$       |                                                |                                                |                                                                                                                                                                                                                                                                                                                                     |     |     |     |     |           |           |     |           |     |
| cardona-qft-methods_F02966 | 281       | 0.574     | $\bar{L}$                                      | $\bar{L}$                                      | gives the $\mathfrak{su}(4)$ algebra, expressed in<br><table><tr><td><math>L</math></td><td><math>P</math></td><td><math>Q</math></td></tr><tr><td><math>K</math></td><td><math>\bar{L}</math></td><td><math>\bar{S}</math></td></tr><tr><td><math>S</math></td><td><math>\bar{Q}</math></td><td><math>R</math></td></tr></table> . | $L$ | $P$ | $Q$ | $K$ | $\bar{L}$ | $\bar{S}$ | $S$ | $\bar{Q}$ | $R$ |
| $L$                        | $P$       | $Q$       |                                                |                                                |                                                                                                                                                                                                                                                                                                                                     |     |     |     |     |           |           |     |           |     |
| $K$                        | $\bar{L}$ | $\bar{S}$ |                                                |                                                |                                                                                                                                                                                                                                                                                                                                     |     |     |     |     |           |           |     |           |     |
| $S$                        | $\bar{Q}$ | $R$       |                                                |                                                |                                                                                                                                                                                                                                                                                                                                     |     |     |     |     |           |           |     |           |     |
| cardona-qft-methods_F02967 | 281       | 0.967     | $\mathfrak{su}(2,2)$                           | $\mathfrak{su}(2,2)$                           |                                                                                                                                                                                                                                                                                                                                     |     |     |     |     |           |           |     |           |     |
| cardona-qft-methods_F02968 | 281       | 0.999     | $\mathfrak{psu}(2,2 4)$                        | $\mathfrak{psu}(2,2 4)$                        |                                                                                                                                                                                                                                                                                                                                     |     |     |     |     |           |           |     |           |     |
| cardona-qft-methods_F02969 | 281       | 0.887     | $\bar{Q}$                                      | $\bar{Q}$                                      |                                                                                                                                                                                                                                                                                                                                     |     |     |     |     |           |           |     |           |     |
| cardona-qft-methods_F02970 | 282       | 1.000     | $Q_{a \, \alpha}$                              | $Q_{a\alpha}$                                  | <i>Integrability and the AdS/CFT correspondence: 271</i>                                                                                                                                                                                                                                                                            |     |     |     |     |           |           |     |           |     |

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| cardona-qft-methods_F02971 | 282  | ■ 1.000 | $\bar{Q}_{\dot{\alpha}}^a$                            | $\bar{Q}_{\dot{\alpha}}^a$                              | <i>Integrability and the AdS/CFT correspondence</i> 271                                                                                                                               |
| cardona-qft-methods_F02972 | 282  | ■ 1.000 | $a \in \{1, \ldots, \mathcal{N}=4\}$                  | $a \in \{1, \ldots, \mathcal{N} = 4\}$                  | This algebra is called a <i>super algebra</i> and it is a symmetry of the $\mathcal{N} = 4$                                                                                           |
| cardona-qft-methods_F02973 | 282  | ■ 1.000 | $\{ \}^{7,18,19}$                                     | 7,18,19                                                 | <b>5.3. Local composite operators and the spect</b>                                                                                                                                   |
| cardona-qft-methods_F02974 | 282  | ■ 0.998 | $\mathcal{N}=4$ S Y M                                 | $\mathcal{N} = 4SYM$                                    | in a gauge in                                                                                                                                                                         |
| cardona-qft-methods_F02975 | 282  | ■ 0.999 | $\mathcal{O}$                                         | $\mathcal{O}$                                           | dimension has to be calculated perturbatively:                                                                                                                                        |
| cardona-qft-methods_F02976 | 282  | ■ 1.000 | $\Delta=\Delta_0+\gamma$                              | $\Delta = \Delta_0 + \gamma$                            | $\mathcal{N} = 4$ SYM amounts to finding all sets of eigenstates, $\mathcal{O}$ , and then scaling dimensions, $\Delta$ . In other words one would like to diagonalise the dilatation |
| cardona-qft-methods_F02977 | 282  | ■ 1.000 | $\mathcal{O}(x)$                                      | $\mathcal{O}(x)$                                        |                                                                                                                                                                                       |
| cardona-qft-methods_F02978 | 282  | ■ 1.000 | $\mathcal{N} =$                                       | $\mathcal{N} =$                                         |                                                                                                                                                                                       |
| cardona-qft-methods_F02979 | 283  | ■ 1.000 | $N=\infty$                                            | $N = \infty$                                            | the diagonalisation problem of a Heisenberg magnet, where the analogous set consists of the eigenfunction $ \psi\rangle$ and its energy, $E$ . The single trace                       |
| cardona-qft-methods_F02980 | 283  | ■ 1.000 | $\mathcal{O} =$                                       | $\mathcal{O} =$                                         | and thus one has $\Delta_0 = j$ and $\gamma = 0$ .                                                                                                                                    |
| cardona-qft-methods_F02981 | 283  | ■ 1.000 | $\operatorname{Tr}\left(Z^j\right)$                   | $\operatorname{Tr}\left(Z^j\right)$                     | $\langle (\bar{Z}^j)(x)(Z^j)(y) \rangle = \frac{1}{(x-y)^{2j}}$                                                                                                                       |
| cardona-qft-methods_F02982 | 283  | ■ 1.000 | $\left \uparrow^j\right\rangle$                       | $ \uparrow^j\rangle$                                    | For a non-trivial example take the operator $\mathcal{O} = \operatorname{Tr}(Z^{j-1} X)$ this situation                                                                               |
| cardona-qft-methods_F02983 | 283  | ■ 1.000 | $\Delta_0=j$                                          | $\Delta_0 = j$                                          | corresponds to the same Heisenberg:                                                                                                                                                   |
| cardona-qft-methods_F02984 | 283  | ■ 1.000 | $\gamma=0$                                            | $\gamma = 0$                                            | corresponds to the same Heisenberg:                                                                                                                                                   |
| cardona-qft-methods_F02985 | 283  | ■ 1.000 | $\mathcal{O}=\operatorname{Tr}\left(Z^{j-1} X\right)$ | $\mathcal{O} = \operatorname{Tr}\left(Z^{j-1} X\right)$ | by                                                                                                                                                                                    |
| cardona-qft-methods_F02986 | 283  | ■ 1.000 | $\left \uparrow^{j-1} \downarrow\right\rangle$        | $ \uparrow^{j-1}\downarrow\rangle$                      | Again, more detailed lectures on AdS/CFT integrability are found in. <sup>7</sup>                                                                                                     |
| cardona-qft-methods_F02987 | 283  | ■ 0.879 | $\boldsymbol{AdS_5} \times \boldsymbol{S^5}$          | $AdS_5 \times S^5$                                      |                                                                                                                                                                                       |
| cardona-qft-methods_F02988 | 284  | ■ 1.000 | $A d s_5$                                             | $AdS_5$                                                 |                                                                                                                                                                                       |
| cardona-qft-methods_F02989 | 284  | ■ 1.000 | $S_5$                                                 | $S_5$                                                   | <i>Integrability and the AdS/CFT correspondence</i> 213                                                                                                                               |
| cardona-qft-methods_F02990 | 284  | ■ 0.985 | $(-,+,+,+,-)$                                         | $(-,+,+,+,-)$                                           | constructed by embedding the strings on the 2D worldsheet into a space                                                                                                                |
| cardona-qft-methods_F02991 | 284  | ■ 0.535 | $(+,+,+,+,+)$                                         | $(+,+,+,+,+)$                                           | which is constructed by the direct product of Lorentzian hyperboloid Anti de Sitter and $S^5$                                                                                         |
| cardona-qft-methods_F02992 | 284  | ■ 0.562 | $A d S_5=\mathrm{SO}(4,2) / \mathrm{SO}(1,4)$         | $AdS_5 = \mathrm{SO}(4,2)/\mathrm{SO}(1,4)$             | observation the background space can be written as $AdS_5 = \mathrm{SO}(4,2)/\mathrm{SO}(1,4)$ and $S^5 = \mathrm{SO}(6)/\mathrm{SO}(5)$ . With this one can construct                |
| cardona-qft-methods_F02993 | 284  | ■ 0.718 | $S^5=\operatorname{SO}(6) / \operatorname{SO}(5)$     | $S^5 = \mathrm{SO}(6)/\mathrm{SO}(5)$                   | $\mathfrak{psu}(2,2 4) = \mathfrak{su}(2,2 4) \oplus \mathfrak{u}(1) \oplus \mathfrak{su}(2,2 4)_{\text{fermions}}$ $\widehat{\mathrm{PSU}}(2,2 4)$                                   |
| cardona-qft-methods_F02994 | 284  | ■ 0.935 | $\operatorname{PSU}(2,2 \mid 4)$                      | $\mathrm{PSU}(2,2 \mid 4)$                              | $\mathfrak{psu}(2,2 4) = \left  \begin{array}{c c} \mathfrak{u}(2,2) & \text{supercharges} \\ \hline \text{supercharges} & \mathfrak{u}(4) \end{array} \right $                       |
| cardona-qft-methods_F02995 | 284  | ■ 1.000 | $M \in\{0,1, \ldots, 5\}, N \in\{1, \ldots, 6\}$      | $M \in \{0,1, \ldots, 5\}, N \in \{1, \ldots, 6\}$      |                                                                                                                                                                                       |
| cardona-qft-methods_F02996 | 284  | ■ 1.000 | $1 / \alpha^{\prime}$                                 | $1/\alpha'$                                             |                                                                                                                                                                                       |
| cardona-qft-methods_F02997 | 284  | ■ 0.918 | $(+,+)$                                               | $(+,+)$                                                 |                                                                                                                                                                                       |

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| cardona-qft-methods_F02998 | 284  | ■ 0.918 | $\partial_0=\partial_\tau, \partial_1=\partial_\sigma$ | $\partial_0 = \partial_\tau, \partial_1 = \partial_\sigma$ |                                                                                                                                                                                                                                                                                         |
| cardona-qft-methods_F02999 | 284  | ■ 1.000 | $\alpha^\prime$                                        | $\alpha'$                                                  |                                                                                                                                                                                                                                                                                         |
| cardona-qft-methods_F03000 | 284  | ■ 0.999 | $\alpha^\prime \rightarrow 0$                          | $\alpha' \rightarrow 0$                                    |                                                                                                                                                                                                                                                                                         |
| cardona-qft-methods_F03001 | 285  | ■ 0.848 | $\mathcal{N}=4 \mathrm{SYM}$                           | $\mathcal{N} = 4\mathrm{SYM}$                              | in 1994 Maldacena made the remarkable observation that a type IIB string theory on an $AdS_5 \times S^5$ background is strictly equivalent to $\mathcal{N} = 4$ SYM. <sup>2</sup>                                                                                                       |
| cardona-qft-methods_F03002 | 285  | ■ 1.000 | $\{ \ }^{3,4}$                                         | 3,4                                                        | see e.g. <sup>3,4</sup> for explanations accessible to students. Note that this means a                                                                                                                                                                                                 |
| cardona-qft-methods_F03003 | 285  | ■ 0.996 | $\lambda=g_{\mathrm{YM}}^2 N^2$                        | $\lambda = g_{\mathrm{YM}} N^2$                            | side there is the radius of $AdS_5$ (or $S^5$ respectively), $R$ and the string                                                                                                                                                                                                         |
| cardona-qft-methods_F03004 | 285  | ■ 0.996 | $N-1$                                                  | $N - 1$                                                    | side there is the radius of $AdS_5$ (or $S^5$ respectively), $R$ and the string                                                                                                                                                                                                         |
| cardona-qft-methods_F03005 | 285  | ■ 0.915 | $g_{\mathrm{YM}}$                                      | $g_{\mathrm{YM}}$                                          | tension $\alpha'$ , which can also be connected to the string coupling constant $g$ . The $AdS/CFT$ correspondence now states that these parameters are                                                                                                                                 |
| cardona-qft-methods_F03006 | 285  | ■ 1.000 | $g_{\mathrm{s}}$                                       | $g_s$                                                      | Comparing the SYM action with the one of the string theory $\sigma$ -model one                                                                                                                                                                                                          |
| cardona-qft-methods_F03007 | 285  | ■ 1.000 | $N \rightarrow \infty$                                 | $N \rightarrow \infty$                                     |                                                                                                                                                                                                                                                                                         |
| cardona-qft-methods_F03008 | 285  | ■ 1.000 | $g_{\mathrm{s}} \rightarrow 0$                         | $g_s \rightarrow 0$                                        |                                                                                                                                                                                                                                                                                         |
| cardona-qft-methods_F03009 | 286  | ■ 1.000 | $\phi_1, \phi_2, \phi_3$                               | $\phi_1, \phi_2, \phi_3$                                   | $Y_1 + iY_2 = r_1 e^{i\varphi_1}, \quad Y_3 + iY_4 = r_2 e^{i\varphi_2}, \quad Y_5 + iY_6 = r_3 e^{i\varphi_3},$                                                                                                                                                                        |
| cardona-qft-methods_F03010 | 286  | ■ 1.000 | $J_1, J_2, J_3$                                        | $J_1, J_2, J_3$                                            | which is strikingly similar to the parametrisation of 3 complex scalars in                                                                                                                                                                                                              |
| cardona-qft-methods_F03011 | 286  | ■ 0.852 | $\frac{s}{6} \simeq \frac{s}{u(4)}$                    | $\mathfrak{so}(6) \simeq \mathfrak{su}(4)$                 | the $U(1)$ case. The three angles $\varphi_1, \varphi_2, \varphi_3$ , are associated with the string angular momenta $J_1, J_2, J_3$ and correspond to the 3 <i>Cartan generators</i> of                                                                                                |
| cardona-qft-methods_F03012 | 286  | ■ 1.000 | $t, \alpha_1, \alpha_2$                                | $t, \alpha_1, \alpha_2$                                    | This time the three angles $t, \alpha_1, \alpha_2$ give the three conserved quantities $L_z, S_z$ and $S_y$ and they come from the three <i>Cartan generators</i> of $\mathfrak{so}(2,4) \simeq$                                                                                        |
| cardona-qft-methods_F03013 | 286  | ■ 1.000 | $S_1$                                                  | $S_1$                                                      | $\mathfrak{su}(2,2)$ . In the table from section 5.2 they belong to                                                                                                                                                                                                                     |
| cardona-qft-methods_F03014 | 286  | ■ 1.000 | $\mathfrak{so}(2,4) \simeq$                            | $\mathfrak{so}(2,4) \simeq$                                | $\mathfrak{su}(2,2)$ . In the table from section 5.2 they belong to                                                                                                                                                                                                                     |
| cardona-qft-methods_F03015 | 286  | ■ 0.474 | $k \ k \ k$                                            | $kkk$                                                      | $\begin{array}{ c c } \hline k & k \\ \hline k & k \\ \hline \end{array} \quad \bar{L}$ .<br>erator one has<br>t that there are<br>ey prediction $o$ .                                                                                                                                  |
| cardona-qft-methods_F03016 | 286  | ■ 0.954 | $30 \ \mathrm{mid} \ 32$                               | $30 \mid 32$                                               |                                                                                                                                                                                                                                                                                         |
| cardona-qft-methods_F03017 | 287  | ■ 0.514 | $\frac{s}{2} \ \mathrm{mid} \ 2$                       | $\mathfrak{su}(2 \mid 2)$                                  | e organised in a $8 \times 8$ matrix:<br><br><div> <div> <div><math>\mathfrak{su}(2 2)</math></div> <div>create<br/><math>8_B + 8_F</math><br/>magnons</div> </div> <div> <div>destroy<br/><math>8 + 8</math></div> <div><math>\overline{\mathfrak{su}(2 2)}</math></div> </div> </div> |

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| cardona-qft-methods_F03018 | 287  | 0.514 | $8_{\mathrm{B}}+8_{\mathrm{F}}$                  | $8_B + 8_F$                              | <p>e organised in a <math>8 \times 8</math> matri:</p>                                                                                                    |
| cardona-qft-methods_F03019 | 287  | 0.514 | $\overline{\mathfrak{s} \mathfrak{u}(2 \mid 2)}$ | $\overline{\mathfrak{su}(2 \mid 2)}$     | <p>e organised in a <math>8 \times 8</math> matri:</p>                                                                                                    |
| cardona-qft-methods_F03020 | 287  | 1.000 | $\dot{A}$                                        | $\dot{A}$                                | ite states, e.g.: $Z \sim A A + Y Y$ or $F \sim D D$ .                                                                                                    |
| cardona-qft-methods_F03021 | 287  | 1.000 | $\bar{Z} \sim X \bar{X} + Y \bar{Y}$             | $\bar{Z} \sim X \bar{X} + Y \bar{Y}$     | References                                                                                                                                                |
| cardona-qft-methods_F03022 | 287  | 1.000 | $F \sim D D$                                     | $F \sim D D$                             | References                                                                                                                                                |
| cardona-qft-methods_F03023 | 288  | 1.000 | $\mathrm{AdS}_5 \times \mathrm{S}^5$             | $\mathrm{AdS}_5 \times \mathrm{S}^5$     |                                                                                                                                                           |
| cardona-qft-methods_F03024 | 289  | 0.747 | $\left(\mathrm{ple}^1 - {}^4\right)$             | $\mathrm{ple}^1 - {}^4$                  |                                                                                                                                                           |
| cardona-qft-methods_F03025 | 290  | 1.000 | $(2,0)$                                          | $(2,0)$                                  | Compactifications of string theory and generalized geometries 219                                                                                         |
| cardona-qft-methods_F03026 | 290  | 0.991 | $\{ \}^{\{5\}} - \{ \}^{\{7\}}$                  | ${}^5{}_7$                               | Model and Einstein’s gravity are embedded as low-energy limits in string                                                                                  |
| cardona-qft-methods_F03027 | 290  | 0.422 | $\{ \}^{\{8\}} - , ^{\{13\}}$                    | ${}^8_{13}$                              | duction of D-branes, compactifications of the type-II string theory involving orientifolds and intersecting D-branes became the center of attention (same |
| cardona-qft-methods_F03028 | 291  | 0.502 | $\{ \}^{\{19\}} - \{ \}^{\{21\}}$                | ${}^{19}_{21}$                           | string compactifications, T-duality among others. In particular, mirror sym-                                                                              |
| cardona-qft-methods_F03029 | 292  | 0.999 | $(0 \leq \sigma \leq \pi$                        | $(0 \leq \sigma \leq \pi$                | for an open string and $0 \leq \sigma \leq 2\pi$ for a closed string with 0 and $2\pi$ identified) and $\pi$                                              |
| cardona-qft-methods_F03030 | 292  | 1.000 | $0 \leq \sigma \leq 2 \pi$                       | $0 \leq \sigma \leq 2\pi$                | The evolution of the worldsheet is given by the “Polyakov” action                                                                                         |
| cardona-qft-methods_F03031 | 292  | 1.000 | $2 \pi$                                          | $2\pi$                                   | The evolution of the worldsheet is given by the “Polyakov” action                                                                                         |
| cardona-qft-methods_F03032 | 292  | 1.000 | $M=0, \ldots, D-1 ; \alpha=\tau, \sigma$         | $M=0, \ldots, D-1 ; \alpha=\tau, \sigma$ |                                                                                                                                                           |
| cardona-qft-methods_F03033 | 292  | 1.000 | $X^{\mathrm{M}}$                                 | $X^M$                                    |                                                                                                                                                           |
| cardona-qft-methods_F03034 | 292  | 1.000 | $X^{\mu}(\tau, \sigma)$                          | $X^{\mu}(\tau, \sigma)$                  |                                                                                                                                                           |
| cardona-qft-methods_F03035 | 292  | 1.000 | $\gamma^{\alpha \beta}$                          | $\gamma^{\alpha \beta}$                  |                                                                                                                                                           |
| cardona-qft-methods_F03036 | 292  | 1.000 | $\eta_{\mathrm{M} \mathrm{N}}$                   | $\eta_{MN}$                              |                                                                                                                                                           |
| cardona-qft-methods_F03037 | 293  | 1.000 | $l_{\mathrm{s}}$                                 | $l_s$                                    |                                                                                                                                                           |
| cardona-qft-methods_F03038 | 293  | 0.999 | $M_{\mathrm{s}}$                                 | $M_s$                                    |                                                                                                                                                           |
| cardona-qft-methods_F03039 | 293  | 0.999 | $c=\hbar=1$                                      | $c=\hbar=1$                              |                                                                                                                                                           |

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| cardona-qft-methods_F03040 | 293  | ■ 1.000 | $\hbar / c$                                                                                     | $\hbar/c$                                                 |                                                                                                                                                                                                  |
| cardona-qft-methods_F03041 | 293  | ■ 1.000 | $\sigma^{\pm}=\tau\pm\sigma$                                                                    | $\sigma^{\pm}=\tau\pm\sigma$                              | By introducing left- and right-moving worldsheet coordinates $\sigma^{\pm}=\tau\pm\sigma$ ,                                                                                                      |
| cardona-qft-methods_F03042 | 293  | ■ 0.999 | $X^{\rm M}(\tau,0)=X^{\rm M}(\tau,2\pi),\,X^{\rm \prime M}(\tau,0)=X^{\rm \prime M}(\tau,2\pi)$ | $X^M(\tau,0)=X^M(\tau,2\pi),X'^M(\tau,0)=X'^M(\tau,2\pi)$ | $X_R^M(\sigma^-)=\frac{1}{2}x^M+\alpha'p^Mp^--i\left(\frac{\alpha'}{2}\right)^{\frac{1}{2}}\sum_n\frac{1}{n}\alpha_n^Me^{-2in\sigma^-}$                                                          |
| cardona-qft-methods_F03043 | 293  | ■ 1.000 | $x^{\rm M}$                                                                                     | $x^M$                                                     | $\mathbf{v}=\frac{1}{2}(\mathbf{v}^L+\mathbf{v}^R),\quad\mathbf{x}=\frac{1}{2}(\mathbf{x}^L+\mathbf{x}^R)\quad(7)$                                                                               |
| cardona-qft-methods_F03044 | 293  | ■ 1.000 | $p^{\rm M}$                                                                                     | $p^M$                                                     | $\mathbf{v}=\frac{1}{2}(\mathbf{v}^L+\mathbf{v}^R),\quad\mathbf{x}=\frac{1}{2}(\mathbf{x}^L+\mathbf{x}^R)\quad(7)$                                                                               |
| cardona-qft-methods_F03045 | 293  | ■ 1.000 | $\alpha_{-n}^{\rm M}=\left(\alpha_n^{\rm M}\right)^*$                                           | $\alpha_{-n}^M=\left(\alpha_n^M\right)^*$                 |                                                                                                                                                                                                  |
| cardona-qft-methods_F03046 | 293  | ■ 1.000 | $\tilde{\alpha}_{-n}^{\rm M}=\left(\tilde{\alpha}_n^{\rm M}\right)^*$                           | $\tilde{\alpha}_{-n}^M=\left(\tilde{\alpha}_n^M\right)^*$ |                                                                                                                                                                                                  |
| cardona-qft-methods_F03047 | 293  | ■ 1.000 | $\gamma_{\alpha\beta}$                                                                          | $\gamma_{\alpha\beta}$                                    |                                                                                                                                                                                                  |
| cardona-qft-methods_F03048 | 294  | ■ 1.000 | $\alpha_n^{\rm M},\,\tilde{\alpha}_n^{\rm M}$                                                   | $\alpha_n^M,\tilde{\alpha}_n^M$                           | $[\alpha_-^P,\alpha_-^Q]=[\tilde{\alpha}_-^P,\tilde{\alpha}_-^Q]=n\delta_{n+m}n^{PQ}\,,\quad[x^P,v^Q]=in^{PQ}\,,\quad(9)$                                                                        |
| cardona-qft-methods_F03049 | 294  | ■ 1.000 | $\alpha_{-n}^{\rm M},\,\tilde{\alpha}_{-n}^{\rm M}$                                             | $\alpha_{-n}^M,\tilde{\alpha}_{-n}^M$                     | $[\alpha_-^P,\alpha_-^Q]=[\tilde{\alpha}_-^P,\tilde{\alpha}_-^Q]=n\delta_{n+m}n^{PQ}\,,\quad[x^P,v^Q]=in^{PQ}\,,\quad(9)$                                                                        |
| cardona-qft-methods_F03050 | 294  | ■ 0.890 | $\mathrm{in}^1-\mathrm{}^3$                                                                     | $\mathrm{in}^1-\mathrm{}^3$                               | $\tilde{L}_0=\frac{1}{2}\left(\sum\alpha_n\cdot\alpha_{-n}-\tilde{\alpha}_n\cdot\tilde{\alpha}_{-n}\right)\quad(11)$                                                                             |
| cardona-qft-methods_F03051 | 294  | ■ 1.000 | $N=\tilde{N}$                                                                                   | $N=\tilde{N}$                                             | $S(n,1)=S(n,1)+1\quad(10)$ <p>and a center of mass momentum <math>k^M=k\cdot k=0</math>. It is not hard to check</p>                                                                             |
| cardona-qft-methods_F03052 | 294  | ■ 1.000 | $N,\,\tilde{N}$                                                                                 | $N,\tilde{N}$                                             | that the norm of the state is positive only if $\xi,\tilde{\xi}$ are space-like vectors. The                                                                                                     |
| cardona-qft-methods_F03053 | 294  | ■ 1.000 | $k^{\rm M},\,k\cdot k=0$                                                                        | $k^M,k\cdot k=0$                                          | $F_{\mu\nu\lambda} = F_{\mu\nu}^0 + F_{\mu\nu}^1 + F_{\mu\nu}^2 + \dots\quad(13)$                                                                                                                |
| cardona-qft-methods_F03054 | 294  | ■ 1.000 | $\xi,\,\tilde{\xi}$                                                                             | $\xi,\tilde{\xi}$                                         | $S_{\rm eff} = S_{\rm eff}^{\rm bos} + S_{\rm eff}^{\rm ferm} + S_{\rm eff}^{\rm grav} + S_{\rm eff}^{\rm gauge} + S_{\rm eff}^{\rm ghost}$                                                      |
| cardona-qft-methods_F03055 | 294  | ■ 1.000 | $\xi\cdot k=\tilde{\xi}\cdot k=0$                                                               | $\xi\cdot k=\tilde{\xi}\cdot k=0$                         | <sup>a</sup> Note that the convention used here for creation (positive modes) and annihilation operators (negative modes) is opposite to the one used most often (for example in <sup>2</sup> ). |
| cardona-qft-methods_F03056 | 294  | ■ 0.999 | $\vec{k}=(k,k,0,\ldots,0)$                                                                      | $\vec{k}=(k,k,0,\ldots,0)$                                |                                                                                                                                                                                                  |
| cardona-qft-methods_F03057 | 294  | ■ 0.999 | $\xi_{\rm M},\,\tilde{\xi}_{\rm M}$                                                             | $\xi_M,\tilde{\xi}_M$                                     |                                                                                                                                                                                                  |
| cardona-qft-methods_F03058 | 294  | ■ 0.999 | $2,\,\ldots,\,D-1$                                                                              | $2,\ldots,D-1$                                            |                                                                                                                                                                                                  |
| cardona-qft-methods_F03059 | 294  | ■ 0.987 | $\operatorname{SO}(D-2)$                                                                        | $\operatorname{SO}(D-2)$                                  |                                                                                                                                                                                                  |
| cardona-qft-methods_F03060 | 294  | ■ 0.987 | $\xi_{\rm M\,N}\equiv\xi_{\rm M}\,\tilde{\xi}_{\rm N}$                                          | $\xi_{MN}\equiv\xi_M\tilde{\xi}_N$                        |                                                                                                                                                                                                  |
| cardona-qft-methods_F03061 | 295  | ■ 1.000 | $\xi_{\rm M\,N}$                                                                                | $\xi_{MN}$                                                | $\xi_{MN}=\frac{1}{2}(\xi_{MN}-\xi_{NM}).$                                                                                                                                                       |
| cardona-qft-methods_F03062 | 295  | ■ 1.000 | $\xi_{\rm t}$                                                                                   | $\xi_t$                                                   | The state corresponding to the polarization $\xi_{MN}$ is a massless state of spin 0, the graviton. The state corresponding to the index $\xi$ is the dilaton, which                             |
| cardona-qft-methods_F03063 | 295  | ■ 0.968 | $-\frac{4}{\alpha'}$                                                                            | $-\frac{4}{\alpha'}$                                      | is $-\frac{1}{\alpha'}$ ). The appearance of this tachyon means that the bosonic string is unstable and will condensate to the true vacuum of the theory. We will see                            |
| cardona-qft-methods_F03064 | 295  | ■ 1.000 | $G_{\rm M\,N}(X)$                                                                               | $G_{MN}(X)$                                               | $\omega=\int\omega_X\omega_Y\omega_Z\quad(14)$                                                                                                                                                   |
| cardona-qft-methods_F03065 | 295  | ■ 1.000 | $G_{\rm M\,N}=$                                                                                 | $G_{MN}=$                                                 | we find that the additional terms correspond to inserting a graviton emis-                                                                                                                       |

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| cardona-qft-methods_F03066 | 295  | ■ 1.000 | $\eta_{\{M\ N\}}+h_{\{M\ N\}}(X)$                                                 | $\eta_{MN} + h_{MN}(X)$                                                 | sion vertex operator with $h_{MN} \propto \xi_{MN}^s$ . Thus, a background metric is                                                                                                                                                                                   |
| cardona-qft-methods_F03067 | 295  | ■ 1.000 | $h_{\{M\ N\}} \propto \xi_{\{M\ N\}}^s$                                           | $h_{MN} \propto \xi_{MN}^s$                                             |                                                                                                                                                                                                                                                                        |
| cardona-qft-methods_F03068 | 296  | ■ 1.000 | $\hbar$                                                                           | $\hbar$                                                                 |                                                                                                                                                                                                                                                                        |
| cardona-qft-methods_F03069 | 296  | ■ 0.986 | $e^{\{-\Phi\} \chi}$                                                              | $e^{-\Phi\chi}$                                                         | in the path integral (29), the resulting amplitudes are weighted by a factor $e^{-\Phi\chi}$ . On a 2-dimensional surface with $h$ handles, $b$ boundaries and $c$ cross-                                                                                              |
| cardona-qft-methods_F03070 | 296  | ■ 1.000 | $\chi=2-2\ h-b-c$                                                                 | $\chi = 2 - 2h - b - c$                                                 | caps, the Euler number is $\chi = 2 - 2h - b - c$ . An emission and reabsorption                                                                                                                                                                                       |
| cardona-qft-methods_F03071 | 296  | ■ 1.000 | $\delta\chi=-2$                                                                   | $\delta\chi = -2$                                                       | tree level closed string diagram, the amplitudes are weighted by $e^{2\Phi}$ . This                                                                                                                                                                                    |
| cardona-qft-methods_F03072 | 296  | ■ 1.000 | $e^{\{2\} \Phi}$                                                                  | $e^{2\Phi}$                                                             | means that the closed string coupling is not a free parameter, but it is given by the VEV of one of the background fields: $a_s = e^{\langle\Phi\rangle}$ . Therefore, the only                                                                                        |
| cardona-qft-methods_F03073 | 296  | ■ 0.992 | $g_{\{s\}}=e^{\{\langle\Phi\rangle\}}$                                            | $g_s = e^{\langle\Phi\rangle}$                                          | scale $(1/\sqrt{\alpha'})$ . However, without the introduction of fluxes the VEV of the dilaton is undetermined, as the action does not contain a potential term for                                                                                                   |
| cardona-qft-methods_F03074 | 296  | ■ 1.000 | $\left(1/\sqrt{\alpha'}\right)$                                                   | $\left(1/\sqrt{\alpha'}\right)$                                         | fluxes, as the latter generate a potential whose minimum determines the dilaton VEV                                                                                                                                                                                    |
| cardona-qft-methods_F03075 | 297  | ■ 1.000 | $H_{\{M\ P\ Q\}}$                                                                 | $H_{MPQ}$                                                               | these beta-functions to zero, we obtain <i>spacetime</i> equations for the fields.                                                                                                                                                                                     |
| cardona-qft-methods_F03076 | 297  | ■ 1.000 | $B_{\{P\ Q\}}$                                                                    | $B_{PQ}$                                                                | these beta-functions to zero, we obtain <i>spacetime</i> equations for the fields.                                                                                                                                                                                     |
| cardona-qft-methods_F03077 | 297  | ■ 1.000 | $H_{\{M\ P\ Q\}}=\partial_{[M} B_{\{P\ Q\}}$                                      | $H_{MPQ} = \partial_{[M} B_{PQ]}$                                       | these beta-functions to zero, we obtain <i>spacetime</i> equations for the fields.                                                                                                                                                                                     |
| cardona-qft-methods_F03078 | 297  | ■ 1.000 | $1/\sqrt{\alpha'}$                                                                | $1/\sqrt{\alpha'}$                                                      | $-\frac{2(D-26)}{2\pi\alpha'} + O(\alpha') \Big] \ . \qquad (21)$                                                                                                                                                                                                      |
| cardona-qft-methods_F03079 | 297  | ■ 1.000 | $D \neq 26$                                                                       | $D \neq 26$                                                             |                                                                                                                                                                                                                                                                        |
| cardona-qft-methods_F03080 | 297  | ■ 0.992 | $\left(\Phi=V_M X^{\{M\}}, V ^2=(D-26)/\alpha'\right)$                            | $\Phi = V_M X^M,  V ^2 = (D - 26)/\alpha'$                              |                                                                                                                                                                                                                                                                        |
| cardona-qft-methods_F03081 | 298  | ■ 1.000 | $\sigma=0,\ \pi$                                                                  | $\sigma = 0, \pi$                                                       | One first remark is that the massless closed string spectrum does not                                                                                                                                                                                                  |
| cardona-qft-methods_F03082 | 298  | ■ 1.000 | $\partial_{\sigma} X^{\{M\}}(\tau,\ 0)=\partial_{\sigma} X^{\{M\}}(\tau,\ \pi)=0$ | $\partial_{\sigma} X^M(\tau, 0) = \partial_{\sigma} X^M(\tau, \pi) = 0$ | contain a regular (one-form) gauge field. There is however a gauge field in the spectrum of massless open strings. The solution to the (worldsheet)                                                                                                                    |
| cardona-qft-methods_F03083 | 298  | ■ 1.000 | $A_{\{M\}}$                                                                       | $A_M$                                                                   | $\int_{\partial\Sigma} d\tau A_M \partial_{\tau} X^M \qquad (25)$                                                                                                                                                                                                      |
| cardona-qft-methods_F03084 | 298  | ■ 1.000 | $F=d\ A$                                                                          | $F = dA$                                                                | sigma-model action then depends on a $p+1$ dim. vector plus $D-p-1$ scalars on the boundary of the open string, similar to (25). It turns out that similar                                                                                                             |
| cardona-qft-methods_F03085 | 298  | ■ 1.000 | $g_{\{o\}}=e^{\{\Phi/2\}}$                                                        | $g_o = e^{\Phi/2}$                                                      | on the boundary of the open string, similar to (25), it turns out that similar to the closed string, the VEV of these scalars determines the position of                                                                                                               |
| cardona-qft-methods_F03086 | 298  | ■ 1.000 | $g_{\{s\}}$                                                                       | $g_s$                                                                   | the Dp-brane.                                                                                                                                                                                                                                                          |
| cardona-qft-methods_F03087 | 298  | ■ 0.997 | $p+1$                                                                             | $p + 1$                                                                 | Chan-Paton factors and the quantum numbers turn out to form the adjoint of a non-Abelian gauge group. In the simplest case, the Chan-Paton fac-                                                                                                                        |
| cardona-qft-methods_F03088 | 298  | ■ 0.997 | $\mathrm{D}\ p$                                                                   | $Dp$                                                                    | Chan-Paton factors and the quantum numbers turn out to form the adjoint of a non-Abelian gauge group. In the simplest case, the Chan-Paton fac-                                                                                                                        |
| cardona-qft-methods_F03089 | 298  | ■ 0.829 | $D-p-1$                                                                           | $D - p - 1$                                                             | tors give rise to a $U(n)$ gauge group. Such non-Abelian gauge groups can                                                                                                                                                                                              |
| cardona-qft-methods_F03090 | 298  | ■ 1.000 | $U(n)$                                                                            | $U(n)$                                                                  |                                                                                                                                                                                                                                                                        |
| cardona-qft-methods_F03091 | 299  | ■ 1.000 | $\Psi^{\{M\}}$                                                                    | $\Psi^M$                                                                | $\mathfrak{L} = \frac{1}{2} \int d^2\sigma \dots \left[ \frac{1}{2} \partial_{\alpha} Y^M \tilde{\partial}_{\dot{\alpha}} Y^N \pm \partial_{\alpha}^M \tilde{\partial}_{\dot{\alpha}} N \pm \tilde{\partial}_{\dot{\alpha}}^M \partial_{\alpha} N \right] \qquad (27)$ |
| cardona-qft-methods_F03092 | 299  | ■ 1.000 | $\tilde{\Psi}^{\{M\}}$                                                            | $\tilde{\Psi}^M$                                                        | $\mathfrak{L} = \frac{1}{2} \int d^2\sigma \dots \left[ \frac{1}{2} \partial_{\alpha} Y^M \tilde{\partial}_{\dot{\alpha}} Y^N \pm \partial_{\alpha}^M \tilde{\partial}_{\dot{\alpha}} N \pm \tilde{\partial}_{\dot{\alpha}}^M \partial_{\alpha} N \right] \qquad (27)$ |



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| cardona-qft-methods_F03119 | 301  | ■ 1.000 | $C_{\{4\}}$                                                                                                                          | $C_4$                                                      |       |
| cardona-qft-methods_F03120 | 301  | ■ 1.000 | $d\,C_{\{4\}}=* \,d\,C_{\{4\}}$                                                                                                      | $dC_4 = *dC_4$                                             |       |
| cardona-qft-methods_F03121 | 301  | ■ 0.988 | $\lambda^A(\mathrm{\sim A}=1,2)$                                                                                                     | $\lambda^A(\,A=1,2)$                                       |       |
| cardona-qft-methods_F03122 | 301  | ■ 0.984 | $\mathbf{56}, \Psi^{A,M}$                                                                                                            | <b>56</b> , $\Psi^{A,M}$                                   |       |
| cardona-qft-methods_F03123 | 302  | ■ 1.000 | $S^{\{1\}}$                                                                                                                          | $S^1$                                                      |       |
| cardona-qft-methods_F03124 | 302  | ■ 1.000 | $1\,/\,R$                                                                                                                            | $1/R$                                                      |       |
| cardona-qft-methods_F03125 | 302  | ■ 1.000 | $X^{\{M\}}(\tau,2\,\pi)=$                                                                                                            | $X^M(\tau,2\pi)=$                                          |       |
| cardona-qft-methods_F03126 | 302  | ■ 1.000 | $X^{\{M\}}(\tau,0)+2\,\pi\,m\,R$                                                                                                     | $X^M(\tau,0)+2\pi mR$                                      |       |
| cardona-qft-methods_F03127 | 302  | ■ 0.943 | $\alpha^{\{\prime\}}$ / R!                                                                                                           | $\alpha'/R!$                                               |       |
| cardona-qft-methods_F03128 | 303  | ■ 1.000 | $X_{\{L\}}^{\{M\}}+X_{\{R\}}^{\{M\}}$                                                                                                | $X_L^M+X_R^M$                                              |       |
| cardona-qft-methods_F03129 | 303  | ■ 1.000 | $X_{\{L\}}^{\{M\}}-X_{\{R\}}^{\{M\}}$                                                                                                | $X_L^M-X_R^M$                                              |       |
| cardona-qft-methods_F03130 | 303  | ■ 0.978 | $\psi^{\{M\}}$                                                                                                                       | $\psi^M$                                                   |       |
| cardona-qft-methods_F03131 | 303  | ■ 0.978 | $\bar{\psi}^{\{M\}}$                                                                                                                 | $\bar{\psi}^M$                                             |       |
| cardona-qft-methods_F03132 | 303  | ■ 1.000 | $O(n,n,\,\mathbb{Z})$                                                                                                                | $O(n,n,\mathbb{Z})$                                        |       |
| cardona-qft-methods_F03133 | 303  | ■ 1.000 | $O(8,8)$                                                                                                                             | $O(8,8)$                                                   |       |
| cardona-qft-methods_F03134 | 303  | ■ 0.960 | $G\,\mathfrak{l}(8)$                                                                                                                 | $Gl(8)$                                                    |       |
| cardona-qft-methods_F03135 | 303  | ■ 1.000 | $O(8)$                                                                                                                               | $O(8)$                                                     |       |
| cardona-qft-methods_F03136 | 303  | ■ 1.000 | $\{ \, \}^{\{\text{e} \, \}}$                                                                                                        | $\mathbf{e}$                                               |       |
| cardona-qft-methods_F03137 | 303  | ■ 0.995 | $\mathrm{D}(p\pm1)$                                                                                                                  | $D(p\pm1)$                                                 |       |
| cardona-qft-methods_F03138 | 303  | ■ 1.000 | $O(10,10)$                                                                                                                           | $O(10,10)$                                                 |       |
| cardona-qft-methods_F03139 | 304  | ■ 0.998 | $T_{\{p\}=8\,\pi^{\{7\}}\,\alpha^{\{\prime\,7/2\}}\left(4\pi^2\alpha^{\{2\}}\right.\left.\alpha^{\{\prime\,7/2\}}\right)^{-p\,/\,2}$ | $T_p=8\pi^7\alpha'^{7/2}\left(4\pi^2\alpha'\right)^{-p/2}$ |       |
| cardona-qft-methods_F03140 | 304  | ■ 0.989 | $S\,U(3)\,\times\,S\,U(2)\,\times$                                                                                                   | $SU(3)\times SU(2)\times$                                  |       |
| cardona-qft-methods_F03141 | 305  | ■ 1.000 | $O(n,\,n)$                                                                                                                           | $O(n,n)$                                                   |       |
| cardona-qft-methods_F03142 | 305  | ■ 1.000 | $\{ \, \}^{\{19,20\}}$                                                                                                               | $19,20$                                                    |       |
| cardona-qft-methods_F03143 | 305  | ■ 1.000 | $\{ \, \}^{\{22,23\}}$                                                                                                               | $22,23$                                                    |       |
| cardona-qft-methods_F03144 | 305  | ■ 0.998 | $T^{\{*\}}$                                                                                                                          | $T^*$                                                      |       |
| cardona-qft-methods_F03145 | 305  | ■ 1.000 | $T\,\oplus\,T^{\{*\}}$                                                                                                               | $T\oplus T^*$                                              |       |

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| cardona-qft-methods_F03146 | 305  | ■ 1.000 | X+\zeta                                                                             | $X + \zeta$                                                                             |                                                                                                                                                                                                                                                                                                          |
| cardona-qft-methods_F03147 | 306  | ■ 1.000 | n=6                                                                                 | $n = 6$                                                                                 |                                                                                                                                                                                                                                                                                                          |
| cardona-qft-methods_F03148 | 306  | ■ 1.000 | 0(6,6)                                                                              | $O(6,6)$                                                                                | The transformations that preserve this metric are $O(n,n)$ , where $n$ is the                                                                                                                                                                                                                            |
| cardona-qft-methods_F03149 | 306  | ■ 1.000 | U_{\alpha}                                                                          | $U_\alpha$                                                                              | also use such two-form shifts $b_{(\alpha\beta)}$ to fiber $T^*$ over $T$ . In total, the patching                                                                                                                                                                                                       |
| cardona-qft-methods_F03150 | 306  | ■ 1.000 | U_{\beta}                                                                           | $U_\beta$                                                                               | also use such two-form shifts $b_{(\alpha\beta)}$ to fiber $T^*$ over $T$ . In total, the patching                                                                                                                                                                                                       |
| cardona-qft-methods_F03151 | 306  | ■ 0.460 | \mathrm{Gl}(6)                                                                      | $\mathrm{Gl}(6)$                                                                        | reads                                                                                                                                                                                                                                                                                                    |
| cardona-qft-methods_F03152 | 306  | ■ 0.460 | a_{(\alpha\beta)}                                                                   | $a_{(\alpha\beta)}$                                                                     | reads                                                                                                                                                                                                                                                                                                    |
| cardona-qft-methods_F03153 | 306  | ■ 0.998 | b_{(\alpha\beta)}                                                                   | $b_{(\alpha\beta)}$                                                                     | $x_{(\alpha)} + \xi_{(\alpha)} = a_{(\alpha\beta)} \cdot x_{(\beta)} + \left[ a_{(\alpha\beta)}^{-t} \xi_{(\beta)} - \iota_{a_{(\alpha\beta)} x_{(\beta)}} b_{(\alpha\beta)} \right], \quad (4)$                                                                                                         |
| cardona-qft-methods_F03154 | 306  | ■ 1.000 | a^{-T}=\left(a^{-1}\right)^T                                                        | $a^{-T} = \left(a^{-1}\right)^T$                                                        | on any triple intersection $U_\alpha \cap U_\beta \cap U_\gamma$ and $y_{\alpha\beta\gamma} := u$ is a $U(1)$ element. This is analogous to the patching of a $U(1)$ bundle, except that the transi-                                                                                                     |
| cardona-qft-methods_F03155 | 306  | ■ 1.000 | b_{(\alpha\beta)}=-\mathrm{d}\Lambda_{(\alpha\beta)}                                | $b_{(\alpha\beta)} = -\mathrm{d}\Lambda_{(\alpha\beta)}$                                | tion “functions” are one-forms, $\Lambda_{(\alpha\beta)}$ . Formally it is called the “structure” of a gauge                                                                                                                                                                                             |
| cardona-qft-methods_F03156 | 306  | ■ 1.000 | \Lambda_{(\alpha\beta)}                                                             | $\Lambda_{(\alpha\beta)}$                                                               | tion “functions” are one-forms, $\Lambda_{(\alpha\beta)}$ . Formally it is called the “structure” of a gauge                                                                                                                                                                                             |
| cardona-qft-methods_F03157 | 306  | ■ 1.000 | U_{\alpha} \cap U_{\beta} \cap U_{\gamma}                                           | $U_\alpha \cap U_\beta \cap U_\gamma$                                                   | structure $\mathcal{J}$ , which is a map of $E$ to itself that squares to $-\mathbb{I}_{2d}$ ( $d$ is real the dimension of the manifold). This is analogous to an almost complex                                                                                                                        |
| cardona-qft-methods_F03158 | 306  | ■ 1.000 | g_{\alpha\beta\gamma}:=\mathrm{i}\alpha                                             | $g_{\alpha\beta\gamma} := \mathrm{i}\alpha$                                             | structure $\mathcal{J}$ , which is a map of $E$ to itself that squares to $-\mathbb{I}_{2d}$ ( $d$ is real the dimension of the manifold). This is analogous to an almost complex                                                                                                                        |
| cardona-qft-methods_F03159 | 306  | ■ 1.000 | \mathcal{J}                                                                         | $\mathcal{J}$                                                                           | $\mathcal{J}_1 = \begin{pmatrix} \cdot & \varpi \\ 0 & -I^t \end{pmatrix}, \quad (6)$                                                                                                                                                                                                                    |
| cardona-qft-methods_F03160 | 306  | ■ 1.000 | -\mathbb{I}_{2d}                                                                    | $-\mathbb{I}_{2d}$                                                                      | $\mathcal{J}_1 = \begin{pmatrix} \cdot & \varpi \\ 0 & -I^t \end{pmatrix}, \quad (6)$                                                                                                                                                                                                                    |
| cardona-qft-methods_F03161 | 306  | ■ 1.000 | I_m^n                                                                               | $I_m^n$                                                                                 |                                                                                                                                                                                                                                                                                                          |
| cardona-qft-methods_F03162 | 306  | ■ 1.000 | -\mathbb{I}_d                                                                       | $-\mathbb{I}_d$                                                                         |                                                                                                                                                                                                                                                                                                          |
| cardona-qft-methods_F03163 | 306  | ■ 1.000 | \mathcal{J}^t \mathcal{G} \mathcal{J} = \mathcal{G}                                 | $\mathcal{J}^t \mathcal{G} \mathcal{J} = \mathcal{G}$                                   |                                                                                                                                                                                                                                                                                                          |
| cardona-qft-methods_F03164 | 306  | ■ 1.000 | \mathcal{G}                                                                         | $\mathcal{G}$                                                                           |                                                                                                                                                                                                                                                                                                          |
| cardona-qft-methods_F03165 | 307  | ■ 1.000 | I^2=-\mathbb{I}_d                                                                   | $I^2 = -\mathbb{I}_d$                                                                   |                                                                                                                                                                                                                                                                                                          |
| cardona-qft-methods_F03166 | 307  | ■ 1.000 | \mathcal{J}^2=-\mathbb{I}_{2d}, \mathcal{J}^t \mathcal{G} \mathcal{J} = \mathcal{G} | $\mathcal{J}^2 = -\mathbb{I}_{2d}, \mathcal{J}^t \mathcal{G} \mathcal{J} = \mathcal{G}$ |                                                                                                                                                                                                                                                                                                          |
| cardona-qft-methods_F03167 | 307  | ■ 1.000 | J_{mn}                                                                              | $J_{mn}$                                                                                | with $I_m{}^n$ a regular algebraic tensor field                                                                                                                                                                                                                                                          |
| cardona-qft-methods_F03168 | 307  | ■ 1.000 | \pi_{\pm}=\frac{1}{2}(\mathbb{I}_d\pm\mathrm{i}I)                                   | $\pi_{\pm} = \frac{1}{2}(\mathbb{I}_d \pm \mathrm{i}I)$                                 | Given an almost complex structure $I_m{}^n$ , one can build holomorphic and antiholomorphic projectors $\pi_{\pm} = \frac{1}{2}(\mathbb{I} \mp \mathrm{i}I)$ . Correspondingly, projectors                                                                                                               |
| cardona-qft-methods_F03169 | 307  | ■ 1.000 | \Pi_{\pm}=                                                                          | $\Pi_{\pm} =$                                                                           | and antiholomorphic projectors $\pi_{\pm} = \frac{1}{2}(\mathbb{I}_d \pm \mathrm{i}I)$ . Correspondingly, projectors can be built out of a generalized almost complex structure, $\Pi_{\pm} =$                                                                                                           |
| cardona-qft-methods_F03170 | 307  | ■ 1.000 | \frac{1}{2}(\mathbb{I}_{2d}\pm\mathrm{i}\mathcal{J})                                | $\frac{1}{2}(\mathbb{I}_{2d} \pm \mathrm{i}\mathcal{J})$                                | $\frac{1}{2}(\mathbb{I}_{2d} \pm \mathrm{i}\mathcal{J})$ . There is an integrability condition for generalized almost complex structures, analogous to the holomorphic condition for usual almost                                                                                                        |
| cardona-qft-methods_F03171 | 307  | ■ 0.999 | \pi_{\mp}[\pi_{\pm}X,\pi_{\pm}Y]=0                                                  | $\pi_{\mp}[\pi_{\pm}X, \pi_{\pm}Y] = 0$                                                 | integrability condition reads exactly the same, with $\pi$ and $X$ replaced respectively                                                                                                                                                                                                                 |
| cardona-qft-methods_F03172 | 307  | ■ 1.000 | \Pi                                                                                 | $\Pi$                                                                                   | $[\pi + \mathcal{S}(\pi + \eta)\mathcal{C} + \mathcal{P}(\pi + \eta)\mathcal{C} + \mathcal{S}X\eta - \mathcal{S}Y\mathcal{S} - 2\mathcal{S}(\pi X\eta - \eta Y\mathcal{S})] = 0$<br>on $\mathcal{C}$ and $\mathcal{C}^{\perp}$ and the integrability condition for generalized almost complex structures |

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| cardona-qft-methods_F03173 | 307  | ■ 1.000 | $T \oplus T^* M$                                                                                   | $TM \oplus T^*M$                                                                                   | on the $i$ -eigenspaces of $\mathcal{J}$ . In case these co                                                                                                                                       |
| cardona-qft-methods_F03174 | 307  | ■ 1.000 | $\mathcal{J}_1$                                                                                    | $\mathcal{J}_1$                                                                                    | ized complex structure is partially complex, partially symplectic. Explic-                                                                                                                        |
| cardona-qft-methods_F03175 | 307  | ■ 1.000 | $\mathcal{J}_2$                                                                                    | $\mathcal{J}_2$                                                                                    | ized complex structure is partially complex, partially symplectic. Explic-                                                                                                                        |
| cardona-qft-methods_F03176 | 307  | ■ 1.000 | $dJ=0$                                                                                             | $dJ = 0$                                                                                           | points or planes).<br>On this generalized tangent bundle $E$ we can also define a positive                                                                                                        |
| cardona-qft-methods_F03177 | 307  | ■ 1.000 | $\mathbb{C}^k \times \left( \mathbb{R}^{d-2k}, J \right)$                                          | $\mathbb{C}^k \times \left( \mathbb{R}^{d-2k}, J \right)$                                          |                                                                                                                                                                                                   |
| cardona-qft-methods_F03178 | 307  | ■ 1.000 | $J=\mathrm{d} x^{2k+1} \wedge \mathrm{d} x^{2k+2}+\ldots+\mathrm{d} x^{d-1} \wedge \mathrm{d} x^d$ | $J=\mathrm{d} x^{2k+1} \wedge \mathrm{d} x^{2k+2}+\ldots+\mathrm{d} x^{d-1} \wedge \mathrm{d} x^d$ |                                                                                                                                                                                                   |
| cardona-qft-methods_F03179 | 307  | ■ 1.000 | $k \leq d / 2$                                                                                     | $k \leq d/2$                                                                                       |                                                                                                                                                                                                   |
| cardona-qft-methods_F03180 | 308  | ■ 1.000 | $\mathcal{J}_i, i=1,2$                                                                             | $\mathcal{J}_i, i=1,2$                                                                             | If we have now a pair of <i>commuting</i> generalized almost complex structu                                                                                                                      |
| cardona-qft-methods_F03181 | 308  | ■ 0.998 | $\mathcal{J}_i$                                                                                    | $\mathcal{J}_i$                                                                                    | alized metric.                                                                                                                                                                                    |
| cardona-qft-methods_F03182 | 308  | ■ 1.000 | $+i$                                                                                               | $+i$                                                                                               | naturally from the worldsheet point of view," while that of pure spinors                                                                                                                          |
| cardona-qft-methods_F03183 | 308  | ■ 1.000 | $\Phi_{\pm}$                                                                                       | $\Phi_{\pm}$                                                                                       | arises on the spacetime side, as we discuss in Section 4.2.                                                                                                                                       |
| cardona-qft-methods_F03184 | 309  | ■ 1.000 | $e^{-B} \Phi =$                                                                                    | $e^{-B} \Phi =$                                                                                    | $\Phi_{\pm} \rightarrow e^{-B} \Phi_{\pm} \equiv \Phi_{\pm}^D \quad (14)$                                                                                                                         |
| cardona-qft-methods_F03185 | 309  | ■ 0.995 | $\left(1-B \wedge+\frac{1}{2} B \wedge B \wedge+\ldots\right) \Phi$                                | $\left(1-B \wedge+\frac{1}{2} B \wedge B \wedge+\ldots\right) \Phi$                                | where on the polyform associated to the spinor, the action is $e^{-B} \Phi =$                                                                                                                     |
| cardona-qft-methods_F03186 | 309  | ■ 1.000 | $\Phi^D$                                                                                           | $\Phi^D$                                                                                           | $(1-B \wedge+\frac{1}{2} B \wedge B \wedge+\ldots) \Phi$ . We will refer to $\Phi$ as naked pure spinor, while $\Phi^D$ will be called dressed pure spinor. The former lives in the spinor bundle |
| cardona-qft-methods_F03187 | 309  | ■ 1.000 | $\eta^1=\eta^2 \equiv \eta$                                                                        | $\eta^1=\eta^2 \equiv \eta$                                                                        | For the particular case $\eta^1=\eta^2 \equiv \eta$ , the pure spinors in (11) reduce to                                                                                                          |
| cardona-qft-methods_F03188 | 309  | ■ 1.000 | $\eta$                                                                                             | $\eta$                                                                                             | not integrable), such that there is a spinor that is invariant under parallel                                                                                                                     |
| cardona-qft-methods_F03189 | 309  | ■ 1.000 | $SO(6)$                                                                                            | $SO(6)$                                                                                            | We can decompose other $SO(6)$ representations, such as the vector <b>6</b> , into four <b>15</b> and three form (self dual and anti self dual) <b>10</b> + <b>10*</b> in rep-                    |
| cardona-qft-methods_F03190 | 309  | ■ 0.891 | $\mathbf{4} \rightarrow \mathbf{3}+\mathbf{1}$                                                     | $\mathbf{4} \rightarrow \mathbf{3}+\mathbf{1}$                                                     | resentation <b>10</b> and three form (non dual and anti non dual) <b>10</b> + <b>10*</b> in rep-                                                                                                  |
| cardona-qft-methods_F03191 | 309  | ■ 0.985 | $\mathbf{10}+\mathbf{10}^*$                                                                        | $\mathbf{10}+\mathbf{10}^*$                                                                        | representations of $SU(3)$ . This yields                                                                                                                                                          |
| cardona-qft-methods_F03192 | 309  | ■ 0.649 | $SU(3)$                                                                                            | $SU(3)$                                                                                            |                                                                                                                                                                                                   |
| cardona-qft-methods_F03193 | 310  | ■ 0.695 | $(J, \rho)$                                                                                        | $(J, \rho)$                                                                                        | five-forms), which implies the first equation in (14).<br>One can actually define the $SU(3)$ structure in terms of a pair of real                                                                |
| cardona-qft-methods_F03194 | 310  | ■ 0.695 | $\rho=\operatorname{Re} \Omega$                                                                    | $\rho=\operatorname{Re} \Omega$                                                                    | five-forms), which implies the first equation in (14).<br>One can actually define the $SU(3)$ structure in terms of a pair of real                                                                |
| cardona-qft-methods_F03195 | 310  | ■ 0.952 | $GL(6, \mathbb{R})$                                                                                | $GL(6, \mathbb{R})$                                                                                | stable. <sup>59</sup> This means that they live in an open orbit under the action of general transformations $GL(6, \mathbb{R})$ in the tangent space at each point. A sta-                       |
| cardona-qft-methods_F03196 | 310  | ■ 0.994 | $Sp(6, \mathbb{R})$                                                                                | $Sp(6, \mathbb{R})$                                                                                | ble two-form $J$ then defines a $Sp(6, \mathbb{R})$ structure while a stable real form $\rho$                                                                                                     |
| cardona-qft-methods_F03197 | 310  | ■ 0.585 | $SL(3, \mathbb{C})$                                                                                | $SL(3, \mathbb{C})$                                                                                | defines a $SL(3, \mathbb{C})$ structure. Together they define an $SU(3)$ structure provided the embeddings of $Sp(6, \mathbb{R})$ and $GL(6, \mathbb{R})$ in $GL(6, \mathbb{R})$ are compatible.  |
| cardona-qft-methods_F03198 | 310  | ■ 0.997 | $J \wedge \rho=0$                                                                                  | $J \wedge \rho=0$                                                                                  | satisfying (17) also defines an $SO(6)$ structure and hence a metric.                                                                                                                             |
| cardona-qft-methods_F03199 | 310  | ■ 0.997 | $SU(3) \subset SO(6)$                                                                              | $SU(3) \subset SO(6)$                                                                              | satisfying (17) also defines an $SO(6)$ structure and hence a metric.                                                                                                                             |

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| cardona-qft-methods_F03200 | 310  | 0.999 | $\eta^{\{2\}}=\left(v+\mathrm{i}\mathrm{v}^{\prime}\right)_{\mathrm{m}}\gamma^{\mathrm{m}}\eta^1$<br>$\gamma^{\mathrm{m}}\eta^{\{1\}}$      | $\eta^2 = (v + \mathrm{i}v')_m \gamma^m \eta^1$                                  | $j$ and $\omega$ are the (1,1) and (2,0)-forms on the local four dimensional space<br>orthogonal to $\sigma$ and $\sigma^*$ . $\mathcal{K}_\pm$ describes therefore a generalized almost complex                                                                       |
| cardona-qft-methods_F03201 | 310  | 0.632 | $\left(\eta^1,\eta^2\right)$                                                                                                                | $(\eta^1, \eta^2)$                                                               |                                                                                                                                                                                                                                                                        |
| cardona-qft-methods_F03202 | 310  | 0.859 | $(1,1)$                                                                                                                                     | $(1,1)$                                                                          |                                                                                                                                                                                                                                                                        |
| cardona-qft-methods_F03203 | 310  | 0.992 | $\mathrm{v}^{\{\prime\}}\cdot\Phi_{+}$                                                                                                      | $v'.\Phi_+$                                                                      |                                                                                                                                                                                                                                                                        |
| cardona-qft-methods_F03204 | 310  | 1.000 | $\Phi_{-}$                                                                                                                                  | $\Phi_-$                                                                         |                                                                                                                                                                                                                                                                        |
| cardona-qft-methods_F03205 | 311  | 0.712 | $\Leftrightarrow \exists$                                                                                                                   | $\Leftrightarrow \exists$                                                        | Integrability condition for the generalized complex structure corre-                                                                                                                                                                                                   |
| cardona-qft-methods_F03206 | 311  | 0.712 | $\mathrm{d}\Phi=(\mathrm{v}\llcorner\mathrm{corner}+\mathrm{zeta}\wedge)\Phi$                                                               | $\mathrm{d}\Phi = (v_{\mathrm{L}} + \zeta \wedge) \Phi$                          | Integrability condition for the generalized complex structure corre-                                                                                                                                                                                                   |
| cardona-qft-methods_F03207 | 311  | 0.997 | $\Leftrightarrow \exists \Phi$                                                                                                              | $\Leftrightarrow \exists \Phi$                                                   | exists:                                                                                                                                                                                                                                                                |
| cardona-qft-methods_F03208 | 311  | 0.997 | $\mathrm{d}\Phi=0$                                                                                                                          | $\mathrm{d}\Phi = 0$                                                             | exists:                                                                                                                                                                                                                                                                |
| cardona-qft-methods_F03209 | 311  | 1.000 | $W_{\{5\}}$                                                                                                                                 | $W_5$                                                                            | if the integrable pure spinor has rank $k$ , then the manifold looks locally like                                                                                                                                                                                      |
| cardona-qft-methods_F03210 | 311  | 1.000 | $\exists f$                                                                                                                                 | $\exists f$                                                                      | if the integrable pure spinor has rank $k$ , then the manifold looks locally like                                                                                                                                                                                      |
| cardona-qft-methods_F03211 | 311  | 1.000 | $\Phi=\mathrm{e}^{-f}\Omega$                                                                                                                | $\Phi = e^{-f} \Omega$                                                           | if the integrable pure spinor has rank $k$ , then the manifold looks locally like                                                                                                                                                                                      |
| cardona-qft-methods_F03212 | 311  | 1.000 | $\mathrm{d}\text{-}H\wedge$                                                                                                                 | $d - H \wedge$                                                                   | $[\mathcal{A} \mp \zeta, \mathcal{I} \mp \eta]H = [\mathcal{A} \mp \zeta, \mathcal{I} \mp \eta]C \mp \mathcal{K} \mathcal{V} \mathcal{H} \ , \qquad (24)$                                                                                                              |
| cardona-qft-methods_F03213 | 311  | 0.980 | $\mathrm{d}\Phi^D=0$                                                                                                                        | $\mathrm{d}\Phi^D = 0$                                                           |                                                                                                                                                                                                                                                                        |
| cardona-qft-methods_F03214 | 312  | 1.000 | $\eta_{\mu\nu}$                                                                                                                             | $\eta_{\mu\nu}$                                                                  | solutions, instead of solving the equations of motion directly, one can try                                                                                                                                                                                            |
| cardona-qft-methods_F03215 | 312  | 0.840 | $\left\langle Q_{\epsilon}\chi\right\rangle =\left\langle \delta_{\epsilon}\chi\right\rangle =0$<br>$\delta_{\epsilon}\chi\right\rangle =0$ | $\langle Q_{\epsilon} \chi \rangle = \langle \delta_{\epsilon} \chi \rangle = 0$ | non supersymmetric solutions of preserving the supersymmetric structure<br>by a supersymmetry-breaking source. As we will see, supersymmetry en-                                                                                                                       |
| cardona-qft-methods_F03216 | 313  | 0.703 | $\langle \delta_{\epsilon}\chi \rangle =0$                                                                                                  | $\langle \delta_{\epsilon} \chi \rangle = 0$                                     | symmetry. A nonzero vacuum expectation value (vev) for a fermionic field<br>would break this symmetry. Therefore, the desired background should only                                                                                                                   |
| cardona-qft-methods_F03217 | 313  | 1.000 | $\psi_M^A, \ A=1,2$                                                                                                                         | $\psi_M^A, A = 1, 2$                                                             | supersymmetry transformation involves bosonic fields only). As shown in<br>(36) the fermionic fields in type II theories are two gravitinos $\psi^A$ , $A = 1, 2$                                                                                                      |
| cardona-qft-methods_F03218 | 313  | 1.000 | $\lambda^A$                                                                                                                                 | $\lambda^A$                                                                      | (36) the fermionic fields in type II theories are two gravitinos $\psi^A$ , $A = 1, 2$<br>and two dilatinos $\lambda^A$ . Their supersymmetry transformations in the string                                                                                            |
| cardona-qft-methods_F03219 | 313  | 1.000 | $M=0,\ldots,9,\psi_M$                                                                                                                       | $M = 0, \dots, 9, \psi_M$                                                        | $2 \times 2$ matrices $\mathcal{P}$ and $\mathcal{P}_n$ are different in IIA and IIB: for IIA $\mathcal{P} = \Gamma_{11}$ and                                                                                                                                          |
| cardona-qft-methods_F03220 | 313  | 1.000 | $\psi_M=$                                                                                                                                   | $\psi_M =$                                                                       | $2 \times 2$ matrices $\mathcal{P}$ and $\mathcal{P}_n$ are different in IIA and IIB: for IIA $\mathcal{P} = \Gamma_{11}$ and                                                                                                                                          |
| cardona-qft-methods_F03221 | 313  | 0.769 | $\binom{\psi_M^1}{\psi_M^2}$                                                                                                                | $\begin{pmatrix} \psi_M^1 \\ \psi_M^2 \end{pmatrix}$                             | $\mathcal{P}_n = \Gamma_{11}^{(n/2)} \sigma^1$ , while for IIB $\mathcal{P} = -\sigma^3$ , $\mathcal{P}_n = \sigma^1$ for $\frac{n+1}{2}$ even and $\mathcal{P}_n = i\sigma^2$<br>for $\frac{n+1}{2}$ odd. A slash means a contraction with gamma matrices in the form |
| cardona-qft-methods_F03222 | 313  | 0.994 | $\mathcal{P}_{\mathrm{n}}$                                                                                                                  | $\mathcal{P}_n$                                                                  | democratic formulation of IIA: for the RR fields, who actually considers<br>all RR potentials ( $C_1...C_9$ in IIA, and $C_0, C_2...C_{10}$ in IIB, instead of only                                                                                                    |
| cardona-qft-methods_F03223 | 313  | 0.994 | $\mathcal{P}=\Gamma_{11}$                                                                                                                   | $\mathcal{P} = \Gamma_{11}$                                                      | democratic formulation of IIA: for the RR fields, who actually considers<br>all RR potentials ( $C_1...C_9$ in IIA, and $C_0, C_2...C_{10}$ in IIB, instead of only                                                                                                    |
| cardona-qft-methods_F03224 | 313  | 1.000 | $\mathcal{P}_{\mathrm{n}}=\Gamma_{11}^{\left(n\text{ / }2\right)}\sigma^1$                                                                  | $\mathcal{P}_n = \Gamma_{11}^{(n/2)} \sigma^1$                                   | those in Eq.(35), imposing a self-duality constraint on their field strengths                                                                                                                                                                                          |
| cardona-qft-methods_F03225 | 313  | 1.000 | $\mathcal{P}=-\sigma^3, \mathcal{P}_{\mathrm{n}}=\sigma^1$                                                                                  | $\mathcal{P} = -\sigma^3, \mathcal{P}_n = \sigma^1$                              | those in Eq.(35), imposing a self-duality constraint on their field strengths                                                                                                                                                                                          |

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| cardona-qft-methods_F03226 | 313  | ■ 1.000 | $\frac{n+1}{2}$                                                                                                   | $\frac{n+1}{2}$                                                                    | those in Eq.(35), imposing a self-duality constraint on their field strengths                                                                    |
| cardona-qft-methods_F03227 | 313  | ■ 1.000 | $\mathrm{cal{P}}_n=\mathrm{i}\,\sigma^2$                                                                          | $\mathcal{P}_n = \mathrm{i}\sigma^2$                                               | those in Eq.(35), imposing a self-duality constraint on their field strengths                                                                    |
| cardona-qft-methods_F03228 | 313  | ■ 0.670 | $\mathrm{scr{F}}_n=\frac{1}{n!}\,F_{P_1\dots P_N}\,\Gamma^{P_1\dots P_N}\,\Gamma^{\mathrm{P}_1\dots\mathrm{P}_N}$ | $\mathcal{F}_n = \frac{1}{n!}F_{P_1\dots P_N}\Gamma^{P_1\dots P_N}$                | $F^{(10)} = \mathrm{d}C - H \wedge C + m\,e^B = \hat{F} - H \wedge C \quad (4)$                                                                  |
| cardona-qft-methods_F03229 | 313  | ■ 0.670 | $H_{\mathrm{M}}\equiv\frac{1}{2}\,H_{\mathrm{M}\,\mathrm{N}\,\mathrm{P}}\,\Gamma^{\mathrm{N}\,\mathrm{P}}$        | $H_M \equiv \frac{1}{2}H_{MNP}\Gamma^{NP}$                                         | $F^{(10)} = \mathrm{d}C - H \wedge C + m\,e^B = \hat{F} - H \wedge C \quad (4)$                                                                  |
| cardona-qft-methods_F03230 | 313  | ■ 0.919 | $\mathrm{C}_{\mathrm{1}}\,\ldots\,\mathrm{C}_{\mathrm{9}}$                                                        | $C_1 \dots C_9$                                                                    | $\mathrm{d}C + m e^B$ , and $m \equiv F_0^{(10)} = \hat{F}_0$ is the mass parameter of IIA. These RR                                             |
| cardona-qft-methods_F03231 | 313  | ■ 0.919 | $\mathrm{C}_{\mathrm{0}},\,\mathrm{C}_{\mathrm{2}}\,\ldots\,\mathrm{C}_{\mathrm{10}}$                             | $C_0, C_2 \dots C_{10}$                                                            | $\mathrm{d}C + m e^B$ , and $m \equiv F_0^{(10)} = \hat{F}_0$ is the mass parameter of IIA. These RR                                             |
| cardona-qft-methods_F03232 | 313  | ■ 1.000 | $\mathrm{F}^{\mathrm{(10)}}$                                                                                      | $F^{(10)}$                                                                         |                                                                                                                                                  |
| cardona-qft-methods_F03233 | 313  | ■ 1.000 | $\mathrm{hat{F}}=$                                                                                                | $\hat{F} =$                                                                        |                                                                                                                                                  |
| cardona-qft-methods_F03234 | 313  | ■ 0.999 | $\mathrm{d}\,C+m\,\mathrm{e}^{\mathrm{B}}$                                                                        | $\mathrm{d}C + m e^B$                                                              |                                                                                                                                                  |
| cardona-qft-methods_F03235 | 313  | ■ 0.999 | $m\,\mathrm{equiv}\,\mathrm{F}_{\mathrm{0}}^{\mathrm{(10)}}=\mathrm{hat{F}}_{\mathrm{0}}$                         | $m \equiv F_0^{(10)} = \hat{F}_0$                                                  |                                                                                                                                                  |
| cardona-qft-methods_F03236 | 313  | ■ 1.000 | $\mathrm{S}\,0(n)$                                                                                                | $SO(n)$                                                                            |                                                                                                                                                  |
| cardona-qft-methods_F03237 | 314  | ■ 1.000 | $\nabla_{\mathrm{M}}\epsilon=0$                                                                                   | $\nabla_M \epsilon = 0$                                                            | When no fluxes are present, demanding zero VEV for the gravitino variation (9) requires the existence of a covariantly constant spinor on the    |
| cardona-qft-methods_F03238 | 314  | ■ 0.998 | $\Gamma^{\mu}=\gamma^{\mu}\otimes\mathbf{1},\,\Gamma^m=\gamma_5\otimes\gamma^m$                                   | $\Gamma^\mu = \gamma^\mu \otimes \mathbf{1}, \Gamma^m = \gamma_5 \otimes \gamma^m$ | gamma's), and written the four dimensional metric as $e^{2A}\tilde{g}_{\mu\nu}$ . This yields                                                    |
| cardona-qft-methods_F03239 | 314  | ■ 1.000 | $\mathrm{e}^{\mathrm{2}\,\mathrm{A}}\,\mathrm{tilde{g}}_{\mu\nu}$                                                 | $e^{2A}\tilde{g}_{\mu\nu}$                                                         | the following integrability condition <sup>38</sup>                                                                                              |
| cardona-qft-methods_F03240 | 314  | ■ 0.968 | $\gamma_{\mu\nu}$                                                                                                 | $\gamma_{\mu\nu}$                                                                  | $\epsilon_{\mathrm{IIA}}^{\pm}=\xi_+^{\pm}\otimes\eta_+^{\pm}+\xi_-^{\pm}\otimes\eta_-^{\pm}\,,\quad (10)$                                       |
| cardona-qft-methods_F03241 | 314  | ■ 0.968 | $\nabla_{\mathrm{m}}\,A\,\nabla^{\mathrm{m}}\,A=0$                                                                | $\nabla_m A \nabla^m A = 0$                                                        | $\epsilon_{\mathrm{IIA}}^{\pm}=\xi_+^{\pm}\otimes\eta_+^{\pm}+\xi_-^{\pm}\otimes\eta_-^{\pm}\,,\quad (10)$                                       |
| cardona-qft-methods_F03242 | 314  | ■ 0.998 | $\gamma_{11}\epsilon_{\mathrm{IIA}}^1=\epsilon_{\mathrm{IIA}}^1$                                                  | $\gamma_{11}\epsilon_{\mathrm{IIA}}^1 = \epsilon_{\mathrm{IIA}}^1$                 | variation, Eq.(2), we get the following condition                                                                                                |
| cardona-qft-methods_F03243 | 314  | ■ 0.998 | $\gamma_{11}\epsilon_{\mathrm{IIA}}^2=-\epsilon_{\mathrm{IIA}}^2$                                                 | $\gamma_{11}\epsilon_{\mathrm{IIA}}^2 = -\epsilon_{\mathrm{IIA}}^2$                | variation, Eq.(2), we get the following condition                                                                                                |
| cardona-qft-methods_F03244 | 314  | ■ 1.000 | $\mathrm{x_i}_{-}^{\mathrm{1,2}}=\mathrm{left}(\mathrm{x_i}_{+}^{\mathrm{1,2}}\mathrm{right})^{\mathrm{*}}$       | $\xi_{-}^{1,2} = \left(\xi_{+}^{1,2}\right)^{*}$                                   | $\nabla_m \eta_{\pm}^A = 0\,. \quad (12)$                                                                                                        |
| cardona-qft-methods_F03245 | 314  | ■ 1.000 | $\mathrm{eta}_{-}^{\mathrm{A}}=\mathrm{left}(\mathrm{eta}_{+}^{\mathrm{A}}\mathrm{right})^{\mathrm{*}}$           | $\eta_{-}^A = \left(\eta_{+}^A\right)^{*}$                                         | $\nabla_m \eta_{\pm}^A = 0\,. \quad (12)$                                                                                                        |
| cardona-qft-methods_F03246 | 314  | ■ 1.000 | $\mathrm{eta}^{\mathrm{1}}=\mathrm{eta}^{\mathrm{2}}$                                                             | $\eta^1 = \eta^2$                                                                  |                                                                                                                                                  |
| cardona-qft-methods_F03247 | 315  | ■ 0.677 | $\operatorname{Hol}(\nabla)\,\subseteq\,\operatorname{SU}(3)\,.\mathrm{^{\mathrm{h}}}$                            | $\mathrm{Hol}(\nabla) \subseteq \mathrm{SU}(3)^{\mathrm{h}}$                       |                                                                                                                                                  |
| cardona-qft-methods_F03248 | 315  | ■ 1.000 | $\mathrm{x_i}^{\mathrm{1}}$                                                                                       | $\xi^1$                                                                            | persymmetric string theory background in the absence of fluxes requires therefore Calabi-Yau 3-folds. When there is one covariantly constant in- |
| cardona-qft-methods_F03249 | 315  | ■ 1.000 | $\mathrm{x_i}^{\mathrm{2}}$                                                                                       | $\xi^2$                                                                            | persymmetric string theory background in the absence of fluxes requires therefore Calabi-Yau 3-folds. When there is one covariantly constant in- |
| cardona-qft-methods_F03250 | 315  | ■ 1.000 | $\mathrm{mathbb{N}}=2$                                                                                            | $\mathbb{N} = 2$                                                                   | ternal spinor, the internal gravitino equation tells us that there are two                                                                       |
| cardona-qft-methods_F03251 | 315  | ■ 1.000 | $\mathrm{K}3\,\times\,T^{\mathrm{2}}$                                                                             | $\mathrm{K}3 \times T^2$                                                           | Internal manifolds with a further reduced holonomy group allow for more                                                                          |

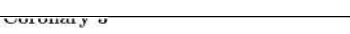
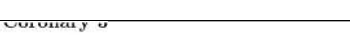
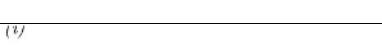
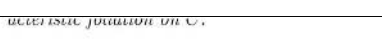
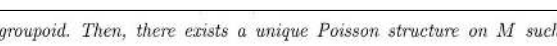
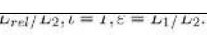
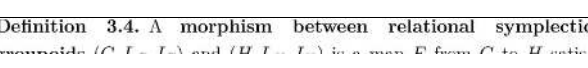
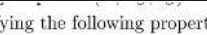
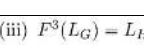
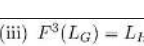
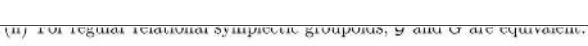
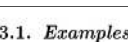
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| cardona-qft-methods_F03252 | 315  | ■ 1.000 | $\mathbb{N}=4$                                                               | $\mathbb{N} = 4$                                                             | preserves $\mathbb{N} = 4$ supersymmetry. Compactifications on $T^6$ preserve all the                                                                                |
| cardona-qft-methods_F03253 | 315  | ■ 1.000 | $T^6$                                                                        | $T^6$                                                                        | preserves $\mathbb{N} = 4$ supersymmetry. Compactifications on $T^6$ preserve all the                                                                                |
| cardona-qft-methods_F03254 | 315  | ■ 0.826 | $\mathbb{N}=8$                                                               | $\mathbb{N} = 8$                                                             | supersymmetries (32, or $\mathbb{N} = 8$ in four dimensions).                                                                                                        |
| cardona-qft-methods_F03255 | 315  | ■ 1.000 | $\mathbb{N}=1$                                                               | $\mathbb{N} = 1$                                                             | these moduli, which can take any value. Furthermore, massless fields lead                                                                                            |
| cardona-qft-methods_F03256 | 315  | ■ 1.000 | $F_{\{n\}}, H_{\{3\}}$                                                       | $F_n, H_3$                                                                   |                                                                                                                                                                      |
| cardona-qft-methods_F03257 | 316  | ■ 1.000 | $\mathcal{N}=2$                                                              | $\mathcal{N} = 2$                                                            | implications of the first condition.                                                                                                                                 |
| cardona-qft-methods_F03258 | 316  | ■ 1.000 | $SO(d)$                                                                      | $SO(d)$                                                                      | reviewed the topological condition, which amounts to the requirement that the manifold has $SU(3)$ structure. As for the differential condition, in the              |
| cardona-qft-methods_F03259 | 317  | ■ 1.000 | $c_1=0$                                                                      | $c_1 = 0$                                                                    |                                                                                                                                                                      |
| cardona-qft-methods_F03260 | 317  | ■ 1.000 | $\delta \Psi_m =$                                                            | $\delta \Psi_m =$                                                            | <i>geometry</i>                                                                                                                                                      |
| cardona-qft-methods_F03261 | 317  | ■ 1.000 | $\delta \lambda = 0$                                                         | $\delta \lambda = 0$                                                         | As we reviewed in the previous sections, as a result of demanding $\delta \Psi_m = \delta \lambda = 0$ supersymmetry imposes differential conditions on the internal |
| cardona-qft-methods_F03262 | 318  | ■ 0.924 | $SU(3) \times SU(3)$                                                         | $SU(3) \times SU(3)$                                                         | Specializing to the pure $SU(3)$ structure case, i.e. for $\Phi_{\pm}$ given by                                                                                      |
| cardona-qft-methods_F03263 | 318  | ■ 1.000 | $6-2k$                                                                       | $6 - 2k$                                                                     | Eq.(19), and looking at (19), we see that the Generalized Calabi-Yau manifold is complex <sup>1</sup> in IIB and (twisted) symplectic in IIA. For the gen-           |
| cardona-qft-methods_F03264 | 318  | ■ 0.942 | $(13,14)$                                                                    | $(13,14)$                                                                    | to satisfy Eqs. (13, 14) for $\tilde{\Phi}_{\pm} = \Phi_{\pm}$ , $F_{IIA} = F_{IIB} = A = 0$ . <sup>36</sup>                                                         |
| cardona-qft-methods_F03265 | 318  | ■ 0.942 | $\tilde{\Phi}_{\pm} = \Phi_{\pm}, F_{IIA} = F_{IIB} = A = 0$ . <sup>36</sup> | $\tilde{\Phi}_{\pm} = \Phi_{\pm}, F_{IIA} = F_{IIB} = A = 0$ . <sup>36</sup> | to satisfy Eqs. (13, 14) for $\tilde{\Phi}_{\pm} = \Phi_{\pm}$ , $F_{IIA} = F_{IIB} = A = 0$ . <sup>36</sup>                                                         |
| cardona-qft-methods_F03266 | 318  | ■ 1.000 | $T^6, K3 \times T^2$                                                         | $T^6, K3 \times T^2$                                                         | complex structure and Kähler deformations of the metric, which are all                                                                                               |
| cardona-qft-methods_F03267 | 318  | ■ 1.000 | $CY_3$                                                                       | $CY_3$                                                                       | complex structure and Kähler deformations of the metric, which are all                                                                                               |
| cardona-qft-methods_F03268 | 318  | ■ 0.999 | $2h^{2,1}$                                                                   | $2h^{2,1}$                                                                   | that open string ones can also stabilize moduli. We showed that within the                                                                                           |
| cardona-qft-methods_F03269 | 318  | ■ 0.999 | $h^{1,1}$                                                                    | $h^{1,1}$                                                                    | that open string ones can also stabilize moduli. We showed that within the                                                                                           |
| cardona-qft-methods_F03270 | 318  | ■ 0.731 | $(6,6)$                                                                      | $(6,6)$                                                                      |                                                                                                                                                                      |
| cardona-qft-methods_F03271 | 318  | ■ 1.000 | ${}^i H$                                                                     | ${}^i H$                                                                     |                                                                                                                                                                      |
| cardona-qft-methods_F03272 | 318  | ■ 1.000 | $(d-H \wedge) \Omega = 0$                                                    | $(d - H \wedge) \Omega = 0$                                                  |                                                                                                                                                                      |
| cardona-qft-methods_F03273 | 318  | ■ 1.000 | $d \Omega = 0$                                                               | $d \Omega = 0$                                                               |                                                                                                                                                                      |
| cardona-qft-methods_F03274 | 322  | ■ 1.000 | $\mathcal{N}=(2,2)$                                                          | $\mathcal{N} = (2,2)$                                                        |                                                                                                                                                                      |
| cardona-qft-methods_F03275 | 322  | ■ 0.976 | $\mathcal{D}=10$                                                             | $\mathcal{D} = 10$                                                           | formulations of $D = 10$ supersymmetry and D8 - D8 domain walls," Class. Quant. Grav. <b>18</b> , 2359 (2001) [arXiv:hep-th/0102223]                                 |
| cardona-qft-methods_F03276 | 322  | ■ 0.978 | $\mathcal{N}=1$                                                              | $\mathcal{N} = 1$                                                            | SU(3) Holonomy," Nucl. Phys. B <b>268</b> , 79 (1986).<br>D. B. Green, E. H. Kilduff, D. J. Raine and G. G. Ross                                                     |
| cardona-qft-methods_F03277 | 326  | ■ 1.000 | $(G, M)$                                                                     | $(G, M)$                                                                     |                                                                                                                                                                      |
| cardona-qft-methods_F03278 | 326  | ■ 1.000 | $(A, M)$                                                                     | $(A, M)$                                                                     |                                                                                                                                                                      |

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| cardona-qft-methods_F03279 | 326  | ■ 1.000 | A=T^{*} M                                                                                           | $A=T^{*}M$                                                                                             |                                                                                                                                                                                   |
| cardona-qft-methods_F03280 | 327  | ■ 1.000 | (M, \Pi)                                                                                            | $(M, \Pi)$                                                                                             | if such a Lie groupoid $G$ exists.                                                                                                                                                |
| cardona-qft-methods_F03281 | 328  | ■ 1.000 | { }^{2,5}                                                                                           | 2,5                                                                                                    | from geometric quantization, what is expected as the quantization of a relational symplectic groupoid is an algebra with a special element, $\mathfrak{p}$                        |
| cardona-qft-methods_F03282 | 328  | ■ 1.000 | \Pi \in \Gamma(T M \wedge T M)                                                                      | $\Pi \in \Gamma(TM \wedge TM)$                                                                         |                                                                                                                                                                                   |
| cardona-qft-methods_F03283 | 328  | ■ 1.000 | \{\} :, \mathcal{C}^{\infty}(M) \otimes \mathcal{C}^{\infty}(M) \rightarrow \mathcal{C}^{\infty}(M) | $\{ \} :, \mathcal{C}^{\infty}(M) \otimes \mathcal{C}^{\infty}(M) \rightarrow \mathcal{C}^{\infty}(M)$ |                                                                                                                                                                                   |
| cardona-qft-methods_F03284 | 329  | ■ 1.000 | T \Sigma                                                                                            | $T\Sigma$                                                                                              | The space of fields for this theory is denoted with $\Phi$ and corresponds to                                                                                                     |
| cardona-qft-methods_F03285 | 329  | ■ 0.968 | (X, \eta)                                                                                           | $(X, \eta)$                                                                                            | the space of vector bundle morphisms between $T\Sigma$ and $T^{*}M$ . This space can be parametrized by the pair $(X, n)$ , where $X$ is a $\mathcal{C}^{k+1}$ -man from $\Sigma$ |
| cardona-qft-methods_F03286 | 329  | ■ 0.968 | \mathcal{C}^{k+1}                                                                                   | $\mathcal{C}^{k+1}$                                                                                    | the space of vector bundle morphisms between $T\Sigma$ and $T^{*}M$ . This space can be parametrized by the pair $(X, n)$ , where $X$ is a $\mathcal{C}^{k+1}$ -man from $\Sigma$ |
| cardona-qft-methods_F03287 | 329  | ■ 1.000 | \eta \in \Gamma^k(\Sigma, T^{*}\Sigma \otimes X^{*}T^{*}M)                                          | $\eta \in \Gamma^k(\Sigma, T^{*}\Sigma \otimes X^{*}T^{*}M)$                                           | to $M$ and $\eta \in \Gamma^k(\Sigma, T^{*}\Sigma \otimes X^{*}T^{*}M)$ , where $k \in \{0, 1, \dots\}$ denotes the                                                               |
| cardona-qft-methods_F03288 | 329  | ■ 1.000 | k \in \{0, 1, \cdots\}                                                                              | $k \in \{0, 1, \dots\}$                                                                                | to $M$ and $\eta \in \Gamma^k(\Sigma, T^{*}\Sigma \otimes X^{*}T^{*}M)$ , where $k \in \{0, 1, \dots\}$ denotes the                                                               |
| cardona-qft-methods_F03289 | 329  | ■ 1.000 | \Pi^{\#}                                                                                            | $\Pi^{\#}$                                                                                             | $\Omega^1(\Sigma, X^{*}(T^{*}M))$ , respectively.                                                                                                                                 |
| cardona-qft-methods_F03290 | 329  | ■ 1.000 | T^{*} M \rightarrow T M                                                                             | $T^{*}M \rightarrow TM$                                                                                | $\Omega^1(\Sigma, X^{*}(T^{*}M))$ , respectively.                                                                                                                                 |
| cardona-qft-methods_F03291 | 329  | ■ 1.000 | d X                                                                                                 | $dX$                                                                                                   | $T_x^{*}M$ , for all $x \in M$ .                                                                                                                                                  |
| cardona-qft-methods_F03292 | 329  | ■ 1.000 | \Omega^1 \left( \Sigma, X^{*}(T M) \right)                                                          | $\Omega^1(\Sigma, X^{*}(TM))$                                                                          | $T_x^{*}M$ , for all $x \in M$ .                                                                                                                                                  |
| cardona-qft-methods_F03293 | 329  | ■ 1.000 | \Omega^1 \left( \Sigma, X^{*} \left( T^{*} M \right) \right)                                        | $\Omega^1(\Sigma, X^{*}(T^{*}M))$                                                                      | The integrand, called the Lagra                                                                                                                                                   |
| cardona-qft-methods_F03294 | 329  | ■ 1.000 | \langle                                                                                             | $\langle$                                                                                              | to this action, the corresponding variational problem $\delta S = 0$ induces the following space                                                                                  |
| cardona-qft-methods_F03295 | 329  | ■ 1.000 | \rangle denotes the pairing between \Omega^1 \left( \Sigma, X^{*}(T M) \right)                      | $\rangle denotes the pairing between \Omega^1(\Sigma, X^{*}(TM))$                                      | to this action, the corresponding variational problem $\delta S = 0$ induces the following space                                                                                  |
| cardona-qft-methods_F03296 | 329  | ■ 1.000 | \mathcal{L}                                                                                         | $\mathcal{L}$                                                                                          | $\delta S = \int \frac{\partial \mathcal{L}}{\partial X} \delta X + \frac{\partial \mathcal{L}}{\partial \dot{x}} \delta \dot{x} + \text{boundary terms.}$                        |
| cardona-qft-methods_F03297 | 329  | ■ 1.000 | \delta S=0                                                                                          | $\delta S = 0$                                                                                         | The partial variations correspond to:                                                                                                                                             |
| cardona-qft-methods_F03298 | 330  | ■ 1.000 | { }^6 \Phi_{\partial}                                                                               | ${}^6\Phi_{\partial}$                                                                                  | <i>Groupoids and Poisson sigma models with boundary</i> 319                                                                                                                       |
| cardona-qft-methods_F03299 | 330  | ■ 1.000 | p: \Phi \rightarrow \Phi_{\partial}                                                                 | $p: \Phi \rightarrow \Phi_{\partial}$                                                                  | $\Phi_{\Sigma} := T\text{vector bundle}$                                                                                                                                          |
| cardona-qft-methods_F03300 | 330  | ■ 1.000 | C_{\Pi}                                                                                             | $C_{\Pi}$                                                                                              | $L_{\Sigma} := p(EL)$ .                                                                                                                                                           |
| cardona-qft-methods_F03301 | 330  | ■ 1.000 | \Phi_{\partial}                                                                                     | $\Phi_{\partial}$                                                                                      | $L_{\Sigma} := p(EL)$ .                                                                                                                                                           |
| cardona-qft-methods_F03302 | 330  | ■ 1.000 | L_{\Sigma'}                                                                                         | $L_{\Sigma'}$                                                                                          | Finally, we define $C_{\Pi}$ as the set of fields in $\Phi_{\partial}$ which c                                                                                                    |
| cardona-qft-methods_F03303 | 330  | ■ 1.000 | \Sigma^{\prime}:=\partial \Sigma \times[0, \varepsilon]                                             | $\Sigma' := \partial \Sigma \times [0, \varepsilon]$                                                   | Finally, we define $C_{\Pi}$ as the set of fields in $\Phi_{\partial}$ which c                                                                                                    |
| cardona-qft-methods_F03304 | 330  | ■ 1.000 | T^{*}(P M)                                                                                          | $T^{*}(PM)$                                                                                            | to a field in $L_{\Sigma'}$ , with $\Sigma' := \partial \Sigma \times [0, \varepsilon]$ , for some $\varepsilon$ .                                                                |
| cardona-qft-methods_F03305 | 330  | ■ 1.000 | S(X, \eta)                                                                                          | $S(X, \eta)$                                                                                           | <b>Proposition 2.2.</b> <sup>2,3</sup> If $C_{\Pi}$ is a coisotropic submanifold of $\Phi_{\partial}$ , then $\Pi$ is a                                                           |

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| cardona-qft-methods_F03306 | 330  | ■ 0.988 | $(A, \rho)$                                                                                   | $(A, \rho)$                                                                                   | $[X, fY]_A = f[X, Y] + \rho_*(X)(f)Y, \forall X, Y \in \Gamma(A), f \in \mathcal{C}^\infty(M).$                          |
| cardona-qft-methods_F03307 | 330  | ■ 1.000 | $[, ]_A$                                                                                      | $[, ]_A$                                                                                      | that is defined for exact forms by                                                                                       |
| cardona-qft-methods_F03308 | 330  | ■ 1.000 | $\Gamma(A)$                                                                                   | $\Gamma(A)$                                                                                   | that is defined for exact forms by                                                                                       |
| cardona-qft-methods_F03309 | 330  | ■ 1.000 | $\rho_*$                                                                                      | $\rho_*$                                                                                      | that is defined for exact forms by                                                                                       |
| cardona-qft-methods_F03310 | 330  | ■ 1.000 | $\Gamma(A) \rightarrow \mathfrak{X}(M)$                                                       | $\Gamma(A) \rightarrow \mathfrak{X}(M)$                                                       | $[df, dg] := d\{f, g\}, \forall f, g \in \mathcal{C}^\infty(M),$                                                         |
| cardona-qft-methods_F03311 | 330  | ■ 0.516 | $[, ]_{T^*M}$                                                                                 | $[, ]_{T^*M}$                                                                                 |                                                                                                                          |
| cardona-qft-methods_F03312 | 331  | ■ 1.000 | $\Pi^\# : T^*M \rightarrow TM$                                                                | $\Pi^\# : T^*M \rightarrow TM$                                                                |                                                                                                                          |
| cardona-qft-methods_F03313 | 331  | ■ 1.000 | $\Lambda^\bullet A^*$                                                                         | $\Lambda^\bullet A^*$                                                                         | 5.20. $A$ , $C$ and $\mathcal{A}$ are $\mathcal{A}$ -modules                                                             |
| cardona-qft-methods_F03314 | 331  | ■ 1.000 | $A^*$                                                                                         | $A^*$                                                                                         | whereas for general 1-forms it is recovered by Leibniz and the anchor map                                                |
| cardona-qft-methods_F03315 | 331  | ■ 1.000 | $\delta_A$                                                                                    | $\delta_A$                                                                                    | whereas for general 1-forms it is recovered by Leibniz and the anchor map                                                |
| cardona-qft-methods_F03316 | 331  | ■ 1.000 | $f \in \mathcal{C}^\infty(M)$                                                                 | $f \in \mathcal{C}^\infty(M)$                                                                 | define a morphism<br>$A^*$ is the dual bundle                                                                            |
| cardona-qft-methods_F03317 | 331  | ■ 1.000 | $X, Y \in \Gamma(A), \alpha \in \Gamma(A^*)$                                                  | $X, Y \in \Gamma(A), \alpha \in \Gamma(A^*)$                                                  | $\langle X, \alpha \rangle = \alpha(X)$<br>$\langle Y, \alpha \rangle = \alpha(Y)$                                       |
| cardona-qft-methods_F03318 | 331  | ■ 0.998 | $\langle \cdot, \cdot \rangle$ is the natural pairing between $\Gamma(A)$ and $\Gamma(A^*)$ . | $\langle \cdot, \cdot \rangle$ is the natural pairing between $\Gamma(A)$ and $\Gamma(A^*)$ . | where $\langle \cdot, \cdot \rangle$ is the natural pairing between $\Gamma(A)$ and $\Gamma(A^*)$ .                      |
| cardona-qft-methods_F03319 | 331  | ■ 0.998 | $\Gamma(A^*)$                                                                                 | $\Gamma(A^*)$                                                                                 | where $\langle \cdot, \cdot \rangle$ is the natural pairing between $\Gamma(A)$ and $\Gamma(A^*)$ .                      |
| cardona-qft-methods_F03320 | 331  | ■ 1.000 | $\varphi : A \rightarrow B$                                                                   | $\varphi : A \rightarrow B$                                                                   | $\delta_A \varphi^* = \varphi^* \delta_B.$                                                                               |
| cardona-qft-methods_F03321 | 331  | ■ 1.000 | $\gamma(A)$                                                                                   | $\gamma(A)$                                                                                   | Lie algebroid structure on the left is given by the Lie bracket of vector fields                                         |
| cardona-qft-methods_F03322 | 331  | ■ 1.000 | $\Gamma(B)$                                                                                   | $\Gamma(B)$                                                                                   | Lie algebroid structure on the left is given by the Lie bracket of vector fields                                         |
| cardona-qft-methods_F03323 | 331  | ■ 1.000 | $T(\partial\Sigma)$                                                                           | $T(\partial\Sigma)$                                                                           | As it was mentioned before, it can be proven that this space is a                                                        |
| cardona-qft-methods_F03324 | 331  | ■ 1.000 | $T^*PM$                                                                                       | $T^*PM$                                                                                       | following data <sup>a</sup>                                                                                              |
| cardona-qft-methods_F03325 | 331  | ■ 0.866 | $M^4$                                                                                         | $M^4$                                                                                         | and unit map respectively, such that the following axioms hold (denoted by $\mathcal{A}$ and $\mathcal{B}$ respectively) |
| cardona-qft-methods_F03326 | 331  | ■ 1.000 | $G \rightrightarrows M$                                                                       | $G \rightrightarrows M$                                                                       | (A.2) If $g \in G_{(x,y)}$ and $h \in G_{(y,z)}$ then $\mu(g,h) \in G_{(x,z)}$                                           |
| cardona-qft-methods_F03327 | 331  | ■ 1.000 | $s, t, \iota, \mu$                                                                            | $s, t, \iota, \mu$                                                                            |                                                                                                                          |
| cardona-qft-methods_F03328 | 331  | ■ 0.578 | $G_{(x,y)} := s^{-1}(x) \cap t^{-1}(y)$                                                       | $G_{(x,y)} := s^{-1}(x) \cap t^{-1}(y)$                                                       |                                                                                                                          |
| cardona-qft-methods_F03329 | 331  | ■ 1.000 | $s \circ \varepsilon = t \circ \varepsilon = id_M$                                            | $s \circ \varepsilon = t \circ \varepsilon = id_M$                                            |                                                                                                                          |
| cardona-qft-methods_F03330 | 331  | ■ 1.000 | $g \in G_{(x,y)}$                                                                             | $g \in G_{(x,y)}$                                                                             |                                                                                                                          |
| cardona-qft-methods_F03331 | 331  | ■ 1.000 | $h \in G_{(y,z)}$                                                                             | $h \in G_{(y,z)}$                                                                             |                                                                                                                          |
| cardona-qft-methods_F03332 | 331  | ■ 1.000 | $\mu(g, h) \in G_{(x,z)}$                                                                     | $\mu(g, h) \in G_{(x,z)}$                                                                     |                                                                                                                          |

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| cardona-qft-methods_F03333 | 331  | 0.647 | $\{ \}^{\mathrm{a}} G \times_{(s, t)} G$                                                                         | ${}^{\mathrm{a}} G \times_{(s, t)} G$                                                                            |                                                                                                                                                                                                                                                                                              |
| cardona-qft-methods_F03334 | 332  | 1.000 | $\mu \left( \varepsilon \circ s \times id_G \right) = \mu \left( id_G \times \varepsilon \circ t \right) = id_G$ | $\mu \left( \varepsilon \circ s \times id_G \right) = \mu \left( id_G \times \varepsilon \circ t \right) = id_G$ |                                                                                                                                                                                                                                                                                              |
| cardona-qft-methods_F03335 | 332  | 1.000 | $\mu \left( id_G \times i \right) = \varepsilon \circ t$                                                         | $\mu \left( id_G \times i \right) = \varepsilon \circ t$                                                         |                                                                                                                                                                                                                                                                                              |
| cardona-qft-methods_F03336 | 332  | 1.000 | $\mu \left( i \times id_G \right) = \varepsilon \circ s$                                                         | $\mu \left( i \times id_G \right) = \varepsilon \circ s$                                                         |                                                                                                                                                                                                                                                                                              |
| cardona-qft-methods_F03337 | 332  | 1.000 | $\mu \left( \mu \times id_G \right) = \mu \left( id_G \times \mu \right)$                                        | $\mu \left( \mu \times id_G \right) = \mu \left( id_G \times \mu \right)$                                        | (A.3) $\mu(\varepsilon \circ s \times id_G) = \mu(id_G$                                                                                                                                                                                                                                      |
| cardona-qft-methods_F03338 | 332  | 1.000 | $G \times G \times \bar{G}$                                                                                      | $G \times G \times \bar{G}$                                                                                      | is a lagrangian submanifold of $G \times G \times \bar{G}$ , where $\bar{G}$ denotes the sign reversed                                                                                                                                                                                       |
| cardona-qft-methods_F03339 | 332  | 1.000 | $\bar{G}$                                                                                                        | $\bar{G}$                                                                                                        | is a lagrangian submanifold of $G \times G \times \bar{G}$ , where $\bar{G}$ denotes the sign reversed                                                                                                                                                                                       |
| cardona-qft-methods_F03340 | 332  | 0.984 | $\underline{C_{\Pi}}$                                                                                            | $\underline{C_{\Pi}}$                                                                                            | of the characteristic foliation), if it is smooth, is a symplectic groupoid over $\mathfrak{A}$                                                                                                                                                                                              |
| cardona-qft-methods_F03341 | 332  | 1.000 | $\mathcal{A} = T^* M$                                                                                            | $\mathcal{A} = T^* M$                                                                                            | subset of the cotangent bundle of $M$ . In general, the dual space of a smooth foliation is not a smooth finite dimensional manifold and in this pa                                                                                                                                          |
| cardona-qft-methods_F03342 | 332  | 0.999 | $\{ \}^{\text{Ext}}$                                                                                             | $\text{Ext}$                                                                                                     |                                                                                                                                                                                                                                                                                              |
| cardona-qft-methods_F03343 | 333  | 0.905 | $\{ \}^{\text{E x t}}$                                                                                           | $\text{Ext}$                                                                                                     |                                                                                                                                                                                                                                                                                              |
| cardona-qft-methods_F03344 | 333  | 1.000 | $L: \mathcal{M} \rightarrowtail \mathcal{N}$                                                                     | $L: \mathcal{M} \rightarrowtail \mathcal{N}$                                                                     | — $\mathcal{M}$ and $\mathcal{N}$ are symplectic manifolds, $L$ is an immersed canonical relation between two symplectic                                                                                                                                                                     |
| cardona-qft-methods_F03345 | 333  | 1.000 | $\mathcal{N}$                                                                                                    | $\mathcal{N}$                                                                                                    | <b>Definition 9.1.</b> $\text{Sym}^{\text{can}}$ is a category in which the objects are symplectic manifolds and the morphisms are immersed canonical relations. <sup>b</sup> Recall                                                                                                         |
| cardona-qft-methods_F03346 | 333  | 0.755 | $\overline{\mathcal{M}} \times \mathcal{N}$                                                                      | $\overline{\mathcal{M}} \times \mathcal{N}$                                                                      | that $L: \mathcal{M} \rightarrowtail \mathcal{N}$ is an immersed canonical relation between two symplectic                                                                                                                                                                                   |
| cardona-qft-methods_F03347 | 333  | 0.755 | $\{ \}^{\text{c}}$                                                                                               | ${}^{\text{c}}$                                                                                                  | that $L: \mathcal{M} \rightarrowtail \mathcal{N}$ is an immersed canonical relation between two symplectic                                                                                                                                                                                   |
| cardona-qft-methods_F03348 | 333  | 0.755 | $\left( \text{Sym}^{\text{Ext}} \right)^{\text{op}} \rightarrow$                                                 | $\left( \text{Sym}^{\text{Ext}} \right)^{\text{op}} \rightarrow$                                                 | that $L: \mathcal{M} \rightarrowtail \mathcal{N}$ is an immersed canonical relation between two symplectic                                                                                                                                                                                   |
| cardona-qft-methods_F03349 | 333  | 0.755 | $^{\text{Ext}}$                                                                                                  | $\text{Ext}$                                                                                                     | that $L: \mathcal{M} \rightarrowtail \mathcal{N}$ is an immersed canonical relation between two symplectic                                                                                                                                                                                   |
| cardona-qft-methods_F03350 | 333  | 1.000 | $f: A \rightarrowtail B$                                                                                         | $f: A \rightarrowtail B$                                                                                         | manifolds $\mathcal{M}$ and $\mathcal{N}$ . By definition if $L$ is an immersed Lagrangian submanifold of $\overline{\mathcal{M}} \times \mathcal{N}$ , $\text{Sym}^{\text{Ext}}$ carries an involution $\dagger: (\text{Sym}^{\text{Ext}})^{\text{op}} \rightarrow \text{Sym}^{\text{Ext}}$ |
| cardona-qft-methods_F03351 | 333  | 1.000 | $f^{\dagger} := \{(b, a) \in B \times A \mid (a, b) \in f\}$                                                     | $f^{\dagger} := \{(b, a) \in B \times A \mid (a, b) \in f\}$                                                     | that is the identity in object                                                                                                                                                                                                                                                               |
| cardona-qft-methods_F03352 | 333  | 0.863 | $(\mathcal{G}, L, I)$                                                                                            | $(\mathcal{G}, L, I)$                                                                                            | where                                                                                                                                                                                                                                                                                        |
| cardona-qft-methods_F03353 | 333  | 1.000 | $\mathcal{G}^3$                                                                                                  | $\mathcal{G}^3$                                                                                                  | satisfying the following axioms                                                                                                                                                                                                                                                              |
| cardona-qft-methods_F03354 | 333  | 1.000 | $1 \ L$                                                                                                          | $1L$                                                                                                             | <b>Lemma 9.1.</b> $\mathcal{G}$ is a groupoid and $L$ and $I$ can be regarded as well as canonical relations                                                                                                                                                                                 |
| cardona-qft-methods_F03355 | 333  | 1.000 | $(x, y, z) \in L$                                                                                                | $(x, y, z) \in L$                                                                                                | <b>Lemma 9.1.</b> $\mathcal{G}$ is a groupoid and $L$ and $I$ can be regarded as well as canonical relations                                                                                                                                                                                 |
| cardona-qft-methods_F03356 | 333  | 1.000 | $(y, z, x) \in L$                                                                                                | $(y, z, x) \in L$                                                                                                | <b>Lemma 9.1.</b> $\mathcal{G}$ is a groupoid and $L$ and $I$ can be regarded as well as canonical relations                                                                                                                                                                                 |
| cardona-qft-methods_F03357 | 333  | 0.995 | $2 \ I$                                                                                                          | $2I$                                                                                                             | $\bar{G} \times \bar{G} \rightarrowtail G$ and                                                                                                                                                                                                                                               |
| cardona-qft-methods_F03358 | 333  | 0.995 | $I^2 = Id$                                                                                                       | $I^2 = Id$                                                                                                       | $\bar{G} \times \bar{G} \rightarrowtail G$ and                                                                                                                                                                                                                                               |

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| cardona-qft-methods_F03359 | 333  | ■ 1.000 | $\mathcal{G} \times \mathcal{G} \rightarrow \overline{\mathcal{G}}$                                                                                                                                                      | $\mathcal{G} \times \mathcal{G} \rightharpoonup \overline{\mathcal{G}}$                                                                                                                                                 |                                                                                                                                                                                                           |
| cardona-qft-methods_F03360 | 333  | ■ 1.000 | $L_{\text{rel}}$                                                                                                                                                                                                         | $L_{\text{rel}}$                                                                                                                                                                                                        |                                                                                                                                                                                                           |
| cardona-qft-methods_F03361 | 333  | ■ 1.000 | $\mathcal{G} \times \mathcal{G}, I$                                                                                                                                                                                      | $\mathcal{G} \times \mathcal{G}, I$                                                                                                                                                                                     |                                                                                                                                                                                                           |
| cardona-qft-methods_F03362 | 333  | ■ 0.996 | $\overline{\mathcal{G}} \rightarrow \mathcal{G}$                                                                                                                                                                         | $\overline{\mathcal{G}} \rightharpoonup \mathcal{G}$                                                                                                                                                                    |                                                                                                                                                                                                           |
| cardona-qft-methods_F03363 | 333  | ■ 0.996 | $I_{\text{rel}}$                                                                                                                                                                                                         | $I_{\text{rel}}$                                                                                                                                                                                                        |                                                                                                                                                                                                           |
| cardona-qft-methods_F03364 | 333  | ■ 0.995 | $\overline{L_{\text{rel}}}$                                                                                                                                                                                              | $\overline{L_{\text{rel}}}$                                                                                                                                                                                             |                                                                                                                                                                                                           |
| cardona-qft-methods_F03365 | 333  | ■ 0.995 | $\overline{I_{\text{rel}}}$                                                                                                                                                                                              | $\overline{I_{\text{rel}}}$                                                                                                                                                                                             |                                                                                                                                                                                                           |
| cardona-qft-methods_F03366 | 333  | ■ 1.000 | $\omega^{\#}: T^*M \rightarrow TM$                                                                                                                                                                                       | $\omega^{\#}: T^*M \rightarrow TM$                                                                                                                                                                                      |                                                                                                                                                                                                           |
| cardona-qft-methods_F03367 | 334  | ■ 0.999 | $Id: \mathcal{G} \rightarrow \mathcal{G}$                                                                                                                                                                                | $Id: \mathcal{G} \rightarrow \mathcal{G}$                                                                                                                                                                               | induces canonical relations                                                                                                                                                                               |
| cardona-qft-methods_F03368 | 334  | ■ 0.999 | $Id_{\text{rel}}$                                                                                                                                                                                                        | $Id_{\text{rel}}$                                                                                                                                                                                                       | induces canonical relations                                                                                                                                                                               |
| cardona-qft-methods_F03369 | 334  | ■ 1.000 | $\mathcal{G} \rightarrow \mathcal{G}$                                                                                                                                                                                    | $\mathcal{G} \rightharpoonup \mathcal{G}$                                                                                                                                                                               |                                                                                                                                                                                                           |
| cardona-qft-methods_F03370 | 334  | ■ 1.000 | $\overline{Id_{\text{rel}}}: \overline{\mathcal{G}} \rightarrow \overline{\mathcal{G}}$                                                                                                                                  | $\overline{Id_{\text{rel}}}: \overline{\mathcal{G}} \rightharpoonup \overline{\mathcal{G}}$                                                                                                                             |                                                                                                                                                                                                           |
| cardona-qft-methods_F03371 | 334  | ■ 0.927 | $3 I_{\text{rel}} \circ L_{\text{rel}} = \bar{L}_{\text{rel}} \circ \bar{T}_{\text{rel}} \circ (\overline{I_{\text{rel}}} \circ \overline{I_{\text{rel}}}) : \mathcal{G} \times \mathcal{G} \rightharpoonup \mathcal{G}$ | $3I_{\text{rel}} \circ L_{\text{rel}} = \bar{L}_{\text{rel}} \circ \bar{T}_{\text{rel}} \circ (\overline{I_{\text{rel}}} \circ \overline{I_{\text{rel}}}) : \mathcal{G} \times \mathcal{G} \rightharpoonup \mathcal{G}$ | The identity map $Id: \mathcal{G} \rightarrow \mathcal{G}$ as a relation will be denoted by $\underline{\underline{Id}}$ .                                                                                |
| cardona-qft-methods_F03372 | 334  | ■ 0.997 | $\overline{I_{\text{rel}}} \circ L_3 = \overline{L_3} \circ \overline{T_{\text{rel}}} \circ (\overline{I_{\text{rel}}} \times \overline{I_{\text{rel}}})$                                                                | $\overline{I_{\text{rel}}} \circ L_3 = \overline{L_3} \circ \overline{T_{\text{rel}}} \circ (\overline{I_{\text{rel}}} \times \overline{I_{\text{rel}}})$                                                               | <ul style="list-style-type: none"> <li>• <b>A.4</b> <math>L_3 \circ (L_3 \times Id) = L_3 \circ (Id \times L_3) : \mathcal{G}^3 \rightharpoonup \mathcal{G}</math> is an immersed submanifold.</li> </ul> |
| cardona-qft-methods_F03373 | 334  | ■ 1.000 | $4 L_3 \circ (L_3 \times Id) = L_3 \circ (Id \times L_3) : \mathcal{G}^3 \rightharpoonup \mathcal{G}$                                                                                                                    | $4L_3 \circ (L_3 \times Id) = L_3 \circ (Id \times L_3) : \mathcal{G}^3 \rightharpoonup \mathcal{G}$                                                                                                                    | The fact that the composition is Lagrangian follows from the fact that,                                                                                                                                   |
| cardona-qft-methods_F03374 | 334  | ■ 0.999 | $(Id \times L_3)$                                                                                                                                                                                                        | $(Id \times L_3)$                                                                                                                                                                                                       | tomatical if we start with a finite dimensional symplectic manifold $\mathcal{G}$ .                                                                                                                       |
| cardona-qft-methods_F03375 | 334  | ■ 0.999 | $(L_3 \times Id)$                                                                                                                                                                                                        | $(L_3 \times Id)$                                                                                                                                                                                                       | tomatical if we start with a finite dimensional symplectic manifold $\mathcal{G}$ .                                                                                                                       |
| cardona-qft-methods_F03376 | 334  | ■ 1.000 | $* \rightarrow \mathcal{G} \times \mathcal{G}$                                                                                                                                                                           | $* \rightharpoonup \mathcal{G} \times \mathcal{G}$                                                                                                                                                                      | Let $L_1 := L_3 \circ L_I : * \rightharpoonup \mathcal{G}$ . From the definitions above we get $L_{\text{rel}}$ .                                                                                         |
| cardona-qft-methods_F03377 | 334  | ■ 1.000 | $L_I$                                                                                                                                                                                                                    | $L_I$                                                                                                                                                                                                                   | Let $L_1 := L_3 \circ L_I : * \rightharpoonup \mathcal{G}$ . From the definitions above we get $L_{\text{rel}}$ .                                                                                         |
| cardona-qft-methods_F03378 | 334  | ■ 1.000 | $5 L_3 \circ L_I$                                                                                                                                                                                                        | $5L_3 \circ L_I$                                                                                                                                                                                                        | <b>Corollary 3.2.</b>                                                                                                                                                                                     |
| cardona-qft-methods_F03379 | 334  | ■ 1.000 | $L_1 := L_3 \circ L_I : * \rightarrow \mathcal{G}$                                                                                                                                                                       | $L_1 := L_3 \circ L_I : * \rightharpoonup \mathcal{G}$                                                                                                                                                                  | $I(L_1) = \overline{L_1}$ ,                                                                                                                                                                               |
| cardona-qft-methods_F03380 | 334  | ■ 1.000 | $L_1$                                                                                                                                                                                                                    | $L_1$                                                                                                                                                                                                                   |                                                                                                                                                                                                           |
| cardona-qft-methods_F03381 | 335  | ■ 1.000 | $6 L_3 \circ (L_1 \times Id)$                                                                                                                                                                                            | $6L_3 \circ (L_1 \times Id)$                                                                                                                                                                                            |                                                                                                                                                                                                           |
| cardona-qft-methods_F03382 | 335  | ■ 1.000 | $\overline{\mathcal{G}} \times \mathcal{G}$                                                                                                                                                                              | $\overline{\mathcal{G}} \times \mathcal{G}$                                                                                                                                                                             |                                                                                                                                                                                                           |

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| cardona-qft-methods_F03383 | 335  | ■ 1.000 | $L_2$                                                                                             | $L_2$                                                                                             |    |
| cardona-qft-methods_F03384 | 335  | ■ 1.000 | $* \rightarrowtail \mathcal{G}$                                                                   | $* \rightarrowtail \mathcal{G}$                                                                   |    |
| cardona-qft-methods_F03385 | 335  | ■ 1.000 | $\mathcal{G}_{\text{rel}}$                                                                        | $\mathcal{G}_{\text{rel}}$                                                                        |    |
| cardona-qft-methods_F03386 | 335  | ■ 0.989 | $7L_2 \circ \mathcal{G}_{\text{rel}}$                                                             | $7L_2 \circ \mathcal{G}_{\text{rel}}$                                                             |    |
| cardona-qft-methods_F03387 | 335  | ■ 1.000 | $C := L_2 \circ \mathcal{G}_{\text{rel}}$                                                         | $C := L_2 \circ \mathcal{G}_{\text{rel}}$                                                         |    |
| cardona-qft-methods_F03388 | 335  | ■ 0.999 | $\underline{L_1} = L_1 / L_2$                                                                     | $\underline{L_1} = L_1 / L_2$                                                                     |                                                                                       |
| cardona-qft-methods_F03389 | 335  | ■ 1.000 | $\underline{L_1}$                                                                                 | $\underline{L_1}$                                                                                 |                                                                                       |
| cardona-qft-methods_F03390 | 336  | ■ 1.000 | $9S := \{(c, [l]) \in C \times M : \exists l \in [l], g \in \mathcal{G} \mid (l, c, g) \in L_3\}$ | $9S := \{(c, [l]) \in C \times M : \exists l \in [l], g \in \mathcal{G} \mid (l, c, g) \in L_3\}$ |                                                                                       |
| cardona-qft-methods_F03391 | 336  | ■ 1.000 | $\mathcal{G} \times M$                                                                            | $\mathcal{G} \times M$                                                                            |                                                                                       |
| cardona-qft-methods_F03392 | 336  | ■ 0.655 | $\mathcal{G}, L, I$                                                                               | $\mathcal{G}, L, I$                                                                               |    |
| cardona-qft-methods_F03393 | 336  | ■ 1.000 | $G := C / L_2$                                                                                    | $G := C / L_2$                                                                                    |    |
| cardona-qft-methods_F03394 | 336  | ■ 1.000 | $s := S / L_2, t := T / L_2, \mu :=$                                                              | $s := S / L_2, t := T / L_2, \mu :=$                                                              |    |
| cardona-qft-methods_F03395 | 336  | ■ 0.898 | $L_{\text{rel}} / L_2, \iota = I, \varepsilon = L_1 / L_2$                                        | $L_{\text{rel}} / L_2, \iota = I, \varepsilon = L_1 / L_2$                                        |    |
| cardona-qft-methods_F03396 | 336  | ■ 0.998 | $(G, L_G, I_G)$                                                                                   | $(G, L_G, I_G)$                                                                                   |  |
| cardona-qft-methods_F03397 | 336  | ■ 0.998 | $(H, L_H, I_H)$                                                                                   | $(H, L_H, I_H)$                                                                                   |  |
| cardona-qft-methods_F03398 | 336  | ■ 1.000 | $G \times \bar{H}$                                                                                | $G \times \bar{H}$                                                                                |  |
| cardona-qft-methods_F03399 | 336  | ■ 1.000 | $F \circ I_G = I_H \circ F$                                                                       | $F \circ I_G = I_H \circ F$                                                                       |  |
| cardona-qft-methods_F03400 | 336  | ■ 1.000 | $F^3(L_G) = L_H$                                                                                  | $F^3(L_G) = L_H$                                                                                  |  |
| cardona-qft-methods_F03401 | 336  | ■ 1.000 | $F: G \rightarrow$                                                                                | $F: G \rightarrow$                                                                                |  |
| cardona-qft-methods_F03402 | 336  | ■ 0.998 | $F^\dagger$                                                                                       | $F^\dagger$                                                                                       |  |
| cardona-qft-methods_F03403 | 336  | ■ 0.965 | $\mathcal{C}^k$                                                                                   | $\mathcal{C}^k$                                                                                   |                                                                                       |
| cardona-qft-methods_F03404 | 337  | ■ 0.909 | $(G, \omega)$                                                                                     | $(G, \omega)$                                                                                     |  |
| cardona-qft-methods_F03405 | 337  | ■ 0.998 | $G_{(1)} = G, G_{(2)}$                                                                            | $G_{(1)} = G, G_{(2)}$                                                                            |                                                                                       |
| cardona-qft-methods_F03406 | 337  | ■ 0.998 | $G \times_{(s,t)} G, G_{(3)} = G \times_{(s,t)}$                                                  | $G \times_{(s,t)} G, G_{(3)} = G \times_{(s,t)}$                                                  |                                                                                       |
| cardona-qft-methods_F03407 | 337  | ■ 0.615 | $(G \times_{(s,t)} G)$                                                                            | $(G \times_{(s,t)} G)$                                                                            |                                                                                       |
| cardona-qft-methods_F03408 | 337  | ■ 0.998 | $G_{(n)}$                                                                                         | $G_{(n)}$                                                                                         |                                                                                       |
| cardona-qft-methods_F03409 | 337  | ■ 0.998 | $G^n$                                                                                             | $G^n$                                                                                             |                                                                                       |

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|----------------------------|------|-------|--------------------------------------------------------------------------------------|----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| cardona-qft-methods_F03410 | 338  | 1.000 | $P_{\{n\}}: \underline{G_{\{n\}}} \rightarrow G^n$                                   | $P_n: \underline{G_{(n)}} \rightarrow G^n$                                       | there exists a natural symplectic groupoid structure on $\underline{G_{(n)}}$ coming from the natural decomposition of the associated structure on $G^n$                          |
| cardona-qft-methods_F03411 | 338  | 1.000 | $G_{\{i\}}$                                                                          | $G_{(i)}$                                                                        | is given by $P_i \circ P_i^\dagger$ .                                                                                                                                             |
| cardona-qft-methods_F03412 | 338  | 1.000 | $G_{\{j\}}, \forall i, j \geq 1$                                                     | $G_{(j)}, \forall i, j \geq 1$                                                   | is given by $P_i \circ P_i^\dagger$ .                                                                                                                                             |
| cardona-qft-methods_F03413 | 338  | 1.000 | $P_{\{i\}} \circ P_{\{j\}}^{\dagger}$                                                | $P_i \circ P_j^\dagger$                                                          |                                                                                                                                                                                   |
| cardona-qft-methods_F03414 | 338  | 0.966 | $\sim$                                                                               | $\sim$                                                                           | groupoid (which is the reduction to the boundary of the solutions of the Euler-Lagrange equations in the bulk)                                                                    |
| cardona-qft-methods_F03415 | 339  | 1.000 | $(\overline{\{X, \eta\}}) = \{ (X, \eta) \mid X \equiv X_0, \eta \in \ker \Pi^\# \}$ | $(\overline{X, \eta}) = \{ (X, \eta) \mid X \equiv X_0, \eta \in \ker \Pi^\# \}$ |                                                                                                                                                                                   |
| cardona-qft-methods_F03416 | 339  | 1.000 | $N \left( \Gamma_0 \left( T^* P M \right) \right)$                                   | $N \left( \Gamma_0 \left( T^* P M \right) \right)$                               | <b>Theorem 3.2.</b> <sup>3</sup> If the Poisson manifold $M$ is integrable, then there                                                                                            |
| cardona-qft-methods_F03417 | 339  | 1.000 | $\overline{L_1} \cap N \left( \Gamma_0 \left( T^* P M \right) \right)$               | $\overline{L_1} \cap N \left( \Gamma_0 \left( T^* P M \right) \right)$           | <b>Theorem 3.2.</b> <sup>3</sup> If the Poisson manifold $M$ is integrable, then there                                                                                            |
| cardona-qft-methods_F03418 | 339  | 1.000 | $L_1 \cap N \left( \Gamma_0 \left( T^* P M \right) \right), L_2 \cap$                | $L_1 \cap N \left( \Gamma_0 \left( T^* P M \right) \right), L_2 \cap$            |                                                                                                                                                                                   |
| cardona-qft-methods_F03419 | 339  | 1.000 | $N \left( \Gamma_0 \left( T^* P M \right) \right)^2$                                 | $N \left( \Gamma_0 \left( T^* P M \right) \right)^2$                             | <b>Theorem 3.3.</b> <sup>3</sup> If $M$ is integrable, then $L_1 \cap N \left( \Gamma_0 \left( T^* P M \right) \right), L_2 \cap$                                                 |
| cardona-qft-methods_F03420 | 339  | 1.000 | $L_3 \cap N \left( \Gamma_0 \left( T^* P M \right) \right)^3$                        | $L_3 \cap N \left( \Gamma_0 \left( T^* P M \right) \right)^3$                    | <b>Theorem 3.3.</b> <sup>3</sup> If $M$ is integrable, then $L_1 \cap N \left( \Gamma_0 \left( T^* P M \right) \right), L_2 \cap$                                                 |
| cardona-qft-methods_F03421 | 339  | 0.971 | $\rangle and an asso-$                                                               | $\rangle and an asso-$                                                           |                                                                                                                                                                                   |
| cardona-qft-methods_F03422 | 339  | 1.000 | $\{ \}^{\{-}}$                                                                       | $-$                                                                              | $e := \sum m(e_i, \bar{e}_i)$                                                                                                                                                     |
| cardona-qft-methods_F03423 | 339  | 1.000 | $\left\{ e_i \right\}$                                                               | $\{e_i\}$                                                                        | is an orthogonal projection.                                                                                                                                                      |
| cardona-qft-methods_F03424 | 340  | 0.998 | $\left( H, m, \langle, \rangle_H \right)$                                            | $(H, m, \langle, \rangle_H)$                                                     |                                                                                                                                                                                   |
| cardona-qft-methods_F03425 | 342  | 1.000 | $LM$                                                                                 | $LM$                                                                             | having total degree $-n$ .                                                                                                                                                        |
| cardona-qft-methods_F03426 | 342  | 1.000 | $-n$                                                                                 | $-n$                                                                             | section one we rev                                                                                                                                                                |
| cardona-qft-methods_F03427 | 343  | 1.000 | $[* / G]$                                                                            | $[*/G]$                                                                          | quotient orbifolds over the rationals and the construction of <sup>3</sup> of the string                                                                                          |
| cardona-qft-methods_F03428 | 343  | 1.000 | $H^{\{*\}}(L \times B \times G)$                                                     | $H^*(LBG)$                                                                       | topology of quotient orbifolds over the in to the case of $[*/G]$ and give geometric                                                                                              |
| cardona-qft-methods_F03429 | 343  | 1.000 | $H_{\{p\}}(M) \cong \left( H^{\{p\}}(M) \right)^*$                                   | $H_p(M) \cong (H^p(M))^*$                                                        |                                                                                                                                                                                   |
| cardona-qft-methods_F03430 | 343  | 1.000 | $H_{\{*+n\}}(M)$                                                                     | $H_{*+n}(M)$                                                                     |                                                                                                                                                                                   |
| cardona-qft-methods_F03431 | 344  | 1.000 | $\sigma \in H_p(M), \tau \in H_q(M)$                                                 | $\sigma \in H_p(M), \tau \in H_q(M)$                                             | ring structure can be interpreted in terms of intersection theory because                                                                                                         |
| cardona-qft-methods_F03432 | 344  | 0.958 | $\{ \}^{\{\mathrm{a}\}} P^{\{p\}} \subseteq M$                                       | ${}^a P^p \subseteq M$                                                           | when two homology classes $\sigma \in H_p(M), \tau \in H_q(M)$ can be represented by submanifolds $\Sigma, DP \subset M$ and $Q\Omega \subset M$ respectively then the product is |
| cardona-qft-methods_F03433 | 344  | 0.958 | $Q^{\{q\}} \subseteq M$                                                              | $Q^q \subseteq M$                                                                | when two homology classes $\sigma \in H_p(M), \tau \in H_q(M)$ can be represented by submanifolds $\Sigma, DP \subset M$ and $Q\Omega \subset M$ respectively then the product is |
| cardona-qft-methods_F03434 | 344  | 1.000 | $P \cap Q$                                                                           | $P \cap Q$                                                                       | represented by the tranverse intersection $P \cap Q$ .                                                                                                                            |
| cardona-qft-methods_F03435 | 344  | 0.951 | $\Delta: M \rightarrow M \times M$                                                   | $\Delta: M \rightarrow M \times M$                                               | is surjective compact map, since the submanifold containing $\Delta$ is compact which has a tubular neighborhood $\nu_\Delta$ homeomorphic to the total space                     |

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| cardona-qft-methods_F03436 | 344  | ■ 1.000 | $\nu_{\Delta}$                                                                                            | $\nu_{\Delta}$                                                                 | of the normal bundle, which is $TM$ . Consider the projection map that                                                                                                                     |
| cardona-qft-methods_F03437 | 344  | ■ 1.000 | $(M \times M) / \nu_{\Delta}^c$                                                                           | $(M \times M) / \nu_{\Delta}^c$                                                | $\tau_{\Delta} : M \times M \rightarrow M^{TM}$                                                                                                                                            |
| cardona-qft-methods_F03438 | 344  | ■ 1.000 | $M^{\mathrm{T} M}$                                                                                        | $M^{TM}$                                                                       | $\tau_{\Delta} : M \times M \rightarrow M^{TM}$                                                                                                                                            |
| cardona-qft-methods_F03439 | 344  | ■ 1.000 | $e \colon P \rightarrowtail Q$                                                                            | $e : P \rightarrow Q$                                                          | <sup>a</sup> We say that a homology class $\sigma \in H_p(M)$ can be represented by submanifold if there                                                                                   |
| cardona-qft-methods_F03440 | 344  | ■ 1.000 | $Q \rightarrowtail P^{\nu(e)}$                                                                            | $Q \rightarrow P^{\nu(e)}$                                                     | the Thom-Pontrjagin class of $P$ under the inclusion map.                                                                                                                                  |
| cardona-qft-methods_F03441 | 344  | ■ 1.000 | $\nu(e) \rightarrowtail P$                                                                                | $\nu(e) \rightarrow P$                                                         |                                                                                                                                                                                            |
| cardona-qft-methods_F03442 | 344  | ■ 1.000 | $e_! \colon H_*(Q) \rightarrow H_{*-\dim \nu(e)}(P)$                                                      | $e_! : H_*(Q) \rightarrow H_{*-\dim \nu(e)}(P)$                                |                                                                                                                                                                                            |
| cardona-qft-methods_F03443 | 344  | ■ 1.000 | $\sigma \in H_p(M)$                                                                                       | $\sigma \in H_p(M)$                                                            |                                                                                                                                                                                            |
| cardona-qft-methods_F03444 | 345  | ■ 1.000 | $\operatorname{Map}(8, M)$                                                                                | $\operatorname{Map}(8, M)$                                                     | Let $\operatorname{Map}(8, M)$ be the space of piecewise smooth maps from the figure                                                                                                       |
| cardona-qft-methods_F03445 | 345  | ■ 0.991 | $e \colon v \colon L M \rightarrowtail M$                                                                 | $ev : LM \rightarrow M$                                                        | of the concatenation of paths. Concatenation is a natural operation to give to the inverse image of the normal bundle to $M \rightarrowtail M \times M$ . Thus we have a                   |
| cardona-qft-methods_F03446 | 345  | ■ 0.570 | $\operatorname{Map}(8, M) \rightarrowtail L M \times L M$                                                 | $\operatorname{Map}(8, M) \rightarrow LM \times LM$                            | Thom-Pontrjagin collapse map $\tau_{\Delta} : LM \times LM \rightarrow \operatorname{Map}(8, M)^{ev^*(TM)}$ .                                                                              |
| cardona-qft-methods_F03447 | 345  | ■ 1.000 | $M \rightarrowtail M \times M$                                                                            | $M \rightarrow M \times M$                                                     | the map induced by the Thom-Pontrjagin collapse map, together with the                                                                                                                     |
| cardona-qft-methods_F03448 | 345  | ■ 0.927 | $\tau_{\Delta} \colon L M \times L M \rightarrowtail \operatorname{Map}(8, M)^{\wedge v^*(\mathrm{T} M)}$ | $\tau_{\Delta} : LM \times LM \rightarrow \operatorname{Map}(8, M)^{ev^*(TM)}$ | Thom isomorphism and the map induced by the concatenation map gives the Chas-Sullivan loop product                                                                                         |
| cardona-qft-methods_F03449 | 345  | ■ 0.745 | $\operatorname{Map}(8, M) \rightarrowtail L M$                                                            | $\operatorname{Map}(8, M) \rightarrow LM$                                      | $H^*(LM) \otimes H^*(LM) \rightarrowtail H^*(LM \times LM)$                                                                                                                                |
| cardona-qft-methods_F03450 | 345  | ■ 1.000 | $\Omega M \times \Omega M \rightarrowtail \Omega M$                                                       | $\Omega M \times \Omega M \rightarrow \Omega M$                                | $\Omega M \times \Omega M \rightarrowtail \Omega M$                                                                                                                                        |
| cardona-qft-methods_F03451 | 346  | ■ 1.000 | $L M^{\mathrm{-T} M}$                                                                                     | $LM^{-TM}$                                                                     |                                                                                                                                                                                            |
| cardona-qft-methods_F03452 | 346  | ■ 1.000 | $\mathrm{-T} M \rightarrowtail M$                                                                         | $-TM \rightarrow M$                                                            | <sup>space</sup> The Poincare duality identification is realized at the level of stable ho-                                                                                                |
| cardona-qft-methods_F03453 | 346  | ■ 1.000 | $M^{\mathrm{-T} M} \simeq F(\mathrm{M}_+, S)$                                                             | $M^{-TM} \simeq F(M_+, S)$                                                     | $H_p(M^{-TM}) \cong H_{p+n}(M)$                                                                                                                                                            |
| cardona-qft-methods_F03454 | 346  | ■ 1.000 | $M^{\mathrm{-T} M}$                                                                                       | $M^{-TM}$                                                                      | $H_p(M^{-TM}) \cong H_{p+n}(M)$                                                                                                                                                            |
| cardona-qft-methods_F03455 | 346  | ■ 0.999 | $\mathrm{M}_+^{\mathrm{b}}$                                                                               | $M_+^{\mathrm{b}}$                                                             | and the Atiyah duality $M^{-TM} \simeq F(M_+, S)$ , that identifies $M^{-TM}$ as the Spanier-Whithead dual of $M$ b. Therefore $H^*(M^{-TM}) \simeq$                                       |
| cardona-qft-methods_F03456 | 346  | ■ 0.999 | $H_*\bigl(\mathrm{M}^{\mathrm{-T} M}\bigr) \cong$                                                         | $H_*\bigl(M^{-TM}\bigr) \cong$                                                 | and the Atiyah duality $M^{-TM} \simeq F(M_+, S)$ , that identifies $M^{-TM}$ as the Spanier-Whithead dual of $M$ b. Therefore $H^*(M^{-TM}) \simeq$                                       |
| cardona-qft-methods_F03457 | 346  | ■ 1.000 | $H_*\bigl(F\bigl(\mathrm{F}\bigl(\mathrm{M}_+, S\bigr)\bigr)\bigr) \cong H^{\mathrm{-*}}(\mathrm{M})$     | $H_*\bigl(F(M_+, S)\bigr) \cong H^{-*}(M)$                                     | as the Spanier-Whithead dual of $M$ b. Therefore $H^*(M^{-TM}) \simeq$                                                                                                                     |
| cardona-qft-methods_F03458 | 346  | ■ 1.000 | $e \colon M \hookrightarrowtail \mathbb{R}^k$                                                             | $e : M \hookrightarrow \mathbb{R}^k$                                           | given an embedding $e : M \hookrightarrow \mathbb{R}^k$ , let $\nu(e) \rightarrowtail M$ the normal bundle of the embedding and denote by $M^{\nu(e)}$ its Thom space. Define the spectrum |
| cardona-qft-methods_F03459 | 346  | ■ 1.000 | $\nu(e) \rightarrowtail M$                                                                                | $\nu(e) \rightarrow M$                                                         | given an embedding $e : M \hookrightarrow \mathbb{R}^k$ , let $\nu(e) \rightarrowtail M$ the normal bundle of the embedding and denote by $M^{\nu(e)}$ its Thom space. Define the spectrum |
| cardona-qft-methods_F03460 | 346  | ■ 0.999 | $M^{\nu(e)}$                                                                                              | $M^{\nu(e)}$                                                                   | $M^{-TM} := \Sigma^{-k} M^{\nu(e)}$                                                                                                                                                        |
| cardona-qft-methods_F03461 | 346  | ■ 0.894 | $M \times M \rightarrowtail M^{\mathrm{T} M}$                                                             | $M \times M \rightarrow M^{TM}$                                                | that is associative up to homotopy.                                                                                                                                                        |
| cardona-qft-methods_F03462 | 346  | ■ 0.894 | $\mathrm{-2} \ \mathrm{T} \ M$                                                                            | $-2TM$                                                                         | that is associative up to homotopy.                                                                                                                                                        |

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| cardona-qft-methods_F03463 | 346  | 0.997 | $\operatorname{Map}(8, M) \rightarrow$                                  | $\operatorname{Map}(8, M) \rightarrow$             | Composing these maps we get a map of spectra<br>$\operatorname{rev}^*(-TM) \rightarrow \operatorname{rev}^*(-TM) \rightarrow \operatorname{rev}^* \dots \operatorname{rev}^*(-2TM)$ |
| cardona-qft-methods_F03464 | 346  | 1.000 | $e^{v^*}(-TM)$                                                          | $ev^*(-TM)$                                        | $LM \rightarrow \dots \rightarrow LM \xrightarrow{\quad} (LM \times LM) \xrightarrow{\quad} \dots$                                                                                  |
| cardona-qft-methods_F03465 | 346  | 0.481 | $\operatorname{Map}(8, M)^{e^{v^*}}(-TM) \rightarrow LM^{e^{v^*}}(-TM)$ | $Map(8, M)^{ev^*}(-TM) \rightarrow LM^{ev^*}(-TM)$ | $\operatorname{Map}(8, M)^{ev^*}(-TM) \rightarrow LM^{ev^*}(-TM)$                                                                                                                   |
| cardona-qft-methods_F03466 | 346  | 0.998 | $e^{v^*}(-2TM)$                                                         | $ev^*(-2TM)$                                       | $\operatorname{rev}^* \dots \operatorname{rev}^*(-2TM)$                                                                                                                             |
| cardona-qft-methods_F03467 | 346  | 0.999 | $X_+$                                                                   | $X_+$                                              |                                                                                                                                                                                     |
| cardona-qft-methods_F03468 | 346  | 1.000 | $F\left(X_+, Y\right)$                                                  | $F\left(X_+, Y\right)$                             |                                                                                                                                                                                     |
| cardona-qft-methods_F03469 | 347  | 0.716 | $1.2^{\left(7\right)}$                                                  | $1.2^{\left(7\right)}$                             | 336 <i>A. Angel</i>                                                                                                                                                                 |
| cardona-qft-methods_F03470 | 347  | 0.811 | $LM^{e^{v^*}}(-TM)$                                                     | $LM^{ev^*}(-TM)$                                   | that is associative up to homotopy.                                                                                                                                                 |
| cardona-qft-methods_F03471 | 347  | 0.999 | $1.3^{\left(\mathbf{7}\right)}$                                         | $1.3^{\left(7\right)}$                             |                                                                                                                                                                                     |
| cardona-qft-methods_F03472 | 348  | 0.979 | $\bar{U}, G, U$                                                         | $\bar{U}, G, U$                                    | <i>orbifold chart</i> on $X$ is a triple $(\bar{U}, G, U)$ , where $\bar{U}$ is a connected manifold,                                                                               |
| cardona-qft-methods_F03473 | 348  | 0.979 | $\bar{U}$                                                               | $\bar{U}$                                          | <i>orbifold chart</i> on $X$ is a triple $(\bar{U}, G, U)$ , where $\bar{U}$ is a connected manifold,                                                                               |
| cardona-qft-methods_F03474 | 348  | 1.000 | $\bar{U} / G$                                                           | $\bar{U} / G$                                      |                                                                                                                                                                                     |
| cardona-qft-methods_F03475 | 348  | 0.910 | $\bar{U}_{\alpha}, G_{\alpha}, U_{\alpha}$                              | $\bar{U}_{\alpha}, G_{\alpha}, U_{\alpha}$         | $X$ is just an equivalence class of orbifold atlases, where two atlases are equivalent if there is a zig-zag of common refinements.                                                 |
| cardona-qft-methods_F03476 | 348  | 1.000 | $[M / G]$                                                               | $[M / G]$                                          | Given a point $x \in X$ take a chart $(\bar{U}, G, U)$ around $x$ . let $\bar{\pi}$                                                                                                 |
| cardona-qft-methods_F03477 | 348  | 0.936 | $x \in X$                                                               | $x \in X$                                          | $\operatorname{rev}^* \dots \operatorname{rev}^*(-TM) \rightarrow \operatorname{rev}^* \dots \operatorname{rev}^*(-2TM)$                                                            |
| cardona-qft-methods_F03478 | 348  | 0.936 | $\bar{x} \in \bar{U}$                                                   | $\bar{x} \in \bar{U}$                              | $\operatorname{rev}^* \dots \operatorname{rev}^*(-TM) \rightarrow \operatorname{rev}^* \dots \operatorname{rev}^*(-2TM)$                                                            |
| cardona-qft-methods_F03479 | 348  | 1.000 | $G_x$                                                                   | $G_x$                                              | $\operatorname{rev}^* \dots \operatorname{rev}^*(-TM) \rightarrow \operatorname{rev}^* \dots \operatorname{rev}^*(-2TM)$                                                            |
| cardona-qft-methods_F03480 | 348  | 1.000 | $\mathbb{C}^2 - \{0\}$                                                  | $\mathbb{C}^2 - \{0\}$                             | <i>homotopy space</i> of the original desc. For example for a good quotient $\mathcal{X} = [M/G]$ with $G$ a finite group acting on $M$ , $B\mathcal{X}$ is homotopy equivalent     |
| cardona-qft-methods_F03481 | 348  | 1.000 | $\lambda\left(z_1, z_2\right)=$                                         | $\lambda\left(z_1, z_2\right)=$                    | <i>homotopy space</i> of the original desc. For example for a good quotient $\mathcal{X} = [M/G]$ with $G$ a finite group acting on $M$ , $B\mathcal{X}$ is homotopy equivalent     |
| cardona-qft-methods_F03482 | 348  | 0.999 | $\left(\lambda^2 z, \lambda z\right)$                                   | $\left(\lambda^2 z, \lambda z\right)$              |                                                                                                                                                                                     |
| cardona-qft-methods_F03483 | 348  | 0.999 | $\mathcal{X}$                                                           | $\mathcal{X}$                                      |                                                                                                                                                                                     |
| cardona-qft-methods_F03484 | 348  | 0.999 | $B \mathcal{X}$                                                         | $B\mathcal{X}$                                     |                                                                                                                                                                                     |
| cardona-qft-methods_F03485 | 348  | 1.000 | $\mathcal{X}=[M / G]$                                                   | $\mathcal{X}=[M / G]$                              |                                                                                                                                                                                     |
| cardona-qft-methods_F03486 | 348  | 1.000 | $M, B \mathcal{X}$                                                      | $M, B\mathcal{X}$                                  |                                                                                                                                                                                     |
| cardona-qft-methods_F03487 | 349  | 1.000 | $E G \times _G M$                                                       | $EG \times_G M$                                    |                                                                                                                                                                                     |
| cardona-qft-methods_F03488 | 349  | 1.000 | $E G \rightarrow B G$                                                   | $EG \rightarrow BG$                                |                                                                                                                                                                                     |
| cardona-qft-methods_F03489 | 349  | 0.993 | $ \mathcal{X} $                                                         | $ \mathcal{X} $                                    | to the homotopy orbit space $EG \times_G M$ , where $EG \rightarrow BG$ is the universal                                                                                            |

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| cardona-qft-methods_F03490 | 349  | ■ 0.993 | $B \mathcal{\rightarrow}  \mathcal{X} $                                                                                                                                                    | $B\mathcal{X} \rightarrow  \mathcal{X} $                                                                     | to the homotopy orbit space $EG \times_G M$ , where $EG \rightarrow BG$ is the universal                                                                                                                                                                                           |
| cardona-qft-methods_F03491 | 349  | ■ 1.000 | $\left[P_{\{G\}} M / G\right]$                                                                                                                                                             | $[P_G M / G]$                                                                                                | group, Lupercio and Uribe <sup>10</sup> constructed the loop groupoid $[F_G M / G]$ as the natural model for the free loop space of the orbifold, whose homotonic                                                                                                                  |
| cardona-qft-methods_F03492 | 349  | ■ 1.000 | $M \times_{\{G\}} E G$                                                                                                                                                                     | $M \times_G EG$                                                                                              | $P_G M \times_G EG \simeq I$                                                                                                                                                                                                                                                       |
| cardona-qft-methods_F03493 | 349  | ■ 1.000 | $[X / G]$                                                                                                                                                                                  | $[X / G]$                                                                                                    | whose space of objects is                                                                                                                                                                                                                                                          |
| cardona-qft-methods_F03494 | 349  | ■ 1.000 | $X \times G$                                                                                                                                                                               | $X \times G$                                                                                                 | $\mathcal{P}_\bullet \mathcal{X} := \{ \mid \mathcal{P} \mathcal{X} \vee f(n) \} \quad \text{with} \quad \mathcal{P} \mathcal{X} := \{ f \cdot (n-1) \hookrightarrow \mathcal{X}(f(n)) \}$                                                                                         |
| cardona-qft-methods_F03495 | 349  | ■ 1.000 | $s(x, g)=x$                                                                                                                                                                                | $s(x, g) = x$                                                                                                | $\mathcal{P}_\bullet \mathcal{X} := \{ \mid \mathcal{P} \mathcal{X} \vee f(n) \} \quad \text{with} \quad \mathcal{P} \mathcal{X} := \{ f \cdot (n-1) \hookrightarrow \mathcal{X}(f(n)) \}$                                                                                         |
| cardona-qft-methods_F03496 | 349  | ■ 1.000 | $t(x, g)=x \cdot g$                                                                                                                                                                        | $t(x, g) = xg$                                                                                               | $\mathcal{P}_\bullet \mathcal{X} := \{ \mid \mathcal{P} \mathcal{X} \vee f(n) \} \quad \text{with} \quad \mathcal{P} \mathcal{X} := \{ f \cdot (n-1) \hookrightarrow \mathcal{X}(f(n)) \}$                                                                                         |
| cardona-qft-methods_F03497 | 349  | ■ 1.000 | $[\mathbf{R} / \mathbf{Z}]$                                                                                                                                                                | $[\mathbf{R} / \mathbf{Z}]$                                                                                  | and which are endowed with the $G$ -action                                                                                                                                                                                                                                         |
| cardona-qft-methods_F03498 | 349  | ■ 1.000 | $\left[P_{\{G\}} X / G\right]$                                                                                                                                                             | $[P_G X / G]$                                                                                                |                                                                                                                                                                                                                                                                                    |
| cardona-qft-methods_F03499 | 350  | ■ 1.000 | $e_{\{t\}}: P_{\{g\}} M \rightarrowtail M$                                                                                                                                                 | $e_t : P_g M \rightarrowtail M$                                                                              |                                                                                                                                                                                                                                                                                    |
| cardona-qft-methods_F03500 | 350  | ■ 1.000 | $t: e_{\{t\}}(f):=f(t)$                                                                                                                                                                    | $t : e_t(f) := f(t)$                                                                                         |                                                                                                                                                                                                                                                                                    |
| cardona-qft-methods_F03501 | 350  | ■ 1.000 | $e_{\{0\}} \circ \pi_1$                                                                                                                                                                    | $e_0 \circ \pi_1$                                                                                            | where the downward arrow $e_0 \circ \pi_1$ is the evaluation at zero of the first map. The natural bundle of the above horizontal map becomes isomor                                                                                                                               |
| cardona-qft-methods_F03502 | 350  | ■ 1.000 | $\Delta_g$                                                                                                                                                                                 | $\Delta_g$                                                                                                   | $P_g M^{-e_0^* TM} \wedge P_h M^{-e_0^* TM} \rightarrow P_g M : {}_1 \times_0 P_h M^{-\pi_1^* e_0^* TM}$ . The concatenation                                                                                                                                                       |
| cardona-qft-methods_F03503 | 350  | ■ 0.820 | $P_{\{g\}} M \times P_{\{h\}} M \rightarrowtail P_{\{g\}} M : \{ \}_1 \times_0 P_{\{h\}} M^{\pi_1^*} e_0^* N_g$                                                                            | $P_g M \times P_h M \rightarrowtail P_g M : {}_1 \times_0 P_h M^{\pi_1^*} e_0^* N_g$                         | of paths in $P_g M : {}_1 \times_0 P_h M$ produces a map $P_g M : {}_1 \times_0 P_h M \rightarrowtail P_{gh} M$ which                                                                                                                                                              |
| cardona-qft-methods_F03504 | 350  | ■ 0.806 | $P_{\{g\}} M^{\wedge\{-e_{\{0\}}^*\} T M} \wedge P_{\{h\}} M^{\wedge\{-e_{\{0\}}^*\} T M} \rightarrowtail P_{\{g\}} M : \{ \}_1 \times_0 P_{\{h\}} M^{\wedge\{-\pi_1^* e_{\{0\}}^*\} T M}$ | $P_g M^{-e_0^* TM} \wedge P_h M^{-e_0^* TM} \rightarrowtail P_g M : {}_1 \times_0 P_h M^{-\pi_1^*} e_0^* TM$ | induces the map of spectra $P_{gh} M : {}_1 \times_0 P_h M^{\wedge\{-\pi_1^* e_{\{0\}}^*\} T M} \rightarrowtail P_{gh} M^{\wedge\{-e_{\{0\}}^*\} T M}$ that defines the map $\mu_{g,h} : P_g M^{-e_0^* TM} \wedge P_h M^{-e_0^* TM} \rightarrowtail P_{gh} M^{-e_0^* TM}$ . Define |
| cardona-qft-methods_F03505 | 350  | ■ 0.991 | $P_{\{g\}} M : \{ \}_1 \times_0 P_{\{h\}} M$                                                                                                                                               | $P_g M : {}_1 \times_0 P_h M$                                                                                | $\mathcal{P}_\bullet M^{-TM} := \{ \mid \mathcal{P} M^{-TM} \}$                                                                                                                                                                                                                    |
| cardona-qft-methods_F03506 | 350  | ■ 0.991 | $P_{\{g\}} M : \{ \}_1 \times_0 P_{\{h\}} M \rightarrowtail P_{gh} M$                                                                                                                      | $P_g M : {}_1 \times_0 P_h M \rightarrowtail P_{gh} M$                                                       | $\mathcal{P}_\bullet M^{-TM} := \{ \mid \mathcal{P} M^{-TM} \}$                                                                                                                                                                                                                    |
| cardona-qft-methods_F03507 | 350  | ■ 0.926 | $P_{\{g\}} M : \{ \}_1 \times_0 P_{\{h\}} M^{\wedge\{-\pi_1^* e_{\{0\}}^*\} T M} \rightarrowtail P_{gh} M^{\wedge\{-e_{\{0\}}^*\} T M}$                                                    | $P_g M : {}_1 \times_0 P_h M^{-\pi_1^*} e_0^* TM \rightarrowtail P_{gh} M^{-e_0^* TM}$                       | $\wedge_{g \in G} g^{*} \mathcal{A} = \varinjlim_{g \in G} \wedge g^{*} \mathcal{A}$                                                                                                                                                                                               |
| cardona-qft-methods_F03508 | 350  | ■ 1.000 | $\mu_{g,h} : P_{\{g\}} M^{\wedge\{-e_{\{0\}}^*\} T M} \wedge P_{\{h\}} M^{\wedge\{-e_{\{0\}}^*\} T M} \rightarrowtail P_{gh} M^{\wedge\{-e_{\{0\}}^*\} T M}$                               | $\mu_{g,h} : P_g M^{-e_0^* TM} \wedge P_h M^{-e_0^* TM} \rightarrowtail P_{gh} M^{-e_0^* TM}$                | which by assembling the maps $\mu_{g,h}$ define a map that is associative up to                                                                                                                                                                                                    |
| cardona-qft-methods_F03509 | 350  | ■ 1.000 | $\mu_{g,h}$                                                                                                                                                                                | $\mu_{g,h}$                                                                                                  | in the homotopy category, taking homology, we get a $G$ -module ring $H^*(P_\bullet M^{-TM}, \mathbb{Z})$ , the induced ring structure on the invariant set                                                                                                                        |
| cardona-qft-methods_F03510 | 350  | ■ 0.546 | $3.1 \left( \{ \}^{\wedge \mathbf{14}} \right) . P_{\{G\}} M^{\wedge \{-T M\}}$                                                                                                            | $3.1 \left( \mathbf{14} \right) . P_G M^{-TM}$                                                               | This ring was called in <sup>14</sup> the orbifold string topology ring.                                                                                                                                                                                                           |
| cardona-qft-methods_F03511 | 350  | ■ 1.000 | $H_{\{*\}} \left( P_{\{G\}} M^{\wedge \{-T M\}} ; \mathbb{Z} \right)$                                                                                                                      | $H_* \left( P_G M^{-TM}; \mathbb{Z} \right)$                                                                 | In the case that $M$ is simply connected, Angel, Backelin and Uribe in <sup>3</sup>                                                                                                                                                                                                |
| cardona-qft-methods_F03512 | 351  | ■ 1.000 | $C^{\{*\}}(M ; \mathbb{Q}) \# G$                                                                                                                                                           | $C^*(M; \mathbb{Q}) \# G$                                                                                    |                                                                                                                                                                                                                                                                                    |
| cardona-qft-methods_F03513 | 351  | ■ 1.000 | $C^{\{*\}}(M) \setminus \# G$                                                                                                                                                              | $C^*(M) \# G$                                                                                                | there is an isomorphism of rings                                                                                                                                                                                                                                                   |
| cardona-qft-methods_F03514 | 351  | ■ 1.000 | $C^{\{*\}}(X) \setminus \# G$                                                                                                                                                              | $C^*(X) \# G$                                                                                                | $(C^*(X) \# G)^e$ -module, together with an explicit diagonal map. It is shown                                                                                                                                                                                                     |
| cardona-qft-methods_F03515 | 351  | ■ 1.000 | $\left( C^{\{*\}}(X) \setminus \# G \right)^e$                                                                                                                                             | $(C^*(X) \# G)^e$                                                                                            | in <sup>3</sup> that if we consider $C^*(X) \# G$ as a $G$ -module-dg-ring by the action $a \cdot (x \otimes h) := a(x) \otimes ah a^{-1}$ , then we have that                                                                                                                     |

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| cardona-qft-methods_F03516 | 351  | ■ 1.000 | $g \cdot (x \otimes h) \mapsto g(x) \otimes gh^{-1}$                                                                                                                                                    | $g \cdot (x \otimes h) \mapsto g(x) \otimes ghg^{-1}$                                                                                                  | <b>Theorem 3.2.</b> For any space $X$ there are isomor-                                                                                                                                                                     |
| cardona-qft-methods_F03517 | 351  | ■ 0.951 | $H H^*\left(C^*(X) \rtimes G, C^*(X) \rtimes G\right) \cong \operatorname{Ext}_{\mathbb{Z} G}^*\left(\mathbb{Z}, \mathcal{R} \operatorname{Hom}_{C^*(X)^e}\left(C^*(X), C^*(X) \rtimes G\right)\right)$ | $HH^*(C^*(X) \# G, C^*(X) \# G) \cong \operatorname{Ext}_{\mathbb{Z} G}^*(\mathbb{Z}, \mathcal{R} \operatorname{Hom}_{C^*(X)^e}(C^*(X), C^*(X) \# G))$ | Using a cosimplicial description for the space $P_G X$ , in it is proved that                                                                                                                                               |
| cardona-qft-methods_F03518 | 351  | ■ 0.984 | $P_{\{G\}} X, \operatorname{in}^3$                                                                                                                                                                      | $P_G X, \operatorname{in}^3$                                                                                                                           | that induces the equivariant isomorphism                                                                                                                                                                                    |
| cardona-qft-methods_F03519 | 351  | ■ 1.000 | $P_{\{G\}} M$                                                                                                                                                                                           | $P_G M$                                                                                                                                                |                                                                                                                                                                                                                             |
| cardona-qft-methods_F03520 | 352  | ■ 1.000 | $E \ G$                                                                                                                                                                                                 | $EG$                                                                                                                                                   | where by abuse of notation we                                                                                                                                                                                               |
| cardona-qft-methods_F03521 | 352  | ■ 1.000 | $E \ G_{\{1\}} \subset \cdots \subset E \ G_{\{n\}} \subset E \ G_{\{n+1\}} \subset \cdots \subset E \ G$                                                                                               | $EG_1 \subset \cdots \subset EG_n \subset EG_{n+1} \subset \cdots \subset EG$                                                                          | $(f, \lambda) \in P_G M \times_G EG_n$ also by $e_t$ . Let us call $M_n = M \times_G EG_n$ .                                                                                                                                |
| cardona-qft-methods_F03522 | 352  | ■ 1.000 | $E \ G \rightarrow$                                                                                                                                                                                     | $EG \rightarrow$                                                                                                                                       | take the trivial spectra formed by pulling back the negative of the tangent bundle of $M_n$ under $e_n$                                                                                                                     |
| cardona-qft-methods_F03523 | 352  | ■ 1.000 | $B \ G$                                                                                                                                                                                                 | $BG$                                                                                                                                                   |                                                                                                                                                                                                                             |
| cardona-qft-methods_F03524 | 352  | ■ 1.000 | $(f, \lambda) \in P_{\{G\}} M \times_{\{G\}} E \ G_{\{n\}}$                                                                                                                                             | $(f, \lambda) \in P_G M \times_G EG_n$                                                                                                                 | mology of this pro-spectrum                                                                                                                                                                                                 |
| cardona-qft-methods_F03525 | 352  | ■ 1.000 | $e_{\{t\}}$                                                                                                                                                                                             | $e_t$                                                                                                                                                  | mology of this pro-spectrum                                                                                                                                                                                                 |
| cardona-qft-methods_F03526 | 352  | ■ 1.000 | $M_{\{n\}} = M \times_{\{G\}} E \ G_{\{n\}}$                                                                                                                                                            | $M_n = M \times_G EG_n$                                                                                                                                | mology of this pro-spectrum                                                                                                                                                                                                 |
| cardona-qft-methods_F03527 | 352  | ■ 1.000 | $M_{\{n\}}$                                                                                                                                                                                             | $M_n$                                                                                                                                                  |                                                                                                                                                                                                                             |
| cardona-qft-methods_F03528 | 352  | ■ 1.000 | $e_{\{0\}}$                                                                                                                                                                                             | $e_0$                                                                                                                                                  |                                                                                                                                                                                                                             |
| cardona-qft-methods_F03529 | 352  | ■ 1.000 | $E \ G_{\{n\}} \subset E \ G_{\{n+1\}} \subset \cdots \subset E \ G$                                                                                                                                    | $EG_n \subset EG_{n+1} \subset \cdots \subset EG$                                                                                                      |                                                                                                                                                                                                                             |
| cardona-qft-methods_F03530 | 352  | ■ 1.000 | $\mathbb{L}\left(M \times_{\{G\}} E \ G_{\{n\}} \times_{\{G\}} E \ G_{\{n+1\}} \times \cdots \times_{\{G\}} E \ G_{\{n+k\}}\right)$                                                                     | $\mathbb{L}\left(M \times_G EG\right)^{-T(M \times_G EG)}$                                                                                             |                                                                                                                                                                                                                             |
| cardona-qft-methods_F03531 | 353  | ■ 0.548 | $3.5 \left( \mathbf{3} \right)$                                                                                                                                                                         | $3.5 \left( \mathbf{3} \right)$                                                                                                                        | is the orbifold string topology ring of the orbifold $[M/G]$ that in the simi                                                                                                                                               |
| cardona-qft-methods_F03532 | 353  | ■ 1.000 | $M = *$                                                                                                                                                                                                 | $M = *$                                                                                                                                                | When $M = *$ the orbifold string topology of $[M/G]$ is precisely the non-                                                                                                                                                  |
| cardona-qft-methods_F03533 | 353  | ■ 1.000 | $L \ B \ G^{-T \ B \ G}$                                                                                                                                                                                | $LBG^{-TBG}$                                                                                                                                           | ring spectrum $LBG^{-TBG}$ which has been studied in, <sup>1011</sup> and <sup>12</sup> for general                                                                                                                         |
| cardona-qft-methods_F03534 | 353  | ■ 0.919 | $\operatorname{L}[* / G]$                                                                                                                                                                               | $L[* / G]$                                                                                                                                             | $\mathbb{L}[* / G]$ is the groupoid $[G / G]$ that is the translation groupoid associated to the action of $G$ on itself by conjugation, since $G$ is a discrete group we                                                   |
| cardona-qft-methods_F03535 | 353  | ■ 0.919 | $\left[G^{ad} / G\right]$                                                                                                                                                                               | $[G^{ad} / G]$                                                                                                                                         | $\mathbb{L}[* / G]$ is the groupoid $[G / G]$ that is the translation groupoid associated to the action of $G$ on itself by conjugation, since $G$ is a discrete group we                                                   |
| cardona-qft-methods_F03536 | 353  | ■ 1.000 | $C(g)$                                                                                                                                                                                                  | $C(g)$                                                                                                                                                 | ring product described in section 3.1 is just the product of the group ring. Approximate $EG$ by $EC_1 \hookrightarrow \cdots \hookrightarrow EC_2 \hookrightarrow EC_3 \hookrightarrow \cdots \hookrightarrow EC_n$ finite |
| cardona-qft-methods_F03537 | 353  | ■ 1.000 | $H_*\left(P_{\{G\}} M^{-T \ M} ; \mathbb{Q}\right) \cong \mathbb{Q} G$                                                                                                                                  | $H_*\left(P_G M^{-TM} ; \mathbb{Q}\right)^G \cong \mathbb{Q} G$                                                                                        | ring spectrum $LBG^{-TBG}$ is therefore the inverse system formed by the                                                                                                                                                    |
| cardona-qft-methods_F03538 | 353  | ■ 1.000 | $B \ G_{\{n\}} := E \ G_{\{n\}} / G$                                                                                                                                                                    | $BG_n := EG_n / G$                                                                                                                                     | $\mathbb{L}(g) \times (L(G_n) \cup (g))$ . Since any model of $EG$ is also a model of $LBG$ for any subgroup $H \subset G$ we can take $EG_1 \subset \cdots \subset EG_n \subset EG_{n+1} \subset \cdots$ to                |
| cardona-qft-methods_F03539 | 353  | ■ 1.000 | $\left(G^{ad} \times_{\{G\}} E \ G_{\{n\}}\right)^{-T \ B \ G_{\{n\}}}$                                                                                                                                 | $\left(G^{ad} \times_G EG_n\right)^{-TBG_n}$                                                                                                           | spectrum $LBG^{-TBG}$ is for                                                                                                                                                                                                |
| cardona-qft-methods_F03540 | 353  | ■ 0.999 | $G^{ad} \times_{\{G\}} E \ G_{\{n\}}$                                                                                                                                                                   | $G^{ad} \times_G EG_n$                                                                                                                                 | $\bigsqcup_{(n)} BC(g)_n^{-T \ B \ G / n}$                                                                                                                                                                                  |

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| cardona-qft-methods_F03541 | 353  | ■ 1.000 | $\bigcup_{(g)} E G_n / C(g)$                                                     | $\bigsqcup_{(g)} EG_n / C(g)$                                                 | The homology of $LBG^{-TBG}$ is therefore the inverse limit of                                                                                                                                          |
| cardona-qft-methods_F03542 | 353  | ■ 1.000 | $T\left(G^{\mathrm{ad}} \times_G E G_n\right)$                                   | $T\left(G^{\mathrm{ad}} \times_G EG_n\right)$                                 | The homology of $LBG^{-TBG}$ is therefore the inverse limit of                                                                                                                                          |
| cardona-qft-methods_F03543 | 353  | ■ 1.000 | $\bigcup_{(g)} T\left(E G_n / C(g)\right)$                                       | $\bigsqcup_{(g)} T\left(EG_n / C(g)\right)$                                   | $\bigoplus_{(g)} H_*\left((BC(g)_n)^{-TBC(g)_n}\right)$ , which is by Atiyah Duality the inverse system of $\mathcal{O}_{H^*(BC(g)_n)}^*$ that is just $\mathcal{O}_{H^*(BC(g))}^*$ . To get the direct |
| cardona-qft-methods_F03544 | 353  | ■ 1.000 | $E H$                                                                            | $EH$                                                                          | $\bigoplus_{(g)} H_*\left((BC(g)_n)^{-TBC(g)_n}\right)$ , which is by Atiyah Duality the inverse system of $\mathcal{O}_{H^*(BC(g)_n)}^*$ that is just $\mathcal{O}_{H^*(BC(g))}^*$ . To get the direct |
| cardona-qft-methods_F03545 | 353  | ■ 1.000 | $H \subseteq G$                                                                  | $H \subseteq G$                                                               | tem of $\bigoplus_{(g)} H^*(BC(g)_n)$ , that is just $\bigoplus_{(g)} H^*(BC(g))$ . To get the ring structure we have to understand the map induced by the correspondence                               |
| cardona-qft-methods_F03546 | 353  | ■ 1.000 | $E G_1 \subseteq \cdots \subseteq E G_n \subseteq E G_{n+1} \subseteq \cdots$    | $EG_1 \subset \cdots \subset EG_n \subset EG_{n+1} \subset \cdots$            | tem of $\bigoplus_{(g)} H^*(BC(g)_n)$ , that is just $\bigoplus_{(g)} H^*(BC(g))$ . To get the ring structure we have to understand the map induced by the correspondence                               |
| cardona-qft-methods_F03547 | 353  | ■ 1.000 | $E C(g)$                                                                         | $EC(g)$                                                                       | $EC(g) = EC(g) \cap EC(h) = EC(gh) = EC(h) \cap EC(g)$                                                                                                                                                  |
| cardona-qft-methods_F03548 | 353  | ■ 1.000 | $\bigoplus_{(g)} H_*\left(\left(BC(g)_n\right)^{-TBC(g)_n}\right)$               | $\bigoplus_{(g)} H_*\left((BC(g)_n)^{-TBC(g)_n}\right)$                       |                                                                                                                                                                                                         |
| cardona-qft-methods_F03549 | 353  | ■ 1.000 | $\bigoplus_{(g)} H^*(BC(g)_n)$                                                   | $\bigoplus_{(g)} H^*(BC(g)_n)$                                                |                                                                                                                                                                                                         |
| cardona-qft-methods_F03550 | 353  | ■ 1.000 | $\bigoplus_{(g)} H^*(BC(g))$                                                     | $\bigoplus_{(g)} H^*(BC(g))$                                                  |                                                                                                                                                                                                         |
| cardona-qft-methods_F03551 | 354  | ■ 1.000 | $k_n = \operatorname{dim}\left(E G_n\right)$                                     | $k_n = \dim\left(EG_n\right)$                                                 | First the Thom-Pontrjagin construction applied to $\Delta$                                                                                                                                              |
| cardona-qft-methods_F03552 | 354  | ■ 0.979 | $H_*^{\mathrm{pro}}\left(\mathbb{L} B G^{-T B G}\right)$                         | $H_*^{\mathrm{pro}}\left(\mathbb{L} B G^{-T B G}\right)$                      | classes in $H^*(BC(g))$ and $H^*(BC(h))$ , pulling them back to $H^*(B(U(g) \cap C(h)))$ and then pushing them forward to $H^*(BC(ah))$ . This procedure is                                             |
| cardona-qft-methods_F03553 | 354  | ■ 1.000 | $H^*(B C(g))$                                                                    | $H^*(BC(g))$                                                                  | what is known as the pull-push formalism. For a finite abelian group, this                                                                                                                              |
| cardona-qft-methods_F03554 | 354  | ■ 1.000 | $H^*(B C(h))$                                                                    | $H^*(BC(h))$                                                                  | what is known as the pull-push formalism. For a finite abelian group, this                                                                                                                              |
| cardona-qft-methods_F03555 | 354  | ■ 1.000 | $H^*(B(C(g) \cap C(h)))$                                                         | $H^*(B(C(g) \cap C(h)))$                                                      | what is known as the pull-push formalism. For a finite abelian group, this                                                                                                                              |
| cardona-qft-methods_F03556 | 354  | ■ 0.835 | $C(h))$                                                                          | $C(h))$                                                                       | is just $H^*(BG) \otimes \mathbb{Z} G$ .<br>For a finite group $G$ theorem 3.5 gives an isomorphism of rings                                                                                            |
| cardona-qft-methods_F03557 | 354  | ■ 0.835 | $H^*(B C(g h))$                                                                  | $H^*(BC(gh))$                                                                 | is just $H^*(BG) \otimes \mathbb{Z} G$ .<br>For a finite group $G$ theorem 3.5 gives an isomorphism of rings                                                                                            |
| cardona-qft-methods_F03558 | 354  | ■ 1.000 | $H^*(B G) \otimes \mathbb{Z} G$                                                  | $H^*(BG) \otimes \mathbb{Z} G$                                                | The reverse                                                                                                                                                                                             |
| cardona-qft-methods_F03559 | 354  | ■ 1.000 | $L B G^{-\mathrm{ad}}$                                                           | $LBG^{-\mathrm{ad}}$                                                          | There is a product on the cohomology of $LBG$ that can be described by a coproduct at the level of spectra defined using a transfer map of                                                              |
| cardona-qft-methods_F03560 | 354  | ■ 1.000 | $L B G$                                                                          | $LBG$                                                                         | the multiplication of the group on the suspension spectrum of $LBG$ . This                                                                                                                              |
| cardona-qft-methods_F03561 | 354  | ■ 1.000 | $L B G \cong E G \times_G G$                                                     | $LBG \cong EG \times_G G$                                                     |                                                                                                                                                                                                         |
| cardona-qft-methods_F03562 | 354  | ■ 1.000 | $\mu: G \times G \rightarrow G$                                                  | $\mu: G \times G \rightarrow G$                                               |                                                                                                                                                                                                         |
| cardona-qft-methods_F03563 | 354  | ■ 1.000 | $m_!: G_+ \rightarrow G \wedge G_+$                                              | $m_!: G_+ \rightarrow G \wedge G_+$                                           |                                                                                                                                                                                                         |
| cardona-qft-methods_F03564 | 354  | ■ 1.000 | $E G \times_G G \rightarrow E G \times_G (G \times G)$                           | $EG \times_G G \rightarrow EG \times_G (G \times G)$                          |                                                                                                                                                                                                         |
| cardona-qft-methods_F03565 | 354  | ■ 1.000 | $E G \times_G (G \times G) \rightarrow (E G \times_G G) \times (E G \times_G G)$ | $EG \times_G (G \times G) \rightarrow (EG \times_G G) \times (EG \times_G G)$ |                                                                                                                                                                                                         |
| cardona-qft-methods_F03566 | 354  | ■ 1.000 | $E G \times_G G \rightarrow$                                                     | $EG \times_G G \rightarrow$                                                   |                                                                                                                                                                                                         |

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| cardona-qft-methods_F03567 | 355  | 0.887 | $\left(E \times_G G \times G\right) \times \left(E \times_G G \times G\right)$                                      | $(EG \times_G G) \times (EG \times_G G)$                                    |                                                                                                                                                                                |
| cardona-qft-methods_F03568 | 355  | 0.989 | $\operatorname{Map}(\Theta, B \times G) \rightarrow L \times B \times G$                                            | $\operatorname{Map}(\Theta, BG) \rightarrow LBG$                            | topically finite fiber Dwyer" has defined a transfer map on homology and cohomology. Using this transfer map we obtain a map $H^*(\operatorname{Map}(\Theta, BG)) \rightarrow$ |
| cardona-qft-methods_F03569 | 355  | 0.989 | $\Omega B \times G$                                                                                                 | $\Omega BG$                                                                 | topically finite fiber Dwyer" has defined a transfer map on homology and cohomology. Using this transfer map we obtain a map $H^*(\operatorname{Map}(\Theta, BG)) \rightarrow$ |
| cardona-qft-methods_F03570 | 355  | 0.927 | $H^*(\operatorname{Map}(\Theta, B \times G)) \rightarrow$                                                           | $H^*(\operatorname{Map}(\Theta, BG)) \rightarrow$                           | $\operatorname{Map}(\Theta, BG) \otimes H^*(LBG) \rightarrow H^*(\operatorname{Map}(\Theta, BG))$                                                                              |
| cardona-qft-methods_F03571 | 355  | 0.751 | $H^*(\operatorname{Map}(\Theta, B \times G)) \rightarrow H^*(L \times B \times G)$                                  | $H^*(\operatorname{Map}(\Theta, BG)) \rightarrow H^*(LBG)$                  | string topology of $[*/G]$ .                                                                                                                                                   |
| cardona-qft-methods_F03572 | 355  | 0.915 | $H^*(L \times B \times G) \otimes H^*(L \times B \times G) \rightarrow H^*(\operatorname{Map}(\Theta, B \times G))$ | $H^*(LBG) \otimes H^*(LBG) \rightarrow H^*(\operatorname{Map}(\Theta, BG))$ |                                                                                                                                                                                |
| cardona-qft-methods_F03573 | 357  | 1.000 | $\mathbb{P}_k^{n-1}$                                                                                                | $\mathbb{P}_k^{n-1}$                                                        | graphs in order to define irreducible hypersurfaces in general.                                                                                                                |
| cardona-qft-methods_F03574 | 358  | 0.999 | $\{ \}^{1,4}$                                                                                                       | $1,4$                                                                       | abelian group generated by isomorphi-                                                                                                                                          |
| cardona-qft-methods_F03575 | 358  | 1.000 | $\mathcal{V}_k$                                                                                                     | $\mathcal{V}_k$                                                             | We start off with a characteristic zero field $k$ and the the category of                                                                                                      |
| cardona-qft-methods_F03576 | 358  | 1.000 | $K_0(\mathcal{V}_k)$                                                                                                | $K_0(\mathcal{V}_k)$                                                        | algebraic varieties $V_k$ defined over it. Then we look at the abelian group $K_0(\mathcal{V}_k)$ of isomorphism classes $[X]$ of varieties $X$ over the field $k$ with the    |
| cardona-qft-methods_F03577 | 358  | 1.000 | $Y \subset X$                                                                                                       | $Y \subset X$                                                               | that the Grothendieck class of the empty set is zero,                                                                                                                          |
| cardona-qft-methods_F03578 | 358  | 1.000 | $X=Y$                                                                                                               | $X = Y$                                                                     | that the Grothendieck class of the empty set is zero,                                                                                                                          |
| cardona-qft-methods_F03579 | 358  | 1.000 | $\chi: \mathcal{V}_k \rightarrow R$                                                                                 | $\chi: \mathcal{V}_k \rightarrow R$                                         | Thus, an additive invariant is the same as a ring homomorphism $\chi$ from the Grothendieck ring of varieties to the ring $R$ . Some interesting addition                      |
| cardona-qft-methods_F03580 | 358  | 1.000 | $\chi(X)=\chi(Y)$                                                                                                   | $\chi(X) = \chi(Y)$                                                         | For the following discussion, let $L = [\mathbb{A}_k^1]$ be                                                                                                                    |
| cardona-qft-methods_F03581 | 358  | 1.000 | $\chi(X \setminus Y)=\chi(X)-\chi(Y)$                                                                               | $\chi(X \setminus Y) = \chi(X) - \chi(Y)$                                   | Grothendieck ring of varieties of the one dimensiona the product (3) of $K_0(\mathcal{V}_k)$ we have that the Grothen                                                          |
| cardona-qft-methods_F03582 | 358  | 1.000 | $\chi(X \times Y)=\chi(X) \chi(Y)$                                                                                  | $\chi(X \times Y) = \chi(X)\chi(Y)$                                         | spaces $\mathbb{A}_k^n$ are given in                                                                                                                                           |
| cardona-qft-methods_F03583 | 358  | 1.000 | $\mathbb{L}=\left[\mathbb{A}_k^1\right]$                                                                            | $\mathbb{L} = [\mathbb{A}_k^1]$                                             |                                                                                                                                                                                |
| cardona-qft-methods_F03584 | 358  | 1.000 | $\mathbb{A}_k^1$                                                                                                    | $\mathbb{A}_k^1$                                                            |                                                                                                                                                                                |
| cardona-qft-methods_F03585 | 358  | 1.000 | $\mathbb{A}_k^n$                                                                                                    | $\mathbb{A}_k^n$                                                            |                                                                                                                                                                                |
| cardona-qft-methods_F03586 | 359  | 1.000 | $k^\times$                                                                                                          | $k^\times$                                                                  |                                                                                                                                                                                |
| cardona-qft-methods_F03587 | 359  | 1.000 | $X \setminus (X \cap Y)=(X \cup Y) \setminus Y$                                                                     | $X \setminus (X \cap Y) = (X \cup Y) \setminus Y$                           | $[X] - [X \cap Y] = [X \setminus (X \cap Y)] = [(X \cup Y) \setminus Y] = [X \cup Y] - [Y]$                                                                                    |
| cardona-qft-methods_F03588 | 359  | 1.000 | $\mathbb{P}_k^{n+1} \setminus \mathbb{A}_k^{n+1} \simeq \mathbb{P}_k^n$                                             | $\mathbb{P}_k^{n+1} \setminus \mathbb{A}_k^{n+1} \simeq \mathbb{P}_k^n$     | <b>3.1. Graph Hypersurfaces</b>                                                                                                                                                |
| cardona-qft-methods_F03589 | 359  | 0.992 | $t_1, t_2, \ldots, t_n$                                                                                             | $t_1, t_2, \ldots, t_n$                                                     |                                                                                                                                                                                |
| cardona-qft-methods_F03590 | 359  | 0.992 | $Z[t_1, t_2, \ldots, t_n]$                                                                                          | $Z[t_1, t_2, \ldots, t_n]$                                                  |                                                                                                                                                                                |
| cardona-qft-methods_F03591 | 360  | 1.000 | $E(T)$                                                                                                              | $E(T)$                                                                      |                                                                                                                                                                                |
| cardona-qft-methods_F03592 | 360  | 1.000 | $v-1$                                                                                                               | $v-1$                                                                       | where $T$ runs through all spanning trees of $\Gamma$ and $E(T)$ is the set of all                                                                                             |
| cardona-qft-methods_F03593 | 360  | 1.000 | $\Psi_\Gamma(t)$                                                                                                    | $\Psi_\Gamma(t)$                                                            | edges of $\Gamma$ . For a graph with $v$ vertices, the pigeon hole principle gives that all                                                                                    |

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| cardona-qft-methods_F03594 | 360  | 1.000 | n-v+1                                                                            | $n - v + 1$                                                                      | edges of $\mathcal{L}$ .<br>For a graph with $v$ vertices, the pigeon hole principle gives that all                                                                                                     |
| cardona-qft-methods_F03595 | 360  | 1.000 | $X_{\Gamma}$                                                                     | $X_{\Gamma}$                                                                     | Since $\Psi_{\Gamma}(t)$ is homogeneous, $X_{\Gamma}$ is well defined as a projective variety.                                                                                                          |
| cardona-qft-methods_F03596 | 360  | 1.000 | $\Gamma^{\vee}$                                                                  | $\Gamma^{\vee}$                                                                  | ing:                                                                                                                                                                                                    |
| cardona-qft-methods_F03597 | 360  | 1.000 | $\mathbb{S}^2$                                                                   | $\mathbb{S}^2$                                                                   | (iii) if two                                                                                                                                                                                            |
| cardona-qft-methods_F03598 | 360  | 0.997 | $\Psi_{\Gamma^{\vee}}(t)$                                                        | $\Psi_{\Gamma^{\vee}}(t)$                                                        |                                                                                                                                                                                                         |
| cardona-qft-methods_F03599 | 360  | 0.820 | $\{ \}^{1,2}$                                                                    | $1,2$                                                                            | $\mathcal{C} : \mathbb{P}_k^{n-1} \rightarrow \mathbb{P}_k^{n-1}$ (3)                                                                                                                                   |
| cardona-qft-methods_F03600 | 360  | 0.820 | $X_{\Gamma^{\vee}}$                                                              | $X_{\Gamma^{\vee}}$                                                              | $\mathcal{C} : \mathbb{P}_k^{n-1} \rightarrow \mathbb{P}_k^{n-1}$ (3)                                                                                                                                   |
| cardona-qft-methods_F03601 | 361  | 1.000 | $\mathbb{P}_k^{n-1} \times \mathbb{P}_k^{n-1}$                                   | $\mathbb{P}_k^{n-1} \times \mathbb{P}_k^{n-1}$                                   |                                                                                                                                                                                                         |
| cardona-qft-methods_F03602 | 361  | 1.000 | $\mathcal{G}(\mathcal{C})$                                                       | $\mathcal{G}(\mathcal{C})$                                                       |                                                                                                                                                                                                         |
| cardona-qft-methods_F03603 | 361  | 1.000 | $\pi_1$                                                                          | $\pi_1$                                                                          | 950 D. Morales-Almaraz                                                                                                                                                                                  |
| cardona-qft-methods_F03604 | 361  | 1.000 | $\pi_2$                                                                          | $\pi_2$                                                                          | 950 D. Morales-Almaraz                                                                                                                                                                                  |
| cardona-qft-methods_F03605 | 361  | 0.902 | $(t_1 : t_2 : \cdots : t_{2n})$                                                  | $(t_1 : t_2 : \cdots : t_{2n})$                                                  | where we used the coordinates $(t_1 : t_2 : \cdots : t_{2n})$ for $\mathbb{P}_k^{n-1} \times \mathbb{P}_k^{n-1}$ .                                                                                      |
| cardona-qft-methods_F03606 | 361  | 1.000 | $\pi_i$                                                                          | $\pi_i$                                                                          |                                                                                                                                                                                                         |
| cardona-qft-methods_F03607 | 361  | 1.000 | $\pi_2 : \mathcal{G}(\mathcal{C}) \rightarrow \mathbb{P}_k^{n-1}$                | $\pi_2 : \mathcal{G}(\mathcal{C}) \rightarrow \mathbb{P}_k^{n-1}$                |                                                                                                                                                                                                         |
| cardona-qft-methods_F03608 | 362  | 1.000 | $\mathcal{G}(C)$                                                                 | $\mathcal{G}(C)$                                                                 | Grothendieck ring class of known and known maps                                                                                                                                                         |
| cardona-qft-methods_F03609 | 362  | 0.998 | $X_{\Gamma^{\vee}}$                                                              | $X_{\Gamma^{\vee}}$                                                              | <b>Proof.</b> Since $X_{\Gamma^{\vee}} = \Sigma_n = \mathbb{P}_k^{n-1} \setminus (k^{\times})^{n-1}$ , the result follows from the                                                                      |
| cardona-qft-methods_F03610 | 362  | 0.998 | $\Sigma_n$                                                                       | $\Sigma_n$                                                                       | <b>Proof.</b> Since $X_{\Gamma^{\vee}} = \Sigma_n = \mathbb{P}_k^{n-1} \setminus (k^{\times})^{n-1}$ , the result follows from the                                                                      |
| cardona-qft-methods_F03611 | 362  | 1.000 | $(k^{\times})^{n-1}$                                                             | $(k^{\times})^{n-1}$                                                             |                                                                                                                                                                                                         |
| cardona-qft-methods_F03612 | 362  | 1.000 | $\mathbb{T}^{n-1}$                                                               | $\mathbb{T}^{n-1}$                                                               |                                                                                                                                                                                                         |
| cardona-qft-methods_F03613 | 362  | 1.000 | $X_{\Gamma^{\vee}} = \Sigma_n = \mathbb{P}_k^{n-1} \setminus (k^{\times})^{n-1}$ | $X_{\Gamma^{\vee}} = \Sigma_n = \mathbb{P}_k^{n-1} \setminus (k^{\times})^{n-1}$ |                                                                                                                                                                                                         |
| cardona-qft-methods_F03614 | 363  | 0.996 | $t_1, t_2, \dots, t_{n-1}$                                                       | $t_1, t_2, \dots, t_{n-1}$                                                       | Notice that choosing any $n-1$ points for $t_1, t_2, \dots, t_{n-1}$ in $\mathcal{L}$ gives a unique value for $t_n$ and hence we can identify $\mathcal{L}$ with $\mathbb{P}^{n-2}$ . Thus we find the |
| cardona-qft-methods_F03615 | 363  | 1.000 | $t_n$                                                                            | $t_n$                                                                            | Grothendieck class of $\mathcal{L}$ to be given by                                                                                                                                                      |
| cardona-qft-methods_F03616 | 363  | 1.000 | $\mathbb{P}_k^{n-2}$                                                             | $\mathbb{P}_k^{n-2}$                                                             | Grothendieck class of $\mathcal{L}$ to be given by                                                                                                                                                      |
| cardona-qft-methods_F03617 | 363  | 0.997 | $(n-1)$                                                                          | $(n-1)$                                                                          | (11)<br>where $\hat{t}_i$ means that the variable $t_i$ is omitted from the product. In order to                                                                                                        |
| cardona-qft-methods_F03618 | 363  | 1.000 | $t_i$                                                                            | $t_i$                                                                            |                                                                                                                                                                                                         |
| cardona-qft-methods_F03619 | 363  | 1.000 | $\hat{t}_i$                                                                      | $\hat{t}_i$                                                                      | a result that tells us how to find the Grothendieck class of a hyperplane                                                                                                                               |

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| cardona-qft-methods_ F03620 | 363  | ■ 1.000 | $X_{\backslash \Gamma} \backslash \Sigma_n$                                              | $X_\Gamma \backslash \Sigma_n$                                                | section of a given class in the Grothendieck ring.                                                                                                                                              |
| cardona-qft-methods_ F03621 | 363  | ■ 0.930 | $X_{\backslash \Gamma^\vee} \backslash \Sigma_n$                                         | $X_{\Gamma^\vee} \backslash \Sigma_n$                                         | <b>Lemma 4.1.</b> $I$                                                                                                                                                                           |
| cardona-qft-methods_ F03622 | 363  | ■ 1.000 | $X_{\backslash \Gamma} \backslash \Sigma$                                                | $X_\Gamma \backslash \Sigma$                                                  | $\mathcal{H}: g(\mathbb{T}) \mapsto \frac{g(\frac{\mathbb{T}+1}{2})-g(\frac{\mathbb{T}-1}{2})}{\mathbb{T}+1} \tag{13}$                                                                          |
| cardona-qft-methods_ F03623 | 363  | ■ 1.000 | $\mathcal{L} \cap \Sigma_n$                                                              | $\mathcal{L} \cap \Sigma_n$                                                   | $\mathcal{H}: g(\mathbb{T}) \mapsto \frac{g(\frac{\mathbb{T}+1}{2})-g(\frac{\mathbb{T}-1}{2})}{\mathbb{T}+1} \tag{13}$                                                                          |
| cardona-qft-methods_ F03624 | 363  | ■ 1.000 | $C=g(\mathbb{T})$                                                                        | $C=g(\mathbb{T})$                                                             |                                                                                                                                                                                                 |
| cardona-qft-methods_ F03625 | 364  | ■ 1.000 | $g(\mathbb{T})=\left[P^n\right]=\frac{(\mathbb{T}-1)^n-1}{\mathbb{T}}$                   | $g(\mathbb{T})=[\mathbb{P}^n]=\frac{(\mathbb{T}-1)^n-1}{\mathbb{T}}$          |                                                                                                                                                                                                 |
| cardona-qft-methods_ F03626 | 364  | ■ 1.000 | $\left[\Sigma_n\right]=\frac{(\mathbb{T}+1)^n-1-\mathbb{T}^n}{\mathbb{T}}=g(\mathbb{T})$ | $[\Sigma_n]=\frac{(\mathbb{T}+1)^n-1-\mathbb{T}^n}{\mathbb{T}}=g(\mathbb{T})$ | Since $[\Sigma_n]=\frac{(\mathbb{T}+1)^n-1-\mathbb{T}^n}{\mathbb{T}}=n(\mathbb{T})$ , using the previous result gives that                                                                      |
| cardona-qft-methods_ F03627 | 364  | ■ 1.000 | $\mathcal{S}_n$                                                                          | $\mathcal{S}_n$                                                               | This is the singularity subvariety of the divisor of singular normal crossings                                                                                                                  |
| cardona-qft-methods_ F03628 | 364  | ■ 1.000 | $\Sigma_{n-1}$                                                                           | $\Sigma_{n-1}$                                                                | given by                                                                                                                                                                                        |
| cardona-qft-methods_ F03629 | 365  | ■ 0.995 | $\left[X_{\backslash \Gamma^\vee}\right]$                                                | $[X_{\Gamma^\vee}]$                                                           |                                                                                                                                                                                                 |
| cardona-qft-methods_ F03630 | 365  | ■ 0.989 | $\left[X_{\backslash \Gamma^\vee} \backslash \Sigma\right]$                              | $[X_{\Gamma^\vee} \backslash \Sigma]$                                         | <b>Proof.</b> Writing $[X_{\Gamma^\vee}]$ as                                                                                                                                                    |
| cardona-qft-methods_ F03631 | 365  | ■ 0.989 | $\left[\mathcal{S}_n\right]$                                                             | $[\mathcal{S}_n]$                                                             | <b>Proof.</b> Writing $[X_{\Gamma^\vee}]$ as                                                                                                                                                    |
| cardona-qft-methods_ F03632 | 366  | ■ 1.000 | $p(t)$                                                                                   | $p(t)$                                                                        | $t_j$ . Since the coefficient of $x$ in $q(x)$ is non-zero for our initial choice of $t_j$ ,                                                                                                    |
| cardona-qft-methods_ F03633 | 366  | ■ 1.000 | $q(t)$                                                                                   | $q(t)$                                                                        | $t_j$ . Since the coefficient of $x$ in $q(x)$ is non-zero for our initial choice of $t_j$ ,                                                                                                    |
| cardona-qft-methods_ F03634 | 366  | ■ 1.000 | $t_i=x$                                                                                  | $t_i=x$                                                                       | $t_j$ . Since the coefficient of $x$ in $q(x)$ is non-zero for our initial choice of $t_j$ ,                                                                                                    |
| cardona-qft-methods_ F03635 | 366  | ■ 1.000 | $t_j, j \neq i$                                                                          | $t_j, j \neq i$                                                               | by continuity of $q_j$ it remains non-zero in a neighborhood of these initial values. Hence the coefficient of $x$ in $p(x)$ is zero for a neighborhood of $t_i$ ,                              |
| cardona-qft-methods_ F03636 | 366  | ■ 1.000 | $\Psi_\Gamma(x)$                                                                         | $\Psi_\Gamma(x)$                                                              | by continuity of $q_j$ it remains non-zero in a neighborhood of these initial values. Hence the coefficient of $x$ in $p(x)$ is zero for a neighborhood of $t_i$ ,                              |
| cardona-qft-methods_ F03637 | 366  | ■ 1.000 | $p(x)$                                                                                   | $p(x)$                                                                        | and since $p(t)$ is a polynomial, $p(x)$ must be constant for all $t_j$ .                                                                                                                       |
| cardona-qft-methods_ F03638 | 366  | ■ 1.000 | $q(x)$                                                                                   | $q(x)$                                                                        | and since $p(t)$ is a polynomial, $p(x)$ must be constant for all $t_j$ .                                                                                                                       |
| cardona-qft-methods_ F03639 | 366  | ■ 1.000 | $t_j$                                                                                    | $t_j$                                                                         | and since $p(t)$ is a polynomial, $p(x)$ must be constant for all $t_j$ .                                                                                                                       |
| cardona-qft-methods_ F03640 | 367  | ■ 1.000 | $P \cap Q = \emptyset, P \cup Q$                                                         | $P \cap Q = \emptyset, P \cup Q$                                              | Since $P \cap Q = \emptyset$ , $P \cup Q$ are all the variables appearing in $\Psi_\Gamma$ , and no                                                                                             |
| cardona-qft-methods_ F03641 | 367  | ■ 1.000 | $\Psi_\Gamma$                                                                            | $\Psi_\Gamma$                                                                 | Since $P \cap Q = \emptyset$ , $P \cup Q$ are all the variables appearing in $\Psi_\Gamma$ , and no                                                                                             |
| cardona-qft-methods_ F03642 | 368  | ■ 1.000 | $\tilde{m} g$                                                                            | $\tilde{m} g$                                                                 | called <i>integrable</i> , and any cyclic vector $\nu$ enables one to see them as the action                                                                                                    |
| cardona-qft-methods_ F03643 | 368  | ■ 0.960 | $\mathbb{I} \pi$                                                                         | $\mathbb{I} \pi$                                                              | of $m g$ by derivations on the algebra of representative functions $g \mapsto \langle \xi, \pi(g) \eta \rangle$ , which are naturally defined on the homogeneous space $M = G/H$ , where $H$ is |
| cardona-qft-methods_ F03644 | 368  | ■ 1.000 | $g \mapsto \langle \xi, \pi(g) \eta \rangle$                                             | $g \mapsto \langle \xi, \pi(g) \eta \rangle$                                  | on a homogeneous manifold, one can show how to use the properties of the universal enveloping algebra $\mathcal{E}(\tilde{m} g)$ to generalize this to representations which                    |
| cardona-qft-methods_ F03645 | 368  | ■ 1.000 | $M = G/H$                                                                                | $M = G/H$                                                                     | are not necessarily integrable. This provides a first step towards a geometric                                                                                                                  |
| cardona-qft-methods_ F03646 | 368  | ■ 1.000 | $\mathcal{E}(\tilde{m} g)$                                                               | $\mathcal{E}(\tilde{m} g)$                                                    |                                                                                                                                                                                                 |

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| cardona-qft-methods_F03647 | 368  | ■ 1.000 | $\mathbb{C}[G]$                                                                                          | $\mathbb{C}[G]$                                                                                               |                                                                                                                                                                                |
| cardona-qft-methods_F03648 | 368  | ■ 1.000 | $g \otimes h \mapsto gh$                                                                                 | $g \otimes h \mapsto gh$                                                                                      |                                                                                                                                                                                |
| cardona-qft-methods_F03649 | 368  | ■ 1.000 | $g \mapsto g^{-1}$                                                                                       | $g \mapsto g^{-1}$                                                                                            |                                                                                                                                                                                |
| cardona-qft-methods_F03650 | 368  | ■ 1.000 | $\bar{g}=g$                                                                                              | $\bar{g} = g$                                                                                                 |                                                                                                                                                                                |
| cardona-qft-methods_F03651 | 368  | ■ 1.000 | $\mathcal{E}=\mathcal{E}(\tilde{m}g)$                                                                    | $\mathcal{E} = \mathcal{E}(\tilde{m}g)$                                                                       |                                                                                                                                                                                |
| cardona-qft-methods_F03652 | 369  | ■ 1.000 | $\mathcal{T}(\tilde{m}g)=\left(\bigoplus_{n\in\mathbb{N}}\tilde{m}g^{\otimes n}\right)\otimes\mathbb{C}$ | $\mathcal{T}(\tilde{m}g) = \left(\bigoplus_{n\in\mathbb{N}} \tilde{m}g^{\otimes n}\right) \otimes \mathbb{C}$ |                                                                                                                                                                                |
| cardona-qft-methods_F03653 | 369  | ■ 1.000 | $\left(X_1\cdots X_n\right)^{\dagger}=(-1)^nX_n\cdots X_1$                                               | $(X_1\cdots X_n)^{\dagger} = (-1)^n X_n\cdots X_1$                                                            | $XY-YX=[X,Y],\quad X,Y\in\tilde{m}g.$                                                                                                                                          |
| cardona-qft-methods_F03654 | 369  | ■ 1.000 | $\overline{X_1\cdots X_n}=X_1\cdots X_n$                                                                 | $\overline{X_1\cdots X_n} = X_1\cdots X_n$                                                                    | It comes equipped with the involution and complex conjugation obtained                                                                                                         |
| cardona-qft-methods_F03655 | 369  | ■ 1.000 | $\mathcal{E}_{\mathbb{R}}=\mathcal{E}(\tilde{m}g)$                                                       | $\mathcal{E}_{\mathbb{R}} = \mathcal{E}_{\mathbb{R}}(\tilde{m}g)$                                             | by linear extension from $(A_1\cdots A_n)^{\dagger}=(-1)^nA_n\cdots A_1$ , and anti-linear extension from $X_1\cdots X_n=X_1\cdots X_n$ , respectively. The subalgebra of real |
| cardona-qft-methods_F03656 | 369  | ■ 1.000 | $\xi\in\mathcal{H}$                                                                                      | $\xi\in\mathcal{H}$                                                                                           | of $\pi(g)\xi$ , and consequently we can also think of $\mathcal{H}$ as a $\mathbb{C}[G]$ -module. In                                                                          |
| cardona-qft-methods_F03657 | 369  | ■ 1.000 | $g\xi$                                                                                                   | $g\xi$                                                                                                        | of $\pi(g)\xi$ , and consequently we can also think of $\mathcal{H}$ as a $\mathbb{C}[G]$ -module. In                                                                          |
| cardona-qft-methods_F03658 | 369  | ■ 1.000 | $\pi(g)\xi$                                                                                              | $\pi(g)\xi$                                                                                                   | what follows we will use interchangeably the languages of representation and module theory.                                                                                    |
| cardona-qft-methods_F03659 | 369  | ■ 1.000 | $C(G)$                                                                                                   | $C(G)$                                                                                                        | $gf(x)=f(g^{-1}x),\quad f\in C(M).$                                                                                                                                            |
| cardona-qft-methods_F03660 | 369  | ■ 1.000 | $\nu\in\mathcal{H}$                                                                                      | $\nu\in\mathcal{H}$                                                                                           | $e_{\xi}:g\in G\mapsto\langle J\xi,g\nu\rangle\in\mathbb{C}.$                                                                                                                  |
| cardona-qft-methods_F03661 | 369  | ■ 0.693 | $\mathcal{H}=\overline{\mathbb{C}[G]\nu}$                                                                | $\mathcal{H} = \overline{\mathbb{C}[G]\nu}$                                                                   | Since $ge_{\xi}=e_{g\xi}$ , we see that $\xi\mapsto e_{\xi}$ exposes $\mathcal{H}$ as a submodule of $C(G)$ .                                                                  |
| cardona-qft-methods_F03662 | 369  | ■ 1.000 | $ge_{\xi}=e_{g\xi}$                                                                                      | $ge_{\xi} = e_{g\xi}$                                                                                         | to modules admitting an anti-unitary $J$ as above.                                                                                                                             |
| cardona-qft-methods_F03663 | 369  | ■ 1.000 | $\xi\mapsto e_{\xi}$                                                                                     | $\xi\mapsto e_{\xi}$                                                                                          | to modules admitting an anti-unitary $J$ as above.                                                                                                                             |
| cardona-qft-methods_F03664 | 369  | ■ 1.000 | $g\mapsto\langle g\nu,\xi\rangle$                                                                        | $g\mapsto\langle g\nu,\xi\rangle$                                                                             |                                                                                                                                                                                |
| cardona-qft-methods_F03665 | 369  | ■ 0.996 | $\{e_{\xi}\mid\xi\in\mathcal{H}\}$                                                                       | $\{e_{\xi}\mid\xi\in\mathcal{H}\}$                                                                            |                                                                                                                                                                                |
| cardona-qft-methods_F03666 | 369  | ■ 1.000 | $e_{\xi}$                                                                                                | $e_{\xi}$                                                                                                     |                                                                                                                                                                                |
| cardona-qft-methods_F03667 | 369  | ■ 1.000 | $gH$                                                                                                     | $gH$                                                                                                          |                                                                                                                                                                                |
| cardona-qft-methods_F03668 | 370  | ■ 1.000 | $\mathcal{H}^{\infty}=\mathcal{H}^{\infty}(\pi)\subseteq\mathcal{H}$                                     | $\mathcal{H}^{\infty} = \mathcal{H}^{\infty}(\pi) \subseteq \mathcal{H}$                                      |                                                                                                                                                                                |
| cardona-qft-methods_F03669 | 370  | ■ 0.964 | $g\in G\mapsto\pi(g)\xi\in\mathcal{H}$                                                                   | $g\in G\mapsto\pi(g)\xi\in\mathcal{H}$                                                                        |                                                                                                                                                                                |
| cardona-qft-methods_F03670 | 370  | ■ 1.000 | $X\in\tilde{m}g$                                                                                         | $X\in\tilde{m}g$                                                                                              |                                                                                                                                                                                |
| cardona-qft-methods_F03671 | 370  | ■ 1.000 | $\xi\in\mathcal{H}^{\infty}$                                                                             | $\xi\in\mathcal{H}^{\infty}$                                                                                  |                                                                                                                                                                                |
| cardona-qft-methods_F03672 | 370  | ■ 1.000 | $\exp:\tilde{m}g\rightarrow G$                                                                           | $\exp:\tilde{m}g\rightarrow G$                                                                                | $\mathbb{D}\pi(X)\xi=X\xi=\lim_{t\rightarrow 0}\frac{1}{t}\left(\exp(tX)\xi-\xi\right),$                                                                                       |

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| cardona-qft-methods_F03673 | 370  | ■ 1.000 | $\mathbb{D}\pi$                                                     | $\mathbb{D}\pi$                                                     | where $\exp : \mathfrak{m}g \rightarrow G$ is the exponential map. This way, the unitary repre-                                                                                                              |
| cardona-qft-methods_F03674 | 370  | ■ 1.000 | $\mathcal{H}^{\infty}$                                              | $\mathcal{H}^{\infty}$                                              | sentation $\pi$ of $G$ gives rise to a representation $\mathbb{D}\pi$ of $\mathfrak{m}g$ by unbounded, closed self-adjoint operators. These operators can be seen as $\mathfrak{C}^*$ -modules. We           |
| cardona-qft-methods_F03675 | 370  | ■ 1.000 | $\left\{e_{\xi} \mid \xi \in \mathcal{H}^{\infty}\right\}$          | $\{e_{\xi} \mid \xi \in \mathcal{H}^{\infty}\}$                     | it with $\chi \in \mathfrak{g} \mid \xi \in \mathcal{H}^{\infty}$ . Now, the action of $\mathfrak{m}g$ is by derivation along the vector fields which generate the flows $gH \in M \mapsto e^{tX}gH \in M$ . |
| cardona-qft-methods_F03676 | 370  | ■ 1.000 | $gH \in M \mapsto \epsilon^{\epsilon^t X}gH \in M$                  | $gH \in M \mapsto \epsilon^{tX}gH \in M$                            | It will be helpful to be more precise regarding the notion of representa-                                                                                                                                    |
| cardona-qft-methods_F03677 | 370  | ■ 1.000 | $\left.X \mapsto \mathbb{D}\pi(X)\right _{\mathcal{F}}$             | $X \mapsto \mathbb{D}\pi(X) _{\mathcal{F}}$                         | <b>Remark 1.2.</b> Recall that an unbounded operator is <i>closed</i> if its graph                                                                                                                           |
| cardona-qft-methods_F03678 | 370  | ■ 1.000 | $\mathcal{F} \subseteq \mathcal{H}^{\infty}(\pi)$                   | $\mathcal{F} \subseteq \mathcal{H}^{\infty}(\pi)$                   | <b>Remark 1.2.</b> Recall that an unbounded operator is <i>closed</i> if its graph                                                                                                                           |
| cardona-qft-methods_F03679 | 370  | ■ 0.998 | $\left.\mathbb{D}\pi(X)\right _{\mathcal{F}}$                       | $\mathbb{D}\pi(X) _{\mathcal{F}}$                                   | is closed, and is <i>closable</i> if the closure of its graph is the graph of another                                                                                                                        |
| cardona-qft-methods_F03680 | 370  | ■ 1.000 | $\mathbb{D}\pi(X)$                                                  | $\mathbb{D}\pi(X)$                                                  | operator, called its closure. An operator is <i>closable</i> if and only if its graph is densely defined; thus, the operators $\mathbb{D}\pi$                                                                |
| cardona-qft-methods_F03681 | 370  | ■ 1.000 | $\left.\mathbb{D}\pi(X)\right _{\mathcal{F}}$                       | $\mathbb{D}\pi(X) _{\mathcal{F}}$                                   | Let $\mathcal{F}$ be a complex vector space sitting densely inside a Hilbert space                                                                                                                           |
| cardona-qft-methods_F03682 | 370  | ■ 1.000 | $\mathfrak{a}\mathcal{H}^{\infty}$                                  | $\mathfrak{a}\mathcal{H}^{\infty}$                                  |                                                                                                                                                                                                              |
| cardona-qft-methods_F03683 | 371  | ■ 1.000 | $\mathcal{L}^{\dagger}(\mathcal{F})$                                | $\mathcal{L}^{\dagger}(\mathcal{F})$                                |                                                                                                                                                                                                              |
| cardona-qft-methods_F03684 | 371  | ■ 1.000 | $\mathcal{F} \rightarrow \mathcal{F}$                               | $\mathcal{F} \rightarrow \mathcal{F}$                               | 360 R Varma Le-Bert                                                                                                                                                                                          |
| cardona-qft-methods_F03685 | 371  | ■ 1.000 | $\mathcal{F} \subseteq \operatorname{dom}(A^*)$                     | $\mathcal{F} \subseteq \operatorname{dom}(A^*)$                     | 360 R Varma Le-Bert                                                                                                                                                                                          |
| cardona-qft-methods_F03686 | 371  | ■ 1.000 | $A^{\dagger} = A^* _{\mathcal{F}}$                                  | $A^{\dagger} = A^* _{\mathcal{F}}$                                  |                                                                                                                                                                                                              |
| cardona-qft-methods_F03687 | 371  | ■ 1.000 | $\mathcal{C}^*(\mathcal{F})$                                        | $\mathcal{C}^*(\mathcal{F})$                                        | $\mathcal{F} \rightarrow \mathcal{F}$ such that $\mathcal{F} \subseteq \operatorname{dom}(A^*)$ . It comes equipped with the involution $A^{\dagger} = A^* _{\mathcal{F}}$ .                                 |
| cardona-qft-methods_F03688 | 371  | ■ 1.000 | $A _{\mathcal{F}} \in \mathcal{L}^{\dagger}(\mathcal{F})$           | $A _{\mathcal{F}} \in \mathcal{L}^{\dagger}(\mathcal{F})$           | $\mathcal{C}^*(\mathcal{F})$ is the $C^*$ -algebra generated by $\{A _{\mathcal{F}} \mid A \in \mathcal{A}\}$ .                                                                                              |
| cardona-qft-methods_F03689 | 371  | ■ 1.000 | $\pi : \mathcal{A} \rightarrow \mathcal{L}^{\dagger}(\mathcal{F})$  | $\pi : \mathcal{A} \rightarrow \mathcal{L}^{\dagger}(\mathcal{F})$  | Recall that there is a natural order defined on any involutive algebra $\mathcal{A}$ by $A \leq B$ if $B - A$ is a sum of squares.                                                                           |
| cardona-qft-methods_F03690 | 371  | ■ 1.000 | $\nu \in \mathcal{F}$                                               | $\nu \in \mathcal{F}$                                               | normalized by $\omega(1) = 1$ . We have introduced all this no                                                                                                                                               |
| cardona-qft-methods_F03691 | 371  | ■ 1.000 | $\mathcal{F} = \mathcal{A}\nu$                                      | $\mathcal{F} = \mathcal{A}\nu$                                      | normalized by $\omega(1) = 1$ . We have introduced all this no                                                                                                                                               |
| cardona-qft-methods_F03692 | 371  | ■ 1.000 | $A^{\dagger}A$                                                      | $A^{\dagger}A$                                                      | any vector $\nu \in \mathcal{F}$ with $\ \nu\  = 1$ by the formula $\omega(A) = \langle A\nu, \nu \rangle = \langle A^{\dagger}A\nu, \nu \rangle = \langle A^{\dagger}A, 1 \rangle$ .                        |
| cardona-qft-methods_F03693 | 371  | ■ 1.000 | $\mathcal{A}^{\dagger}$                                             | $\mathcal{A}^{\dagger}$                                             | The GNS construction allows for a recovery of both $\pi _{\mathcal{A}\nu}$ (modulo unitary conjugation) and $\nu$ (modulo a phase factor) from $\omega$ , thus establishing the                              |
| cardona-qft-methods_F03694 | 371  | ■ 1.000 | $\omega \in \mathcal{A}^{\dagger}$                                  | $\omega \in \mathcal{A}^{\dagger}$                                  | The GNS construction allows for a recovery of both $\pi _{\mathcal{A}\nu}$ (modulo unitary conjugation) and $\nu$ (modulo a phase factor) from $\omega$ , thus establishing the                              |
| cardona-qft-methods_F03695 | 371  | ■ 1.000 | $\ \nu\  = 1$                                                       | $\ \nu\  = 1$                                                       |                                                                                                                                                                                                              |
| cardona-qft-methods_F03696 | 371  | ■ 0.999 | $\left.\pi\right _{\mathcal{A}\nu}$                                 | $\pi _{\mathcal{A}\nu}$                                             |                                                                                                                                                                                                              |
| cardona-qft-methods_F03697 | 372  | ■ 1.000 | $\mathcal{I} = \{A \in \mathcal{A} \mid \omega(A^{\dagger}A) = 0\}$ | $\mathcal{I} = \{A \in \mathcal{A} \mid \omega(A^{\dagger}A) = 0\}$ |                                                                                                                                                                                                              |
| cardona-qft-methods_F03698 | 372  | ■ 1.000 | $\mathcal{I}$                                                       | $\mathcal{I}$                                                       | On the commutator subalgebra and the states vanishing on $\mathcal{I}$                                                                                                                                       |
| cardona-qft-methods_F03699 | 372  | ■ 1.000 | $\mathcal{F} = \mathcal{A}/\mathcal{I}$                             | $\mathcal{F} = \mathcal{A}/\mathcal{I}$                             | One can guarantee that every $\omega$ has a unique representative $\omega _{\mathcal{F}}$ .                                                                                                                  |



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| cardona-qft-methods_F03726 | 374  | ■ 1.000 | $E, F \in \mathcal{E}$                                           | $E, F \in \mathcal{E}$                                           | The main properties of $\mathcal{E}^\dagger$ are given in the next two propositions.                                                 |
| cardona-qft-methods_F03727 | 374  | ■ 1.000 | $\tilde{m}g \rightarrow \operatorname{der}(\mathcal{E}^\dagger)$ | $\tilde{m}g \rightarrow \operatorname{der}(\mathcal{E}^\dagger)$ | On the other hand, given $E \in \mathcal{E}$ ,                                                                                       |
| cardona-qft-methods_F03728 | 374  | ■ 0.999 | $\operatorname{der}(\mathcal{E}^\dagger)$                        | $\operatorname{der}(\mathcal{E}^\dagger)$                        | $\Delta(X^\dagger E) = \Delta(X^\dagger)\Delta(E) = (X^\dagger \otimes 1 + 1 \otimes X^\dagger)\Delta(E)$                            |
| cardona-qft-methods_F03729 | 374  | ■ 0.999 | $\delta: \mathcal{E}^\dagger \rightarrow \mathcal{E}^\dagger$    | $\delta: \mathcal{E}^\dagger \rightarrow \mathcal{E}^\dagger$    | $\Delta(X^\dagger E) = \Delta(X^\dagger)\Delta(E) = (X^\dagger \otimes 1 + 1 \otimes X^\dagger)\Delta(E)$                            |
| cardona-qft-methods_F03730 | 374  | ■ 1.000 | $a, b \in \mathcal{E}^\dagger$                                   | $a, b \in \mathcal{E}^\dagger$                                   |                                                                                                                                      |
| cardona-qft-methods_F03731 | 375  | ■ 1.000 | $X_1, \ldots, X_n$                                               | $X_1, \ldots, X_n$                                               | $= (Xa \otimes b + a \otimes Xb) \circ \Delta = (Xa)b + a(Xb). \quad \square$                                                        |
| cardona-qft-methods_F03732 | 375  | ■ 1.000 | $\mathcal{E}^\dagger \cong$                                      | $\mathcal{E}^\dagger \cong$                                      | $= (Xa \otimes b + a \otimes Xb) \circ \Delta = (Xa)b + a(Xb). \quad \square$                                                        |
| cardona-qft-methods_F03733 | 375  | ■ 1.000 | $\mathbb{C}[\![x_1, \ldots, x_n]\!]$                             | $\mathbb{C}[\![x_1, \ldots, x_n]\!]$                             | <b>Prop 2.2.</b> Let $X_1, \ldots, X_n$ be a basis of $\tilde{m}g$ . On                                                              |
| cardona-qft-methods_F03734 | 375  | ■ 1.000 | $1 \in \mathcal{E}$                                              | $1 \in \mathcal{E}$                                              | $a^\dagger(E + \int F) = \overline{a(E)} + \int \overline{a(F)}, \quad E, F \in \mathcal{E}_{\mathbb{R}}.$                           |
| cardona-qft-methods_F03735 | 375  | ■ 1.000 | $\chi_0$                                                         | $\chi_0$                                                         | $a^\dagger(E + \int F) = \overline{a(E)} + \int \overline{a(F)}, \quad E, F \in \mathcal{E}_{\mathbb{R}}.$                           |
| cardona-qft-methods_F03736 | 375  | ■ 1.000 | $a \in \mathcal{E}^\dagger$                                      | $a \in \mathcal{E}^\dagger$                                      |                                                                                                                                      |
| cardona-qft-methods_F03737 | 375  | ■ 1.000 | $E \in \mathcal{E}, (Ea)^\dagger = \bar{E}a^\dagger$             | $E \in \mathcal{E}, (Ea)^\dagger = \bar{E}a^\dagger$             |                                                                                                                                      |
| cardona-qft-methods_F03738 | 375  | ■ 1.000 | $E, F \in \mathcal{E}_{\mathbb{R}}$                              | $E, F \in \mathcal{E}_{\mathbb{R}}$                              |                                                                                                                                      |
| cardona-qft-methods_F03739 | 375  | ■ 1.000 | $(Ea)^\dagger(F) = \overline{a(E^\dagger F)} = Ea^\dagger(F)$    | $(Ea)^\dagger(F) = \overline{a(E^\dagger F)} = Ea^\dagger(F)$    |                                                                                                                                      |
| cardona-qft-methods_F03740 | 376  | ■ 1.000 | $J\nu = \nu$                                                     | $J\nu = \nu$                                                     | vector $\nu \in \mathcal{F}$ , where $\tilde{m}g$ is a real Lie algebra and $\mathcal{F}$ is a dense subspace                        |
| cardona-qft-methods_F03741 | 376  | ■ 1.000 | $\{e_\xi \mid \xi \in \mathcal{F}\}$                             | $\{e_\xi \mid \xi \in \mathcal{F}\}$                             | $M = \int_{\mathbb{R}} \nu \in M^\dagger \mid \nu(a\bar{b}) = \nu(a)\nu(b), \quad \nu(a^\dagger) = \overline{\nu(a)}, \quad \forall$ |
| cardona-qft-methods_F03742 | 376  | ■ 1.000 | $\chi_0 \in M$                                                   | $\chi_0 \in M$                                                   | not smooth                                                                                                                           |
| cardona-qft-methods_F03743 | 376  | ■ 1.000 | $\xi \in \mathcal{H} \backslash \mathcal{F}$                     | $\xi \in \mathcal{H} \backslash \mathcal{F}$                     | for $\mathcal{A}$ is not necessarily finitely generated.                                                                             |
| cardona-qft-methods_F03744 | 376  | ■ 1.000 | $E \in \mathcal{E}, Ee_\xi$                                      | $E \in \mathcal{E}, Ee_\xi$                                      | <b>Prop 2.4.</b> One has that $Ee_\xi = e_{E\xi}$ , for all $\xi \in \mathcal{F}$ and $E \in \mathcal{E}$ . In particular,           |
| cardona-qft-methods_F03745 | 376  | ■ 1.000 | $e_\eta$                                                         | $e_\eta$                                                         | $\mathcal{A}$ is a sub                                                                                                               |
| cardona-qft-methods_F03746 | 376  | ■ 1.000 | $\eta \in \mathcal{H}$                                           | $\eta \in \mathcal{H}$                                           | $\mathcal{A}$ is a sub                                                                                                               |
| cardona-qft-methods_F03747 | 376  | ■ 1.000 | $Ee_\xi = e_{E\xi}$                                              | $Ee_\xi = e_{E\xi}$                                              | algebra of $\mathcal{E}^\dagger$ generated by the orbit of the state                                                                 |
| cardona-qft-methods_F03748 | 376  | ■ 1.000 | $\xi \in \mathcal{F}$                                            | $\xi \in \mathcal{F}$                                            | algebra of $\mathcal{E}^\dagger$ generated by the orbit of the state                                                                 |
| cardona-qft-methods_F03749 | 376  | ■ 1.000 | $F \in \mathcal{E}$                                              | $F \in \mathcal{E}$                                              | $\mathcal{A}$                                                                                                                        |
| cardona-qft-methods_F03750 | 376  | ■ 1.000 | $\omega \in \mathcal{E}^\dagger$                                 | $\omega \in \mathcal{E}^\dagger$                                 |                                                                                                                                      |
| cardona-qft-methods_F03751 | 377  | ■ 1.000 | $\mathcal{E}^\dagger \cong \mathbb{C}[\![x_1, \ldots, x_n]\!]$   | $\mathcal{E}^\dagger \cong \mathbb{C}[\![x_1, \ldots, x_n]\!]$   | 306 R. Vargas Le-Bert                                                                                                                |

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| cardona-qft-methods_F03752 | 377  | ■ 1.000 | $\chi_{\{0\}}(a)=a(1)$                                        | $\chi_0(a) = a(1)$                                                | <p>recall that the elements of <math>\mathcal{A}</math> can be seen as formal power series around <math>\chi_0</math>, where <math>\chi_0(a) = a(1)</math>. Here, we have only the derivation <math>X</math>, which will be,</p> |
| cardona-qft-methods_F03753 | 377  | ■ 1.000 | $\omega \cong \cos (x)$                                       | $\omega \cong \cos(x)$                                            | <p>In the previous example the algebra <math>\mathcal{A}</math> was finitely generated, but non-</p>                                                                                                                             |
| cardona-qft-methods_F03754 | 377  | ■ 1.000 | $X \, \omega$                                                 | $X\omega$                                                         | <p>integrable representations are necessarily infinite dimensional. Thus, an important problem is to find an adequate topology for it. Then, assuming a</p>                                                                      |
| cardona-qft-methods_F03755 | 377  | ■ 1.000 | $\mathcal{A} \cong \mathbb{C}[u, v] /\left(u^2+v^2-1 \right)$ | $\mathcal{A} \cong \mathbb{C}[u, v] / \left(u^2 + v^2 - 1\right)$ | <p>suitable topology has been defined, an obvious question a</p>                                                                                                                                                                 |